Lattice Green functions and the Bałaban variational problem

Exponential decay, regularity and the background field uniformly in the number of renormalization steps, and the small-field renormalization group for three-dimensional Yang–Mills theory

First version, October 2026

Abstract

In Bałaban's renormalization group approach to lattice Yang–Mills theory, every step requires the minimizer of the Wilson action subject to prescribed block averages (the background field), and all estimates rest on the exponential decay and the local regularity of a small family of linear operators: Green functions with an averaging term, the minimizer map, the constrained (fluctuation) Green function, and the inverses of averaged Green functions. We give a self-contained treatment of these operators and of the variational problem on a single torus, after an arbitrary number k of block-averaging steps, with bounds uniform in k and in the size of the torus. On this basis we carry out the small-field renormalization group for pure Yang–Mills theory in three dimensions.

Part I treats the linearization at the trivial background $U = 1$. It covers the scalar operators, which enter through gauge transformations and gauge fixing, and the gauge-field operators in Bałaban's Landau-type gauge. The proofs use Poincaré-type coercivity estimates, the Combes–Thomas method, local elliptic regularity on the fine lattice, and a single elementary Fourier lower bound; the coercivity of the gauge-field operator is proved in position space. Part II extends the results by perturbation to complex backgrounds that are close to 1 in a global gauge, including Bałaban's gauge-invariant extension via the slice projection. Part III replaces the global hypothesis by Bałaban's local regularity. It constructs Dirichlet operators on unions of blocks, proves coercivity by an IMS localization that reduces it to the trivial background, proves locality in the domain and in the background, and establishes Bałaban's random-walk expansion for the scalar Green function; for the gauge-field Green function the expansion is completed in factorized form in Part IX.

Part IV constructs Bałaban's nonlinear averaging, the multiscale axial gauge and the gauge-adapted averaging. It shows that the linearized averaging is exactly gauge covariant and satisfies the hypotheses of Parts II and III, and it proves Bałaban's axial-to-Landau gauge-fixing theorem. Part V solves the variational problem relative to a regular reference configuration: existence, uniqueness and minimality of the background field, second-order and Hölder bounds, the Euler–Lagrange equation, analyticity in the coarse field, and exponential decay of the derivative. Part VI carries out the induction over the number of steps. The regularity constants are kept independent of k by a local improvement: the relative theorem is applied once more, with a reference configuration that is spliced to the trivial one near a given point. This proves Bałaban's variational theorem on the torus, uniformly in k, without nested domains. As a consequence, every real configuration with small curvature and current is locally regular, so the hypothesis of Part III follows from gauge-invariant conditions. Part VII begins the small-field renormalization group for pure Yang–Mills theory in three dimensions. It writes one renormalization group step exactly as a Gaussian integral around the background field, and proves the linear input for its analysis: coercivity and decay of the fluctuation covariance, analyticity and tree decay of the coarse effective action, local determinant densities, and a bound on the curvature of background fields with a constant independent of the blocking factor. Part VIII localizes

the dependence on the background. With Dirichlet approximants and telescoping, the fluctuation covariance, the coarse Hessian and the Gaussian determinant become sums of local, gauge-invariant pieces with exponential decay. The determinant becomes a constant plus normalized activities of second order in the curvature. The nonlinear interaction is localized in the same way. This uses interior pointwise bounds with derivatives for the Dirichlet operators, proved by an absorption argument at all scales for a local elliptic system obtained by treating two scalar potentials of the gauge-fixing projection as unknowns. This localization is of radius type. Part IX constructs random walks with weakening parameters for all operators of the step, in L^2 norms, giving the volume-type localization needed for polymer expansions of the covariance, the coarse Hessian and the determinant. It identifies the pointwise control of the weakened operators as the remaining input for the background map, proves interior estimates with L^p data for the Dirichlet propagators, and with them pointwise $C^{1,\beta}$ bounds for the weakened propagators $\mathbf{G}(s)$, $\mathbf{H}(s)$ on the torus, and for $\mathbf{C}(s)$, using a lattice trace inequality on boxes. These are the linear inputs of the weakened background map, which is constructed as the fixed point of a weakened contraction, analytic in the weakening parameters and factorizing over polymers.

Part X carries out the small-field renormalization group. The activities must be analytic in the background, so the results of Part IX, and Bałaban's multiscale averaged backgrounds of Part IV, are first extended to complex references. The ingredients of one step become weakened step quantities, and the fluctuation integral, with a cutoff on the white noise, is evaluated by a Mayer and polymer expansion. An activity norm on a graded complex domain makes normalized gauge-invariant activities irrelevant in the sense of power counting: by complex gauge invariance and semisimplicity, on the domain of the next scale such an activity is smaller than its supremum by a factor quadratic in the ratio of the curvature, the current and the complex part to the size of the domain. This gives the bound for one step of the flow. The iteration, with a vacuum energy counterterm, gives the small-field flow (Theorem K): for semisimple G in three dimensions, the effective densities keep their form at every scale, with activities of order $(p_k g_k)^{2-2\epsilon}$, where g_k is the dimensionless coupling at scale k, p_k grows logarithmically in g_k^{-1}, and $0 < \epsilon < \frac{1}{4}$. The partition function of the small-field model is an explicit product of normalizations times a factor between $e^{-c|\mathbb{T}|}$ and $e^{c|\mathbb{T}|}$ on the final lattice $\mathbb{T}$, with c independent of the lattice spacing. The large-field stage is not treated.

Contents

1 Introduction **8**
- 1.1 Where the operators come from . . . 8
- 1.2 What this paper does . . . 9
- 1.3 Main results . . . 9
- 1.4 Methods . . . 15
- 1.5 What is not done here . . . 16

Part I: The trivial background **16**

2 Lattices, fields and averaging operators **16**
- 2.1 Lattices and blocks . . . 17
- 2.2 Fields, derivatives and the Hodge identity . . . 17
- 2.3 Directional convolutions . . . 18
- 2.4 Averaging operators . . . 20
- 2.5 Restriction of product convolutions . . . 21

3 Poincaré inequalities and scalar coercivity **21**

4 Two Fourier estimates **23**
4.1 Fourier transform and Bloch decomposition . . . 23
4.2 The lower bound . . . 24
4.3 Biharmonic extensions . . . 24

5 Operators with exponential decay and the Combes–Thomas method **25**
5.1 Block decay . . . 25
5.2 Conjugation by exponential weights . . . 26
5.3 The Combes–Thomas method . . . 26

6 Local regularity on the fine lattice **28**
6.1 The Green function of the fine lattice . . . 28
6.2 The regularity estimate . . . 28

7 Scalar Green functions **32**

8 Averaged scalar operators, minimizers and the projection R **33**
8.1 The averaged Green function and its inverse . . . 33
8.2 The minimizer map . . . 34
8.3 The effective action . . . 34
8.4 The constrained Green function . . . 35
8.5 The projection R . . . 35

9 Gauge fields: Bałaban's gauge and coercivity **36**
9.1 The quadratic action and its gauge invariance . . . 36
9.2 Bałaban's gauge . . . 36
9.3 The averaging nearly separates exact and co-closed fields . . . 37
9.4 Coercivity . . . 38

10 The gauge-field Green function **39**

11 The gauge-field minimizer and the fluctuation operators **40**
11.1 Identities . . . 40
11.2 The averaged gauge Green function . . . 40
11.3 The minimizer map . . . 40
11.4 The gauge-field effective action . . . 41
11.5 The constrained Green function . . . 41

Part II: Backgrounds **43**

12 Background fields: setting and hypotheses **43**
12.1 Lie algebra valued fields . . . 44
12.2 Backgrounds and the hypothesis (B) . . . 44
12.3 Covariant derivatives and curvature . . . 45
12.4 The quadratic form of the action . . . 46
12.5 Averaging operators with a background . . . 47

13 Perturbation lemmas **48**
13.1 Complex fields . . . 48
13.2 Admissible perturbations . . . 48
13.3 Pointwise admissibility and the bootstrap . . . 49

13.4 Building blocks . . . 50

14 Scalar operators with a background **50**
14.1 The gauge-fixing projection . . . 52

15 The gauge-field Green function with a background **52**
15.1 The slice projection . . . 52
15.2 The operator and its inverse . . . 53

16 Minimizer and fluctuation operators with a background **55**
16.1 Analyticity and gauge covariance . . . 56
16.2 What the variational problem needs . . . 57

Part III: Domains, local regularity and random walks **58**

17 Domains, partitions of unity and two general lemmas **58**
17.1 Domains and field spaces . . . 59
17.2 Partitions of unity at scale M . . . 59
17.3 An algebraic comparison formula . . . 61

18 Regular backgrounds and scalar operators on domains **61**
18.1 Local regularity . . . 61
18.2 Scalar operators on domains . . . 63

19 Gauge-field operators at regular backgrounds **66**
19.1 Local potential terms and operators on domains . . . 66
19.2 Coercivity from local information . . . 67
19.3 Green functions at regular backgrounds . . . 69
19.4 The averaged Green function, the minimizer and the fluctuation operator . . . 70
19.5 Comparison of domains and locality in the background . . . 72
19.6 Complex perturbations . . . 74

20 Bałaban's random-walk expansion **75**
20.1 The parametrix . . . 75

Part IV: Nonlinear averaging, axial gauge and the Landau chart **78**

21 Configurations and the nonlinear averaging **78**
21.1 Scales and contours . . . 78
21.2 Configurations, curvature and lattice Stokes . . . 79
21.3 The one-step averaging . . . 80
21.4 Regularity of averaged configurations . . . 81

22 The axial gauge and the gauge-adapted averaging **82**
22.1 The reference configuration and relative configurations . . . 82
22.2 The axial gauge . . . 82
22.3 The gauge-adapted averaging and the correspondence theorem . . . 83
22.4 Expansion of the modified average . . . 84
22.5 Expansion of the gauge-adapted averaging . . . 86
22.6 The linearized averages . . . 88

23 The Wilson action near a regular configuration **89**
23.1 The action and the current . . . 89
23.2 Expansion . . . 90

24 The axial-to-Landau chart **91**
24.1 Linear input . . . 92
24.2 The theorem . . . 92
24.3 Axial configurations are small . . . 93
24.4 The a priori bound . . . 94
24.5 One scale of gauge transformation . . . 94
24.6 Proof of Theorem 24.2 . . . 96

Part V: The variational problem **97**

25 The variational theorem **97**
25.1 The problem . . . 97
25.2 Statement . . . 97
25.3 Reduction to the Landau chart . . . 98

26 Solution in the Landau chart **99**
26.1 Linearizing the constraint . . . 99
26.2 The functional . . . 100
26.3 The equation and its solution . . . 101
26.4 Convexity and minimality . . . 102

27 The Euler–Lagrange equation and regularity of the minimizer **104**
27.1 Covariance of the gauge-adapted averaging . . . 104
27.2 The Euler–Lagrange equation . . . 105
27.3 The constructed point is critical . . . 105
27.4 Second-order bounds and regularity . . . 105
27.5 Analyticity and decay . . . 107
27.6 Proof of Theorem 25.1 . . . 108

Part VI: The induction over the number of steps **108**

28 Local gauges and splicing **108**
28.1 Regularity on a box . . . 109
28.2 Gauges on large cubes . . . 109
28.3 A gauge adapted to the coarse axial gauge . . . 111
28.4 Splicing . . . 112

29 Localized bounds and the local improvement **113**
29.1 Localized bounds for the relative problem . . . 113
29.2 The local improvement . . . 115

30 The induction over the number of steps **117**
30.1 The trivial extension . . . 117
30.2 Change of units . . . 117
30.3 The theorem . . . 118
30.4 Consequences . . . 119

Part VII: One step of the renormalization group **120**

31 The renormalization group transformation **120**
31.1 The model and the transformation . . . 121
31.2 Curvature of background fields . . . 122
31.3 The background of a step . . . 127

32 Gauge fixing and fluctuation coordinates **128**
32.1 The one-step axial gauge . . . 128
32.2 Elimination of the constraint . . . 129
32.3 The exact formula . . . 130

33 The coarse effective action **132**
33.1 Gradient and decay of derivatives . . . 132
33.2 The Hessian at the base configuration . . . 134
33.3 Dependence on the base configuration . . . 135

34 The fluctuation covariance and the step **135**
34.1 The quadratic form of the step . . . 135
34.2 Coercivity . . . 136
34.3 The fluctuation covariance . . . 138
34.4 The determinant . . . 139
34.5 One step as a Gaussian integral . . . 140

Part VIII: Localization of the background dependence **142**

35 Dirichlet operators for the averaged problem **142**
35.1 Definitions . . . 142
35.2 Bounds . . . 143
35.3 Interior pointwise bounds . . . 145

36 Interior regularity for Dirichlet operators **146**
36.1 Absorption at all scales . . . 146
36.2 The local system . . . 148
36.3 A consequence of gauge invariance . . . 150
36.4 Proof of Theorem 35.2 . . . 151

37 Localization of the Hessian and of the fluctuation covariance **153**
37.1 The telescoping principle . . . 153
37.2 The coarse Hessian as an operator of Part II . . . 154
37.3 Dirichlet Hessians . . . 154
37.4 The fluctuation operator on domains . . . 155
37.5 Local expansion of the covariance . . . 157

38 Localization and normalization of the determinant **158**
38.1 Local densities . . . 158
38.2 Normalized background activities . . . 159

39 Localization of the background map and of the interaction **165**
39.1 Comparison of fixed points . . . 165
39.2 Dirichlet background maps . . . 165
39.3 Localization of the interaction . . . 168
39.4 Summary . . . 169

Part IX: Weakened random walks **170**

40 Weakenable families **170**
40.1 Decay norms . . . 171
40.2 Weakenable families . . . 172

41 Weakened inverses by nested walks **174**
41.1 Seam norms . . . 174
41.2 The walk . . . 175
41.3 The levels . . . 178
41.4 The one-loop source in polymer form . . . 183

42 Pointwise norms: what is missing **185**

43 Interior estimates with L^p data **187**
43.1 Two inequalities on the fine lattice . . . 187
43.2 The flat estimates . . . 188
43.3 Scalar equations . . . 189
43.4 The local system . . . 191
43.5 Consequences for the Dirichlet operators . . . 192

44 Regularity tiers and step bounds **193**
44.1 Tiers and tier norms . . . 194
44.2 Interior factors . . . 195
44.3 The scalar levels on domains: lifting . . . 198
44.4 Step bounds for the top-level walks . . . 199
44.5 The constrained operators . . . 202
44.6 What this gives, and what remains . . . 203

45 The weakened background map **204**
45.1 The current at critical backgrounds . . . 205
45.2 The weakened equation . . . 205
45.3 Bounds . . . 206
45.4 The theorem . . . 207
45.5 Remarks . . . 209

Part X: The small-field flow **210**

46 Complex references **210**
46.1 Averaged backgrounds at complex references . . . 211
46.2 Extension of the results of Part IX . . . 217

47 The weakened step quantities **221**
47.1 The localization lemma . . . 223
47.2 The interaction pieces . . . 224
47.3 The Jacobians . . . 225
47.4 The composed activities . . . 226
47.5 The Hessian away from critical backgrounds . . . 228

48 The cluster expansion **229**
48.1 The small-field step . . . 230
48.2 Localization in the white noise . . . 231
48.3 The Mayer and polymer expansion . . . 232
48.4 The order g and the size of the pure sources . . . 234
48.5 The step in polymer form . . . 235

49 The activity norm and one step of the flow **236**
49.1 Polymers, domains and the norm . . . 236
49.2 Reblocking . . . 238
49.3 The linear part . . . 238
49.4 The sources . . . 239
49.5 One step . . . 241

50 The iteration and the small-field flow **243**
50.1 Scales and the torus . . . 243
50.2 The first step . . . 244
50.3 The vacuum energy . . . 246
50.4 The small-field flow . . . 247

Conclusion **248**

51 Outlook **248**

A Decay of the lattice Green function and its derivatives **251**

1 Introduction

1.1 Where the operators come from

Bałaban's construction of lattice Yang–Mills theory [3, 4, 5, 6, 7, 8, 9] is a renormalization group (RG) analysis. A gauge field on a fine lattice of spacing $\eta = L^{-k}$ is averaged over blocks of side 1, and this averaging is performed in k steps, each enlarging the blocks by a factor L. After k steps one has to integrate over all fine configurations whose block averages equal a given coarse configuration. Following the usual Gaussian strategy, one expands the action around its minimizer subject to the averaging constraint. This minimizer is called the *background field* [8]. The fluctuations around it are then Gaussian to leading order, with an explicit covariance.

Every estimate in this programme rests on properties of a small family of *linear* operators:

- Green functions of the form (Laplace-type operator $+ a\, Q^*Q)^{-1}$, where Q is the k-step averaging operator;
- the *minimizer map* H, which sends a coarse configuration B to the fine configuration of least action with average B;

- the *constrained Green function*, i.e. the covariance of the fluctuation field, which has vanishing averages;
- inverses of averaged Green functions, such as $(QGQ^*)^{-1}$, which live on the coarse lattice.

The properties needed are *exponential decay* on the unit scale and *local regularity*: bounds on values, first derivatives and Hölder quotients of first derivatives. They must be uniform in the number k of RG steps, which means uniform in the ratio η^{-1} between the unit scale and the lattice scale, and uniform in the size of the volume.

For the full non-abelian problem these operators depend on the background field U. Bałaban obtained them by perturbation from the case $U = 1$ [3], through random-walk expansions [4, 7]. The case $U = 1$ is the foundation of the whole analysis. It is also the case in which all the scales are already present, together with the gauge structure that distinguishes gauge fields from scalar fields.

1.2 What this paper does

We give a complete, self-contained treatment of these operators at the trivial background $U = 1$. The setting is a single torus with k-fold averaging everywhere, i.e. without the nested domains Ω_j of [8]. The paper is meant to be readable by a PhD student who knows linear algebra, Fourier series and the basics of elliptic equations. All lattice computations are written out.

Let us make precise what "$U = 1$" means. The gauge field is $U = e^{\mathrm{i}\eta A}$ with A taking values in the Lie algebra $\mathfrak{g}$ of the gauge group. Expanding the Wilson action to second order around $U = 1$ gives $\frac{1}{2}\|dA\|^2$, and infinitesimal gauge transformations act as $A \mapsto A + d\lambda$. The non-abelian structure enters only at higher orders. At $U = 1$ every operator acts componentwise in a basis of $\mathfrak{g}$. Hence it suffices to treat real-valued fields, and this is what we do. This is *not* abelian gauge theory. It is the linearization of non-abelian theory at its trivial background. The operators obtained are those which Bałaban perturbs around in [7, 8].

The operators. The precise definitions are given in Sections 2–9. Here is an overview. We use a fine torus $\mathbb{T}_\eta$ of spacing η and a unit torus $\mathbb{T} \subset \mathbb{T}_\eta$, the scalar block average Q_0, and the gauge-field average $\mathbf{Q}$ of [3]. The averaged gauge field $(\mathbf{Q}A)_\mu$ is an average of A_μ over the block and along lines of unit length in direction μ. It satisfies $\mathbf{Q}\, d = d_1 Q_0$, where d_1 is the unit-lattice gradient (gauge covariance). We study:

(i) the scalar Green functions $G_\sigma = (-\Delta + aQ_\sigma^* Q_\sigma)^{-1}$. Here Q_σ is either the block average Q_0 or one of the "tube" averages $Q_1, \dots, Q_d$ that make up $\mathbf{Q}$;

(ii) the operators derived from them: $(Q_\sigma G_\sigma Q_\sigma^*)^{-1}$, the minimizer map H_σ, the constrained Green function C_σ, and the effective action. We also study the projection R which defines Bałaban's Landau-type gauge;

(iii) the gauge-field operator $\mathbf{\Delta}_a = d^*d + dRd^* + a\mathbf{Q}^*\mathbf{Q}$ and its inverse $\mathbf{G}$;

(iv) the gauge-field minimizer map $\mathbf{H} = \mathbf{G}\mathbf{Q}^*(\mathbf{Q}\mathbf{G}\mathbf{Q}^*)^{-1}$, the effective action $\mathbf{D}$, and the constrained Green function $\mathbf{C}$, which is the operator appearing in the Euler–Lagrange equation of the variational problem [8, (110)–(111)].

1.3 Main results

Throughout, $d \geq 2$, $L \geq 2$ and $k \geq 1$ are integers, $\eta = L^{-k}$, and the unit torus has side $n \geq 8$. The parameter $a > 0$ in front of the averaging term is fixed. Bałaban takes $a = 1$. Every constant in Part I depends only on d, a and a Hölder exponent $\alpha \in (0, 1)$. In Part II it depends

in addition on $\dim \mathfrak{g}$ and on the decay rate γ_W of the potential term W. In particular, it does not depend on L, k or n. The statements about pointwise regularity assume $d \geq 3$. This is only for convenience in Section 6, and $d = 3$ is the case of interest.

For a fine-lattice field u (scalar or vector-valued) and a set X we write

$$|u|_X = \sup_{x \in X} |u(x)|, \qquad |\partial u|_X = \max_{\mu} \sup_{x \in X} |\partial_\mu u(x)|, \qquad [v]_{\alpha, X} = \sup_{x \neq x' \in X} \frac{|v(x) - v(x')|}{|x - x'|^\alpha}.$$

Here ∂_μ is the forward lattice derivative, and the maximum also runs over components. $B(y)$ is the unit block with corner $y \in \mathbb{T}$, and $|y - y'|$ is the distance on the torus.

Theorem A (Coercivity; Corollary 3.4, Theorem 9.6)**.** *There is $c > 0$ such that for all scalar fields f and all $\sigma \in \{0, \dots, d\}$,*

$$\|\partial f\|^2 + a\|Q_\sigma f\|^2 \;\geq\; c\,(\|f\|^2 + \|\partial f\|^2),$$

and for all gauge fields A,

$$\langle A, \mathbf{\Delta}_a A \rangle = \|dA\|^2 + \|Rd^*A\|^2 + a\|\mathbf{Q}A\|^2 \;\geq\; c\,(\|A\|^2 + \|\partial A\|^2).$$

Theorem B (Scalar Green functions; Theorem 7.1)**.** *There are $C, \gamma > 0$ such that for all $y, y' \in \mathbb{T}$ and all f supported in $B(y')$:*

$$|G_\sigma f|_{B(y)} + |\partial G_\sigma f|_{B(y)} + [\partial G_\sigma f]_{\alpha, B(y)} \leq C e^{-\gamma |y - y'|}\, |f|_{B(y')},$$
$$|G_\sigma \partial^*_\mu f|_{B(y)} + [G_\sigma \partial^*_\mu f]_{\alpha, B(y)} \leq C e^{-\gamma |y - y'|}\, |f|_{B(y')}.$$

Corresponding L^2 bounds hold for G_σ, ∂G_σ, $G_\sigma \partial^*$ and $\partial G_\sigma \partial^*$ between blocks.

Theorem C (Averaged scalar operators; Section 8)**.** *The operator $Q_\sigma G_\sigma Q_\sigma^*$ on the unit torus satisfies $c \leq Q_\sigma G_\sigma Q_\sigma^* \leq C$. Its inverse has a kernel bounded by $Ce^{-\gamma|y-y'|}$. The minimizer map H_σ satisfies*

$$|H_\sigma g|_{B(y)} + |\partial H_\sigma g|_{B(y)} + [\partial H_\sigma g]_{\alpha, B(y)} \leq C \sum_{y'} e^{-\gamma |y - y'|} |g(y')|,$$

and the derivative bounds hold with $|g(y')|$ replaced by the unit-lattice gradient $|d_1 g(y')|$. The constrained Green function C_σ satisfies the bounds of Theorem B. The effective action $\mathcal{D}_\sigma$, defined by $\langle g, \mathcal{D}_\sigma g \rangle = \min\{\|\partial f\|^2 : Q_\sigma f = g\}$, satisfies $c\|d_1 g\|^2 \leq \langle g, \mathcal{D}_\sigma g \rangle \leq C\|d_1 g\|^2$ and has an exponentially decaying kernel. The projection R is I minus an operator with exponential decay.

Theorem D (Gauge-field Green function; Theorem 10.1)**.** *The operator $\mathbf{G} = \mathbf{\Delta}_a^{-1}$ satisfies, for J supported in $B(y')$,*

$$|\mathbf{G}J|_{B(y)} + |\partial \mathbf{G} J|_{B(y)} + [\partial \mathbf{G} J]_{\alpha, B(y)} \leq C e^{-\gamma |y - y'|}\, |J|_{B(y')},$$

together with L^2 block bounds for $\mathbf{G}$, $\partial \mathbf{G}$, $\mathbf{G}\partial^*$ and $\partial \mathbf{G} \partial^*$.

Theorem E (Gauge-field minimizer and fluctuation operators; Section 11)**.** *One has $c \leq \mathbf{Q}\mathbf{G}\mathbf{Q}^* \leq C$, and $(\mathbf{Q}\mathbf{G}\mathbf{Q}^*)^{-1}$ has an exponentially decaying kernel. The minimizer map $\mathbf{H}$ sends B to the minimizer of $\|dA\|^2$ subject to $\mathbf{Q}A = B$ and $Rd^*A = 0$, and it satisfies*

$$|\mathbf{H}B|_{B(y)} + |\partial \mathbf{H} B|_{B(y)} + [\partial \mathbf{H} B]_{\alpha, B(y)} \leq C \sum_{y'} e^{-\gamma |y - y'|} |B(y')|.$$

The gauge-field effective action satisfies $c\|d_1 B\|^2 \leq \langle B, \mathbf{D} B \rangle = \|d\mathbf{H}B\|^2 \leq C\|d_1 B\|^2$. The constrained Green function $\mathbf{C} = \mathbf{G} - \mathbf{G}dRd^*\mathbf{G} - \mathbf{H}\mathbf{Q}\mathbf{G}$ is the Green function of $\|dA\|^2$ on $\{\mathbf{Q}A = 0,\ Rd^*A = 0\}$, and, for J supported in $B(y')$,

$$|\mathbf{C}J|_{B(y)} + |\partial \mathbf{C} J|_{B(y)} + [\mathbf{C}J]_{\alpha, B(y)} \leq C e^{-\gamma |y - y'|}\, |J|_{B(y')}.$$

These are, at $U = 1$ and on a single torus, the bounds that Bałaban states in [3, Sect. F] and [7, Theorems 3.12, 3.13] and uses in [8].

Backgrounds (Part II). In Part II the fields take values in the complexified Lie algebra $\mathfrak{g}^c$. The background $U = e^{i\eta A}$ satisfies $|A|, |\partial A|, [\partial A]_\alpha \le \varepsilon$ in a global gauge (hypothesis (B)). Hypothesis (B) is restrictive. It forces all Wilson loops to obey a perimeter bound, so the background must be close to flat on scales larger than $O(1)$ (Section 12). The operators now involve parallel transport. We use a simplified averaging $\mathbf{Q}(U)$, the main term of Bałaban's linearized averaging. It transforms exactly under gauge transformations, but it intertwines the covariant derivatives only up to a defect: $\mathbf{Q}(U)D_U = D_V Q'(U) + E(U)$ with $E(U) = O(\varepsilon)$. For Bałaban's full averaging $E = 0$.

Theorem F (Backgrounds; Sections 14–16)**.** *Let U satisfy (B) with parameter ε, and let $W \in \mathcal{W}(\varepsilon, \gamma_W)$ in this gauge (Section 12). For ε small enough, with constants independent of L, k, n and U:*

(i) $G'(U) = (D^T D + aQ'^T Q')^{-1}$ satisfies the bounds of Theorem B. The inverse $\Gamma'(U)$ of $Q'G'Q'^T$, the minimizer map $H'(U)$, the constrained Green function $C'(U)$ and the projection $R(U)$ satisfy the decay and pointwise bounds of Theorem C. Not claimed: the bounds in terms of $|d_1 g|$ and the effective-action bounds (Theorem 14.1, Proposition 14.2).

(ii) The gauge-field Green function $\mathbf{G}(U)$ satisfies the bounds of Theorem D (Theorem 15.4).

(iii) The inverse of $\mathbf{Q}_\pi \mathbf{G} \mathbf{Q}_\pi^T$, the minimizer map $\mathbf{H}(U)$ and the constrained Green function $\mathbf{C}(U)$ satisfy the decay and pointwise bounds of Theorem E. For every J, $\mathbf{C}(U)J$ is the unique $A \in \mathcal{X}(U)$ with $q_U(A', A) = \mathrm{b}(A', J)$ for all $A' \in \mathcal{X}(U)$. For real U, real symmetric W and real data B, $\mathbf{H}(U)B$ is the unique minimizer of $q_U(A, A)$ among real A with $\mathbf{Q}(U)A = B$ and $R(U)D_U^T A = 0$. Not claimed: the effective-action bounds (Theorem 16.1).

Here $\mathrm{b}(A, A') = \eta^d \sum_{x,\mu} \mathrm{tr}\,(A_\mu(x) A'_\mu(x))$ is the bilinear form, $q_U(A, A') = \mathrm{b}(D_U A, D_U A') + \mathrm{b}(A, WA')$ is the quadratic form, with the small potential term W, $\mathcal{X}(U) = \{A : \mathbf{Q}(U)A = 0,\ R(U)D_U^T A = 0\}$, and $\mathbf{Q}_\pi = \mathbf{Q}(U)\Pi(U)$. For real U, real symmetric W and real A, $q_U(A, A) = \|D_U A\|^2 + \langle A, WA\rangle$. The projection $\Pi(U)$ onto Bałaban's gauge slice $\{R(U)D_U^T A = 0\}$ along the residual gauge directions is Bałaban's gauge-invariant extension [7, (3.118)–(3.119)]. All these operators depend analytically on A and W and transform covariantly under G-valued gauge transformations, with W transformed accordingly. The identities $\mathbf{G}DR = DG'R$, $RD^T\mathbf{G} = RG'D^T$ and $\mathbf{Q}_\pi \mathbf{G} DR = 0$ hold exactly.

The operators are covariant under G-valued gauge transformations u, with W replaced by $R(u)WR(u)^{-1}$; the latter need not belong to $\mathcal{W}(\varepsilon, \gamma_W)$ in the new gauge. Hence Theorem F transfers to every background gauge equivalent to one satisfying (B), as follows (Proposition 16.3): the bounds on values and the L^2 block bounds hold with ∂ replaced by the covariant derivative, the identities hold, and the Hölder bounds hold in the gauge in which (B) holds. The comparisons with the operators at $U = 1$ proved in Sections 14–16 hold only in that gauge; they fail for a pure gauge $U = 1^u$.

The proofs are perturbative: every operator is written as its $U = 1$ counterpart plus a perturbation of a specific structure (Section 13), and the three tools of Part I are shown to be stable under such perturbations. For the pointwise bounds a further regularity estimate is needed, a lattice Schauder estimate for divergence-form equations (Theorem 6.3(d)).

Local regularity (Part III). In Part III the global hypothesis (B) is replaced by a local one. A real background is (M, ε)-*regular* if on every cube of side of order M there is a gauge in which (B) holds; the gauges on different cubes are unrelated. This is the form of Bałaban's regularity conditions on the background.

Theorem G (Regular backgrounds; Sections 18–20 and 41)**.** *There is M_0 such that for $M \ge M_0$ and (M, ε)-regular backgrounds with ε small (independent of M) the following hold.*

(i) *The statements of Theorem F remain true, with local potentials $W \in \mathcal{W}_{\mathrm{loc}}(\varepsilon)$ (Section 19). The comparisons with the operators at $U = 1$ of Sections 14–16 need a global gauge, fail for a pure gauge $U = 1^u$, and are not claimed. The scalar operators need only $(1, \varepsilon)$-regularity. Sup norms of derivatives are taken with the covariant derivative, and Hölder quotients in local gauges.*

(ii) *The operators whose inverses are $G'(U)$ and $\mathbf{G}(U)$ have Dirichlet versions on arbitrary unions X of blocks. These depend only on the background on the bonds of $\hat{X}$, the set of points at ∞-distance at most 1 from X, and they are coercive (the gauge-field version with a mass term near the boundary and Dirichlet conditions for all components of the gauge field). Their inverses, the inverse of the averaged scalar Green function on X, and $\mathbf{1}_X - R_X$, with R_X the Dirichlet version of the projection R, have exponential decay uniformly in the domain.*

(iii) Locality in the background: *if two regular backgrounds agree on a region, the propagators, the minimizer map and the fluctuation operator agree deep inside it up to errors that are exponentially small in the distance to the complement.*

(iv) *Bałaban's random-walk expansion converges for M large: for the scalar Green function $G'(U)$ as a sum of products of local propagators (Theorem 20.2(a)); for the gauge-field Green function $\mathbf{G}(U)$ in the factorized form of Section 41, with the domains Λ_j^v of Section 40 for a fixed $\rho \geq 15$, so that (41.1) holds with $w = 10M$, for M large depending on ρ, on tori of side $n \geq C(\rho + C_v)M$, with C_v the margin of level (L3) there (Theorem 20.2(b)). Under the same hypotheses, $\Gamma'(U)$, $H'(U)$ and $R(U)$ are expanded at levels (L1) and (L2) there. Section 41 is written for $E = 0$; the additional terms for $E \neq 0$ are treated in the proof of Theorem 20.2(b).*

(v) *All results extend analytically to complex perturbations $Ue^{\mathrm{i}\eta\delta A}$ of a regular background by a field δA that is small in the local gauges, for potentials W that depend analytically on the background, with coercivity replaced by* Re*-coercivity.*

The key new step is a local proof of coercivity (Section 19). An IMS localization at scale M reduces it to cubes. On each cube one works in the local gauge, perturbs around $U = 1$, and uses the coercivity of Part I. Fourier analysis is used only at $U = 1$.

The nonlinear problem (Parts IV and V). Bałaban's averaging of a gauge field is nonlinear and is performed one scale at a time. Its k-fold composition $U \mapsto U^{(k)}$ is covariant with respect to the values of gauge transformations at the points of $\mathbb{T}$. So the residual gauge group of the variational problem is $\mathfrak{G}_0 = \{u : u|_{\mathbb{T}} = 1\}$. Part IV constructs the bridge to the linear theory. Following [5, 6], we introduce the multiscale axial gauge and the gauge-adapted averaging $\mathcal{Q}$. We show (Proposition 22.7, Corollary 22.8) that the linearization of $\mathcal{Q}$ at a regular reference U_0 is an averaging pair $(Q'_k(U_0), \mathbf{Q}_k(U_0))$ to which Parts II and III apply, with *exact* covariance ($E = 0$). We then prove Bałaban's axial-to-Landau theorem [6, Theorem 2] on a single torus.

Theorem H (Axial-to-Landau chart; Theorem 24.2)**.** *Let U_0 be (M, ε)-regular with current at most ε, where ε is small. Every real configuration U with small curvature and current, and with k-fold averages close to those of U_0, has exactly one small representative $e^{\mathrm{i}\eta A}$ relative to U_0 that lies in Bałaban's Landau gauge $R(U_0)D_{U_0}^T A = 0$ and is related to the axial representative of U by the correspondence transformation. It obeys $|A| + |\nabla A| + |\Delta_{U_0} A| + [\nabla A]_\alpha \leq C(\varepsilon +$ curvature $+$ current $+$ difference of averages$)$, uniformly in k, n and η. The curvature and the current in this bound are those of U.*

Theorem I (The variational problem relative to a regular reference; Theorem 25.1)**.** *Let U_0 be (M,ε)-regular with current at most ε, and let V be a real configuration on $\mathbb{T}$ with $V(U_0^{(k)})^{-1} = e^{\mathrm{i}B}$, $|B| \le \beta$. If $\varepsilon+\beta$ is small, then the Wilson action restricted to $\{U : U^{(k)} = V\}$ has exactly one critical $\mathfrak{G}_0$-orbit among configurations of small curvature and current. The orbit consists of the minimizers of the action among these configurations. Its Landau representative relative to U_0 satisfies*

$$|A_*| + |\nabla A_*| + |\Delta_{U_0} A_*| + [\nabla A_*]_\alpha \le C(\varepsilon+\beta), \qquad A_* = \mathbf{H}B - \mathbf{C}J_{U_0} + O((\varepsilon+\beta)^2).$$

The minimizer is again regular, it depends analytically on B, and the derivative with respect to B decays exponentially.

This is the content of Bałaban's variational theorem [8, Theorem 1] for one step k of his induction, given the reference configuration U_0, on a single torus. Along the way we reprove Bałaban's results on the averaging [5] and on gauge fixing [6] in the form needed.

The induction (Part VI). Bałaban proves his theorem by induction over k. The reference at step k is the solution of the $(k-1)$-step problem for the trivial extension of V; it has exactly the prescribed averages. Theorem I then gives the solution at step k. Its regularity bound is worse by a power of L, because units change by a factor L. Bałaban restores the bound by a local analysis on nested domains [8, Sect. F]. We stay on the torus and use the uniformity of Theorem I in the reference. Near a given point we replace the reference by a configuration that is trivial, in a local gauge, near the point and equals the solution far away. Relative to it, the solution near the point is determined by the coarse field in an axial gauge centred at the point, which is controlled by the curvature of V alone. The errors are exponentially small in the splicing radius, or quadratic.

Theorem J (Bałaban's variational theorem; Theorem 30.3, Corollary 30.4)**.** *There are a_1, B_3, n_0 such that the following holds on tori of side $n \ge n_0$, for every k and every real configuration V on $\mathbb{T}$ whose curvature is at most $\varepsilon_1 \le a_1$. The Wilson action on $\{U : U^{(k)} = V\}$ has exactly one critical $\mathfrak{G}_0$-orbit among configurations of small curvature and current. It consists of the minimizers of the action among these configurations, which have curvature and current at most $B_3\varepsilon_1$ and are $(M, B_3\varepsilon_1)$-regular. The constants do not depend on k, n or η. Moreover, every real configuration whose curvature and current are at most θ, with θ small, is $(M, C\theta)$-regular.*

The last statement is the torus version of Bałaban's derivation of local regularity from gauge-invariant conditions [6, Sect. F]. So the hypothesis of Part III holds for all configurations of Bałaban's small-field class.

One renormalization group step (Part VII). The renormalization group transformation integrates over configurations on $\mathbb{T}^{(k)}$ with prescribed averages on $\mathbb{T}^{(k+1)}$. We write it exactly as a Gaussian integral:

- the background of the step is the $(k+1)$-step background field (Lemma 31.6);
- the gauge is fixed by a one-step axial gauge, and the constraint is eliminated on one bond per coarse bond, with local Jacobians;
- the action becomes a function of the coarse fluctuation through the coarse effective action $V \mapsto S(U_k(V))$, whose gradient is the Lagrange multiplier and whose derivatives decay along trees (Proposition 33.1).

The fluctuation covariance is the inverse of a bounded operator on the unit lattice. It is coercive on the one-step constraint set, by a Poincaré inequality and an IMS localization, and has exponential decay (Theorem 34.4, Corollary 34.5). The background-dependent determinant is a sum of local, gauge-invariant densities (Proposition 34.7). Finally, the curvature of background fields is bounded by $(C_* + \delta)$ times the curvature of the coarse configuration, with C_* independent of L (Proposition 31.5). This makes the small-field sets of consecutive steps compatible for L large.

Localization (Parts VIII and IX). The renormalization group needs the background dependence of the step as a sum of local pieces. Part VIII obtains this with Dirichlet approximants and telescoping, using interior pointwise bounds with derivatives for the Dirichlet operators (Theorem 35.2). The pieces are of *radius type*: a piece depending on a ball of radius m is bounded by $e^{-\gamma m}$. A polymer expansion needs *volume type*, a small factor for every cube of the dependence region. Part IX provides it, as Bałaban and Dimock do, by random walks with weakening parameters $s \in \mathbb{D}_\kappa$. It constructs them for the covariance, its square root, the coarse Hessian and the determinant in L^2 norms (Section 41), proves pointwise $C^{1,\beta}$ bounds for the weakened propagators (Sections 43 and 44), and builds the weakened background map $A_*(s; B)$, analytic in s and factorizing over polymers (Theorem 45.5).

The small-field flow (Part X). The activities are analytic functions on a complex domain, so Part X first extends Part IX, and the averaged backgrounds of Part IV, to complex references (Section 46). Then it follows Dimock's scheme [15]:

- the step quantities (interaction, Jacobians, old activities composed with the background map, Hessian) become functions of the weakening parameters that factorize over polymers (Section 47);
- with a cutoff on the white noise, the fluctuation integral is a Mayer and polymer expansion (Section 48). By parity in the coupling there is no term of order g; the pure sources are of order $g^2 p_0^6$;
- the activity norm lives on a graded complex domain. On the domain of the next scale, a normalized gauge-invariant activity is smaller than its supremum by a factor quadratic in the ratio of the curvature, the current and the complex part to the size of the domain (Lemma 38.5). So the linear part of the flow is bounded by a constant times $L^{-\epsilon}$, which is beaten by the growth $L^{1-\epsilon}$ per step of the bound $h_k^{2-2\epsilon}$ (Section 49);
- the steps are iterated, and the vacuum energy is fixed by a counterterm (Section 50).

The finest lattice has spacing L^{-N} in physical units, and g is the coupling, so the dimensionless couplings are $g_k^2 = g^2 L^{k-N}$. Fix $\epsilon \in (0, \frac{1}{4})$, put $h_k = p_k g_k$ with p_k a slowly growing function of g_k^{-1} (Sections 49 and 50), and let $0 < \beta' < 2 - 2\epsilon$. The activity norm $\| \cdot \|_{k,\kappa_n}$ and the constant $K_{\mathcal{E}}$ are those of Section 49.

Theorem K (Small-field flow; Theorem 50.4, Proposition 50.3, Corollary 50.5)**.** *Let $d = 3$ and G compact, connected and semisimple. There are L, M, $g_* > 0$ and $n_{\min}$ with the following property, for a torus of physical side ℓ with $\ell g^2/g_*^2 \geq n_{\min}$ and $L^{-N} \leq g_*^2/(Lg^2)$, such that ℓL^N and M are powers of L. Let K be the largest integer with $g_K \leq g_*$. There is a counterterm E_0, fixed by $\varepsilon_K = 0$ and with $|E_0| \leq C g_0^{\beta'}$ per lattice site, such that the small-field steps map $\rho_0 = \exp(-g_0^{-2} S - E_0)$ to densities*

$$\rho_k(V) = \mathcal{Z}_k \exp\Big(-\frac{1}{g_k^2} S(U_k(V)) - \varepsilon_k |\mathbb{T}^{(k)}| + \sum_X \mathcal{E}_k(X, U_k(V)) \Big), \qquad k \leq K,$$

on the small-field sets. Here $\mathcal{Z}_k$ is an explicit product of normalizations, $|\varepsilon_k| \le C g_k^{\beta'}$, and the $\mathcal{E}_k(X,\cdot)$ are polymer activities that are invariant under real gauge transformations, normalized at $U = 1$, real for real configurations, and analytic on the complex domains $\mathcal{U}_k(X)$, which contain the background fields $U_k(V)$ of all small-field V; moreover $\|\mathcal{E}_k\|_{k,\kappa_n} \le K_{\mathcal{E}} h_k^{2-2\epsilon}$. The partition function of the resulting small-field model is $\mathcal{Z}_K I_K$, with $e^{-c|\mathbb{T}^{(K)}|} \le I_K \le e^{c|\mathbb{T}^{(K)}|}$. The constants C, $K_{\mathcal{E}}$, c do not depend on N or k.

Boundary conditions. The verification of 31 October 2026 found that an earlier version of the gauge-field Dirichlet operators, with relative boundary conditions, did not support the weakened walks on proper domains. The operators now carry Dirichlet conditions for all components (Section 19, Remark 41.5). This leaves the torus operators unchanged, and with them Theorems F, H–J and the torus statements of Theorem G.

The small-field model is a modification of the lattice theory: at every step the integration is restricted to small fields and the rest is discarded (Remark 50.6). Controlling the discarded part is the large-field stage, the analogue of [16, 17], which is not treated here. So Theorem K is the analogue of the small-field analysis of [15]; it says nothing yet about the partition function of the lattice theory itself. Ultraviolet stability bounds for the partition function of the full three-dimensional theory, large fields included, were obtained by Bałaban [9], in an outline that refers to his four-dimensional work for the details. Theorem K writes out the small-field part of such an analysis in Dimock's scheme, on the operator theory of Parts I–IX.

Reading guide. For the small-field flow the main line is Parts I–VII, Section 35 and the interior estimates of Section 36, Section 37 up to Proposition 37.6, Section 38 except Proposition 38.1 and Corollary 38.8, Part IX and Part X. The radius-type results of Part VIII (Theorem 37.7, Corollary 37.8, Proposition 38.1, Corollary 38.8, and Section 39) are superseded on the main line by Parts IX and X. Each superseded part is marked as such where it appears.

1.4 Methods

Four tools are used. Each is introduced in its own section, before it is used.

1. Coercivity (Section 3 and Section 9). A Poincaré inequality on unit blocks shows that the averaging term controls exactly those modes that the Laplacian does not control. This gives the bounds of Theorem A for scalars. For gauge fields the situation is subtler, because $\|dA\|^2 + a\|\mathbf{Q}A\|^2$ is invariant under gauge transformations $A \mapsto A + d\lambda$ with $Q_0\lambda = 0$. The gauge-fixing term $\|Rd^*A\|^2$ removes this degeneracy. We prove the coercivity in position space. We split A into a gradient part and a co-closed part, and show that the averaging operator nearly separates them: $d_1^*\mathbf{Q}A$ differs from an average of d^*A by the unit-lattice divergence of a quantity bounded by $\|\partial A\|$ (Lemma 9.3). Bałaban's original proof [3] is a computation in momentum space.

2. The Combes–Thomas method (Section 5). If an operator is coercive and conjugation by $e^{\alpha\rho}$, for Lipschitz ρ, perturbs it by $O(\alpha)$, then its inverse decays exponentially [11]. This works for the fine-lattice operators G_σ and $\mathbf{G}$, which contain the large term $-\Delta$ of norm $\sim \eta^{-2}$, because the perturbation is controlled by the quadratic form itself. It also works for the coarse operators $(QGQ^*)^{-1}$. Bałaban and [19] use Fourier analysis or random-walk expansions instead. We use neither: on a single torus at $U = 1$, Combes–Thomas suffices, and it also survives perturbation in U.

3. Local regularity (Section 6). Exponential decay is first obtained in L^2 between unit blocks. To pass to pointwise bounds on values and derivatives, uniformly in η, we prove a discrete interior regularity estimate. If $-\Delta u = F$ holds on a cube of side 4 (with u defined on a slightly larger cube), then u, ∂u and the Hölder quotients of ∂u on the concentric cube of side 2 are bounded by $|F|$ and the L^2 norm of u. The proof uses a cutoff and the decay of the Green function of the fine lattice $\eta\mathbb{Z}^d$ and its derivatives. The latter is proved in Appendix A.

4. One Fourier lower bound (Section 4). Averaged Green functions QGQ^* must be bounded *below* uniformly in k. We deduce this from the operator inequality $G \geq (-\Delta + 2a)^{-1}$ and an elementary computation in momentum space. The same computation gives one more estimate needed in the gauge-field coercivity proof.

1.5 What is not done here

The variational theorem is proved on a single torus with the small-field condition everywhere (Theorem J), and the renormalization group is carried out for small fields only (Theorem K). We do not treat:

- the large-field stage: the nested domains Ω_j of [8], large fields, and with them the partition function and the correlation functions of the lattice theory itself, as opposed to the small-field model of Theorem K;
- gauges on cubes of size up to $c\,\varepsilon_1^{-1}$ with bounds linear in the size [8, (9), (10)]; we obtain polynomial bounds (Remark 30.6);
- gauge groups with abelian factors in Part X: the second-order bound for normalized activities (Lemma 38.5) uses semisimplicity;
- dimensions other than 3 from Part VII on.

Section 16.2 lists the input of the variational theorem and where it is provided.

Relation to [19]. The paper [19] gives a self-contained account of exponential decay for the *scalar* operators of Bałaban's approach on general domains with Neumann boundary conditions, using the method of images and analytic continuation of Fourier transforms. Here the domain is a torus, the methods are different, and the emphasis is on gauge fields. The scalar part (Sections 7–8) is included because the gauge-field operators are built from scalar ones: gauge transformations are scalar fields, and the projection R is defined in terms of G_0.

Status. This is a first version, written with the assistance of Claude (Anthropic). It has not been checked by a human expert.

Part I: The trivial background

2 Lattices, fields and averaging operators

This section fixes notation and collects the elementary identities used throughout. Nothing here is deep, but the identities of Lemma 2.4 and the restriction estimate of Lemma 2.7 are used repeatedly.

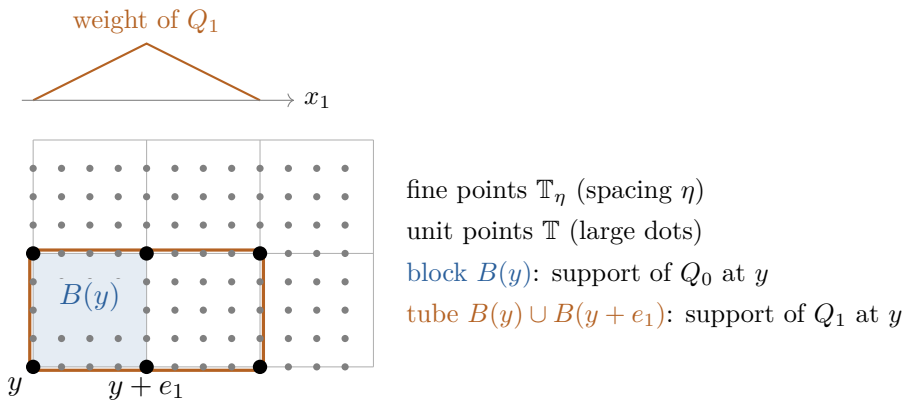


Figure 1: The two lattices for $d = 2$, $N = 4$. The block average $(Q_0 f)(y)$ is the mean of f over $B(y)$. The tube average $(Q_1 f)(y)$ is the mean over $B(y)$ of the averages of f along unit segments in direction e_1. Its weight is the triangle shown above the tube.

2.1 Lattices and blocks

Fix integers $d \geq 2$, $L \geq 2$, $k \geq 1$ and $n \geq 8$, and put

$$N = L^k, \qquad \eta = \frac{1}{N}.$$

The *unit torus* and the *fine torus* are

$$\mathbb{T} = (\mathbb{Z}/n\mathbb{Z})^d, \qquad \mathbb{T}_\eta = (\eta\mathbb{Z}/n\mathbb{Z})^d.$$

Thus $\mathbb{T}_\eta$ is a torus of side n with lattice spacing η, and $\mathbb{T} \subset \mathbb{T}_\eta$ is the sublattice of points with integer coordinates. We write $e_1, \dots, e_d$ for the unit coordinate vectors, and $|x - x'|$ for the Euclidean distance on the torus (the minimum over representatives). Since $n \geq 8$, every subset of a cube of side at most 7 can be identified with a subset of $\eta\mathbb{Z}^d$, and we do this tacitly. On cubes of side at most $4 \leq n/2$ the torus distance coincides with the distance in $\eta\mathbb{Z}^d$. On larger cubes it can only be smaller, which makes Hölder quotients computed with the torus distance larger; this direction is harmless wherever such quotients appear as hypotheses.

For $y \in \mathbb{T}$ the *block* with corner y is

$$B(y) = \{x \in \mathbb{T}_\eta : \ y_\mu \leq x_\mu < y_\mu + 1, \ \mu = 1, \dots, d\}.$$

It contains N^d fine points, the blocks partition $\mathbb{T}_\eta$, and we write $y(x)$ for the corner of the block containing x. The only role of L is that $N = L^k$. Our constants never depend on L, and the reader may think of N as an arbitrary positive integer.

Remark 2.1 (Relation to Bałaban's notation). In [3] the fine lattice is T_η, the unit lattice is $T_1^{(k)}$, and the fields have been rescaled so that the averaging blocks have side 1. Our $\mathbb{T}_\eta$ and $\mathbb{T}$ are these lattices, with the torus side denoted by n. All of Bałaban's operators after k renormalization steps live on this pair of lattices.

2.2 Fields, derivatives and the Hodge identity

A *scalar field* is a function $f : \mathbb{T}_\eta \to \mathbb{R}$. A *gauge field* (a 1-form) is a d-tuple $A = (A_1, \dots, A_d)$ of scalar fields. Here $A_\mu(x)$ is the value of A on the oriented bond $(x, x + \eta e_\mu)$, and the value on the reversed bond is $-A_\mu(x)$. A 2-form is a collection $F = (F_{\mu\nu})$ with $F_{\nu\mu} = -F_{\mu\nu}$. Here $F_{\mu\nu}(x)$ is the value on the plaquette spanned by x, ηe_μ and ηe_ν. The inner products are

$$\langle f, g \rangle = \eta^d \sum_{x \in \mathbb{T}_\eta} f(x) g(x), \qquad \langle A, B \rangle = \sum_\mu \langle A_\mu, B_\mu \rangle, \qquad \langle F, E \rangle = \sum_{\mu < \nu} \langle F_{\mu\nu}, E_{\mu\nu} \rangle,$$

and $\|\cdot\|$ denotes the corresponding norms. For a set $X \subset \mathbb{T}_\eta$, $\|f\|_X$ is the norm of $\mathbf{1}_X f$. Unit-lattice fields are defined in the same way on $\mathbb{T}$, with inner product $\sum_{y\in\mathbb{T}}$ and no factor η^d. With these normalizations a block has volume $\eta^d N^d = 1$, so constant fields have comparable norms on both lattices.

The forward and backward derivatives are

$$(\partial_\mu f)(x) = \frac{f(x+\eta e_\mu) - f(x)}{\eta}, \qquad (\partial_\mu^* f)(x) = \frac{f(x-\eta e_\mu) - f(x)}{\eta}.$$

Then ∂_μ^* is the adjoint of ∂_μ. All ∂_μ and ∂_ν^* commute with one another and with translations of $\mathbb{T}_\eta$. The exterior derivative and its adjoint are

$$(df)_\mu = \partial_\mu f, \qquad d^*A = \sum_\mu \partial_\mu^* A_\mu,$$

$$(dA)_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu, \qquad (d^*F)_\nu = \sum_\mu \partial_\mu^* F_{\mu\nu}.$$

One checks directly that d^* is the adjoint of d and that $d(df) = 0$. The (non-positive) Laplacian is $\Delta = -\sum_\mu \partial_\mu^*\partial_\mu = -d^*d$ on scalar fields. It acts on gauge fields componentwise. We use the shorthand

$$\|\partial f\|^2 = \sum_\mu \|\partial_\mu f\|^2 = \langle f, -\Delta f\rangle, \qquad \|\partial A\|^2 = \sum_{\mu,\nu} \|\partial_\nu A_\mu\|^2.$$

Gauge fields of the form $d\lambda$ are called *exact*, and those with $d^*A = 0$ *co-closed*.

Lemma 2.2 (Hodge identities)**.** *For every gauge field A,*

$$(d^*d + dd^*)A = -\Delta A \quad \textit{(componentwise)}, \qquad \|dA\|^2 + \|d^*A\|^2 = \|\partial A\|^2.$$

Moreover $-\Delta$ is invertible on scalar fields with mean zero, and every gauge field decomposes orthogonally as $A = d\lambda + A'$ with $d^*A' = 0$ and $\lambda = (-\Delta)^{-1}d^*A$. The same statements hold on the unit torus, with ∂_μ replaced by the unit difference $D_\mu g(y) = g(y+e_\mu) - g(y)$.

Proof. For the first identity,

$$(d^*dA)_\mu = \sum_\nu \partial_\nu^*(\partial_\nu A_\mu - \partial_\mu A_\nu), \qquad (dd^*A)_\mu = \partial_\mu \sum_\nu \partial_\nu^* A_\nu,$$

and the mixed terms cancel because ∂_μ and ∂_ν^* commute. Taking the inner product with A gives the second identity. Next, $-\Delta$ is symmetric and $\langle f, -\Delta f\rangle = \|\partial f\|^2$. This vanishes only for constant f on the connected torus. Hence $-\Delta$ is invertible on the orthogonal complement of the constants. Since d^*A has mean zero ($\langle 1, d^*A\rangle = \langle d1, A\rangle = 0$), λ is well defined. Then $d^*(A - d\lambda) = d^*A + \Delta\lambda = 0$, and $\langle d\lambda, A'\rangle = \langle \lambda, d^*A'\rangle = 0$. The proof on $\mathbb{T}$ is identical. □

On the unit torus we write d_1 for the exterior derivative built from D_μ, and d_1^* for its adjoint, with $(D_\mu^* g)(y) = g(y - e_\mu) - g(y)$.

2.3 Directional convolutions

The flat block and tube averages introduced in this section are obtained by restricting products of one-dimensional averages to the unit lattice, so we introduce a convenient language for those products.

Definition 2.3. Let $\mu \in \{1, \dots, d\}$ and let $\kappa : \eta\mathbb{Z} \to \mathbb{R}$ vanish outside an interval of length at most 6. The *directional convolution* in direction μ with kernel κ is

$$(Kf)(x) = \sum_{s\in\eta\mathbb{Z}} \eta\,\kappa(s)\, f(x + se_\mu),$$

where $x + se_\mu$ is taken modulo the torus. Products of directional convolutions in different directions are called *product convolutions*.

Directional convolutions commute with each other and with all translations of $\mathbb{T}_\eta$. The adjoint of K has kernel $s \mapsto \kappa(-s)$. The composition of two convolutions in the same direction has kernel $(\kappa_1 \star \kappa_2)(s) = \sum_t \eta\, \kappa_1(t)\kappa_2(s-t)$. Moreover $|\kappa_1 \star \kappa_2| \le \|\kappa_1\|_\infty \sum_t \eta |\kappa_2(t)|$, and the support lengths add. The box averages a_μ, a_μ^* and the triangle average c_μ have nonnegative kernels of mass 1, bounded by 1. Hence every composition of such averages, possibly composed with one further convolution whose kernel is bounded by M, has a kernel bounded by 1 (respectively M). Its support length is the sum of the support lengths.

We use the following operators in direction μ:

- the forward and backward *box averages* a_μ, a_μ^*, with kernels $\mathbf{1}_{[0,1)}$ and $\mathbf{1}_{(-1,0]}$:

$$(a_\mu f)(x) = \eta \sum_{j=0}^{N-1} f(x + j\eta e_\mu), \qquad (a_\mu^* f)(x) = \eta \sum_{j=0}^{N-1} f(x - j\eta e_\mu);$$

- the unit difference $D_\mu f(x) = f(x + e_\mu) - f(x)$ and the shift $(\tau_\mu f)(x) = f(x + (1-\eta)e_\mu)$;
- the *triangle average* $c_\mu = a_\mu a_\mu^*$.

Lemma 2.4 (One-dimensional identities)**.** *For each μ:*

(a) $D_\mu = a_\mu \partial_\mu = \partial_\mu a_\mu$ *and* $D_\mu^* = a_\mu^* \partial_\mu^*$;

(b) $a_\mu = \tau_\mu a_\mu^*$;

(c) c_μ *has kernel* $c(s) = (1 - |s|)_+$, *and* $\sum_s \eta\, c(s) = 1$;

(d) $1 - c_\mu = e_\mu \partial_\mu^* \partial_\mu$, *where* e_μ *has a kernel* k *with* $0 \le k \le \frac{1}{2}$, $k(s) = 0$ *for* $|s| \ge 1$, *and* $|k(s) - k(s')| \le |s - s'|$. *Consequently* $e_\mu \partial_\mu$ *is a directional convolution with kernel* $k_1(s) = (k(s-\eta) - k(s))/\eta$, *which satisfies* $|k_1| \le 1$ *and vanishes outside* $[-1, 1+\eta]$;

(e) $a_\mu - 1 = g_\mu \partial_\mu$, *where* g_μ *has kernel* $(1 - \eta - s)\mathbf{1}_{[0,1)}(s) \in [0, 1]$.

Proof. All identities are between convolution operators, so it suffices to check them on $\eta\mathbb{Z}$, i.e. for functions of one variable t. The torus versions follow by periodization.

(a) $(a\partial f)(t) = \eta \sum_{j=0}^{N-1} \frac{f(t+(j+1)\eta) - f(t+j\eta)}{\eta} = f(t+1) - f(t)$, since $N\eta = 1$. Taking adjoints gives the second identity.

(b) $(\tau a^* f)(t) = \eta \sum_{j=0}^{N-1} f(t + 1 - \eta - j\eta) = \eta \sum_{j'=0}^{N-1} f(t + j'\eta)$, with $j' = N - 1 - j$.

(c) The kernel of aa^* at $s \in \eta\mathbb{Z}$ is $\eta \cdot \#\{t \in \eta\mathbb{Z} : t \in [0,1),\ s - t \in (-1, 0]\}$. For $0 \le s < 1$ the admissible t are $s, s+\eta, \dots, 1-\eta$, which gives $\eta \cdot (1-s)/\eta = 1 - s$. The case $s < 0$ is symmetric. Finally $\sum_s \eta\, c(s) = \eta(1 + 2\sum_{j=1}^{N-1}(1 - j\eta)) = \eta(2N - 1 - (N-1)) = 1$.

(d) Let $-\Delta_1 h(t) = \eta^{-2}(2h(t) - h(t+\eta) - h(t-\eta))$ denote the one-dimensional operator $\partial^*\partial$. A direct check shows $-\Delta_1(-|t|/2) = \eta^{-1}\delta_{t,0}$. Put

$$k(t) = -\frac{|t|}{2} + \frac{1}{2} \sum_s \eta\, |t - s|\, c(s).$$

Then $-\Delta_1 k = \eta^{-1}\delta_{\cdot,0} - c$. Hence the convolution e with kernel k satisfies $e\,\partial^*\partial f = (-\Delta_1 k) \star f = f - cf$. If $|t| \ge 1$, then $|t - s| = |t| - \operatorname{sgn}(t)s$ for all s in the support of c. Since c has mass 1 and mean 0, this gives $k(t) = 0$. For $|t| < 1$, the triangle inequality and $\sum \eta c(s)(t-s) = t$ give $k(t) \ge 0$, while $\sum_s \eta |t-s| c(s) \le |t| + 1$ gives $k(t) \le \frac{1}{2}$. Each of the two terms defining k is Lipschitz with constant $\frac{1}{2}$. The formula for the kernel of $e\partial$ follows by summation by parts: $\sum_s \eta k(s) \frac{f(t+s+\eta) - f(t+s)}{\eta} = \sum_s \eta \frac{k(s-\eta) - k(s)}{\eta} f(t+s)$.

(e) $(af)(t) - f(t) = \eta \sum_{j=0}^{N-1} \sum_{i=0}^{j-1} \eta\, \partial f(t + i\eta) = \sum_{i=0}^{N-1} \eta\, \eta(N - 1 - i)\, \partial f(t + i\eta)$, and $\eta(N - 1 - i) = 1 - \eta - i\eta$. □

2.4 Averaging operators

Let ι denote restriction to the unit torus, $(\iota f)(y) = f(y)$ for $y \in \mathbb{T}$. Its adjoint is $(\iota^* g)(x) = \eta^{-d} g(x)$ if $x \in \mathbb{T}$ and 0 otherwise. Put $b = a_1 a_2 \cdots a_d$.

Definition 2.5 (Averaging operators)**.** The *block average* Q_0 and the *tube averages* Q_μ, $\mu = 1, \dots, d$, map scalar fields on $\mathbb{T}_\eta$ to scalar fields on $\mathbb{T}$:

$$(Q_0 f)(y) = (\iota b f)(y) = \eta^d \sum_{x \in B(y)} f(x),$$

$$(Q_\mu f)(y) = (\iota a_\mu b f)(y) = \eta^d \sum_{x \in B(y)} \eta \sum_{j=0}^{N-1} f(x + j\eta e_\mu).$$

The *gauge-field average* maps gauge fields on $\mathbb{T}_\eta$ to gauge fields on $\mathbb{T}$:

$$(\mathbf{Q}A)_\mu = Q_\mu A_\mu, \qquad \text{i.e.} \qquad (\mathbf{Q}A)_\mu(y) = \eta^d \sum_{x \in B(y)} \eta \sum_{j=0}^{N-1} A_\mu(x + j\eta e_\mu).$$

We write Q_σ, $\sigma \in \{0, 1, \dots, d\}$, when a statement holds for all of them, and $\mathcal{K}_\sigma$ for the product convolution with $Q_\sigma = \iota \mathcal{K}_\sigma$ ($\mathcal{K}_0 = b$, $\mathcal{K}_\mu = a_\mu b$).

The gauge-field average is the linearization of Bałaban's averaging at $U = 1$. In [3, (1.11), (1.18)] it is written as $(Q_k A)_b = \sum_{x \in B_k(b_-)} \eta^{d+1} A([x, x(b)])$, where $A([x, x(b)])$ is the sum of A over the unit segment from x to $x + e_\mu$. This is exactly $(\mathbf{Q}A)_\mu(y)$. Bałaban's original averaging [3, (1.8)] also contains contour terms. These vanish in the axial gauge, and dropping them is the "second point of view" of [3, (1.13)], which we adopt.

Lemma 2.6 (Properties of the averages)**.** *(a)* $(Q_\sigma f)(y) = \eta^d \sum_x w_\sigma(y, x) f(x)$ *with* $0 \le w_\sigma \le 1$, $\eta^d \sum_x w_\sigma(y, x) = 1$ *and* $\sum_y w_\sigma(y, x) = 1$. *Moreover* $w_0(y, x) = \mathbf{1}_{B(y)}(x)$, *and* $w_\mu(y, \cdot)$ *vanishes outside* $B(y) \cup B(y + e_\mu)$.

(b) $Q_\sigma 1 = 1$ *and* $Q_\sigma^* 1 = 1$. *Further,* $(Q_0^* g)(x) = g(y(x))$, $Q_0 Q_0^* = I$, *and* $Q_0^* Q_0$ *is the orthogonal projection onto fields that are constant on each block.*

(c) $|(Q_\sigma f)(y)|^2 \le \eta^d \sum_x w_\sigma(y, x) |f(x)|^2$. *Hence* $\|Q_\sigma\| \le 1$, $\|\mathbf{Q}\| \le 1$, *and* $|(Q_\sigma^* g)(x)| \le \max\{|g(y)| : w_\sigma(y, x) \neq 0\}$, *where at most two* y *occur.*

(d) (Gauge covariance.) $D_\mu Q_0 = Q_\mu \partial_\mu$, *i.e.* $\mathbf{Q}\, d = d_1 Q_0$. *More generally,* $D_\mu Q_\sigma = \iota\, a_\mu \mathcal{K}_\sigma \partial_\mu$.

(e) (Curvature.) $d_1 \mathbf{Q} = \mathbf{Q}^{(2)} d$, *where* $(\mathbf{Q}^{(2)} F)_{\mu\nu} = \iota\, a_\mu a_\nu b\, F_{\mu\nu}$.

Proof. (a) For Q_0 this is the definition. For Q_μ, the coefficient of $f(x)$ in $(Q_\mu f)(y)$ is $\eta^d \cdot \eta \,\#\{j \in \{0, \dots, N-1\} : x - j\eta e_\mu \in B(y)\}$. So $w_\mu(y, x) \le \eta N = 1$, and $w_\mu(y, x) \neq 0$ only if $x \in B(y) \cup B(y + e_\mu)$. The normalization $\eta^d \sum_x w_\sigma = Q_\sigma 1 = 1$ is clear. The identity $\sum_y w_\sigma(y, x) = 1$ is $Q_\sigma^* 1 = 1$, which follows from $(b^* \iota^* g)(x) = g(y(x))$ and $a_\mu^* 1 = 1$.

(b) This follows from (a). Moreover $(Q_0 Q_0^* g)(y) = \eta^d \sum_{x \in B(y)} g(y) = g(y)$. Hence $Q_0^* Q_0$ is a self-adjoint idempotent, and its range consists of the block-constant fields.

(c) The first bound is Cauchy–Schwarz with the probability weights $\eta^d w_\sigma(y, \cdot)$. Summing over y and using $\sum_y w_\sigma(y, x) = 1$ gives $\|Q_\sigma f\|^2 \le \|f\|^2$. (Below we sometimes use the weaker bound $\|Q_\sigma\|^2 \le 2$.) The bound on $Q_\sigma^* g$ follows from $(Q_\sigma^* g)(x) = \sum_y w_\sigma(y, x) g(y)$ and $\sum_y w_\sigma(y, x) = 1$.

(d) ι commutes with unit translations, so $D_\mu Q_\sigma = \iota D_\mu \mathcal{K}_\sigma = \iota \mathcal{K}_\sigma a_\mu \partial_\mu$ by Lemma 2.4(a). For $\sigma = 0$ this equals $\iota a_\mu b\, \partial_\mu = Q_\mu \partial_\mu$.

(e) $(d_1 \mathbf{Q} A)_{\mu\nu} = D_\mu Q_\nu A_\nu - D_\nu Q_\mu A_\mu = \iota\, a_\nu b\, a_\mu \partial_\mu A_\nu - \iota\, a_\mu b\, a_\nu \partial_\nu A_\mu$, by (a) of Lemma 2.4. □

Part (d) is the reason for the particular form of $\mathbf{Q}$. A gauge transformation $A \mapsto A + d\lambda$ changes the average by the unit-lattice gradient of the block average of λ:

$$\mathbf{Q}(A + d\lambda) = \mathbf{Q}A + d_1 Q_0 \lambda.$$

In particular, the gauge transformations that preserve all averages are those with $Q_0\lambda$ constant. Modulo constants, these are those with $Q_0\lambda = 0$. This is the residual gauge freedom of the variational problem, cf. [3, (1.13), (1.20)].

2.5 Restriction of product convolutions

The averaging operators restrict a smoothed field to the unit lattice. The following estimate is used whenever we need to bound such restrictions in ℓ^2.

Lemma 2.7 (Restriction lemma)**.** *Let $\mathcal{K} = K_1 \cdots K_d$ be a product convolution, where K_μ has a kernel bounded by M_μ and vanishing outside an interval of length $\ell_\mu \le 6$. Then for every scalar field f,*

$$\|\iota \mathcal{K} f\|_{\ell^2(\mathbb{T})} \le \prod_{\mu=1}^{d} M_\mu(\ell_\mu + 1)\ \|f\|.$$

Proof. Write $(\mathcal{K}f)(y) = \sum_t \eta^d \kappa(t) f(y+t)$ with $\kappa(t) = \prod_\mu \kappa_\mu(t_\mu)$. By Cauchy–Schwarz,

$$|(\mathcal{K}f)(y)|^2 \le \Big(\sum_t \eta^d |\kappa(t)|\Big)\Big(\sum_t \eta^d |\kappa(t)|\,|f(y+t)|^2\Big).$$

The first factor is at most $\prod_\mu M_\mu(\ell_\mu + \eta)$. Summing the second factor over $y \in \mathbb{T}$ gives $\sum_x \eta^d |f(x)|^2 \sum_y |\kappa(x-y)|$. For fixed x and μ, at most $\ell_\mu + 1$ integers y_μ satisfy $\kappa_\mu(x_\mu - y_\mu) \neq 0$. Since $\ell_\mu + 1 \le 7 < n$, no point is counted twice on the torus. Hence $\sum_y |\kappa(x-y)| \le \prod_\mu M_\mu(\ell_\mu + 1)$. □

When several convolutions act in the same direction, they are first merged into one directional convolution, using the composition rule above. For example, Q_σ itself, $\iota\, a_\mu a_\nu b$, and $\iota\, b\, g_\mu$ satisfy the hypotheses with constants depending only on d. A shift τ_μ composed with a convolution only moves the support of the kernel, so it does not change the bound.

3 Poincaré inequalities and scalar coercivity

The Laplacian $-\Delta$ on $\mathbb{T}_\eta$ has eigenvalues as small as $(2\pi/n)^2$. So it controls a field only up to modes that vary on the scale of the whole torus. The averaging term $a\|Q_\sigma f\|^2$ supplies the missing control on the unit scale. The precise statement is a Poincaré inequality on unit blocks, uniform in η.

Lemma 3.1 (Block Poincaré inequality)**.** *For every block $B = B(y)$ and every scalar field f,*

$$\|f - (Q_0 f)(y)\|_B^2 \le \frac{d}{2}\,\eta^d \sum_{b \subset B} |\partial f(b)|^2.$$

Here the sum runs over all fine bonds $b = (x, x + \eta e_\mu)$ with both endpoints in B, and $\partial f(b) = \partial_\mu f(x)$. Consequently, for every scalar field f,

$$\|f - Q_0^* Q_0 f\|^2 \le \frac{d}{2}\|\partial f\|^2, \qquad \|f\|^2 \le \frac{d}{2}\|\partial f\|^2 + \|Q_0 f\|^2. \tag{3.1}$$

Proof. Let $\bar f = (Q_0 f)(y)$ be the mean of f over B. Since B has N^d points and volume 1,

$$\|f - \bar f\|_B^2 = \eta^d \sum_{x \in B} |f(x) - \bar f|^2 = \frac{\eta^{2d}}{2} \sum_{x,x' \in B} |f(x) - f(x')|^2.$$

For $x, x' \in B$ we join x' to x by the lattice path that first moves in direction 1 from x'_1 to x_1, then in direction 2 from x'_2 to x_2, and so on. It stays in B and has at most $d(N-1)$ steps. Then $f(x) - f(x')$ is a sum of at most dN terms $\pm\eta\,\partial f(b)$, and by Cauchy–Schwarz

$$|f(x) - f(x')|^2 \le dN\eta^2 \sum_{b \in \mathrm{path}(x',x)} |\partial f(b)|^2 = d\eta \sum_{b \in \mathrm{path}(x',x)} |\partial f(b)|^2.$$

A bond $b = (z, z + \eta e_\mu)$ lies on the μ-th segment of the path only if $z_\nu = x_\nu$ for $\nu < \mu$ and $z_\nu = x'_\nu$ for $\nu > \mu$. The remaining free coordinates are x'_ν for $\nu < \mu$, x_ν for $\nu > \mu$, and the pair (x_μ, x'_μ). Hence at most $N^{\mu-1} \cdot N^{d-\mu} \cdot N^2 = N^{d+1}$ pairs (x, x') use b. Therefore

$$\|f - \bar f\|_B^2 \le \frac{\eta^{2d}}{2}\, d\eta\, N^{d+1} \sum_{b \subset B} |\partial f(b)|^2 = \frac{d}{2}\, \eta^d \sum_{b \subset B} |\partial f(b)|^2,$$

using $\eta N = 1$. Summing over blocks gives the first inequality in (3.1), since bonds between different blocks only add to the right-hand side. The second follows from the orthogonal decomposition $f = (f - Q_0^* Q_0 f) + Q_0^* Q_0 f$ and $\|Q_0^* Q_0 f\| = \|Q_0 f\|$ (Lemma 2.6(b)). □

Lemma 3.2 (Comparison of averages)**.** *For $\mu = 1, \dots, d$ and every scalar field f,*

$$\|Q_\mu f - Q_0 f\| \le 3 \cdot 2^d \, \|\partial_\mu f\|, \qquad \|D_\nu Q_\sigma f\| \le 6^d \, \|\partial_\nu f\| \quad (\sigma = 0, \dots, d).$$

Proof. By Lemma 2.4(e), $Q_\mu f - Q_0 f = \iota\, b(a_\mu - 1) f = \iota\, b\, g_\mu\, \partial_\mu f$. In direction μ the operator $a_\mu g_\mu$ has a kernel bounded by $1 + \eta \le 2$ and supported in an interval of length 2. In every other direction the kernel is $\mathbf{1}_{[0,1)}$. Lemma 2.7 gives the constant $2 \cdot 3 \cdot 2^{d-1}$. Similarly, $D_\nu Q_\sigma f = \iota\, a_\nu \mathcal{K}_\sigma \partial_\nu f$ by Lemma 2.6(d). This is a product of at most three box averages in each direction, with kernels bounded by 1 and supported in intervals of length at most 3. Lemma 2.7 gives a factor at most 4 in direction ν and at most 3 in each other direction. □

Corollary 3.3 (Poincaré inequality for all averages)**.** *There is C_P, depending only on d, such that for every $\sigma \in \{0, \dots, d\}$ and every scalar field f,*

$$\|f\|^2 \le C_P(\|\partial f\|^2 + \|Q_\sigma f\|^2).$$

For gauge fields, $\|A\|^2 \le C_P(\|\partial A\|^2 + \|\mathbf{Q} A\|^2)$.

Proof. By (3.1) and Lemma 3.2, $\|f\|^2 \le \frac{d}{2}\|\partial f\|^2 + 2\|Q_\sigma f\|^2 + 2 \cdot 9 \cdot 4^d \|\partial f\|^2$. Apply this to each component A_μ with $\sigma = \mu$ and sum over μ. □

Corollary 3.4 (Scalar coercivity)**.** *Let $a > 0$ and $c_0 = \min(1, a)/(2 + 2C_P)$. For every σ and every scalar field f,*

$$\langle f, (-\Delta + a Q_\sigma^* Q_\sigma) f \rangle = \|\partial f\|^2 + a\|Q_\sigma f\|^2 \ge c_0(\|f\|^2 + \|\partial f\|^2).$$

In particular $-\Delta + a Q_\sigma^* Q_\sigma$ is invertible, and its inverse G_σ satisfies $0 < G_\sigma \le c_0^{-1}$.

Proof. $\|\partial f\|^2 + a\|Q_\sigma f\|^2 \ge \min(1, a)(\frac{1}{2}\|\partial f\|^2 + \frac{1}{2}(\|\partial f\|^2 + \|Q_\sigma f\|^2))$. Then use Corollary 3.3 on the last bracket. □

This is the scalar half of Theorem A. The next lemma is used in the gauge-field part. It concerns fields with vanishing block averages, i.e. generators of the residual gauge transformations.

Lemma 3.5 (Fields with vanishing averages)**.** *If $Q_0\lambda = 0$, then*

$$\|\lambda\| \le \sqrt{d/2}\,\|\partial\lambda\|, \qquad \|\partial\lambda\| \le \sqrt{d/2}\,\|\Delta\lambda\|.$$

In particular $-\Delta$ is injective on $\mathcal{N}(Q_0) = \{\lambda : Q_0\lambda = 0\}$.

Proof. The first inequality is (3.1) with $Q_0\lambda = 0$. For the second, $\|\partial\lambda\|^2 = \langle\lambda, -\Delta\lambda\rangle \le \|\lambda\|\|\Delta\lambda\| \le \sqrt{d/2}\,\|\partial\lambda\|\,\|\Delta\lambda\|$. □

4 Two Fourier estimates

Fourier analysis plays only a small role in this paper. Its main use is to show that averaged Green functions such as $Q_\sigma G_\sigma Q_\sigma^*$ are bounded *below* uniformly in η. Upper bounds and decay are proved in position space. We also need one estimate on "biharmonic extensions" in the gauge-field coercivity proof.

4.1 Fourier transform and Bloch decomposition

The dual of the fine torus is $\mathbb{T}_\eta^* = \big((2\pi/n)\mathbb{Z}/(2\pi/\eta)\mathbb{Z}\big)^d$. We represent it by the momenta p with $-\pi/\eta < p_\mu \le \pi/\eta$. For a scalar field f we put

$$\hat f(p) = \eta^d \sum_{x\in\mathbb{T}_\eta} e^{-\mathrm{i}p\cdot x} f(x), \qquad f(x) = n^{-d} \sum_{p\in\mathbb{T}_\eta^*} e^{\mathrm{i}p\cdot x}\hat f(p), \qquad \|f\|^2 = n^{-d}\sum_p |\hat f(p)|^2.$$

On the unit torus, $\hat g(q) = \sum_{y\in\mathbb{T}} e^{-\mathrm{i}q\cdot y} g(y)$ for $q \in \mathbb{T}^*$, which we represent by $-\pi < q_\mu \le \pi$, and $\|g\|^2 = n^{-d}\sum_q |\hat g(q)|^2$.

Every $p \in \mathbb{T}_\eta^*$ can be written uniquely as $p = q + 2\pi l$ with $q \in \mathbb{T}^*$ and $l \in \mathbb{Z}^d$. For fixed q exactly N^d vectors l occur. This is the *Bloch decomposition*.

A directional convolution with kernel κ in direction μ acts in momentum space as multiplication by $\tilde\kappa(p_\mu)$, where $\tilde\kappa(\theta) = \sum_s \eta\,\kappa(s)e^{\mathrm{i}\theta s}$. Its adjoint has symbol $\overline{\tilde\kappa}$. In particular

$$\widehat{\partial_\mu f}(p) = \frac{e^{\mathrm{i}\eta p_\mu} - 1}{\eta}\,\hat f(p), \qquad \widehat{-\Delta f}(p) = \hat\Delta(p)\hat f(p), \quad \hat\Delta(p) = \sum_\mu \frac{4}{\eta^2}\sin^2\frac{\eta p_\mu}{2},$$

and a_μ has symbol $\alpha(p_\mu)$, with

$$\alpha(\theta) = \eta\sum_{j=0}^{N-1} e^{\mathrm{i}j\eta\theta} = \eta\,\frac{e^{\mathrm{i}\theta}-1}{e^{\mathrm{i}\eta\theta}-1}, \qquad |\alpha(\theta)| = \frac{|\sin(\theta/2)|}{N|\sin(\eta\theta/2)|}.$$

Restriction to the unit lattice and its adjoint satisfy the *Poisson summation formula*:

$$\widehat{\iota f}(q) = \sum_l \hat f(q + 2\pi l), \qquad \widehat{\iota^* g}(p) = \hat g(q) \quad \text{for } p = q + 2\pi l. \tag{4.1}$$

The first identity follows by inserting the inversion formula for f into $\sum_y e^{-\mathrm{i}q\cdot y} f(y)$ and using $\sum_{y\in\mathbb{T}} e^{\mathrm{i}(p-q)\cdot y} = n^d \mathbf{1}[p - q \in 2\pi\mathbb{Z}^d]$. The second is immediate from the definition of ι^*.

Lemma 4.1 (Symbol bounds)**.** *Let $|\theta| \le \pi/\eta$.*

(a) $\frac{4}{\pi^2}\theta^2 \le \frac{4}{\eta^2}\sin^2\frac{\eta\theta}{2} \le \theta^2$. *Hence* $\frac{4}{\pi^2}|p|^2 \le \hat\Delta(p) \le |p|^2$ *on the fundamental domain.*

(b) $|\alpha(\theta)| \le 1$, *and* $|\alpha(\theta)| \ge 2/\pi$ *if* $|\theta| \le \pi$.

(c) If $\theta = q + 2\pi l$ *with* $|q| \le \pi$ *and* $l \in \mathbb{Z} \setminus \{0\}$, *then* $|\theta| \ge \pi|l|$ *and* $|\alpha(\theta)| \le |q|/(2|l|)$.

Proof. (a) uses $\frac{2}{\pi}t \le \sin t \le t$ for $0 \le t \le \pi/2$. (b) The first bound is the triangle inequality. For $|\theta| \le \pi$, $|\sin(\theta/2)| \ge |\theta|/\pi$ and $N|\sin(\eta\theta/2)| \le |\theta|/2$. (c) $|\theta| \ge 2\pi|l| - |q| \ge \pi|l|$. Moreover $|\sin(\theta/2)| = |\sin(q/2)| \le |q|/2$, and $N|\sin(\eta\theta/2)| \ge N\frac{2}{\pi}\frac{\eta|\theta|}{2} = |\theta|/\pi \ge |l|$. □

4.2 The lower bound

Lemma 4.2 (Fourier lower bound)**.** *Let* K *be a translation-invariant operator on scalar fields of* $\mathbb{T}_\eta$ *with symbol* $\hat{K}(p) \ge 0$ *on* $\mathbb{T}^*_\eta$, *and let* $\mathcal{K}$ *be a product convolution with symbol* $\kappa(p)$. *Represent each* $q \in \mathbb{T}^*$ *by* $-\pi < q_\mu \le \pi$; *this is a point of* $\mathbb{T}^*_\eta$. *Then for every field* g *on* $\mathbb{T}$,

$$\langle \mathcal{K}^*\iota^* g, K\mathcal{K}^*\iota^* g\rangle \;\ge\; m\,\|g\|^2, \qquad m = \min_{q\in\mathbb{T}^*} \hat{K}(q)|\kappa(q)|^2.$$

Proof. By (4.1), $h = \mathcal{K}^*\iota^* g$ has $\hat{h}(q + 2\pi l) = \overline{\kappa(q + 2\pi l)}\,\hat{g}(q)$. Therefore

$$\langle h, Kh\rangle = n^{-d}\sum_q |\hat{g}(q)|^2 \sum_l \hat{K}(q + 2\pi l)|\kappa(q + 2\pi l)|^2 \;\ge\; n^{-d}\sum_q |\hat{g}(q)|^2\,\hat{K}(q)|\kappa(q)|^2,$$

where we kept only the term $l = 0$. The claim follows from Parseval on $\mathbb{T}$. □

Corollary 4.3. *For every* $\sigma \in \{0, \dots, d\}$ *and* $a > 0$,

$$Q_\sigma(-\Delta + 2a)^{-1}Q^*_\sigma \;\ge\; c_F, \qquad c_F = \Big(\frac{2}{\pi}\Big)^{2d+2}\frac{1}{d\pi^2 + 2a}.$$

Proof. Apply Lemma 4.2 with $K = (-\Delta + 2a)^{-1}$ and $\mathcal{K} = \mathcal{K}_\sigma$. The symbol of $\mathcal{K}_\sigma$ is a product of at most $d + 1$ factors $\alpha(p_\nu)$, each of modulus at least $2/\pi$ on $[-\pi, \pi]^d$, which contains the representatives of $\mathbb{T}^*$. Moreover $\hat{\Delta}(p) \le |p|^2 \le d\pi^2$ there. □

4.3 Biharmonic extensions

The following estimate is used in Section 9. Given a unit-lattice field ω with mean zero, the field $\lambda = (-\Delta)^{-2}Q^*_0\omega$ is the smoothest (in the sense of $\|\Delta\lambda\|$) fine field with prescribed block averages $Q_0\lambda$, which then have mean zero. We will see that this is exactly the gradient part of a gauge field in Bałaban's gauge. Its energy is controlled by the unit-lattice gradient of its averages.

Lemma 4.4 (Biharmonic extensions)**.** *There is* C_B, *depending only on* d, *with the following property. Let* ω *be a field on* $\mathbb{T}$ *with mean zero, and let* $\lambda = (-\Delta)^{-2}Q^*_0\omega$ *(the inverse taken on mean-zero fields). Then*

$$\|\partial\lambda\| \le C_B\,\|d_1 Q_0\lambda\|.$$

Proof. Let $\beta(p) = \prod_\nu \alpha(p_\nu)$ be the symbol of b. By (4.1), $\hat{\lambda}(p) = \overline{\beta(p)}\,\hat{\omega}(q)\,\hat{\Delta}(p)^{-2}$ for $p = q + 2\pi l \ne 0$, and $\hat{\lambda}(0) = 0$. For $j \in \{2, 3\}$ and $q \in \mathbb{T}^* \setminus \{0\}$ put $S_j(q) = \sum_l |\beta(q+2\pi l)|^2\hat{\Delta}(q+2\pi l)^{-j}$. Then

$$\|\partial\lambda\|^2 = \langle\lambda, -\Delta\lambda\rangle = n^{-d}\sum_{q\ne 0} |\hat{\omega}(q)|^2 S_3(q).$$

The term $q = 0$ is absent, because $\hat{\omega}(0) = 0$, and because for $l \ne 0$ some factor $\alpha(2\pi l_\nu)$ vanishes. Similarly, $g = Q_0\lambda = \iota b\lambda$ has $\hat{g}(q) = \sum_l \beta(q + 2\pi l)\hat{\lambda}(q + 2\pi l) = \hat{\omega}(q)S_2(q)$, and

$$\|d_1 g\|^2 = n^{-d}\sum_{q\ne 0} \hat{\Delta}_1(q)\,S_2(q)^2\,|\hat{\omega}(q)|^2, \qquad \hat{\Delta}_1(q) = \sum_\nu 4\sin^2\frac{q_\nu}{2}.$$

It therefore suffices to show $S_3(q) \le C_B^2 \hat\Delta_1(q) S_2(q)^2$ for $q \in [-\pi,\pi]^d \setminus \{0\}$.

By Lemma 4.1, keeping only the term $l = 0$ in S_2, we get $S_2(q) \ge (2/\pi)^{2d}|q|^{-4}$, and $\hat\Delta_1(q) \ge \frac{4}{\pi^2}|q|^2$. Hence

$$\hat\Delta_1(q) S_2(q)^2 \ge \frac{4}{\pi^2}\Big(\frac{2}{\pi}\Big)^{4d}|q|^{-6}.$$

In S_3, the term $l = 0$ is at most $(\pi/2)^6|q|^{-6}$. For $l \neq 0$, Lemma 4.1(a),(c) gives $\hat\Delta(q+2\pi l) \ge \frac{4}{\pi^2}|q+2\pi l|^2 \ge 4|l|^2 \ge 4$ and $|\beta(q+2\pi l)|^2 \le \prod_{\nu: l_\nu \neq 0} \frac{\pi^2}{4l_\nu^2}$. Hence

$$\sum_{l\neq 0} \frac{|\beta(q+2\pi l)|^2}{\hat\Delta(q+2\pi l)^3} \le \frac{1}{64}\prod_{\nu=1}^{d}\Big(1 + 2\sum_{m\ge 1}\frac{\pi^2}{4m^2}\Big) = \frac{1}{64}\Big(1+\frac{\pi^4}{12}\Big)^d.$$

Since $|q|^2 \le d\pi^2$, this constant is at most $\frac{1}{64}(1+\pi^4/12)^d(d\pi^2)^3|q|^{-6}$, and the lemma follows. □

5 Operators with exponential decay and the Combes–Thomas method

Exponential decay will first be established in L^2, between unit blocks. This section develops the bookkeeping for this notion and the basic mechanism that produces it: if K is coercive and conjugation by $e^{\alpha\rho}$ changes K only by $O(\alpha)$, then K^{-1} decays exponentially.

5.1 Block decay

We work with four kinds of field spaces: scalar fields and gauge fields on $\mathbb{T}_\eta$, and scalar fields and gauge fields on $\mathbb{T}$. On each of them we have the *localization projections* $\mathbf{1}_y$, $y \in \mathbb{T}$. On the fine lattice, $\mathbf{1}_y$ is multiplication by the indicator of $B(y)$ (for gauge fields: of the bonds $(x, x+\eta e_\mu)$ with $x \in B(y)$). On the unit lattice, $\mathbf{1}_y$ keeps the value(s) at y. In all cases $\sum_y \mathbf{1}_y = I$.

Definition 5.1. A linear operator K between two such spaces has *decay* (C,γ) if

$$\|\mathbf{1}_y K \mathbf{1}_{y'}\| \le C e^{-\gamma|y-y'|} \qquad \text{for all } y, y' \in \mathbb{T}.$$

For operators between unit-lattice spaces, $\|\mathbf{1}_y K \mathbf{1}_{y'}\|$ is the absolute value (or matrix norm) of the kernel $K(y,y')$, so decay is a pointwise kernel bound. We put

$$S_\gamma = \sum_{z\in\mathbb{Z}^d} e^{-\gamma|z|}.$$

Then $\sum_{y'\in\mathbb{T}} e^{-\gamma|y-y'|} \le S_\gamma$ for every y, uniformly in n, because distinct y' have distinct nearest representatives of $y'-y$ in $\mathbb{Z}^d$.

Lemma 5.2 (Calculus of decay)**.** *(a) If K has decay (C,γ), then $\|K\| \le C S_\gamma$.*

(b) If K_1 and K_2 have decay (C_1,γ) and (C_2,γ), then K_1K_2 has decay $(C_1C_2S_{\gamma/2}, \gamma/2)$.

(c) If K has decay (C,γ) then K^* has decay (C,γ). Sums behave in the obvious way, and decay (C,γ) implies decay (C,γ') for $\gamma' \le \gamma$.

(d) If $\mathbf{1}_y K \mathbf{1}_{y'} = 0$ for $|y-y'| > r$ and $\|K\| \le M$, then K has decay $(Me^{\gamma r},\gamma)$ for every $\gamma > 0$.

Proof. (a) $\|\mathbf{1}_y K f\| \le \sum_{y'} C e^{-\gamma|y-y'|}\|\mathbf{1}_{y'} f\|$. The matrix $(e^{-\gamma|y-y'|})$ has row and column sums at most S_γ, so by the Schur test it defines an operator on $\ell^2(\mathbb{T})$ of norm at most S_γ. Hence $\|Kf\|^2 = \sum_y \|\mathbf{1}_y K f\|^2 \le C^2 S_\gamma^2 \sum_{y'} \|\mathbf{1}_{y'} f\|^2$.

(b) $\mathbf{1}_y K_1 K_2 \mathbf{1}_{y'} = \sum_{y''} \mathbf{1}_y K_1 \mathbf{1}_{y''} K_2 \mathbf{1}_{y'}$. By the triangle inequality $|y-y''| + |y''-y'| \ge \frac12|y-y'| + \frac12|y-y''|$, so the norm is at most $C_1C_2 e^{-\gamma|y-y'|/2}\sum_{y''} e^{-\gamma|y-y''|/2}$.

(c) and (d) are immediate. □

5.2 Conjugation by exponential weights

A function $\rho : \mathbb{T}_\eta \to \mathbb{R}$ is 1-*Lipschitz* if $|\rho(x) - \rho(x')| \le |x - x'|$ for all x, x'. A typical example is $\rho(x) = \operatorname{dist}(x, B(y'))$, the torus distance to a block. For $\alpha \in \mathbb{R}$, multiplication by $e^{\alpha\rho}$ acts on all four field spaces. On unit-lattice fields it multiplies the value at y by $e^{\alpha\rho(y)}$, which is consistent because $\mathbb{T} \subset \mathbb{T}_\eta$. For an operator K we write

$$K_\alpha = e^{\alpha\rho} K e^{-\alpha\rho}.$$

Then $(K_1K_2)_\alpha = (K_1)_\alpha(K_2)_\alpha$, $(K^*)_\alpha = (K_{-\alpha})^*$, and $(K^{-1})_\alpha = (K_\alpha)^{-1}$. Since ι commutes with multiplication operators, $(\iota)_\alpha = \iota$.

Lemma 5.3 (Conjugation of decaying operators)**.** *Let K have decay (C, γ) and let ρ be 1-Lipschitz. For $|\alpha| \le \gamma/2$,*

$$\|K_\alpha - K\| \le C\, c_1(\gamma)\, |\alpha|, \qquad c_1(\gamma) = e^{\gamma\sqrt{d}}(2\sqrt{d} + 4/\gamma) S_{\gamma/4}.$$

Proof. Fix y, y', put $X = \mathbf{1}_y K \mathbf{1}_{y'}$, and write $e^{\alpha\rho} = e^{\alpha\rho(y)} E$ on the support of $\mathbf{1}_y$. Here E is multiplication by $e^{\alpha(\rho - \rho(y))}$. Since points of $B(y)$ are within distance $\sqrt{d}$ of y, $\|E - 1\| \le |\alpha|\sqrt{d}\, e^{|\alpha|\sqrt{d}}$ and $\|E\| \le e^{|\alpha|\sqrt{d}}$. Define E' similarly at y'. Then $\mathbf{1}_y(K_\alpha - K)\mathbf{1}_{y'} = sEXE'^{-1} - X$ with $s = e^{\alpha(\rho(y) - \rho(y'))}$. Since $|s - 1| \le |\alpha||y - y'| e^{|\alpha||y-y'|}$, the decomposition

$$sEXE'^{-1} - X = (s-1)EXE'^{-1} + (E-1)XE'^{-1} + X(E'^{-1} - 1)$$

gives

$$\|\mathbf{1}_y(K_\alpha - K)\mathbf{1}_{y'}\| \le C|\alpha|\, e^{2|\alpha|\sqrt{d}}(|y - y'| + 2\sqrt{d}) e^{-(\gamma - |\alpha|)|y-y'|}.$$

For $|\alpha| \le \gamma/2$ we have $e^{2|\alpha|\sqrt{d}} \le e^{\gamma\sqrt{d}}$ and $(t + 2\sqrt{d})e^{-\gamma t/2} \le (2\sqrt{d} + 4/\gamma)e^{-\gamma t/4}$. Hence $K_\alpha - K$ has decay $(C|\alpha| e^{\gamma\sqrt{d}}(2\sqrt{d} + 4/\gamma), \gamma/4)$, and Lemma 5.2(a) concludes. □

Lemma 5.4 (Conjugated derivatives)**.** *Let ρ be 1-Lipschitz and $|\alpha| \le 1$. Then $(\partial_\mu)_\alpha = \partial_\mu + Y$ and $(\partial_\mu^*)_\alpha = \partial_\mu^* + Y'$, where Y, Y' are operators of norm at most $e|\alpha|$. On gauge fields the same holds componentwise.*

Proof. $(e^{\alpha\rho}\partial_\mu e^{-\alpha\rho} u)(x) = \partial_\mu u(x) + \eta^{-1}(e^{\alpha(\rho(x) - \rho(x + \eta e_\mu))} - 1) u(x + \eta e_\mu)$. Moreover $|e^t - 1| \le |t| e^{|t|}$ with $|t| \le |\alpha|\eta$. The computation for ∂_μ^* is the same. □

5.3 The Combes–Thomas method

We give two versions. The first is for bounded operators, typically on the unit lattice. The second is for fine-lattice operators containing $-\Delta$, whose norm is of order η^{-2}.

Lemma 5.5 (Combes–Thomas, bounded version)**.** *Let K act on one of the four field spaces, have decay (C, γ), and satisfy $\langle f, Kf\rangle \ge c\|f\|^2$ for all (real) fields f. Put $\gamma' = \min\{\gamma/2,\ c/(2Cc_1(\gamma))\}$. Then K is invertible and K^{-1} has decay $(2c^{-1}e^{2\gamma'\sqrt{d}}, \gamma')$.*

Proof. Fix y', let $\rho(x) = \operatorname{dist}(x, B(y'))$ and $\alpha = \gamma'$. By Lemma 5.3, $\langle f, K_\alpha f\rangle \ge c\|f\|^2 - \frac{c}{2}\|f\|^2$. Hence K_α is invertible with $\|K_\alpha^{-1}\| \le 2/c$. Here we use that in finite dimensions $\langle f, Xf\rangle \ge \delta\|f\|^2$ implies $\|Xf\| \ge \delta\|f\|$. Now $K^{-1} = e^{-\alpha\rho} K_\alpha^{-1} e^{\alpha\rho}$. On the support of $\mathbf{1}_{y'}$ we have $\rho = 0$. On the support of $\mathbf{1}_y$ we have $\rho \ge |y - y'| - 2\sqrt{d}$. Therefore $\|\mathbf{1}_y K^{-1} \mathbf{1}_{y'}\| \le e^{-\alpha(|y-y'| - 2\sqrt{d})}\, 2/c$. □

Lemma 5.6 (Neumann series)**.** *Let $\mathfrak{N}$ have decay (C, γ) and $\|\mathfrak{N}\| \le \theta < 1$. Then $(I + \mathfrak{N})^{-1}$ has decay $(2(1-\theta)^{-1}e^{2\gamma'\sqrt{d}}, \gamma')$ with $\gamma' = \min\{\gamma/2, (1-\theta)/(2Cc_1(\gamma))\}$.*

Proof. As above, with $\|\mathfrak{R}_\alpha\| \le \theta + (1-\theta)/2$, so that $\|(I+\mathfrak{R}_\alpha)^{-1}\| \le 2/(1-\theta)$. □

This lemma is the form in which exponential decay is used in the variational problem, cf. [8, (68)–(71)].

For the fine-lattice operators we write ∂ for the collection $(\partial_\mu)_\mu$ acting on scalar fields, or componentwise on gauge fields, and ∂^* for its adjoint.

Lemma 5.7 (Combes–Thomas, form version)**.** *Let $\mathfrak{A}$ act on scalar or gauge fields on $\mathbb{T}_\eta$. Suppose that:*

(i) $\langle u, \mathfrak{A}u\rangle \ge c\,(\|u\|^2 + \|\partial u\|^2)$ for all u;

(ii) there are $C_1, \alpha_1 > 0$ such that for every 1-Lipschitz ρ, every $|\alpha| \le \alpha_1$ and every u,

$$|\langle u, (\mathfrak{A}_\alpha - \mathfrak{A})u\rangle| \le C_1|\alpha|\,(\|u\|^2 + \|\partial u\|^2).$$

Let $\gamma_0 = \min\{\alpha_1, 1/3, c/(2C_1)\}$. Then the four operators $\mathfrak{A}^{-1}$, $\partial\mathfrak{A}^{-1}$, $\mathfrak{A}^{-1}\partial^*$ and $\partial\mathfrak{A}^{-1}\partial^*$ have decay (C_0, γ_0), with $C_0 = 2(1+\sqrt{d})^2 c^{-1} e^{2\gamma_0\sqrt{d}}$.

Proof. Let $|\alpha| \le \gamma_0$. By (i) and (ii), $\langle u, \mathfrak{A}_\alpha u\rangle \ge \frac{c}{2}(\|u\|^2 + \|\partial u\|^2)$, so $\mathfrak{A}_\alpha$ is invertible. Let $u = \mathfrak{A}_\alpha^{-1}h$. Then $\frac{c}{2}(\|u\|^2 + \|\partial u\|^2) \le \|u\|\|h\|$. This gives $\|u\| \le \frac{2}{c}\|h\|$ and then $\|\partial u\|^2 \le \frac{2}{c}\|u\|\|h\| \le \frac{4}{c^2}\|h\|^2$. If instead $h = \partial^* g$, then $\frac{c}{2}(\|u\|^2 + \|\partial u\|^2) \le \langle \partial u, g\rangle \le \|\partial u\|\|g\|$, which gives $\|\partial u\| \le \frac{2}{c}\|g\|$ and $\|u\| \le \frac{2}{c}\|g\|$. Thus

$$\|\mathfrak{A}_\alpha^{-1}\|,\ \|\partial\mathfrak{A}_\alpha^{-1}\|,\ \|\mathfrak{A}_\alpha^{-1}\partial^*\|,\ \|\partial\mathfrak{A}_\alpha^{-1}\partial^*\| \le \frac{2}{c}.$$

Now fix y', let $\rho = \operatorname{dist}(\cdot, B(y'))$ and $\alpha = \gamma_0$. By Lemma 5.4, $\partial e^{-\alpha\rho} = e^{-\alpha\rho}(\partial + Y)$ and $e^{\alpha\rho}\partial^* = (\partial^* + Y'^*)e^{\alpha\rho}$, where Y' comes from $(\partial)_{-\alpha}$. Here $Y = (Y_\mu)_\mu$ and Y' consist of d operators, each of norm at most $e\alpha$ by Lemma 5.4, so $\|Y\|, \|Y'\| \le \sqrt{d}\,e\alpha$. Hence, for instance,

$$\mathbf{1}_y \partial\mathfrak{A}^{-1}\partial^*\mathbf{1}_{y'} = \mathbf{1}_y e^{-\alpha\rho}(\partial + Y)\mathfrak{A}_\alpha^{-1}(\partial^* + Y'^*)\mathbf{1}_{y'},$$

where we used $e^{\alpha\rho}\mathbf{1}_{y'} = \mathbf{1}_{y'}$. The middle operator has norm at most $\frac{2}{c}(1+\sqrt{d}\,e\alpha)^2 \le \frac{2}{c}(1+\sqrt{d})^2$, because $e\alpha \le 1$, and $\|\mathbf{1}_y e^{-\alpha\rho}\| \le e^{-\alpha(|y-y'|-2\sqrt{d})}$. The other three operators are treated in the same way. □

The hypothesis (ii) is verified with the help of the following observation.

Lemma 5.8 (Conjugation of quadratic forms)**.** *Let $X = X_0 - Z$, where $X_0 = \sum_j c_j\,\partial_{\mu_j}^{(*)}$ is a finite linear combination of the operators ∂_μ, ∂_μ^* (acting componentwise on scalar or gauge fields, with values in the appropriate form degree) with scalar coefficients $\sum_j |c_j| \le M$, and Z is bounded with $\|Z\| \le M$ and $\|Z_\alpha - Z\| \le M|\alpha|$ for $|\alpha| \le \alpha_1 \le 1$. Then for $|\alpha| \le \alpha_1$,*

$$|\langle u, ((X^*X)_\alpha - X^*X)u\rangle| \le C_M|\alpha|\,(\|u\|^2 + \|\partial u\|^2),$$

with C_M depending only on M and d.

Proof. By Lemma 5.4, applied to each term of X_0, $X_\alpha = X + W_\alpha$ with $\|W_\alpha\| \le (eM + M)|\alpha|$. Also $\|X_0 u\| \le M\|\partial u\|$. Since $(X^*X)_\alpha = (X_{-\alpha})^* X_\alpha$,

$$\langle u, ((X^*X)_\alpha - X^*X)u\rangle = \langle W_{-\alpha}u, Xu\rangle + \langle Xu, W_\alpha u\rangle + \langle W_{-\alpha}u, W_\alpha u\rangle.$$

Finally, $\|Xu\| \le M(\|\partial u\| + \|u\|)$. □

6 Local regularity on the fine lattice

The Combes–Thomas method gives L^2 bounds between unit blocks. To obtain bounds on values, derivatives and Hölder quotients that are uniform in η, we use an interior regularity estimate for the lattice Laplacian. This is the lattice analogue of the classical fact that a solution of $-\Delta u = F$ with bounded F is $C^{1,\alpha}$ in the interior for every $\alpha < 1$. In this section $d \geq 3$.

6.1 The Green function of the fine lattice

Let g_1 be the Green function of $-\Delta$ on the infinite unit lattice $\mathbb{Z}^d$:

$$g_1(x) = \frac{1}{(2\pi)^d} \int_{[-\pi,\pi]^d} \frac{e^{\mathrm{i}p\cdot x}}{\sum_\mu 4\sin^2(p_\mu/2)}\, dp,$$

which is well defined for $d \geq 3$ and satisfies $-\Delta g_1 = \delta_0$. On $\eta\mathbb{Z}^d$ put $g_\eta(x) = \eta^{2-d} g_1(x/\eta)$. Then the convolution

$$(\mathcal{G}h)(x) = \sum_{x' \in \eta\mathbb{Z}^d} \eta^d\, g_\eta(x - x')\, h(x')$$

satisfies $-\Delta\mathcal{G}h = h$ and $\mathcal{G}(-\Delta h) = h$ for every finitely supported h on $\eta\mathbb{Z}^d$. For the second identity, sum by parts. Since $\mathcal{G}$ is a convolution, it commutes with all lattice derivatives: $\partial_\mu \mathcal{G}h = (\partial_\mu g_\eta) \star h$, and similarly for ∂_μ^*.

Proposition 6.1 (Decay of derivatives)**.** *For every word ∂^β of length $|\beta| \leq 3$ in the operators $\partial_\mu, \partial_\mu^*$,*

$$|\partial^\beta g_\eta(x)| \leq C_\beta\, (|x| + \eta)^{2-d-|\beta|}, \qquad x \in \eta\mathbb{Z}^d,$$

with C_β depending only on d and β.

Proof. By scaling, $\partial^\beta g_\eta(x) = \eta^{2-d-|\beta|}(\nabla^\beta g_1)(x/\eta)$, where ∇^β is the corresponding word of unit-lattice differences. By Theorem A.1 in Appendix A, $|\nabla^\beta g_1(z)| \leq C_\beta(1 + |z|)^{2-d-|\beta|}$. □

We also need two elementary lattice sums.

Lemma 6.2 (Lattice sums)**.** *Let $x \in \eta\mathbb{Z}^d$ and $R \geq \eta$.*

(a) For $0 \leq s < d$: $\sum_{|x'-x|\leq R} \eta^d (|x - x'| + \eta)^{-s} \leq C_{d,s} R^{d-s}$.

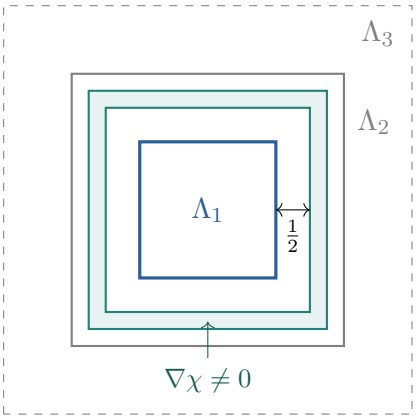
(b) For $\sqrt{d}\,\eta \leq r \leq R$: $\sum_{r<|x'-x|\leq R} \eta^d |x - x'|^{-d} \leq C_d\,(1 + \log(R/r))$.

(c) For $s > d$ and $r \geq \sqrt{d}\,\eta$: $\sum_{|x'-x|>r} \eta^d |x - x'|^{-s} \leq C_{d,s}\, r^{d-s}$.

Proof. Let $W(x') = x' + [-\eta/2, \eta/2)^d$. For $z \in W(x')$ we have $|z-x|+\eta \leq (1+\sqrt{d})(|x-x'|+\eta)$. Hence each term in (a) is at most $(1 + \sqrt{d})^s \int_{W(x')} (|z - x| + \eta)^{-s} dz$. The cubes $W(x')$ are disjoint and contained in the ball of radius $R + \sqrt{d}\eta \leq (1 + \sqrt{d})R$, which gives (a). In (b), $|x - x'| > r \geq \sqrt{d}\eta$ implies $\frac{1}{2}|x - x'| \leq |z - x| \leq \frac{3}{2}|x - x'|$ for $z \in W(x')$. Hence the sum is at most $(3/2)^d \int_{r/2\leq|z-x|\leq 3R/2} |z - x|^{-d} dz = (3/2)^d |S^{d-1}| \log(3R/r)$. The same comparison gives (c). □

6.2 The regularity estimate

For $z \in \mathbb{R}^d$ and $r > 0$ let $\Lambda_r = \{x \in \mathbb{T}_\eta : |x - z|_\infty \leq r\}$. Since $n \geq 8$, Λ_3 can be identified with a subset of $\eta\mathbb{Z}^d$.



$\chi = 1$ inside the inner green square,
$\chi = 0$ outside the outer one.
The commutator supports lie in the green annulus enlarged by at most 2η.
For $\eta \le \frac{1}{16}$ this enlarged annulus is at distance $\ge \frac{1}{4}$ from $\Lambda_{1+2\eta}$.

Figure 2: Geometry of the proof of Theorem 6.3.

Theorem 6.3 (Interior regularity)**.** *Let $d \ge 3$ and $\alpha \in (0,1)$. There is C, depending only on d and α, such that the following holds for every lattice spacing $\eta \le 1$, in particular $\eta = 1/N$ with $N \ge 1$, and all z; for a spacing not of the form $1/N$, Λ_r is the set of $x \in \eta\mathbb{Z}^d$ with $|x - z|_\infty \le r$. Let $u, F : \Lambda_3 \to \mathbb{R}$ and $g = (g_\mu) : \Lambda_3 \to \mathbb{R}^d$ satisfy*

$$-\Delta u = F + \sum_\mu \partial_\mu^* g_\mu \qquad \text{at every point of } \Lambda_2.$$

(a) If $g = 0$: $|u|_{\Lambda_1} + |\partial u|_{\Lambda_1} + [\partial u]_{\alpha,\Lambda_1} \le C(|F|_{\Lambda_2} + \|u\|_{\Lambda_2})$.

(b) In general: $|u|_{\Lambda_1} + [u]_{\alpha,\Lambda_1} \le C(|F|_{\Lambda_2} + |g|_{\Lambda_3} + \|u\|_{\Lambda_2})$.

(c) If $g = 0$: $\max_{\mu,\nu} |\partial_\mu\partial_\nu u|_{\Lambda_1} \le C(|F|_{\Lambda_2} + [F]_{\alpha,\Lambda_2} + \|u\|_{\Lambda_2})$.

(d) (Divergence form with Hölder data.) $|u|_{\Lambda_1} + |\partial u|_{\Lambda_1} + [\partial u]_{\alpha,\Lambda_1} \le C(|F|_{\Lambda_2} + |g|_{\Lambda_3} + [g]_{\alpha,\Lambda_3} + \|u\|_{\Lambda_2})$.

Proof. Step 0: coarse lattices. Let $\eta > \frac{1}{16}$ (for $\eta = 1/N$: $N < 16$). Then $|u(x)| \le \eta^{-d/2}\|u\|_{\Lambda_2} \le 16^{d/2}\|u\|_{\Lambda_2}$ for $x \in \Lambda_2$, and $[v]_{\alpha,X} \le 2\eta^{-\alpha}|v|_X$. A first difference at a point of Λ_1 involves only points of $\Lambda_{1+\eta} \subset \Lambda_2$ and costs a factor at most $2/\eta \le 32$. This proves (a), (b) and (d). For (c) with $\eta \le \frac{1}{2}$, a second difference at a point of Λ_1 involves only points of $\Lambda_{1+2\eta} \subset \Lambda_2$ and costs a factor at most $4/\eta^2$. If $\eta > \frac{1}{2}$ (for $\eta = 1/N$: $N = 1$), the difference $\partial_\mu\partial_\mu u(x)$ involves $u(x + 2\eta e_\mu)$, which may lie outside Λ_2. Put $v = x + \eta e_\mu$. Since $g = 0$ in (c), the equation at $v \in \Lambda_2$ gives

$$\eta^2\partial_\mu\partial_\mu u(x) = (2d-2)\,u(v) - \sum_{\nu\neq\mu} \big(u(v+\eta e_\nu) + u(v-\eta e_\nu)\big) - \eta^2 F(v),$$

and all points on the right lie in Λ_2. The mixed differences $\partial_\mu\partial_\nu u(x)$, $\mu \neq \nu$, involve only points of Λ_2. We assume $\eta \le \frac{1}{16}$ from now on.

Step 1: cutoff. Fix $\chi \in C^\infty(\mathbb{R}^d)$ with $0 \le \chi \le 1$, $\chi = 1$ on $\{|x - z|_\infty \le \frac{3}{2}\}$ and $\chi = 0$ on $\{|x - z|_\infty \ge \frac{7}{4}\}$. Its lattice differences up to order three are bounded by constants depending only on d. Let $v = \chi u$, extended by zero to $\eta\mathbb{Z}^d$. Write $S_\mu u(x) = u(x + \eta e_\mu)$. The discrete Leibniz rules

$$\partial_\mu(\chi u) = \chi\,\partial_\mu u + (\partial_\mu\chi)S_\mu u, \qquad \partial_\mu^*(\varphi w) = \varphi\,\partial_\mu^* w + (\partial_\mu^*\varphi)S_\mu^{-1} w,$$

and $S_\mu^{-1}\partial_\mu u = -\partial_\mu^* u$ give, after a short computation,

$$-\Delta(\chi u) = \chi(-\Delta u) + \sum_\mu (\partial_\mu^* m_\mu + n_\mu), \qquad \begin{aligned} m_\mu &= (\partial_\mu\chi)S_\mu u - (\partial_\mu^*\chi)u, \\ n_\mu &= (\partial_\mu^*\partial_\mu^*\chi)S_\mu^{-1} u. \end{aligned} \tag{6.1}$$

The functions m_μ, n_μ vanish outside the annulus $A = \Lambda_{7/4+2\eta} \setminus \Lambda_{3/2-2\eta}$, where the differences of χ (up to second order, involving points at distance up to 2η) can be non-zero. They involve u only at points of Λ_2, and $|m_\mu| + |n_\mu| \le C|u|$ at neighbouring points. Similarly $\chi\,\partial_\mu^* g_\mu = \partial_\mu^*(\chi g_\mu) - (\partial_\mu^*\chi) S_\mu^{-1} g_\mu$. Hence

$$-\Delta v = \Phi_1 + \Phi_2 + \sum_\mu \partial_\mu^*(\chi g_\mu + m_\mu), \qquad \Phi_1 = \chi F - \sum_\mu (\partial_\mu^*\chi) S_\mu^{-1} g_\mu, \quad \Phi_2 = \sum_\mu n_\mu.$$

All terms vanish outside Λ_2. Since v has finite support,

$$v = \mathcal{G}\Phi_1 + \mathcal{G}\Phi_2 + \sum_\mu (\partial_\mu^* g_\eta) \star (\chi g_\mu) + \sum_\mu (\partial_\mu^* g_\eta) \star m_\mu. \tag{6.2}$$

For $x \in \Lambda_{1+2\eta}$ we have $v(x) = u(x)$. So u, ∂u and $\partial\partial u$ on Λ_1 can be computed from (6.2).

Step 2: the terms supported in the annulus. Points of $\Lambda_{1+2\eta}$ have distance at least $\frac12 - 4\eta \ge \frac14$ from A. By Proposition 6.1, every derivative of order at most three of g_η is bounded by a constant on $\{|x| \ge \frac14\}$. Therefore, for $x \in \Lambda_{1+2\eta}$ and $|\beta| \le 2$,

$$|\partial^\beta \mathcal{G}\Phi_2(x)| + \sum_\mu |\partial^\beta (\partial_\mu^* g_\eta) \star m_\mu(x)| \le C \sum_{x' \in A} \eta^d \Big(|\Phi_2(x')| + \sum_\mu |m_\mu(x')| \Big) \le C\|u\|_{\Lambda_2},$$

by Cauchy–Schwarz. For $|\beta| \le 1$ the derivatives of order one more are also bounded (they involve at most three derivatives of g_η), so these terms and their first derivatives are Lipschitz on Λ_1; for (c) only the sup bounds with $|\beta| = 2$ are needed. A Lipschitz function on Λ_1 is Hölder: join x, x' by a monotone lattice path inside Λ_1, of length at most $\sqrt{d}|x - x'| \le 2d$.

Step 3: the near terms in (a) and (b). Here Φ_1 and χg are bounded by $C(|F|_{\Lambda_2} + |g|_{\Lambda_3})$ and supported in Λ_2. By Lemma 6.2(a) with $s = d-2$ and $s = d-1$, for $x \in \Lambda_{1+2\eta}$,

$$|\mathcal{G}\Phi_1(x)| + |\partial\mathcal{G}\Phi_1(x)| + |(\partial^* g_\eta) \star (\chi g)(x)| \le C(|F|_{\Lambda_2} + |g|_{\Lambda_3}).$$

This proves the sup bounds in (a) and (b), and shows that $\mathcal{G}\Phi_1$ is Lipschitz on Λ_1. It remains to estimate the Hölder quotient of a convolution $k \star h$ with $|h| \le M$ supported in Λ_2. Here $k = \partial_\mu g_\eta$ in (a), and $k = \partial_\mu^* g_\eta$ in (b). In both cases $|k(y)| \le C(|y| + \eta)^{1-d}$, and every first difference of k is bounded by $C(|y| + \eta)^{-d}$. Let $x, x' \in \Lambda_1$, $\delta = |x - x'| \ge \eta$. Split the sum over x'' at $|x - x''| = 4\sqrt{d}\,\delta$. On the near part, by Lemma 6.2(a) with $s = d-1$ (note $|x' - x''| \le (4\sqrt{d}+1)\delta$),

$$\sum_{|x-x''| \le 4\sqrt{d}\delta} \eta^d (|k(x - x'')| + |k(x' - x'')|) \le C\delta,$$

On the far part we join x to x' by a monotone lattice path of at most $\sqrt{d}\delta/\eta$ steps. Every point w of the path, and its neighbours, satisfies $|w - x''| \ge \frac12 |x - x''|$. Hence $|k(x-x'') - k(x'-x'')| \le C\sqrt{d}\,\delta\,|x - x''|^{-d}$, and by Lemma 6.2(b)

$$\sum_{4\sqrt{d}\delta < |x-x''| \le 5\sqrt{d}} \eta^d |k(x - x'') - k(x' - x'')| \le C\delta(1 + \log(1/\delta)) \le C_\alpha \delta^\alpha.$$

Here we may assume $\delta \le 1$, the case $\delta > 1$ being covered by the sup bound. Multiplying by M proves the Hölder bounds in (a) and (b).

Step 4: second derivatives, (c). Fix $x \in \Lambda_1$ and let $\tilde\chi \in C_c^\infty$, with $0 \le \tilde\chi \le 1$, be equal to 1 on $\{|x' - z|_\infty \le \frac74\}$ and supported in $\{|x' - z|_\infty < \frac{15}{8}\}$. Write $\chi F(x'') = [\chi F(x'') - \chi F(x)\tilde\chi(x'')] + \chi F(x)\tilde\chi(x'')$. Then

$$\partial_\mu\partial_\nu \mathcal{G}(\chi F)(x) = \sum_{x''} \eta^d (\partial_\mu\partial_\nu g_\eta)(x - x'') [\chi F(x'') - \chi F(x)\tilde\chi(x'')] + \chi F(x)\, \mathcal{G}(\partial_\mu\partial_\nu\tilde\chi)(x),$$

because convolution commutes with differences. The last term is bounded by $C|F|_{\Lambda_2}$, since $\partial_\mu\partial_\nu\tilde\chi$ is bounded and supported in Λ_2: a second difference at x' involves only points within ∞-distance $2\eta \le \frac18$ of x', and $\frac{15}{8}+\frac18 = 2$. In the sum, the term $x''=x$ vanishes. For $x''\in\Lambda_{7/4}$ the bracket is at most $[\chi F]_{\alpha,\Lambda_2}|x-x''|^\alpha$. For $x''\notin\Lambda_{7/4}$ we have $|x-x''|\ge\frac34$, and the bracket is at most $|F|_{\Lambda_2}$. Using $|\partial_\mu\partial_\nu g_\eta(y)|\le C|y|^{-d}$ for $y\neq0$ and Lemma 6.2(a) with $s=d-\alpha$, the sum is at most $C([\chi F]_{\alpha,\Lambda_2}+|F|_{\Lambda_2})$. Finally $[\chi F]_{\alpha,\Lambda_2}\le[F]_{\alpha,\Lambda_2}+C|F|_{\Lambda_2}$. The second derivatives of $\mathcal{G}\Phi_2$ and of the m-terms were estimated in Step 2.

Step 5: divergence form with Hölder data, (d). By Steps 2 and 3, all terms of (6.2) except $w=\sum_\mu(\partial_\mu^* g_\eta)\star G_\mu$, $G_\mu=\chi g_\mu$, already satisfy the bounds of (d); note that Φ_1 is bounded by $C(|F|_{\Lambda_2}+|g|_{\Lambda_3})$. The field G vanishes outside Λ_2, $|G|\le|g|_{\Lambda_3}$, and $[G]_\alpha := [G]_{\alpha,\eta\mathbb{Z}^d}\le [g]_{\alpha,\Lambda_3}+C|g|_{\Lambda_3}$. Here $[G]_{\alpha,\eta\mathbb{Z}^d}$ is the Hölder quotient over all pairs of points. For $z,z'\in\Lambda_3$ we use $|\chi(z)g(z)-\chi(z')g(z')|\le|g(z)-g(z')|+|\chi(z)-\chi(z')|\,|g(z')|$. If $z\in\Lambda_3$ and $z'\notin\Lambda_3$, then $G(z')=\chi(z')=0$ and $|G(z)|=|\chi(z)-\chi(z')|\,|g(z)|$; if both points lie outside Λ_3, the difference is zero. The bounded Lipschitz norm of χ gives the stated bound. We have $\partial_\nu w=\sum_\mu K_{\nu\mu}\star G_\mu$ with $K=K_{\nu\mu}=\partial_\nu\partial_\mu^* g_\eta$. By Proposition 6.1, $|K(y)|\le C|y|^{-d}$ for $y\neq0$ and every first difference of K at y is bounded by $C(|y|+\eta)^{-d-1}$.

Box cancellation. Let B be a *lattice box*, i.e. the set of lattice points z with $|z-c|_\infty\le r$, and let ∂B be the set of points of B with a lattice neighbour outside B. Let v be a lattice point with $\operatorname{dist}_\infty(v,\partial B)\ge s\ge\eta$. Write $h=\partial_\mu^* g_\eta$, so that $K(v-z)$ is a difference quotient of $h(v-z)$ in the variable z_ν. Summing over $z\in B$ telescopes along the lines parallel to e_ν, so that $\sum_{z\in B}\eta^d K(v-z)$ equals η^{d-1} times a sum of at most $2(2r/\eta+1)^{d-1}$ values $h(v-z')$. Here z' is either a point of ∂B or a lattice point just outside B, one lattice step from a point of ∂B. So $|v-z'|_\infty+\eta\ge s$; if $v\notin B$, then $z'=v$ is possible. The set of points z' depends only on B, not on v. By Proposition 6.1, $|h(y)|\le C(|y|+\eta)^{1-d}$, so each value is at most Cs^{1-d}. Since $\eta\le s$, the number of values times η^{d-1} is at most $2(2r+\eta)^{d-1}\le 2^d(r+s)^{d-1}$. Hence

$$\Big|\sum_{z\in B}\eta^d K(v-z)\Big|\le C\,(r/s+1)^{d-1}. \tag{6.3}$$

Sup bound. In this step z denotes a summation variable; the centre of the cubes Λ_r is fixed. Let Q_0 be the lattice box of radius $\frac52$ with the same centre as Λ_1; it contains the support of G, and ∂Q_0 has ∞-distance at least $\frac32-\eta$ from Λ_1. For $x\in\Lambda_1$, $\sum_z\eta^d K(x-z)G(z)=\sum_{z\in Q_0}\eta^d K(x-z)(G(z)-G(x))+G(x)\sum_{z\in Q_0}\eta^d K(x-z)$. The first sum is at most $C[G]_\alpha\sum_{z\in Q_0\setminus\{x\}}\eta^d|x-z|^{\alpha-d}\le C[G]_\alpha$ by Lemma 6.2(a), the second at most $C|G|$ by (6.3).

Hölder bound. Let $x,x'\in\Lambda_1$ and $\delta=|x-x'|_\infty$; we may assume $\eta\le\delta\le\frac18$. With $S(v)=\sum_{z\in Q_0}\eta^d K(v-z)$,

$$(K\star G)(x)-(K\star G)(x')=\sum_{z\in Q_0}\eta^d\big(K(x-z)-K(x'-z)\big)\big(G(z)-G(x)\big)+G(x)\big(S(x)-S(x')\big).$$

By the telescoping argument, $S(v)$ is η^{d-1} times a sum of $O(\eta^{1-d})$ values $h(v-z')$, over a set of points z' near ∂Q_0 that does not depend on v. These points are at distance ≥1 from Λ_1, where h has bounded derivatives. Hence $|S(x)-S(x')|\le C\delta$. Let $B=\{z:|z-x|_\infty\le 2\delta\}$. On B,

$$\sum_{z\in B}\eta^d|K(x-z)||G(z)-G(x)|\le C[G]_\alpha\sum_{0<|z-x|\le2\sqrt{d}\delta}\eta^d|x-z|^{\alpha-d}\le C[G]_\alpha\,\delta^\alpha,$$

and

$$\sum_{z\in B}\eta^d K(x'-z)(G(z)-G(x))=\sum_{z\in B}\eta^d K(x'-z)(G(z)-G(x'))+(G(x')-G(x))\sum_{z\in B}\eta^d K(x'-z).$$

The first term on the right is at most $C[G]_\alpha \delta^\alpha$, since $|z - x'| \le 3\sqrt{d}\delta$ on B. The second is at most $[G]_\alpha(\sqrt{d}\delta)^\alpha \cdot C$ by (6.3), because x' has ∞-distance at least δ from ∂B. Outside B we have $|x - z|_\infty > 2\delta$; a monotone path from x to x' stays at ∞-distance at least $\frac{1}{2}|x - z|_\infty$ from z, so $|K(x - z) - K(x' - z)| \le C\delta|x - z|^{-d-1}$. Let $r = \max(2\delta, \sqrt{d}\,\eta)$. By Lemma 6.2(c),

$$\sum_{|x-z|>r} \eta^d \, C\delta|x - z|^{-d-1} \, [G]_\alpha |x - z|^\alpha \le C[G]_\alpha \, \delta \cdot r^{\alpha-1} \le C[G]_\alpha \, \delta^\alpha.$$

If $r > 2\delta$, the remaining points with $2\delta < |x - z|_\infty$ and $|x - z| \le \sqrt{d}\eta$ are at most C in number. Each contributes at most $\eta^d \cdot C\eta^{-d} \cdot [G]_\alpha(\sqrt{d}\eta)^\alpha \le C[G]_\alpha\delta^\alpha$. Collecting the terms gives $[\partial w]_{\alpha,\Lambda_1} \le C([G]_\alpha + |G|)$, and (d) follows. □

Remark 6.4. The restriction $d \ge 3$ is used to have a Green function on $\eta\mathbb{Z}^d$ that decays at infinity. For $d = 2$ one would replace $-\Delta$ by $-\Delta + 1$ and move the extra term u to the right-hand side. This requires two additional arguments not given here: uniform-in-η derivative bounds for the Green function of $-\Delta_\eta + 1$ (with a logarithm for $|\beta| = 0$), and an $L^2 \to L^\infty$ step, since the sup norm of u then appears among the data. We do not pursue this.

7 Scalar Green functions

We now combine the three tools for the simplest operators,

$$G_\sigma = (-\Delta + aQ_\sigma^* Q_\sigma)^{-1}, \qquad \sigma \in \{0, 1, \dots, d\}.$$

For $\sigma = 0$ this is the scalar Green function of [2] and [19] at the k-th step on a single torus. In [3] it is denoted G'_k. The tube versions $\sigma \ge 1$ are the componentwise building blocks of the gauge-field average $\mathbf{Q}$. They are needed in Section 11.

Theorem 7.1 (Scalar Green functions)**.** *Let $a > 0$. There are $C, \gamma > 0$, depending only on d and a, such that for every σ:*

(a) (L^2 decay) The operators G_σ, ∂G_σ, $G_\sigma\partial^*$ and $\partial G_\sigma \partial^*$ have decay (C, γ).

(b) (Pointwise bounds, $d \ge 3$) Let $\alpha \in (0, 1)$. There is C_α, depending only on d, a, α, such that for all y, y' and every f supported in $B(y')$,

$$|G_\sigma f|_{B(y)} + |\partial G_\sigma f|_{B(y)} + [\partial G_\sigma f]_{\alpha,B(y)} \le C_\alpha e^{-\gamma|y-y'|}|f|_{B(y')},$$
$$|G_\sigma \partial_\mu^* f|_{B(y)} + [G_\sigma \partial_\mu^* f]_{\alpha,B(y)} \le C_\alpha e^{-\gamma|y-y'|}|f|_{B(y')}.$$

By linearity, (b) implies for arbitrary bounded f

$$|G_\sigma f|_{B(y)} + |\partial G_\sigma f|_{B(y)} + [\partial G_\sigma f]_{\alpha,B(y)} \le C_\alpha \sum_{y'} e^{-\gamma|y-y'|}|f|_{B(y')}. \tag{7.1}$$

The Hölder quotient is only taken inside a single block, so it is subadditive under such decompositions. Hence (7.1) follows from the triangle inequality.

Proof. (a) We apply Lemma 5.7 to $\mathfrak{A} = -\Delta + aQ_\sigma^* Q_\sigma$. Hypothesis (i) is Corollary 3.4. For (ii), write $-\Delta = d^*d$ and apply Lemma 5.8 with $X = d$, $Z = 0$. This gives $|\langle u, ((-\Delta)_\alpha + \Delta)u\rangle| \le C|\alpha|(\|u\|^2 + \|\partial u\|^2)$ for $|\alpha| \le 1$. The operator $Q_\sigma^* Q_\sigma$ has norm at most 2 and satisfies $\mathbf{1}_y Q_\sigma^* Q_\sigma \mathbf{1}_{y'} = 0$ for $|y - y'| > 2$. By Lemma 5.2(d) it has decay $(2e^2, 1)$, so by Lemma 5.3, $\|(Q_\sigma^* Q_\sigma)_\alpha - Q_\sigma^* Q_\sigma\| \le C|\alpha|$ for $|\alpha| \le \frac{1}{2}$. Lemma 5.7 gives (a).

(b) Let $u = G_\sigma f$ with f supported in $B(y')$. Then

$$-\Delta u = F, \qquad F = f - aQ_\sigma^* Q_\sigma u.$$

Fix y, let z be the centre of $B(y)$, and let $\Lambda_r = \Lambda_r(z)$ as in Section 6. Then $B(y) \subset \Lambda_1$. Let Y be the set of y'' with $B(y'')$ within distance 4 of Λ_3. It contains at most C points, all with $|y'' - y| \le C$. By Lemma 2.6(c) and (a),

$$|F|_{\Lambda_2} + \|u\|_{\Lambda_2} \le |f|_{\Lambda_2} + C \max_{y'' \in Y} \|u\|_{B(y'')} \le |f|_{\Lambda_2} + C \max_{y'' \in Y} e^{-\gamma|y''-y'|} \|f\|.$$

Here $\|f\| \le |f|_{B(y')}$ because $B(y')$ has volume 1. Moreover $|f|_{\Lambda_2} = 0$ unless $|y - y'| \le C$. Since $|y'' - y'| \ge |y - y'| - C$, the right-hand side is at most $Ce^{-\gamma|y-y'|}|f|_{B(y')}$. Theorem 6.3(a) gives the first bound. For the second, $u = G_\sigma \partial_\mu^* f$ satisfies $-\Delta u = \partial_\mu^* f - aQ_\sigma^* Q_\sigma u$. Its L^2 norms on blocks are controlled by the decay of $G_\sigma \partial^*$ from part (a), and Theorem 6.3(b) applies with $g = f e_\mu$. □

Remark 7.2 (Cubes instead of blocks). Nothing in the proof of (b) uses the fact that $B(y)$ is a block. For every $z \in \mathbb{R}^d$ the same argument, applied to the cube $\Lambda_1(z)$, gives the bounds of (b) on $\Lambda_1(z)$, with $|y - y'|$ replaced by $\operatorname{dist}(z, B(y'))$. Consequently the Hölder bounds also hold on cubes $\Lambda_2(z)$, with a larger constant. Indeed, two points $x, x' \in \Lambda_2(z)$ with $|x - x'|_\infty \le \frac{1}{2}$ lie in a common cube $\Lambda_1(z_i)$ with $z_i \in z + \frac{1}{2}\mathbb{Z}^d$, $|z_i - z|_\infty \le \frac{9}{4}$, and for $|x - x'|_\infty > \frac{1}{2}$ the Hölder quotient is bounded by 4 times the supremum. The same remark applies to all pointwise bounds proved below.

Remark 7.3 (L^2 regularity). On the torus, $\sum_{\mu,\nu} \|\partial_\mu \partial_\nu u\|^2 = \|\Delta u\|^2$, because the differences commute. Hence $u = G_\sigma f$ satisfies $\sum_{\mu\nu} \|\partial_\mu \partial_\nu u\|^2 \le (\|f\| + 2a\|u\|)^2 \le (1 + 2a/c_0)^2 \|f\|^2$. This is the analogue of [3, Prop. 1.1]. Pointwise, second derivatives of $G_\sigma f$ are *not* bounded uniformly in η for merely bounded f. The Hölder bound on $\partial G_\sigma f$ is the substitute.

8 Averaged scalar operators, minimizers and the projection R

Throughout this section $\sigma \in \{0, \dots, d\}$ is fixed, $G = G_\sigma$, $Q = Q_\sigma$, and C, γ denote constants depending only on d and a (and on α in Hölder bounds). Their values may change from line to line.

8.1 The averaged Green function and its inverse

Proposition 8.1. *The operator QGQ^* on fields on $\mathbb{T}$ satisfies*

$$c_F \le QGQ^* \le 2c_0^{-1},$$

with c_F from Corollary 4.3. Both QGQ^* and $\Gamma = (QGQ^*)^{-1}$ have decay (C, γ), i.e. $|\Gamma(y, y')| \le Ce^{-\gamma|y-y'|}$.

Proof. The upper bound follows from $\|Q\|^2 \le 2$ and $G \le c_0^{-1}$. For the lower bound, $aQ^*Q \le 2a$ gives $-\Delta + aQ^*Q \le -\Delta + 2a$. Inversion reverses inequalities between positive definite operators: if $0 < X \le Y$, then $\langle h, Y^{-1}h\rangle = \|Y^{-1/2}X^{1/2}X^{-1/2}h\|^2 \le \|X^{-1/2}h\|^2$, because $\|X^{1/2}Y^{-1/2}\| \le 1$. Hence $G \ge (-\Delta + 2a)^{-1}$ and $QGQ^* \ge Q(-\Delta + 2a)^{-1}Q^* \ge c_F$ by Corollary 4.3.

For $\sigma = 0$, $\mathbf{1}_y Q_0 \mathbf{1}_{y'} = 0$ unless $y' = y$; for $\sigma \ge 1$, $\mathbf{1}_y Q_\sigma \mathbf{1}_{y'} = 0$ unless $y' \in \{y, y + e_\sigma\}$. Thus Q, viewed as a map from fine to unit fields, satisfies $\mathbf{1}_y Q \mathbf{1}_{y'} = 0$ for $|y - y'| > 1$. So it has decay $(\sqrt{2}e^\gamma, \gamma)$ for every γ, and the same holds for Q^*. With Theorem 7.1(a) and Lemma 5.2(b), QGQ^* has decay. Lemma 5.5 then gives the decay of Γ. □

8.2 The minimizer map

Definition 8.2. The *minimizer map* is $H = GQ^*\Gamma = GQ^*(QGQ^*)^{-1}$. It maps fields on $\mathbb{T}$ to fields on $\mathbb{T}_\eta$.

Lemma 8.3 (Variational characterization)**.** *For every field g on $\mathbb{T}$, Hg is the unique minimizer of $\|\partial f\|^2$ subject to $Qf = g$. In particular $QH = I$, H does not depend on a, and $H1 = 1$.*

Proof. $QHg = QGQ^*\Gamma g = g$. Let $Qf = g$ and write $f = Hg + h$ with $Qh = 0$. With $\mathfrak{A} = G^{-1}$,

$$\langle f, \mathfrak{A} f\rangle = \langle Hg, \mathfrak{A} Hg\rangle + 2\langle h, Q^*\Gamma g\rangle + \langle h, \mathfrak{A} h\rangle = \langle Hg, \mathfrak{A} Hg\rangle + \langle h, \mathfrak{A} h\rangle,$$

since $\langle h, Q^*\Gamma g\rangle = \langle Qh, \Gamma g\rangle = 0$. Thus Hg is the unique minimizer of $\langle f, \mathfrak{A} f\rangle$ on $\{Qf = g\}$. On this set $\langle f, \mathfrak{A} f\rangle = \|\partial f\|^2 + a\|g\|^2$, so Hg also minimizes $\|\partial f\|^2$, which does not involve a. The constant field 1 satisfies $Q1 = 1$ and $\|\partial 1\| = 0$, so $H1 = 1$. □

For a field g on $\mathbb{T}$ write $|d_1 g(y)| = \max_\mu \max\left(|D_\mu g(y)|, |D_\mu g(y - e_\mu)|\right)$, the largest unit difference over the $2d$ bonds containing y.

Theorem 8.4 (Bounds on the minimizer map)**.** *(a) H and ∂H have decay (C, γ).*

(b) ($d \geq 3$) For every field g on $\mathbb{T}$ and every y,

$$|Hg|_{B(y)} + |\partial Hg|_{B(y)} + [\partial Hg]_{\alpha, B(y)} \leq C \sum_{y'} e^{-\gamma|y-y'|} |g(y')|,$$
$$|\partial Hg|_{B(y)} + [\partial Hg]_{\alpha, B(y)} \leq C \sum_{y'} e^{-\gamma|y-y'|} |d_1 g(y')|.$$

(c) $\|\partial Hg\| \leq C\|d_1 g\|$.

Proof. (a) follows from Proposition 8.1, Theorem 7.1(a) and Lemma 5.2.

(b) $Hg = Gf$ with $f = Q^*\Gamma g$. By Lemma 2.6(c), $|f|_{B(y'')} \leq \max|\Gamma g|$ over at most two points near y'', which is bounded by $C\sum_{y'} e^{-\gamma|y''-y'|}|g(y')|$. Inserting this in (7.1) and using Lemma 5.2(b) for the convolution of exponentials gives the first bound. For the second, $H1 = 1$ gives $\partial Hg = \partial H(g - g(y))$, so we may replace $|g(y')|$ by $|g(y') - g(y)|$. Join y to y' by a monotone path on $\mathbb{T}$ with at most $\sqrt{d}|y - y'|$ steps. Every step starts at a point y_b with $|y_b - y| \leq \sqrt{d}|y - y'|$. Then

$$\sum_{y'} e^{-\gamma|y-y'|}|g(y') - g(y)| \leq \sum_{y'} e^{-\gamma|y-y'|} \sum_{b \in \text{path}} |d_1 g(y_b)| \leq S_{\gamma/2} \sum_{y_b} e^{-\gamma|y_b - y|/(2\sqrt{d})} |d_1 g(y_b)|,$$

after exchanging the sums and splitting $e^{-\gamma|y-y'|} \leq e^{-\gamma|y-y'|/2} e^{-\gamma|y_b - y|/(2\sqrt{d})}$. Each y_b is counted at most d times per y'; this factor is absorbed in C.

(c) The second bound in (b), without the Hölder term and in ℓ^2, follows by the same argument from the L^2 decay in (a). Indeed, $\|\partial Hg\|_{B(y)} \leq \sum_{y'} \|\mathbf{1}_y \partial H \mathbf{1}_{y'}\| \, |g(y') - g(y)|$. Summing the squares over y and using the Schur test as in Lemma 5.2(a) gives the claim. □

8.3 The effective action

Proposition 8.5 (Effective action)**.** *Let $\mathcal{D} = \Gamma - a$. Then for every field g on $\mathbb{T}$,*

$$\langle g, \mathcal{D} g\rangle = \|\partial Hg\|^2 = \min\{\|\partial f\|^2 : Qf = g\},$$

and

$$6^{-2d}\|d_1 g\|^2 \leq \langle g, \mathcal{D} g\rangle \leq C\|d_1 g\|^2.$$

Moreover $\mathcal{D}1 = 0$ and $\mathcal{D}$ has decay (C, γ).

Proof. $\langle Hg, G^{-1}Hg\rangle = \langle \Gamma g, QGQ^*\Gamma g\rangle = \langle g, \Gamma g\rangle$. On the other hand it equals $\|\partial Hg\|^2 + a\|QHg\|^2 = \|\partial Hg\|^2 + a\|g\|^2$. The upper bound is Theorem 8.4(c). For the lower bound, $D_\mu g = D_\mu QHg$, so Lemma 3.2 gives $\|D_\mu g\| \le 6^d \|\partial_\mu Hg\|$. Since $H1 = 1$ and $\partial 1 = 0$, $\langle 1, \mathcal{D}1\rangle = 0$ with $\mathcal{D} \ge 0$, hence $\mathcal{D}1 = 0$. Decay follows from Proposition 8.1. □

In Bałaban's language, $\frac{1}{2}\langle g, \mathcal{D}g\rangle$ is the quadratic action after k RG steps with δ-function averaging (the scalar analogue of the gauge-field action of [3, (1.65)–(1.67)]). The two-sided bound says that it is comparable to the unit-lattice Dirichlet form, uniformly in k.

8.4 The constrained Green function

Proposition 8.6 (Constrained Green function)**.** *Let $C_\sigma = G - HQG = G - GQ^*\Gamma QG$. Then:*

(a) C_σ is symmetric, $0 \le C_\sigma \le G$, $QC_\sigma = 0$ and $C_\sigma Q^* = 0$;

(b) for every f, $u = C_\sigma f$ is the unique field with $Qu = 0$ and $\langle \partial v, \partial u\rangle = \langle v, f\rangle$ for all v with $Qv = 0$. In particular C_σ does not depend on a, and $-\Delta C_\sigma f - f \in \operatorname{ran} Q^*$;

(c) $G = C_\sigma + H(QGQ^*)H^*$;

(d) C_σ, ∂C_σ, $C_\sigma\partial^*$ and $\partial C_\sigma \partial^*$ have decay (C, γ). For $d \ge 3$ the pointwise bounds of Theorem 7.1(b) hold with G_σ replaced by C_σ.

Proof. (a) $C_\sigma = G^{1/2}(I - \Pi)G^{1/2}$ with $\Pi = G^{1/2}Q^*\Gamma QG^{1/2}$, an orthogonal projection. The identities $QC_\sigma = 0 = C_\sigma Q^*$ are direct.

(b) We have $(-\Delta + aQ^*Q)u = f - Q^*\Gamma QGf$, and $Q^*Qu = 0$. So $-\Delta u = f - Q^*\omega$ with $\omega = \Gamma QGf$, and $\langle \partial v, \partial u\rangle = \langle v, f\rangle - \langle Qv, \omega\rangle = \langle v, f\rangle$. Uniqueness holds because $\|\partial v\| > 0$ for $v \ne 0$ with $Qv = 0$.

(c) $H(QGQ^*)H^* = GQ^*\Gamma QG$.

(d) $C_\sigma f = Gf - H(QGf)$, where QGf is a unit-lattice field whose values decay away from the support of f. Combine Theorem 7.1 and Theorem 8.4. For $C_\sigma\partial^*$, the field $QG\partial^* f$ is bounded by the L^2 decay of $G\partial^*$. □

Remark 8.7. Probabilistically, C_σ is the covariance of the Gaussian field with covariance G conditioned on $Q\phi = 0$. Identity (c) is the decomposition of ϕ into the fluctuation field and H applied to the block averages. Iterating it over the RG steps gives the renormalization group formula for Green functions of [19, Sect. 2.5].

8.5 The projection R

From now on $\sigma = 0$. Bałaban's gauge condition involves the orthogonal projection onto the space

$$\mathcal{W} = -\Delta\,\mathcal{N}(Q_0) = \{-\Delta\nu :\ Q_0\nu = 0\}$$

of scalar fields, cf. [3, (1.38)–(1.44)]. By Lemma 3.5, $-\Delta$ is injective on $\mathcal{N}(Q_0)$. Since Q_0 is onto, $\dim \mathcal{W} = \dim \mathcal{N}(Q_0) = n^d(N^d - 1)$. Let

$$\mathcal{V} = \operatorname{ran}(G_0Q_0^*), \qquad M = Q_0G_0^2Q_0^*, \qquad P_{\mathcal{V}} = G_0Q_0^*M^{-1}Q_0G_0.$$

Proposition 8.8 (The projection R)**.** *(a) $M \ge c_F^2$. Hence $P_{\mathcal{V}}$ is well defined; it is the orthogonal projection onto $\mathcal{V}$, and $\dim \mathcal{V} = n^d$.*

(b) $\mathcal{V} = \mathcal{W}^\perp$. Thus $R := I - P_{\mathcal{V}}$ is the orthogonal projection onto $\mathcal{W}$, and R does not depend on a.

(c) $\mathcal{V}$ consists of the constants plus the fields $(-\Delta)^{-1}Q_0^*\omega$ with $\sum_y \omega(y) = 0$. Here $(-\Delta)^{-1}$ is the inverse on mean-zero fields.

(d) For $h \in \mathcal{W}$: $G_0 h = C_0 h$, $Q_0 G_0 h = 0$ and $-\Delta G_0 h = h$.

(e) $P_{\mathcal{V}}$, $\partial P_{\mathcal{V}}$ and $P_{\mathcal{V}} \partial^*$ have decay (C, γ). In particular $\|P_{\mathcal{V}} d^*\| \le C$ on gauge fields.

Proof. (a) By Cauchy–Schwarz and Proposition 8.1, $\|G_0 Q_0^* \omega\| \, \|Q_0^* \omega\| \ge \langle Q_0^* \omega, G_0 Q_0^* \omega \rangle \ge c_F \|\omega\|^2$. Moreover $\|Q_0^* \omega\| = \|\omega\|$, so $\langle \omega, M\omega \rangle = \|G_0 Q_0^* \omega\|^2 \ge c_F^2 \|\omega\|^2$. Thus $G_0 Q_0^*$ is injective, and $T(T^*T)^{-1}T^*$ with $T = G_0 Q_0^*$ is the orthogonal projection onto $\operatorname{ran} T$.

(b) For $\nu \in \mathcal{N}(Q_0)$,

$$\langle G_0 Q_0^* \omega, -\Delta \nu \rangle = \langle \omega, Q_0 G_0 (-\Delta + a Q_0^* Q_0) \nu \rangle = \langle \omega, Q_0 \nu \rangle = 0.$$

So $\mathcal{V} \perp \mathcal{W}$, and $\dim \mathcal{V} + \dim \mathcal{W} = n^d N^d$ is the dimension of the whole space.

(c) If $\varphi = G_0 Q_0^* \omega$, then $-\Delta \varphi = Q_0^* (\omega - a Q_0 \varphi)$. The left side has mean zero, so $\omega' = \omega - a Q_0 \varphi$ has mean zero and $\varphi - \bar{\varphi} = (-\Delta)^{-1} Q_0^* \omega'$. Hence $\mathcal{V}$ is contained in the space described, which has dimension $(n^d - 1) + 1 = n^d$.

(d) Let $h = -\Delta \nu$ with $Q_0 \nu = 0$. Then $(-\Delta + a Q_0^* Q_0)\nu = h$, so $G_0 h = \nu$. The other statements follow, and $C_0 h = G_0 h - H_0 Q_0 G_0 h = \nu$.

(e) Compositions of operators with decay (Lemma 5.2), using Theorem 7.1(a), Proposition 8.1 and Lemma 5.5 for M^{-1}. □

Remark 8.9. Bałaban defines $R = I - \Delta^{-1} Q_0^* (Q_0 \Delta^{-2} Q_0^*)^{-1} Q_0 \Delta^{-1}$ on mean-zero fields [3, (1.38)]. It involves the non-local operator Δ^{-1}. He then observes that the same projection is given by the formula with G_0 [3, (1.44)]. Part (c) is the equivalence of the two descriptions. The G_0-formula is the one that makes the exponential decay of $I - R$ evident.

9 Gauge fields: Bałaban's gauge and coercivity

9.1 The quadratic action and its gauge invariance

The Wilson action of a lattice gauge field $U = e^{i\eta A}$ near $U = 1$ is $\frac{1}{2}\|dA\|^2$ plus terms of order three and higher in A ([3, (1.3)], [8, (26)]). The quadratic part is invariant under gauge transformations

$$A \mapsto A + d\lambda,$$

with λ an arbitrary scalar field. By Lemma 2.6(d), $\mathbf{Q}(A + d\lambda) = \mathbf{Q}A + d_1 Q_0 \lambda$. Hence the constraint $\mathbf{Q}A = B$ is preserved exactly when $d_1 Q_0 \lambda = 0$, that is, when $Q_0 \lambda$ is constant. Constant λ act trivially, so the residual gauge group may be taken to be $\mathcal{N}(Q_0)$. Consequently the variational problem

$$\text{minimize } \|dA\|^2 \quad \text{subject to} \quad \mathbf{Q}A = B$$

has, at best, a minimizer that is unique up to $A \mapsto A + d\lambda$ with $\lambda \in \mathcal{N}(Q_0)$. A gauge condition is needed to select one representative.

9.2 Bałaban's gauge

Bałaban uses the condition

$$R d^* A = 0, \tag{9.1}$$

with R the orthogonal projection onto $\mathcal{W} = -\Delta \mathcal{N}(Q_0)$ (Proposition 8.8). He calls it the Landau gauge [3, Sect. D], [8, (21)].

Lemma 9.1 (Gauge fixing)**.** *For every gauge field A there is exactly one $\lambda \in \mathcal{N}(Q_0)$ such that $A + d\lambda$ satisfies (9.1). Moreover $\|d\lambda\| \le \sqrt{d/2}\, \|R d^* A\|$, and $d(A + d\lambda) = dA$, $\mathbf{Q}(A + d\lambda) = \mathbf{Q}A$.*

Proof. For $\lambda \in \mathcal{N}(Q_0)$ we have $d^*d\lambda = -\Delta\lambda \in \mathcal{W}$, hence $Rd^*d\lambda = -\Delta\lambda$. The condition $Rd^*(A + d\lambda) = 0$ becomes $-\Delta\lambda = -Rd^*A$. The right-hand side lies in $\mathcal{W}$, and $-\Delta$ is a bijection from $\mathcal{N}(Q_0)$ onto $\mathcal{W}$. The bound is Lemma 3.5. □

By Proposition 8.8(c), condition (9.1) says that d^*A is a constant plus $(-\Delta)^{-1}Q_0^*\omega$ for some unit-lattice ω. So d^*A is the potential of a block-constant charge density. It is a $C^{1,\alpha}$ function on the unit scale, not a rough one. By contrast, the gauge that minimizes $\|A\|$ on the orbit would require $d^*A \in \operatorname{ran} Q_0^*$, a field with jumps at block boundaries. This regularity is one reason for Bałaban's choice.

Remark 9.2 (Origin of the operator $\mathbf{\Delta}_a$). Bałaban arrives at (9.1) by a Faddeev–Popov argument [3, (1.21)–(1.39)]. It replaces the axial gauge fixing by the density $\exp(-\frac{1}{2}\|Rd^*A\|^2)$ and then takes the Landau limit. Adding the gauge-fixing term and the averaging term to the action gives the quadratic form

$$\langle A, \mathbf{\Delta}_a A\rangle = \|dA\|^2 + \|Rd^*A\|^2 + a\|\mathbf{Q}A\|^2, \qquad \mathbf{\Delta}_a = d^*d + dRd^* + a\mathbf{Q}^*\mathbf{Q} = -\Delta - dP_{\mathcal{V}}d^* + a\mathbf{Q}^*\mathbf{Q},$$

where we used Lemma 2.2 and $R = I - P_{\mathcal{V}}$. Up to the term $dP_{\mathcal{V}}d^*$, which has exponential decay, this is a Feynman-type gauge: the principal part is the componentwise Laplacian. On the constraint set $\{\mathbf{Q}A = B,\ Rd^*A = 0\}$ the form reduces to $\|dA\|^2 + a\|B\|^2$.

9.3 The averaging nearly separates exact and co-closed fields

The heart of the coercivity proof is the following identity. On the unit lattice, gradients $d_1 g$ and co-closed fields are orthogonal. We show that $\mathbf{Q}$ maps fine co-closed fields to unit fields that are co-closed up to a divergence controlled by derivatives.

Lemma 9.3. *There is C_W, depending only on d, with the following property. For every gauge field A on $\mathbb{T}_\eta$,*

$$d_1^*\mathbf{Q}A = \iota\, b\, \bar{c}\, d^*A + d_1^*\mathsf{W}A, \qquad \bar{c} = c_1 c_2 \cdots c_d,$$

where $\mathsf{W}A$ is a gauge field on $\mathbb{T}$ with $\|\mathsf{W}A\| \le C_W\|\partial A\|$.

Proof. By Lemma 2.4(a), $D_\mu^* = a_\mu^*\partial_\mu^*$, and all operators involved commute. Hence

$$d_1^*\mathbf{Q}A = \sum_\mu D_\mu^* \iota\, a_\mu b A_\mu = \iota \sum_\mu b\, c_\mu\, \partial_\mu^* A_\mu.$$

For fixed μ, the telescoping identity $1 - x_1\cdots x_m = \sum_i x_1\cdots x_{i-1}(1 - x_i)$ for commuting x_i gives

$$c_\mu - \bar{c} = c_\mu\Big(1 - \prod_{\nu\neq\mu} c_\nu\Big) = \sum_{\nu\neq\mu} c_\mu \Pi_{\mu\nu}(1 - c_\nu), \qquad \Pi_{\mu\nu} = \prod_{\nu'<\nu,\ \nu'\neq\mu} c_{\nu'}.$$

By Lemma 2.4(d), $1 - c_\nu = e_\nu\partial_\nu^*\partial_\nu$. By Lemma 2.4(a),(b), $a_\nu\partial_\nu^* = \tau_\nu a_\nu^*\partial_\nu^* = \tau_\nu D_\nu^*$, so that $b\,\partial_\nu^* = \big(\prod_{\nu'\neq\nu} a_{\nu'}\big)\tau_\nu D_\nu^*$. Since D_ν^* commutes with ι,

$$\iota\, b\,(c_\mu - \bar{c})\,\partial_\mu^* A_\mu = \sum_{\nu\neq\mu} D_\nu^*\, \iota\Big(\prod_{\nu'\neq\nu} a_{\nu'}\Big)\tau_\nu\, c_\mu \Pi_{\mu\nu}\,(e_\nu\partial_\nu)\,\partial_\mu^* A_\mu.$$

Summing over μ and collecting the coefficient of D_ν^* defines

$$(\mathsf{W}A)_\nu = \sum_{\mu\neq\nu} \iota\Big(\prod_{\nu'\neq\nu} a_{\nu'}\Big)\tau_\nu\, c_\mu \Pi_{\mu\nu}\,(e_\nu\partial_\nu)\,\partial_\mu^* A_\mu,$$

and the identity follows. Each operator acting on $\partial_\mu^* A_\mu$ is ι composed with a product convolution. In direction ν it is $\tau_\nu e_\nu \partial_\nu$, whose kernel is bounded by 1 and supported in an

interval of length at most 3 (Lemma 2.4(d)). In every other direction it is $a_{\nu'}$ times at most one triangle average $c_{\nu'}$, whose kernel is an average of c and therefore bounded by 1, with support of length at most 3. By Lemma 2.7 the norm is at most $4^d\|\partial_\mu^* A_\mu\| = 4^d\|\partial_\mu A_\mu\|$. Hence, by Cauchy–Schwarz over $\mu \neq \nu$, $\|\mathsf{W}A\| \le 4^d d\,\|\partial A\|$. □

Lemma 9.4 (Separation)**.** *There is $C_\star$, depending only on d, such that for every co-closed gauge field B' on $\mathbb{T}_\eta$ and every scalar field g on $\mathbb{T}$,*

$$\|B'\|^2 + \|d_1 g\|^2 \le C_\star(\|dB'\|^2 + \|\mathbf{Q}B' + d_1 g\|^2).$$

Proof. Expand the square:

$$\|\mathbf{Q}B' + d_1 g\|^2 = \|\mathbf{Q}B'\|^2 + \|d_1 g\|^2 + 2\langle d_1^*\mathbf{Q}B', g\rangle.$$

By Lemma 9.3 and $d^*B' = 0$, $\langle d_1^*\mathbf{Q}B', g\rangle = \langle \mathsf{W}B', d_1 g\rangle$. This is bounded in absolute value by $C_W\|\partial B'\|\|d_1 g\| \le \frac14\|d_1 g\|^2 + C_W^2\|\partial B'\|^2$. By Lemma 2.2, $\|\partial B'\| = \|dB'\|$. Hence

$$\|\mathbf{Q}B' + d_1 g\|^2 \ge \|\mathbf{Q}B'\|^2 + \tfrac12\|d_1 g\|^2 - 2C_W^2\|dB'\|^2.$$

On the other hand, Corollary 3.3 gives $\|B'\|^2 \le C_P(\|dB'\|^2 + \|\mathbf{Q}B'\|^2)$. Therefore

$$\begin{aligned}\|B'\|^2 + \|d_1 g\|^2 &\le C_P\|dB'\|^2 + (C_P+2)(\|\mathbf{Q}B'\|^2 + \tfrac12\|d_1 g\|^2)\\ &\le (C_P + 2C_W^2(C_P+2))\|dB'\|^2 + (C_P+2)\|\mathbf{Q}B' + d_1 g\|^2.\end{aligned}$$ □

9.4 Coercivity

Lemma 9.5 (Gradient part in Bałaban's gauge)**.** *Let λ be a scalar field with mean zero and $R(-\Delta\lambda) = 0$. Then $\|d\lambda\| \le C_B\|d_1 Q_0\lambda\|$, with C_B from Lemma 4.4.*

Proof. $-\Delta\lambda$ lies in $\mathcal{W}^\perp = \mathcal{V}$ and has mean zero. By Proposition 8.8(c), $-\Delta\lambda = (-\Delta)^{-1}Q_0^*\omega$ with ω of mean zero. Since λ has mean zero, $\lambda = (-\Delta)^{-2}Q_0^*\omega$, and Lemma 4.4 applies. □

Theorem 9.6 (Coercivity of the gauge-field operator)**.** *There is $c_g > 0$, depending only on d and a, such that for every gauge field A on $\mathbb{T}_\eta$*

$$\langle A, \mathbf{\Delta}_a A\rangle = \|dA\|^2 + \|Rd^*A\|^2 + a\|\mathbf{Q}A\|^2 \;\ge\; c_g(\|A\|^2 + \|\partial A\|^2).$$

Proof. Step 1 (gauge fixing). Let $\lambda_1 \in \mathcal{N}(Q_0)$ be given by Lemma 9.1 and $B = A + d\lambda_1$. Then $Rd^*B = 0$, $dB = dA$, $\mathbf{Q}B = \mathbf{Q}A$, and $\|d\lambda_1\| \le \sqrt{d/2}\|Rd^*A\|$.

Step 2 (Hodge decomposition). By Lemma 2.2, $B = d\lambda + B'$ with $\lambda = (-\Delta)^{-1}d^*B$ of mean zero and $d^*B' = 0$. Then $R(-\Delta\lambda) = Rd^*B = 0$. By Lemma 9.5, $\|d\lambda\| \le C_B\|d_1 g\|$ with $g = Q_0\lambda$.

Step 3 (separation). By gauge covariance, $\mathbf{Q}B = \mathbf{Q}B' + d_1 g$, and $dB' = dB = dA$. Lemma 9.4 gives

$$\|B'\|^2 + \|d_1 g\|^2 \le C_\star(\|dA\|^2 + \|\mathbf{Q}A\|^2).$$

Step 4. Since $A = -d\lambda_1 + d\lambda + B'$,

$$\|A\| \le \sqrt{d/2}\,\|Rd^*A\| + (C_B+1)\sqrt{2C_\star}\,(\|dA\|^2 + \|\mathbf{Q}A\|^2)^{1/2}.$$

Hence $\|A\|^2 \le C'(\|dA\|^2 + \|Rd^*A\|^2 + \|\mathbf{Q}A\|^2)$.

Step 5 (derivatives). By Lemma 2.2, $\|\partial A\|^2 = \|dA\|^2 + \|d^*A\|^2$. Moreover $\|d^*A\| \le \|Rd^*A\| + \|P_{\mathcal{V}}d^*A\| \le \|Rd^*A\| + C\|A\|$ by Proposition 8.8(e). Combining with Step 4 proves the theorem, with c_g depending on $\min(1,a)$ and the constants above. □

Corollary 9.7. *On the space $\mathcal{X} = \{A : \mathbf{Q}A = 0,\ Rd^*A = 0\}$ one has $\|dA\|^2 \ge c_g\|A\|^2$.*

This completes the proof of Theorem A.

10 The gauge-field Green function

Let $\mathbf{G} = \mathbf{\Delta}_a^{-1}$ with $\mathbf{\Delta}_a = d^*d + dRd^* + a\mathbf{Q}^*\mathbf{Q}$. By Theorem 9.6 it is well defined and $0 < \mathbf{G} \le c_g^{-1}$. In [3] this operator is denoted G_k (with $a = 1$). In [7, 8] its covariant version G_1, $G(U)$ is the basic propagator.

Theorem 10.1 (Gauge-field Green function)**.** *There are $C, \gamma > 0$, depending only on d and a, such that:*

(a) $\mathbf{G}$, $\partial\mathbf{G}$, $\mathbf{G}\partial^*$ *and* $\partial\mathbf{G}\partial^*$ *have decay* (C, γ)*;*

(b) *(*$d \ge 3$*) for* $\alpha \in (0, 1)$ *there is* C_α *such that for all* y, y' *and every gauge field* J *supported in* $B(y')$*,*

$$|\mathbf{G}J|_{B(y)} + |\partial\mathbf{G}J|_{B(y)} + [\partial\mathbf{G}J]_{\alpha,B(y)} \le C_\alpha e^{-\gamma|y-y'|}\,|J|_{B(y')}.$$

As in Remark 7.2, the bounds of (b) also hold on arbitrary cubes $\Lambda_1(z)$ and, for the Hölder quotient, on cubes $\Lambda_2(z)$.

Proof. (a) We verify the hypotheses of Lemma 5.7 for $\mathfrak{A} = \mathbf{\Delta}_a$ acting on gauge fields. Hypothesis (i) is Theorem 9.6. For (ii), write

$$\mathbf{\Delta}_a = d^*d + (Rd^*)^*(Rd^*) + a\mathbf{Q}^*\mathbf{Q},$$

which holds because R is an orthogonal projection. The first term is X^*X with $X = d$ (from gauge fields to 2-forms). Lemma 5.8 applies with $Z = 0$. The second term is X^*X with $X = d^* - Z$ and $Z = P_{\mathcal{V}}d^*$. By Proposition 8.8(e), Z is bounded and has decay. By Lemma 5.3, $\|Z_\alpha - Z\| \le C|\alpha|$ for $|\alpha| \le \gamma/2$. So Lemma 5.8 applies again. The last term is bounded and local, as in the proof of Theorem 7.1.

(b) Let $u = \mathbf{G}J$. Since $d^*d + dd^* = -\Delta$ componentwise (Lemma 2.2) and $dRd^* = dd^* - dP_{\mathcal{V}}d^*$,

$$-\Delta u = F, \qquad F = J - a\mathbf{Q}^*\mathbf{Q}u + dP_{\mathcal{V}}d^*u.$$

Each component of u satisfies a scalar equation of the form treated in Theorem 6.3(a). Fix y and let Λ_r be centred at the centre of $B(y)$. As in the proof of Theorem 7.1, part (a) gives

$$\|u\|_{\Lambda_2} + |J - a\mathbf{Q}^*\mathbf{Q}u|_{\Lambda_2} \le Ce^{-\gamma|y-y'|}|J|_{B(y')}.$$

For the last term, write $dP_{\mathcal{V}}d^*u = (dG_0)\,Q_0^*\omega$ with $\omega = M^{-1}Q_0G_0d^*u$. The operator $M^{-1}Q_0G_0d^*\mathbf{G}$ has decay by Proposition 8.8 and part (a). Hence $|\omega(y'')| \le Ce^{-\gamma|y''-y'|}\|J\|$, with a smaller γ. By (7.1) for G_0 and Lemma 5.2(b),

$$|dP_{\mathcal{V}}d^*u|_{B(y'')} = |\partial G_0Q_0^*\omega|_{B(y'')} \le C\sum_{y'''} e^{-\gamma|y''-y'''|}|\omega(y''')| \le Ce^{-\gamma|y''-y'|}|J|_{B(y')}.$$

Thus $|F|_{\Lambda_2} \le Ce^{-\gamma|y-y'|}|J|_{B(y')}$, and Theorem 6.3(a) applied to each component concludes the proof. □

By linearity, for an arbitrary bounded gauge field J,

$$|\mathbf{G}J|_{B(y)} + |\partial\mathbf{G}J|_{B(y)} + [\partial\mathbf{G}J]_{\alpha,B(y)} \le C_\alpha \sum_{y'} e^{-\gamma|y-y'|}|J|_{B(y')}. \tag{10.1}$$

This completes the proof of Theorem D.

Remark 10.2. The operator $\mathbf{\Delta}_a$ is not of the form "Laplacian plus a local term": $dP_{\mathcal{V}}d^*$ is non-local. It is nevertheless exponentially decaying, and it is a bounded operator up to the two derivatives, which is all that Combes–Thomas and the regularity estimate need. In the covariant case this term becomes $D_UP_{\mathcal{V}}(U)D_U^*$, and the same structure persists [7].

11 The gauge-field minimizer and the fluctuation operators

11.1 Identities

The following identities express how $\mathbf{G}$ interacts with gauge directions. They reduce the gauge part of every operator below to scalar operators. Recall from Proposition 8.8(d) that for h in $\mathcal{W} = \operatorname{ran} R$, $\nu = G_0 h$ is the unique field with $Q_0\nu = 0$ and $-\Delta\nu = h$.

Proposition 11.1. *For all scalar fields h:*

$$\mathbf{G}dRh = dG_0Rh, \qquad \mathbf{Q}\mathbf{G}dR = 0, \qquad Rd^*\mathbf{G}dR = R.$$

Taking adjoints, $Rd^*\mathbf{G} = RG_0d^*$ and $Rd^*\mathbf{G}\mathbf{Q}^* = 0$.

Proof. Let $\nu = G_0Rh$, so $Q_0\nu = 0$ and $-\Delta\nu = Rh$. Then

$$\mathbf{\Delta}_a d\nu = d^*dd\nu + dRd^*d\nu + a\mathbf{Q}^*\mathbf{Q}d\nu = 0 + dR(-\Delta\nu) + a\mathbf{Q}^*d_1Q_0\nu = dRh.$$

Hence $\mathbf{G}dRh = d\nu$. Further, $\mathbf{Q}d\nu = d_1Q_0\nu = 0$ and $Rd^*d\nu = R(-\Delta\nu) = Rh$. □

These are Bałaban's identities [3, (1.95), (1.97)] and [7, (3.152)].

11.2 The averaged gauge Green function

Proposition 11.2. *$c_F \le \mathbf{Q}\mathbf{G}\mathbf{Q}^* \le 2c_g^{-1}$. Both $\mathbf{Q}\mathbf{G}\mathbf{Q}^*$ and $\mathbf{\Gamma} = (\mathbf{Q}\mathbf{G}\mathbf{Q}^*)^{-1}$ have decay (C, γ).*

Proof. For every gauge field, $\|Rd^*A\| \le \|d^*A\|$ and $\|\mathbf{Q}A\|^2 \le 2\|A\|^2$. With Lemma 2.2, $\mathbf{\Delta}_a \le -\Delta + 2a$, hence $\mathbf{G} \ge (-\Delta + 2a)^{-1}$ as in Proposition 8.1. The operator $-\Delta$ acts componentwise and $(\mathbf{Q}A)_\mu = Q_\mu A_\mu$, so $\mathbf{Q}(-\Delta + 2a)^{-1}\mathbf{Q}^*$ is the direct sum of the operators $Q_\mu(-\Delta + 2a)^{-1}Q_\mu^*$. Each of these is at least c_F by Corollary 4.3. The decay statements follow as in Proposition 8.1, using Theorem 10.1(a) and Lemma 5.5. □

11.3 The minimizer map

Theorem 11.3 (Gauge-field minimizer)**.** *Let $\mathbf{H} = \mathbf{G}\mathbf{Q}^*\mathbf{\Gamma}$.*

(a) $\mathbf{Q}\mathbf{H} = I$ *and* $Rd^*\mathbf{H} = 0$.

(b) For every gauge field B on $\mathbb{T}$, $\mathbf{H}B$ is the unique minimizer of $\|dA\|^2$ on $\{A : \mathbf{Q}A = B,\ Rd^*A = 0\}$. Moreover $\|d\mathbf{H}B\|^2 = \min\{\|dA\|^2 : \mathbf{Q}A = B\}$, and $\mathbf{H}$ does not depend on a. If B is constant, then $\mathbf{H}B = B$ (the constant fine field).

(c) $\mathbf{H}$ and $\partial\mathbf{H}$ have decay (C, γ).

(d) ($d \ge 3$) For every B and y,

$$|\mathbf{H}B|_{B(y)} + |\partial\mathbf{H}B|_{B(y)} + [\partial\mathbf{H}B]_{\alpha,B(y)} \le C_\alpha \sum_{y'} e^{-\gamma|y-y'|}|B(y')|.$$

Proof. (a) $\mathbf{Q}\mathbf{H} = \mathbf{Q}\mathbf{G}\mathbf{Q}^*\mathbf{\Gamma} = I$, and $Rd^*\mathbf{H} = 0$ by Proposition 11.1.

(b) Let $\mathbf{Q}A = B$ and $Rd^*A = 0$. Then $h = A - \mathbf{H}B$ lies in $\mathcal{X}$. As in Lemma 8.3,

$$\langle A, \mathbf{\Delta}_a A\rangle = \langle \mathbf{H}B, \mathbf{\Delta}_a \mathbf{H}B\rangle + \langle h, \mathbf{\Delta}_a h\rangle,$$

because $\langle h, \mathbf{\Delta}_a\mathbf{H}B\rangle = \langle h, \mathbf{Q}^*\mathbf{\Gamma}B\rangle = 0$. On the constraint set $\langle A, \mathbf{\Delta}_a A\rangle = \|dA\|^2 + a\|B\|^2$. Moreover $\langle h, \mathbf{\Delta}_a h\rangle = \|dh\|^2 > 0$ for $h \ne 0$ by Corollary 9.7. This proves the first claim and the independence of a. If only $\mathbf{Q}A = B$ is assumed, Lemma 9.1 provides a gauge transform

of A that satisfies $Rd^*A = 0$ and has the same dA and $\mathbf{Q}A$, which gives the second claim. A constant fine field B satisfies $\mathbf{Q}B = B$, $dB = 0$ and $d^*B = 0$, so it is the minimizer.

(c) Composition of operators with decay.

(d) $\mathbf{H}B = \mathbf{G}J$ with $J = \mathbf{Q}^*\mathbf{\Gamma}B$. By Lemma 2.6(c) and Proposition 11.2, $|J|_{B(y'')} \le C\sum_{y'} e^{-\gamma|y''-y'|}|B(y')|$. Apply (10.1). □

Remark 11.4 (Comparison with [8]). In [8, (45)–(46)] the operator H is defined exactly by (b), in the covariant setting, and the bounds $|HB| \le B_0|B|$ and $|\nabla HB| \le B_0|B|$ are quoted from [7, Theorem 3.12]. Part (d) is their $U = 1$ version on a single torus, with the additional Hölder bound.

11.4 The gauge-field effective action

Proposition 11.5. *Let* $\mathbf{D} = \mathbf{\Gamma} - a$. *Then* $\langle B, \mathbf{D}B\rangle = \|d\mathbf{H}B\|^2$, $\mathbf{D}$ *has decay* (C, γ), *and*

$$c_D\|d_1B\|^2 \le \langle B, \mathbf{D}B\rangle \le C_D\|d_1B\|^2,$$

where d_1B *is the unit-lattice curl and* $c_D, C_D > 0$ *depend only on* d. *(The operators* $\mathbf{H}$ *and* H_μ *do not depend on* a, *so we may take* $a = 1$ *when estimating them.)*

Proof. The identity is proved as in Proposition 8.5, using $Rd^*\mathbf{H}B = 0$.

Lower bound. By Lemma 2.6(e), $d_1B = d_1\mathbf{Q}\mathbf{H}B = \mathbf{Q}^{(2)}d\mathbf{H}B$. Each component of $\mathbf{Q}^{(2)}$ is $\iota a_\mu a_\nu b$, which by Lemma 2.7 has norm at most $9\cdot 2^{d-2}$. Hence $\|d_1B\| \le 9\cdot 2^{d-2}\|d\mathbf{H}B\|$.

Upper bound. Decompose B on the unit torus as $B = d_1h + B''$ with $d_1^*B'' = 0$ (Lemma 2.2 on $\mathbb{T}$). Then $d_1B'' = d_1B$, and by the unit-lattice Hodge identity $\sum_{\mu,\nu}\|D_\nu B''_\mu\|^2 = \|d_1B''\|^2 = \|d_1B\|^2$. Define a fine gauge field A by

$$A_\mu = \partial_\mu Q_0^*h + H_\mu B''_\mu,$$

where H_μ is the scalar minimizer map for the tube average Q_μ (Section 8 with $\sigma = \mu$). Then $\mathbf{Q}A = d_1Q_0Q_0^*h + (Q_\mu H_\mu B''_\mu)_\mu = d_1h + B'' = B$. By Theorem 11.3(b), Lemma 2.2 and Theorem 8.4(c),

$$\|d\mathbf{H}B\|^2 \le \|dA\|^2 = \big\|d(H_\mu B''_\mu)_\mu\big\|^2 \le \sum_\mu \|\partial H_\mu B''_\mu\|^2 \le C\sum_\mu \|d_1B''_\mu\|^2 = C\|d_1B\|^2.$$

Here $d_1B''_\mu$ is the unit gradient of the scalar field B''_μ. The decay of $\mathbf{D}$ follows from Proposition 11.2. □

This is the $U = 1$ form of [3, (1.66)–(1.67)]: after k steps the gauge-field action is comparable to the unit-lattice Wilson action $\|d_1B\|^2$, uniformly in k. Note how the upper bound uses the tube averages Q_μ. This is why the scalar theory was developed for all σ.

11.5 The constrained Green function

The Euler–Lagrange equation of the variational problem is posed on the tangent space

$$\mathcal{X} = \{A :\ \mathbf{Q}A = 0,\ Rd^*A = 0\}$$

of the constraint set, cf. [8, (100)–(111)]. The relevant Green function is defined as follows.

Definition 11.6. $\mathbf{C} = \mathbf{G} - \mathbf{G}dRd^*\mathbf{G} - \mathbf{H}\mathbf{Q}\mathbf{G} = \mathbf{G} - dG_0RG_0d^* - \mathbf{G}\mathbf{Q}^*\mathbf{\Gamma}\mathbf{Q}\mathbf{G}$.

The two expressions agree by Proposition 11.1: $\mathbf{G}dRd^*\mathbf{G} = (\mathbf{G}dR)(Rd^*\mathbf{G}) = dG_0R\,RG_0d^*$. In [7, 8], $\mathbf{C}$ is the operator $\mathfrak{G} = G_1\mathfrak{P}^*$ of [7, (3.148)–(3.153)]. Here $\mathfrak{P}$ is the orthogonal projection onto $\mathcal{X}$ in the inner product $\langle\cdot, \mathbf{\Delta}_a\cdot\rangle$.

Theorem 11.7 (Constrained Green function). *(a)* $\mathbf{C}$ *is symmetric,* $\mathbf{Q}\mathbf{C} = 0$ *and* $Rd^*\mathbf{C} = 0$.

(b) For every gauge field J, $A = \mathbf{C}J$ *is the unique element of* $\mathcal{X}$ *such that*

$$\langle d\,\delta A, dA\rangle = \langle \delta A, J\rangle \qquad \text{for all } \delta A \in \mathcal{X}.$$

In particular $\mathbf{C}$ *does not depend on* a, *and* $\langle J, \mathbf{C}J\rangle \le c_g^{-1}\|J\|^2$.

(c) $\mathbf{C}$ *and* $\partial\mathbf{C}$ *have decay* (C, γ).

(d) ($d \ge 3$) *For all* y, y' *and every* J *supported in* $B(y')$,

$$|\mathbf{C}J|_{B(y)} + |\partial\mathbf{C}J|_{B(y)} + [\mathbf{C}J]_{\alpha,B(y)} \le C_\alpha e^{-\gamma|y-y'|}\,|J|_{B(y')}.$$

Proof. (a) Each of the three terms is symmetric. By Proposition 8.8(d), $Q_0G_0R = 0$, so $\mathbf{Q}dG_0RG_0d^* = d_1Q_0G_0RG_0d^* = 0$. Together with $\mathbf{Q}\mathbf{H} = I$ this gives $\mathbf{Q}\mathbf{C} = \mathbf{Q}\mathbf{G} - 0 - \mathbf{Q}\mathbf{G} = 0$. Next,

$$Rd^*\mathbf{C} = RG_0d^* - R(-\Delta)G_0RG_0d^* - 0 = RG_0d^* - RRG_0d^* = 0,$$

using Proposition 11.1 and $-\Delta G_0R = R$ (Proposition 8.8(d)).

(b) Let $\delta A \in \mathcal{X}$. By (a), $\mathbf{C}J \in \mathcal{X}$, so $\langle \delta A, \mathbf{\Delta}_a\mathbf{C}J\rangle = \langle d\,\delta A, d\mathbf{C}J\rangle$. On the other hand, the computation in the proof of Proposition 11.1 shows $\mathbf{\Delta}_a dG_0Rh = dRh$. Hence

$$\mathbf{\Delta}_a\mathbf{C}J = J - dRG_0d^*J - \mathbf{Q}^*\mathbf{\Gamma}\mathbf{Q}\mathbf{G}J,$$

and the last two terms are orthogonal to $\mathcal{X}$. Uniqueness and the bound follow from Corollary 9.7.

(c) The first and third terms are compositions of operators with decay, also after applying ∂: $\mathbf{G}$, $\partial\mathbf{G}$, $\mathbf{H}$, $\partial\mathbf{H}$, $\mathbf{Q}$. For the middle term, Proposition 11.1 and $R^2 = R$ give $dG_0RG_0d^* = \mathbf{G}\,dR\,G_0d^*$. On scalar fields $\partial_\mu = -S_\mu\partial_\mu^*$, with S_μ the shift by η. So $d = \partial^*T$ with $(Th)_{\mu\nu} = -\delta_{\mu\nu}S_\mu h$, a local operator of norm $\sqrt{d}$, and $\partial\mathbf{G}d = (\partial\mathbf{G}\partial^*)T$ has decay by Theorem 10.1(a) and Lemma 5.2(b),(d). Together with the decay of $R = I - P_{\mathcal{V}}$ and $G_0\partial^*$, this gives the decay of $\partial(dG_0RG_0d^*)$ without second derivatives of G_0.

(d) Write $\mathbf{C}J = \mathbf{G}J - d\sigma - \mathbf{H}(\mathbf{Q}\mathbf{G}J)$ with $\sigma = G_0RG_0d^*J$. The first and third terms are controlled by Theorems 10.1 and 11.3: the unit field $\mathbf{Q}\mathbf{G}J$ decays away from y' by Theorem 10.1(a). For σ we use the scalar regularity theory. By Proposition 8.8(d), $\sigma \in \mathcal{N}(Q_0)$ and

$$-\Delta\sigma = h, \qquad h = RG_0d^*J = G_0d^*J - P_{\mathcal{V}}G_0d^*J.$$

There is no averaging term in this equation, and h is Hölder continuous. Indeed, G_0d^*J is bounded and Hölder on cubes $\Lambda_2(z)$ by Theorem 7.1(b) and Remark 7.2. Moreover $P_{\mathcal{V}}G_0d^*J = G_0Q_0^*\omega$ with $\omega = M^{-1}Q_0G_0G_0d^*J$, which is bounded with bounded first derivatives. All these bounds carry the factor $e^{-\gamma\,\mathrm{dist}(z,B(y'))}|J|_{B(y')}$, and so does $\|\sigma\|_{\Lambda_2(z)}$, because G_0, R and $G_0\partial^*$ have decay and $\Lambda_2(z)$ meets a bounded number of blocks. Theorem 6.3(a) and (c) now bound $|\partial\sigma|$, its Hölder quotient, and $|\partial\partial\sigma|$ on $\Lambda_1(z)$. This controls $|d\sigma|$ and $|\partial d\sigma|$. The Hölder bound on $\mathbf{C}J$ follows from the bound on $|\partial\mathbf{C}J|$. □

Remark 11.8 (Use in the variational problem). In [8, (110)–(111)], the critical points of the action on the constraint set, after the linearizing change of variables, are the solutions of

$$A_1 + \mathbf{C}J + \mathbf{C}\,\tfrac{\delta V}{\delta A}(A_1 + \mathbf{H}B) = 0$$

(in the covariant version). The solution is found by the contraction mapping principle. This requires exactly the bounds $|\mathbf{C}J|, |\partial\mathbf{C}J| \le C\sup|J|$ in the weighted norms, i.e. part (d), together with the bounds on $\mathbf{H}$ in Theorem 11.3. Bałaban notes [8, Sect. E] that the regularity theorem for $\mathfrak{G}$ [7, Theorem 3.13] gives no bounds on second-order derivatives, and he supplies the needed second-order bounds by a separate argument. Part (d) likewise controls only $\mathbf{C}J$ and its first derivatives.

Remark 11.9 (A renormalization-group decomposition). Combining the definitions,

$$\mathbf{G} = \mathbf{C} + dG_0RG_0d^* + \mathbf{H}(\mathbf{Q}\mathbf{G}\mathbf{Q}^*)\mathbf{H}^*.$$

The three terms are, respectively, the fluctuation covariance on $\mathcal{X}$, a pure-gauge part, and the part generated by the averages. This is the gauge-field analogue of Proposition 8.6(c).

This completes the proof of Theorem E.

Part II: Backgrounds

12 Background fields: setting and hypotheses

In this part of the paper we extend the results of Sections 7–11 to operators depending on a non-trivial background field U. These are the operators $G(U)$, $H(U)$, $\mathfrak{G}(U)$, $R(U)$, $G'(U)$ of [7], which enter the variational problem [8]. Two features distinguish this part from the case $U = 1$.

- The fields take values in the (complexified) Lie algebra, and the operators involve parallel transport. Everything is proved for complex backgrounds near real ones. Analyticity in the background then comes for free.
- The gauge directions no longer decouple from the action. Bałaban handles this by replacing the quadratic form by its gauge-invariant extension, built with the projection $\Pi(U)$ onto the gauge slice along the residual gauge directions [7, (3.118)–(3.119)]. We use the same construction (Section 15).
- We work with a simplified averaging operator: only the main term of Bałaban's linearized averaging. For it, the covariance identity $\mathbf{Q}(U)D_U = D_VQ'(U)$, which is exact for Bałaban's full operator [7, (3.114)–(3.115)], holds only up to a small "defect" $E(U)$. We correct for the defect by one additional term in the averaging part of the operator. With this correction all exact identities of Section 11 are restored.

The global small-field hypothesis. We assume that the background is close to 1 *in a single global gauge* (hypothesis (B) below). Bałaban's conditions are local. They consist of a gauge-invariant bound on the curvature [8, (2)] and, for the propagators, of the requirement that on each cube of size M there be a gauge in which U is close to 1, with bounds proportional to M [7, (3.35)–(3.36)]. Hypothesis (B) is a condition on the background U as given; a real gauge transformation can destroy it. The existence of a G-valued gauge transformation that carries U to a background satisfying (B) is a condition on the real gauge orbit of U. All operators below are defined covariantly, and they transform covariantly under G-valued gauge transformations (Proposition 16.3). Hence their bounds hold for every background that is gauge equivalent, by a G-valued gauge transformation, to one satisfying (B), with derivatives replaced by covariant derivatives and Hölder quotients taken in the gauge in which (B) holds. The comparisons with the operators at $U = 1$ are different: they hold in that gauge, and they transfer only as comparisons with the transformed reference 1^u. Apart from these comparisons, the special gauge is only a device in the proofs. For G^c-valued gauge transformations the operator $R(u)$ need not be an isometry, and the bounds do not transfer.

How restrictive (B) is. Hypothesis (B) is much stronger than Bałaban's local conditions, and it is not a local condition. The basic constraint is an isoperimetric bound. Let S be a planar lattice region and ∂S its boundary contour. If (B) holds, then $|U(\partial S) - 1| \le e^{\varepsilon|\partial S|} - 1$,

where $|\partial S|$ is the length of ∂S in unit lengths. So all Wilson loops obey a perimeter bound with constant ε.

A background whose field strength is of size ε and *coherent* over a region of side ℓ has $U(\partial S) \approx e^{\mathrm{i}\varepsilon\ell^2}$ for squares S of side ℓ. In the abelian case, for $\varepsilon\ell^2 < \pi$, the perimeter bound then forces $\ell \lesssim 4$. Thus (B) essentially requires the background to be close to flat on all scales larger than $O(1)$. Field strengths of size ε are allowed only if they average out on the unit scale. Backgrounds with non-zero flux through a 2-cycle of the torus are the extreme case. In the abelian case, every gauge with $\eta|A| < \pi/4$ has vanishing total flux, because each plaquette angle is then the principal one and the flux is a sum of differences. So non-zero flux forces $|A| \geq \pi/(4\eta)$ somewhere, whatever the size of the torus.

Flat backgrounds, on the other hand, can satisfy (B). For a real flat background, if the holonomies h_μ around the cycles lie in a common maximal torus of G (always the case for $U(\mathsf{N})$ and $SU(\mathsf{N})$), one writes $h_\mu = e^{\mathrm{i}\theta_\mu}$ with commuting θ_μ and spreads them evenly, $A_\mu = \theta_\mu/n$. This gives (B) once $n \geq C_G/\varepsilon$. For other groups (e.g. $SO(3)$) commuting holonomies need not lie in a common torus.

Coherent field strengths of size ε over many blocks are natural inputs for the variational problem and typical configurations in the renormalization group. Hypothesis (B) excludes them. Section 51 discusses the consequences and the route back to local conditions.

12.1 Lie algebra valued fields

Let $G \subset U(\mathsf{N})$ be a compact connected Lie group, and let $\mathfrak{g}$ be the real Lie algebra of hermitian matrices X with $e^{\mathrm{i}tX} \in G$ for every $t \in \mathbb{R}$, following the convention of [3, 7]. Connectedness is used through $\det R(g) = 1$ on $\mathfrak{g}$, for $g \in G$ and for $g \in G^c$ (Propositions 32.5 and 47.6). For connected G this holds: $\det R(e^{\mathrm{i}X}) = e^{\mathrm{i}\,\mathrm{tr}\,\mathrm{ad}\,X} = 1$, since $\mathrm{ad}\,X$ is antisymmetric for the trace form, and G and G^c are generated by exponentials. For disconnected groups it can fail: on $(SU(2) \times SU(2)) \rtimes \mathbb{Z}_2$ the exchange of the factors has $\det R = -1$. Let $\mathfrak{g}^c = \mathfrak{g} + \mathrm{i}\mathfrak{g}$ be its complexification. On $\mathfrak{g}^c$ we use the bilinear form $\mathrm{b}(X, Y) = \mathrm{tr}(XY)$ and the Hilbert–Schmidt norm $|X| = (\mathrm{tr}\, X^*X)^{1/2}$; on $\mathfrak{g}$ the two are related by $\mathrm{b}(X, X) = |X|^2$. Then $|[X, Y]| \leq 2|X||Y|$, and for $Z \in \mathfrak{g}^c$ the operator $R(e^Z)Y = e^Z Y e^{-Z} = e^{\mathrm{ad}\,Z}Y$ satisfies

$$\|R(e^Z) - 1\| \leq e^{2|Z|} - 1. \tag{12.1}$$

Scalar fields, gauge fields and 2-forms are now $\mathfrak{g}^c$-valued. The inner products of Section 2 are replaced by

$$\mathrm{b}(f, g) = \eta^d \sum_x \mathrm{tr}\,(f(x)g(x)) \quad \text{(bilinear)}, \qquad \langle f, g\rangle = \eta^d \sum_x \mathrm{tr}\,(f(x)^* g(x)) \quad \text{(hermitian)},$$

with the analogous expressions for gauge fields, 2-forms and unit-lattice fields; norms $\|\cdot\|$ always refer to $\langle\cdot,\cdot\rangle$. The *transpose* X^T of a linear operator is taken with respect to b. If X maps $\mathfrak{g}$-valued fields to $\mathfrak{g}$-valued fields, X^T is the complex-linear extension of its adjoint on real fields.

Choosing an orthonormal basis (t_a) of $\mathfrak{g}$, every $\mathfrak{g}^c$-valued field is a $\dim \mathfrak{g}$-tuple of complex scalar fields. All operators of Part I act componentwise and are real. Hence every result of Sections 2–11 holds for $\mathfrak{g}^c$-valued fields. The L^2 statements hold with the same constants. For the pointwise statements, the kernel arguments of Section 6 go through with Hilbert–Schmidt norms; applying them componentwise instead costs a factor depending on $\dim \mathfrak{g}$. In particular, for real symmetric K the coercivity bounds of Theorem A hold for $\mathrm{Re}\langle u, Ku\rangle$ with complex u.

12.2 Backgrounds and the hypothesis (B)

A *background* is an assignment of $U(b) \in G^c$ to the oriented bonds b of $\mathbb{T}_\eta$, with $U(-b) = U(b)^{-1}$. We write $U_\mu(x) = U(x, x + \eta e_\mu)$, and $U(\Gamma)$ for the ordered product along a contour Γ. A

gauge transformation is a map $u : \mathbb{T}_\eta \to G^c$; it is *real* if u takes values in G. It acts by $U^u(b) = u(b_-)U(b)u(b_+)^{-1}$ on backgrounds and by $\lambda \mapsto R(u)\lambda$ on scalar and gauge fields.

Fix $\alpha \in (0,1)$ and $\varepsilon > 0$.

Definition 12.1 (Hypothesis (B))**.** A background satisfies (B) with parameter ε if $U_\mu(x) = e^{i\eta A_\mu(x)}$ for a $\mathfrak{g}^c$-valued gauge field A with

$$|A|_{\mathbb{T}_\eta} \le \varepsilon, \qquad |\partial A|_{\mathbb{T}_\eta} \le \varepsilon, \qquad \sup_z [\partial A]_{\alpha,\Lambda_3(z)} \le \varepsilon.$$

These bounds have the form of Bałaban's regularity conditions on the background in a cube [7, (3.35)–(3.36)], and of the regularity bounds [8, (9)] that the minimizers satisfy, but here they are imposed in one global gauge. The set of A satisfying (B) with strict inequalities is open and convex. The uniform bounds and the analyticity statements below hold on the smallness domains specified in the respective results. Operators that involve W are analytic in (A, W); they are analytic in A alone for fixed W or for an analytic choice $W = W(A)$.

12.3 Covariant derivatives and curvature

For scalar fields,

$$(D_\mu\lambda)(x) = \frac{R(U_\mu(x))\lambda(x+\eta e_\mu) - \lambda(x)}{\eta}, \qquad (D\lambda)_\mu = D_\mu\lambda,$$

and for gauge fields $(DA)_{\mu\nu} = D_\mu A_\nu - D_\nu A_\mu$, where D_μ acts on the scalar field A_ν. This is [7, (3.3), (3.4)]. We write $D = D_U$ when the background must be indicated. The transposes D^T play the role of d^*. At $U = 1$ we recover d and d^*.

Lemma 12.2 (Expansion of covariant derivatives)**.** *Assume (B) with $\varepsilon \le 1$. Then $D_\mu = \partial_\mu + T_\mu$ with $(T_\mu\lambda)(x) = \tau_\mu(x)\lambda(x+\eta e_\mu)$, where $\tau_\mu(x) = \eta^{-1}(R(e^{i\eta A_\mu(x)}) - 1)$ is a linear map of $\mathfrak{g}^c$ satisfying*

$$|\tau_\mu| \le 2e^2\varepsilon, \qquad |\partial_\nu\tau_\mu| \le C\varepsilon, \qquad [\partial_\nu\tau_\mu]_{\alpha,\Lambda_3(z)} \le C\varepsilon.$$

Consequently, on scalar fields and componentwise on gauge fields,

$$D^TD = -\Delta + \sum_\mu \partial_\mu^*(a_\mu\,\cdot\,) + b, \tag{12.2}$$

where a_μ and b are multiplication operators composed with shifts by at most η, with $|a_\mu| + |\partial a_\mu| + |b| \le C\varepsilon$ and $[\partial a_\mu]_{\alpha,\Lambda_3(z)} + [b]_{\alpha,\Lambda_3(z)} \le C\varepsilon$. The same normal form holds for $D^TD + DD^T$ on gauge fields, which equals $-\Delta$ componentwise when $U = 1$ (Lemma 2.2).

Proof. The formula for D_μ is immediate. The bounds on τ follow from (12.1). Moreover $\tau_\mu(x) = \phi(\eta A_\mu(x))\,\mathrm{i}\,\mathrm{ad}\,A_\mu(x)$, where $\phi(Z) = (e^{\mathrm{i}\,\mathrm{ad}\,Z} - 1)/(\mathrm{i}\,\mathrm{ad}\,Z)$ is an entire function of Z; differentiating gives the bounds on $\partial\tau$ and its Hölder quotient. Next, $D_\mu^T = \partial_\mu^* + T_\mu^T$ with $(T_\mu^T w)(x) = \tau_\mu(x-\eta e_\mu)^T w(x-\eta e_\mu)$. Write

$$D_\mu^TD_\mu = \partial_\mu^*\partial_\mu + \partial_\mu^*T_\mu + T_\mu^T\partial_\mu + T_\mu^TT_\mu.$$

The second term is already in divergence form with $a_\mu u = \tau_\mu S_\mu u$. For the third, $T_\mu^T\partial_\mu u = -\tilde\tau_\mu\,\partial_\mu^* u$ with $\tilde\tau_\mu(x) = \tau_\mu(x-\eta e_\mu)^T$, and the discrete Leibniz rule gives $\tilde\tau_\mu\partial_\mu^* u = \partial_\mu^*(\tilde\tau_\mu u) - (\partial_\mu^*\tilde\tau_\mu)S_\mu^{-1}u$. The last term is bounded. Collecting terms gives (12.2). The computation for gauge fields is the same, term by term. □

On scalar fields $DD \neq 0$ in general. Let $\mathfrak{F} = D \circ D$ (scalar fields to 2-forms) denote the *curvature operator*.

Lemma 12.3 (Curvature operator)**.** $(\mathfrak{F}\lambda)_{\mu\nu}(x) = c_{\mu\nu}(x)\,\lambda(x+\eta e_\mu+\eta e_\nu)$ *with*

$$c_{\mu\nu}(x) = \eta^{-2}\Big(R(U_\mu(x)U_\nu(x+\eta e_\mu)) - R(U_\nu(x)U_\mu(x+\eta e_\nu))\Big).$$

Under (B) with $\varepsilon \le 1$, $|c_{\mu\nu}| \le C\varepsilon$ *and* $[c_{\mu\nu}]_{\alpha,\Lambda_3(z)} \le C\varepsilon$.

Proof. Expanding $D_\mu D_\nu \lambda(x)$ gives four terms. After antisymmetrization in (μ,ν), the three terms $-R(U_\mu(x))\lambda(x+\eta e_\mu)$, $-R(U_\nu(x))\lambda(x+\eta e_\nu)$ and $\lambda(x)$ cancel. This leaves the formula. Write $c = \eta^{-2}R(X_2)(R(X_2^{-1}X_1)-1)$. Then $X_2^{-1}X_1 = e^{-\mathrm{i}a'}e^{-\mathrm{i}b}e^{\mathrm{i}a}e^{\mathrm{i}b'}$, where

$$a = \eta A_\mu(x), \quad a' = \eta A_\mu(x+\eta e_\nu), \quad b = \eta A_\nu(x), \quad b' = \eta A_\nu(x+\eta e_\mu).$$

Let $\Psi(a,a',b,b') = e^{-\mathrm{i}a'}e^{-\mathrm{i}b}e^{\mathrm{i}a}e^{\mathrm{i}b'} - 1$, an entire function. Then

$$\Psi(a,a',b,b') = \big[\Psi(a,a',b,b') - \Psi(a,a,b,b)\big] + \Psi(a,a,b,b).$$

By Hadamard's lemma the bracket equals $(a'-a)\cdot\Psi_1 + (b'-b)\cdot\Psi_2$, with Ψ_i analytic functions of (a,a',b,b'). Moreover $a'-a = \eta^2\partial_\nu A_\mu(x)$ and $b'-b = \eta^2\partial_\mu A_\nu(x)$. The group commutator $\Psi(a,a,b,b)$ vanishes when $a=0$ or $b=0$, so it equals $\sum_{i,j} a_i b_j \Psi_{ij}(a,b)$ with analytic Ψ_{ij}. Hence $\eta^{-2}\Psi$ is a sum of products of ∂A, of $A\otimes A$, and of analytic functions of the four arguments $\eta A(\cdot)$. Each argument is bounded by $\eta\varepsilon \le 1$ and Lipschitz in x with constant $\eta\varepsilon$, and $|A\otimes A| \le \varepsilon^2 \le \varepsilon$. Each factor is therefore bounded by $C\varepsilon$ or C, and is Hölder on $\Lambda_3(z)$ with seminorm $C\varepsilon$. Finally $(R(X_2^{-1}X_1)-1)Y = [\Psi,Y](1+\Psi)^{-1}$ for every Y, and $(1+\Psi)^{-1} = X_1^{-1}X_2$ is a product of four exponentials of arguments of size at most $\eta\varepsilon \le 1$, hence bounded and Lipschitz in x. Also $R(X_2)$ is Lipschitz. □

At leading order $c_{\mu\nu} \approx \mathrm{i}\,\mathrm{ad}\,F_{\mu\nu}$, where F is the curvature on the unit scale.

12.4 The quadratic form of the action

Definition 12.4 (Admissible potential terms)**.** For $\varepsilon,\gamma > 0$, let $\mathcal{W}(\varepsilon,\gamma)$ be the set of linear operators W on gauge fields such that $W^T = W$ and, for all y, y' and all u,

$$\|\mathbf{1}_y W \mathbf{1}_{y'}\| \le \varepsilon e^{-\gamma|y-y'|}, \qquad |Wu|_{\Lambda_1(c_y)} + [Wu]_{\alpha,\Lambda_1(c_y)} \le \varepsilon \sum_{y'} e^{-\gamma|y-y'|}\big(|u|_{\Lambda_1(c_{y'})} + [u]_{\alpha,\Lambda_1(c_{y'})}\big).$$

Here and below c_y is the centre of the block $B(y)$.

The quadratic form that replaces $\|dA\|^2$ is

$$q_U(A,A') = \mathrm{b}(DA, DA') + \mathrm{b}(A, WA'), \qquad W \in \mathcal{W}(\varepsilon,\gamma).$$

Two potentials motivate this definition. For $W = \Delta'(U)$, the curvature term of [7, (3.10)], $q_U(A,A)$ is the second-order term of the Wilson action at U. It is expected to be a local operator whose coefficients are, up to bounded analytic factors, the curvature coefficients $c_{\mu\nu}$ of Lemma 12.3. The linearizing transformation of the variational problem [8, (79)], [7, (3.127)–(3.128)] produces a further quadratic term $-2\langle HC^{(2)}(A), J\rangle$, proportional to the source J. We do not verify that these two terms belong to $\mathcal{W}$, and nothing below uses it. The results of Part II are stated for general W. Later parts use $W = 0$, except for the potential W_2 of Section 37, for which only L^2 statements are used; they are proved there.

12.5 Averaging operators with a background

For $y \in \mathbb{T}$ and $x \in B(y)$, let $\Gamma_{y,x}$ be Bałaban's contour from y to x inside $B(y)$ [3, (1.7)]: it changes the coordinates in the order $d, d-1, \dots, 1$. For $x_j = x + j\eta e_\mu$, let $[x, x_j]$ be the straight contour. Define

$$(Q'(U)\lambda)(y) = \eta^d \sum_{x \in B(y)} R\big(U(\Gamma_{y,x})\big)\lambda(x),$$

$$(\mathbf{Q}(U)A)_\mu(y) = \eta^d \sum_{x \in B(y)} \eta \sum_{j=0}^{N-1} R\big(U(\Gamma_{y,x} \cup [x, x_j])\big)A_\mu(x_j).$$

Both average after parallel transport to the corner y. At $U = 1$ they reduce to Q_0 and $\mathbf{Q}$. $Q'(U)$ is the linear scalar averaging of [7, (3.18), (3.19)]. $\mathbf{Q}(U)$ is the main term [5, (125)] of the linearization [5, (124)] of the one-step gauge-field averaging; the remaining small terms of [5, (124)] are dropped. On the unit lattice let $V(y, y+e_\mu) = U([y, y+e_\mu])$ (straight contour), and $(D_V g)_\mu(y) = R(V(y, y+e_\mu))g(y+e_\mu) - g(y)$.

Lemma 12.5 (Properties of the averages)**.** *Assume (B) with $\varepsilon \le 1$.*

(a) $Q'(U)Q'(U)^T = I$, *and* $(Q'(U)^T g)(x) = R\big(U(\Gamma_{y(x),x})\big)^{-1} g(y(x))$.

(b) $Q'(U) - Q_0$ *and* $\mathbf{Q}(U) - \mathbf{Q}$ *act within distance* 2 *(that is,* $\mathbf{1}_y(\cdot)\mathbf{1}_{y'} = 0$ *for* $|y - y'| > 2$*), have block norms at most* $C\varepsilon$*, and satisfy the pointwise bounds* $|(Q'(U) - Q_0)\lambda(y)| \le C\varepsilon|\lambda|_{B(y)}$, $|(\mathbf{Q}(U) - \mathbf{Q})A(y)| \le C\varepsilon|A|_{B(y)\cup B(y+e_\mu)}$. *The same holds for the transposes.*

(c) (Covariance defect.) $\mathbf{Q}(U)D_U = D_V Q'(U) + E(U)$, *where*

$$(E(U)\lambda)_\mu(y) = \eta^d \sum_{x' \in B(y+e_\mu)} \Big[R\big(U(\Gamma_{y,x'-e_\mu} \cup [x'-e_\mu, x'])\big) - R\big(V(y, y+e_\mu)U(\Gamma_{y+e_\mu,x'})\big)\Big]\lambda(x').$$

In particular $E(U)$ acts within distance 2, $|E(U)\lambda(y)| \le C\varepsilon|\lambda|_{B(y+e_\mu)}$, *and* $\|E(U)\| \le C\varepsilon$. *Moreover* $|E(U)^T\omega|_\infty \le C\varepsilon|\omega|_\infty$*: by the formula, $(E(U)^T\omega)(x')$ is a sum of d terms, one for each μ, each the transpose of a bracket of norm at most $C\varepsilon$ applied to $\omega_\mu(y)$ with $x' \in B(y+e_\mu)$.*

(d) (Gauge covariance.) *For every gauge transformation u, $Q'(U^u)R(u) = R(u|_{\mathbb{T}})Q'(U)$, $\mathbf{Q}(U^u)R(u) = R(u|_{\mathbb{T}})\mathbf{Q}(U)$, $D_{U^u}R(u) = R(u)D_U$ and $E(U^u)R(u) = R(u|_{\mathbb{T}})E(U)$.*

Proof. (a) uses $R(X)^T = R(X^{-1})$ with respect to b, together with the fact that the blocks partition $\mathbb{T}_\eta$. (b) Every contour involved has at most $(d+1)N$ bonds, so $|U(\Gamma) - 1| \le e^{(d+1)\varepsilon} - 1$ and similarly for $U(\Gamma)^{-1}$, by submultiplicativity. Then $R(g)Y - Y = (g-1)Yg^{-1} + Y(g^{-1} - 1)$ gives $\|R(g) - 1\| \le |g-1||g^{-1}| + |g^{-1} - 1| \le C\varepsilon$ for $g = U(\Gamma)$. (c) Along each line the sum over j telescopes:

$$\eta \sum_{j=0}^{N-1} R\big(U(\Gamma \cup [x, x_j])\big)(D_\mu\lambda)(x_j) = R\big(U(\Gamma \cup [x, x+e_\mu])\big)\lambda(x+e_\mu) - R(U(\Gamma))\lambda(x),$$

with $\Gamma = \Gamma_{y,x}$. Summing over $x \in B(y)$, the second terms give $Q'(U)\lambda(y)$, and the first terms differ from $R(V)Q'(U)\lambda(y+e_\mu)$ exactly by $E(U)\lambda$. Both transports in the bracket are within $C\varepsilon$ of 1, by the bound in (b). (d) is the usual transformation law of contour products, $U^u(\Gamma) = u(\Gamma_-)U(\Gamma)u(\Gamma_+)^{-1}$. □

Remark 12.6 (Which averaging?). Only properties (a)–(d) of $Q'(U)$ and $\mathbf{Q}(U)$ are used below; the identity in (a) supplies a right inverse of $Q'(U)$, and the formula for $Q'(U)^T$ in (a) is used, with the bounds of (b), for pointwise bounds and trial fields. Bałaban's k-step averaging is a composition of nonlinear one-step averagings with intermediate backgrounds [7, (3.15)], linearized with the full formula [5, (124)]. For it the covariance identity $\mathbf{Q}(U)D_U = D_V Q'(U)$ is exact [7, (3.114)–(3.115)], i.e. $E = 0$; the identities [7, (3.124), (3.152)] are consequences. Our model keeps only the main term [5, (125)], and the defect $E(U)$ is the price of this simplification. Our treatment does not require $E = 0$. Its real statements apply to every averaging with properties (i)–(iv) of Remark 19.12, which correspond to (a)–(d) above; the analytic statements need in addition property (v) of that remark. In Part IV we construct Bałaban's averaging and show that its linearization has these properties with $E = 0$ (Proposition 22.7, Corollary 22.8).

13 Perturbation lemmas

Under hypothesis (B), every operator of Part II will be written as the corresponding operator of Part I plus a perturbation of a specific structure. This section shows that the three tools of Part I (Combes–Thomas, inversion with decay, local regularity) are stable under such perturbations.

13.1 Complex fields

Lemmas 5.5, 5.6 and 5.7 hold for operators on complex ($\mathfrak{g}^c$-valued) fields that are not necessarily symmetric, provided the lower bounds are imposed on $\mathrm{Re}\langle u, Ku\rangle$ for all complex fields u (not only real ones). The proofs are unchanged: $\mathrm{Re}\langle u, K_\alpha u\rangle \geq \delta\|u\|^2$ implies $\|K_\alpha u\| \geq \delta\|u\|$, and in the proof of Lemma 5.7 one uses $|\langle u, h\rangle| \geq \mathrm{Re}\langle u, h\rangle$. We use these lemmas in this form without further comment.

13.2 Admissible perturbations

Let $\mathfrak{D}$ be the set of operators consisting of the identity and all ∂_μ, ∂_μ^*, D_μ, D_μ^T, acting on scalar fields, on gauge fields (componentwise), or between gauge fields and 2-forms. Under (B) with $\varepsilon \leq 1$, every $X \in \mathfrak{D}$ satisfies $\|Xu\| \leq 2\|\partial u\| + C\|u\|$ and $\|X^*u\| \leq 2\|\partial u\| + C\|u\|$. By Lemma 5.4 and Lemma 12.2, $X_\alpha - X$ is bounded by $C|\alpha|$ for $|\alpha| \leq 1$ and 1-Lipschitz ρ.

Definition 13.1 (Admissible perturbations)**.** An operator δ is an *admissible perturbation of size ε and rate γ* if $\delta = \sum_{i=1}^m X_i M_i Y_i$ with $m \leq m_0$, $X_i, Y_i \in \mathfrak{D}$, and each M_i has decay (ε, γ). Here m_0 is a fixed number depending only on d.

Lemma 13.2 (Stability of Combes–Thomas)**.** *Let $\mathfrak{A}_0$ satisfy the hypotheses of Lemma 5.7 with constants c, C_1, α_1, and let δ be admissible of size ε and rate γ. There is $\varepsilon_1 > 0$, depending only on c, γ, m_0 and d, such that for $\varepsilon \leq \varepsilon_1$ the operator $\mathfrak{A} = \mathfrak{A}_0 + \delta$ satisfies these hypotheses with constants $c/2$, $C_1 + 1$ and $\min(\alpha_1, \gamma/2, 1)$. In particular $\mathfrak{A}^{-1}$, $\partial\mathfrak{A}^{-1}$, $\mathfrak{A}^{-1}\partial^*$ and $\partial\mathfrak{A}^{-1}\partial^*$ have decay (C, γ_0) with C, γ_0 depending only on $c, C_1, \alpha_1, \gamma, m_0, d$. The same holds with ∂ replaced by D.*

Proof. By Lemma 5.2(a), $|\langle u, X_i M_i Y_i u\rangle| \leq \|X_i^* u\|\, \varepsilon S_\gamma\, \|Y_i u\| \leq C\varepsilon(\|u\|^2 + \|\partial u\|^2)$. This gives (i) with $c/2$ if $C\varepsilon \leq c/2$. For (ii), note that $(XMY)_\alpha - XMY$ is a sum of three terms, each containing one of the differences $X_\alpha - X$, $M_\alpha - M$, $Y_\alpha - Y$. The first and last have norm at most $C|\alpha|$, and $\|M_\alpha - M\| \leq \varepsilon c_1(\gamma)|\alpha|$ by Lemma 5.3. Every term is therefore bounded in form sense by $C\varepsilon|\alpha|(\|u\|^2 + \|\partial u\|^2)$. Replacing ∂ by D in the conclusions costs a bounded operator of size $C\varepsilon$, by Lemma 12.2. □

Lemma 13.3 (Perturbed inverses)**.** *In the situation of Lemma 13.2, let r and r_0 be decay rates of $\mathfrak{A}^{-1}$ and $\mathfrak{A}_0^{-1}$, respectively (with derivatives), and set $\gamma_0 = \min(\gamma, r, r_0)$. Then $\mathfrak{A}^{-1} - \mathfrak{A}_0^{-1} = -\mathfrak{A}^{-1}\delta\mathfrak{A}_0^{-1}$. This operator, as well as its compositions with ∂ on the left and with ∂^* on the right, has decay $(C\varepsilon, \gamma_0/4)$. The same statements hold, with ∂ replaced by 1, for bounded operators $K = K_0 + \delta$ on unit-lattice fields, where K_0 has decay, $\mathrm{Re}\langle f, K_0 f\rangle \geq c\|f\|^2$, and δ has decay (ε, γ) with $\varepsilon S_\gamma \leq c/2$. In this bounded case, set $\gamma_0 = \min(\gamma, r_K, r_{K_0})$, where r_K, r_{K_0} are chosen decay rates of K^{-1}, K_0^{-1}.*

Proof. The identity is algebraic. Each term $\mathfrak{A}^{-1}X_iM_iY_i\mathfrak{A}_0^{-1}$ is a product of the operators $\mathfrak{A}^{-1}X_i$ and $Y_i\mathfrak{A}_0^{-1}$ and of M_i, which has decay (ε, γ). Lemma 13.2 gives decay of $\mathfrak{A}^{-1}\partial^*$ and $\partial\mathfrak{A}_0^{-1}$. The remaining combinations reduce to these: $\partial_\mu = -\partial_\mu^* S_\mu$ and $\partial_\mu^* = -S_\mu^{-1}\partial_\mu$, where the shifts $S_\mu^{\pm 1}$ by η are local isometries, and $D_\mu = \partial_\mu + T_\mu$ with T_μ local and bounded. All three factors have decay at the common rate γ_0, by Lemma 5.2(c). Apply Lemma 5.2(b) twice; each application halves the rate. In the bounded case use Lemma 5.5 for K. □

13.3 Pointwise admissibility and the bootstrap

For pointwise bounds we need perturbations whose action is controlled locally. For a field u on $\mathbb{T}_\eta$ and $y \in \mathbb{T}$ put

$$N_y^0(u) = |u|_{\Lambda_1(c_y)}, \qquad N_y^1(u) = |u|_{\Lambda_1(c_y)} + |\partial u|_{\Lambda_1(c_y)}.$$

Definition 13.4 (Pointwise admissibility)**.** Let $i \in \{0, 1\}$. A linear operator δ is *pointwise admissible of type i*, with constants $(\varepsilon, C_\delta, \gamma)$, if for every field u there are fields F_u, g_u with $\delta u = F_u + \sum_\mu \partial_\mu^* g_{u,\mu}$ and, for every y,

$$|F_u|_{\Lambda_2(c_y)} + |g_u|_{\Lambda_3(c_y)} + i\,[g_u]_{\alpha,\Lambda_3(c_y)} \leq \sum_{y'} e^{-\gamma|y-y'|}\Big(\varepsilon N_{y'}^i(u) + C_\delta\|u\|_{B(y')}\Big).$$

The smallness is required only for the terms controlled by N^i. Terms controlled by local L^2 norms, such as $dP_{\mathcal{V}}d^*$ in Section 10, may be large.

Lemma 13.5 (Bootstrap)**.** *Let $d \geq 3$, and let δ be pointwise admissible of type i with constants $(\varepsilon, C_\delta, \gamma)$. There is $\varepsilon_* > 0$, depending only on d, α and γ, such that the following holds if $\varepsilon \leq \varepsilon_*$; it does not depend on C_δ, M or γ_2. Let*

$$-\Delta u = J + \delta u \qquad \text{on } \mathbb{T}_\eta,$$

with $J = J_F + \sum_\mu \partial_\mu^* J_{g,\mu}$. Suppose that for some y_0, $M > 0$ and $\gamma_2 > 0$, and for all y,

$$\|u\|_{B(y)} + |J_F|_{\Lambda_2(c_y)} + |J_g|_{\Lambda_3(c_y)} + i\,[J_g]_{\alpha,\Lambda_3(c_y)} \leq Me^{-\gamma_2|y-y_0|}.$$

Let $\gamma' = \frac{1}{2}\min(\gamma, \gamma_2)$. Then, with C depending only on $d, \alpha, \gamma, \gamma_2, C_\delta$,

$$\begin{aligned} &\textit{type } 1: && |u|_{B(y)} + |\partial u|_{B(y)} + [\partial u]_{\alpha,B(y)} \leq CMe^{-\gamma'|y-y_0|},\\ &\textit{type } 0: && |u|_{B(y)} + [u]_{\alpha,B(y)} \leq CMe^{-\gamma'|y-y_0|}. \end{aligned}$$

Proof. We give the proof for type 1; type 0 is identical, with Theorem 6.3(b) in place of (d). The lattice is finite, so

$$\mathcal{N} = \max_y e^{\gamma'|y-y_0|}N_y^1(u) < \infty.$$

Fix y and apply Theorem 6.3(d) on the cubes $\Lambda_r(c_y)$, with $F = J_F + F_u$ and $g = J_g + g_u$:

$$N_y^1(u) + [\partial u]_{\alpha,\Lambda_1(c_y)} \leq C(|F|_{\Lambda_2} + |g|_{\Lambda_3} + [g]_{\alpha,\Lambda_3} + \|u\|_{\Lambda_2}).$$

The cube $\Lambda_2(c_y)$ meets at most C blocks, all within distance C of y, so $\|u\|_{\Lambda_2} \le CMe^{-\gamma_2|y-y_0|}$. By the definition of pointwise admissibility and $N^1_{y'}(u) \le e^{-\gamma'|y'-y_0|}\mathcal{N}$,

$$|F_u|+|g_u|+[g_u]_\alpha \le \sum_{y'} e^{-\gamma|y-y'|}\Big(\varepsilon e^{-\gamma'|y'-y_0|}\mathcal{N}+C_\delta M e^{-\gamma_2|y'-y_0|}\Big) \le S_{\gamma/2}\, e^{-\gamma'|y-y_0|}(\varepsilon\mathcal{N}+C_\delta M),$$

where we used $\gamma|y-y'|+\gamma'|y'-y_0| \ge \gamma'|y-y_0|+\frac{\gamma}{2}|y-y'|$. Multiplying by $e^{\gamma'|y-y_0|}$ and maximizing over y gives $\mathcal{N} \le C_0 M + C_{\mathrm{reg}} S_{\gamma/2}\,\varepsilon\,\mathcal{N}$. Here C_{reg} is the constant of Theorem 6.3, which depends only on d, α, and C_0 depends on $d, \alpha, \gamma, \gamma_2, C_\delta$. For $\varepsilon \le \varepsilon_* = 1/(2C_{\mathrm{reg}}S_{\gamma/2})$ this yields $\mathcal{N} \le 2C_0 M$, and the Hölder bound follows from the same inequality. □

13.4 Building blocks

The perturbations in Sections 14–16 are assembled from the following pieces.

Lemma 13.6 (Building blocks)**.** *Assume (B) with $\varepsilon \le 1$.*

(a) The differential parts $D^TD+\Delta$ (on scalar fields) and $D^TD+DD^T+\Delta$ (on gauge fields) are admissible of size $C\varepsilon$ and pointwise admissible of both types with constants $(C\varepsilon, 0, 1)$.

(b) Every $W \in \mathcal{W}(\varepsilon,\gamma)$ is admissible of size ε and pointwise admissible of type 1 with constants $(C\varepsilon, 0, \gamma)$, since $[u]_{\alpha,\Lambda_1} \le C(|u|+|\partial u|)_{\Lambda_1}$.

(c) The operators $Q'(U)^TQ'(U)-Q_0^*Q_0$ and $\mathbf{Q}(U)^T\mathbf{Q}(U)-\mathbf{Q}^*\mathbf{Q}$ are admissible of size $C\varepsilon$ and pointwise admissible of both types with constants $(C\varepsilon,0,1)$. The operators $Q_0^*Q_0$ and $\mathbf{Q}^*\mathbf{Q}$ themselves are pointwise admissible with constants $(0,C,1)$.

(d) If K is an operator with values in fine fields such that $u \mapsto Ku$ satisfies $|Ku|_{B(y)} + |\partial Ku|_{B(y)} + [\partial Ku]_{\alpha,B(y)} \le C_K \sum_{y'} e^{-\gamma|y-y'|}\|u\|_{B(y')}$, then $\partial_\mu K$ is pointwise admissible with constants $(0, C\,C_K, \gamma)$ (it contributes only to F_u).

Proof. (a) By (12.2), $(D^TD+\Delta)u = \sum_\mu \partial_\mu^*(a_\mu u) + bu$, and $[a_\mu u]_\alpha \le C\varepsilon(|u|+|\partial u|)$ on each cube. For type 0 the Hölder quotient is not needed. Admissibility in the sense of Definition 13.1 follows from the same formula: $\partial^* \circ a_\mu \circ 1$ and $1 \circ b \circ 1$ with a_μ, b local of size $C\varepsilon$. (b) is the definition of $\mathcal{W}$. (c) By Lemma 12.5(b), the differences are local, bounded by $C\varepsilon$, and their values are bounded pointwise by $C\varepsilon|u|$ on neighbouring blocks. They contribute only to F_u. The second statement follows from $|Q_0^*Q_0u| \le \max|Q_0u| \le \|u\|$ on blocks (Lemma 2.6). (d) is immediate. □

14 Scalar operators with a background

Throughout Sections 14–16 we assume (B) with parameter ε, fix $a > 0$ and $\alpha \in (0,1)$, and let C, γ denote constants depending only on d, a, α, $\dim \mathfrak{g}$ and, from Section 15 on, the decay rate $\gamma_W > 0$ of the potential term W. The rate γ_W is fixed first. The statement "for $\varepsilon \le \varepsilon_0$" means for ε below a threshold depending on the same quantities. It cannot be chosen uniformly as $\gamma_W \to 0$ (Remark 15.2). Pointwise statements assume $d \ge 3$.

The scalar operators are built from

$$\mathfrak{A}'(U) = D^TD + a\,Q'(U)^TQ'(U), \qquad \mathfrak{A}'(1) = -\Delta + aQ_0^*Q_0.$$

At real U, $\langle\lambda, \mathfrak{A}'(U)\lambda\rangle = \|D\lambda\|^2 + a\|Q'(U)\lambda\|^2$. This is the operator Δ'_a of [7, (3.24)] on a single torus. Its inverse $G'(U)$ is [7, G', (3.25)].

Theorem 14.1 (Scalar operators with a background)**.** *For $\varepsilon \le \varepsilon_0$:*

(a) $\mathfrak{A}'(U) = \mathfrak{A}'(1) + \delta'$, *where* δ' *is admissible of size* $C\varepsilon$ *and pointwise admissible of both types with constants* $(C\varepsilon, 0, 1)$. *Hence* $G'(U) = \mathfrak{A}'(U)^{-1}$ *exists and is analytic in* A. *Moreover* G', DG', $G'D^T$, $DG'D^T$ *have decay* (C, γ), *and* $G'(U) - G_0$ *composed with derivatives has decay* $(C\varepsilon, \gamma)$.

(b) *For* f *supported in* $B(y')$,

$$|G'f|_{B(y)} + |\partial G'f|_{B(y)} + [\partial G'f]_{\alpha,B(y)} \le Ce^{-\gamma|y-y'|}|f|_{B(y')},$$
$$|G'D_\mu^T f|_{B(y)} + [G'D_\mu^T f]_{\alpha,B(y)} \le Ce^{-\gamma|y-y'|}|f|_{B(y')}.$$

If f *is merely bounded and Hölder, then*

$$|\partial G'D_\mu^T f|_{B(y)} + [\partial G'D_\mu^T f]_{\alpha,B(y)} \le C\sum_{y'} e^{-\gamma|y-y'|}(|f|_{\Lambda_3(c_{y'})} + [f]_{\alpha,\Lambda_3(c_{y'})}).$$

(c) $Q'G'Q'^T$ *differs from* $Q_0G_0Q_0^*$ *by an operator with decay* $(C\varepsilon, \gamma)$. *Its inverse* $\Gamma'(U)$ *exists and has decay* (C, γ).

(d) *The minimizer map* $H'(U) = G'Q'^T\Gamma'$ *satisfies* $Q'H' = I$. *For real* U, $H'g$ *is the unique minimizer of* $\|D\lambda\|^2$ *subject to* $Q'(U)\lambda = g$. *Moreover* $|H'g|_{B(y)} + |\partial H'g|_{B(y)} + [\partial H'g]_{\alpha,B(y)} \le C\sum_{y'} e^{-\gamma|y-y'|}|g(y')|$.

(e) *The constrained Green function* $C'(U) = G' - H'Q'G'$ *satisfies* $Q'C' = 0 = C'Q'^T$. *For every* f, $C'f$ *is the unique* $\nu \in \mathcal{N}(Q'(U))$ *with* $\mathrm{b}(D\nu', D\nu) = \mathrm{b}(\nu', f)$ *for all* $\nu' \in \mathcal{N}(Q'(U))$. *The operators* C', DC', $C'D^T$ *and* $DC'D^T$ *have decay* (C, γ), $C'(U) - C'(1)$ *composed with derivatives has decay* $(C\varepsilon, \gamma)$, *and the pointwise bounds of (b) hold with* G' *replaced by* C'.

Proof. (a) $\delta' = (D^TD + \Delta) + a(Q'^TQ' - Q_0^*Q_0)$, so Lemma 13.6(a),(c) applies. The operator $\mathfrak{A}'(1)$ satisfies the hypotheses of Lemma 5.7 by the proof of Theorem 7.1. Lemmas 13.2 and 13.3 give the decay statements. Analyticity holds because $\mathfrak{A}'(U)$ is a polynomial in the entries of $R(e^{i\eta A})$, and its inverse exists with a uniform bound on the open set where (B) holds.

(b) Let $u = G'f$. Then $-\Delta u = J - \delta' u - aQ_0^*Q_0u$ with $J = f$. The term $aQ_0^*Q_0u$ is pointwise admissible with constants $(0, C, 1)$, and $\|u\|_{B(y)} \le Ce^{-\gamma|y-y'|}\|f\|$ by (a). Lemma 13.5 of type 1 gives the first bound. For $u = G'D_\mu^T f$ we have $J = \partial_\mu^* f + T_\mu^T f$, with $J_g = fe_\mu$ bounded. Lemma 13.5 of type 0 applies. For bounded Hölder f that is not localized, split $f = \sum_{y'} \chi_{y'} f$ with a smooth partition of unity $(\chi_{y'})$ subordinate to the cubes $\Lambda_1(c_{y'})$, with bounded derivatives. Then $[\chi_{y'} f]_\alpha \le C(|f| + [f]_\alpha)_{\Lambda_3(c_{y'})}$. Apply Lemma 13.5 of type 1 to each piece, with $J_g = \chi_{y'} fe_\mu$, and sum; this explains the Λ_3 norms on the right-hand side.

(c) $Q'G'Q'^T - Q_0G_0Q_0^* = (Q' - Q_0)G'Q'^T + Q_0(G' - G_0)Q'^T + Q_0G_0(Q'^T - Q_0^*)$. By Lemma 12.5(b) and (a), this has decay $(C\varepsilon, \gamma)$. Since $Q_0G_0Q_0^* \ge c_F$, the bounded version of Lemma 13.3 applies.

(d) $Q'H' = Q'G'Q'^T\Gamma' = I$. For real U the variational characterization is proved exactly as in Lemma 8.3, with $\langle f, \mathfrak{A}'f\rangle = \|Df\|^2 + a\|g\|^2$ on $\{Q'f = g\}$. The pointwise bound follows from (b) applied to $f = Q'^T\Gamma'g$, whose values on $B(y'')$ are bounded by $(1 + C\varepsilon)|\Gamma'g(y'')|$, by Lemma 12.5(a),(b). The factor $1 + C\varepsilon$ appears because $R(U(\Gamma))^{-1}$ need not be an isometry for complex U.

(e) The identities are algebraic. If $u = C'f$, then $\mathfrak{A}'u = f - Q'^T\Gamma'Q'G'f$ and $Q'u = 0$. Hence $D^TDu = f - Q'^T\omega$, which gives the characterization. For uniqueness, let $\nu \in \mathcal{N}(Q')$ with $\mathrm{b}(D\nu', D\nu) = 0$ for all $\nu' \in \mathcal{N}(Q')$. Then $D^TD\nu \in \mathcal{N}(Q')^{\perp_\mathrm{b}} = \operatorname{ran} Q'^T$, say $D^TD\nu = Q'^T\omega$. Hence $\mathfrak{A}'\nu = Q'^T\omega$, i.e. $\nu = G'Q'^T\omega$. Then $0 = Q'\nu = Q'G'Q'^T\omega$ gives $\omega = 0$ by (c), and so $\nu = 0$. The bounds follow from $C'f = G'f - H'(Q'G'f)$ and from (a), (c) and (d); at $U = 1$, $C'(1)$ is the operator C_σ of Proposition 8.6 with $\sigma = 0$. □

14.1 The gauge-fixing projection

Let

$$\mathcal{W}(U) = D^T D\,\mathcal{N}(Q'(U)), \qquad \mathcal{V}(U) = \operatorname{ran}\,(G'(U)Q'(U)^T), \qquad M'(U) = Q'G'^2Q'^T,$$

$$P_{\mathcal{V}}(U) = G'Q'^T M'^{-1}Q'G', \qquad R(U) = I - P_{\mathcal{V}}(U).$$

Proposition 14.2 (The projection $R(U)$)**.** *For $\varepsilon \le \varepsilon_0$:*

(a) $M'(U)$ is invertible and M'^{-1} has decay (C,γ).

(b) $R(U)^2 = R(U) = R(U)^T$, $\operatorname{ran} R(U) = \mathcal{W}(U)$ and $\ker R(U) = \mathcal{V}(U)$. In particular $R(U)$ does not depend on a. For real U, it is the orthogonal projection onto $\mathcal{W}(U)$, as in [7, (3.21), (3.25)].

(c) For $h \in \mathcal{W}(U)$: $G'h = C'h \in \mathcal{N}(Q'(U))$ and $D^T DG'h = h$. Equivalently, $Q'G'R = 0$ and $D^T DG'R = R$.

(d) $P_{\mathcal{V}}$, $DP_{\mathcal{V}}$, $P_{\mathcal{V}}D^T$ and $DP_{\mathcal{V}}D^T$ have decay (C,γ), and $P_{\mathcal{V}}(U) - P_{\mathcal{V}}(1)$ has decay $(C\varepsilon,\gamma)$.

(e) Pointwise: $P_{\mathcal{V}}h = G'Q'^T\omega$ with $|\omega(y)| \le C\sum_{y'} e^{-\gamma|y-y'|}\|h\|_{B(y')}$. Hence $|P_{\mathcal{V}}h|_{B(y)} + |\partial P_{\mathcal{V}}h|_{B(y)} + [\partial P_{\mathcal{V}}h]_{\alpha,B(y)} \le C\sum_{y'} e^{-\gamma|y-y'|}\|h\|_{B(y')}$. The same holds for $P_{\mathcal{V}}D^T u$, with $\|h\|$ replaced by $\|u\|$.

Proof. (a) $M'(U) - M'(1)$ has decay $(C\varepsilon,\gamma)$ by Theorem 14.1, and $M'(1) = M \ge c_F^2$ (Proposition 8.8). Apply Lemma 13.3.

(b) With $T = G'Q'^T$ we have $T^T = Q'G'$, because $\mathfrak{A}'^T = \mathfrak{A}'$, and $P_{\mathcal{V}} = T(T^TT)^{-1}T^T$. Hence $P_{\mathcal{V}}^2 = P_{\mathcal{V}} = P_{\mathcal{V}}^T$, $\operatorname{ran} P_{\mathcal{V}} = \mathcal{V}$, and $\ker P_{\mathcal{V}} = \ker(Q'G')$. For $\nu \in \mathcal{N}(Q')$,

$$Q'G'D^T D\nu = Q'G'(\mathfrak{A}' - aQ'^TQ')\nu = Q'\nu = 0,$$

so $\mathcal{W} \subset \ker(Q'G')$. The map D^TD is injective on $\mathcal{N}(Q')$: if $D^TD\nu = 0$ with $Q'\nu = 0$, then $\mathfrak{A}'\nu = 0$, so $\nu = 0$. Since Q' is onto ($Q'Q'^T = I$), $\dim\mathcal{W} = \dim\mathcal{N}(Q')$. On the other hand $Q'G'$ is onto, because $Q'G'Q'^T$ is invertible, so $\dim\ker(Q'G') = \dim\mathcal{N}(Q')$ as well. Therefore $\mathcal{W} = \ker P_{\mathcal{V}} = \operatorname{ran} R$. Finally $\mathrm{b}(G'Q'^T\omega, D^TD\nu) = \mathrm{b}(\omega, Q'G'D^TD\nu) = 0$, so $\mathcal{V}$ is the b-annihilator of $\mathcal{W}$, which does not involve a.

(c) If $h = D^TD\nu$ with $\nu \in \mathcal{N}(Q')$, then $\mathfrak{A}'\nu = h$, so $G'h = \nu$. Moreover $C'h = G'h - H'Q'G'h = \nu$.

(d) and (e) are compositions of operators with decay, using Theorem 14.1(b) for the pointwise statement. □

15 The gauge-field Green function with a background

15.1 The slice projection

The residual gauge directions are $D\mathcal{N}(Q'(U))$, and Bałaban's gauge slice is

$$\mathcal{S}(U) = \{A :\ R(U)D^TA = 0\}.$$

At $U = 1$ the identities $dd = 0$ and $\mathbf{Q}d = d_1Q_0$ made the slice and the gauge directions interact trivially with the action and the averaging. With a background these identities fail by the curvature operator $\mathfrak{F} = DD$ and the defect $E(U)$. Following Bałaban [7, (3.118)–(3.119)], we correct for this with the projection onto the slice along the gauge directions. Let

$$S(U) = G'(U)R(U)D^T, \qquad \Pi(U) = I - DS(U).$$

Lemma 15.1 (Slice projection)**.** *For $\varepsilon \le \varepsilon_0$:*

(a) $S\,D\nu = \nu$ for $\nu \in \mathcal{N}(Q'(U))$, and $\operatorname{ran} S \subset \mathcal{N}(Q'(U))$. Hence $SDS = S$ and $\Pi^2 = \Pi$.

(b) $\Pi D\nu = 0$ for $\nu \in \mathcal{N}(Q'(U))$, and $RD^T\Pi = 0$. Thus Π is the projection onto $\mathcal{S}(U)$ along $D\mathcal{N}(Q'(U))$, and $\Pi A = A$ for $A \in \mathcal{S}(U)$.

(c) $D\Pi = D - \mathfrak{F}S$ and $\mathbf{Q}(U)\Pi = \mathbf{Q}(U) - E(U)S$.

(d) At $U = 1$: $\Pi(1) = I - dG_0Rd^*$, $d\Pi(1) = d$ and $\mathbf{Q}\Pi(1) = \mathbf{Q}$.

Proof. (a) For $\nu \in \mathcal{N}(Q')$ we have $D^TD\nu \in \mathcal{W}$, so $RD^TD\nu = D^TD\nu$ and $G'D^TD\nu = \nu$ by Proposition 14.2(c). Also $\operatorname{ran} S \subset \operatorname{ran} G'R \subset \mathcal{N}(Q')$ by the same proposition.

(b) The first claim follows from (a). Next, $RD^T\Pi = RD^T - RD^TDG'RD^T = RD^T - R\,RD^T = 0$, by $D^TDG'R = R$.

(c) $D\Pi = D - DDS$. Moreover $\mathbf{Q}DS = (D_VQ' + E)S = ES$, because $Q'S = Q'G'RD^T = 0$.

(d) At $U = 1$, $\mathfrak{F} = 0$ and $E = 0$. □

15.2 The operator and its inverse

Let $W \in \mathcal{W}(\varepsilon, \gamma_W)$ and $\mathfrak{B} = D^TD + W$, so that $q_U(A, A') = \mathrm{b}(A, \mathfrak{B}A')$. Define

$$\mathfrak{A}_\pi(U) = \Pi^T\mathfrak{B}\,\Pi + DR\,D^T + a\,\Pi^T\mathbf{Q}(U)^T\mathbf{Q}(U)\,\Pi.$$

It is transpose-symmetric. By Lemma 15.1(d), at $U = 1$ and $W = 0$ it reduces to the operator $\mathbf{\Delta}_a$ of Section 9. The first term is Bałaban's gauge-invariant extension of the form q_U [7, (3.118)–(3.120), (3.128)]. Bałaban's averaging term is $a\,\mathbf{Q}^T\mathbf{Q}$ rather than $a\,\Pi^T\mathbf{Q}^T\mathbf{Q}\Pi$. The two coincide when $E = 0$, which holds for his averaging. With $E \neq 0$ the version with Π is needed, since it is what makes $\mathbf{Q}_\pi\mathbf{G}DR = 0$ in Theorem 15.4(d). The difference is the term $\mathcal{K}_2$ below, which is the only genuinely new ingredient.

Remark 15.2 (The threshold depends on γ_W). The threshold ε_0 below cannot be chosen uniformly as $\gamma_W \to 0$. Let $U = 1$, fix μ and $t \in \mathfrak{g}$ with $|t| = 1$, let e be the constant gauge field with $e_\mu = t$ and $e_\nu = 0$ for $\nu \neq \mu$, and let $W = -aP$, where P is the orthogonal projection onto the multiples of e. Since $\|e\|^2 = n^d$ and every block has volume 1, $\|\mathbf{1}_yW\mathbf{1}_{y'}\| = an^{-d}$. Moreover Wu is constant, with $|Wu| \le an^{-d}\sum_{y'}|u|_{B(y')}$. Torus distances are at most $dn/2$, so $W \in \mathcal{W}(\varepsilon_n, 1/n)$ with $\varepsilon_n = ae^{d/2}n^{-d}$. On the other hand $de = 0$ and $d^*e = 0$, so $\Pi(1)e = e$ by Lemma 15.1(d). Also $\mathbf{Q}^*\mathbf{Q}e = e$, because $Q_\mu 1 = 1$ and $Q_\mu^* 1 = 1$ (Lemma 2.6(b)). Hence $\mathfrak{A}_\pi(1)e = \Pi(1)^T(-ae + ae) = 0$, although $\varepsilon_n \to 0$ as $n \to \infty$.

Lemma 15.3 (Identities)**.** *(a) $\mathfrak{A}_\pi D\nu = D(D^TD\nu)$ for $\nu \in \mathcal{N}(Q'(U))$. Consequently $\mathfrak{A}_\pi DG'Rh = DRh$ for every scalar field h.*

(b) For $A, A' \in \mathcal{S}(U)$: $\mathrm{b}(A, \mathfrak{A}_\pi A') = q_U(A, A') + a\,\mathrm{b}(\mathbf{Q}A, \mathbf{Q}A')$.

Proof. (a) $\Pi D\nu = 0$ kills the first and last terms, and $RD^TD\nu = D^TD\nu$. Apply this with $\nu = G'Rh$, for which $D^TD\nu = Rh$. (b) For $A \in \mathcal{S}$, $\Pi A = A$ and $RD^TA = 0$. □

Theorem 15.4 (Gauge-field Green function with a background)**.** *For $\varepsilon \le \varepsilon_0$:*

(a) $\mathfrak{A}_\pi(U) = \mathbf{\Delta}_a + \delta_\pi$, where δ_π is admissible of size $C\varepsilon$ and pointwise admissible of type 1 with constants $(C\varepsilon, C, \gamma)$.

(b) $\mathbf{G}(U) = \mathfrak{A}_\pi(U)^{-1}$ exists and is analytic in A and W. The operators $\mathbf{G}$, $D\mathbf{G}$, $\mathbf{G}D^T$, $D\mathbf{G}D^T$ have decay (C, γ), and $\mathbf{G}(U) - \mathbf{G}$ (with derivatives) has decay $(C\varepsilon, \gamma)$.

(c) For all y, y' and J supported in $B(y')$,

$$|\mathbf{G}J|_{B(y)} + |\partial\mathbf{G}J|_{B(y)} + [\partial\mathbf{G}J]_{\alpha,B(y)} \le Ce^{-\gamma|y-y'|}|J|_{B(y')}.$$

(d) $\mathbf{G}DR = DG'R$, $RD^T\mathbf{G} = RG'D^T$, *and* $\mathbf{Q}_\pi\mathbf{G}DR = 0$, *where* $\mathbf{Q}_\pi = \mathbf{Q}(U)\Pi$.

Proof. (a) Using $DRD^T = DD^T - DP_{\mathcal{V}}D^T$ and $\mathbf{\Delta}_a = -\Delta - dP_{\mathcal{V}}(1)d^* + a\mathbf{Q}^*\mathbf{Q}$, we get

$$\delta_\pi = (D^TD + DD^T + \Delta) + W - (DP_{\mathcal{V}}(U)D^T - dP_{\mathcal{V}}(1)d^*) + a(\mathbf{Q}^T\mathbf{Q} - \mathbf{Q}^*\mathbf{Q}) + \mathcal{K}_1 + \mathcal{K}_2,$$

with $\mathcal{K}_1 = \Pi^T\mathfrak{B}\Pi - \mathfrak{B}$ and $\mathcal{K}_2 = a(\Pi^T\mathbf{Q}^T\mathbf{Q}\Pi - \mathbf{Q}^T\mathbf{Q})$. By Lemma 15.1(c), $\mathfrak{B}D = D^T\mathfrak{F} + WD$ and $D^TD^T = \mathfrak{F}^T$. Hence

$$\mathcal{K}_1 = -\mathfrak{B}DS - (\mathfrak{B}DS)^T + S^T(\mathfrak{F}^T\mathfrak{F} + D^TWD)S, \qquad \mathfrak{B}DS = D^T\mathfrak{F}S + WDS,$$
$$\mathcal{K}_2 = -a(\mathbf{Q}^TES + (\mathbf{Q}^TES)^T) + aS^TE^TES.$$

Admissibility. Every term is of the form XMY with $X, Y \in \mathfrak{D}$ and M a composition of decaying operators containing at least one factor of size $C\varepsilon$: $\mathfrak{F}$ or W or E, the differences in Lemma 12.5(b), $T = D - d$, or $P_{\mathcal{V}}(U) - P_{\mathcal{V}}(1)$. For example $D^T\mathfrak{F}S = D^T(\mathfrak{F}G'R)D^T$, and

$$DP_{\mathcal{V}}(U)D^T - dP_{\mathcal{V}}(1)d^* = D\big(P_{\mathcal{V}}(U) - P_{\mathcal{V}}(1)\big)D^T + TP_{\mathcal{V}}(1)D^T + dP_{\mathcal{V}}(1)T^T.$$

The decay of the factors comes from Theorem 14.1 and Proposition 14.2.

Pointwise admissibility. The first, second and fourth terms are covered by Lemma 13.6(a)–(c). By Proposition 14.2(e) and Lemma 13.6(d), $DP_{\mathcal{V}}(U)D^Tu$ and $dP_{\mathcal{V}}(1)d^*u$ contribute to F_u with a bound $C\sum e^{-\gamma|y-y'|}\|u\|_{B(y')}$. For $\mathcal{K}_1, \mathcal{K}_2$ we use the following facts. Write $Su = G'D^Tu - G'P_{\mathcal{V}}D^Tu$. By Theorem 14.1(b) and Proposition 14.2(e),

$$|Su| + [Su]_\alpha \le C\sum e^{-\gamma|\cdot|}(|u| + \|u\|), \qquad |DSu| + [DSu]_\alpha \le C\sum e^{-\gamma|\cdot|}(|u| + [u]_\alpha + \|u\|).$$

Moreover $S^T = DRG'$, and $DRG'f = DG'f - DP_{\mathcal{V}}G'f$ is bounded and Hölder whenever f is bounded, again by Theorem 14.1(b). With these:

- $D^T\mathfrak{F}Su = \sum_\mu \partial_\mu^* g_\mu$ + (bounded) with $g = \mathfrak{F}Su$. By Lemma 12.3, $|g| + [g]_\alpha \le C\varepsilon(|Su| + [Su]_\alpha)$.
- $WDSu$ and its Hölder quotient are bounded by $\varepsilon(|DSu| + [DSu]_\alpha)$, by the definition of $\mathcal{W}$.
- $(\mathfrak{B}DS)^Tu = DRG'(\mathfrak{F}^TDu + D^TWu)$. Here $\mathfrak{F}^TDu$ is bounded by $C\varepsilon(|u| + |\partial u|)$. Moreover $DG'D^T(Wu)$ is bounded by $C(|Wu| + [Wu]_\alpha)$ by the Hölder part of Theorem 14.1(b), and $DP_{\mathcal{V}}G'D^T(Wu)$ by $C\|Wu\|$ on nearby blocks, by Proposition 14.2(e).
- $S^T\mathfrak{F}^T\mathfrak{F}Su = DRG'(\mathfrak{F}^T\mathfrak{F}Su)$, and the terms of $\mathcal{K}_2$ are $\mathbf{Q}^TESu$, $DRG'(E^T\mathbf{Q}u)$ and $DRG'(E^TESu)$. They are of the form DRG'(bounded) or $\mathbf{Q}^T$(bounded), with bounds containing a factor ε.
- $S^TD^TWDSu = DRG'D^Tf$ with $f = WDSu$. Here a derivative of f appears, so boundedness of f is not enough. By the second item, f is bounded and Hölder, with a factor ε. Write $DRG'D^Tf = DG'D^Tf - DP_{\mathcal{V}}G'D^Tf$. The first term is bounded by $C(|f| + [f]_\alpha)$ by the Hölder part of Theorem 14.1(b), and the second by $C\|f\|$ on nearby blocks, by Proposition 14.2(e).

All bounds are summed with exponential weights, local in the sense of Definition 13.4. Using $[u]_\alpha \le C(|u| + |\partial u|)$ on cubes, this proves pointwise admissibility of type 1.

(b) The operator $\mathbf{\Delta}_a$ satisfies the hypotheses of Lemma 5.7 (proof of Theorem 10.1). Apply Lemmas 13.2 and 13.3. Analyticity follows as in Theorem 14.1.

(c) Let $u = \mathbf{G}J$. Then $-\Delta u = J + dP_{\mathcal{V}}(1)d^*u - a\mathbf{Q}^*\mathbf{Q}u - \delta_\pi u$, where the middle terms are pointwise admissible with constants $(0, C, \gamma)$, with $\gamma > 0$ chosen no larger than 1, the rate in part (a), and a rate of Proposition 14.2(e) at $U = 1$. We fix this rate before reducing ε_0 to

satisfy the threshold of Lemma 13.5. The L^2 decay of u is given by (b). Apply Lemma 13.5 of type 1.

(d) The first identity follows from Lemma 15.3(a), and the second by transposition, since $\mathbf{G}^T = \mathbf{G}$, $G'^T = G'$ and $R^T = R$. The third follows from $\mathbf{Q}_\pi \mathbf{G} D R = \mathbf{Q}\Pi D G' R = 0$, by Lemma 15.1(b), because $G'R$ maps into $\mathcal{N}(Q')$. □

16 Minimizer and fluctuation operators with a background

We keep the assumptions of Section 15 and write $\mathbf{Q}_\pi = \mathbf{Q}(U)\Pi$. Let

$$\mathcal{X}(U) = \{A : \ \mathbf{Q}(U)A = 0, \ R(U)D^T A = 0\} = \mathcal{S}(U) \cap \ker \mathbf{Q}_\pi,$$

the tangent space of the constraint set in Bałaban's gauge [8, (100)].

Theorem 16.1 (Minimizer and constrained Green function)**.** *For $\varepsilon \le \varepsilon_0$:*

(a) $\mathbf{Q}_\pi \mathbf{G} \mathbf{Q}_\pi^T$ *differs from* $\mathbf{Q}\mathbf{G}\mathbf{Q}^*$ *by an operator with decay* $(C\varepsilon, \gamma)$*. Its inverse* $\mathbf{\Gamma}_\pi$ *exists and has decay* (C, γ)*.*

(b) The minimizer map $\mathbf{H}(U) = \mathbf{G}\mathbf{Q}_\pi^T \mathbf{\Gamma}_\pi$ *satisfies* $\mathbf{Q}(U)\mathbf{H} = I$*,* $RD^T\mathbf{H} = 0$ *and* $\Pi\mathbf{H} = \mathbf{H}$*. For real* U*, real symmetric* W *and real data* B*,* $\mathbf{H}B$ *is real, and it is the unique minimizer of* $q_U(A, A)$ *among real* A *with* $\mathbf{Q}(U)A = B$ *and* $R(U)D^T A = 0$*.* $\mathbf{H}$ *and* $\partial\mathbf{H}$ *have decay* (C, γ)*, and*

$$|\mathbf{H}B|_{B(y)} + |\partial\mathbf{H}B|_{B(y)} + [\partial\mathbf{H}B]_{\alpha,B(y)} \le C \sum_{y'} e^{-\gamma|y-y'|} |B(y')|.$$

(c) The constrained Green function

$$\mathbf{C}(U) = \mathbf{G} - DG'RG'D^T - \mathbf{H}\mathbf{Q}_\pi\mathbf{G}$$

is transpose-symmetric and maps into $\mathcal{X}(U)$*. For every* J*,* $\mathbf{C}J$ *is the unique* $A \in \mathcal{X}(U)$ *with*

$$q_U(A', A) = \mathrm{b}(A', J) \qquad \text{for all } A' \in \mathcal{X}(U).$$

(d) $\mathbf{C}$ *and* $\partial\mathbf{C}$ *have decay* (C, γ)*, and for* J *supported in* $B(y')$*,*

$$|\mathbf{C}J|_{B(y)} + |\partial\mathbf{C}J|_{B(y)} + [\mathbf{C}J]_{\alpha,B(y)} \le Ce^{-\gamma|y-y'|} |J|_{B(y')}.$$

Proof. (a) $\mathbf{Q}_\pi - \mathbf{Q} = (\mathbf{Q}(U) - \mathbf{Q}) - E(U)S$, and $ES = EG'RD^T$ has decay $(C\varepsilon, \gamma)$. Together with Theorem 15.4(b), every term of the difference has decay $(C\varepsilon, \gamma)$. Since $\mathbf{Q}\mathbf{G}\mathbf{Q}^* \ge c_F$ (Proposition 11.2), Lemma 13.3 applies.

(b) $\mathbf{Q}_\pi\mathbf{H} = I$. By Theorem 15.4(d), $RD^T\mathbf{H} = (\mathbf{Q}_\pi\mathbf{G}DR)^T\mathbf{\Gamma}_\pi = 0$. Hence $\mathbf{H}B \in \mathcal{S}$, so $\Pi\mathbf{H} = \mathbf{H}$ and $\mathbf{Q}\mathbf{H} = \mathbf{Q}_\pi\mathbf{H} = I$.

For the minimization, let U, W and B be real, let $A \in \mathcal{S}$ be real with $\mathbf{Q}A = B$, and put $h = A - \mathbf{H}B \in \mathcal{X}(U)$. All operators are real, so $\mathbf{H}B$ and h are real. Then $\mathrm{b}(h, \mathfrak{A}_\pi \mathbf{H}B) = \mathrm{b}(\mathbf{Q}\Pi h, \mathbf{\Gamma}_\pi B) = 0$. Lemma 15.3(b) gives

$$q_U(A, A) = q_U(\mathbf{H}B, \mathbf{H}B) + q_U(h, h), \qquad q_U(h, h) = \langle h, \mathfrak{A}_\pi h\rangle \ge \tfrac{c_g}{2}\|h\|^2,$$

for real fields. The bound on $q_U(h, h)$ is the coercivity of $\mathfrak{A}_\pi$ from Lemma 13.2.

Decay is a composition. For the pointwise bound, $\mathbf{H}B = \mathbf{G}J$ with $J = \mathbf{Q}_\pi^T\mathbf{\Gamma}_\pi B$. Now

$$\mathbf{Q}_\pi^T = \Pi^T\mathbf{Q}^T = \mathbf{Q}^T - DRG'(D^T\mathbf{Q}^T), \qquad D^T\mathbf{Q}(U)^T = (\mathbf{Q}D)^T = Q'^T D_V^T + E^T$$

by Lemma 12.5(c). So $D^T\mathbf{Q}^T\omega$ is bounded on each block by $C\max|\omega|$ over neighbouring points. By Theorem 14.1(b), DRG' maps bounded fields to bounded fields. Hence $|J|_{B(y'')} \le C\sum e^{-\gamma|y''-y'|}|B(y')|$, and the sum form of Theorem 15.4(c) concludes.

(c) Symmetry is clear. Since $\mathbf{Q}_\pi DG'R = \mathbf{Q}\Pi DG'R = 0$,

$$\mathbf{Q}_\pi\mathbf{C} = \mathbf{Q}_\pi\mathbf{G} - 0 - \mathbf{Q}_\pi\mathbf{G} = 0.$$

By Theorem 15.4(d) and $D^TDG'R = R$,

$$RD^T\mathbf{C} = RG'D^T - R\,RG'D^T - 0 = 0.$$

So $\mathbf{C}J \in \mathcal{S}\cap\ker\mathbf{Q}_\pi = \mathcal{X}$. For $A'\in\mathcal{X}$, Lemma 15.3(b) gives $q_U(A',\mathbf{C}J) = \mathrm{b}(A',\mathfrak{A}_\pi\mathbf{C}J)$. By Lemma 15.3(a) with $h = G'D^TJ$,

$$\mathfrak{A}_\pi\mathbf{C}J = J - DR\,G'D^TJ - \mathbf{Q}_\pi^T\mathbf{\Gamma}_\pi\mathbf{Q}_\pi\mathbf{G}J.$$

The last two terms are b-orthogonal to $\mathcal{X}$: $\mathrm{b}(A',DRh) = \mathrm{b}(RD^TA',h) = 0$ and $\mathrm{b}(A',\mathbf{Q}_\pi^T\omega) = \mathrm{b}(\mathbf{Q}A',\omega) = 0$.

For uniqueness, suppose $A\in\mathcal{X}$ and $q_U(A',A) = 0$ for all $A'\in\mathcal{X}$. Then $\mathfrak{A}_\pi A$ annihilates $\mathcal{X} = \ker(RD^T)\cap\ker\mathbf{Q}_\pi$, so $\mathfrak{A}_\pi A = DR\theta + \mathbf{Q}_\pi^T\omega$ for some θ,ω. Hence $A = DG'R\theta + \mathbf{G}\mathbf{Q}_\pi^T\omega$. Applying RD^T gives $0 = R\theta$, so $A = \mathbf{G}\mathbf{Q}_\pi^T\omega$. Applying $\mathbf{Q}_\pi$ gives $0 = \mathbf{\Gamma}_\pi^{-1}\omega$, so $\omega = 0$.

(d) For the decay, the first and third terms of $\mathbf{C}$ are compositions of operators with decay. For the middle term, Theorem 15.4(d) and $R^2 = R$ give $DG'RG'D^T = \mathbf{G}\,DR\,G'D^T$. Hence $\partial DG'RG'D^T = (\partial\mathbf{G}D)\,R\,(G'D^T)$, and $\partial\mathbf{G}D$ has decay: on scalar fields $D_\mu = -S_\mu\partial_\mu^* + T_\mu$, with S_μ the shift by η, so $\partial\mathbf{G}D$ is $\partial\mathbf{G}\partial^*$ up to local bounded factors. No second derivatives of G' are needed. For the pointwise bounds write $\mathbf{C}J = \mathbf{G}J - D\sigma - \mathbf{H}(\mathbf{Q}_\pi\mathbf{G}J)$ with $\sigma = G'RG'D^TJ$. The first and third terms are controlled by Theorem 15.4(c) and part (b), since $\mathbf{Q}_\pi\mathbf{G}J$ is a bounded unit-lattice field decaying away from y'.

By Proposition 14.2(c), $\sigma\in\mathcal{N}(Q'(U))$ and $D^TD\sigma = h := RG'D^TJ$. By Theorem 14.1(b) and Proposition 14.2(e), h is bounded and Hölder on cubes, with bounds $Ce^{-\gamma\,\mathrm{dist}}|J|_{B(y')}$. The local L^2 norms of σ decay by composition. First, $-\Delta\sigma = h - (D^TD+\Delta)\sigma$, and Lemma 13.5 of type 1 bounds $|\sigma|$, $|\partial\sigma|$ and $[\partial\sigma]_\alpha$ on blocks. Next, by (12.2) and the Leibniz rule,

$$(D^TD+\Delta)\sigma = \sum_\mu (a_\mu\partial_\mu^*\sigma + (\partial_\mu^*a_\mu)S_\mu^{-1}\sigma) + b\sigma.$$

The right-hand side is bounded and Hölder on $\Lambda_2(z)$, with constants $C\varepsilon$ times the bounds just obtained. This uses the Hölder bounds on cubes (Remark 7.2) and $[\partial a_\mu]_\alpha + [b]_\alpha \le C\varepsilon$ from Lemma 12.2; this uses the Hölder condition on ∂A in (B), as does the curvature bound used in the pointwise admissibility argument of Theorem 15.4. Theorem 6.3(c) now bounds $|\partial\partial\sigma|$, and hence $|\partial D\sigma| \le |\partial\partial\sigma| + |\partial(T\sigma)|$. The Hölder bound on $\mathbf{C}J$ follows from the bound on $|\partial\mathbf{C}J|$. □

Remark 16.2 (Comparison with [7]). Bałaban's operators G_1, H_1 and $\mathfrak{G}$ [7, (3.128), (3.129), (3.153)] correspond to $\mathbf{G}(U)$, $\mathbf{H}(U)$ and $\mathbf{C}(U)$. His gauge-invariant extension Δ_π, with correction Δ'_π, corresponds to $\Pi^T\mathfrak{B}\Pi$, i.e. to the term $\mathcal{K}_1$. Our W plays the role of his curvature term Δ' [7, (3.10)] and of the term $-2\langle HC^{(2)}(A),J\rangle$ of [7, (3.128)]; we do not use that these terms belong to $\mathcal{W}$ (Section 12). For his averaging $E = 0$, so there is no $\mathcal{K}_2$.

16.1 Analyticity and gauge covariance

Proposition 16.3. *(a) All operators of Sections 14–16 are analytic functions of A on the open set where (B) holds with $\varepsilon < \varepsilon_0$, and of W in the ball of $\mathcal{W}(\varepsilon_0,\gamma_W)$. All bounds*

are uniform there. By the Cauchy estimates, on the smaller set where (B) holds with $\varepsilon \le \varepsilon_0/2$, the derivatives of order m with respect to A, in directions δA obeying (B) with $\varepsilon = 1$, satisfy the decay and pointwise bounds of the operators themselves, with constants multiplied by $m!\,(C/\varepsilon_0)^m$. The factor ε in the comparisons with $U = 1$ is not preserved. If $W = W(A)$ is an analytic function of A with values in the ball of $\mathcal{W}(\varepsilon_0, \gamma_W)$, the operators are analytic functions of A; otherwise only the joint analyticity in (A, W) is claimed.

(b) *Let $u : \mathbb{T}_\eta \to G$ be a gauge transformation, and suppose $W(U^u) = R(u)W(U)R(u)^{-1}$. Then*

$$\begin{aligned} G'(U^u) &= R(u)G'(U)R(u)^{-1}, & R(U^u) &= R(u)R(U)R(u)^{-1}, \\ \Pi(U^u) &= R(u)\Pi(U)R(u)^{-1}, & \mathbf{G}(U^u) &= R(u)\mathbf{G}(U)R(u)^{-1}, \\ \mathbf{H}(U^u) &= R(u)\mathbf{H}(U)R(u|_{\mathbb{T}})^{-1}, & \mathbf{C}(U^u) &= R(u)\mathbf{C}(U)R(u)^{-1}. \end{aligned}$$

Consequently, let U satisfy (B), let $W \in \mathcal{W}(\varepsilon, \gamma_W)$ in this gauge, and let u be G-valued. At U^u, with the potential $R(u)WR(u)^{-1}$, the L^2 block bounds and the pointwise bounds of Sections 14–16 hold with ∂ replaced by D_{U^u}, and the Hölder bounds hold after transforming back by $R(u)^{-1}$, i.e. in the gauge in which (B) holds. A bound comparing an operator with its value at $U = 1$ transfers only as a comparison with the transformed reference: for example, $G'(U^u) - G'(1^u) = R(u)(G'(U) - G'(1))R(u)^{-1}$ has decay $(C\varepsilon, \gamma)$. It does not transfer as a comparison with $G'(1)$.

Proof. (a) Each operator is built from $R(e^{i\eta A})$ by sums, products and inverses. The inverses exist, with uniform bounds, on the indicated set, so the operators are analytic there. (b) follows from Lemma 12.5(d) and the definitions, which are built from D, Q', $\mathbf{Q}$, W by gauge-covariant operations. For G-valued u, $R(u)$ is an isometry, and it acts pointwise, so it commutes with the localization projections $\mathbf{1}_y$. The derivative bounds transfer because $D_{U^u}R(u) = R(u)D_U$ (Lemma 12.5(d)) and $D_U = \partial + T$ with T local of size $C\varepsilon$ in the gauge of (B) (Lemma 12.2). For the last claim, let G be non-abelian, and let x and $z = x + \eta e_\mu$ lie in different blocks with $R(u(x)u(z)^{-1}) \neq 1$. The coefficient of $\lambda(z)$ in $(\mathfrak{A}'(1^u)\lambda)(x)$ is $-\eta^{-2}R(u(x)u(z)^{-1})$, and in $(\mathfrak{A}'(1)\lambda)(x)$ it is $-\eta^{-2}$, because $Q'^T Q'$ has no matrix elements between different blocks (Lemma 12.5(a)). So $G'(1^u) \neq G'(1)$, although 1^u is gauge equivalent to 1, which satisfies (B) for every $\varepsilon > 0$. □

16.2 What the variational problem needs

We list the input of Bałaban's proof of the variational theorem [8], specialized to a single torus with the small-field condition everywhere, and indicate where it is provided in this paper.

Linear input (Parts II and III).

- The minimizer map $H(U_0)$ and its bounds [8, (45), (46)]: Theorem 16.1(b), and Theorem 19.9 at regular backgrounds.
- The fluctuation operator $\mathfrak{G} = G_1\mathfrak{P}^*$ and the bounds of [7, Theorem 3.13] used in [8, (110)–(120)]: Theorem 16.1(c),(d) and Theorem 19.9. In Part V the additional quadratic terms of [8, (79)] are not put into the operator; they are treated as part of the nonlinearity.
- The Neumann series for operators with exponential decay [8, (68)–(71)]: Lemma 5.6.
- The scalar operators $G'(U_0)$, $H'(U_0)$ and the projection $R(U_0)$ entering the Landau gauge condition [8, (21)] and the gauge-fixing theorem [6, (1.91)–(1.100)]: Theorem 14.1, Proposition 14.2, Theorem 18.2.

- The Feynman-gauge operator $G(U_0) = \mathfrak{A}_F^{-1} = (D^T D + DRD^T + a\mathbf{Q}^T\mathbf{Q})^{-1}$ of [6, (1.58)]: Lemma 24.1(c).

Nonlinear input (Part IV).

- The nonlinear averaging, the regularity of averaged configurations [5, Props. 1, 2]: Section 21.
- The axial gauge, the modified averaging and its composition [5, Sect. C], the expansion $\mathcal{Q}(e^{\mathrm{i}\eta A}) = e^{\mathrm{i}(\mathbf{Q}A + C(A))}$ with quadratic remainder [5, Props. 3–5]: Section 22. That the linearization of Bałaban's averaging satisfies $E = 0$ [7, (3.114)–(3.115)] is Proposition 22.7(b).
- The expansion of the Wilson action [8, (25)–(40)]: Proposition 23.3.
- The passage from the axial gauge to the Landau gauge [6, Theorem 2]: Theorem 24.2.

The variational theorem relative to a reference (Part V). The linearizing transformation, the contraction, uniqueness, minimality, the second-order bounds, analyticity and the expansion [8, Sects. C–E, G]: Theorem 25.1.

The induction (Part VI). The trivial extension and the induction over k [8, Sect. A, (11)–(14)], and the local improvement of the regularity constants [8, Sect. F]: Lemma 30.1, Lemma 29.2 and Theorem 30.3. The derivation of local regularity from gauge-invariant conditions on the curvature [6, Sect. F, Prop. 6]: Corollary 30.4. Analyticity around the background field [8, Sect. G]: Corollary 30.5.

Not provided. The nested domains Ω_j and large fields, and gauges on cubes of size of order ε_1^{-1} with linear bounds (Section 51).

Part III: Domains, local regularity and random walks

17 Domains, partitions of unity and two general lemmas

Part II needs the background to be close to 1 in a single global gauge. Bałaban's conditions are local: on every cube there is a gauge in which the background is close to 1. To work with local conditions we need operators that live on bounded regions and depend only on the background in a neighbourhood of width 1 of these regions, together with a mechanism that turns local information into global estimates. This part provides both.

- *Dirichlet operators* on unions of blocks (this section and Section 18). Their decay and regularity follow as in Part I, because the Poincaré inequality, Combes–Thomas and local regularity are local.
- *Coercivity from local information* (Section 19). The scalar operators are coercive block by block. For the gauge-field operator we use an IMS localization at a large scale M: on each cube of size M we work in the local gauge, perturb around $U = 1$, and use the coercivity of Part I. Near the boundary of a domain we add a mass term. This makes the local operators coercive without any further gauge-theoretic input. The gauge-field operators also get Dirichlet conditions for all components, through the divergence at the exterior points next to the domain.
- *Comparison of domains and locality in the background* (Sections 18, 19). If two backgrounds agree on a region, the corresponding operators agree deep inside it up to

exponentially small errors.

- *Bałaban's random-walk expansion* (Section 20), which writes the scalar Green function as a convergent sum of products of local propagators. For the gauge-field Green function the walk is completed in Section 41.

Unlike Part II, no Fourier analysis is used for backgrounds $U \neq 1$.

17.1 Domains and field spaces

Definition 17.1 (Domains)**.** A *domain* is a non-empty union of blocks $X = \bigcup_{y \in X_1} B(y)$, $X_1 \subset \mathbb{T}$. We write:

- $\ell^2_0(X)$ for the scalar fields vanishing outside X;
- $\ell^2_1(X)$ for the gauge fields vanishing on all bonds with no endpoint in X;
- $\ell^2(X_1)$ and $\ell^2(\mathcal{B}_1(X))$ for unit-lattice scalar fields supported in X_1 and unit-lattice gauge fields supported on unit bonds with an endpoint in X_1.

In all cases $\mathbf{1}_X$ denotes the orthogonal projection onto the corresponding space. For an operator $\mathfrak{A}$ on torus fields, $\mathfrak{A}_X = \mathbf{1}_X \mathfrak{A} \mathbf{1}_X$, regarded as an operator on $\ell^2(X)$, is its *Dirichlet restriction* to X. We write $\hat{X}$ for the set of points at ∞-distance at most 1 from X. The torus $\mathbb{T}_\eta$ is itself a domain, with $\hat{\mathbb{T}}_\eta = \mathbb{T}_\eta$.

The exterior derivative maps $\ell^2_0(X)$ into $\ell^2_1(X)$: if λ vanishes outside X, then $d\lambda$ vanishes on bonds without an endpoint in X. The adjoint of $d : \ell^2_0(X) \to \ell^2_1(X)$ is $d^*_X = \mathbf{1}_X d^*$. The same holds for the covariant derivative D and $D^T_X = \mathbf{1}_X D^T$. On $\ell^2_1(X)$ we use the norm

$$\|\!|A|\!\|^2_X = \|A\|^2 + \|DA\|^2 + \|D^T_X A\|^2,$$

and on $\ell^2_0(X)$ the norm $\|\!|\lambda|\!\|^2_X = \|\lambda\|^2 + \|D\lambda\|^2$. For $X = \mathbb{T}_\eta$ and $U = 1$ these are equivalent to $\|A\|^2 + \|\partial A\|^2$ and $\|\lambda\|^2 + \|\partial\lambda\|^2$, by Lemma 2.2. Fields in $\ell^2(X)$ are extended by zero, so DA includes the values on bonds and plaquettes that cross the boundary of X.

Remark 17.2 (Why Dirichlet restrictions). For a local operator such as $-\Delta + aQ^*_\sigma Q_\sigma$, the quadratic form of $\mathfrak{A}_X$ is the restriction of the torus form to fields supported in X. Coercivity is therefore inherited. Moreover $\mathfrak{A}_X$ depends only on the coefficients of $\mathfrak{A}$ inside $\hat{X}$. For operators built from inverses, such as the projection R, the Dirichlet version is defined by applying the construction to Dirichlet restrictions of local operators (Sections 18, 19). It is not the Dirichlet restriction of the torus operator. This is what makes the local operators depend only on the background in $\hat{X}$. For the gauge-field operator, the local part is the Dirichlet restriction of the torus operator $D^T D + DD^T$. This needs the divergence at the exterior points next to X, which $D^T_X = \mathbf{1}_X D^T$ omits (Section 19).

17.2 Partitions of unity at scale M

Let $M \geq 1$ be an integer, and assume $n \geq 8M$. (If $n < 8M$, the torus is covered by a single cube of the size considered below. The local hypothesis of Section 18 then gives the global hypothesis (B) with parameter 2ε, and Part II applies. In this case we use the trivial partition: a single index $j = 0$, with any point z_0 of the torus as its centre, $\chi_0 = 1$ and $K_0 = \Lambda_0 = \mathbb{T}_\eta$ in place of the sets defined below. Lemmas 17.3–17.6 hold for it trivially, since every commutator with χ_0 vanishes.) Let $M' = n/\lceil n/M \rceil \in [M/2, M]$, and let z_j run over the points of the grid $M'\mathbb{Z}^d$ on the torus.

Lemma 17.3 (Partition of unity)**.** *There are smooth functions χ_j on the torus with the following properties:*

(a) $\sum_j \chi_j^2 = 1$;

(b) χ_j *vanishes outside* $K_j = \{x : |x - z_j|_\infty \le M\}$;

(c) $|\partial^\beta \chi_j| \le C_\beta M^{-|\beta|}$ *for all lattice derivatives;*

(d) every point lies in at most 3^d *of the sets* K_j.

The constants depend only on d.

Proof. Let ϕ be a smooth function on $\mathbb{R}$ with $\phi = 1$ on $[-\frac{1}{2}, \frac{1}{2}]$, $\phi = 0$ outside $(-1, 1)$ and $0 \le \phi \le 1$. Put $\phi_j(x) = \prod_\mu \phi((x_\mu - z_{j,\mu})/M)$, periodized. Then $\sum_k \phi_k^2 \ge 1$, since every point is within ∞-distance $M'/2 \le M/2$ of some z_k. Take $\chi_j = \phi_j / (\sum_k \phi_k^2)^{1/2}$. Property (c) follows from $M' \ge M/2$. For (d), $n \ge 8M$ gives $\lceil n/M \rceil \le n/M + 1 \le 9n/(8M)$, so $M' \ge 8M/9 > 2M/3$, and an interval of length $2M$ contains at most three points of $M'\mathbb{Z}$. □

For each j let Λ_j be the union of the blocks that meet $\{x : |x - z_j|_\infty \le 3M\}$. Thus K_j lies at ∞-distance at least $2M - 1$ from $\mathbb{T}_\eta \setminus \Lambda_j$.

Lemma 17.4 (IMS localization)**.** *Let* $\mathfrak{A}$ *be a real symmetric operator on* $\ell_1^2(X)$ *(or* $\ell_0^2(X)$*). Then*

$$\mathfrak{A} = \sum_j \chi_j \mathfrak{A} \chi_j + \frac{1}{2} \sum_j [\chi_j, [\chi_j, \mathfrak{A}]].$$

Proof. Expand $[\chi, [\chi, \mathfrak{A}]] = \chi^2 \mathfrak{A} - 2\chi \mathfrak{A} \chi + \mathfrak{A} \chi^2$, sum over j, and use $\sum_j \chi_j^2 = 1$. □

Here χ_j acts on scalar fields by multiplication, on gauge fields by $(\chi_j A)(b) = \chi_j(b_-) A(b)$, and on 2-forms by $(\chi_j F)(p) = \chi_j(p_-) F(p)$, with p_- the base point of the plaquette p. The double commutators are small whenever $\mathfrak{A}$ is built from derivatives and operators with exponential decay.

Lemma 17.5 (Double commutators)**.** *Let* $\mathfrak{A} = \sum_i X_i^T M_i Y_i$ *be a finite sum of operators on* $\ell_1^2(X)$*. Here each* X_i *and each* Y_i *is* 1*, the derivative* D *from gauge fields to* 2*-forms, or the divergence* D_X^T*, and each* M_i *has decay* (C_M, γ)*, as an operator between the corresponding field spaces. Then*

$$\Big| \Big\langle A, \sum_j [\chi_j, [\chi_j, \mathfrak{A}]] A \Big\rangle \Big| \le \frac{C\, C_M}{M^2} \, |\!|\!| A |\!|\!|_X^2,$$

with C *depending only on* d*,* γ *and the number of terms.*

Proof. Derivatives. Let $Z \in \{D, D_X^T\}$. The value of Zg at a point or a plaquette is a sum of transported values of g at distance at most 2η, divided by η. So $[\chi_j, Z]$ is a sum of multiplications by differences of χ_j over distances at most 2η, divided by η, composed with shifts and unitary transports. The coefficients of $[\chi_j, [\chi_j, Z]]$ are the squares of these differences, divided by η. By Lemma 17.3(c),(d),

$$\|[\chi_j, Z]\| \le C M^{-1}, \qquad \sum_j \|[\chi_j, Z] g\|^2 \le C M^{-2} \|g\|^2, \qquad \Big\| \sum_j [\chi_j, [\chi_j, Z]] \Big\| \le C \eta M^{-2}.$$

Decaying operators. Let K have decay (C_M, γ), fix blocks y, y', and write $\chi_j = \chi_j(c_y) + \psi_j$. Constants commute with K, so $\mathbf{1}_y [\chi_j, K] \mathbf{1}_{y'} = \mathbf{1}_y [\psi_j, K] \mathbf{1}_{y'}$ and $\mathbf{1}_y [\chi_j, [\chi_j, K]] \mathbf{1}_{y'} = \mathbf{1}_y [\psi_j, [\psi_j, K]] \mathbf{1}_{y'}$. Moreover $|\psi_j| \le C M^{-1}$ on $B(y)$ and $|\psi_j| \le C M^{-1}(1 + |y - y'|)$ on $B(y')$, and only boundedly many j have $\chi_j \ne 0$ on $B(y) \cup B(y')$. Hence $\sum_j [\chi_j, [\chi_j, K]]$ has decay $(C C_M M^{-2}, \gamma/2)$, and its norm is at most $C C_M M^{-2}$ by Lemma 5.2(a). In the same way $\sum_j \|[\chi_j, K] g\|^2 \le C C_M^2 M^{-2} \|g\|^2$. Decay is a bound on blocks, so no bound on the individual entries of K is used. For operators between unit-lattice spaces, where decay is a kernel bound,

the argument is the kernel bound $\sum_j(\chi_j(x)-\chi_j(x'))^2|K(x,x')| \le CM^{-2}|x-x'|^2|K(x,x')|$ together with the Schur test.

The form. Since $[\chi_j,\cdot\,]$ is a derivation,

$$[\chi_j,[\chi_j,X^TMY]] = [\chi_j,[\chi_j,X^T]]MY + X^T[\chi_j,[\chi_j,M]]Y + X^TM[\chi_j,[\chi_j,Y]] \\ + 2[\chi_j,X^T][\chi_j,M]Y + 2[\chi_j,X^T]M[\chi_j,Y] + 2X^T[\chi_j,M][\chi_j,Y].$$

Pair with A, sum over j, and use the Cauchy–Schwarz inequality in the sums over j. Every term is bounded by $CC_MM^{-2}(\|A\|+\|XA\|+\|YA\|)^2$. For example, the fourth term gives $2\sum_j|\langle[\chi_j,X]A,[\chi_j,M]YA\rangle| \le 2(\sum_j\|[\chi_j,X]A\|^2)^{1/2}(\sum_j\|[\chi_j,M]YA\|^2)^{1/2} \le CC_MM^{-2}\|A\|\,\|YA\|$. Since $\|XA\|,\|YA\| \le \|\!|A|\!\|_X$ for $X,Y\in\{1,D,D_X^T\}$, the claim follows. No derivative of M_i enters. □

Lemma 17.6 (Localized norms)**.** *For every $A\in\ell^2_1(X)$,*

$$\sum_j\|\chi_jA\|^2=\|A\|^2, \qquad \sum_j\|\!|\chi_jA|\!\|_X^2 \ge \tfrac12\|\!|A|\!\|_X^2 - CM^{-2}\|A\|^2.$$

Proof. The first identity is Lemma 17.3(a). For the second, the Leibniz rule gives $D(\chi_jA)=\chi_j^\sharp DA + (\text{differences of }\chi_j)\cdot A$, where $\chi_j^\sharp$ denotes χ_j evaluated at shifted points. Hence $|D(\chi_jA)|^2 \ge \frac12(\chi_j^\sharp)^2|DA|^2 - C|\partial\chi_j|^2|A|^2$. Summing over j and using $\sum_j(\chi_j^\sharp)^2=1$ (the partition identity at the shifted point) gives the bound for D. The term D_X^T is treated in the same way. □

17.3 An algebraic comparison formula

Lemma 17.7 (Comparison of inverses)**.** *Let $X'\subset X$ be domains, and let $\mathfrak{A}$ and $\mathfrak{A}'$ be invertible operators on $\ell^2(X)$ and $\ell^2(X')$. Define $\Delta=\mathfrak{A}\mathbf{1}_{X'}-\mathbf{1}_{X'}\mathfrak{A}'\mathbf{1}_{X'}$, as an operator from $\ell^2(X')$ to $\ell^2(X)$. Then*

$$\mathbf{1}_{X'}\mathfrak{A}^{-1}\mathbf{1}_{X'}-\mathfrak{A}'^{-1} = -\mathbf{1}_{X'}\mathfrak{A}^{-1}\Delta\,\mathfrak{A}'^{-1}.$$

Proof. For $f\in\ell^2(X')$ let $u=\mathfrak{A}'^{-1}f\in\ell^2(X')$. Then $\mathfrak{A}u=\mathbf{1}_{X'}\mathfrak{A}'u+\Delta u=f+\Delta u$, so $u=\mathfrak{A}^{-1}f+\mathfrak{A}^{-1}\Delta u$. Apply $\mathbf{1}_{X'}$. □

In applications, Δ is the sum of two kinds of terms. First, terms that couple X' across its boundary; for scalar second-order local operators these include "cut-bond" terms of the form $\mathbf{1}_{X\setminus X'}D^T\mathbf{1}_{\mathrm{cut}}D\mathbf{1}_{X'}$. On gauge-field spaces the complementary projection is $\mathbf{1}_X-\mathbf{1}_{X'}$, which need not equal $\mathbf{1}_{X\setminus X'}$. Second, differences of domain-dependent ingredients, including mass profiles and non-local terms, which are supported near the cut or exponentially small away from $X\setminus X'$. Derivative factors may occur at one or both ends of these terms. The cut-bond terms, however, contain one-sided derivatives at the cut, which are not bounded uniformly in η, so the sharp-cut form of the identity is not used for the bounds. The comparisons are proved with a smooth cutoff instead (Theorem 18.2(e)); this gives exponential smallness of the difference away from $X\setminus X'$, uniformly in η.

18 Regular backgrounds and scalar operators on domains

18.1 Local regularity

Fix $\alpha\in(0,1)$. Backgrounds in this part are real (G-valued); complex perturbations are treated at the end of Section 19.

Definition 18.1 (Regular backgrounds)**.** Let $M \geq 1$ be an integer and $\varepsilon \in (0,1]$. A real background U is (M,ε)*-regular* if the following holds for every $z \in \mathbb{T}_\eta$. On the cube $\hat{\Lambda} = \{x : |x - z|_\infty \leq 4M + 2\}$ there is a G-valued gauge transformation u such that $U^u = e^{i\eta A}$ on all bonds with both endpoints in $\hat{\Lambda}$, with

$$|A| \leq \varepsilon, \qquad |\partial A| \leq \varepsilon, \qquad [\partial A]_{\alpha,\Lambda_3(z')} \leq \varepsilon \quad \text{whenever } \Lambda_3(z') \subset \hat{\Lambda}.$$

Every background satisfying (B) with parameter ε is (M,ε)-regular for every M. Regularity is a local condition: the gauges on different cubes are unrelated. In Bałaban's framework it is a consequence of gauge-invariant local conditions on the curvature. On cubes of size M one obtains gauges with bounds proportional to M [7, (3.35)–(3.36)], via the axial gauge and a change to a Landau-type gauge [6, Theorem 2]. In Parts II–V we take regularity as a hypothesis. Part VI shows that it follows from Bałaban's gauge-invariant conditions: the background fields are regular (Theorem 30.3), and so is every real configuration with small curvature and current (Corollary 30.4).

Two consequences are used repeatedly. If U is $(1,\varepsilon)$-regular, then on every block and its neighbours the covariant derivative is, in a suitable gauge, $\partial + T$ with $\|T\| \leq C\varepsilon$ (Lemma 12.2). The curvature operator $\mathfrak{F} = DD$ satisfies $|(\mathfrak{F}\lambda)_{\mu\nu}(x)| \leq C\varepsilon|\lambda(x + \eta e_\mu + \eta e_\nu)|$; this bound is gauge covariant, and it holds by Lemma 12.3 in the local gauge. Hence $|\mathfrak{F}\lambda|_\infty \leq C\varepsilon|\lambda|_\infty$ and $\|\mathfrak{F}\lambda\| \leq C\varepsilon\|\lambda\|$.

Change of local gauge. Let u_1, u_2 be gauges in which (B) holds with parameter ε on a box Λ, with $U^{u_i} = e^{i\eta A_i}$ there; for instance, Λ is the intersection of two cubes of Definition 18.1. Put $L = R(u_2 u_1^{-1})$, so that $R(u_2)f = L\,R(u_1)f$ for every field f. For a bond $(x, x + \eta e_\mu)$ of Λ let $V_i(x) = R(e^{i\eta A_{i,\mu}(x)})$. Since $U^{u_2} = (U^{u_1})^{u_2 u_1^{-1}}$, we have $V_2(x)L(x + \eta e_\mu) = L(x)V_1(x)$, hence

$$\partial_\mu L(x) = V_2(x)^{-1}\Big(L(x)\,\frac{V_1(x) - 1}{\eta} - \frac{V_2(x) - 1}{\eta}\,L(x)\Big).$$

The matrices $L(x)$ and $V_i(x)$ are orthogonal. Since $|A_i|, |\partial A_i| \leq \varepsilon$, the factors $(V_i - 1)/\eta$ are bounded by $C\varepsilon$ and change by at most $C\eta\varepsilon$ between neighbouring points. So $|\partial L| \leq C\varepsilon$. The second difference of L is bounded by $C\varepsilon$ wherever its complete stencil lies in Λ. Hence $|\partial L(x) - \partial L(x')| \leq C\varepsilon|x - x'|$ on a common lifted box, provided the relevant coordinate paths and stencils lie in that lift, with distance measured there. For a field f put $\tilde{f}_i = R(u_i)f$. The product rule $\partial_\mu \tilde{f}_2(x) = \partial_\mu L(x)\,\tilde{f}_1(x + \eta e_\mu) + L(x)\,\partial_\mu \tilde{f}_1(x)$ gives, on every common lifted box $Q \subset \Lambda$ of side at most 8, with all relevant stencils and coordinate paths contained in that lift, sup norms over Q, and distances measured in the lift,

$$|\partial \tilde{f}_2| \leq |\partial \tilde{f}_1| + C\varepsilon|f|, \qquad [\tilde{f}_2]_{\alpha,Q} \leq [\tilde{f}_1]_{\alpha,Q} + C\varepsilon|f|,$$
$$[\partial \tilde{f}_2]_{\alpha,Q} \leq [\partial \tilde{f}_1]_{\alpha,Q} + C\varepsilon(|f| + |\partial \tilde{f}_1|).$$

For gauge fields the same holds, with $L(b_-)$ in place of $L(x)$. So Hölder quotients of a field and of its first differences, computed in two local gauges, differ by at most $C\varepsilon$ times the sup norms of the field and of its first differences. For the applications below, fix a gauge on $\Lambda_4(c_y)$ and consider data on $\Lambda_3(c_y)$ and source pieces supported strictly inside $\Lambda_1(c_{y'})$. The required Hölder bounds transfer between these local gauges with a fixed larger constant. To prove this, use radius-one cubes on a half-unit grid. A pair at torus ∞-distance less than $\frac{1}{2}$, together with its required stencils, fits in a common such cube; its coordinate paths stay inside the good-gauge cubes. If $n = 8$, $\Lambda_4(c_y)$ is the whole torus. If $n \geq 9$, every omitted coordinate interval has width at least one, so such a close pair cannot cross it. The other quotients are bounded by four times the supremum. For a source piece supported inside $\Lambda_1(c_{y'})$, pairs with a point outside $\Lambda_3(c_{y'})$ are controlled by this fixed support margin. No assertion on an arbitrary proper periodic box without such a collar is used.

18.2 Scalar operators on domains

For a domain X and a regular background U define, on $\ell^2_0(X)$ and on $\ell^2(X_1)$:

$$\begin{aligned} \mathfrak{A}'_X &= \mathbf{1}_X(D^TD + aQ'(U)^TQ'(U))\mathbf{1}_X, & G'_X &= (\mathfrak{A}'_X)^{-1}, & \Gamma'_X &= (Q'G'_XQ'^T)^{-1}, \\ H'_X &= G'_XQ'^T\Gamma'_X, & C'_X &= G'_X - H'_XQ'G'_X, & M'_X &= Q'(G'_X)^2Q'^T, \\ P_{\mathcal{V},X} &= G'_XQ'^T(M'_X)^{-1}Q'G'_X, & R_X &= \mathbf{1}_X - P_{\mathcal{V},X}, & S_X &= G'_XR_XD_X^T. \end{aligned}$$

Here $Q' = Q'(U)$ maps $\ell^2_0(X)$ to $\ell^2(X_1)$, because the block averages of a field supported in X vanish on blocks outside X. All these operators depend only on the restriction of U to bonds in $\hat{X}$. They are covariant under gauge transformations on $\hat{X}$, as in Proposition 16.3. For $X = \mathbb{T}_\eta$ they are the operators of Section 14. Let $\mathcal{N}_X = \{\lambda \in \ell^2_0(X) : Q'\lambda = 0\}$ and $\mathcal{W}_X = D_X^TD\mathcal{N}_X$.

Theorem 18.2 (Scalar operators at regular backgrounds)**.** *There are $\varepsilon_0, c, C, \gamma > 0$, depending only on d, a, α and $\dim\mathfrak{g}$, such that the following holds for every $(1,\varepsilon)$-regular background with $\varepsilon \le \varepsilon_0$ and every domain X.*

(a) Coercivity: $\langle\lambda, \mathfrak{A}'_X\lambda\rangle \ge c\|\lambda\|^2_X$ *for* $\lambda \in \ell^2_0(X)$.

(b) Decay: G'_X, DG'_X, $G'_XD_X^T$ *and* $DG'_XD_X^T$ *have decay* (C,γ).

(c) Lower bounds: $Q'G'_XQ'^T \ge c$ *and* $M'_X \ge c^2$. *Moreover* Γ'_X *and* $(M'_X)^{-1}$ *have decay* (C,γ).

(d) The projection: R_X *is the orthogonal projection onto* $\mathcal{W}_X$. *For* $h \in \mathcal{W}_X$ *one has* $G'_Xh \in \mathcal{N}_X$ *and* $D_X^TDG'_Xh = h$; *equivalently* $Q'G'_XR_X = 0$ *and* $D_X^TDG'_XR_X = R_X$. *The operators* $P_{\mathcal{V},X}$, $DP_{\mathcal{V},X}$, $P_{\mathcal{V},X}D_X^T$, $DP_{\mathcal{V},X}D_X^T$, S_X *and* DS_X *have decay* (C,γ).

(e) Comparison of domains: *let* $X' \subset X$, *let* $\Sigma = \Sigma(X,X')$ *be the set of unit points of* X_1 *at distance at most* 2 *from* $X \setminus X'$ *(the* seam*), and let* F_X *be any of the operators in (b)–(d), with* $F_{X'}$ *its analogue on* X', *both extended by zero. Then for all unit points* y, y',

$$\|\mathbf{1}_y(F_X - F_{X'})\mathbf{1}_{y'}\| \le C\sum_{w\in\Sigma} e^{-\gamma(|y-w|+|w-y'|)}.$$

The bound covers the blocks of the initial points of all bonds on which the gauge-field versions DG', $G'D^T$, $DG'D^T$, $DP_{\mathcal{V}}$, $P_{\mathcal{V}}D^T$, $DP_{\mathcal{V}}D^T$, S and DS act or take values, including the bonds that cross $\partial X'$, whose initial points may lie outside X'_1. In particular $\|\mathbf{1}_y(F_X - F_{X'})\mathbf{1}_{y'}\| \le C'e^{-\gamma(\operatorname{dist}(y,\Sigma)+\operatorname{dist}(y',\Sigma))/2}$, and $\|\mathbf{1}_K(F_X - F_{X'})\| \le C'e^{-\gamma\operatorname{dist}(K,\Sigma)/2}$ for every set K of blocks, by a Schur bound.

(f) Locality in the background: *if U and U' are $(1,\varepsilon)$-regular and coincide on the bonds of $\hat{X}'$, $X' \subset X$, then $F_X(U) - F_X(U')$ satisfies the same bound.*

(g) Pointwise bounds ($d \ge 3$, $X = \mathbb{T}_\eta$): *the bounds of Theorem 14.1(b), (d) and Proposition 14.2(e) hold. Here $|\partial\cdot|$ is replaced by $|D\cdot|$ in the sup norms, and the Hölder quotients of $\partial(\cdot)$ on $B(y)$ are computed in a gauge in which (B) holds near $B(y)$.*

Proof. (a) Fix a block $B = B(y) \subset X$ and a gauge u in which (B) holds on a neighbourhood of B. Put $\tilde{\lambda} = R(u)\lambda$, so $|\tilde{\lambda}| = |\lambda|$. On bonds with both endpoints in B, $|D_U\lambda| = |(\partial + T)\tilde{\lambda}|$ with $\|T\| \le C\varepsilon$, where T only involves values in B. By Lemma 3.1, $\|\tilde{\lambda} - (Q_0\tilde{\lambda})(y)\|^2_B \le \frac{d}{2}\|\partial\tilde{\lambda}\|^2_{\mathrm{in}(B)}$, where $\mathrm{in}(B)$ denotes the bonds inside B. Moreover $R(U(\Gamma_{y,x}))\lambda(x) = R(u(y))^{-1}R(U^u(\Gamma_{y,x}))\tilde{\lambda}(x)$, and $|U^u(\Gamma_{y,x}) - 1| \le C\varepsilon$. Hence $|(Q'\lambda)(y)| \ge |(Q_0\tilde{\lambda})(y)| - C\varepsilon\|\lambda\|_B$. Combining,

$$\|\lambda\|^2_B \le d\,\|\partial\tilde{\lambda}\|^2_{\mathrm{in}(B)} + 2|(Q_0\tilde{\lambda})(y)|^2 \le 2d\,\|D\lambda\|^2_{\mathrm{in}(B)} + 4|(Q'\lambda)(y)|^2 + C\varepsilon^2\|\lambda\|^2_B.$$

Summing over the blocks of X and absorbing the last term for $C\varepsilon^2 \le \frac{1}{2}$ gives $\|\lambda\|^2 \le C(\|D\lambda\|^2 + \|Q'\lambda\|^2)$. This implies (a), as in Corollary 3.4, since $\langle \lambda, \mathfrak{A}'_X \lambda\rangle = \|D\lambda\|^2 + a\|Q'\lambda\|^2$.

(b) Apply Lemma 5.7, with ∂ replaced by D. Its proof uses ∂ only through Lemma 5.4. For D the same computation gives $(D_\mu)_\alpha = D_\mu + Y$ with $\|Y\| \le e|\alpha|$, since $R(U)$ is an isometry. Conjugation preserves $\ell^2_0(X)$. Hypothesis (ii) follows as in Lemma 5.8, because $Q'^T Q'$ is block-diagonal and bounded.

(c) Let $g \in \ell^2(X_1)$ and $\psi(x) = \prod_\mu \sin^2(\pi x_\mu)$ for $x \in B(0)$. Then $m = \eta^d \sum_{x\in B(0)} \psi(x) = 2^{-d}$ for $N \ge 2$. Define $\varphi(x) = m^{-1}\psi(x-y)\, R(U(\Gamma_{y,x}))^{-1} g(y)$ for $x \in B(y)$, $y \in X_1$. Then $Q'\varphi = g$ and $\langle Q'^T g, \varphi\rangle = \|g\|^2$. In the local gauge, φ is $m^{-1}\psi$ times a transport of $g(y)$ along $\Gamma_{y,x}$. The contours $\Gamma_{y,x}$ and $\Gamma_{y,x+\eta e_\mu}$ consist of at most dN pairs of parallel bonds at distance η, plus at most one further bond. So, by (B), the transports at neighbouring points differ by at most $C\eta\varepsilon$. Across the faces of blocks, ψ is of order η^2. Hence $\|D\varphi\| \le C\|g\|$, and $\langle \varphi, \mathfrak{A}'_X \varphi\rangle \le C\|g\|^2$. Since $\mathfrak{A}'_X$ is positive, Cauchy–Schwarz in the inner product $\langle \cdot, \mathfrak{A}'_X \cdot\rangle$ gives

$$\langle Q'^T g, G'_X Q'^T g\rangle \ge \frac{\langle Q'^T g, \varphi\rangle^2}{\langle \varphi, \mathfrak{A}'_X \varphi\rangle} \ge \frac{\|g\|^2}{C}.$$

Then $\|G'_X Q'^T g\|\|g\| \ge \langle Q'^T g, G'_X Q'^T g\rangle$ and $\|Q'^T g\| = \|g\|$ give $M'_X \ge c^2$. Both operators have decay, by (b) and the locality of Q'. Lemma 5.5 gives the decay of the inverses.

(d) The proof of Proposition 14.2(b),(c) is purely algebraic. It uses $\mathfrak{A}'^T_X = \mathfrak{A}'_X$, the invertibility of $\mathfrak{A}'_X$ (for the injectivity of $D^T_X D$ on $\mathcal{N}_X$), and the surjectivity of Q' and $Q'G'_X$. All of these hold on X. The decay statements are compositions of (b) and (c).

(e) We use a smooth cutoff instead of the sharp cut, which would produce boundary terms. Let χ be a smooth function, at unit scale, of a mollified torus distance to $X \setminus X'$, with $|\partial^k \chi| \le C$ for $k \le 2$. It vanishes at distance at most 1 and equals 1 at distance at least 3. In particular it vanishes on $X \setminus X'$ and on the blocks of X' adjacent to $X \setminus X'$, and it is smooth also across ∂X. Thus $1-\chi$ and $\partial\chi$, restricted to X', and the one-mesh enlargements needed for their commutator stencils, are supported in blocks $B(y)$ whose unit labels satisfy $\mathrm{dist}(y, \Sigma) \le R_d$, where one may take $R_d = 6 + 3\sqrt{d}$. This is a distance between unit labels, and R_d depends only on the fixed dimension. This changes only the constant in the seam bound.

Let $f \in \ell^2_0(X')$ and $u = G'_X f$. The field χu lies in $\ell^2_0(X')$ and vanishes near the cut. On such fields $\mathfrak{A}'_X$ and $\mathfrak{A}'_{X'}$ agree: $D^T D(\chi u)$ vanishes at the points of $X \setminus X'$, the values outside X are projected out, and $Q'^T Q'$ is block-local. Hence $\mathfrak{A}'_{X'}(\chi u) = \chi f + [\mathfrak{A}'_X, \chi]u$, and

$$G'_X f - G'_{X'} f = (1-\chi) G'_X f - G'_{X'}(1-\chi) f + G'_{X'}[\mathfrak{A}'_X, \chi] G'_X f$$

as fields on $\mathbb{T}_\eta$. The commutator is $[D^T D, \chi] = D^T[D, \chi] + [D^T, \chi]D$, a sum of first-order operators with bounded coefficients supported where $\partial\chi \ne 0$, plus the block-local $a[Q'^T Q', \chi]$. Since $G'_{X'}$ contains the projection $\mathbf{1}_{X'}$, $G'_{X'} D^T$ acts as $G'_{X'} D^T_{X'}$. By the decay of G'_X, DG'_X and $G'_{X'} D^T_{X'}$ from (b), each term has a kernel bounded by $C\sum_{w\in\Sigma} e^{-\gamma(|y-w|+|w-y'|)}$, for all unit points y, y': the identity holds as an identity of torus fields, and the decay bounds of (b) hold at all blocks, also at those of the initial points of the bonds crossing $\partial X'$. If an input or output block has its label in $(X \setminus X')_1$, that label belongs to Σ. The individual decay bounds of both F_X and $F_{X'}$ then bound their difference by the corresponding seam summand. For gauge-field input or output, $F_{X'}$ need not vanish there: a bond crossing $\partial X'$ can have its initial point in such a block. For DG' apply D to the identity; $G'D^T$ follows by transposition. For $DG'D^T$ use the analogous identity with the cutoff on the input,

$$G'_X D^T_X - G'_{X'} D^T_{X'} = (1-\chi) G'_X D^T_X - G'_{X'} D^T_{X'}(1-\chi) - G'_{X'}[D^T, \chi] + G'_{X'}[\mathfrak{A}'_X, \chi] G'_X D^T_X,$$

in which every factor is bounded, and apply D to it. An earlier version used a sharp-cut identity with $G'_X D^T_X$ in place of $G'_X \mathbf{1}_{X\setminus X'} D^T \mathbf{1}_{\mathrm{cut}}$. That identity is not correct: it adds the

term $G'_X \mathbf{1}_{X'} D^T \mathbf{1}_{\mathrm{cut}} D G'_{X'}$. The correct sharp-cut identity contains the factor $G'_X \mathbf{1}_{X \setminus X'} D^T \mathbf{1}_{\mathrm{cut}}$, which is of order $\eta^{-1/2}$. For the composite operators in (c) and (d), write the difference of two products as a telescoping sum of products in which exactly one factor is replaced by its difference. Composing a kernel of the form $\sum_{w \in \Sigma} e^{-\gamma(|y-w|+|w-y'|)}$ with kernels that decay exponentially gives a kernel of the same form, with a smaller γ. For the unit-lattice inverses use Lemma 17.7 on $\ell^2(X_1)$.

(f) $F_{X'}(U) = F_{X'}(U')$, because these operators depend only on U on $\hat{X}'$. Write $F_X(U) - F_X(U') = [F_X(U) - F_{X'}(U)] - [F_X(U') - F_{X'}(U')]$ and apply (e) twice.

(g) In a gauge in which (B) holds on $\Lambda_4(c_y)$, the equation $\mathfrak{A}' u = f$ takes on $\Lambda_3(c_y)$ the form treated in Theorem 14.1(b). Lemma 18.3 below therefore applies, of type 1 for $G'f$ and of type 0 for $G' D_\mu^T f$ with bounded f, with the L^2 decay from (b). The perturbation is local and block-local, so its small inputs are bounded by the invariant norms $\bar{N}_z^i$ of that lemma. For bounded Hölder f split f with a partition of unity at unit scale, as in the proof of Theorem 14.1(b). Here the Hölder quotient of f on $\Lambda_3(c_{y'})$ is computed in a gauge in which (B) holds on $\Lambda_4(c_{y'})$. The piece $\chi_{y'} f$ is supported in $\Lambda_1(c_{y'})$. On $\Lambda_3(c_y)$ it is moved to the gauge of $\Lambda_4(c_y)$ by the change of local gauge after Definition 18.1, in the collared form stated at the end of that paragraph. This costs only a larger constant, so Lemma 18.3 applies to each piece. The statements for H', C', $P_{\mathcal{V}}$ and S follow from those for G' exactly as in Section 14. □

Lemma 18.3 (Bootstrap in local gauges)**.** *Let $d \geq 3$ and $i \in \{0, 1\}$, and let U be $(1, \varepsilon)$-regular. For each $y \in \mathbb{T}$ let u_y be a real gauge in which (B) holds on $\Lambda_4(c_y)$, and $\tilde{u}_y = R(u_y)u$. Let ∇ be the covariant gradient: $\nabla v = Dv$ for scalar fields, and ∇v has the components $D_\nu v_\mu$ for gauge fields. Put $\bar{N}_z^0(v) = |v|_{\Lambda_1(c_z)}$ and $\bar{N}_z^1(v) = |v|_{\Lambda_1(c_z)} + |\nabla v|_{\Lambda_1(c_z)}$; these norms are invariant under real gauge transformations. Suppose that for each y the field $\tilde{u}_y$ satisfies on $\Lambda_2(c_y)$ an equation $-\Delta \tilde{u}_y = J_y + \delta_y \tilde{u}_y$, with $J_y = J_{F,y} + \sum_\mu \partial_\mu^* J_{g,y,\mu}$ and, for some y_0, $M > 0$, $\gamma_2 > 0$ and all y,*

$$\|u\|_{B(y)} + |J_{F,y}|_{\Lambda_2(c_y)} + |J_{g,y}|_{\Lambda_3(c_y)} + i\,[J_{g,y}]_{\alpha,\Lambda_3(c_y)} \leq M e^{-\gamma_2 |y-y_0|}.$$

Suppose moreover that for every field v there are fields $F_y[v]$ and $g_y[v]$ with $\delta_y R(u_y) v = F_y[v] + \sum_\mu \partial_\mu^* g_y[v]_\mu$ on $\Lambda_2(c_y)$ and

$$|F_y[v]|_{\Lambda_2(c_y)} + |g_y[v]|_{\Lambda_3(c_y)} + i\,[g_y[v]]_{\alpha,\Lambda_3(c_y)} \leq \sum_z e^{-\gamma|y-z|} \big(\varepsilon\, \bar{N}_z^i(v) + C_\delta \|v\|_{B(z)}\big).$$

Let $\gamma' = \frac{1}{2}\min(\gamma, \gamma_2)$. If $\varepsilon \leq \varepsilon_*$, then for $i = 1$

$$|u|_{B(y)} + |\nabla u|_{B(y)} + [\partial \tilde{u}_y]_{\alpha,B(y)} \leq C M e^{-\gamma'|y-y_0|},$$

and for $i = 0$, $|u|_{B(y)} + [\tilde{u}_y]_{\alpha,B(y)} \leq C M e^{-\gamma'|y-y_0|}$. For gauge fields the bound on $|\nabla u|$ contains that on $|Du|$.

Proof. Put $\mathcal{N} = \max_y e^{\gamma'|y-y_0|} \bar{N}_y^i(u)$. The lattice is finite, so $\mathcal{N} < \infty$, and $\mathcal{N}$ is invariant under real gauge transformations. On the stencils based in $\Lambda_3(c_y)$, $R(u_y)$ is orthogonal, and by (B) on $\Lambda_4(c_y)$ the transports of U^{u_y} differ from 1 by at most $C\eta\varepsilon$; so there $|\tilde{u}_y| = |u|$, $|\partial \tilde{u}_y| \leq |\nabla u| + C\varepsilon |u|$ and $|\nabla u| \leq |\partial \tilde{u}_y| + C\varepsilon |u|$. Apply Theorem 6.3(d) for $i = 1$, and (b) for $i = 0$, to the equation on $\Lambda_2(c_y)$, with $F = J_{F,y} + F_y[u]$ and $g = J_{g,y} + g_y[u]$ on $\Lambda_3(c_y)$. The cube $\Lambda_2(c_y)$ meets at most C blocks. As in the proof of Lemma 13.5, the two hypotheses give

$$\mathcal{N} \leq (1 + C\varepsilon) C_0 M + (1 + C\varepsilon)^2 C_{\mathrm{reg}} S_{\gamma/2}\, \varepsilon\, \mathcal{N}.$$

So $\varepsilon_* = \min\{1, [2(1+C)^2 C_{\mathrm{reg}} S_{\gamma/2}]^{-1}\}$ suffices. It does not depend on M, C_δ or γ_2. The Hölder bound follows from the same local estimate. The small inputs of δ_y are the gauge-invariant

norms $\bar{N}_z^i$; ordinary differences of $R(u_y)v$ at blocks far from y, where the gauge u_y is not controlled, do not enter. The Hölder hypotheses on J_y refer to the gauge u_y; data with Hölder bounds in another local gauge are moved to it by the change of local gauge after Definition 18.1. For $i = 0$ only values and block L^2 norms enter, and these are invariant under real gauge transformations; no Hölder bound on the divergence data is used. □

19 Gauge-field operators at regular backgrounds

19.1 Local potential terms and operators on domains

In this part we restrict the potential term to local operators, so that the Dirichlet versions on a domain X depend only on the background on the bonds of $\hat{X}$ (Definition 17.1).

Definition 19.1. $\mathcal{W}_{\mathrm{loc}}(\varepsilon)$ is the set of maps $U \mapsto W(U)$ with the following properties: $W(U)^T = W(U)$; $W(U)$ is real for real U, i.e. it maps real fields to real fields; $W(U^u) = R(u)W(U)R(u)^{-1}$; $(W(U)A)(b)$ depends only on the values of A, and of U, on bonds of plaquettes that contain b; and in every gauge in which (B) holds on a cube, the coefficients of W are bounded by ε and have Hölder quotients bounded by ε on that cube.

$\mathcal{W}^{\mathrm{an}}_{\mathrm{loc}}(\varepsilon)$ is the set of $W \in \mathcal{W}_{\mathrm{loc}}(\varepsilon)$ that extend holomorphically to the complex perturbations U_δ of Section 19.6, for $\delta \le \delta_W$ with some $\delta_W > 0$, with $W(U_\delta) - W(U)$ a local operator of size $C\delta$ of the same form.

The conditions defining $\mathcal{W}_{\mathrm{loc}}(\varepsilon)$ are spatial; they do not imply analytic dependence on the background. They suffice for the real results of this section. The results on complex perturbations (Proposition 19.11) are stated for $W \in \mathcal{W}^{\mathrm{an}}_{\mathrm{loc}}(\varepsilon)$. The only local potential used in the later parts is $W = 0$. The curvature term $\Delta'(U)$ of the Wilson action motivates these classes; we do not verify that it belongs to them (Section 12), and nothing below uses it. The potential W_2 and its Dirichlet versions W_2^X (Sections 37 and 41) are not local in this sense: $(W_2^X A)(b)$ depends on A only within distance 3 of b, but on U on all of $\hat{X}$, through ω_0^X. For them only existence, Re-coercivity and decay are taken from Proposition 19.11. Their comparison of domains is Proposition 35.1(d), which allows different potentials on the two domains, together with Proposition 37.4(d).

In a global gauge in which (B) holds, every $W \in \mathcal{W}_{\mathrm{loc}}(\varepsilon)$ belongs to $\mathcal{W}(C_\gamma\varepsilon, \gamma)$ for each fixed $\gamma > 0$. At a regular background the corresponding Hölder estimates hold in each local gauge in which (B) holds; the L^2 block-decay estimates hold in every gauge.

For a domain X let $\mathfrak{B}_X = \mathbf{1}_X(D^TD + W)\mathbf{1}_X$ and $\Pi_X = \mathbf{1}_X - DS_X$ on $\ell^2_1(X)$. Define

$$\mathfrak{A}^X = \Pi_X^T \mathfrak{B}_X \Pi_X + DR_X D_X^T + a\,\Pi_X^T \mathbf{Q}^T \mathbf{Q}\,\Pi_X \qquad \text{on } \ell^2_1(X).$$

Here $\mathbf{Q} = \mathbf{Q}(U)$, and all operators are restricted to $\ell^2_1(X)$. Let $J(X) = \{j : \Lambda_j \subset X\}$ and $m_X = \sum_{j \notin J(X)} \chi_j^2$ (as a multiplication operator on $\ell^2_1(X)$). For $\kappa \ge 0$ put

$$\mathfrak{A}^{X,\kappa} = \mathfrak{A}^X + \mathfrak{A}^{\mathrm{ext}}_X + \kappa\, m_X, \qquad \mathfrak{A}^{\mathrm{ext}}_X = D\mathbf{1}_{\mathbb{T}_\eta \setminus X} D^T \quad \text{on } \ell^2_1(X).$$

In Part IX, where κ denotes the weakening rate, this mass parameter is written κ_m. For $X = \mathbb{T}_\eta$ every j belongs to $J(\mathbb{T}_\eta)$, so $m_{\mathbb{T}_\eta} = 0$, and $\mathfrak{A}^{\mathbb{T}_\eta}$ is the operator $\mathfrak{A}_\pi(U)$ of Section 15. Its scalar ingredients are now those of Section 18. If $n < 8M$, the partition is the trivial one of Section 17; then $J(X)$ is empty and $m_X = 1$ for every proper domain X. Put $c_{\mathrm{seam}}(d) = \max(10, 5\sqrt{d})$, so that $c_{\mathrm{seam}}(d) = 10$ for $d \le 4$, and take $M_0 \ge 2$. For $M \ge M_0$ and a proper domain the mass term is supported in blocks at Euclidean distance at most $c_{\mathrm{seam}}(d)M$ from $\mathbb{T}_\eta \setminus X$. It plays the role of a boundary condition that is harmless for the gauge structure: it makes the local operators coercive near the boundary, where nothing else does.

Dirichlet conditions for all components. In $\mathfrak{A}_X^{\mathrm{ext}}$, D^T is the torus divergence of the zero extension, so $\langle A, \mathfrak{A}_X^{\mathrm{ext}} A\rangle = \|\mathbf{1}_{\mathbb{T}_\eta \setminus X} D^T A\|^2$. At a point $x \notin X$, $(D^T A)(x)$ involves only the bonds from x into X. So $\mathfrak{A}_X^{\mathrm{ext}}$ acts only on the bonds crossing ∂X, and $\mathfrak{A}_{\mathbb{T}_\eta}^{\mathrm{ext}} = 0$. With this term the divergence part of the form is the full $\|D^T A\|^2$. The local part of $\mathfrak{A}^{X,\kappa}$ is then the Dirichlet restriction $\mathbf{1}_X(D^T D + DD^T + W + a\mathbf{Q}^T\mathbf{Q})\mathbf{1}_X$ of the torus operator: all components of the gauge field satisfy Dirichlet conditions. Without $\mathfrak{A}_X^{\mathrm{ext}}$ the normal components on the bonds crossing ∂X would be free; these are relative boundary conditions, as for 1-forms in the continuum. The local commutators of the walks of Section 41 would then not be bounded uniformly in η (Remark 41.5). On $\ell_1^2(X)$ we use, besides $\|\!| \cdot |\!\|_X$, the norm

$$\|\!| A |\!\|_{X,+}^2 = \|\!| A |\!\|_X^2 + \|\mathbf{1}_{\mathbb{T}_\eta \setminus X} D^T A\|^2 = \|A\|^2 + \|DA\|^2 + \|D^T A\|^2,$$

with the torus operators applied to the zero extension.

The operators $\mathfrak{A}^{X,\kappa}$ are transpose-symmetric, real for real U, depend only on U on $\hat{X}$ (the crossing bonds lie in $\hat{X}$), and are gauge covariant.

Lemma 19.2 (Identities on domains)**.** *For a regular background and a domain X, the statements of Lemmas 15.1 and 15.3 hold with Π, S, R, G', $\mathcal{N}(Q')$, $\mathfrak{A}_\pi$ replaced by Π_X, S_X, R_X, G'_X, $\mathcal{N}_X$, $\mathfrak{A}^X$, and with D^T replaced by D_X^T.*

Proof. The proofs of Lemmas 15.1 and 15.3 use only the identities $Q'G'R = 0$, $D^T D G' R = R$, $\mathbf{Q}D = D_V Q' + E$ and the definitions. The first two hold on X by Theorem 18.2(d). The third holds for all fields. □

The identities hold for $\mathfrak{A}^X$. For $\mathfrak{A}^{X,\kappa}$ on a proper domain, the analogue of Lemma 15.3(a) holds only up to the terms $\kappa m_X D\nu$ and $\mathfrak{A}_X^{\mathrm{ext}} D\nu$. These are supported in blocks at Euclidean distance at most $c_{\mathrm{seam}}(d)M$ from $\mathbb{T}_\eta \setminus X$, respectively on the bonds crossing ∂X.

Lemma 19.3 (The full gradient on domains)**.** *Let U be real with curvature at most ε; (M,ε)-regular backgrounds have curvature at most $C\varepsilon$ (Lemma 12.3). Let ∇ be the covariant gradient, with components $D_\nu A_\mu$. Then for every domain X and every real $A \in \ell_1^2(X)$, extended by zero,*

$$\|\nabla A\|^2 \le \|\!| A |\!\|_{X,+}^2 + C\varepsilon \|A\|^2.$$

For complex A the same holds, by splitting into real and imaginary parts. At the complex perturbations U_δ of Proposition 19.11, $\|\nabla_{U_\delta} A\|^2 \le (1 + C\delta)\|\!| A |\!\|_{X,+}^2 + C(\varepsilon + \delta)\|A\|^2$, with the norm taken at U_δ.

Proof. The covariant Hodge inequality $\big|\|Dh\|^2 + \|D^T h\|^2 - \|\nabla h\|^2\big| \le C\varepsilon\|h\|^2$ holds for every gauge field h on the torus (Lemma 26.4). Its proof is a summation by parts and uses only the curvature bound. Apply it to the zero extension of A: its torus norms $\|DA\|$ and $\|D^T A\|$ are those in $\|\!| A |\!\|_{X,+}$. At U_δ the operators D, D^T and ∇ change by local operators of size $C\delta$. □

This is the gain from the Dirichlet conditions. With $\|\!| \cdot |\!\|_X$ alone, the full gradient is not controlled up to ∂X, because the exterior divergence is missing.

19.2 Coercivity from local information

Lemma 19.4 (Gårding inequality)**.** *There are ε_0, C_1 such that for $(1,\varepsilon)$-regular U with $\varepsilon \le \varepsilon_0$, every domain X and every real $B \in \ell_1^2(X)$,*

$$\langle B, \mathfrak{A}^X B\rangle \ge \tfrac{1}{2}\|\!| B |\!\|_X^2 - C_1\|B\|^2.$$

Proof. For real B,

$$\langle B, \mathfrak{A}^X B\rangle = \|D\Pi_X B\|^2 + \langle \Pi_X B, W\Pi_X B\rangle + \|R_X D_X^T B\|^2 + a\|\mathbf{Q}\Pi_X B\|^2.$$

By Lemma 19.2, $D\Pi_X B = DB - \mathfrak{F}S_X B$, where $\|\mathfrak{F}S_X B\|$ is at most $C\varepsilon\|B\|$. Moreover $|\langle \Pi_X B, W\Pi_X B\rangle| \le C\varepsilon\|B\|^2$, because Π_X is bounded by Theorem 18.2(d). Finally, $\|R_X D_X^T B\| \ge \|D_X^T B\| - \|P_{\mathcal{V},X} D_X^T B\|$, and the last term is at most $C\|B\|$. Use $(s-t)^2 \ge \frac{1}{2}s^2 - t^2$. □

The Gårding inequality controls everything except $\|B\|$ itself. That control comes from the averaging and the gauge fixing, and this is where Part I enters.

Lemma 19.5 (Local coercivity)**.** *There is M_0 with the following property. Let $M \ge M_0$, and let U be (M,ε)-regular with $\varepsilon \le \varepsilon_0$, where ε_0 does not depend on M. Let X be a domain and $j \in J(X)$. Then every real $B \in \ell^2_1(X)$ vanishing on bonds that do not start in K_j satisfies*

$$\langle B, \mathfrak{A}^X B\rangle \ge \tfrac{c_g}{2}\|\!|B|\!\|_X^2,$$

with c_g the constant of Theorem 9.6.

Proof. Step 1: from X to Λ_j. Compare $\mathfrak{A}^X$ with $\mathfrak{A}^{\Lambda_j}$ on such B. The local parts agree, since B and its derivatives are supported inside Λ_j. The quadratic forms differ through $\Pi_X - \Pi_{\Lambda_j} = -D(S_X - S_{\Lambda_j})$ and $R_X - R_{\Lambda_j} = P_{\mathcal{V},\Lambda_j} - P_{\mathcal{V},X}$. By Theorem 18.2(e), applied to DS and to $DP_{\mathcal{V}}D^T$, and since K_j has distance at least $2M-3$ from the seam $\Sigma(X,\Lambda_j)$, these differences, acting on fields supported in K_j, are bounded by $Ce^{-\gamma M}$. On blocks meeting $X \setminus \Lambda_j$, differentiated operators on Λ_j can have values on crossing bonds. Applied to fields supported in K_j, both these operators and the operators on X are of size $Ce^{-\gamma M}$ on such blocks, by their individual decay bounds. So

$$|\langle B, (\mathfrak{A}^X - \mathfrak{A}^{\Lambda_j})B\rangle| \le Ce^{-\gamma M}\|\!|B|\!\|_X^2.$$

Step 2: the local gauge. Let u be a gauge in which (B) holds on $\hat\Lambda_j$, and $\tilde B = R(u)B$. By covariance, $\langle B, \mathfrak{A}^{\Lambda_j}(U)B\rangle = \langle \tilde B, \mathfrak{A}^{\Lambda_j}(U^u)\tilde B\rangle$, where $\mathfrak{A}^{\Lambda_j}(U^u)$ carries the potential $W(U^u) = R(u)W(U)R(u)^{-1}$. Here and in Step 3, $\mathfrak{A}^{\Lambda_j}(1)$ is the operator at $U=1$ with the potential $W=0$. Since U^u satisfies (B) on $\hat\Lambda_j$, the proof of Theorem 15.4(a) applies on the domain Λ_j without change: all operators are Dirichlet operators on Λ_j, and their scalar ingredients are controlled by Theorem 18.2. That proof puts the potential into the perturbation: δ_π contains W, and $\mathcal{K}_1$ contains the terms with W. The potential enters only through $\langle \Pi_{\Lambda_j}\tilde B, W(U^u)\Pi_{\Lambda_j}\tilde B\rangle$; since $\|W(U^u)\| = \|W(U)\| \le C\varepsilon$ and Π_{Λ_j} is bounded (Theorem 18.2(d)), this is at most $C\varepsilon\|\tilde B\|^2$. So the proof shows that $\mathfrak{A}^{\Lambda_j}(U^u) - \mathfrak{A}^{\Lambda_j}(1)$ is an admissible perturbation of size $C\varepsilon$, with C independent of M. Hence $|\langle \tilde B, (\mathfrak{A}^{\Lambda_j}(U^u) - \mathfrak{A}^{\Lambda_j}(1))\tilde B\rangle| \le C\varepsilon\|\!|\tilde B|\!\|^2_{\Lambda_j,1}$. Here the subscript 1 indicates the norm at $U=1$. The admissible perturbation contains componentwise differences, so this bound needs $\|\partial\tilde B\|$. Since $\tilde B$ vanishes on the bonds that do not start in K_j, which has distance at least $2M-1$ from $\mathbb{T}_\eta \setminus \Lambda_j$, the exterior divergence of $\tilde B$ vanishes. So $\|\!|\tilde B|\!\|_{\Lambda_j,1}$ is the torus norm of $\tilde B$ at $U=1$, which controls $\|\partial\tilde B\|$ by Lemma 2.2.

Step 3: $U=1$ and $W=0$. At $U=1$ we have $d\Pi_{\Lambda_j}(1) = d$ and $\mathbf{Q}\Pi_{\Lambda_j}(1) = \mathbf{Q}$, as in Lemma 15.1(d), and $d^*_{\Lambda_j}\tilde B = d^*\tilde B$. The operator $\mathfrak{A}^{\Lambda_j}(1)$ has no potential term; the potential is part of the perturbation of Step 2. Therefore

$$\langle \tilde B, \mathfrak{A}^{\Lambda_j}(1)\tilde B\rangle - \langle \tilde B, \mathbf{\Delta}_a\tilde B\rangle = \langle \tilde B, d(P_{\mathcal{V}}(1) - P_{\mathcal{V},\Lambda_j}(1))d^*\tilde B\rangle,$$

where $\mathbf{\Delta}_a$ is the torus operator of Part I. By Theorem 18.2(e) with $X = \mathbb{T}_\eta$, $X' = \Lambda_j$, this is bounded by $Ce^{-\gamma M}\|\tilde B\|^2$. By Theorem 9.6 and Lemma 2.2, $\langle \tilde B, \mathbf{\Delta}_a\tilde B\rangle \ge c_g\|\!|\tilde B|\!\|^2_{\Lambda_j,1}$.

Conclusion. The norms $\|\tilde{B}\|_{\Lambda_j,1}$ and $\|B\|_X$ differ by a factor $1+C\varepsilon$, because $D_{U^u} = \partial + T$ with $\|T\| \le C\varepsilon$. Collecting the three steps,

$$\langle B, \mathfrak{A}^X B\rangle \ge (c_g - Ce^{-\gamma M} - C\varepsilon)(1+C\varepsilon)^{-1}\|B\|_X^2.$$

The constants C and γ do not depend on M. Choose M_0 with $Ce^{-\gamma M_0} \le \frac{c_g}{8}$, and ε_0, not larger than the thresholds of Theorems 18.2 and 15.4, with $C\varepsilon_0 \le \min(\frac{c_g}{8}, \frac{1}{2})$. Then the right side is at least $\frac{c_g}{2}\|B\|_X^2$ for $M \ge M_0$ and $\varepsilon \le \varepsilon_0$, and ε_0 does not depend on M. □

Theorem 19.6 (Coercivity at regular backgrounds)**.** *There are M_0, κ, $c_* > 0$ and $\varepsilon_0 > 0$, depending only on d, a, α, $\dim \mathfrak{g}$, such that the following holds for every $M \ge M_0$, every (M,ε)-regular background with $\varepsilon \le \varepsilon_0$ and every domain X. In particular ε_0 does not depend on M:*

$$\langle A, \mathfrak{A}^{X,\kappa} A\rangle \ge c_* \|A\|_{X,+}^2 \qquad \text{for all real } A \in \ell_1^2(X).$$

In particular, for $X = \mathbb{T}_\eta$ the operator $\mathfrak{A}_\pi(U)$ of Part II is coercive under the local hypothesis of regularity, without hypothesis (B).

Proof. Since $\langle A, \mathfrak{A}_X^{\mathrm{ext}} A\rangle = \|\mathbf{1}_{\mathbb{T}_\eta \setminus X} D^T A\|^2$, it suffices to show $\langle A, (\mathfrak{A}^X + \kappa m_X)A\rangle \ge c_*' \|A\|_X^2$, and to take $c_* = \min(c_*', 1)$. Let $\kappa = C_1 + \frac{1}{2}$, with C_1 from Lemma 19.4. By Lemma 17.4 applied to $\mathfrak{A}^X$, and since $\langle A, m_X A\rangle = \sum_{j \notin J(X)} \|\chi_j A\|^2$,

$$\begin{aligned}\langle A, (\mathfrak{A}^X + \kappa m_X)A\rangle = &\sum_{j \in J(X)} \langle \chi_j A, \mathfrak{A}^X \chi_j A\rangle + \sum_{j \notin J(X)} \Big(\langle \chi_j A, \mathfrak{A}^X \chi_j A\rangle + \kappa \|\chi_j A\|^2\Big) \\ &+ \frac{1}{2}\sum_j \langle A, [\chi_j, [\chi_j, \mathfrak{A}^X]]A\rangle.\end{aligned}$$

For $j \in J(X)$, Lemma 19.5 bounds the term below by $\frac{c_g}{2}\|\chi_j A\|_X^2$. For $j \notin J(X)$, Lemma 19.4 gives at least $\frac{1}{2}\|\chi_j A\|_X^2$. The operator $\mathfrak{A}^X$ is a finite sum of terms $X^T M Y$ as in Lemma 17.5. Indeed, $\Pi_X = \mathbf{1}_X - DS_X$ and $D\Pi_X = D - \mathfrak{F}S_X$ (Lemma 19.2) give

$$\Pi_X^T \mathfrak{B}_X \Pi_X = D^T D - D^T \mathfrak{F} S_X - (D^T \mathfrak{F} S_X)^T + (\mathfrak{F} S_X)^T \mathfrak{F} S_X + \Pi_X^T W \Pi_X,$$

and $DR_X D_X^T = DD_X^T - DP_{\mathcal{V},X} D_X^T$. The local terms $D^T D$ and DD_X^T have $M = 1$. The terms $(\mathfrak{F}S_X)^T \mathfrak{F} S_X$, $\Pi_X^T W \Pi_X$, $DP_{\mathcal{V},X} D_X^T$ and $a\Pi_X^T \mathbf{Q}^T \mathbf{Q} \Pi_X$ have $X = Y = 1$ and an M with decay, by Theorem 18.2(d). The term $D^T \mathfrak{F} S_X$ has $X = D$, $M = \mathfrak{F}S_X$ and $Y = 1$, and its transpose has $X = 1$ and $Y = D$. No derivative of the coefficients of $\mathfrak{F}$ or W enters. Hence the double commutators are bounded by $CM^{-2}\|A\|_X^2$. With Lemma 17.6,

$$\langle A, (\mathfrak{A}^X + \kappa m_X)A\rangle \ge \tfrac{1}{2}\min(c_g, 1)\Big(\tfrac{1}{2}\|A\|_X^2 - CM^{-2}\|A\|^2\Big) - CM^{-2}\|A\|_X^2 \ge c_*' \|A\|_X^2$$

for $M \ge M_0$.

If $n < 8M$, the partition is trivial (Section 17), and the double commutators vanish. For $X = \mathbb{T}_\eta$, the proof of Lemma 19.5 applies with $j = 0$ and $\Lambda_0 = \mathbb{T}_\eta$, in the global gauge: Step 1 and the error of Step 3 are absent. So $\langle A, \mathfrak{A}^{\mathbb{T}_\eta} A\rangle \ge \frac{c_g}{2}\|A\|_{\mathbb{T}_\eta}^2$. For a proper domain, $m_X = 1$, and Lemma 19.4 with $\kappa = C_1 + \frac{1}{2}$ gives $\langle A, (\mathfrak{A}^X + \kappa m_X)A\rangle \ge \frac{1}{2}\|A\|_X^2$. By the last display $c_*' \le \frac{1}{4}\min(c_g, 1)$, so both bounds are at least $c_*'\|A\|_X^2$, and the theorem holds with the same κ and c_*. □

19.3 Green functions at regular backgrounds

Theorem 19.7 (Gauge-field Green functions at regular backgrounds)**.** *Let $M \ge M_0$ and let U be (M,ε)-regular with $\varepsilon \le \varepsilon_0$.*

(a) *For every domain X, $\mathbf{G}^X = (\mathfrak{A}^{X,\kappa})^{-1}$ exists, and the following operators have decay (C,γ), uniformly in X:*

- *$\mathbf{G}^X$, $D\mathbf{G}^X$, $\mathbf{G}^X D^T$, $D\mathbf{G}^X D^T$, with D the derivative from gauge fields to 2-forms;*
- *$D^T\mathbf{G}^X$, $D^T\mathbf{G}^X D^T$, $\mathbf{G}^X D$, $D\mathbf{G}^X D$ and $D^T\mathbf{G}^X D$, with D^T on the left the torus divergence of the zero extension (which contains $D_X^T = \mathbf{1}_X D^T$), and D on the right the gradient of torus scalar fields followed by the projection onto $\ell^2_1(X)$; in $D^T\mathbf{G}^X D^T$ the right factor is the codifferential of 2-forms, and in $D\mathbf{G}^X D$ the left factor is the derivative into 2-forms;*
- *$\nabla\mathbf{G}^X$, $\nabla\mathbf{G}^X D^T$ and $\nabla\mathbf{G}^X D$, with ∇ the covariant gradient.*

(b) *For $X = \mathbb{T}_\eta$, the operator $\mathbf{G}(U) = \mathfrak{A}_\pi(U)^{-1}$ has the decay of (a) and satisfies Theorem 15.4(c), (d): the pointwise bounds ($d \geq 3$) and the identities. In the pointwise bounds $|\partial\,\cdot\,|$ is replaced by $|\nabla\,\cdot\,|$, with ∇ the covariant gradient of (a), and Hölder quotients are taken in local gauges. The comparison of Theorem 15.4(b) with the operator at $U = 1$ is not claimed: it needs a global gauge in which (B) holds. For a pure gauge $U = 1^u$ one has $\mathbf{G}(U) = R(u)\mathbf{G}(1)R(u)^{-1}$, which is not close to $\mathbf{G}(1)$ unless $R(u)$ is nearly constant.*

Proof. (a) Theorem 19.6 gives hypothesis (i) of Lemma 5.7, with the norm $\|\!|\!|\cdot|\!|\!|_{X,+}$; the proof of that lemma goes through for this norm. For hypothesis (ii) note that conjugating $\mathbf{1}_{\mathbb{T}_\eta\setminus X}D^T$ by $e^{\alpha\rho}$ changes it by an operator of norm at most $C|\alpha|$. Since $\|\!|\!|A|\!|\!|^2_{X,+} = \|A\|^2 + \|DA\|^2 + \|D^TA\|^2$, Combes–Thomas in this norm controls all blocks of $\mathcal{D}\mathbf{G}^X\mathcal{D}^T$ with $\mathcal{D} = (D, D^T)$. This gives the operators of the first two items; $\mathbf{G}^X D$ and $D\mathbf{G}^X D$ are transposes of $D^T\mathbf{G}^X$ and $D^T\mathbf{G}^X D^T$. For the last item apply Lemma 19.3 to φv, where $v = \mathbf{G}^X g$, $\mathbf{G}^X D^T g$ or $\mathbf{G}^X D g$ and φ is a cutoff at unit scale around a block. The commutators of D, D^T with φ are bounded, so $\|\nabla v\|_{B(y)}$ is bounded by $\|v\| + \|Dv\| + \|D^Tv\|$ near $B(y)$, and these decay. Hypothesis (ii) follows from the structure of $\mathfrak{A}^X$ as a sum of terms X^TMY with decaying M, by Lemmas 5.3 and 5.8. The mass term commutes with $e^{\alpha\rho}$.

(b) Decay is (a) with $X = \mathbb{T}_\eta$. The identities follow from Lemma 19.2. For the pointwise bounds, write the equation $\mathfrak{A}_\pi u = J$ on $\Lambda_3(c_y)$ in a gauge in which (B) holds on $\Lambda_4(c_y)$. The local parts of $\mathfrak{A}_\pi$ have the normal form of Lemma 12.2. The non-local parts ($DP_{\mathcal{V}}D^T$ and the terms involving S) were shown in the proof of Theorem 15.4(a) to be pointwise admissible. That proof used only the pointwise bounds of the scalar operators, which hold at regular backgrounds by Theorem 18.2(g). These scalar bounds hold on the cubes $\Lambda_1(z)$, in their local gauges (Remark 7.2). By the change of local gauge after Definition 18.1, they hold, with a larger constant, in the gauge of $\Lambda_4(c_y)$ on $\Lambda_3(c_y)$. The small terms of the non-local parts involve only values, Du and Hölder quotients in local gauges, so they are bounded by the invariant norms $|u| + |\nabla u|$ on the cubes $\Lambda_1(c_{y'})$, with weights $e^{-\gamma|y-y'|}$. Lemma 18.3 concludes. □

19.4 The averaged Green function, the minimizer and the fluctuation operator

At $U = 1$ the lower bound on $\mathbf{QGQ}^*$ came from Fourier analysis. At regular backgrounds we use explicit trial fields instead.

Lemma 19.8 (Trial fields for the gauge average)**.** *Let U be $(1,\varepsilon)$-regular with $\varepsilon \leq \varepsilon_0$. There is a linear map $b \mapsto \Phi_b$ from unit-lattice gauge fields to fine gauge fields such that $\mathbf{Q}(U)\Phi_b = b$, $\|\!|\!|\Phi_b|\!|\!|_{\mathbb{T}_\eta} \leq C\|b\|$, and Φ_b has exponential decay away from the support of b. Moreover there is $\Phi_b^\pi \in \mathcal{S}(U)$ with $\mathbf{Q}_\pi\Phi_b^\pi = b$ and $\|\!|\!|\Phi_b^\pi|\!|\!|_{\mathbb{T}_\eta} \leq C\|b\|$.*

Proof. Fix a unit bond (y,μ). For $N \geq 16$ let $\phi(t) = \sin^2(\pi(t - \frac{1}{2}))$ for $t \in [\frac{1}{2}, \frac{3}{2}]$ and $\phi(t) = 0$ otherwise. Let $\psi_\perp(x) = \prod_{\nu\neq\mu}\sin^2(\pi(x_\nu - y_\nu))$ for $x \in B(y)$ and 0 otherwise; this function

vanishes on the transverse faces of $B(y)$. For a fine point x with $0 \le x_\nu - y_\nu < 1$ for $\nu \neq \mu$ and $0 \le x_\mu - y_\mu < 2$, let x_0 be its projection onto the face $\{x_\mu = y_\mu\}$ of $B(y)$, so that $x = x_0 + j\eta e_\mu$ with $0 \le j < 2N$. Let $\Gamma^{(\mu)}_{y,x} = \Gamma_{y,x_0} \cup [x_0, x]$. This canonical contour is single-valued in x. Define the μ-component

$$\Phi_{(y,\mu)}(x) = \psi_\perp(x_0)\, \phi(x_\mu - y_\mu)\, R(U(\Gamma^{(\mu)}_{y,x}))^{-1},$$

acting on a vector $\beta \in \mathfrak{g}^c$; all other components are 0. In $\mathbf{Q}(U)$ the value at x is transported along $\Gamma_{y,x'} \cup [x', x]$, for each base point $x' \in B(y)$ on the line of x with $0 \le x_\mu - x'_\mu < 1$. This contour and the canonical one have the same endpoints, lie in $B(y) \cup B(y + e_\mu)$, and have at most CN bonds. Since U is $(1, \varepsilon)$-regular, (B) holds in some gauge on a cube containing $B(y) \cup B(y + e_\mu)$. There every bond variable is $e^{i\eta A}$ with $|A| \le \varepsilon$, so both transports are within $C\varepsilon$ of 1, and so is the product of the one with the inverse of the other. This statement is gauge covariant. Hence

$$(\mathbf{Q}(U)\Phi_{(y,\mu)}\beta)(y,\mu) = m_\perp \sigma_0 \beta + r_{(y,\mu)}\beta, \qquad \|r_{(y,\mu)}\| \le C\varepsilon,$$

with $m_\perp = 2^{1-d}$ and $\sigma_0 = \sum_t \eta\, w(t)\phi(t)$, where $w(t)$ is the weight of the tube average along the line (Lemma 2.6). With exact cancellation of the transports the error term would vanish; it does not, since the transports depend on the base point x' when the background is curved. The only other unit bonds where $\mathbf{Q}(U)\Phi_{(y,\mu)}\beta$ does not vanish are $(y \pm e_\mu, \mu)$. There the value has norm at most $m_\perp \sigma_\pm (1 + C\varepsilon)|\beta|$, with $\sigma_\pm$ the corresponding weighted sums. The tube weights of the bonds $(y + me_\mu, \mu)$, $m \in \mathbb{Z}$, add up to 1 at every bond in direction μ (Lemma 2.6(a)), so $\sigma_0 + \sigma_+ + \sigma_- = \sum_t \eta\, \phi(t) = \frac{1}{2}$. For $t \in \eta\mathbb{Z}$ with $0 \le t \le 2 - 2\eta$ the weight of (y, μ) is $w(t) = 1 - |t - 1 + \eta|$. Since $\phi(1 + s) = \phi(1 - s)$ and $\phi(1) = 1$, pairing the samples $t = 1 \pm m\eta$ shows that $\sigma_0 = \sum_t \eta\, (1 - |t-1|)\phi(t) - \eta^2$. With $\phi(1 + m\eta) = \cos^2(\pi m\eta)$, summing the arithmetico-geometric series gives

$$\sigma_0 = \frac{3}{8} + \frac{\eta^2}{2\sin^2(\pi\eta)} - \eta^2 \quad (N \text{ even}), \qquad \sigma_0 = \frac{3}{8} + \frac{\eta^2}{8} + \frac{\eta^2}{8\sin^2(\pi\eta/2)} - \eta^2 \quad (N \text{ odd}).$$

Since $\sin s \le s$, for $N \ge 16$ we get $\sigma_0 \ge \frac{3}{8} + \frac{1}{2\pi^2} - \eta^2 > \frac{2}{5}$ and $\sigma_+ + \sigma_- = \frac{1}{2} - \sigma_0 < \frac{1}{10}$. The continuum values are $\frac{3}{8} + \frac{1}{2\pi^2}$ and $\frac{1}{8} - \frac{1}{2\pi^2}$, and the errors are $O(\eta^2)$. For $N < 16$ we replace ϕ by the indicator of the single bond crossing the face between $B(y)$ and $B(y + e_\mu)$. Then $\sigma_\pm = 0$ and $\sigma_0 = \eta \ge \frac{1}{16}$, and derivatives cost at most a factor $2/\eta \le 32$.

Thus $\mathbf{Q}(U) \sum_{(y,\mu)} \Phi_{(y,\mu)}\beta_{(y,\mu)} = T_U \beta$. The diagonal of T_U is $m_\perp\sigma_0 + O(\varepsilon)$, and the sum of the off-diagonal entries of a row is at most $m_\perp(\sigma_+ + \sigma_-)(1 + C\varepsilon)$. Since $\sigma_0 - \sigma_+ - \sigma_- \ge \frac{1}{5}$, resp. $\sigma_0 \ge \frac{1}{16}$ and $\sigma_\pm = 0$ for $N < 16$, T_U is diagonally dominant uniformly in $\varepsilon \le \varepsilon_0$, for ε_0 small. So T_U^{-1} is bounded with exponentially decaying kernel, by the Neumann series around the diagonal. Put $\Phi_b = \sum \Phi_{(y,\mu)}(T_U^{-1}b)_{(y,\mu)}$.

The derivative bound follows as in the proof of Theorem 18.2(c). In a local gauge the canonical transports vary by $O(\eta\varepsilon)$ between neighbouring points. For a step in direction μ the contour grows by one bond. For a step ηe_ν, $\nu \neq \mu$, the two canonical contours agree up to the segment in direction ν, where their lengths differ by one bond, and then run parallel at distance η. So, together with that bond, they bound a ladder of at most CN plaquettes, each with holonomy $O(\eta^2\varepsilon)$ in the gauge where (B) holds. The holonomy of the loop is a product of conjugates of these plaquette holonomies. Since U is real, $|g_1 \cdots g_m - 1| \le \sum_i |g_i - 1|$ and $|hgh^{-1} - 1| = |g - 1|$ for G-valued g_i, g, h. So the holonomy of the loop is $O(N\eta^2\varepsilon) = O(\eta\varepsilon)$. For $N \ge 16$, moreover ϕ and $\psi_\perp$ are smooth on their supports, ϕ vanishes at $t = \frac{1}{2}$ and $t = \frac{3}{2}$, and $\psi_\perp$ is $O(\eta^2)$ next to the transverse faces, so $\Phi_{(y,\mu)}$ jumps by $O(\eta^2)$ across the boundary of its support. For $N < 16$ use instead the bound $2/\eta \le 32$ for the indicator profile. For the slice version, let $L : b \mapsto \Phi_b$. Then $\mathbf{Q}_\pi L = \mathbf{Q}\Pi L = I - ESL$, by Lemma 15.1(c), and

$\|ESL\| \le C\varepsilon \le \frac{1}{2}$. Put $\Phi_b^\pi = \Pi L(I - ESL)^{-1} b$; then $\mathbf{Q}_\pi \Phi_b^\pi = b$, since $\mathbf{Q}_\pi \Pi = \mathbf{Q}_\pi$. Finally $\Pi = I - DS$ is bounded in $\|\!|\!| \cdot |\!|\!|$, because DS is bounded, $D(DS) = \mathfrak{F}S$, and $D^T DS = RD^T$ by Theorem 18.2(d). □

Theorem 19.9 (Minimizer and fluctuation operator at regular backgrounds)**.** *Let $M \ge M_0$ and let U be (M, ε)-regular with $\varepsilon \le \varepsilon_0$. Then $\mathbf{Q}_\pi \mathbf{G} \mathbf{Q}_\pi^T \ge c$. Its inverse $\mathbf{\Gamma}_\pi$ has decay (C, γ). Theorem 16.1(b)–(d) hold for $\mathbf{H}(U)$ and $\mathbf{C}(U)$, with $|\partial \cdot|$ replaced by $|\nabla \cdot|$ (the covariant gradient of Theorem 19.7(a)) in the pointwise bounds and Hölder quotients taken in local gauges. The comparison of Theorem 16.1(a) with the operator $\mathbf{Q}\mathbf{G}\mathbf{Q}^*$ at $U = 1$ is not claimed; the lower bound above takes its place.*

Proof. Since $\mathfrak{A}_\pi$ is positive (Theorem 19.6), the minimum of $\langle A, \mathfrak{A}_\pi A \rangle$ over $\{A : \mathbf{Q}_\pi A = b\}$ equals $\langle b, (\mathbf{Q}_\pi \mathbf{G} \mathbf{Q}_\pi^T)^{-1} b \rangle$. With the trial field of Lemma 19.8 this is at most $\langle \Phi_b^\pi, \mathfrak{A}_\pi \Phi_b^\pi \rangle \le C |\!|\!| \Phi_b^\pi |\!|\!|^2 \le C \|b\|^2$. Hence $(\mathbf{Q}_\pi \mathbf{G} \mathbf{Q}_\pi^T)^{-1} \le C$. Decay of the inverse follows from Lemma 5.5. The rest of the proof of Theorem 16.1 is algebraic, or uses only the bounds now available: Theorems 19.7 and 18.2, and Lemma 18.3 in place of Lemma 13.5. □

19.5 Comparison of domains and locality in the background

Theorem 19.10 (Locality)**.** *Let $M \ge M_0$ and $\varepsilon \le \varepsilon_0$.*

(a) Let $X' \subset X$ be domains, U an (M, ε)-regular background, and Σ_M the set of unit points of X within distance $c_{\text{seam}}(d)M$ of $X \setminus X'$. For all unit points y, y', with the operators extended by zero,

$$\|\mathbf{1}_y (\mathbf{G}^X - \mathbf{G}^{X'}) \mathbf{1}_{y'}\| \le C \sum_{w \in \Sigma_M} e^{-\gamma(|y-w| + |w-y'|)}.$$

The same holds for $P\mathbf{G}^X Q - P\mathbf{G}^{X'} Q$ with $P \in \{1, D, D^T, \nabla\}$ and $Q \in \{1, D^T, D\}$, as in Theorem 19.7(a). The bound covers the blocks of the initial points of the bonds of $\ell_1^2(X')$ that cross $\partial X'$, which may lie outside X_1', and, for $P = D^T$, the torus divergence, the output points within distance η of X'.

(b) Let U and U' be (M, ε)-regular backgrounds that coincide on the bonds of $\hat{X}'$. Then each of the torus operators $\mathbf{G}$, $D\mathbf{G}$, $\mathbf{G}D^T$, $\mathbf{\Gamma}_\pi$, $\mathbf{H}$, $\mathbf{C}$ satisfies

$$\begin{aligned} \|\mathbf{1}_y (F(U) - F(U')) \mathbf{1}_{y'}\| &\le C \sum_{w \in \Sigma_M} e^{-\gamma(|y-w| + |w-y'|)} \\ &\le C' e^{CM} e^{-\frac{\gamma}{2}(\operatorname{dist}(y, \mathbb{T}_\eta \setminus X') + \operatorname{dist}(y', \mathbb{T}_\eta \setminus X'))}, \end{aligned}$$

where Σ_M is defined with $X = \mathbb{T}_\eta$.

Proof. (a) We use a smooth cutoff. A sharp cut would produce one-sided derivatives at the cut, which are not bounded uniformly in η. Let χ be a smooth function, at unit scale, of a mollified torus distance to $X \setminus X'$, with $|\partial^k \chi| \le C$ for $k \le 2$. It vanishes at distance at most 1 and equals 1 at distance at least 3. Gauge fields are multiplied by $(\chi A)(b) = \chi(b_-) A(b)$. Then $\chi(b_-) = 0$ for every bond b of $\ell_1^2(X)$ that is not in $\ell_1^2(X')$. So χ maps $\ell_1^2(X)$ into $\ell_1^2(X')$, and $P'\chi = \chi P$, with P, P' the projections onto $\ell_1^2(X)$, $\ell_1^2(X')$.

Write $\mathfrak{A}^{X,\kappa} = PLP + \mathfrak{N}^X + \kappa m_X$. Here $L = D^T D + DD^T + W + a\mathbf{Q}^T\mathbf{Q}$ is the torus operator: its local part includes $\mathfrak{A}_X^{\text{ext}}$. And $\mathfrak{N}^X$ is the non-local part, built from $P_{\mathcal{V},X}$, S_X and R_X. The same holds on X' with the same L. Let $u = \mathbf{G}^X g$. Then $P'LP'(\chi u) = \chi P L u + P'[L, \chi]u$, so

$$\mathfrak{A}^{X',\kappa}(\chi u) = \chi g + P'[L, \chi]u + (\mathfrak{N}^{X'} - P'\mathfrak{N}^X P')\chi u + P'[\mathfrak{N}^X, \chi]u + \kappa(m_{X'} - m_X)\chi u.$$

Since $\chi u \in \ell^2_1(X')$, this gives, as torus fields,

$$\mathbf{G}^X g - \mathbf{G}^{X'} g = (1-\chi)u - \mathbf{G}^{X'}(1-\chi)g \\ + \mathbf{G}^{X'}\Big(P'[L,\chi] + (\mathfrak{N}^{X'} - P'\mathfrak{N}^X P')\chi + P'[\mathfrak{N}^X,\chi] + \kappa(m_{X'} - m_X)\chi\Big)u.$$

Each term has the seam form:

- $1-\chi$ is supported within distance 3 of $X \setminus X'$.
- $[L,\chi]$ is a first-order operator with bounded coefficients, supported where $\partial\chi \neq 0$. There are no boundary terms on $\partial X \cap \partial X'$: both local parts are restrictions of the same torus operator, and the projections agree there. $\mathbf{G}^{X'}P'[L,\chi]u$ is bounded by $\mathbf{G}^{X'}D^T$, $\mathbf{G}^{X'}D$ and $\|\!|u|\!\|_{X,+}$ (Theorem 19.7(a)).
- $\mathfrak{N}^{X'} - P'\mathfrak{N}^X P'$ is a sum of products in which one scalar factor is replaced by its difference between X and X'. It is of seam form by Theorem 18.2(e), with the derivatives standing outside, where they are absorbed by $\mathbf{G}^{X'}$ and by u.
- $\mathfrak{N}^X$ is a finite sum of terms $X_i^T M_i Y_i$ as in Lemma 17.5, with $X_i, Y_i \in \{1, D, D_X^T\}$ and M_i with decay (proof of Theorem 19.6). Some terms carry a derivative at an end, for instance $-D^T\mathfrak{F}S_X$ and its transpose, so we do not bound the kernel of $[\mathfrak{N}^X,\chi]$ alone. We use
$$[X_i^T M_i Y_i, \chi] = [X_i^T,\chi]M_i Y_i + X_i^T[M_i,\chi]Y_i + X_i^T M_i[Y_i,\chi].$$
The commutators of D and D^T with χ are bounded and supported where $\partial\chi \neq 0$. If M_i has decay (C,γ), then $\|\mathbf{1}_y[M_i,\chi]\mathbf{1}_{y'}\| \le 2Ce^{-\gamma|y-y'|}$, and it vanishes unless y or y' lies within distance $3+\sqrt{d}$ of $X \setminus X'$; so $[M_i,\chi]$ has the seam form. The derivative X_i^T on the left is absorbed by $\mathbf{G}^{X'}$, since $\mathbf{G}^{X'}D^T$ and $\mathbf{G}^{X'}D$ decay (Theorem 19.7(a)). The derivative Y_i on the right falls on u, and $D\mathbf{G}^X$, $D^T\mathbf{G}^X$ decay. With P on the left and Q on the right, as below, the products $P\mathbf{G}^{X'}X_i^T$ and $Y_i\mathbf{G}^X Q$ also decay.
- $m_{X'} - m_X$ is supported in blocks at Euclidean distance at most $c_{\mathrm{seam}}(d)M$ from $X \setminus X'$, inside Σ_M. Indeed their labels are at ∞-distance at most $4M+2$ from a block of $X \setminus X'$, and $\sqrt{d}(4M+2) \le c_{\mathrm{seam}}(d)M$ for $M \ge M_0 \ge 2$.

Composing with the decay of $\mathbf{G}^X$ and $\mathbf{G}^{X'}$ (Theorem 19.7(a)) gives the bound. The identity holds as an identity of torus fields, and these decay bounds hold at all blocks, so the bound holds for all unit points y, y', including the blocks of the initial points of the bonds crossing $\partial X'$. For the derivatives, apply $P \in \{D, D^T, \nabla\}$ to the identity, using Lemma 19.3 for ∇. For $Q \in \{D^T, D\}$ on the right, move the cutoff onto the input as in the proof of Theorem 18.2(e); this produces the commutators $[Q,\chi]$, which are bounded.

An earlier version used the sharp-cut identity of Lemma 17.7. As written, that argument was incomplete: its cut terms contain one-sided derivatives at the cut. With the previous, relative, boundary conditions the cutoff argument also left boundary terms on $\partial X \cap \partial X'$, where only the interior divergence was present.

(b) Apply (a) with $X = \mathbb{T}_\eta$ to U and to U', and note that $\mathbf{G}^{X',\kappa}(U) = \mathbf{G}^{X',\kappa}(U')$. For $\mathbf{\Gamma}_\pi$, $\mathbf{H}$, $\mathbf{C}$ use the telescoping argument of Theorem 18.2(e), together with Lemma 17.7 on the unit lattice for $\mathbf{\Gamma}_\pi$. The factor e^{CM} comes from the width $c_{\mathrm{seam}}(d)M$ of Σ_M. □

This is the property needed to localize the effective actions in the renormalization group. If the background is changed far away, the propagators deep inside change by an exponentially small amount. The proof does not compare covariant derivatives of different backgrounds, so the η^{-1} problem discussed in Section 51 does not arise. It compares each background with a Dirichlet operator that does not see the change.

19.6 Complex perturbations

In the variational problem and in the renormalization group, the fields are expanded around a real regular background U, and the deviation is small in a global gauge relative to U. We therefore consider $U_\delta(b) = U(b)\, e^{i\eta\, \delta A(b)}$, where δA is $\mathfrak{g}^c$-valued with $|\delta A|, |\partial \delta A|, [\partial \delta A]_\alpha \le \delta$, the derivatives being taken in the local gauges of U.

Proposition 19.11. *Let $M \ge M_0$, $\varepsilon \le \varepsilon_0$, let U be (M,ε)-regular, and let $W \in \mathcal{W}^{\mathrm{an}}_{\mathrm{loc}}(\varepsilon)$. There is $\delta_0 \in (0, \delta_W]$ such that for $\delta \le \delta_0$ all operators of Sections 18 and 19 exist at U_δ, depend analytically on δA, and satisfy the same bounds, with coercivity replaced by* Re*-coercivity. The* Re*-coercivity constants and the decay rates do not depend on $\delta \le \delta_0$: by Lemma 13.2 the operators at U_δ satisfy the hypotheses of Lemma 5.7 with the constants $c/2$, $C_1 + 1$ and $\min(\alpha_1, \gamma/2, 1)$, where c, C_1, α_1 are those at U and γ is the rate of the perturbation. Theorem 19.10 holds for pairs U_δ, $U'_{\delta'}$ that coincide on $\hat{X}'$. The L^2 statements (existence,* Re*-coercivity, decay, comparison of domains and locality) also hold, with $\hat{X}$ replaced by $X^{(r_W+1)}$, for potentials of range $r_W \le 3$ as in Section 35 that are real for real U, covariant, of norm at most ε, and extend holomorphically to the U_δ with $W(U_\delta) - W(U)$ of norm at most $C\delta$ and the same range. Here the range bounds the dependence on A and on U. The pointwise statements are not claimed for such potentials.*

Proof. $D_{U_\delta} = D_U + T_\delta$ with T_δ local and of size $C\delta$ in the sense of Lemma 12.2. Similarly, the averages change by local operators of size $C\delta$, and so do W, by the definition of $\mathcal{W}^{\mathrm{an}}_{\mathrm{loc}}$, and $\mathfrak{A}^{\mathrm{ext}}_X$, in the form $\|\!| \cdot |\!\|_{X,+}$. Hence every operator at U_δ is the corresponding operator at U plus an admissible perturbation of size $C\delta$. Lemmas 13.2 and 13.3, with $\mathfrak{A}_0$ the operator at U, give existence and the bounds. For the gauge-field operators on domains, Lemma 5.7 is used in the norm $\|\!| \cdot |\!\|_{X,+}$, as in Theorem 19.7(a). The constants of the statement, and the rate $\gamma_0/4$ of the differences in Lemma 13.3, do not depend on $\delta \le \delta_0$. The lower bounds of Theorem 18.2(c) and Theorem 19.9 persist by perturbation. Analyticity follows as in Proposition 16.3. The bootstrap of Lemma 18.3 applies with ε replaced by $\varepsilon + C\delta$; we take $\varepsilon_0 \le \frac{1}{2}\varepsilon_*$ and $C\delta_0 \le \frac{1}{2}\varepsilon_*$, so that $\varepsilon + C\delta \le \varepsilon_*$. Locality follows as in Theorem 19.10, since the Dirichlet operators on X' at U_δ depend only on U_δ on $\hat{X}'$. For the potentials of range $r_W \le 3$, existence, Re-coercivity and decay use W only through its reality, its norm, its range in A, its covariance and the size of $W(U_\delta) - W(U)$; comparison of domains uses the same properties, and locality uses in addition its range in U, as in Section 35. □

Remark 19.12 (General averaging operators). The proofs of Parts II and III use the averaging pair $(Q'(U), \mathbf{Q}(U))$ only through the following properties, required in every local gauge. Property (v) is used only for the analytic statements.

(i) $Q'Q'^T = I$ for real U, and at real U the maps Q' and $\mathbf{Q}$ take real fields to real fields;

(ii) the closeness and locality bounds of Lemma 12.5(b);

(iii) a covariance identity $\mathbf{Q} D_U = D_V Q' + E$, with a real unit-lattice configuration V and a defect E satisfying the bounds of Lemma 12.5(c), including the bound on E^T;

(iv) gauge covariance as in Lemma 12.5(d), and dependence on U only on the bonds of $B(y)$, respectively $B(y) \cup B(y + e_\mu)$;

(v) the pair extends holomorphically to the complex backgrounds of the analytic statements (in Part II the complex A with (B), in Section 19.6 the U_δ), and (i)–(iii) hold there, uniformly, with the bilinear transpose in (i) and with V replaced by its holomorphic continuation in (iii).

The explicit transports along the contours $\Gamma_{y,x}$ enter only in two places: the trial fields of Theorem 18.2(c) and of Lemma 19.8. For a general pair use the scalar block profiles and the

raw gauge tube profiles from those proofs, constructed in the local gauges. For the scalar profiles the trial-average map is $I + O(\varepsilon)$. For the gauge profiles, let L_0 be the raw tube map and let $T_{\text{model}} = \mathbf{Q}_{\text{model}} L_0$. By (ii), $T_{\text{general}} - T_{\text{model}}$ is a local operator of size $O(\varepsilon)$ in the block norm. The diagonal row and column margin of Lemma 19.8 persists, so its Neumann series gives the exact right inverse $L_0 T_{\text{general}}^{-1}$; then apply the slice correction from that lemma. The resulting exact gauge trial fields have exponential localization, rather than support in a single tube. Hence the real results of Parts II and III hold for every averaging pair with properties (i)–(iv), with constants depending only on upper bounds for the parameters in (i)–(iv), once these are below a fixed threshold. The analytic results (Proposition 16.3(a) and Proposition 19.11) hold if in addition (v) holds.

Properties (i)–(iv) alone do not give analyticity. Let G be non-abelian, and let $h_y(U) \in G$ be close to 1, covariant, depending only on U on the bonds of $B(y)$, and not analytic in U; for instance $h_y(U) = e^{\mathrm{i}\theta |F_y| F_y}$, with F_y the curvature of a fixed plaquette of $B(y)$ transported to y and $\theta > 0$ small. Put $L_y = R(h_y(U))$, let L_1 act by L_y at the initial point y of a unit bond, and replace $V(y,z)$ by $h_y V(y,z) h_z^{-1}$. The pair $(LQ', L_1 \mathbf{Q})$ then has properties (i)–(iv), with the defect $L_1 E$, and the same operator $\mathfrak{A}'$, since L is orthogonal. Its minimizer map is $H' L^{-1}$, which is not analytic.

By Proposition 22.7, Bałaban's linearized averages $(Q'_k(U_0), \mathbf{Q}_k(U_0))$ have properties (i)–(iv) with $E = 0$ and $V = U_0^{(k)}$. Property (v) holds for them at the perturbations U_δ of Section 19.6, for δ small. Holomorphy, $Q' Q'^T = I$ and the closeness bounds are Proposition 46.4(iv)–(vi). The identity (iii) with $E = 0$ holds there by the identity theorem, as in step (iii) of the proof of Proposition 46.7, since for real δA the U_δ are $(M, \varepsilon + C\delta)$-regular. This needs $\varepsilon + C\delta \le \min(\varepsilon_0, c'_3)$. Proposition 46.4 alone does not give $E = 0$ on $\{|A'|_\infty < \delta_{AB}\}$ (Remark 46.6).

20 Bałaban's random-walk expansion

The estimates of Sections 18 and 19 were obtained with Combes–Thomas, IMS localization and comparison of domains. In the renormalization group one also needs an *explicit* representation of the global operators as convergent sums of products of local propagators, each term depending on the background only in a bounded region. This is Bałaban's random-walk expansion [2, 4, 7]. We give it here for the scalar Green function G', with the local propagators of the previous sections. For the gauge-field Green function $\mathbf{G}$ the remainder is kept in factorized form, with the derivatives outside. That walk is constructed in Section 41; part (b) of Theorem 20.2 states what it gives.

20.1 The parametrix

Let $M \ge M_0$ and let U be (M, ε)-regular with $\varepsilon \le \varepsilon_0$. For each j let Λ_j^w be the union of the blocks meeting $\{x : |x - z_j|_\infty \le 9M\}$. If $|z_k - z_j|_\infty \le 2M$, then $\Lambda_k \subset \Lambda_j^w$, so $k \in J(\Lambda_j^w)$. Since the sets K_k with $|z_k - z_j|_\infty \le 2M$ cover K_j, the mass profile $m_{\Lambda_j^w}$ vanishes on K_j. Let $\mathfrak{A}$ be either the scalar operator $\mathfrak{A}'_{\mathbb{T}_\eta}$ or the gauge-field operator $\mathfrak{A}_\pi = \mathfrak{A}^{\mathbb{T}_\eta}$, and define the local propagators

$$G_j^w = \big(\mathfrak{A}'_{\Lambda_j^w}\big)^{-1} \quad \text{(scalar case)}, \qquad G_j^w = \big(\mathfrak{A}^{\Lambda_j^w, \kappa}\big)^{-1} \quad \text{(gauge case)}.$$

They depend on U only on $\hat{\Lambda}_j^w$ and have the bounds of Theorems 18.2(b) and 19.7(a). With the partition of unity $h_j = \chi_j^2$ ($\sum_j h_j = 1$) put

$$K = \sum_j h_j G_j^w, \qquad \mathcal{R}_j = -\Big([\mathfrak{A}, h_j] + h_j \big(\mathfrak{A} \mathbf{1}_{\Lambda_j^w} - \mathbf{1}_{\Lambda_j^w} \mathfrak{A}^{\Lambda_j^w} \mathbf{1}_{\Lambda_j^w} \big) \Big) G_j^w, \qquad \mathcal{R} = \sum_j \mathcal{R}_j.$$

In the scalar case $\mathfrak{A}^{\Lambda_j^w}$ means $\mathfrak{A}'_{\Lambda_j^w}$.

Lemma 20.1. $\mathfrak{A}K = I - \mathcal{R}$.

Proof. $\mathfrak{A}h_j G_j^w = h_j \mathfrak{A} G_j^w + [\mathfrak{A}, h_j] G_j^w$. Since h_j is supported in $K_j \subset \Lambda_j^w$, $h_j m_{\Lambda_j^w} = 0$ and $h_j \mathfrak{A}^{\mathrm{ext}}_{\Lambda_j^w} = 0$, we have $h_j \mathfrak{A}^{\Lambda_j^w,\kappa} G_j^w = h_j$. Hence $h_j \mathfrak{A} G_j^w = h_j + h_j(\mathfrak{A}\mathbf{1}_{\Lambda_j^w} - \mathbf{1}_{\Lambda_j^w}\mathfrak{A}^{\Lambda_j^w}\mathbf{1}_{\Lambda_j^w}) G_j^w$. Sum over j. □

Theorem 20.2 (Random-walk expansion)**.** *There is $M_1 \geq M_0$ such that for $M \geq M_1$ and $\varepsilon \leq \varepsilon_0$ the following holds; in (b), M_1 depends also on ρ.*

(a) The scalar Green function. *Let $\mathfrak{A} = \mathfrak{A}'_{\mathbb{T}_\eta}$, so that $\mathfrak{A}^{-1} = G'$ and $\mathcal{R}_j = -[\mathfrak{A}', h_j] G_j^w$, and let $\hat{K}_j = \{x : \mathrm{dist}_\infty(x, K_j) \leq 1\}$. The operator $\mathcal{R}_j$ vanishes unless it acts from Λ_j^w to $\hat{K}_j$, and $\mathcal{R}_j$ and $\mathcal{R}_j D^T$ have decay (CM^{-1}, γ). Hence $\|\mathcal{R}\| + \|\mathcal{R}D^T\| \leq C/M \leq \frac{1}{2}$, and*

$$G' = K \sum_{n \geq 0} \mathcal{R}^n = \sum_\omega G_\omega, \qquad G_\omega = h_{j_0} G_{j_0}^w \mathcal{R}_{j_1} \cdots \mathcal{R}_{j_n},$$

where $\omega = (j_0, \dots, j_n)$ runs over all finite sequences of indices. The series converges in norm, also after applying D on the left and D^T on the right. The term G_ω vanishes unless $|z_{j_{i+1}} - z_{j_i}|_\infty \leq 10M+2$ for all i. It depends on U only on $\bigcup_i \hat{\Lambda}_{j_i}^w$, and $\|G_\omega\| \leq C(C/M)^n$. For fixed j_0 the sum of $\|G_\omega\|$ over all ω converges. The expansion reproduces the exponential decay of G', with a rate of order $M^{-1} \log M$.

(b) The gauge-field Green function. *Let $\rho \geq 15$, so that (41.1) holds with $w = 10M$, let Λ_j^v be the domains of Section 40 for this ρ, and let $n \geq C(\rho + C_v)M$, with C_v the margin of level (L3) of Section 41. Lemma 41.2 with $X = \mathbb{T}_\eta$ gives $\mathfrak{A}_\pi K^v = I - \mathcal{R}^v$, where $K^v = \sum_j h_j (\mathfrak{A}^{\Lambda_j^v,\kappa})^{-1}$ and $\mathcal{R}^v = D^T \mathcal{A} + \mathcal{B}$, with $\mathcal{A}$, $\mathcal{A}D^T$, $\mathcal{B}$ and $\mathcal{B}D^T$ of norm at most C/M (Lemma 41.3 at $s = 1$). Every D^T in front of $\mathcal{A}$ is absorbed by the factor on its left. The series $K^v \sum_n (\mathcal{R}^v)^n$ converges in norm, also with D on the left and D^T on the right, and it equals $\mathbf{G} = \mathfrak{A}_\pi^{-1}$. This is Theorem 41.4(a) at level (L3), with $X = \mathbb{T}_\eta$ and $s = 1$. The scalar ingredients G', Γ' and $(M')^{-1}$ of $\mathfrak{A}_\pi$ are expanded at levels (L1) and (L2). The expansion reproduces the exponential decay of $\mathbf{G}$, with some rate $\gamma > 0$.*

Proof. (a), the commutator. We have $[\mathfrak{A}', h_j] = D^T[D, h_j] + [D^T, h_j]D + a[Q'^T Q', h_j]$. The first two terms form a first-order operator whose coefficients are differences of h_j, bounded by CM^{-1} and supported where $\partial h_j \neq 0$, as in the proof of Theorem 18.2(e). The last term is block-local, with norm at most CM^{-1}, and it reaches every point of a unit block that meets K_j. So $[\mathfrak{A}', h_j]$ maps into fields supported in $\hat{K}_j$. By the decay of G_j^w, DG_j^w, $G_j^w D^T$ and $DG_j^w D^T$ (Theorem 18.2(b)), $\mathcal{R}_j$ and $\mathcal{R}_j D^T$ have decay (CM^{-1}, γ).

(a), the second term. It vanishes: $\mathfrak{A}'$ is local and h_j is supported at distance $\geq 8M - 1$ from $\mathbb{T}_\eta \setminus \Lambda_j^w$, so $h_j \mathfrak{A}' \mathbf{1}_{\Lambda_j^w} = h_j \mathbf{1}_{\Lambda_j^w} \mathfrak{A}' \mathbf{1}_{\Lambda_j^w}$.

(a), summation. Every point lies in at most 3^d sets K_j, so a unit block meets boundedly many sets $\hat{K}_j$. Hence $\mathcal{R}$ and $\mathcal{R}D^T$ have decay (CM^{-1}, γ), and Lemma 5.2(a) gives $\|\mathcal{R}\| + \|\mathcal{R}D^T\| \leq C/M$. Since $\mathfrak{A}'K = I - \mathcal{R}$ with $\|\mathcal{R}\| \leq \frac{1}{2}$, $K(I - \mathcal{R})^{-1}$ is a right inverse of $\mathfrak{A}'$. It equals G', which exists by coercivity. With derivatives, D falls on K and D^T on the last factor. The operators DK, KD^T and DKD^T are bounded, since $[D, h_j]$ is bounded, and $\mathcal{R}D^T$ is bounded. No derivative on the left of $\mathcal{R}$ is needed.

(a), localization. Since $\Lambda_j^w \subset \{x : |x - z_j|_\infty \leq 9M + 1\}$ and $\hat{K}_j \subset \{x : |x - z_j|_\infty \leq M + 1\}$, $G_{j_i}^w \mathcal{R}_{j_{i+1}}$ vanishes unless $|z_{j_{i+1}} - z_{j_i}|_\infty \leq 10M + 2$. The dependence on U is that of the local propagators. Since the grid spacing M' of Lemma 17.3 satisfies $M' \geq 8M/9$ (proof of that lemma), each step has at most 28^d choices, so the sum of $\|G_\omega\| \leq C(C/M)^n$ over the ω with fixed j_0 converges for M large. Finally $\mathbf{1}_y G_\omega \mathbf{1}_{y'} = 0$ unless $n \geq |y - y'|/(C_d M) - 1$. Hence

$\|\mathbf{1}_y G' \mathbf{1}_{y'}\| \le C \sum_{n \ge |y-y'|/(C_d M)-1} (C/M)^n$, which decays like $e^{-c|y-y'|M^{-1}\log M}$. If the side of $\mathbb{T}$ is less than $8M$, the partition is trivial (Section 17): $\Lambda_0^w = \mathbb{T}_\eta$, $K = G'$ and $\mathcal{R} = 0$. Then (a) holds with the single term G', whose decay is Theorem 18.2(b).

(b). For the parametrix of Lemma 20.1 with $\mathfrak{A} = \mathfrak{A}_\pi$, the argument of (a) gives no bound on $\mathcal{R}_j$ uniform in η: $[\mathfrak{A}_\pi, h_j]$ contains $D^T \mathfrak{F} [S, h_j]$, and the coefficients of $\mathfrak{F}$ are only Hölder continuous. This is why the parametrix K^v of Lemma 41.2 and the factorized remainder are used. Section 41 is written in the conventions of Part VIII, with Bałaban's averaging pair and $E = 0$. This is used only in the formula for the non-local part $\mathfrak{N}^X$ at level (L3). For a pair with $E \neq 0$, $\mathbf{Q}\Pi_X = \mathbf{Q} - (D_V Q' + E) S_X = \mathbf{Q} - E S_X$, since $Q' S_X = Q' G'_X R_X D_X^T = 0$ (Theorem 18.2(d)). So $\mathfrak{N}^X$ contains in addition

$$-a(\mathbf{Q}^T E S_X + S_X^T E^T \mathbf{Q}) + a S_X^T E^T E S_X .$$

Since $S_X = G'_X R_X D_X^T$, these are factorized terms with $X_i^T = 1$, $Y_i \in \{1, D_X^T\}$ and middle factors $\mathbf{Q}^T E G'_X R_X$, $D R_X G'_X E^T \mathbf{Q}$ and $D R_X G'_X E^T E G'_X R_X$. They are bounded and compatible by (L1), (L2), Lemma 40.5(a) and the locality of E (Lemma 12.5(c)). So (A3) holds, and the rest of the proof of Theorem 41.4 is unchanged. The inputs of (L1)–(L3) are Theorems 18.2, 19.7(a), 19.10(a) and Lemma 19.3. Their thresholds and constants do not depend on $M \ge M_0$ (Theorem 19.6). Theorem 19.10(b), whose bound contains e^{CM}, is not used. So (b) holds for $\varepsilon \le \varepsilon_0$, with ε_0 independent of M. This threshold is that of Part III for the model pair, whose averaging parameter is $C_{\mathrm{mod}}\varepsilon$ (Section 31). For Bałaban's pair one needs in addition $\varepsilon \le \varepsilon_{\mathrm{B}}$ (Lemma 31.3(c)); by Lemma 31.2 it may be taken at $M = M_0$, so this threshold does not depend on M either (compare Section 41). □

Remark 20.3 (What the expansion adds). For the bounds of Parts I–VIII the random-walk expansion is not needed; in Part IX its weakened form also gives the bounds on the weakened inverses (Theorem 41.4, Corollary 41.6). Decay comes from Combes–Thomas, and locality in the background from Theorem 19.10. What it adds is a representation of G' and of $\mathbf{G}$ by terms that are local in the background, with explicit smallness: $\|G_\omega\| \le C(C/M)^{|\omega|}$ in (a), and, in (b), $(C/M)^n$ for each of the 2^n regrouped products of bounded factors in $K^v(\mathcal{R}^v)^n$. For $W \in \mathcal{W}^{\mathrm{an}}_{\mathrm{loc}}(\varepsilon)$ and averaging pairs with property (v) of Remark 19.12, the terms are analytic in the complex perturbations of Proposition 19.11. Bałaban's cluster expansions of the fluctuation integrals and of the effective actions are built on this representation [7].

In the gauge case a fully local representation needs walk expansions of the scalar ingredients as well, including the averaged inverses Γ' and $(M')^{-1}$. They are constructed in Section 41, levels (L1)–(L3), with bounds on the sum of the norms of all terms. The averaged operators $\mathbf{\Gamma}_\pi$, $\mathbf{H}$ and $\mathbf{C}$ are expanded there at level (L4), in the setting of Part VIII.

The local propagators used here differ from Bałaban's. Bałaban works with Dirichlet operators and proves their positivity from local estimates in the axial gauge [4, Lemma 2.4]. We add a mass term near the boundary of the localization domain instead, and impose Dirichlet conditions on all components of the gauge field (Section 19). Both additions act only near the boundary: the mass term in blocks at Euclidean distance at most $c_{\mathrm{seam}}(d)M$ from the complement of the domain, and the exterior divergence on the bonds that cross its boundary. So the local expressions of the two operators coincide away from the boundary. Their inverses differ in general, also away from the boundary. The parametrix uses the local expressions only on $\operatorname{supp} h_j$, where they agree with those of the operator being inverted (proofs of Lemmas 20.1 and 41.2), and it uses the local propagators only through their own properties (Lemma 20.1, hypothesis (A2) of Section 41), never through a comparison with Bałaban's.

Part IV: Nonlinear averaging, axial gauge and the Landau chart

21 Configurations and the nonlinear averaging

In Parts I–III the averaging was linear. Bałaban's renormalization group uses a nonlinear block averaging of lattice gauge fields [5]. It is defined one scale at a time, with blocks of L^d points, and the k-step averaging is the composition of k one-step averagings. This part develops the nonlinear input of the variational problem:

- the averaging and the regularity of averaged configurations (this section);
- the axial gauge and the *gauge-adapted averaging*, whose linearization is the averaging operator to which Parts II and III apply (Section 22);
- the expansion of the Wilson action (Section 23);
- the axial-to-Landau chart: every real configuration with small curvature and prescribed averages has a representative in Bałaban's Landau gauge relative to a regular reference configuration, with bounds (Section 24).

Part V uses these results to solve the variational problem relative to a regular reference configuration.

Conventions for Parts IV and V. We assume $d \geq 3$, because the pointwise bounds of Parts II and III are used. From now on constants may depend on L, in contrast to Parts I–III. Unless stated otherwise, $C, c > 0$ depend only on d, L, the group $G \subset U(\mathsf{N})$, including the radius c_G of its logarithm chart (defined below), the averaging parameter a of Part I, the Hölder exponent α, and the integer $M \geq M_0$ of Part III, which we fix once and for all; in Part VI we also assume $M \geq 5L$. The letter a also denotes curvature bounds, as in the classes $\mathfrak{U}_j(a)$ of Definition 21.3; there it is unrelated to the averaging parameter. All Lie algebra valued fields are measured in the units of the unit lattice $\mathbb{T}$. A configuration on a lattice of spacing δ is written $X = e^{\mathrm{i}\delta B}$, and B is called its Lie field.

21.1 Scales and contours

For $j = 0, \dots, k$ let $\delta_j = L^j \eta$ and $\mathbb{T}^{(j)} = (\delta_j \mathbb{Z}/n\mathbb{Z})^d$. Thus $\mathbb{T}^{(0)} = \mathbb{T}_\eta$, $\mathbb{T}^{(k)} = \mathbb{T}$, and $\mathbb{T}^{(j)} \subset \mathbb{T}^{(j-1)}$. For $j \geq 1$ and $y \in \mathbb{T}^{(j)}$ the *block*

$$B_j(y) = \{x \in \mathbb{T}^{(j-1)} : \ y_\mu \leq x_\mu < y_\mu + \delta_j, \ \mu = 1, \dots, d\}$$

has L^d points. The points of $\mathbb{T}_\eta$ in the scale-j block of y form $B^{(j)}(y) = B_1(B_2(\cdots B_j(y)))$; in particular $B^{(k)}(y) = B(y)$ is the block of Part I.

For $x \in B_j(y)$ let $\Gamma^j_{y,x}$ be the contour on $\mathbb{T}^{(j-1)}$ from y to x that changes the coordinates in the order $d, d-1, \dots, 1$ [3, (1.7)]. It stays in $B_j(y)$ and has at most $d(L-1)$ bonds. For a bond $c = (y, y + \delta_j e_\mu)$ of $\mathbb{T}^{(j)}$ and $x \in B_j(y)$ put $x(c) = x + \delta_j e_\mu \in B_j(c_+)$. Let $[x, x(c)]$ be the straight contour of L bonds and $[c] = [c_-, c_+]$. The contour

$$\Gamma^j_{c,x} = \Gamma^j_{c_-,x} \cup [x, x(c)] \cup \big(\Gamma^j_{c_+,x(c)}\big)^{-1}$$

runs from c_- to c_+ through x. Its two contour pieces are translates of each other by $\delta_j e_\mu$.

21.2 Configurations, curvature and lattice Stokes

A *configuration* on $\mathbb{T}^{(j)}$ assigns $Y(b) \in G^c$ to the oriented bonds, with $Y(-b) = Y(b)^{-1}$. It is *real* if it is G-valued. Products along contours are written $Y(\Gamma)$. A gauge transformation $g : \mathbb{T}^{(j)} \to G^c$ acts by $Y^g(b) = g(b_-)Y(b)g(b_+)^{-1}$, so that $Y^g(\Gamma) = g(\Gamma_-)Y(\Gamma)g(\Gamma_+)^{-1}$.

For a matrix h with $|h-1| < 1$ let $\ell(h) = -\mathrm{i}\log h$ be the principal logarithm, defined by the power series, so that $e^{\mathrm{i}\ell(h)} = h$. Norms of matrices are Hilbert–Schmidt norms, as in Part II. We fix $c_G > 0$ so small that ℓ maps $\{g \in G : |g-1| \le c_G\}$ into $\mathfrak{g}$, and $\{g \in G^c : |g-1| \le c_G\}$ into $\mathfrak{g}^c$; all group elements below satisfy such a bound. The radius c_G depends on the group $G \subset U(\mathsf{N})$, not only on N. For $m \ge 7$ the circle $G_m = \{\mathrm{diag}(e^{\mathrm{i}t}, e^{\mathrm{i}mt}) : t \in \mathbb{R}\} \subset U(2)$ contains $g = \mathrm{diag}(e^{2\pi\mathrm{i}/m}, 1)$, with $|g-1| = 2\sin(\pi/m) < 1$ and $\ell(g) = \mathrm{diag}(2\pi/m, 0) \notin \mathfrak{g}_m = \{\mathrm{diag}(s, ms) : s \in \mathbb{R}\}$. So every threshold below c_G depends on G, including the radius c_G of its logarithm chart, and not only on N.

Lemma 21.1 (Elementary estimates)**.** *There is C, depending only on N, such that:*

(a) For $|X| \le \frac{1}{2}$: $|e^{\mathrm{i}X} - 1| \le 2|X|$. For $|h-1| \le \frac{1}{2}$: $|\ell(h)| \le 2|h-1|$.

(b) For G-valued $g_1, \dots, g_m$ with $\sum_i |\ell(g_i)| \le c_G$: $|\ell(g_1 \cdots g_m)| \le \sum_i |\ell(g_i)|$. Moreover $\ell(hgh^{-1}) = h\ell(g)h^{-1}$ and $\ell(g^{-1}) = -\ell(g)$.

(c) For $X_1, \dots, X_m \in \mathfrak{g}^c$ with $s = \sum_i |X_i| \le \frac{1}{4}$: $\ell(e^{\mathrm{i}X_1} \cdots e^{\mathrm{i}X_m}) = \sum_i X_i + \beta$, where β is an analytic function of the X_i with $|\beta| \le Cs^2$.

Proof. (a) and (c) follow from the power series of exp and log; (c) is the Baker–Campbell–Hausdorff formula with the elementary bound on its remainder [5, (30)]. For (b), the Hilbert–Schmidt inner product defines a bi-invariant Riemannian metric on $U(\mathsf{N})$, and the distance from 1 to g is $|\ell(g)|$ when this is smaller than π. The triangle inequality and bi-invariance give $\mathrm{dist}(1, g_1 \cdots g_m) \le \sum_i \mathrm{dist}(1, g_i)$. □

A closed contour Γ based at z is *spanned by m plaquettes* if, in the groupoid of contours (with backtracking cancelled), $\Gamma = \prod_{i=1}^{m} \gamma_i(\partial p_i)^{\pm 1}\gamma_i^{-1}$ for plaquettes p_i and contours γ_i from z to the base point of p_i. Then $Y(\Gamma) = \prod_i Y(\gamma_i)Y(\partial p_i)^{\pm 1}Y(\gamma_i)^{-1}$ for every configuration Y.

Lemma 21.2 (Lattice Stokes)**.** *Let Y be real and let Γ be spanned by $p_1, \dots, p_m$ with $\sum_i |\ell(Y(\partial p_i))| \le c_G$. Then $|\ell(Y(\Gamma))| \le \sum_i |\ell(Y(\partial p_i))|$. On the lattice $\mathbb{T}^{(j-1)}$:*

(i) the loop $\Gamma^j_{c,x} \cup [c]^{-1}$ is spanned by at most dL^2 plaquettes;

(ii) for neighbours $x, x' \in B_j(y)$, the loop $\Gamma^j_{y,x} \cup [x, x'] \cup (\Gamma^j_{y,x'})^{-1}$ is spanned by at most dL plaquettes;

(iii) a closed contour of m bonds contained in a box of side s lattice spacings, at most half the side of the torus, is spanned by at most dsm plaquettes.

Proof. The bound follows from Lemma 21.1(b). (i) Since $\Gamma^j_{c_+,x(c)}$ is the translate of $\Gamma^j_{c_-,x}$ by $\delta_j e_\mu$, the loop bounds the surface swept by $\Gamma^j_{c_-,x}$ under this translation. It consists of at most $d(L-1)\cdot L$ plaquettes. (ii) The two contours agree up to the segment in direction ν, where $x' = x + \delta_{j-1}e_\nu$. After it they run parallel at distance δ_{j-1}, through at most $d(L-1)$ further bonds. (iii) Join every vertex of the contour to its first vertex by the coordinate-ordered path inside the box. Each bond of the contour, together with the two paths to its endpoints, bounds a surface of at most ds plaquettes. □

Definition 21.3 (Curvature and regularity classes)**.** For a plaquette $p = p_{\mu\nu}(x)$, $\mu < \nu$, of $\mathbb{T}^{(j)}$ write $Y(\partial p) = Y(b_1)Y(b_2)Y(b_3)^{-1}Y(b_4)^{-1}$ with $b_1 = (x, x+\delta_j e_\mu)$, $b_2 = (x+\delta_j e_\mu, x+\delta_j e_\mu+\delta_j e_\nu)$, $b_3 = (x+\delta_j e_\nu, x+\delta_j e_\mu+\delta_j e_\nu)$, $b_4 = (x, x+\delta_j e_\nu)$. If $|Y(\partial p) - 1| \le c_G$, the *curvature* is

$F_Y(p) = \delta_j^{-2}\ell(Y(\partial p))$. For $a > 0$ let $\mathfrak{U}_j(a)$ be the set of real configurations on $\mathbb{T}^{(j)}$ with $|F_Y(p)| \le a$ for all plaquettes.

The curvature transforms by $F_{Y^g}(p) = g(x)F_Y(p)g(x)^{-1}$, so $\mathfrak{U}_j(a)$ is invariant under real gauge transformations. The bound $|F_Y| \le a$ is Bałaban's condition $|Y(\partial p) - 1| \le a\delta_j^2$ up to a factor $1 + O(a\delta_j^2)$ [6, (1.7)]. In unit-lattice units it says that the field strength is at most a. For fine configurations we add a condition on the current in Section 23.

21.3 The one-step averaging

Definition 21.4 (Bałaban's averaging [5, (42)])**.** For a configuration Y on $\mathbb{T}^{(j-1)}$ define the configuration $\mathfrak{a}_j(Y)$ on $\mathbb{T}^{(j)}$ by

$$\mathfrak{a}_j(Y)(c) = \exp\Big(\mathrm{i}L^{-d}\sum_{x\in B_j(c_-)} \ell\big(Y(\Gamma^j_{c,x})\,Y([c])^{-1}\big)\Big)\,Y([c]),$$

whenever the logarithms are defined. Put $\mathfrak{a}^{(0)} = \mathrm{id}$, $\mathfrak{a}^{(j)} = \mathfrak{a}_j \circ \mathfrak{a}^{(j-1)}$, and $U^{(j)} = \mathfrak{a}^{(j)}(U)$ for configurations U on $\mathbb{T}_\eta$.

Each loop $Y(\Gamma^j_{c,x})Y([c])^{-1}$ is the holonomy of a small loop based at c_-. The average of their logarithms corrects the straight-line transport $Y([c])$.

Lemma 21.5 (Basic properties)**.** *(a) If $Y \in \mathfrak{U}_{j-1}(b)$ with $db\delta_j^2 \le c_G$, then $\mathfrak{a}_j(Y)$ is defined and real, and*

$$|\ell(Y(\Gamma^j_{c,x})Y([c])^{-1})| \le db\delta_j^2.$$

(b) (Covariance.) *For every real gauge transformation g on $\mathbb{T}^{(j-1)}$, $\mathfrak{a}_j(Y^g) = \mathfrak{a}_j(Y)^{g|_{\mathbb{T}^{(j)}}}$.*

(c) (Locality.) *$\mathfrak{a}_j(Y)(c)$ depends only on Y on the bonds of $B_j(c_-) \cup B_j(c_+)$, i.e. bonds with both endpoints in this set.*

Part (b) also holds for G^c-valued g close to 1, whenever all logarithms are defined: $\ell(ghg^{-1}) = g\ell(h)g^{-1}$ holds for such g and h, by analytic continuation from the real case.

Consequently $\mathfrak{a}^{(k)}(U^u) = \mathfrak{a}^{(k)}(U)^{u|_{\mathbb{T}}}$ for all real gauge transformations u on $\mathbb{T}_\eta$, and $U^{(k)}(c)$ depends only on U on the bonds of $B(c_-) \cup B(c_+)$.

Proof. (a) Lemma 21.2(i) gives the bound, since each plaquette contributes at most $b\delta_{j-1}^2$ and $dL^2\delta_{j-1}^2 = d\delta_j^2$. The average of elements of $\mathfrak{g}$ lies in $\mathfrak{g}$. (b) $Y^g(\Gamma^j_{c,x})Y^g([c])^{-1} = g(c_-)Y(\Gamma^j_{c,x})Y([c])^{-1}g(c_-)^{-1}$, and ℓ commutes with conjugation. Hence the exponential is conjugated by $g(c_-)$, and $Y^g([c]) = g(c_-)Y([c])g(c_+)^{-1}$. (c) is clear from the definition. □

The covariance in (b) is with respect to the values of g at the points of the coarse lattice. Exact covariance with respect to an *average* of g cannot be expected. A blockwise rule ϕ with $\phi(1) = 1$ satisfying $\mathfrak{a}(Y^g) = \mathfrak{a}(Y)^{\phi(g)}$ for all g must be a homomorphism from $G^{B_j(y)}$ to G. For simple G, the nontrivial continuous homomorphisms of this kind are, up to automorphisms, evaluations at a single point. This is why the residual gauge group of the variational problem is

$$\mathfrak{G}_0 = \{u : \mathbb{T}_\eta \to G :\ u(y) = 1 \text{ for } y \in \mathbb{T}\},$$

and not the group of transformations with vanishing block averages. Section 22 shows how Bałaban nevertheless arrives at an averaging whose linearization is covariant with respect to averaged gauge transformations.

21.4 Regularity of averaged configurations

Lemma 21.6 (One step)**.** *There are $c_1, C_1 > 0$ such that, if $Y \in \mathfrak{U}_{j-1}(b)$ with $b\delta_j^2 \le c_1$, then $\mathfrak{a}_j(Y) \in \mathfrak{U}_j(b(1 + C_1 b\delta_j^2))$.*

Proof. Write $\delta = \delta_{j-1}$, $\delta' = \delta_j$. Fix a plaquette p' of $\mathbb{T}^{(j)}$ in the (μ,ν)-plane with corners y, $y_1 = y + \delta' e_\mu$, $y_2 = y_1 + \delta' e_\nu$, $y_3 = y + \delta' e_\nu$. The four averaged bonds of $\partial p'$ depend only on Y in the box $R = B_j(\{y, y_1, y_2, y_3\})$. By Lemma 21.5(b) and the gauge invariance of $|F|$ we may replace Y by $\tilde{Y} = Y^v$, where v is a real gauge transformation on R with $\tilde{Y}(\Gamma'_{y,x}) = 1$ for all $x \in R$. Here $\Gamma'_{y,x}$ is the coordinate-ordered contour from y to x inside R.

Size of $\tilde{Y}$. For a bond $b = (x, x')$ of R, the loop $\Gamma'_{y,x} \cup b \cup (\Gamma'_{y,x'})^{-1}$ is spanned by at most $(d+2)L$ plaquettes, as in Lemma 21.2(ii). Its holonomy is $\tilde{Y}(b)$. Hence $\tilde{Y}(b) = e^{\mathrm{i}\delta A(b)}$ with $|\delta A(b)| \le (d+2) b\delta\delta'$.

Averages. Every contour used for $\partial p'$ has at most $3dL$ bonds in R, so the sum of $|\delta A|$ along it is at most $\sigma = 3d(d+2)b\delta'^2 \le \frac{1}{4}$. By Lemma 21.1(c), $\ell(\tilde{Y}(\gamma)) = \delta A(\gamma) + O(\sigma^2)$ for each such contour, where $A(\gamma) = \sum_{b\in\gamma} \pm A(b)$ is the signed sum. Therefore

$$\mathfrak{a}_j(\tilde{Y})(c) = \exp\Big(\mathrm{i}\delta L^{-d} \sum_x \big(A(\Gamma^j_{c,x}) - A([c])\big) + O(\sigma^2)\Big) e^{\mathrm{i}\delta A([c]) + O(\sigma^2)} = e^{\mathrm{i}\delta\, \overline{A(\Gamma_{c,\cdot})} + O(\sigma^2)},$$

with $\overline{A(\Gamma_{c,\cdot})} = L^{-d} \sum_{x \in B_j(c_-)} A(\Gamma^j_{c,x})$. Multiplying the four averaged bonds of $\partial p'$, again by Lemma 21.1(c),

$$\ell(\mathfrak{a}_j(\tilde{Y})(\partial p')) = \delta L^{-d} \sum_{x \in B_j(y)} \Big(A(\Gamma^j_{c_1,x}) + A(\Gamma^j_{c_2,x+\delta' e_\mu}) - A(\Gamma^j_{c_3,x+\delta' e_\nu}) - A(\Gamma^j_{c_4,x}) \Big) + O(\sigma^2),$$

where $c_1 = (y, y_1)$, $c_2 = (y_1, y_2)$, $c_3 = (y_3, y_2)$, $c_4 = (y, y_3)$. We used the translation invariance of the contours: $B_j(y_1) = B_j(y) + \delta' e_\mu$ and $\Gamma^j_{y_1, x+\delta' e_\mu} = \Gamma^j_{y,x} + \delta' e_\mu$. In the bracket the pieces $\Gamma^j_{y_i,\cdot}$ cancel in pairs. What remains is $A(\partial S_x)$, where S_x is the square of side δ' with corner x in the (μ,ν)-plane. By the abelian Stokes formula $A(\partial S_x) = \sum_{p \subset S_x} A(\partial p)$, a sum over L^2 plaquettes. By Lemma 21.1(c) again, $\delta A(\partial p) = \ell(\tilde{Y}(\partial p)) + O((\delta\delta' b)^2)$, and $|\ell(\tilde{Y}(\partial p))| \le b\delta^2$. Altogether

$$|\ell(\mathfrak{a}_j(\tilde{Y})(\partial p'))| \le L^2 b\delta^2 + CL^2 (b\delta\delta')^2 + C\sigma^2 \le b\delta'^2 (1 + C_1 b\delta'^2). \qquad \square$$

Proposition 21.7 (Regularity of averages [5, Prop. 2])**.** *Possibly after decreasing c_1, so that $2c_1$ satisfies the hypothesis of Lemma 21.6, for $U \in \mathfrak{U}_0(a)$ with $a \le c_1$ and all $j \le k$, $U^{(j)} \in \mathfrak{U}_j(a(1 + Ca)) \subset \mathfrak{U}_j(2a)$.*

Proof. Let $b_0 = a$ and $b_j = b_{j-1}(1 + C_1 b_{j-1}\delta_j^2)$. As long as $b_i \le 2a$ for $i < j$, Lemma 21.6 applies, and

$$b_j \le a \exp\Big(2C_1 a \sum_{i \le j} \delta_i^2\Big) \le a\, e^{3C_1 a},$$

because $\sum_{i\le k} \delta_i^2 \le \sum_{m \ge 0} L^{-2m} \le \frac{4}{3}$. For a small this is at most $a(1 + Ca) \le 2a$, which closes the induction. $\square$

The proposition is local: the bound for a plaquette of $\mathbb{T}^{(j)}$ only uses the curvature of U on the scale-j blocks containing its corners. We also need the size of averaged configurations in a gauge in which U is small.

Lemma 21.8 (Averages in a local gauge)**.** *Let $U \in \mathfrak{U}_0(a)$ with $a \le c_1$. Let Λ be a union of blocks $B(y)$, and suppose $U = e^{\mathrm{i}\eta A}$ on the bonds of Λ with $|A| \le s$, where $s + 4da \le \frac{1}{2} c_G$. Then for every j and every bond c of $\mathbb{T}^{(j)}$ with $B^{(j)}(c_-) \cup B^{(j)}(c_+) \subset \Lambda$ we have $U^{(j)}(c) = e^{\mathrm{i}\delta_j A_j(c)}$ with $|A_j(c)| \le s + 4da$.*

Proof. By Lemma 21.5(a) and Proposition 21.7, the exponent in Definition 21.4 is at most $2da\delta_j^2$. By Lemma 21.1(b), $|\ell(U^{(j)}(c))| \le 2da\delta_j^2 + \sum_{b\in[c]} |\ell(U^{(j-1)}(b))|$. The straight contour $[c]$ has L bonds. Hence $\max|A_j| \le \max|A_{j-1}| + 2da\delta_j$, and $\sum_j \delta_j \le 2$. □

22 The axial gauge and the gauge-adapted averaging

The linear theory of Parts II and III is formulated in Bałaban's Landau gauge, with the residual gauge transformations $\mathcal{N}(Q'(U))$, i.e. those with vanishing *averages*. The nonlinear averaging of Section 21 is covariant with respect to the *values* of gauge transformations at the points of $\mathbb{T}$. Bałaban bridges the two by an intermediate gauge, the axial gauge, and by a modified averaging [5, Sect. C]. This section develops that construction in a form adapted to our needs. The main results are the correspondence theorem (Theorem 22.4), the expansion of the gauge-adapted averaging (Proposition 22.6), and the properties of its linearization (Proposition 22.7). The last one shows that Parts II and III apply to Bałaban's averaging.

22.1 The reference configuration and relative configurations

Fix a real configuration $U_0 \in \mathfrak{U}_0(a)$ on $\mathbb{T}_\eta$ with $a \le c_1$, and put $W_j = U_0^{(j)}$. By Proposition 21.7, $W_j \in \mathfrak{U}_j(2a)$. A *relative configuration* at scale j is a configuration X on $\mathbb{T}^{(j)}$ close to 1; it represents the configuration XW_j (bondwise product). We use the following notation.

- *Relative transports.* $X_W(\Gamma) = (XW_j)(\Gamma)\, W_j(\Gamma)^{-1}$. For $\Gamma = b_1 \cdots b_m$ this is $\prod_i R(W_j(\Gamma_{<i}))X(b_i)$, with $\Gamma_{<i} = b_1 \cdots b_{i-1}$ and $R(Y)Z = YZY^{-1}$.
- *Relative gauge action.* $X^{(g)} = (XW_j)^g W_j^{-1}$, i.e. $X^{(g)}(b) = g(b_-)X(b)\, R(W_j(b))\big(g(b_+)^{-1}\big)$. Then $(X^{(g)})^{(h)} = X^{(hg)}$, and $X_W^{(g)}(\Gamma) = g(\Gamma_-)\, X_W(\Gamma)\, R(W_j(\Gamma))\big(g(\Gamma_+)^{-1}\big)$. The configurations $1^{(g)}$ are the *pure gauges*.
- *Relative averages.* For X at scale $j-1$ put $\overline{X} = \mathfrak{a}_j(XW_{j-1})W_j^{-1}$. By Lemma 21.5(b), $\overline{X^{(g)}} = \overline{X}^{(g|_{\mathbb{T}^{(j)}})}$ for real g. For a fine relative configuration U' put $U'^{[j]} = (U'U_0)^{(j)}W_j^{-1}$, the iterated relative average.
- *Lie fields.* We write $X = e^{\mathrm{i}\delta_j B}$, and $D_{W_j}\lambda(b) = \delta_j^{-1}\big(R(W_j(b))\lambda(b_+) - \lambda(b_-)\big)$. At $j=0$ this is the covariant derivative D_{U_0} of Part II; at $j=k$ it is the unit-lattice derivative D_V with $V = W_k$. To first order, $(e^{\mathrm{i}\delta_j B})^{(e^{\mathrm{i}\lambda})} = e^{\mathrm{i}\delta_j(B - D_{W_j}\lambda)}$. On reversed bonds the relative configuration is $X(-b) = R(W_j(b))^{-1}(X(b)^{-1})$, since $(XW_j)(-b) = (XW_j)(b)^{-1}$; correspondingly $B(-b) = -R(W_j(b))^{-1}B(b)$. Only positively oriented bonds carry independent variables.

22.2 The axial gauge

Definition 22.1 (Axial gauge [6, (1.19)])**.** A relative configuration X at scale $j-1$ is *axial at scale* j if $X_W(\Gamma^j_{y,x}) = 1$ for all $y \in \mathbb{T}^{(j)}$ and $x \in B_j(y)$. A fine relative configuration U' belongs to Ax_j if $U'^{[i-1]}$ is axial at scale i for $i = 1, \dots, j$.

For h on $\mathbb{T}^{(j)}$ let $\mathcal{E}_j(h)$ be the function on $\mathbb{T}^{(j-1)}$ given by $\mathcal{E}_j(h)(x) = R(W_{j-1}(\Gamma^j_{y,x}))^{-1}h(y)$ for $x \in B_j(y)$, and put $\mathcal{E}^{(j)} = \mathcal{E}_1 \circ \cdots \circ \mathcal{E}_j$. This is the extension of h from $\mathbb{T}^{(j)}$ to $\mathbb{T}_\eta$ by parallel transport with the background along the trees. It is multiplicative, $\mathcal{E}^{(j)}(hh') = \mathcal{E}^{(j)}(h)\mathcal{E}^{(j)}(h')$, and it does not change values at the points of $\mathbb{T}^{(j)}$.

Lemma 22.2 (Axial transformations)**.** *(a) Let X be a relative configuration at scale $j-1$ and h a function on $\mathbb{T}^{(j)}$. There is exactly one g on $\mathbb{T}^{(j-1)}$ with $g|_{\mathbb{T}^{(j)}} = h$ and $X^{(g)}$ axial*

at scale j, namely

$$g(x) = R(W_{j-1}(\Gamma^j_{y,x}))^{-1}\big(h(y)\,X_W(\Gamma^j_{y,x})\big), \qquad x \in B_j(y).$$

If X is axial at scale j, this is $g = \mathcal{E}_j(h)$.

(b) Let U' be a real fine relative configuration whose iterated relative averages $U'^{[i]}$, $i < j$, are defined, and h a real function on $\mathbb{T}^{(j)}$. There is exactly one real g on $\mathbb{T}_\eta$ with $g|_{\mathbb{T}^{(j)}} = h$ and $U'^{(g)} \in \mathrm{Ax}_j$. If $U' \in \mathrm{Ax}_j$, this is $g = \mathcal{E}^{(j)}(h)$.

(c) Every orbit of $\mathfrak{G}_0$ in $\mathfrak{U}_0(c_1)$ contains exactly one U with $UU_0^{-1} \in \mathrm{Ax}_k$.

Proof. (a) By the transformation law of relative transports, $X_W^{(g)}(\Gamma^j_{y,x}) = 1$ iff

$$R(W_{j-1}(\Gamma^j_{y,x}))g(x) = g(y)X_W(\Gamma^j_{y,x}).$$

(b) Let g be real. Conjugation by G-valued elements preserves the domain of ℓ, so the relative averages of $U'^{(g)}$ are defined and covariant, $(U'^{(g)})^{[i-1]} = (U'^{[i-1]})^{(g|_{\mathbb{T}^{(i-1)}})}$ (Lemma 21.5(b)). By (a), its axiality at scale i determines $g|_{\mathbb{T}^{(i-1)}}$ uniquely from $g|_{\mathbb{T}^{(i)}}$, and the formula of (a) shows that it is real. Proceed from $i = j$ down to $i = 1$. (c) is (b) with $j = k$, $h = 1$ and $U' = UU_0^{-1}$, whose iterated relative averages are defined by Proposition 21.7. □

22.3 The gauge-adapted averaging and the correspondence theorem

For a relative configuration X at scale $j-1$ let

$$\tau_X(y) = \exp\Big(\mathrm{i}L^{-d}\sum_{x\in B_j(y)} \ell(X_W(\Gamma^j_{y,x}))\Big), \qquad y \in \mathbb{T}^{(j)},$$

be the averaged contour variable. Let g_X be the transformation of Lemma 22.2(a) with $h = \tau_X^{-1}$.

Definition 22.3 (Gauge-adapted averaging)**.** The *modified one-step average* is $\tilde{\mathfrak{a}}_j(X) = \overline{X^{(g_X)}} = \overline{X}^{(\tau_X^{-1})}$, i.e.

$$\tilde{\mathfrak{a}}_j(X)(c) = \tau_X(c_-)^{-1}\,\overline{X}(c)\,R(W_j(c))\big(\tau_X(c_+)\big).$$

The *gauge-adapted averaging* is $\mathcal{Q}_0 = \mathrm{id}$, $\mathcal{Q}_j = \tilde{\mathfrak{a}}_j \circ \mathcal{Q}_{j-1}$, and $\mathcal{Q} = \mathcal{Q}_k$. For a fine relative configuration U_1 put $X_i = \mathcal{Q}_i(U_1)$ and define the *correspondence transformation*

$$\mathsf{w}_0(U_1) = 1, \qquad \mathsf{w}_j(U_1) = \mathcal{E}^{(j-1)}(g_{X_{j-1}})\,\mathsf{w}_{j-1}(U_1).$$

Finally let $\mathfrak{m}_j(g)(y) = g(y)\exp\big(\mathrm{i}L^{-d}\sum_{x\in B_j(y)}\ell(g(y)^{-1}R(W_{j-1}(\Gamma^j_{y,x}))g(x))\big)$ and $\mathfrak{M}^{(j)} = \mathfrak{m}_j \circ \cdots \circ \mathfrak{m}_1$; these are Bałaban's averages of gauge transformations [5, (78)–(80)].

This is Bałaban's construction [5, (58)–(63), (88)–(92)]: the modified average first moves X to the axial gauge at scale j, with a normalization chosen at the coarse points, and then averages. All maps are defined when the configurations involved are sufficiently close to 1; precise domains are given in Lemma 22.5.

Theorem 22.4 (Correspondence)**.** *Let U_1 be a real fine relative configuration for which the maps above are defined. In (b), (e) and (f) the functions h and g are real.*

(a) $U_1^{(\mathsf{w}_j(U_1))} \in \mathrm{Ax}_j$ and $(U_1^{(\mathsf{w}_j(U_1))})^{[j]} = \mathcal{Q}_j(U_1)$.

(b) The transformations g with $U_1^{(g)} \in \mathrm{Ax}_j$ are exactly $g = \mathcal{E}^{(j)}(h)\,\mathsf{w}_j(U_1)$, with h a function on $\mathbb{T}^{(j)}$.

(c) At the points $y \in \mathbb{T}^{(j)}$: $\mathsf{w}_j(U_1)(y) = \tau_{X_{j-1}}(y)^{-1}\tau_{X_{j-2}}(y)^{-1}\cdots\tau_{X_0}(y)^{-1}$.

(d) If X_{j-1} is axial at scale j, then $\mathsf{w}_j(U_1) = \mathsf{w}_{j-1}(U_1)$ *and* $\mathcal{Q}_j(U_1) = \overline{X_{j-1}}$.

(e) (Pure gauges.) *If the maps are defined at $1^{(g)}$, then* $\mathcal{Q}_j(1^{(g)}) = 1^{(\mathfrak{M}^{(j)}(g))}$ *and* $\mathsf{w}_j(1^{(g)}) = \mathcal{E}^{(j)}(\mathfrak{M}^{(j)}(g))\, g^{-1}$.

(f) (Normalization.) $\mathfrak{m}_j(\mathcal{E}_j(h)g) = h\,\mathfrak{m}_j(g)$, *and* $\mathfrak{M}^{(j)}(\mathsf{w}_j(U_1)) = 1$.

Proof. (a), (b) By induction on j; the case $j = 0$ is trivial. All configurations and transformations below are real, so the transformations preserve the domain of ℓ, and the relative averages are covariant. Let $U'' = U_1^{(\mathsf{w}_{j-1})} \in \mathrm{Ax}_{j-1}$, so that $U''^{[j-1]} = X_{j-1} =: X$. By Lemma 22.2(b), the transformations preserving Ax_{j-1} for U'' are $\mathcal{E}^{(j-1)}(h')$ with h' on $\mathbb{T}^{(j-1)}$, and $(U''^{(\mathcal{E}^{(j-1)}(h'))})^{[j-1]} = X^{(h')}$. Hence $U_1^{(g)} \in \mathrm{Ax}_j$ iff $g = \mathcal{E}^{(j-1)}(h')\mathsf{w}_{j-1}$ with $X^{(h')}$ axial at scale j. By Lemma 22.2(a), the latter h' are exactly $\mathcal{E}_j(h'')\,g_X$, h'' on $\mathbb{T}^{(j)}$, because $X^{(g_X)}$ is axial and the transformations preserving its axiality are the $\mathcal{E}_j(h'')$. By multiplicativity $g = \mathcal{E}^{(j)}(h'')\,\mathcal{E}^{(j-1)}(g_X)\,\mathsf{w}_{j-1} = \mathcal{E}^{(j)}(h'')\,\mathsf{w}_j$. For $g = \mathsf{w}_j$,

$$(U_1^{(\mathsf{w}_j)})^{[j]} = \overline{(U''^{(\mathcal{E}^{(j-1)}(g_X))})^{[j-1]}} = \overline{X^{(g_X)}} = \tilde{\mathfrak{a}}_j(X) = \mathcal{Q}_j(U_1).$$

(c) $\mathcal{E}^{(i)}$ does not change values at points of $\mathbb{T}^{(i)} \supset \mathbb{T}^{(j)}$, and $g_{X_i} = \tau_{X_i}^{-1}$ on $\mathbb{T}^{(i+1)}$. (d) If X is axial, then $\tau_X = 1$ and $g_X = \mathcal{E}_j(1) = 1$.

(e) One step: let $X = 1^{(g)}$ at scale $j-1$. Then $X_W(\Gamma_{y,x}) = g(y)R(W(\Gamma_{y,x}))(g(x))^{-1}$. Since 1 is axial and $X^{(g_X)} = 1^{(g_X g)}$, Lemma 22.2(a) gives $g_X g = \mathcal{E}_j(h)$ with $h = (g_X g)|_{\mathbb{T}^{(j)}} = \tau_X^{-1} g$. Using $\ell(k^{-1}) = -\ell(k)$ and $\ell(RkR^{-1}) = R\ell(k)R^{-1}$,

$$\begin{aligned} h(y) &= \exp\Big(-\mathrm{i}L^{-d}\sum_x \ell\big(g(y)R(W(\Gamma_{y,x}))g(x)^{-1}\big)\Big)g(y) \\ &= g(y)\exp\Big(\mathrm{i}L^{-d}\sum_x \ell\big(g(y)^{-1}R(W(\Gamma_{y,x}))g(x)\big)\Big) = \mathfrak{m}_j(g)(y). \end{aligned}$$

Hence $\tilde{\mathfrak{a}}_j(1^{(g)}) = \overline{1^{(\mathcal{E}_j(h))}} = 1^{(h)}$ and $g_X = \mathcal{E}_j(\mathfrak{m}_j(g))g^{-1}$. Iterating, with $X_{j-1} = 1^{(\mathfrak{M}^{(j-1)}(g))}$,

$$\mathsf{w}_j(1^{(g)}) = \mathcal{E}^{(j-1)}(\mathcal{E}_j(\mathfrak{M}^{(j)}g)(\mathfrak{M}^{(j-1)}g)^{-1})\,\mathcal{E}^{(j-1)}(\mathfrak{M}^{(j-1)}g)\,g^{-1} = \mathcal{E}^{(j)}(\mathfrak{M}^{(j)}g)\,g^{-1}.$$

(f) For $g' = \mathcal{E}_j(h)g$, $g'(y)^{-1}R(W(\Gamma_{y,x}))g'(x) = g(y)^{-1}h(y)^{-1}h(y)R(W(\Gamma_{y,x}))g(x)$, which gives the first identity. For the second, by induction and the first identity, $\mathfrak{M}^{(j)}(\mathsf{w}_j) = \mathfrak{m}_j(g_X\,\mathfrak{M}^{(j-1)}(\mathsf{w}_{j-1})) = \mathfrak{m}_j(g_X)$. By the defining relation of g_X, $g_X(y)^{-1}R(W(\Gamma_{y,x}))g_X(x) = X_W(\Gamma_{y,x})$, so $\mathfrak{m}_j(g_X) = \tau_X^{-1}\tau_X = 1$. □

Part (f) shows that $\mathsf{w}_k(U_1)$ is the transformation singled out in [6, (1.29)] by Bałaban's normalization $\mathfrak{M}^{(k)}(g) = 1$. We never need that normalization explicitly: the transformation is given by the recursion, and its values at the coarse points by (c). Parts (c) and (d) use only the definitions and Lemma 22.2(a), so they hold for complex U_1 as well, whenever the maps are defined at U_1.

22.4 Expansion of the modified average

We now expand the modified average around the trivial relative configuration. At scale j write $\delta = \delta_{j-1}$, $\delta' = \delta_j$, $W = W_{j-1}$. For a relative Lie field B at scale $j-1$ and a contour $\gamma = b_1 \cdots b_m$ on $\mathbb{T}^{(j-1)}$ let

$$B^W(\gamma) = \sum_i R(W(\gamma_{<i}))B(b_i),$$

so that $\ell((e^{\mathrm{i}\delta B})_W(\gamma)) = \delta B^W(\gamma) + O((\delta\sum_i |B(b_i)|)^2)$ by Lemma 21.1(c). Define the *transported tube average*

$$(q_j B)(c) = L^{-d} \sum_{x \in B_j(c_-)} L^{-1} R(W(\Gamma^j_{c_-,x})) B^W([x, x(c)])$$

[5, (125)] and the *contour average* $(\gamma_j B)(y) = L^{-d}\sum_{x\in B_j(y)} L^{-1} B^W(\Gamma^j_{y,x})$.

Lemma 22.5 (One-step expansion)**.** *There are $c_2, C > 0$ such that the following holds for $a \le c_2$ and complex B with $|B| \le c_2/\delta'$ on the bonds of $B_j(c_-) \cup B_j(c_+)$ (respectively $B_j(y)$). Put $X = e^{\mathrm{i}\delta B}$. Then*

$$\tilde{\mathfrak{a}}_j(X)(c) = e^{\mathrm{i}\delta'(l_j B + c_j(B))(c)}, \qquad \tau_X(y) = e^{\mathrm{i}\delta'(\gamma_j B + c_j^\tau(B))(y)},$$

where $l_j = q_j + r_j$ is linear and c_j, c_j^τ are analytic, with:

(a) $(l_j B)(c)$ and $c_j(B)(c)$ depend only on B on the bonds of $B_j(c_-) \cup B_j(c_+)$; $\gamma_j B(y)$ and $c_j^\tau(B)(y)$ only on B on the bonds of $B_j(y)$;

(b) $|(q_j B)(c)|^2 \le L^{-d}\sum_x L^{-1}\sum_{b\in[x,x(c)]} |B(b)|^2$, $|(r_j B)(c)| \le Ca\delta'^2 \max|B|$, and $|\gamma_j B| \le d\max|B|$;

(c) $|c_j(B)| + |c_j^\tau(B)| \le C\delta' \max|B|^2$ and $|c_j(B) - c_j(\tilde B)| \le C\delta'(\max|B| + \max|\tilde B|)\max|B - \tilde B|$ for $\max|B|, \max|\tilde B| \le c_2/(2\delta')$, the maxima being over the relevant bonds;

(d) if $W = 1$ on the relevant bonds, then $r_j = 0$ and q_j is the tube average of Part I at scale j;

(e) real B gives $\mathfrak{g}$-valued $l_j B$, $c_j(B)$, $\gamma_j B$, $c_j^\tau(B)$.

Proof. Analyticity. The contours involved have at most $3dL$ bonds, so $\delta\sum|B| \le 3dc_2$. Since W is real, $R(W(\cdot))$ is unitary on $\mathfrak{g}^c$, and all relative transports are products of at most $3dL$ factors $e^{\mathrm{i}\delta R(W(\cdot))B(b)}$. By Lemma 21.1 and Lemma 21.5(a) the arguments of all logarithms in τ_X, $\overline{X}$ and $\tilde{\mathfrak{a}}_j(X)$ are within $C(c_2 + a\delta'^2)$ of 1. Hence, for c_2 small, these are analytic functions of the finitely many variables $B(b)$, bounded by C in the sense that $|\ell(\tilde{\mathfrak{a}}_j(X)(c))| \le C(c_2 + a\delta'^2)$.

Remainders. Let $f(B) = \delta'^{-1}\ell(\tilde{\mathfrak{a}}_j(e^{\mathrm{i}\delta B})(c)) - (l_j B)(c)$, with l_j the derivative at $B = 0$. Then $f(0) = 0$ and $df(0) = 0$, and $t \mapsto f(tB)$ is analytic for $|t| \le c_2/(\delta'\max|B|)$ with $|f| \le C/\delta'$ there. Cauchy's estimate for the second Taylor coefficient gives $|f(B)| \le C\delta'\max|B|^2$ for $\max|B| \le c_2/(2\delta')$. The Lipschitz bound follows from the Cauchy estimate for df. The same argument applies to τ_X.

The linear part. Let $X = e^{\mathrm{i}s\delta B}$ and expand to first order in s. Write $h_x = W(\Gamma^j_{c,x})W([c])^{-1}$, $H_x = \ell(h_x)$ and $E = L^{-d}\sum_x H_x$, so that $|H_x|, |E| \le 2da\delta'^2$ and $W_j(c) = e^{\mathrm{i}E}W([c])$. The loops in Definition 21.4 are

$$(XW)(\Gamma^j_{c,x})(XW)([c])^{-1} = X_W(\Gamma^j_{c,x})\, R(h_x)(X_W([c])^{-1})\, h_x.$$

Let $\mathfrak{f}(\operatorname{ad} H)$ denote the derivative of $Z \mapsto \ell(e^{\mathrm{i}Z}e^{\mathrm{i}H})$ at $Z = 0$, and $\mathfrak{e}(\operatorname{ad} E)$ that of $Z \mapsto \ell(e^{\mathrm{i}(E+Z)}e^{-\mathrm{i}E})$; both equal 1 at 0. The function $\mathfrak{e}$ is entire. The function $\mathfrak{f}$, which is $z \mapsto \mathrm{i}z/(e^{\mathrm{i}z} - 1)$, is analytic for $|z| < 2\pi$ and has poles at the nonzero multiples of 2π; it is used only for $\|\operatorname{ad} H\| \le 2|H|$ small. Then $\ell(\overline{X}(c)) = s\delta[\mathfrak{e}(\operatorname{ad} E)\Xi + R(e^{\mathrm{i}E})B^W([c])] + O(s^2)$, with $\Xi = L^{-d}\sum_x \mathfrak{f}(\operatorname{ad} H_x)(B^W(\Gamma^j_{c,x}) - R(h_x)B^W([c]))$. Moreover $\ell(\tau_X(y)) = s\delta L^{-d}\sum_x B^W(\Gamma^j_{y,x}) + O(s^2)$. Since $\delta/\delta' = L^{-1}$,

$$l_j B(c) = L^{-1}\Big\{\mathfrak{e}(\operatorname{ad} E)\Xi + R(e^{\mathrm{i}E})B^W([c]) - L^{-d}\sum_x B^W(\Gamma^j_{c_-,x}) + R(W_j(c))L^{-d}\sum_{x'\in B_j(c_+)} B^W(\Gamma^j_{c_+,x'})\Big\}.$$

Decompose

$$B^W(\Gamma^j_{c,x}) = B^W(\Gamma^j_{c_-,x}) + R(W(\Gamma^j_{c_-,x}))B^W([x,x(c)]) - R(W(\Gamma^j_{c,x}))B^W(\Gamma^j_{c_+,x(c)}).$$

The last term uses $B^W(\gamma^{-1}) = -R(W(\gamma))^{-1}B^W(\gamma)$. If all factors $\mathfrak{e},\mathfrak{f}$, $R(h_x)$, $R(e^{\mathrm{i}E})$ were 1 and $R(W(\Gamma^j_{c,x})) = R(W_j(c))$, the contour sums would cancel exactly, by the substitution $x' = x(c)$, and only q_jB would remain. The actual differences are controlled by $|\mathfrak{f}(\operatorname{ad} H_x) - 1| \le C|H_x|$, $|\mathfrak{e}(\operatorname{ad} E) - 1| \le C|E|$, $|R(e^{\mathrm{i}E}) - R(h_x)| \le C(|E| + |H_x|)$, and $|R(W(\Gamma^j_{c,x})) - R(W_j(c))| = |R(h_x e^{-\mathrm{i}E}) - 1| \le C(|E| + |H_x|)$. Each of them multiplies a contour sum bounded by $3dL\max|B|$. This gives the bound on $r_j = l_j - q_j$. The bound on q_j is the Cauchy–Schwarz inequality, since $R(W(\cdot))$ is unitary, and the bound on γ_j is clear. (d) and (e) follow from the formulas. □

22.5 Expansion of the gauge-adapted averaging

For a fine field h and a set $S \subset \mathbb{T}_\eta$ let $\|h\|_S$ be the ℓ^2 norm over the bonds starting in S, as in Part III. For a bond c of $\mathbb{T}$ put $S_c = B(c_-) \cup B(c_+)$.

Proposition 22.6 (Expansion of $\mathcal{Q}$)**.** *There are $c_3, C > 0$ such that for $a \le c_3$ and complex fine Lie fields A with $|A|_\infty \le c_3$ the following holds. For $j \le k$, $\mathcal{Q}_j(e^{\mathrm{i}\eta A}) = e^{\mathrm{i}\delta_j B_j}$ with*

$$B_j = \mathbf{Q}_j A + C_j(A), \qquad \mathbf{Q}_j = l_j \cdots l_1,$$

where C_j is analytic, real for real A, and:

(a) $|B_j|_\infty \le 2|A|_\infty$, and $B_j(c)$ depends only on A on the bonds of $B^{(j)}(c_-) \cup B^{(j)}(c_+)$.

(b) $|C_j(A)|_\infty \le C|A|^2_\infty$ and $|C_j(A) - C_j(\tilde A)|_\infty \le C(|A|_\infty + |\tilde A|_\infty)|A - \tilde A|_\infty$.

(c) (Kernels.) *For $j = k$ the linear maps $\mathbf{Q}_k$ and $dC_k(A)$ have kernels, $\mathbf{Q}_k h(c) = \eta^d \sum_b K(c,b)h(b)$ and $dC_k(A)h(c) = \eta^d \sum_b K_A(c,b)h(b)$, supported on bonds b starting in S_c, with*

$$|K(c,b)| \le 1 + Ca, \qquad |K_A(c,b)| \le C|A|_\infty.$$

Moreover $|d^2C_k(A)[h,h](c)| \le C\|h\|^2_{S_c}$. Consequently $|\mathbf{Q}_k h(c)| \le C\|h\|_{S_c}$ and $|dC_k(A)h(c)| \le C|A|_\infty\|h\|_{S_c}$, and the transposes satisfy $|\mathbf{Q}_k^T\omega|_\infty \le C|\omega|_\infty$ and $|dC_k(A)^T\omega|_\infty \le C|A|_\infty|\omega|_\infty$.

(d) The values of the correspondence transformation at the points of $\mathbb{T}$ are $\mathsf{w}_k(e^{\mathrm{i}\eta A})(y) = \exp\big(-\mathrm{i}\,\mathbf{\Gamma}_k A(y) + \mathrm{i}C^{\mathsf{w}}_k(A)(y)\big)$, with

$$\mathbf{\Gamma}_k A = \sum_{i=0}^{k-1} \delta_{i+1}\,\gamma_{i+1}\mathbf{Q}_i A, \qquad |\mathbf{\Gamma}_k A| \le C|A|_\infty, \qquad |C^{\mathsf{w}}_k(A)| \le C|A|^2_\infty,$$

and the analogous Lipschitz bound. At every fine point, $|\ell(\mathsf{w}_k(e^{\mathrm{i}\eta A}))| \le C|A|_\infty$.

Proof. (a), (b) By Lemma 22.5, $B_j = l_jB_{j-1} + c_j(B_{j-1})$ with $|l_jB|_\infty \le (1 + Ca\delta_j^2)|B|_\infty$ and $|c_j(B)|_\infty \le C\delta_j|B|^2_\infty$. By induction, as long as $|B_i|_\infty \le 2|A|_\infty$ for $i < j$,

$$|B_j|_\infty \le |A|_\infty \prod_{i\le j}(1 + Ca\delta_i^2 + 2C\delta_i|A|_\infty) \le |A|_\infty e^{C(a+|A|_\infty)} \le 2|A|_\infty,$$

using $\sum_i \delta_i \le 2$. The hypothesis $|B_{j-1}| \le c_2/\delta_j$ of Lemma 22.5 holds since $\delta_j \le 1$. Moreover $C_j(A) = \sum_{i\le j} l_j \cdots l_{i+1}c_i(B_{i-1})$, so $|C_j(A)| \le 2\sum_i C\delta_i(2|A|)^2 \le C|A|^2$. The Lipschitz bound is proved in the same way from the Lipschitz bound in Lemma 22.5(c). Locality follows from

Lemma 22.5(a) by induction: the bonds of $B_j(c_-) \cup B_j(c_+)$ have both endpoints in this set, and their scale-$(j-1)$ blocks lie in $B^{(j)}(c_-) \cup B^{(j)}(c_+)$.

(c) For $i \le j$, call a linear map K from scale-i fields to scale-j fields *of type* M if $Kv(c') = \delta_i^d \sum_{b'} K(c', b')v(b')$ with $|K(c', b')| \le M\delta_j^{-d}$, and $K(c', b') = 0$ unless both endpoints of b' lie in the union of the scale-j blocks of c'_- and c'_+. This support is preserved under composition, because the scale-j blocks are unions of scale-m blocks for $m \le j$. At $W = 1$ the composition of tube averages from scale i to scale j is the tube average from scale i to scale j. By Lemma 2.6(a), after rescaling, it is of type 1: its segments $[x, x(c')]$ lie in the two blocks, so both endpoints of each of their bonds do.

Transports. For real W, $q_j \cdots q_{i+1}$ is a sum over chains of bonds of the $W = 1$ weights times products of unitary transports. Hence its kernel is bounded in norm by the $W = 1$ kernel, and it is of type 1.

Insertions. Write $l_m = q_m + r_m$. By Lemma 22.5(b), $r_m v(c')$ is a combination of at most C_L values $v(b'')$, with b'' a bond of $B_m(c'_-) \cup B_m(c'_+)$ and coefficients bounded by $Ca\delta_m^2$. Expand $l_j \cdots l_{i+1}$ as a sum over the sets I of scales at which r_m is chosen. For $I = \{m\}$ the term is $(q_j \cdots q_{m+1})\, r_m\, (q_{m-1} \cdots q_{i+1})$. In the normalization above, the left factor has kernel at most $\delta_m^d \delta_j^{-d}$ per scale-m bond, and the right factor at most $\delta_i^d \delta_{m-1}^{-d}$ per scale-i bond. For fixed c', b' at most C_L scale-m bonds contribute, so this term is of type $C_L a\delta_m^2 L^d$. For several insertions, write the term as $A_0 r_{m_1} A_1 \cdots r_{m_p} A_p$, with $m_1 > \cdots > m_p$ and transported tube averages A_q of type 1, and compose from the right. If T is of type M', with the support property above, then $A\, r_m\, T$ is of type $C_L a\delta_m^2 L^d M'$, by the count for one insertion: for fixed output and input bonds at most C_L (indeed $(2d)^2$) pairs of intermediate bonds of scales m and $m-1$ contribute. Hence the term is of type $\prod_{m \in I} C_L a\delta_m^2 L^d$. Each counting factor is attached to one insertion; the composition of the factors A_q alone needs none, by the exactness at $W = 1$. Altogether $l_j \cdots l_{i+1}$ is of type $\prod_m (1 + C_L a\delta_m^2) \le 1 + Ca$. For $i = 0$ and $j = k$ this is the bound on K.

First derivatives. Differentiating the recursion gives $dB_j = (l_j + dc_j(B_{j-1}))\, dB_{j-1}$. By the Cauchy estimate in Lemma 22.5(c), $dc_j(B)v(c')$ is a combination of at most C_L values of v, with coefficients bounded by $C\delta_j |B|_\infty \le 2C\delta_j |A|_\infty$. So dc_j plays the role of r_j, with $a\delta_j^2$ replaced by $C\delta_j |A|_\infty$. The same expansion shows that every product $\prod_m (l_m + dc_m)$ is of type $1 + C(a + |A|_\infty)$. The terms with at least one insertion of some dc_m form a map of type $C|A|_\infty$; for $i = 0$, $j = k$ they make up $dC_k(A)$. This gives the bound on K_A.

Second derivatives. The recursion gives

$$d^2 B_j[h,h] = (l_j + dc_j)\, d^2 B_{j-1}[h,h] + d^2 c_j(B_{j-1})[dB_{j-1}h, dB_{j-1}h].$$

Since dB_{j-1} is of type C from scale 0, the Cauchy–Schwarz inequality gives $|dB_{j-1}h(b')| \le C\delta_{j-1}^{-d/2} \|h\|_{S_{b'}^{(j-1)}}$, where $S_{b'}^{(j-1)} = B^{(j-1)}(b'_-) \cup B^{(j-1)}(b'_+)$. By the Cauchy estimate, the source term at a scale-j bond c' is therefore bounded by $C_L \delta_j \delta_{j-1}^{-d} \|h\|^2_{S_{c'}^{(j)}}$. It is propagated to $\mathbb{T}$ by a map of type C from scale j, so its contribution at c is at most

$$C \sum_{c'} \delta_j^d\, C_L \delta_j \delta_{j-1}^{-d} \|h\|^2_{S_{c'}^{(j)}} \le C_L \delta_j \|h\|^2_{S_c},$$

because each fine bond lies in a bounded number of the sets $S_{c'}^{(j)}$, all of which lie in S_c. Summing over j gives the bound on $d^2 C_k$.

The remaining consequences follow from the kernel bounds. The bounds on $\mathbf{Q}_k h(c)$ and $dC_k(A)h(c)$ follow by the Cauchy–Schwarz inequality, since S_c has volume 2. The bounds on the transposes follow by summing the kernel bounds over c, since each fine bond lies in at most $2d$ sets S_c.

(d) By Theorem 22.4(c) and Lemma 22.5, $\ell(\tau_{X_i}) = \delta_{i+1}(\gamma_{i+1} B_i + c_{i+1}^{\tau}(B_i))$, of size at most $C\delta_{i+1}|A|$. Lemma 21.1(c) applied to the product gives the formula

and the bounds, since $\sum_i \delta_{i+1} \le 2$. At a fine point x, $\mathsf{w}_k = \prod_i \mathcal{E}^{(i)}(g_{X_i})$, and $g_{X_i}(x) = R(W_i(\Gamma))^{-1}(\tau_{X_i}(y)^{-1}(X_i)_W(\Gamma))$ for $x \in B_{i+1}(y)$ has logarithm at most $C\delta_{i+1}|A|$. Conjugation by $R(W)$ does not change the norm. □

22.6 The linearized averages

Let $q'_j\lambda(y) = L^{-d}\sum_{x\in B_j(y)} R(W_{j-1}(\Gamma^j_{y,x}))\lambda(x)$ be the transported block average of scalar fields, and put

$$Q'_k(U_0) = q'_k \cdots q'_1, \qquad \mathbf{Q}_k(U_0) = l_k \cdots l_1.$$

Explicitly, $(Q'_k\lambda)(y) = \eta^d \sum_{x\in B(y)} T(y,x)\lambda(x)$, where

$$T(y,x) = R(W_{k-1}(\Gamma^k_{y,x_{k-1}})) \cdots R(W_0(\Gamma^1_{x_1,x}))$$

is the transport along the multiscale tree, with the averaged backgrounds at each scale. In particular $Q'^T_k = \mathcal{E}^{(k)}$. These are Bałaban's linearized averages [7, (3.15)–(3.19)].

Proposition 22.7 (Linearized averages)**.** *Let $U_0 \in \mathfrak{U}_0(a)$ be real with $a \le c_3$.*

(a) $Q'_k Q'^T_k = I$.

(b) (Exact covariance.) $\mathbf{Q}_k(U_0)D_{U_0} = D_{W_k}Q'_k(U_0)$. *In the notation of Part II, the defect vanishes: $E = 0$, with $V = W_k$.*

(c) At $U_0 = 1$: $Q'_k = Q_0$ and $\mathbf{Q}_k = \mathbf{Q}$, the operators of Part I.

(d) (Locality.) $(\mathbf{Q}_k A)(c)$ *depends only on A and U_0 on the bonds of $B(c_-) \cup B(c_+)$, and $(Q'_k\lambda)(y)$ only on λ and U_0 on $B(y)$.*

(e) (Closeness.) *Suppose that on a cube containing $B(c_-) \cup B(c_+)$ we have $U_0 = e^{\mathrm{i}\eta A_0}$ (in some real gauge) with $|A_0| \le \varepsilon$, and $a \le \varepsilon$, where $\varepsilon \le c_3$. Then, computed in this gauge, the kernels satisfy $|K_{\mathbf{Q}_k(U_0)}(c,b) - K_{\mathbf{Q}}(c,b)| \le C\varepsilon$ and $|K_{Q'_k(U_0)}(y,x) - K_{Q_0}(y,x)| \le C\varepsilon$. Hence the differences are bounded by $C\varepsilon$ from sup norms to sup norms and in the block norms of Lemma 12.5(b), and so are their transposes.*

(f) (Covariance.) *For every real gauge transformation u on $\mathbb{T}_\eta$: $W_k(U_0^u) = W_k(U_0)^{u|_{\mathbb{T}}}$, $Q'_k(U_0^u)R(u) = R(u|_{\mathbb{T}})Q'_k(U_0)$ and $\mathbf{Q}_k(U_0^u)R(u) = R(u|_{\mathbb{T}})\mathbf{Q}_k(U_0)$.*

Proof. (a) $q'^T_j g(x) = R(W_{j-1}(\Gamma^j_{y,x}))^{-1}g(y)$, because the volume factors $L^{-d}\delta_j^d/\delta_{j-1}^d$ equal 1. Hence $q'_j q'^T_j = I$, since $R(W)$ is unitary.

(b) Let λ be real and $g_t = e^{\mathrm{i}t\lambda}$. Then $1^{(g_t)} = e^{-\mathrm{i}t\eta D_{U_0}\lambda + O(t^2)}$ on $\mathbb{T}_\eta$. By Theorem 22.4(e), $\mathcal{Q}_k(1^{(g_t)}) = 1^{(\mathfrak{M}^{(k)}(g_t))}$ on $\mathbb{T}$. Differentiating the definition of $\mathfrak{m}_j$ at $g = 1$ gives $\mathfrak{M}^{(k)}(g_t) = e^{\mathrm{i}tQ'_k\lambda + O(t^2)}$, hence $1^{(\mathfrak{M}^{(k)}(g_t))} = e^{-\mathrm{i}tD_{W_k}Q'_k\lambda + O(t^2)}$. Comparing the terms of first order in t gives $\mathbf{Q}_k D_{U_0}\lambda = D_{W_k}Q'_k\lambda$.

(c) At $U_0 = 1$, q'_j is the block average at scale j, and $l_j = q_j$ is the tube average at scale j by Lemma 22.5(d). Compositions of block averages are block averages. Compositions of tube averages are tube averages: in $(q_2q_1A)(y)$ the shifts $m\delta_1 + m'\delta_0$, $0 \le m, m' < L$, run through $0, \delta_0, \dots, (L^2-1)\delta_0$ exactly once. By induction, $\mathbf{Q}_k$ is the tube average $\mathbf{Q}$ of Definition 2.5.

(d) follows from Lemma 22.5(a) and Lemma 21.5(c).

(e) By (f) we may compute in the given gauge. By Lemma 21.8, $W_j = e^{\mathrm{i}\delta_j A_{0,j}}$ on the relevant bonds with $|A_{0,j}| \le C\varepsilon$. Hence all transports $R(W_{j-1}(\gamma))$ along contours of at most $3dL$ bonds differ from 1 by at most $C\varepsilon\delta_j$. Together with the bound on r_j in Lemma 22.5(b), this shows that $l_j - q_j(1)$ is a combination of at most C_L values with coefficients $C\varepsilon\delta_j$. Telescope:

$$\mathbf{Q}_k(U_0) - \mathbf{Q} = \sum_j l_k \cdots l_{j+1}(l_j - q_j(1))q_{j-1}(1)\cdots q_1(1).$$

By the bookkeeping of Proposition 22.6(c), the kernel of the difference is bounded by $\sum_j C_L\varepsilon\delta_j \le C\varepsilon$. The same argument applies to Q'_k. Kernel bounds give the sup-norm bounds, the block norms (by the Schur test), and the transposes.

(f) follows from Lemma 21.5(b), the definitions, and the transformation law of the relative gauge action. □

Corollary 22.8 (Parts II and III apply to Bałaban's averaging)**.** *Let U_0 be real and (M,ε)-regular with ε small. Replace the averaging pair $(Q'(U),\mathbf{Q}(U))$ by $(Q'_k(U_0),\mathbf{Q}_k(U_0))$, the defect E by 0, and the unit-lattice background V by W_k. Then the real results of Sections 14–19 hold, and so does Theorem 20.2. Proposition 19.11 holds for $\delta \le \min(\delta_0,\delta_{AB}/C)$ with $\varepsilon + C\delta \le \min(\varepsilon_0, c_3')$; at the perturbations U_δ the averages are those of Proposition 46.4.*

Proof. An (M,ε)-regular background has curvature at most $C\varepsilon$, by Lemma 12.3 in the local gauges. Proposition 22.7 gives the properties (a)–(d) of Lemma 12.5, in each local gauge, with ε replaced by $C\varepsilon$. By Remark 19.12, these are the only properties of the averaging used in the real results of Parts II and III. Property (v) of that remark at the U_δ is shown there, from Proposition 46.4(iv)–(vi) and the identity theorem. Proposition 46.4 uses Part IV only at the real reference, so there is no circularity. It controls only the sup norm of the perturbation; the derivative and Hölder bounds on δA are hypotheses of Proposition 19.11. □

From now on $Q' = Q'_k(U_0)$, $\mathbf{Q} = \mathbf{Q}_k(U_0)$, and all operators of Parts II and III (G', R, H', Γ', $\mathbf{G}$, $\mathbf{H}$, $\mathbf{C}$, …) are taken at the background U_0 with this averaging pair. Since $E = 0$, the slice projection satisfies $\mathbf{Q}\Pi = \mathbf{Q}$ (Lemma 15.1(c)), and the correction $\mathcal{K}_2$ of Section 15 vanishes.

23 The Wilson action near a regular configuration

23.1 The action and the current

The Wilson action of a configuration U on $\mathbb{T}_\eta$ is

$$S(U) = \sum_p \eta^{d-4}\Big(\mathsf{N} - \tfrac{1}{2}\operatorname{tr}\big(U(\partial p) + U(\partial p)^{-1}\big)\Big),$$

the sum running over the positively oriented plaquettes. For real U the summand is $\eta^{d-4}\operatorname{Re}\operatorname{tr}(1 - U(\partial p))$. The formula defines an analytic function of G^c-valued configurations. Throughout this section we use the real bilinear pairing $\langle A,B\rangle = \eta^d\sum_b \operatorname{tr}(A(b)B(b))$ on $\mathfrak{g}^c$-valued gauge fields, and $\|A\|^2 = \langle A,A\rangle$ for real fields; the gradient $\nabla\Phi$ of a function Φ of fields is taken with respect to this pairing. For a real configuration U and a fine Lie field A put $\Phi_U(A) = S(e^{\mathrm{i}\eta A}U)$, where $e^{\mathrm{i}\eta A}U$ is the bondwise product.

Definition 23.1 (Current and the class $\mathfrak{U}$)**.** For real U the *current* is $J_U = \nabla\Phi_U(0)$, a real $\mathfrak{g}$-valued gauge field. For $\theta > 0$ let

$$\mathfrak{U}(\theta) = \{U \in \mathfrak{U}_0(\theta) :\ |J_U|_\infty \le \theta\}.$$

Lemma 23.2 (The current)**.** *Let U be real. Put $F^s_U(p) = \eta^{-2}P_{\mathfrak{g}}((U(\partial p) - U(\partial p)^{-1})/2\mathrm{i})$, where $P_{\mathfrak{g}}$ is the orthogonal projection onto $\mathfrak{g}$ with respect to the trace form. Then:*

(a) $J_U = D_U^T F^s_U$;

(b) $D_U^T J_U = 0$, *and* $J_{U^u} = R(u)J_U$ *for real gauge transformations u;*

(c) if $U \in \mathfrak{U}_0(\theta)$, then $|F^s_U - F_U| \le C\theta^3\eta^4$, hence $|J_U - D_U^T F_U| \le C\theta^3\eta^3$.

So $\mathfrak{U}(\theta)$ is Bałaban's class $\mathfrak{U}_k(\{\Omega_j\},\theta)$ for $\Omega_j = \mathbb{T}_\eta$: small curvature and small covariant divergence of the curvature [6, (1.7)–(1.9)], [8, (2)].

Proof. (a) For a plaquette $p = p_{\mu\nu}(x)$ with bonds $b_1,\dots,b_4$ as in Definition 21.3, $H_p = U(\partial p)$, and a fine field A,

$$(e^{\mathrm{i}\eta A}U)(\partial p) = e^{\mathrm{i}\eta a_1}e^{\mathrm{i}\eta a_2}e^{-\mathrm{i}\eta a_3}e^{-\mathrm{i}\eta a_4}\,H_p, \tag{23.1}$$

with $a_1 = A(b_1)$, $a_2 = R(U(b_1))A(b_2)$, $a_3 = R(H_pU(b_4))A(b_3)$, $a_4 = R(H_p)A(b_4)$. To first order the summand of S changes by $\eta^{d-4}\operatorname{Im}\operatorname{tr}\big(\eta(a_1+a_2-a_3-a_4)H_p\big)$. Since $\operatorname{tr}((R(H)X)H) = \operatorname{tr}(XH)$, the factors $R(H_p)$ drop out, and $\eta(a_1+a_2-a_3-a_4)$ may be replaced by $\eta^2(D_UA)(p)$, with $(D_UA)(p) = (D_\mu A_\nu - D_\nu A_\mu)(x)$ as in Part II. Hence the first-order term is $\langle D_UA, F^s_U\rangle$.

(b) S is gauge invariant, and $e^{\mathrm{i}\eta A_t}U = U^{e^{\mathrm{i}t\lambda}}$ with $A_t = -tD_U\lambda + O(t^2)$. Hence $\langle D_U\lambda, J_U\rangle = 0$ for all λ. Covariance is clear.

(c) $F_U(p) = \eta^{-2}\ell(H_p)$ and $(H-H^{-1})/2\mathrm{i} = \sin(\ell(H))$, and $\sin X - X = O(|X|^3)$. The second bound follows because D_U^T applied to a field bounded by s is bounded by $4ds/\eta$. □

23.2 Expansion

For fine fields we use the covariant gradient $\nabla A = \nabla_{U_0}A$, whose components are $D_{U_0,\nu}A_\mu$ for all μ,ν, the sup norm $|A|_\infty$, and $|A|_{C^1} = |A|_\infty + |\nabla A|_\infty$. On the lattice, trivially,

$$\eta|\nabla A|_\infty \le 2|A|_\infty. \tag{23.2}$$

Proposition 23.3 (Expansion of the action)**.** *There are c_5, C such that the following holds. Let U_0 be real and $(1,\varepsilon)$-regular with $\varepsilon \le c_5$, and put $J = J_{U_0}$. For complex fine Lie fields A with $|A|_\infty \le c_5$,*

$$\Phi(A) := S(e^{\mathrm{i}\eta A}U_0) = S(U_0) + \langle A, J\rangle + \tfrac12\|D_{U_0}A\|^2 + \mathcal{N}(A),$$

where $\mathcal{N}$ is analytic, real for real A, and:

(a) $|\nabla\mathcal{N}(A)|_\infty \le C(\varepsilon + |A|_\infty)|A|_{C^1}$;

(b) if $|A|_{C^1}, |A'|_{C^1} \le r \le c_5$, then $|\nabla\mathcal{N}(A) - \nabla\mathcal{N}(A')|_\infty \le C(\varepsilon + r)|A - A'|_{C^1}$;

(c) for real A with $|A|_{C^1} \le c_5$ and all real h: $|d^2\mathcal{N}(A)[h,h]| \le C(\varepsilon + |A|_{C^1})(\|h\|^2 + \|\nabla h\|^2)$;

(d) for real A: $\nabla\Phi(A) = \Psi(A)^T J_{e^{\mathrm{i}\eta A}U_0}$, where $\Psi(A)$ acts bondwise and $|\Psi(A)-1| \le C\eta|A|_\infty$.

Proof. Plaquette variables. Let $Z_p(A) = \ell(e^{\mathrm{i}\eta a_1}e^{\mathrm{i}\eta a_2}e^{-\mathrm{i}\eta a_3}e^{-\mathrm{i}\eta a_4})$, with the a_i of (23.1) for $U = U_0$, and $\zeta_p = \eta^{-2}Z_p$. By Lemma 21.1(c), Z_p is analytic for $|A| \le c_5$ and

$$\zeta_p(A) = (D_{U_0}A)(p) + \rho_p(A), \qquad \rho_p = \rho_p^{(1)} + \rho_p^{(2)},$$

where $\rho_p^{(1)} = \eta^{-1}(a_1+a_2-a_3-a_4) - (D_{U_0}A)(p)$ is linear and $\rho_p^{(2)}$ collects the commutator terms of order two and higher. Since $|R(H_p)-1| \le C\varepsilon\eta^2$, $|\rho_p^{(1)}(A)| \le C\varepsilon\eta|A|$. By the Cauchy estimates, $|\rho_p^{(2)}(A)| \le C|A|^2$, $|d\rho_p^{(2)}(A)h| \le C|A||h|$ and $|d^2\rho_p^{(2)}(A)[h,h]| \le C|h|^2$, with sup norms over the bonds of p.

The summand. With $\mathfrak{s}(Z,H) = \mathsf{N} - \frac12\operatorname{tr}(e^{\mathrm{i}Z}H + H^{-1}e^{-\mathrm{i}Z})$,

$$\mathfrak{s}(Z,H) - \mathfrak{s}(0,H) = \operatorname{tr}\Big(Z\,\tfrac{H-H^{-1}}{2\mathrm{i}}\Big) + \tfrac12\operatorname{tr}\Big(Z^2\,\tfrac{H+H^{-1}}{2}\Big) + \mathfrak{s}_3(Z,H), \qquad |\mathfrak{s}_3| \le C|Z|^3.$$

Summing η^{d-4} times this over p, with $Z = \eta^2\zeta_p$ and $\frac12(H+H^{-1}) = 1 + O(\varepsilon^2\eta^4)$,

$$\Phi(A) - S(U_0) = \langle\zeta, F^s\rangle + \tfrac12\|\zeta\|^2 + \mathcal{R}(A), \qquad \mathcal{R}(A) = \eta^d\sum_p O(\varepsilon^2\eta^4|\zeta_p|^2 + \eta^2|\zeta_p|^3),$$

with $F^s = F^s_{U_0}$ and $\|\zeta\|^2 = \eta^d \sum_p \operatorname{tr} \zeta_p^2$. By the proof of Lemma 23.2(a), $\langle \rho^{(1)}(A), F^s\rangle = 0$, so the linear term is $\langle D_{U_0}A, F^s\rangle = \langle A, J\rangle$. Hence

$$\mathcal{N}(A) = \langle \rho^{(2)}(A), F^s\rangle + \langle D_{U_0}A, \rho(A)\rangle + \tfrac{1}{2}\|\rho(A)\|^2 + \mathcal{R}(A).$$

(a) We bound the gradient of each term. All of them are local, so it suffices to bound them in a gauge in which $U_0 = e^{i\eta A_0}$ near the bond, with $|A_0|, |\partial A_0| \le \varepsilon$.

- $\nabla\langle \rho^{(2)}(A), F^s\rangle = d\rho^{(2)}(A)^T F^s$ is bounded by $C\varepsilon|A|$, since $|F^s| \le C\varepsilon$.
- $\nabla\langle D_{U_0}A, \rho(A)\rangle = D^T_{U_0}\rho(A) + d\rho(A)^T D_{U_0}A$. The second term is bounded by $C(|A| + \varepsilon\eta)|DA|$. For the first, $D^T_{U_0}$ takes differences of ρ between neighbouring plaquettes, divided by η. The function $\rho^{(2)}_p$ vanishes to second order in the values of A on the bonds of p, and its coefficients are analytic functions of the transports $R(U_0(b_1))$, $R(H_pU_0(b_4))$ and $R(H_p)$ of (23.1). On neighbouring parallel plaquettes the bond variables of U_0 differ by at most $\eta^2\varepsilon$, since $|\partial A_0| \le \varepsilon$, and so do the holonomies, up to a constant, since each is $1 + O(\eta^2\varepsilon)$. The field values differ by $\eta|\partial A| \le \eta(|\nabla A| + C\varepsilon|A|)$. So the change of the field values costs $C\eta|A|(|\nabla A| + \varepsilon|A|)$, and the change of the coefficients costs $C\eta^2\varepsilon|A|^2$. The transport in $D^T_{U_0}$ differs from 1 by $O(\eta\varepsilon)$. After division by η, these differences are bounded by $C|A|(|\nabla A| + \varepsilon|A|)$. The differences of $\rho^{(1)}$ are bounded by $C\varepsilon|A| + C\varepsilon\eta|\nabla A|$; here we used $|F_p - F_{p'}| \le 2\varepsilon$ for the curvature factors.
- $\nabla\frac{1}{2}\|\rho\|^2 = d\rho^T\rho$ is bounded by $C(|A| + \varepsilon\eta)(|A|^2 + \varepsilon\eta|A|)$.
- In $\mathcal{R}$, each derivative of ζ_p with respect to a bond variable costs at most $C\eta^{-1}$. Hence $|\nabla\mathcal{R}| \le C(\varepsilon^2\eta^3|\zeta| + \eta|\zeta|^2)$. Finally $|\zeta| \le C(|\nabla A| + \varepsilon\eta|A| + |A|^2)$ and, by (23.2), $\eta|\zeta| \le C|A|$.

(b) The bound (a) holds for complex A in the ball $|A|_{C^1} \le 2r$, since $2r \le 2c_5$ and we may assume (a) on that larger ball, after decreasing c_5. Apply Cauchy's estimate to $t \mapsto \nabla\mathcal{N}(A' + t(A - A'))$, as in Lemma 22.5.

(c) Differentiate the formula for $\mathcal{N}$ twice. The terms $\langle d^2\rho^{(2)}[h,h], F^s\rangle$, $2\langle D_{U_0}h, d\rho[h]\rangle + \langle D_{U_0}A, d^2\rho[h,h]\rangle$, $\|d\rho[h]\|^2 + \langle\rho, d^2\rho[h,h]\rangle$ are bounded by $C(\varepsilon + |A|_{C^1})(\|h\|^2 + \|\nabla h\|^2)$, by the pointwise bounds on ρ and Cauchy–Schwarz. For $d^2\mathcal{R}$ write $\mathcal{R}(A) = \eta^d\sum_p f_p(\zeta_p)$ with $f_p(\zeta) = \frac{1}{2}\operatorname{tr}\big(\zeta^2(\frac{1}{2}(H_p + H_p^{-1}) - 1)\big) + \eta^{-4}\mathfrak{s}_3(\eta^2\zeta, H_p)$. By the Cauchy estimates, $|f_p'(\zeta)| \le C(\varepsilon^2\eta^4|\zeta| + \eta^2|\zeta|^2)$ and $|f_p''(\zeta)| \le C(\varepsilon^2\eta^4 + \eta^2|\zeta|)$. On the bonds of p, $|d\zeta_p[h]| \le C(|\nabla h| + (\varepsilon\eta + |A|)|h|)$ and $|d^2\zeta_p[h,h]| \le C|h|^2$. The chain rule gives two kinds of terms. In $f_p''(\zeta_p)[d\zeta_p[h], d\zeta_p[h]]$ the main term is $\eta^2|\zeta_p||\nabla h|^2 \le C|A||h||\nabla h|$, by (23.2) applied to ζ and to h; the other terms are bounded by $C(\varepsilon + |A|)(|h|^2 + |\nabla h|^2)$, since $\eta|\nabla h| \le 2|h|$. In $f_p'(\zeta_p)\,d^2\zeta_p[h,h]$ we have $|f_p'(\zeta_p)| \le C(\varepsilon^2 + |A|)|A|$, since $\eta|\zeta_p| \le C|A|$. Summing over p with the factor η^d, every bond lying in boundedly many plaquettes, gives the bound.

(d) Bondwise, $e^{i\eta(A+sh)} = e^{i\eta s\Psi(A)h + O(s^2)}e^{i\eta A}$, where $\Psi(A)h(b) = \mathfrak{d}(\eta\operatorname{ad}A(b))h(b)$ for an entire function $\mathfrak{d}$ with $\mathfrak{d}(0) = 1$. Hence $\frac{d}{ds}\Phi(A + sh)|_{s=0} = \langle\Psi(A)h, J_{e^{i\eta A}U_0}\rangle$. □

24 The axial-to-Landau chart

In this section U_0 is a real reference configuration that is (M,ε)-regular and satisfies $|J_{U_0}|_\infty \le \varepsilon$. In particular $U_0 \in \mathfrak{U}(C\varepsilon)$. All operators of Parts II and III are taken at U_0 with the averaging pair $(Q'_k(U_0), \mathbf{Q}_k(U_0))$ (Corollary 22.8). Theorem 24.2 below is Bałaban's [6, Theorem 2] on a single torus: every real configuration with small curvature and current and with averages close to those of U_0 has a unique representative in the Landau gauge relative to U_0, and this representative is small and regular.

24.1 Linear input

Lemma 24.1 (Sup-norm bounds at the reference)**.** *There are ε_*, C such that for $\varepsilon \le \varepsilon_*$:*

(a) $|G'f|_{C^1} \le C|f|_\infty$, $|H'g|_{C^1} \le C|g|_\infty$, *and* $D^T DH'g = Q'^T(\Gamma' - a)g$, *so* $|D^T DH'g|_\infty \le C|g|_\infty$.

(b) $|P_{\mathcal{V}} f|_{C^1} \le C|f|_\infty$ *and* $|P_{\mathcal{V}} D^T A|_{C^1} \le C|A|_\infty$. *In particular* $|Rf|_\infty \le C|f|_\infty$.

(c) The Feynman-gauge operator $\mathfrak{A}_F = D^T D + DRD^T + a\mathbf{Q}^T\mathbf{Q}$ *is invertible, and* $|\mathfrak{A}_F^{-1} f|_{C^1} \le C|f|_\infty$, $|\mathfrak{A}_F^{-1}\mathbf{Q}^T\omega|_{C^1} \le C|\omega|_\infty$.

(d) $|\mathbf{H}B|_{C^1} \le C|B|_\infty$ *and* $|\mathbf{C}f|_{C^1} \le C|f|_\infty$.

Here $|\cdot|_{C^1}$ uses the covariant gradient $\nabla = \nabla_{U_0}$; for scalar fields $\nabla = D_{U_0}$. The same bounds hold with exponential weights, provided $\gamma' \le \gamma/2$: if $|f(x)| \le e^{-\gamma'|x-z|}$ for some z, then $|Kf(x)| + |\nabla Kf(x)| \le Ce^{-\gamma'|x-z|}$ for each operator K above with a C^1 bound, and $|Kf(x)| \le Ce^{-\gamma'|x-z|}$ for $K = R$ and $K = D^T DH'$. For these two only sup bounds hold uniformly in η: at $U_0 = 1$, if f is a unit vector of $\mathfrak{g}$ at one fine point and 0 elsewhere, $|\nabla Rf|$ is of order η^{-1}. The constants ε_* and C may depend on d, L, G, a, α and M_0, but not on $M \ge M_0$, on k, on η or on the side n of $\mathbb{T}$. Indeed, the proof uses the theorems of Parts II and III at U_0, whose constants are uniform in this sense by Lemma 31.3, applied at $M = M_0$ as in Theorem 25.1; an (M, ε)-regular reference is (M_0, ε)-regular (Lemma 31.2).

Proof. By Corollary 22.8, the pointwise bounds of Theorems 18.2(g), 19.7(b) and 19.9 hold. Each has the form $|Kf|_{B(y)} + |\nabla Kf|_{B(y)} \le C\sum_{y'} e^{-\gamma|y-y'|}|f|_{B(y')}$. Summing over y' gives the sup-norm bounds, and the weighted bounds follow because $\sum_{y'} e^{-\gamma|y-y'|}e^{-\gamma'|y'-z|} \le Ce^{-\gamma'|y-z|}$. The identity in (a) follows from $\mathfrak{A}'H' = Q'^T\Gamma'$ and $Q'H' = I$. With it, the plain and weighted sup bounds on $D^T DH'g$ follow from the decay of Γ' and the sup bound on Q'^T, and those on $Rf = f - P_{\mathcal{V}} f$ from the bounds on $P_{\mathcal{V}}$. For (c), $\mathfrak{A}_F = \mathfrak{A}_\pi - \mathcal{K}_1$ with $W = 0$ and $\mathcal{K}_2 = 0$, because $E = 0$. By the proof of Theorem 15.4(a), $\mathcal{K}_1$ is admissible of size $C\varepsilon$ and pointwise admissible of type 1. Lemmas 13.2, 13.3 and 18.3 give the bounds, as for $\mathbf{G}$; in Lemma 18.3 the small inputs of $\mathcal{K}_1$ are bounded by the invariant norms $|v| + |\nabla v|$ on the cubes $\Lambda_1(c_{y'})$, with weights $e^{-\gamma|y-y'|}$. Finally $|\mathbf{Q}^T\omega| \le C|\omega|$ by Proposition 22.7(e). □

24.2 The theorem

Theorem 24.2 (Axial-to-Landau chart)**.** *There are $c_4, r_u, B_1 > 0$, with $B_1 c_4 \le r_u/2$, with the following property. Let U_0 be as above, and let U be real with $U \in \mathfrak{U}(\theta)$. Suppose that $U' = UU_0^{-1} \in \mathrm{Ax}_k$ and $|\ell(U'^{[k]}(c))| \le \beta$ for all bonds c of $\mathbb{T}$. If $\varepsilon + \theta + \beta \le c_4$, there is exactly one fine relative configuration $\tilde{U} = e^{i\eta A}$ with A real and $|A|_\infty \le r_u$ such that*

(i) $R(U_0)D_{U_0}^T A = 0$ *(Bałaban's Landau gauge relative to U_0), and*

(ii) $\tilde{U}^{(\mathsf{w}_k(\tilde{U}))} = U'$.

Moreover $\mathcal{Q}(\tilde{U}) = U'^{[k]}$, and

$$|A|_\infty + |\nabla A|_\infty + |\Delta_{U_0} A|_\infty + \sup_y [\nabla \tilde{A}_y]_{\alpha, B(y)} \le B_1(\varepsilon + \theta + \beta),$$

where $\Delta_{U_0} = D^T D + DD^T$ is the covariant Hodge Laplacian and $\tilde{A}_y$ is A in a local gauge of U_0 near $B(y)$.

Condition (ii) says that $\tilde{U}$ and U' lie in the same gauge orbit, and that the transformation relating them is the correspondence transformation of Theorem 22.4. By Theorem 22.4(f), this transformation satisfies Bałaban's normalization. The bounds are Bałaban's [6, (1.36)–(1.39)] on a single torus.

Corollary 24.3 (Parametrization of the constraint set)**.** *Let V be a real configuration on $\mathbb{T}$ with $|\ell(V(c)W_k(c)^{-1})| \le \beta$ for all c, and let $\varepsilon + \theta + \beta \le c_4$. Let $\mathcal{S}_V$ be the set of real A with $|A|_\infty \le r_u$, $RD^TA = 0$, $e^{i\eta A}U_0 \in \mathfrak{U}(\theta)$ and $\mathcal{Q}(e^{i\eta A}) = VW_k^{-1}$. Then*

$$A \longmapsto \mathsf{U}(A) = ((e^{i\eta A})^{(\mathsf{w}_k(e^{i\eta A}))})U_0$$

is a bijection from $\mathcal{S}_V$ onto the set of axial representatives ($UU_0^{-1} \in \mathrm{Ax}_k$) of the $\mathfrak{G}_0$-orbits in $\mathfrak{B}(V) \cap \mathfrak{U}(\theta)$, where $\mathfrak{B}(V) = \{U : U^{(k)} = V\}$. Moreover $S(\mathsf{U}(A)) = \Phi(A)$.

Proof. $\mathsf{U}(A) = (e^{i\eta A}U_0)^{\mathsf{w}_k}$ is gauge equivalent to $e^{i\eta A}U_0$, so it lies in $\mathfrak{U}(\theta)$ and has the same action. By Theorem 22.4(a), $\mathsf{U}(A)U_0^{-1} \in \mathrm{Ax}_k$ and $\mathsf{U}(A)^{(k)} = \mathcal{Q}(e^{i\eta A})W_k = V$. Conversely, for an axial representative U of an orbit in $\mathfrak{B}(V) \cap \mathfrak{U}(\theta)$ we have $U'^{[k]} = VW_k^{-1}$, and Theorem 24.2 gives exactly one preimage. The orbits have unique axial representatives by Lemma 22.2(c). □

24.3 Axial configurations are small

Lemma 24.4 ([6, Lemma 1, (1.65)])**.** *Let $C_A = 16d$. Let $U_0, U \in \mathfrak{U}_0(\theta)$ be real, with $\theta \le c_1$, and let $U' = UU_0^{-1} \in \mathrm{Ax}_k$. Suppose that $|\ell(U'^{[k]}(c))| \le \beta$ for all bonds c of $\mathbb{T}$, where $\beta + C_A\theta \le \frac{1}{2}c_G$. Then $|\ell(U'^{[j]}(b))| \le \beta + C_A\theta$ for all $j \le k$ and all bonds b of $\mathbb{T}^{(j)}$.*

Proof. One step. Fix j, and let X be a real relative configuration at scale $j-1$, axial at scale j, such that XW and $W = W_{j-1}$ lie in $\mathfrak{U}_{j-1}(2\theta)$, and $|\ell(\overline{X}(c))| \le b'$ for the bonds c of $\mathbb{T}^{(j)}$, where $b' + 12d\theta\delta_j^2 \le \frac{1}{2}c_G$. We claim that $|\ell(X(b))| \le b' + 12d\theta\delta_j^2$ for every bond b of $\mathbb{T}^{(j-1)}$. All configurations below are real. So, by Lemma 21.1(b), the logarithm is subadditive under products and invariant under conjugation, as long as the sums involved are at most c_G. All sums below are at most $b' + 12d\theta\delta_j^2$. Every logarithm is taken of a product of elements whose logarithms are already bounded. Since $X(-b) = R(W(b))^{-1}(X(b)^{-1})$ and $R(W(b))$ is unitary, it suffices to consider positively oriented bonds.

For a bond $b = (x, x')$ inside a block $B_j(y)$, let $\gamma = \Gamma^j_{y,x} \cup b \cup (\Gamma^j_{y,x'})^{-1}$. By axiality, $X_W(\gamma) = R(W(\Gamma^j_{y,x}))X(b)$, while $X_W(\gamma) = (XW)(\gamma)W(\gamma)^{-1}$. By Lemma 21.2(ii), applied to XW and W,

$$|\ell(X(b))| \le |\ell((XW)(\gamma))| + |\ell(W(\gamma))| \le 4d\theta\delta_j\delta_{j-1}.$$

Now let c be a bond of $\mathbb{T}^{(j)}$. For a configuration Y on $\mathbb{T}^{(j-1)}$ and $x \in B_j(c_-)$ write $\lambda_x(Y) = Y(\Gamma^j_{c,x})Y([c])^{-1}$. For $Y = XW$ and $Y = W$, Lemma 21.5(a) gives $|\ell(\lambda_x(Y))| \le 2d\theta\delta_j^2$. Let E_Y be the average of the $\ell(\lambda_x(Y))$ over $x \in B_j(c_-)$. By Definition 21.4 for XW and for W, and since $W_j = \mathfrak{a}_j(W)$,

$$\overline{X}(c) = \mathfrak{a}_j(XW)(c)\, W_j(c)^{-1} = e^{iE_{XW}}\, X_W([c])\, e^{-iE_W}.$$

Since $|E_{XW}|, |E_W| \le 2d\theta\delta_j^2$, this gives $|\ell(X_W([c]))| \le b' + 4d\theta\delta_j^2$. Next, for $x \in B_j(c_-)$,

$$X_W(\Gamma^j_{c,x}) = (XW)(\Gamma^j_{c,x})\, W(\Gamma^j_{c,x})^{-1} = \lambda_x(XW)\, X_W([c])\, \lambda_x(W)^{-1},$$

so $|\ell(X_W(\Gamma^j_{c,x}))| \le b' + 8d\theta\delta_j^2$. By axiality, $X_W(\Gamma^j_{c_-,x}) = 1$ and $X_W(\Gamma^j_{c_+,x(c)}) = 1$, hence $X_W(\Gamma^j_{c,x}) = R(W(\Gamma^j_{c_-,x}))X_W([x, x(c)])$. The segment $[x, x(c)]$ has L bonds. Exactly one of them joins $B_j(c_-)$ to $B_j(c_+)$, and the others lie inside one of the two blocks. Every bond of $\mathbb{T}^{(j-1)}$ that joins two blocks is of this form, for some c and x. Since $X_W([x, x(c)])$ is a product of conjugates of $X(b)$, for the bond b between the blocks, and of $L-1$ factors bounded by the first step,

$$|\ell(X(b))| \le b' + 8d\theta\delta_j^2 + (L-1)\, 4d\theta\delta_j\delta_{j-1} \le b' + 12d\theta\delta_j^2,$$

because $(L-1)\delta_{j-1} \le \delta_j$. This proves the claim. The coefficient of b' is 1.

Iteration. Let $b_j = \max_c |\ell(U'^{[j]}(c))|$, the maximum over the bonds of $\mathbb{T}^{(j)}$. By Proposition 21.7, $U^{(j-1)}, W_{j-1} \in \mathfrak{U}_{j-1}(2\theta)$. The relative configuration $U'^{[j-1]}$ is real and axial at scale j, with relative average $U'^{[j]}$. By the claim, $b_{j-1} \le b_j + 12d\theta\delta_j^2$, as long as $b_j + 12d\theta\delta_j^2 \le \frac{1}{2}c_G$. Since $b_k \le \beta$ and $\sum_i \delta_i^2 \le \frac{4}{3}$, downward induction gives $b_j \le \beta + 12d\theta \sum_{i>j} \delta_i^2 \le \beta + C_A\theta$, and the condition of the claim holds at every step, because $\beta + C_A\theta \le \frac{1}{2}c_G$. □

24.4 The a priori bound

Proposition 24.5 (A priori bound [6, Prop. 3])**.** *There are $r_*, B_0 > 0$ such that the following holds for $\varepsilon \le r_*$. Let $\tilde{U} = e^{\mathrm{i}\eta A}$ with A real, $|A|_\infty \le r_*$ and $RD^TA = 0$. Suppose $|J_{\tilde{U}U_0}|_\infty \le \theta$ and $\mathcal{Q}(\tilde{U}) = e^{\mathrm{i}B}$ with $|B|_\infty \le \beta$. Then*

$$|A|_{C^1} + |\Delta_{U_0}A|_\infty + \sup_y [\nabla \tilde{A}_y]_{\alpha,B(y)} \le B_0(\varepsilon + \theta + \beta).$$

Proof. By Proposition 23.3(d), $|\nabla\Phi(A)|_\infty \le 2\theta$. By Proposition 23.3 and $|J_{U_0}| \le \varepsilon$,

$$D^TDA = \nabla\Phi(A) - J_{U_0} - \nabla\mathcal{N}(A), \qquad |D^TDA|_\infty \le 2\theta + \varepsilon + C(\varepsilon + r_*)|A|_{C^1}.$$

By Proposition 22.6, $\mathcal{Q}(\tilde{U}) = e^{\mathrm{i}(\mathbf{Q}A + C_k(A))}$ with $|C_k(A)| \le C|A|_\infty^2 \le Cr_*|A|_\infty$, and $\mathbf{Q}A + C_k(A)$ is real and small. On each bond, $e^{\mathrm{i}B} = e^{\mathrm{i}(\mathbf{Q}A+C_k(A))}$, so the eigenvalues of B, counted with algebraic multiplicity, differ by multiples of 2π from those of the Hermitian matrix $\mathbf{Q}A + C_k(A)$, which lie in $(-\pi, \pi)$; by Schur triangularization the squared moduli of the eigenvalues of B sum to at most $|B|^2$. So $|\mathbf{Q}A + C_k(A)| \le |B|$ bondwise, and we may replace B by $\mathbf{Q}A + C_k(A)$, still with $|B|_\infty \le \beta$. Then $\mathbf{Q}A = B - C_k(A)$. Since $RD^TA = 0$,

$$\mathfrak{A}_F A = D^TDA + a\mathbf{Q}^T(B - C_k(A)),$$

and Lemma 24.1(c) gives $|A|_{C^1} \le C_F(2\theta + \varepsilon + \beta) + C_FC(\varepsilon + r_*)|A|_{C^1}$. For $C_FC(\varepsilon + r_*) \le \frac{1}{2}$ this gives the bound on $|A|_{C^1}$. Next, $RD^TA = 0$ means $D^TA = P_{\mathcal{V}}D^TA$, so

$$\Delta_{U_0}A = D^TDA + DP_{\mathcal{V}}D^TA,$$

which is bounded by Lemma 24.1(b). Finally, in a gauge in which U_0 is close to 1 on $\Lambda_3(c_y)$, Δ_{U_0} has the normal form $-\Delta + \sum_\mu \partial_\mu^*(a_\mu\,\cdot) + b$ of Lemma 12.2, componentwise. Theorem 6.3(d) gives the Hölder bound. □

The proposition needs only a bound on $|A|_\infty$, not on derivatives. The derivatives are controlled by the gauge condition and the action, because the derivative terms in $\nabla\mathcal{N}$ come with the small factor $\varepsilon + |A|_\infty$. This is why the construction below only has to control the sup norm of the new representative.

24.5 One scale of gauge transformation

For a fine relative configuration $\tilde{U}_1 = e^{\mathrm{i}\eta A_1}$ and a scalar field λ write $\tilde{U}_\lambda = \tilde{U}_1^{(e^{-\mathrm{i}\lambda})} = e^{\mathrm{i}\eta\mathcal{A}(\lambda)}$. A direct computation from the definition of the relative action gives

$$\mathcal{A}(\lambda)(b) = e^{-\mathrm{i}\,\mathrm{ad}\,\lambda(b_-)}A_1(b) + \mathfrak{d}\big(\mathrm{ad}\,\lambda(b_-)\big)D_{U_0}\lambda(b) + \eta\,\mathfrak{b}\big(\lambda(b_-), A_1(b), D_{U_0}\lambda(b)\big), \tag{24.1}$$

where $\mathfrak{d}$ is entire with $\mathfrak{d}(0) = 1$, and $\mathfrak{b}$ is analytic with $|\mathfrak{b}| \le C(|A_1| + |D\lambda|)|D\lambda|$ [6, (1.81)–(1.84)]. For $|\lambda|_{C^1} = |\lambda|_\infty + |D\lambda|_\infty$ small, $|\mathcal{A}(\lambda)|_\infty \le C(|A_1|_\infty + |\lambda|_{C^1})$.

Lemma 24.6 (One scale)**.** *There are $r_0, \rho_0, C_2 > 0$ such that for $\varepsilon \le r_0$ the following holds. Let A_1 be real with $|A_1|_{C^1} \le r \le r_0$. Then there is exactly one real λ with $|\lambda|_{C^1} \le \rho_0$ such that*

$$\textit{(E1)} \quad RD^T\mathcal{A}(\lambda) = 0, \qquad \textit{(E2)} \quad \mathsf{w}_k(\tilde{U}_\lambda) = \mathsf{w}_k(\tilde{U}_1)\,e^{\mathrm{i}\lambda}.$$

It satisfies $|\lambda|_{C^1} \le C_2 r$ and $|\mathcal{A}(\lambda)|_\infty \le C_2 r$.

Proof. Reformulation of (E2). Both $\mathsf{w}_k(\tilde U_\lambda)$ and $\mathsf{w}_k(\tilde U_1)e^{\mathrm{i}\lambda}$ move $\tilde U_\lambda$ into the axial gauge: $\tilde U_\lambda^{(\mathsf{w}_k(\tilde U_1)e^{\mathrm{i}\lambda})} = \tilde U_1^{(\mathsf{w}_k(\tilde U_1))}$. Here A_1 and λ are real, so all configurations and transformations are real, as Theorem 22.4(b) requires. By that theorem the two transformations coincide iff they agree at the points of $\mathbb{T}$. With

$$\Upsilon(\lambda) = \ell\big(\mathsf{w}_k(\tilde U_1)^{-1}\mathsf{w}_k(\tilde U_\lambda)\big) \quad \text{on } \mathbb{T},$$

(E2) is therefore equivalent to $\Upsilon(\lambda) = \lambda|_{\mathbb{T}}$.

Expansion of Υ. Υ is an analytic function of (A_1, λ) on $\{|A_1|_\infty \le 2r_0,\ |\lambda|_{C^1} \le 4(1+C_H)\rho_0\}$, where C_H is the constant in $|H'X|_{C^1} \le C_H|X|_{,}$ with $|\Upsilon| \le C(r_0+\rho_0)$ by Proposition 22.6(d), and $\Upsilon(A_1, 0) = 0$. At $A_1 = 0$, Theorem 22.4(e) gives $\mathsf{w}_k(1^{(g)})|_{\mathbb{T}} = \mathfrak{M}^{(k)}(g)\, g^{-1}|_{\mathbb{T}}$. With $g = e^{-\mathrm{i}\lambda}$, the linear part of $\Upsilon(0,\cdot)$ is $\lambda|_{\mathbb{T}} - Q'_k\lambda$. This uses Theorem 22.4(e) for real λ only: $\Upsilon(0,\cdot)$ is analytic, so its linear part is determined by its values at real λ. Write

$$\Upsilon(A_1, \lambda) = \lambda|_{\mathbb{T}} - Q'\lambda - \mathcal{C}'(\lambda).$$

By the Cauchy estimates, applied separately to $\Upsilon(0,\lambda)$ (quadratic remainder) and to $\Upsilon(A_1,\lambda) - \Upsilon(0,\lambda)$ (which vanishes at $A_1 = 0$ and at $\lambda = 0$),

$$|\mathcal{C}'(\lambda)| \le C(|A_1|_\infty + |\lambda|_{C^1})|\lambda|_{C^1}, \qquad |\mathcal{C}'(\lambda) - \mathcal{C}'(\lambda')| \le C(r+\rho)|\lambda - \lambda'|_{C^1}$$

for $|\lambda|_{C^1}, |\lambda'|_{C^1} \le \rho \le 2(1+C_H)\rho_0$. So (E2) is equivalent to $Q'\lambda + \mathcal{C}'(\lambda) = 0$.

Linearizing (E2). Decompose $\lambda = \lambda' + H'X$ with $\lambda' \in \mathcal{N}(Q')$ and $X = Q'\lambda$. Then (E2) becomes $X = -\mathcal{C}'(\lambda' + H'X)$. Since $|H'X|_{C^1} \le C_H|X|$ (Lemma 24.1(a)), the right-hand side is a contraction in X on $|X|_\infty \le \rho_0$, for $|\lambda'|_{C^1} \le (1+C_H)\rho_0$. Its unique fixed point $X = \mathcal{X}(\lambda')$ satisfies $|\mathcal{X}(\lambda')| \le C(r + |\lambda'|_{C^1})|\lambda'|_{C^1}$ and has Lipschitz constant $C(r+\rho)$ in λ'. This is the analogue of [6, Sect. E].

Solving (E1). Apply D^T to (24.1). By the discrete Leibniz rule,

$$D^T\mathcal{A}(\lambda) = e^{-\mathrm{i}\operatorname{ad}\lambda}D^TA_1 + (1+V(\lambda))D^TD\lambda + \mathfrak{g}_3,$$

where $V(\lambda)$ is multiplication by $\mathfrak{d}(\operatorname{ad}\lambda) - 1$, so $|V| \le C|\lambda|_\infty$, and $|\mathfrak{g}_3| \le C|D\lambda|(|A_1| + |D\lambda|)$. Now $D^TD\lambda = D^TD\lambda' + Q'^T(\Gamma' - a)X$ by Lemma 24.1(a). Moreover $D^TD\lambda' \in \operatorname{ran} R$, so $R(1+V)D^TD\lambda' = (1+RVR)D^TD\lambda'$. Hence (E1) is equivalent to

$$(1+RVR)D^TD\lambda' = -R\mathfrak{h}(\lambda'), \qquad \mathfrak{h} = e^{-\mathrm{i}\operatorname{ad}\lambda}D^TA_1 + (1+V)Q'^T(\Gamma' - a)\mathcal{X}(\lambda') + \mathfrak{g}_3.$$

For $|V|$ small, $1 + RVR$ is invertible on bounded fields (Lemma 24.1(b)), and its inverse maps $\operatorname{ran} R$ into $\operatorname{ran} R$. For $h \in \operatorname{ran} R = \mathcal{W}$ the unique $\lambda' \in \mathcal{N}(Q')$ with $D^TD\lambda' = h$ is $G'h$ (Proposition 14.2(c)). Therefore (E1) and $\lambda' \in \mathcal{N}(Q')$ are equivalent to the fixed point equation

$$\lambda' = \mathcal{T}(\lambda') := -G'(1+RVR)^{-1}R\,\mathfrak{h}(\lambda').$$

We have $|D^TA_1| \le Cr$, so $|\mathfrak{h}|_\infty \le C(r+\rho^2)$ on $|\lambda'|_{C^1} \le \rho$, and $\mathcal{T}$ maps this ball into itself for $\rho = C'r$ and r small. The λ'-dependence of $\mathcal{T}$ enters through V, $\mathcal{X}$, $e^{-\mathrm{i}\operatorname{ad}\lambda}$ and $\mathfrak{g}_3$. Each of these is multiplied by a quantity of size $C(r+\rho)$, or has Lipschitz constant $C(r+\rho)$. Hence $\mathcal{T}$ is a contraction on $|\lambda'|_{C^1} \le (1+C_H)\rho_0$, for r_0 and ρ_0 small. Its unique fixed point, together with $\lambda = \lambda' + H'\mathcal{X}(\lambda')$, is the solution. It is real because all maps preserve reality, and $|\lambda|_{C^1} \le C_2r$. The bound on $\mathcal{A}(\lambda)$ follows from (24.1).

Uniqueness. If λ solves (E1), (E2) with $|\lambda|_{C^1} \le \rho_0$, then $X = Q'\lambda$ and $\lambda' = \lambda - H'X$ satisfy $|X| \le \rho_0$, $|\lambda'|_{C^1} \le (1+C_H)\rho_0$, $X = \mathcal{X}(\lambda')$ and $\lambda' = \mathcal{T}(\lambda')$. Since $\mathcal{T}$ is a strict contraction on this ball, λ', and hence λ, is unique. □

24.6 Proof of Theorem 24.2

By Lemma 24.4, applied with θ replaced by $\theta + C\varepsilon$ (the curvature of U_0 is at most $C\varepsilon$), the hypothesis implies $|\ell(U'^{[j]})| \le \beta_* := \beta + C_A(\theta + C\varepsilon)$ for all j. We prove the following statement $\mathcal{P}(k)$ by induction on k: *under the hypotheses of the theorem, with $|\ell(U'^{[j]})| \le \beta_*$ for all $j \le k$ and $\varepsilon + \theta + \beta_* \le c_4^\sharp$, there is a solution $\tilde U$ of (i), (ii) obeying the bounds with B_0' and β_* in place of B_1 and β.* Here B_0' and $c_4^\sharp$ are fixed in the induction step below. Then c_4 is chosen with $(1 + C_A(1 + C))c_4 \le \min(c_4^\sharp, \frac12 c_G)$ and $(1 + C)c_4 \le c_1$, so that Lemma 24.4 applies; this gives existence with $B_1 = B_0'(1 + C_A(1 + C))$. Finally the uniqueness radius is $r_u = \min(r_*, r_0, \rho_0/(4C_w))$, with ρ_0 from Lemma 24.6 and C_w the constant of the uniqueness step below. We decrease c_4 further until $B_1 c_4 \le r_u/2$ and $C_w B_0 c_4 \le \rho_0/2$, so that the solution lies in the uniqueness ball.

For the induction we use scale covariance. Rescaling all lengths by δ_j^{-1} maps the structure on $(\mathbb{T}_\eta, \mathbb{T}^{(j)})$ to the j-step structure on a torus of side n/δ_j with fine spacing η/δ_j. It multiplies Lie fields by δ_j, gradients by δ_j^2, curvatures by δ_j^2 and currents by δ_j^3. The action changes by the constant factor δ_j^{4-d}. Hence the hypotheses of $\mathcal{P}(k)$ imply those of $\mathcal{P}(k-1)$ for the averaging up to scale $k-1$, in units of δ_{k-1}, with the same β_* and smaller ε, θ. All constants are uniform in k and n.

$k = 0$. There is no averaging: $R = 0$, $\mathsf{w}_0 = 1$, and $\tilde U = U'$. Since $\delta_0 = 1$ in these units, $|A| = |\ell(U')| \le \beta_*$, and the gradient and the Laplacian are bounded by $C|A|$.

From $k-1$ to k. By $\mathcal{P}(k-1)$, in units of δ_{k-1}, there is $\tilde U_1 = e^{\mathrm{i}\eta A_1}$ with $R_{k-1} D^T A_1 = 0$, $\tilde U_1^{(\mathsf{w}_{k-1}(\tilde U_1))} = U'$, and, back in units of $\delta_k = 1$,

$$|A_1|_\infty \le L B_0'(\varepsilon + \theta + \beta_*), \qquad |\nabla A_1|_\infty \le L^2 B_0'(\varepsilon + \theta + \beta_*).$$

Since $U' \in \mathrm{Ax}_k$, the configuration $\mathcal{Q}_{k-1}(\tilde U_1) = U'^{[k-1]}$ is axial at scale k. By Theorem 22.4(d), $\mathsf{w}_k(\tilde U_1) = \mathsf{w}_{k-1}(\tilde U_1)$, so $\tilde U_1^{(\mathsf{w}_k(\tilde U_1))} = U'$. Apply Lemma 24.6 with $r = 2L^2 B_0'(\varepsilon+\theta+\beta_*) \ge |A_1|_{C^1}$; we need $r \le r_0$ and $C_2 r \le r_*$, which holds if $c_4^\sharp \le \min(r_0, r_*/C_2)/(2L^2 B_0')$. We obtain λ such that $\tilde U = \tilde U_\lambda$ is in the Landau gauge at scale k, and

$$\tilde U^{(\mathsf{w}_k(\tilde U))} = \tilde U_1^{(\mathsf{w}_k(\tilde U_1) e^{\mathrm{i}\lambda} e^{-\mathrm{i}\lambda})} = U'.$$

Its Lie field satisfies $|\mathcal{A}(\lambda)|_\infty \le C_2 r$. By Theorem 22.4(a), $\mathcal{Q}(\tilde U) = U'^{[k]}$, so $|B| \le \beta_*$ with $B = \ell(U'^{[k]})$. Moreover $J_{\tilde U U_0}$ is the current of a configuration gauge equivalent to U, so $|J_{\tilde U U_0}| \le \theta$. If $C_2 r \le r_*$, Proposition 24.5 gives the bounds with $B_0' := \max(B_0, C)$, where C is the constant of the case $k = 0$. The constant does not depend on k, which closes the induction.

Uniqueness. Let $\tilde U = e^{\mathrm{i}\eta A}$ and $\tilde U^\sharp = e^{\mathrm{i}\eta A^\sharp}$ both satisfy (i), (ii) and $|A|, |A^\sharp| \le r_u \le r_*$. By Proposition 24.5, both have $|\cdot|_{C^1} \le B_0(\varepsilon + \theta + \beta)$. Let $g = \mathsf{w}_k(\tilde U^\sharp)^{-1} \mathsf{w}_k(\tilde U)$. Then $\tilde U^\sharp = \tilde U^{(g)}$. By Proposition 22.6(d), $|\ell(g)| \le C(|A| + |A^\sharp|)$ pointwise, so $g = e^{-\mathrm{i}\lambda}$ with λ real. We take $C_w \ge 4C$ for this constant C, and $\rho_0 \le \frac12$, as we may in Lemma 24.6. Since $r_u \le \rho_0/(4C_w)$, this gives $|\lambda|_\infty \le 2Cr_u \le \frac12 C_w r_u \le \frac18 \rho_0 \le \frac{1}{16}$. The derivative is bounded by the bond equation. For a bond b put $\xi = \lambda(b_-)$ and $\xi' = R(U_0(b))\lambda(b_+)$, so that $\eta D_{U_0}\lambda(b) = \xi' - \xi$ and $|\xi|, |\xi'| \le \frac14$. The relation $\tilde U^\sharp = \tilde U^{(g)}$ reads $e^{\mathrm{i}\xi'} = e^{-\mathrm{i}\eta A(b)} e^{\mathrm{i}\xi} e^{\mathrm{i}\eta A^\sharp(b)}$. The factors are unitary, so Lemma 21.1(a) gives $|e^{\mathrm{i}\xi'} - e^{\mathrm{i}\xi}| \le |e^{-\mathrm{i}\eta A(b)} - 1| + |e^{\mathrm{i}\eta A^\sharp(b)} - 1| \le 2\eta(|A(b)| + |A^\sharp(b)|)$. Both $e^{\mathrm{i}\xi}$ and $e^{\mathrm{i}\xi'}$ lie in $\{|h - 1| \le \frac12\}$, where ℓ is analytic with bounded derivative, and $\ell(e^{\mathrm{i}\xi}) = \xi$, $\ell(e^{\mathrm{i}\xi'}) = \xi'$, since $|\xi|, |\xi'| < \log 2$: for $|X| < \log 2$ the power series of $\log(1 + w)$ converges absolutely at $w = e^{\mathrm{i}X} - 1$, and its coefficients are those of the scalar identity $-\mathrm{i}\log e^{\mathrm{i}z} = z$. Hence $|D\lambda|_\infty \le C'(|A|_\infty + |A^\sharp|_\infty) \le 2C' r_u$, and we take $C_w \ge 4C'$ too. Thus $|\lambda|_{C^1} \le C_w(r_u + B_0(\varepsilon + \theta + \beta)) \le \rho_0$, by the choice of r_u and c_4, and λ solves (E1), (E2) of Lemma 24.6 with base $\tilde U_1 = \tilde U$. The same holds for $\lambda = 0$. By the uniqueness part of the lemma, $\lambda = 0$ and $\tilde U^\sharp = \tilde U$. □

Part V: The variational problem

25 The variational theorem

25.1 The problem

Let V be a real configuration on $\mathbb{T}$. The *variational problem* is to minimize the Wilson action S on

$$\mathfrak{B}(V) = \{U \in \mathfrak{U}_0(c_1) \text{ real} : \ U^{(k)} = V\},$$

the set of fine configurations of small curvature whose k-fold average is V [8, (3), (5), (6)]. Both S and $\mathfrak{B}(V)$ are invariant under the residual gauge group $\mathfrak{G}_0$. A *critical orbit* is a $\mathfrak{G}_0$-orbit of critical points of S restricted to $\mathfrak{B}(V)$. Its elements are Bałaban's *background fields* (minimal configurations) $U_k(V)$.

Bałaban's Theorem 1 [8] solves this problem by induction over the number k of averaging steps. The k-th step starts from a reference configuration U_0, the solution of the $(k-1)$-step problem for an extension of V. It solves the problem near U_0 by a contraction, and then re-establishes the regularity of the solution with constants independent of k, by a local analysis on cubes. The theorem below is the first part of this scheme: the problem relative to a given regular reference configuration [8, Sects. A–E, G]. The induction over k, with the local improvement of [8, Sect. F], is carried out in Part VI (Theorem 30.3).

25.2 Statement

Throughout this part, U_0 is a real reference configuration that is (M, ε)-regular with $|J_{U_0}|_\infty \le \varepsilon$. As before, $W_k = U_0^{(k)}$, and the operators of Parts II and III are taken at U_0 with the averaging pair $(Q'_k(U_0), \mathbf{Q}_k(U_0))$. We fix once and for all a small constant $\theta_0 > 0$; the uniqueness statements refer to the class $\mathfrak{U}(\theta_0)$.

Theorem 25.1 (The variational problem relative to a regular reference)**.** *Let $d \ge 3$. There are $\theta_0, c_6, C_6, \gamma' > 0$ such that the following holds. Let U_0 be as above, and let V be a real configuration on $\mathbb{T}$ with*

$$V(c)W_k(c)^{-1} = e^{\mathrm{i}B(c)}, \qquad |B|_\infty \le \beta.$$

If $\varepsilon + \beta \le c_6$, then:

(a) (Existence and uniqueness.) *$S|_{\mathfrak{B}(V)}$ has exactly one critical orbit in $\mathfrak{B}(V) \cap \mathfrak{U}(\theta_0)$.*

(b) (Minimality.) *This orbit is the set of minimizers of S on $\mathfrak{B}(V) \cap \mathfrak{U}(\theta_0)$.*

(c) (Landau representative.) *The orbit contains $U_* = \mathsf{U}(A_*) = \left(e^{\mathrm{i}\eta A_*} U_0\right)^{\mathsf{w}_k(e^{\mathrm{i}\eta A_*})}$, where A_* is real, $R(U_0) D_{U_0}^T A_* = 0$, $\mathcal{Q}(e^{\mathrm{i}\eta A_*}) = e^{\mathrm{i}B}$, and*

$$|A_*|_{C^1} + |\Delta_{U_0} A_*|_\infty + \sup_y [\nabla \tilde{A}_{*,y}]_{\alpha, B(y)} \le C_6(\varepsilon + \beta).$$

The proof gives the Hölder bound on the unit cubes $\Lambda_1(c_y)$ in place of $B(y)$ (Theorem 6.3), and, by Remark 7.2, on every cube $\Lambda_1(z)$. This is the form used in the proof of (e) and in Corollary 30.4. To first order, $A_* = \mathbf{H}B - \mathbf{C}J_{U_0} + O((\varepsilon+\beta)^2)$ in $|\cdot|_{C^1}$.

(d) (Euler–Lagrange equation.) *There is a unique unit-lattice field ω with $\nabla\Phi(A_*) = d\mathcal{Q}(A_*)^T \omega$, where $\Phi(A) = S(e^{\mathrm{i}\eta A} U_0)$ and $d\mathcal{Q}(A)$ is the derivative of $A \mapsto \ell(\mathcal{Q}(e^{\mathrm{i}\eta A}))$. It satisfies $|\omega|_\infty \le C_6(\varepsilon + \beta)$.*

(e) (Regularity of the minimizer.) *U_* is $(M, \varepsilon + C_6(\varepsilon+\beta))$-regular and $U_* \in \mathfrak{U}(C_6(\varepsilon+\beta))$.*

(f) (Analyticity and decay.) *$B \mapsto A_*(B)$ extends to an analytic map on complex unit-lattice fields with $\varepsilon + |B|_\infty < c_6$, with $|A_*(B)|_{C^1} \le C_6(\varepsilon + |B|_\infty)$. For complex B the C^1 bound and the decay below are claimed, but not the second-order and Hölder bounds of (c). For real and complex B, if δB is supported on the bond c, then*

$$|\, dA_*(B)\delta B\,(x)| + |\nabla(dA_*(B)\delta B)(x)| \le C_6 e^{-\gamma'|x-c|}|\delta B|.$$

(g) (Covariance.) *For a real v on $\mathbb{T}$, the critical orbit for V^v is the image of the critical orbit for V under any real u with $u|_{\mathbb{T}} = v$.*

Parts (a)–(c) and (e) are Bałaban's Theorem 1 [8], relative to a reference and on a single torus; the regularity bounds are [8, (9), (10)] with the reference-dependent constant. Parts (d) and (f) are [8, (171), (178)] and [8, Prop. 9, (190)]. The constants do not depend on k, n or η. They depend on L. They may be taken independent of $M \ge M_0$, for ε below the threshold at $M = M_0$; this threshold includes ε_{B} of Lemma 31.3(c), taken at $M = M_0$. Indeed, an (M, ε)-regular reference is (M_0, ε)-regular (Lemma 31.2), the torus operators of Parts II and III do not depend on the regularity scale (Step 0 of the proof of Lemma 31.3), and the proof of the regularity statement in (e) uses only local estimates on the cubes of Definition 18.1. The thresholds that depend on M are those of Parts VI–VIII, for instance a_1 of Theorem 30.3 (whose constant B_3 also depends on M), a_δ of Proposition 31.5 and $\varepsilon_L \le \varepsilon_{\mathrm{cmp}}(M'_L)$ with $M'_L \ge M$ (Theorem 34.4); the constants of Part VII do not (Remark 34.6).

In (c), A_* is the Landau representative of the orbit relative to U_0, and $U_* = \mathsf{U}(A_*)$ is its axial representative. The configuration $e^{\mathrm{i}\eta A_*}U_0$ is gauge equivalent to U_*, but it need not lie in $\mathfrak{B}(V)$, since $\mathsf{w}_k(e^{\mathrm{i}\eta A_*})$ need not be 1 on $\mathbb{T}$.

Part (g) does not need (a): for real u with $u|_{\mathbb{T}} = v$, the map $U \mapsto U^u$ is a bijection from $\mathfrak{B}(V) \cap \mathfrak{U}(\theta_0)$ onto $\mathfrak{B}(V^v) \cap \mathfrak{U}(\theta_0)$, because $\mathfrak{a}^{(k)}(U^u) = \mathfrak{a}^{(k)}(U)^{u|_{\mathbb{T}}}$. It preserves S and critical points, and it maps $\mathfrak{G}_0$-orbits to $\mathfrak{G}_0$-orbits, since $u\mathfrak{G}_0 u^{-1} = \mathfrak{G}_0$. The rest of this part proves (a)–(f). The proof has four steps:

1. Reduction to the Landau chart, by Corollary 24.3 (Section 25.3).
2. Linearization of the constraint and solution by contraction (Section 26).
3. Convexity, which gives minimality (Section 26).
4. The Euler–Lagrange equation, which shows that the constructed point is critical on $\mathfrak{B}(V)$ and gives the second-order bounds (Section 27).

25.3 Reduction to the Landau chart

Let $\theta_0 \le c_4/3$ and $\varepsilon + \beta \le c_6 \le c_4/3$, so that Theorem 24.2 and Corollary 24.3 apply with $\theta = \theta_0$. Let

$$\mathcal{M}_V = \{A \text{ real}: \ RD^T A = 0,\ \ell(\mathcal{Q}(e^{\mathrm{i}\eta A})) = B,\ |A|_{C^1} < \rho_1\}, \qquad \rho_1 = 2B_1(\varepsilon + \theta_0 + \beta).$$

Lemma 25.2. *(a) $\mathcal{M}_V$ is a smooth manifold, and $\Phi = S \circ \mathsf{U}$ on $\mathcal{M}_V$, with U as in Corollary 24.3.*

(b) Every U in a critical orbit in $\mathfrak{B}(V) \cap \mathfrak{U}(\theta_0)$ has an axial representative $\mathsf{U}(A)$ with $A \in \mathcal{M}_V$, and A is a critical point of $\Phi|_{\mathcal{M}_V}$.

(c) For every $U \in \mathfrak{B}(V) \cap \mathfrak{U}(\theta_0)$ there is $A \in \mathcal{M}_V$ with $S(U) = \Phi(A)$.

Proof. (a) The map $A \mapsto \ell(\mathcal{Q}(e^{\mathrm{i}\eta A}))$, restricted to the slice $\{RD^T A = 0\}$, has derivative $\mathbf{Q} + dC_k(A)$. This is onto: $\mathbf{QH} = I$, $\mathbf{H}$ maps into the slice, and $|dC_k(A)\mathbf{H}| \le C\rho_1 < 1$ in the

sup norm by Proposition 22.6(b). Hence $\mathcal{M}_V$ is a submanifold. The identity $\Phi = S \circ \mathsf{U}$ holds by gauge invariance of S.

(b), (c) The orbit of U has a unique axial representative (Lemma 22.2(c)). By Corollary 24.3 it equals $\mathsf{U}(A)$ for some $A \in \mathcal{S}_V$, and by Theorem 24.2, $|A|_{C^1} \le B_1(\varepsilon+\theta_0+\beta) < \rho_1$, so $A \in \mathcal{M}_V$. For (b), let $A(t)$ be a smooth curve in $\mathcal{M}_V$ with $A(0) = A$. Since $\rho_1 \le \frac{4}{3}B_1 c_4 \le \frac{2}{3}r_u$, the configurations $\mathsf{U}(A(t))$ are defined, real and smooth in t, and $\mathsf{U}(A(t))^{(k)} = \mathcal{Q}(e^{\mathrm{i}\eta A(t)})W_k = V$. At $t = 0$ the curvature is that of U, at most $\theta_0 \le c_4/3 < c_1$; by continuity, $\mathsf{U}(A(t)) \in \mathfrak{U}_0(c_1)$ for small $|t|$. So $\mathsf{U}(A(t))$ is a curve in $\mathfrak{B}(V)$ through the critical point $\mathsf{U}(A)$, and $\frac{d}{dt}\Phi(A(t))\big|_{t=0} = \frac{d}{dt}S(\mathsf{U}(A(t)))\big|_{t=0} = 0$. □

By Lemma 25.2(b), uniqueness in (a) follows once we show that $\Phi|_{\mathcal{M}_V}$ has at most one critical point. Existence requires the converse: the critical point of $\Phi|_{\mathcal{M}_V}$ that we construct gives a critical point of $S|_{\mathfrak{B}(V)}$. The converse is not a formal consequence of (b), because U involves the gauge transformation w_k, which depends on A. It is proved in Section 27 via the Euler–Lagrange equation.

26 Solution in the Landau chart

We keep the setting of Theorem 25.1. Write $\mathbf{H} = \mathbf{H}(U_0)$ and $\mathbf{C} = \mathbf{C}(U_0)$ for the minimizer map and the constrained Green function of the quadratic form $q(A, A') = \langle DA, DA'\rangle$ (Theorem 19.9, with $W = 0$), and

$$\mathcal{X} = \{A : \ RD^T A = 0, \ \mathbf{Q}A = 0\}$$

for the tangent space of the constraint in Bałaban's gauge. By Theorem 16.1, $\mathbf{QH} = I$, $RD^T\mathbf{H} = 0$, $\mathbf{C}$ maps into $\mathcal{X}$, and for every f, $\mathbf{C}f$ is the unique $A \in \mathcal{X}$ with $q(k, A) = \langle k, f\rangle$ for all $k \in \mathcal{X}$. Moreover $q(k, \mathbf{H}B) = 0$ for $k \in \mathcal{X}$.

26.1 Linearizing the constraint

Lemma 26.1 (Linearizing transformation [8, Sect. C])**.** *There are $r_2, C > 0$ such that for complex A' with $|A'|_{C^1} \le r_2$ the equation $X = C_k(A' - \mathbf{H}X)$ has exactly one solution $X = \mathcal{D}(A')$ with $|X|_\infty \le r_2$. The map $\mathcal{D}$ is analytic, real for real A', and*

$$|\mathcal{D}(A')|_\infty \le C|A'|_\infty^2, \qquad |d\mathcal{D}(A')h(c)| \le C|A'|_\infty \|h\|_{S'_c}, \qquad |d^2\mathcal{D}(A')[h,h](c)| \le C\|h\|_{S'_c}^2,$$

where S'_c is a weighted neighbourhood of c: $\|h\|_{S'_c}^2 = \sum_y e^{-\gamma|y-c_-|}\|h\|_{B(y)}^2$. Moreover $d\mathcal{D}(A')$ has a kernel, $d\mathcal{D}(A')h(c) = \eta^d \sum_b K_{\mathcal{D}}(c,b)h(b)$, with $|K_{\mathcal{D}}(c,b)| \le C|A'|_\infty e^{-\gamma|c_- - b_-|}$; in particular $|d\mathcal{D}(A')^T\omega|_\infty \le C|A'|_\infty|\omega|_\infty$. For every unit-lattice field ω and every fine field h,

$$\big|(d^2\mathcal{D}(A')[h,\cdot])^T\omega\big|_{B(y)} \le C|\omega|_\infty \sum_{y'} e^{-\gamma|y-y'|}|h|_{B(y')} \qquad \text{for all } y \in \mathbb{T}, \tag{26.1}$$

where $|f|_{B(y)}$ is the maximum of $|f(b)|$ over the bonds b starting in $B(y)$, and transposes refer to the bilinear pairings of Section 2: with the factor η^d on $\mathbb{T}_\eta$ and without it on $\mathbb{T}$. The map $A' \mapsto A = A' - \mathbf{H}\mathcal{D}(A')$ satisfies

$$\ell(\mathcal{Q}(e^{\mathrm{i}\eta A})) = \mathbf{Q}A', \qquad RD^T A = RD^T A'.$$

It is a bijection from $\{A' : |A'|_{C^1} < r_2\}$ onto a set containing $\{A : |A|_{C^1} < r_2/2\}$, with analytic inverse.

Proof. By Proposition 22.6(b) and $|\mathbf{H}X|_{C^1} \le C_H|X|_\infty$ (Lemma 24.1(d)),

$$|C_k(A' - \mathbf{H}X) - C_k(A' - \mathbf{H}X')| \le C(|A'| + C_H|X| + C_H|X'|)C_H|X - X'|.$$

So $X \mapsto C_k(A' - \mathbf{H}X)$ is a contraction on $|X| \le r_2$ for r_2 small, mapping this ball into itself. Its fixed point is analytic in A', being a uniform limit of analytic iterates. The bound $|\mathcal{D}| \le C|A'|^2$ follows from $|C_k(A)| \le C|A|^2$.

Derivatives. Put $A = A' - \mathbf{H}\mathcal{D}(A')$, $P = dC_k(A)$, $T = (I + P\mathbf{H})^{-1}$ and $L = I - \mathbf{H}TP$. Differentiating the fixed-point equation once and twice gives

$$d\mathcal{D} = TP, \qquad dA = L, \qquad d^2\mathcal{D}(A')[h,k] = T\, d^2C_k(A)[Lh, Lk].$$

Here T is a unit-lattice operator that is a small perturbation of I with decaying kernel, since $P\mathbf{H}$ has decay $(C|A|_\infty, \gamma)$. Proposition 22.6(c), together with the decay of $\mathbf{H}$ and T, gives the bounds in terms of $\|h\|_{S_c'}$ and the kernel bound.

The mixed bound (26.1). For r_2 small, $|A|_\infty \le (1 + C_H)r_2 \le c_3/2$. By Proposition 22.6(a),(c), the kernel $K_A(c,b)$ of $dC_k(A)$ is an analytic function of A on $\{|A|_\infty \le c_3\}$, bounded there by Cc_3, and $K_A(c,b) = 0$ unless b starts in S_c. Moreover $C_k(A)(c)$ depends only on the values of A on the bonds starting in S_c. Let v be a fine field. By the Cauchy estimate in $t \mapsto K_{A+tv}(c,b)$, the kernel of $d^2C_k(A)[v,\cdot]$, defined by $d^2C_k(A)[v,k](c) = \eta^d \sum_b \partial_v K_A(c,b)k(b)$, satisfies $|\partial_v K_A(c,b)| \le C \max_{S_c}|v|$, with the same support; here $\max_{S_c}$ is over the bonds starting in S_c. The transpose has no factor η^d, so $(d^2C_k(A)[v,\cdot])^T\omega(b) = \sum_c \partial_v K_A(c,b)^T\omega(c)$. Every fine bond starts in S_c for at most $2d$ bonds c of $\mathbb{T}$. Hence

$$\big|(d^2C_k(A)[v,\cdot])^T\omega\big|_{B(y)} \le C|\omega|_\infty \max_{|y'-y|_\infty \le 1} |v|_{B(y')}.$$

Now $(d^2\mathcal{D}(A')[h,\cdot])^T\omega = L^T\big(d^2C_k(A)[Lh,\cdot]\big)^T T^T\omega$. The operators T and T^T have decaying kernels, so $|T^T\omega|_\infty \le C|\omega|_\infty$. The operators P and P^T are local and bounded by $C|A|_\infty$. The operators $\mathbf{H}$ and $\mathbf{H}^T$ satisfy block bounds of the form $|\mathbf{H}X|_{B(y)} \le C\sum_c e^{-\gamma|y-c_-|}|X(c)|$ and $|\mathbf{H}^T f(c)| \le C\sum_{y'} e^{-\gamma|c_- - y'|}|f|_{B(y')}$, by the block form of the pointwise bounds in the proof of Lemma 24.1; the second follows from the first, since $\mathbf{H}^T f(c) = \langle \mathbf{H}\delta_c, f\rangle$ componentwise and a block has volume 1. Hence $|Lh|_{B(y)} \le C\sum_{y'} e^{-\gamma|y-y'|}|h|_{B(y')}$, and $L^T = I - P^TT^T\mathbf{H}^T$ preserves bounds of this form, with a smaller rate. This gives (26.1), after decreasing γ.

The identities follow from $\ell(\mathcal{Q}(e^{i\eta A})) = \mathbf{Q}A + C_k(A)$, $\mathbf{QH} = I$ and $RD^T\mathbf{H} = 0$. The inverse is explicit: at the fixed point $\mathcal{D}(A') = C_k(A)$ with $A = A' - \mathbf{H}\mathcal{D}(A')$, so $A' = A + \mathbf{H}C_k(A)$. Conversely, for $|A|_{C^1} < r_2/2$ the field $A' = A + \mathbf{H}C_k(A)$ satisfies $|A'|_{C^1} < r_2$, and $X = C_k(A)$ is a fixed point for this A'; by uniqueness $\mathcal{D}(A') = C_k(A)$. □

By Lemma 26.1, the manifold $\mathcal{M}_V$ is mapped analytically onto (a subset of) the affine space

$$\mathcal{K}_B = \{A' :\ RD^TA' = 0,\ \mathbf{Q}A' = B\} = \mathbf{H}B + \mathcal{X}.$$

Critical points of $\Phi|_{\mathcal{M}_V}$ correspond to critical points of

$$\mathfrak{F}(A') = \Phi(A' - \mathbf{H}\mathcal{D}(A')) \quad \text{on } \mathcal{K}_B \cap \{|A'|_{C^1} < r_2\},$$

provided $\rho_1 \le r_2/2$, which we assume.

26.2 The functional

By Proposition 23.3, with $J = J_{U_0}$,

$$\mathfrak{F}(A') = S(U_0) + \langle A', J\rangle + \tfrac{1}{2}\|DA'\|^2 + \mathcal{V}(A'), \tag{26.2}$$

$$\mathcal{V}(A') = -\langle \mathbf{H}\mathcal{D}(A'), J\rangle - \langle DA', D\mathbf{H}\mathcal{D}(A')\rangle + \tfrac{1}{2}\|D\mathbf{H}\mathcal{D}(A')\|^2 + \mathcal{N}(A' - \mathbf{H}\mathcal{D}(A')).$$

Lemma 26.2 (Bounds on $\mathcal{V}$ [8, Prop. 4])**.** *There are $r_3 \le r_2$ and C such that for complex A' with $|A'|_{C^1} \le r \le r_3$ (in the proof below, A' on the slice $RD^TA' = 0$; off the slice use $\mathbf{H}^TD^TDA' = (D\mathbf{H})^TDA'$, bounded by $C|DA'|_\infty$ by the kernel decay of $D\mathbf{H}$):*

(a) $|\nabla\mathcal{V}(A')|_\infty \le C(\varepsilon + r)|A'|_{C^1}$, and $|\nabla\mathcal{V}(A') - \nabla\mathcal{V}(A'')|_\infty \le C(\varepsilon + r)|A' - A''|_{C^1}$ if also $|A''|_{C^1} \le r$;

(b) for real A' and real h: $|d^2\mathcal{V}(A')[h,h]| \le C(\varepsilon + r)(\|h\|^2 + \|\nabla h\|^2)$.

Proof. We use two identities. First, $\mathfrak{A}_\pi\mathbf{H} = \mathbf{Q}^T\mathbf{\Gamma}_\pi$, $\Pi\mathbf{H} = \mathbf{H}$ and $\mathbf{Q}\Pi = \mathbf{Q}$ give $\Pi^TD^TD\mathbf{H} = \mathbf{Q}^T(\mathbf{\Gamma}_\pi - a)$. Hence

$$D^TD\mathbf{H} = \mathbf{Q}^T(\mathbf{\Gamma}_\pi - a) + S^T\mathfrak{F}^TD\mathbf{H},$$

since $\Pi = I - DS$ and $D^TD^T = \mathfrak{F}^T$. By Lemma 24.1 and $|\mathfrak{F}| \le C\varepsilon$, $|D^TD\mathbf{H}X|_\infty \le C|X|_\infty$. Second, transposing the first identity, $\mathbf{H}^TD^TD = (\mathbf{\Gamma}_\pi - a)\mathbf{Q} + \mathbf{H}^TD^T\mathfrak{F}S$. On the slice $SA' = G'RD^TA' = 0$, so $\mathbf{H}^TD^TDA' = (\mathbf{\Gamma}_\pi - a)\mathbf{Q}A'$ for $A' \in \mathcal{K}_B$, which is bounded by $C|A'|_\infty$. Moreover $|\mathbf{H}^Tf|_\infty \le C|f|_\infty$, since $\mathbf{H}^T = \mathbf{\Gamma}_\pi\mathbf{Q}\mathbf{G}$ has a decaying kernel.

(a) The gradient is

$$\nabla\mathcal{V} = -d\mathcal{D}^T\mathbf{H}^TJ - D^TD\mathbf{H}\mathcal{D} - d\mathcal{D}^T\mathbf{H}^TD^TDA' + d\mathcal{D}^T\mathbf{H}^TD^TD\mathbf{H}\mathcal{D} + (I - \mathbf{H}d\mathcal{D})^T\nabla\mathcal{N}(A' - \mathbf{H}\mathcal{D}).$$

Each term is bounded using the identities above, $|J| \le \varepsilon$, Lemma 26.1, Proposition 23.3(a), and the kernel bound $|d\mathcal{D}(A')^T\omega|_\infty \le C|A'|_\infty|\omega|_\infty$ of Lemma 26.1. The Lipschitz bound follows from the Cauchy estimate on the complex ball of radius $2r$, as in Proposition 23.3(b).

(b) $d^2\mathcal{V}(A')[h,h]$ consists of three kinds of terms:

- terms containing $d\mathcal{D}[h]$ once or twice, bounded by $C|A'|\|h\|^2$ or $C|A'|\|h\|\|\nabla h\|$ by Lemma 26.1 and the L^2 boundedness of $\mathbf{H}$, $D\mathbf{H}$;
- terms $\langle d^2\mathcal{D}[h,h], \omega'\rangle$ with $\omega' = -\mathbf{H}^T(J + D^TDA + \nabla\mathcal{N}(A))$, $A = A' - \mathbf{H}\mathcal{D}$. By the identities above $|\omega'|_\infty \le C(\varepsilon + r)$, and $\sum_c |d^2\mathcal{D}[h,h](c)| \le C\|h\|^2$ because the weights $e^{-\gamma|y-c_-|}$ are summable;
- $d^2\mathcal{N}(A)[(I - \mathbf{H}d\mathcal{D})h, (I - \mathbf{H}d\mathcal{D})h]$, bounded by Proposition 23.3(c).

Altogether this gives (b). □

26.3 The equation and its solution

Proposition 26.3 ([8, Props. 5, 6])**.** *Let C_H, C_C be constants with $|\mathbf{H}B|_{C^1} \le C_H|B|_\infty$ and $|\mathbf{C}f|_{C^1} \le C_C|f|_\infty$ (Lemma 24.1(d)), and $K = 4C_C(1 + C_H)$. There is $c_7 > 0$ such that, if $\varepsilon + \beta \le c_7$, $\theta_0 \le c_7$ and $\rho_2 = 2\rho_1 + C_H\beta + K(\varepsilon + \beta)$ (with ρ_1 from Section 25.3), the following holds. A point $A' = A_1 + \mathbf{H}B$ with $A_1 \in \mathcal{X}$ and $|A'|_{C^1} < r_3$ is a critical point of $\mathfrak{F}|_{\mathcal{K}_B}$ iff*

$$A_1 = \mathcal{T}(A_1) := -\mathbf{C}(J + \nabla\mathcal{V}(A_1 + \mathbf{H}B)). \tag{26.3}$$

The map $\mathcal{T}$ is a contraction on $\{A_1 \in \mathcal{X} : |A_1|_{C^1} \le \rho_2\}$. Its unique fixed point $A_{1,*}$ satisfies $|A_{1,*}|_{C^1} \le K(\varepsilon + \beta)$, and $A_{1,*} = -\mathbf{C}J + O((\varepsilon + \beta)^2)$. In particular $\mathfrak{F}|_{\mathcal{K}_B}$ has exactly one critical point $A'_* = A_{1,*} + \mathbf{H}B$ with $|A'_* - \mathbf{H}B|_{C^1} \le \rho_2$.

Proof. The tangent space of $\mathcal{K}_B$ is $\mathcal{X}$. By (26.2), A' is critical iff $\langle k, J + D^TDA' + \nabla\mathcal{V}(A')\rangle = 0$ for all $k \in \mathcal{X}$. Since $\langle k, D^TD\mathbf{H}B\rangle = q(k, \mathbf{H}B) = 0$ for $k \in \mathcal{X}$, this is equivalent to $q(k, A_1) = -\langle k, J + \nabla\mathcal{V}(A')\rangle$ for all $k \in \mathcal{X}$, i.e. to (26.3), by the characterization of $\mathbf{C}$.

By Lemma 24.1(d), $|\mathbf{C}f|_{C^1} \le C_C|f|_\infty$. For $|A_1|_{C^1} \le \rho$ we have $|A'|_{C^1} \le \rho + C_H\beta$. By Lemma 26.2(a),

$$|\mathcal{T}(A_1)|_{C^1} \le C_C\varepsilon + C_CC(\varepsilon + \rho + C_H\beta)(\rho + C_H\beta),$$
$$|\mathcal{T}(A_1) - \mathcal{T}(A_1')|_{C^1} \le C_CC(\varepsilon + \rho + C_H\beta)|A_1 - A_1'|_{C^1}.$$

For $\rho \le \rho_2$, where $\rho_2 \le C(\varepsilon+\theta_0+\beta)$, the Lipschitz constant is at most $\frac12$ if c_7 is small. At $A_1 = 0$ the first bound gives $|\mathcal{T}(0)|_{C^1} \le C_C\varepsilon + C_C C(\varepsilon + C_H\beta)C_H\beta \le \frac12 K(\varepsilon+\beta)$ if c_7 is small. Hence, for $K(\varepsilon+\beta) \le \rho \le \rho_2$ and $|A_1|_{C^1} \le \rho$,

$$|\mathcal{T}(A_1)|_{C^1} \le |\mathcal{T}(0)|_{C^1} + \tfrac12 |A_1|_{C^1} \le \rho.$$

With $\rho = \rho_2$, which is at least $K(\varepsilon+\beta)$ by its definition, $\mathcal{T}$ maps the ball of radius ρ_2 into itself, and it is a contraction there. With $\rho = K(\varepsilon+\beta)$, its fixed point lies in the ball of radius $K(\varepsilon+\beta)$. The first-order formula follows from $|\nabla\mathcal{V}(A')| \le C(\varepsilon+\beta)^2$ at the fixed point. □

Put $A_* = A'_* - \mathbf{H}\mathcal{D}(A'_*)$. By Lemma 26.1, A_* is the unique critical point of $\Phi|_{\mathcal{M}_V}$; if $|A|_{C^1} < \rho_1$, then $|A'|_{C^1} \le \rho_1 + C\rho_1^2 \le 2\rho_1$, so $A_1 = A' - \mathbf{H}B$ satisfies $|A_1|_{C^1} \le \rho_2$. Moreover $|A_*|_{C^1} \le K'(\varepsilon+\beta)$ with $K' = K + C_H + 1$, and we assume $c_6 \le B_1\theta_0/K'$, so that $|A_*|_{C^1} < \rho_1$ and $A_* \in \mathcal{M}_V$. Also $A_* = \mathbf{H}B - \mathbf{C}J + O((\varepsilon+\beta)^2)$. This proves the first-order formula in Theorem 25.1(c), and, by Lemma 25.2(b), the uniqueness in (a).

26.4 Convexity and minimality

Lemma 26.4 (Covariant Hodge inequality)**.** *Let U_0 be real with curvature at most ε. For every real gauge field h, $\big|\|Dh\|^2 + \|D^Th\|^2 - \|\nabla h\|^2\big| \le C\varepsilon\|h\|^2$.*

Proof. Let D_ν act on the components h_μ as on scalar fields. Then $\|Dh\|^2 = \|\nabla h\|^2 - \sum_\mu \|D_\mu h_\mu\|^2 - \sum_{\mu\neq\nu}\langle D_\mu h_\nu, D_\nu h_\mu\rangle$ and $\|D^Th\|^2 = \sum_\mu \|D_\mu^T h_\mu\|^2 + \sum_{\mu\neq\nu}\langle D_\mu^T h_\mu, D_\nu^T h_\nu\rangle$. Since $R(U_0)$ is an isometry, $|D_\mu^T f(x)| = |D_\mu f(x-\eta e_\mu)|$, so the diagonal terms cancel. Summation by parts turns the remaining terms into $\sum_{\mu\neq\nu}\langle h_\mu, [D_\mu, D_\nu^T]h_\nu\rangle$. Expanding both products, all terms cancel except those at $x+\eta e_\mu - \eta e_\nu$:

$$[D_\mu, D_\nu^T]f(x) = \eta^{-2}\Big(R(U_\mu(x)U_\nu(x+\eta e_\mu-\eta e_\nu)^{-1}) \\ - R(U_\nu(x-\eta e_\nu)^{-1}U_\mu(x-\eta e_\nu))\Big) f(x+\eta e_\mu-\eta e_\nu).$$

The two transports go around the two halves of a plaquette, so their quotient is conjugate to its holonomy. Hence the operator norm is at most $C\varepsilon$, by Lemma 21.1. □

Lemma 26.5 (Coercivity on the tangent space)**.** *There is $c_X > 0$ such that $\|Dh\|^2 \ge c_X(\|h\|^2 + \|\nabla h\|^2)$ for all real $h \in \mathcal{X}$.*

Proof. For $h \in \mathcal{X}$, $\Pi h = h$, $RD^Th = 0$ and $\mathbf{Q}h = 0$, so $\langle h, \mathfrak{A}_\pi h\rangle = \|Dh\|^2$. By Theorem 19.6, $\langle h, \mathfrak{A}_\pi h\rangle \ge c_*(\|h\|^2 + \|Dh\|^2 + \|D^Th\|^2)$. Finally $\|Dh\|^2 + \|D^Th\|^2 \ge \|\nabla h\|^2 - C\varepsilon\|h\|^2$ by Lemma 26.4. □

Proposition 26.6 (Minimality)**.** *If $\varepsilon+\beta \le c_7$ and $\theta_0 \le c_7$ (after decreasing c_7; Lemma 26.7 below decreases it once more), then $S(U) \ge S(U_*)$ for all $U \in \mathfrak{B}(V)\cap\mathfrak{U}(\theta_0)$, where $U_* = \mathsf{U}(A_*)$. Equality holds iff U lies in the $\mathfrak{G}_0$-orbit of U_*.*

Proof. By Lemma 25.2(c) and Lemma 26.1, $S(U) = \mathfrak{F}(A')$ for some $A' = A_1 + \mathbf{H}B \in \mathcal{K}_B$ with $|A_1|_{C^1} \le \rho_2$. The set $\{A_1 + \mathbf{H}B :\ A_1 \in \mathcal{X},\ |A_1|_{C^1} \le \rho_2\}$ is convex and contains A'_*; on it $|A'|_{C^1} \le \rho_2 + C_H\beta \le r_3$. For $h = A' - A'_* \in \mathcal{X}$ and $A'_t = A'_* + th$,

$$\frac{d^2}{dt^2}\mathfrak{F}(A'_t) = \|Dh\|^2 + d^2\mathcal{V}(A'_t)[h,h] \ge (c_X - C(\varepsilon+\rho_2))(\|h\|^2 + \|\nabla h\|^2),$$

by Lemmas 26.5 and 26.2(b). This is positive for c_7 small and $h \neq 0$. Since A'_* is critical, $\mathfrak{F}(A') > \mathfrak{F}(A'_*)$ unless $A' = A'_*$. Finally $\mathfrak{F}(A'_*) = \Phi(A_*) = S(U_*)$. If $\mathfrak{F}(A') = \mathfrak{F}(A'_*)$, then $A' = A'_*$, and by Corollary 24.3 the axial representative of U is U_*. □

The convexity argument requires L^2 bounds on the second derivatives of the nonlinear averaging, Proposition 22.6(c). These hold because the averaging is built one scale at a time. At each scale the contours have only $O(L)$ bonds, and the field entering them is already an average over the finer scales. For a one-shot nonlinear averaging over blocks of side 1, the quadratic terms would contain integrals over single lattice lines, and the corresponding bounds would fail as $\eta \to 0$.

Proposition 26.6 proves (b) of Theorem 25.1, once we know that U_* lies in $\mathfrak{B}(V) \cap \mathfrak{U}(\theta_0)$ and is critical there. Membership in $\mathfrak{B}(V)$ holds by construction; the remaining two facts are proved in the next section.

At a critical reference the Hessian of the Lagrangian is coercive on all fine fields, not only on $\mathcal{X}$, once the gauge directions are fixed by the Landau and averaging terms. In the four-dimensional note this is used in the one-loop analysis; it is not used below.

Lemma 26.7 (The Lagrangian Hessian on all fields)**.** *In the setting of Theorem 25.1, suppose that $\beta = 0$ and $A_* = 0$, i.e. $V = W_k$ and U_0 is critical on $\mathfrak{B}(V)$. Let ω_0 be the multiplier of Proposition 27.2 at $A_* = 0$, so that $J_{U_0} = \nabla\Phi(0) = \mathbf{Q}_k^T\omega_0$. For real fine fields h put*

$$q_2(h,h) = \|Dh\|^2 + d^2\mathcal{N}(0)[h,h] - \langle \omega_0, d^2C_k(0)[h,h]\rangle,$$
$$\langle h, \mathbf{\Delta}_* h\rangle = q_2(h,h) + \|RD^Th\|^2 + a\|\mathbf{Q}_k h\|^2,$$

with $D = D_{U_0}$, $R = R(U_0)$, $\mathcal{N}$ of Proposition 23.3 and C_k of Proposition 22.6. There is C', depending only on the constants of Propositions 23.3, 22.6 and 27.2, Theorem 18.2 and Lemmas 12.3 and 26.4, such that

$$\langle h, \mathbf{\Delta}_* h\rangle \geq (c_* - C'\varepsilon)(\|h\|^2 + \|Dh\|^2 + \|D^Th\|^2) \qquad \text{for all real } h, \tag{26.4}$$

with c_* of Theorem 19.6; more precisely, $|\langle h, (\mathbf{\Delta}_* - \mathfrak{A}_\pi)h\rangle| \leq C'\varepsilon(\|h\|^2 + \|Dh\|^2 + \|D^Th\|^2)$, with $\mathfrak{A}_\pi$ of Section 15 at U_0. We decrease c_7 so that c_7 is at most the thresholds of Theorems 18.2 and 19.6 and of Proposition 23.3, and $C'c_7 \leq \frac{1}{2}c_*$. Then, for $\varepsilon \leq c_7$, $\langle h, \mathbf{\Delta}_* h\rangle \geq \frac{1}{2}c_*(\|h\|^2 + \|Dh\|^2 + \|D^Th\|^2)$ for all real h, and $q_2(h,h) \geq \frac{1}{2}c_*(\|h\|^2 + \|Dh\|^2)$ for $h \in \mathcal{X}$.

Proof. Since $E = 0$ (Proposition 22.7(b)), and since the potential of Part II is $W = 0$ here, as in the proof of Lemma 26.5, Lemma 15.1(c) gives $\mathbf{Q}_k\Pi = \mathbf{Q}_k$ and $D\Pi = D - \mathfrak{F}S$, with $S = G'RD^T$. So

$$\langle h, \mathfrak{A}_\pi h\rangle = \|Dh - \mathfrak{F}Sh\|^2 + \|RD^Th\|^2 + a\|\mathbf{Q}_k h\|^2,$$

and

$$\langle h, \mathbf{\Delta}_* h\rangle = \langle h, \mathfrak{A}_\pi h\rangle + 2\langle Dh, \mathfrak{F}Sh\rangle - \|\mathfrak{F}Sh\|^2 + d^2\mathcal{N}(0)[h,h] - \langle \omega_0, d^2C_k(0)[h,h]\rangle.$$

Theorems 18.2 and 19.6 apply to the averaging of Part IV, by Corollary 22.8. We bound the last four terms.

- The coefficients $c_{\mu\nu}$ of $\mathfrak{F}$ (Lemma 12.3) change under a real gauge transformation by multiplication with orthogonal matrices on both sides. In a local gauge in which (B) holds, which exists by regularity, $|c_{\mu\nu}| \leq C\varepsilon$. So $|(\mathfrak{F}\lambda)_{\mu\nu}(x)| \leq C\varepsilon|\lambda(x+\eta e_\mu + \eta e_\nu)|$, by the exact formula of Lemma 12.3. Summing over the periodic lattice gives $\|\mathfrak{F}\lambda\| \leq C\varepsilon\|\lambda\|$. Moreover $\|Sh\| \leq C\|h\|$ (Theorem 18.2(d) with $X = \mathbb{T}_\eta$). Hence $|2\langle Dh, \mathfrak{F}Sh\rangle - \|\mathfrak{F}Sh\|^2| \leq C\varepsilon(\|h\|^2 + \|Dh\|^2)$ for $\varepsilon \leq 1$.
- By Proposition 23.3(c) at $A = 0$, $|d^2\mathcal{N}(0)[h,h]| \leq C\varepsilon(\|h\|^2 + \|\nabla h\|^2)$, and by Lemma 26.4, $\|\nabla h\|^2 \leq \|Dh\|^2 + \|D^Th\|^2 + C\varepsilon\|h\|^2$.
- By Proposition 27.2 with $\beta = 0$, $|\omega_0|_\infty \leq C\varepsilon$. By Proposition 22.6(c), $|d^2C_k(0)[h,h](c)| \leq C\|h\|^2_{S_c}$. Every fine bond starts in exactly one block, and every block lies in S_c for $2d$ bonds c, so $\sum_c \|h\|^2_{S_c} = 2d\|h\|^2$. Hence $|\langle \omega_0, d^2C_k(0)[h,h]\rangle| \leq C\varepsilon\|h\|^2$.

By Theorem 19.6 with $X = \mathbb{T}_\eta$, where $m_{\mathbb{T}_\eta} = 0$, $\mathfrak{A}^{\mathbb{T}_\eta} = \mathfrak{A}_\pi$ and $\|\!|h|\!\|^2_{\mathbb{T}_\eta,+} = \|h\|^2 + \|Dh\|^2 + \|D^T h\|^2$, we have $\langle h, \mathfrak{A}_\pi h\rangle \geq c_*(\|h\|^2 + \|Dh\|^2 + \|D^T h\|^2)$. This gives the bound on $\boldsymbol{\Delta}_* - \mathfrak{A}_\pi$ and (26.4). The constants of the cited results do not depend on c_7, θ_0 or c_6, so the additional condition on c_7 is compatible with the order of the constants at the end of the proof of Theorem 25.1. For $h \in \mathcal{X}$, $\mathbf{Q}_k h = 0$ and $RD^T h = 0$, so $\langle h, \boldsymbol{\Delta}_* h\rangle = q_2(h,h)$. □

27 The Euler–Lagrange equation and regularity of the minimizer

Let A_* be the critical point of $\Phi|_{\mathcal{M}_V}$ constructed in Section 26, and $U_* = \mathsf{U}(A_*)$. In this section we show that U_* is a critical point of $S|_{\mathfrak{B}(V)}$, derive its Euler–Lagrange equation, and prove the second-order bounds and the regularity of U_*. The key input is a gauge covariance property of the gauge-adapted averaging.

27.1 Covariance of the gauge-adapted averaging

Lemma 27.1. *Let $\tilde U$ be a small real fine relative configuration and g a real gauge transformation close to* 1.

(a) $\mathcal{Q}(\tilde U) = \big((\tilde U U_0)^{(k)}\big)^{\mathsf{w}_k(\tilde U)|_{\mathbb{T}}}\, W_k^{-1}$.

(b) $\mathcal{Q}(\tilde U^{(g)}) = \mathcal{Q}(\tilde U)^{(h_{\tilde U}(g))}$, *with* $h_{\tilde U}(g) = \big(\mathsf{w}_k(\tilde U^{(g)})\, g\, \mathsf{w}_k(\tilde U)^{-1}\big)|_{\mathbb{T}}$.

(c) $h_1(g) = \mathfrak{M}^{(k)}(g)$.

Proof. (a) By Theorem 22.4(a), $\mathcal{Q}(\tilde U) = (\tilde U^{(\mathsf{w}_k)})^{[k]} = ((\tilde U U_0)^{\mathsf{w}_k})^{(k)} W_k^{-1}$, and $\mathfrak{a}^{(k)}$ is covariant (Lemma 21.5). (b) Both $\mathsf{w}_k(\tilde U^{(g)})$ and $\mathsf{w}_k(\tilde U) g^{-1}$ move $\tilde U^{(g)}$ into the axial gauge. Everything is real, so Theorem 22.4(b) applies, and $\mathsf{w}_k(\tilde U) g^{-1} = \mathcal{E}^{(k)}(h')\, \mathsf{w}_k(\tilde U^{(g)})$ for some h' on $\mathbb{T}$. Hence

$$(\tilde U^{(g)})^{(\mathsf{w}_k(\tilde U^{(g)}))} = \big(\tilde U^{(\mathsf{w}_k(\tilde U))}\big)^{(\mathcal{E}^{(k)}(h')^{-1})},$$

and taking k-fold relative averages gives $\mathcal{Q}(\tilde U^{(g)}) = \mathcal{Q}(\tilde U)^{(h'^{-1})}$ with $h'^{-1} = h_{\tilde U}(g)$. (c) By Theorem 22.4(e), $\mathsf{w}_k(1^{(g)}) = \mathcal{E}^{(k)}(\mathfrak{M}^{(k)}(g)) g^{-1}$, and $\mathsf{w}_k(1) = 1$. □

For a real fine field A with $|A|_{C^1}$ small, and scalar fields λ on $\mathbb{T}_\eta$ and μ on $\mathbb{T}$, define the linear maps

$$\Xi_A \lambda = \tfrac{d}{dt}\Big|_0 \mathcal{A}_t, \qquad e^{\mathrm{i}\eta \mathcal{A}_t} = (e^{\mathrm{i}\eta A})^{(e^{\mathrm{i}t\lambda})},$$
$$\Theta_A \lambda = -\mathrm{i}\tfrac{d}{dt}\Big|_0 h_{e^{\mathrm{i}\eta A}}(e^{\mathrm{i}t\lambda}), \qquad \Lambda_A \mu = \tfrac{d}{dt}\Big|_0 \ell(\mathcal{Q}(e^{\mathrm{i}\eta A})^{(e^{\mathrm{i}t\mu})}).$$

Differentiating Lemma 27.1(b) gives

$$d\mathcal{Q}(A)\, \Xi_A \lambda = \Lambda_A \Theta_A \lambda. \tag{27.1}$$

By (24.1), $\Xi_A \lambda = -D_{U_0}\lambda + \mathrm{i}[\lambda(b_-), A(b)] + \eta\, \mathfrak{b}'$ with $|\mathfrak{b}'| \leq C|A||D\lambda|$. By Lemma 27.1(c) and the proof of Proposition 22.7(b), $\Theta_0 = Q'$. The definitions of Ξ_A, Θ_A, Λ_A make sense for complex A with $|A|_\infty$ small, by the same formulas. By the definition of $h_{\tilde U}$ in Lemma 27.1(b), with $\mathsf{w} = \mathsf{w}_k(e^{\mathrm{i}\eta A})$ evaluated at $y \in \mathbb{T}$,

$$\Theta_A \lambda(y) = R(\mathsf{w}(y))\lambda(y) - \mathrm{i}\,\big(d\mathsf{w}_k(A)[\Xi_A\lambda]\big)(y)\, \mathsf{w}(y)^{-1}.$$

By Proposition 22.6(d), $\mathsf{w}_k(e^{\mathrm{i}\eta A})(y)$ is analytic and bounded for complex A with $|A|_\infty \leq c_3$. So, by the Cauchy estimate, $|d\mathsf{w}_k(A)[h]| \leq C|h|_\infty$ for $|A|_\infty \leq c_3/2$. By (24.1), $\Xi_A\lambda$ is analytic in A with $|\Xi_A\lambda|_\infty \leq C|\lambda|_{C^1}$. Hence Θ_A is analytic in A on $\{|A|_\infty \leq c\}$, with $|\Theta_A\lambda|_\infty \leq C|\lambda|_{C^1}$ there. The Cauchy estimate in A gives $|\Theta_A\lambda - Q'\lambda|_\infty \leq C|A|_\infty|\lambda|_{C^1}$.

27.2 The Euler–Lagrange equation

Proposition 27.2 ([8, (171), (178)])**.** *There is a unique unit-lattice field ω with*

$$\nabla\Phi(A_*) = d\mathcal{Q}(A_*)^T\omega.$$

Moreover $\Lambda_{A_*}^T\omega = 0$, and $|\omega|_\infty \le C(\varepsilon+\beta)$.

Proof. Transversality. Consider the map $\mathcal{O}\lambda = (\Theta_{A_*}\lambda,\, RD^T\Xi_{A_*}\lambda)$. For $A_* = 0$ it is $\mathcal{O}_0\lambda = (Q'\lambda, -RD^TD\lambda)$, which is invertible with inverse $(g,f)\mapsto H'g - G'(f + RD^TDH'g)$, for $f\in\operatorname{ran}R$. Indeed, $\lambda = H'g+\lambda_0$ with $\lambda_0\in\mathcal{N}(Q')$ and $D^TD\lambda_0 = -f - RD^TDH'g\in\operatorname{ran}R$, and $\lambda_0 = G'D^TD\lambda_0$ by Proposition 14.2(c). By Lemma 24.1, the inverse is bounded from sup norms to $|\cdot|_{C^1}$. Moreover $|\mathcal{O}\lambda - \mathcal{O}_0\lambda| \le C|A_*|_{C^1}|\lambda|_{C^1}$ by the bounds above, since $|D^T[\lambda,A_*]| \le C(|D\lambda||A_*| + |\lambda||\nabla A_*|)$. Hence $\mathcal{O}$ is invertible.

Orthogonality. The tangent space of $\mathcal{M}_V$ at A_* is $\ker RD^T\cap\ker d\mathcal{Q}(A_*)$. Let $k\in\ker d\mathcal{Q}(A_*)$, and let $\lambda = \mathcal{O}^{-1}(0,RD^Tk)$. Then $\Theta_{A_*}\lambda = 0$, so $d\mathcal{Q}(A_*)\Xi_{A_*}\lambda = 0$ by (27.1), and $RD^T(k-\Xi_{A_*}\lambda) = 0$. Thus $k-\Xi_{A_*}\lambda$ is tangent to $\mathcal{M}_V$, and $\langle\nabla\Phi(A_*), k-\Xi_{A_*}\lambda\rangle = 0$ since A_* is critical. Moreover $\langle\nabla\Phi(A_*),\Xi_{A_*}\lambda\rangle = 0$, because Φ is gauge invariant: $\Phi(\mathcal{A}_t) = \Phi(A_*)$. Therefore $\nabla\Phi(A_*)\perp\ker d\mathcal{Q}(A_*)$. Since $d\mathcal{Q}(A_*)$ is onto, $\nabla\Phi(A_*) = d\mathcal{Q}(A_*)^T\omega$ for exactly one ω.

The multiplier. For every λ, gauge invariance and (27.1) give $0 = \langle\nabla\Phi(A_*),\Xi\lambda\rangle = \langle\omega,\Lambda\Theta\lambda\rangle = \langle\Lambda^T\omega,\Theta\lambda\rangle$. Since Θ_{A_*} is onto (it is a small perturbation of Q', which has the right inverse H'), $\Lambda^T\omega = 0$.

Bound on ω. For a bond c of $\mathbb{T}$ and $\beta_0\in\mathfrak{g}$ with $|\beta_0| = 1$, let

$$\varphi = \mathbf{H}\big(I + dC_k(A_*)\mathbf{H}\big)^{-1}\delta_c\beta_0,$$

so that $d\mathcal{Q}(A_*)\varphi = \delta_c\beta_0$. By Proposition 22.6 and Lemma 24.1 (weighted version), $|\varphi(x)| + |\nabla\varphi(x)| \le Ce^{-\gamma'|x-c|}$. Then, by Proposition 23.3,

$$\operatorname{tr}(\beta_0\,\omega(c)) = \langle\varphi,\nabla\Phi(A_*)\rangle = \langle\varphi, J+\nabla\mathcal{N}(A_*)\rangle + \langle D\varphi, DA_*\rangle.$$

Hence $|\omega(c)| \le C\big(\varepsilon + C(\varepsilon+\beta)^2 + C(\varepsilon+\beta)\big)$, since $|A_*|_{C^1}\le C(\varepsilon+\beta)$. □

27.3 The constructed point is critical

Proposition 27.3. *U_* is a critical point of $S|_{\mathfrak{B}(V)}$.*

Proof. Let $\mathsf{w}_* = \mathsf{w}_k(e^{i\eta A_*})$ and $U_\flat = e^{i\eta A_*}U_0$, so that $U_* = U_\flat^{\mathsf{w}_*}$. Let $U(t)$ be a smooth curve in $\mathfrak{B}(V)$ with $U(0) = U_*$. For small t write $U(t)^{\mathsf{w}_*^{-1}} = e^{i\eta\mathcal{A}(t)}U_0$ with $\mathcal{A}(0) = A_*$. Then $S(U(t)) = \Phi(\mathcal{A}(t))$. Moreover $(e^{i\eta\mathcal{A}(t)}U_0)^{(k)} = V^{\mathsf{w}_*|_{\mathbb{T}}^{-1}}$ does not depend on t. By Lemma 27.1(a),

$$\mathcal{Q}(e^{i\eta\mathcal{A}(t)}) = V^{g(t)}W_k^{-1} = \mathcal{Q}(e^{i\eta A_*})^{(g(t))},\qquad g(t) = \mathsf{w}_k(e^{i\eta\mathcal{A}(t)})|_{\mathbb{T}}\,\mathsf{w}_*|_{\mathbb{T}}^{-1},$$

with $g(0) = 1$. Differentiating at $t=0$ gives $d\mathcal{Q}(A_*)\mathcal{A}'(0) = \Lambda_{A_*}\mu$ with $\mu = -ig'(0)$. By Proposition 27.2,

$$\tfrac{d}{dt}S(U(t))\big|_{t=0} = \langle\nabla\Phi(A_*),\mathcal{A}'(0)\rangle = \langle\omega,\Lambda_{A_*}\mu\rangle = \langle\Lambda_{A_*}^T\omega,\mu\rangle = 0.$$ □

27.4 Second-order bounds and regularity

Proposition 27.4. *(a)* $|\Delta_{U_0}A_*|_\infty + \sup_y[\nabla\tilde{A}_{*,y}]_{\alpha,B(y)} \le C(\varepsilon+\beta)$.

(b) $U_*\in\mathfrak{U}(C(\varepsilon+\beta))$.

(c) U_* is $(M, \varepsilon + C(\varepsilon+\beta))$-regular.

Proof. (a) By Proposition 23.3 and Proposition 27.2,

$$D^T D A_* = d\mathcal{Q}(A_*)^T \omega - J - \nabla \mathcal{N}(A_*).$$

Here $d\mathcal{Q}(A_*)^T\omega = \mathbf{Q}^T\omega + dC_k(A_*)^T\omega$ is bounded by $C|\omega|$, by the kernel bounds of Proposition 22.6(c). Hence $|D^T D A_*|_\infty \le C(\varepsilon+\beta)$. Since $RD^T A_* = 0$, $\Delta_{U_0} A_* = D^T D A_* + D P_{\mathcal{V}} D^T A_*$, and the second term is bounded by $C|A_*|_\infty$ (Lemma 24.1(b)). The Hölder bound follows as in Proposition 24.5.

(b) U_* is gauge equivalent to $U_\flat = e^{\mathrm{i}\eta A_*} U_0$. By (23.1), $U_\flat(\partial p) = e^{\mathrm{i}\eta^2 \zeta_p(A_*)} U_0(\partial p)$ with $|\zeta_p(A_*)| \le C|A_*|_{C^1}$. Hence $|F_{U_\flat}| \le C\varepsilon + C(\varepsilon+\beta)$. By Proposition 23.3(d), $J_{U_\flat} = \Psi(A_*)^{-T} \nabla \Phi(A_*)$, which is bounded by $C|\omega| \le C(\varepsilon+\beta)$.

(c) Let $\hat\Lambda$ be a cube as in Definition 18.1, and u a real gauge transformation with $U_0^u = e^{\mathrm{i}\eta A_0}$ on $\hat\Lambda$, where $|A_0|, |\partial A_0|, [\partial A_0]_\alpha \le \varepsilon$. Then

$$U_\flat^u = e^{\mathrm{i}\eta R(u) A_*} e^{\mathrm{i}\eta A_0} = e^{\mathrm{i}\eta (A_0 + R(u)A_* + \eta \mathfrak{c})} \quad \text{on } \hat\Lambda,$$

where $\mathfrak{c} = \mathfrak{c}(a,b) = \eta^{-2}\big(\ell(e^{\mathrm{i}\eta a} e^{\mathrm{i}\eta b}) - \eta a - \eta b\big)$ with $a = R(u)A_*$ and $b = A_0$. In this gauge, ∂a differs from $R(u)\nabla A_*$ by at most $C\varepsilon|A_*|$. Since $|A_*|_{C^1} \le K'(\varepsilon+\beta)$ (Section 26), $|a| + |\partial a| \le C(\varepsilon+\beta)$; moreover $|b| + |\partial b| \le 2\varepsilon$.

The term $\eta\mathfrak{c}$. The function $(X,Y) \mapsto \ell(e^{\mathrm{i}X} e^{\mathrm{i}Y}) - X - Y$ is analytic near 0 and vanishes if $X = 0$ or $Y = 0$. So $\mathfrak{c}(a,b) = \mathfrak{m}(\eta a, \eta b)[a,b]$, where $\mathfrak{m}(X,Y)$ is a bilinear map depending analytically on (X,Y). Hence $|\mathfrak{c}| \le C|a||b|$, and the chain rule for lattice differences gives $|\partial \mathfrak{c}| \le C(|a| + |\partial a|)(|b| + |\partial b|) \le C\varepsilon(\varepsilon+\beta)$. So $|\eta\mathfrak{c}| + |\partial(\eta\mathfrak{c})| \le C\eta\varepsilon(\varepsilon+\beta)$. Distinct points of $\mathbb{T}_\eta$ have distance at least η, so the Hölder quotients of $\partial(\eta\mathfrak{c})$ are at most $2\eta^{-\alpha} \cdot C\eta\varepsilon(\varepsilon+\beta) \le C\varepsilon(\varepsilon+\beta)$. The bound (23.2) alone gives only $|\partial(\eta\mathfrak{c})| \le 2|\mathfrak{c}|$, without the factor η.

The term a. Let $\Lambda_3(z') \subset \hat\Lambda$, and let $x \ne x'$ be points of $\Lambda_3(z')$. If $|x - x'|_\infty \ge \frac12$, the Hölder quotient of ∂a at x, x' is at most $4 \sup |\partial a| \le C(\varepsilon+\beta)$. If $|x - x'|_\infty < \frac12$, the points of the two differences lie in a common cube $\Lambda_1(z_i)$ with z_i on a grid of spacing $\frac12$, as in Remark 7.2. Let u'' be a gauge of Definition 18.1 on a cube $\Lambda_{4M+2}(z'') \supset \Lambda_3(z_i)$, with $U_0^{u''} = e^{\mathrm{i}\eta A''}$ and $|A''|, |\partial A''| \le \varepsilon$. By (a), in the form on unit cubes (Theorem 25.1(c) and Remark 7.2), the Hölder quotient of $\partial(R(u'')A_*)$ on $\Lambda_1(z_i)$ is at most $C(\varepsilon+\beta)$. On the box $O = \Lambda_1(z_i) \cap \hat\Lambda$ put $h = u(u'')^{-1}$. Then

$$h(b_+) = e^{-\mathrm{i}\eta A_0(b)}\, h(b_-)\, e^{\mathrm{i}\eta A''(b)} \qquad \text{for the bonds } b \text{ of } O.$$

So $\partial h(b) = \eta^{-1}\big(e^{-\mathrm{i}\eta A_0(b)} h(b_-) e^{\mathrm{i}\eta A''(b)} - h(b_-)\big)$ is linear in $A_0(b)$ and $A''(b)$, with coefficients that are analytic functions of $h(b_-)$, $\eta A_0(b)$ and $\eta A''(b)$, since $e^{\mathrm{i}X} - 1 = X \int_0^1 \mathrm{i} e^{\mathrm{i}tX}\, dt$. Hence $|\partial h| \le C\varepsilon$. One more difference brings in ∂A_0, $\partial A''$ and ∂h, each with a bounded factor, so $|\partial\partial h| \le C\varepsilon$; no factor η^{-1} appears. The transformation h is unitary, so $R(h)$ is an isometry, and no smallness of h itself is used. Since $a = R(h)R(u'')A_*$ on O, the discrete product rule along a coordinate path from x to x' in O gives

$$|\partial a(x) - \partial a(x')| \le \Big([\partial(R(u'')A_*)]_{\alpha, \Lambda_1(z_i)} + C\varepsilon\big(|A_*|_\infty + |\nabla A_*|_\infty\big)\Big) |x - x'|^\alpha \le C(\varepsilon+\beta)|x-x'|^\alpha.$$

All points used here lie in the box spanned by the points of the two differences, which lies in O.

So $A_0 + a + \eta\mathfrak{c}$ satisfies the three bounds of Definition 18.1 on $\hat\Lambda$, with $\varepsilon + C(\varepsilon+\beta)$ in place of ε. Hence $U_\flat$, and therefore U_*, is $(M, \varepsilon + C(\varepsilon+\beta))$-regular. □

27.5 Analyticity and decay

Proposition 27.5. *The map $B \mapsto A_*(B)$ extends analytically to complex B with $\varepsilon + |B|_\infty < c_6$, with the C^1 bound of Theorem 25.1(c). The second-order and Hölder bounds are claimed for real B only. For real and complex B, if δB is supported on the bond c, then $|dA_*(B)\delta B(x)| + |\nabla(dA_*(B)\delta B)(x)| \le Ce^{-\gamma'|x-c|}|\delta B|$.*

Proof. Lemma 26.1, Lemma 26.2(a) and Proposition 26.3 hold verbatim for complex B and complex A_1; reality was used only in the minimality argument. This gives the bound $|A_*(B)|_{C^1} \le C(\varepsilon + |B|)$. The fixed point of the contraction $\mathcal{T}$ is the uniform limit of iterates that are analytic in B, hence analytic. For real B it is the real solution, by uniqueness.

For the decay we first prove a block bound. The argument holds for complex $A' = A_1 + \mathbf{H}B$ with $|A_1|_{C^1} \le \rho_2$. For a fine field h put $a_y(h) = |h|_{B(y)} + |\nabla h|_{B(y)}$, with $|\cdot|_{B(y)}$ as in (26.1). The map $h \mapsto d\nabla\mathcal{V}(A')h$ is obtained by differentiating the formula for $\nabla\mathcal{V}$ in the proof of Lemma 26.2. Its ingredients are the following.

- *The terms with $d^2\mathcal{D}$.* They add up to $(d^2\mathcal{D}(A')[h,\cdot])^T\omega'$, with $\omega' = -\mathbf{H}^T(J + D^TDA + \nabla\mathcal{N}(A))$ and $A = A' - \mathbf{H}\mathcal{D}(A')$, as in the proof of Lemma 26.2(b). There $|\omega'|_\infty \le C(\varepsilon + \rho_2)$, and (26.1) bounds these terms. The ℓ^2 bound on $d^2\mathcal{D}$ in Lemma 26.1 would not suffice: it gives no bound on this fine-lattice field that is pointwise and uniform in η.
- *The local term $d\nabla\mathcal{N}(A)[Lh]$,* with $L = I - \mathbf{H}\,d\mathcal{D}(A')$. The value $\nabla\mathcal{N}(A)(b)$ depends on A only on the bonds of the plaquettes that contain b (proof of Proposition 23.3). So, for b starting in $B(y)$, $d\nabla\mathcal{N}(A)k(b) = d\nabla\mathcal{N}(A)(\chi_y k)(b)$, where χ_y is a smooth function equal to 1 on $\Lambda_{3/2}(c_y)$ and to 0 outside $\Lambda_2(c_y)$; its differences are bounded by an absolute constant. Proposition 23.3(b) gives $|d\nabla\mathcal{N}(A)g|_\infty \le C(\varepsilon + \rho_2)|g|_{C^1}$ for every fine field g. Hence $|d\nabla\mathcal{N}(A)k|_{B(y)} \le C(\varepsilon + \rho_2)\max_{|y'-y|_\infty \le 3} a_{y'}(k)$.
- *The operators $\mathbf{H}$, $D\mathbf{H}$, $\mathbf{H}^T = \mathbf{\Gamma}_\pi\mathbf{Q}\mathbf{G}$ and $(D\mathbf{H})^T$.* They satisfy block bounds with exponential decay, by the block form of the pointwise bounds in the proof of Lemma 24.1, as in the proof of (26.1).
- *The operator $D^TD\mathbf{H}$.* By the first identity in the proof of Lemma 26.2, $D^TD\mathbf{H} = \mathbf{Q}^T(\mathbf{\Gamma}_\pi - a) + S^T\mathfrak{F}^TD\mathbf{H}$, with $S^T = DRG'$ bounded by Lemma 24.1(a),(b). So it satisfies a block bound of the same kind.
- *The operators $d\mathcal{D}$ and $d\mathcal{D}^T$.* Their kernel $K_\mathcal{D}$ is bounded by $C\rho_2 e^{-\gamma|c_- - b_-|}$ (Lemma 26.1).

Every term of $d\nabla\mathcal{V}(A')h$ is a composition of these ingredients, and it contains at least one factor of size $C(\varepsilon + \rho_2)$: ω', $d\mathcal{D}$, $d\mathcal{D}^T$ or $d\nabla\mathcal{N}$. Hence

$$|d\nabla\mathcal{V}(A')h|_{B(y)} \le C(\varepsilon + \rho_2)\sum_{y'} e^{-\gamma|y-y'|}a_{y'}(h) \qquad \text{for all } y \in \mathbb{T}. \tag{27.2}$$

Now let $|f|_c = \sup_x e^{\gamma'|x-c|}|f(x)|$ and $|f|_{c,1} = |f|_c + |\nabla f|_c$. By Lemma 24.1 (weighted version), $|\mathbf{C}f|_{c,1} \le C|f|_c$ and $|\mathbf{H}\delta B|_{c,1} \le C|\delta B|$ for δB supported on c. Block bounds pass to the weighted norms when $\gamma' \le \gamma/2$, by $e^{\gamma'|x-c|} \le e^{\gamma'|x-x'|}e^{\gamma'|x'-c|}$. So (27.2) gives $|d\nabla\mathcal{V}(A')h|_c \le C(\varepsilon + \rho_2)|h|_{c,1}$. Differentiating (26.3),

$$\delta A_1 = -\mathbf{C}\,d\nabla\mathcal{V}(A')[\delta A_1 + \mathbf{H}\delta B].$$

The operator on the right has norm at most $\frac{1}{2}$ in $|\cdot|_{c,1}$. Hence $|\delta A_1|_{c,1} \le C|\delta B|$, and $\delta A_* = (I - \mathbf{H}\,d\mathcal{D})(\delta A_1 + \mathbf{H}\delta B)$ obeys the same bound. □

27.6 Proof of Theorem 25.1

(a) Uniqueness was shown after Proposition 26.3. Existence: $U_* \in \mathfrak{B}(V)$ by construction; it is critical by Proposition 27.3; and it lies in $\mathfrak{U}(\theta_0)$ by Proposition 27.4(b), provided $C(\varepsilon+\beta) \le \theta_0$, i.e. $c_6 \le \theta_0/C$. (b) is Proposition 26.6. (c) follows from Proposition 26.3 and Proposition 27.4(a). (d) is Proposition 27.2. (e) is Proposition 27.4(b), (c). (f) is Proposition 27.5. (g) was noted after the statement. The constants are fixed in the following order. First B_1, C_H, C_C, K, r_2, r_3 and c_7, which do not depend on θ_0. Then θ_0, subject to $\theta_0 \le \min(c_4/3, c_7)$, $\rho_1 \le r_2/2$, $C\rho_1 < 1$ (Lemma 25.2(a)) and $\rho_2 + C_H\beta \le r_3$, all for $\varepsilon+\beta \le \theta_0$; these are conditions on θ_0 alone, since ρ_1 and ρ_2 are then at most $C\theta_0$. For the Dirichlet version of the contraction in Section 39 (Proposition 39.3(a), where $\varepsilon + |B|_\infty \le r_4 \le \theta_0$), θ_0 also satisfies $4C_{\mathrm{map}}S_{\gamma_{\mathrm{map}}/4}(1+8B_1+C_H+K)\theta_0 \le 1$, with $S_\gamma = \sum_{z\in\mathbb{Z}^d} e^{-\gamma|z|}$ as in Section 5. Here C_{map} and γ_{map} are the constant and the rate of the kernel $K(y,y') = C_{\mathrm{map}}(\varepsilon+\rho_2)e^{-\gamma_{\mathrm{map}}|y-y'|/2}$ in the proof of that proposition; they come from Theorem 35.2 and (27.2), and depend neither on θ_0 nor on M. Since $\varepsilon+\rho_2 \le (1+8B_1+C_H+K)\theta_0$ for $\varepsilon+|B|_\infty \le \theta_0$, this gives $\sum_{y'} K(y,y')e^{\gamma'|y-y'|} \le \frac{1}{4}$ for $\gamma' \le \gamma_{\mathrm{map}}/4$. This condition is not used on the main line. Finally $c_6 \le \min\,(c_4/3,\, c_7,\, \theta_0,\, \theta_0/C,\, B_1\theta_0/K',\, \min(\theta_0,c_1)/(20d),\, \min(c_3,c_5)/C_6)$. The condition with $20d$ is used for $k=0$ in Corollary 30.5. The last one is used in Proposition 33.1: there $|A_*(B)|_{C^1} \le C_6(\varepsilon+|B|_\infty) \le \min(c_3,c_5)$, so that Propositions 23.3 and 22.6 apply at $A_*(B)$ for complex B. The constant C_6 of (c)–(f) holds for all $\varepsilon+\beta$ below thresholds fixed before c_6, so it does not depend on c_6. There is no circularity. □

Remark 27.6 (What is used, and what is not). The proof uses Parts II and III only through the pointwise bounds of Lemma 24.1, the identities for $\mathbf{H}$ and $\mathbf{C}$, and the coercivity on $\mathcal{X}$ (Lemma 26.5). All of these hold at (M,ε)-regular backgrounds. The random-walk expansion is not used. The reference U_0 enters only through its regularity and through $|J_{U_0}| \le \varepsilon$. The theorem does not use the induction over k. In Bałaban's scheme the reference comes from that induction, and the bound on the regularity of the minimizer, $\varepsilon + C_6(\varepsilon+\beta)$, has to be improved to a bound depending only on the curvature of V. This is the local improvement of [8, Sect. F], carried out in Part VI (Lemma 29.2).

Part VI: The induction over the number of steps

28 Local gauges and splicing

Theorem 25.1 solves the variational problem relative to a regular reference configuration U_0. It bounds the regularity of the minimizer by $\varepsilon + C_6(\varepsilon+\beta)$, where ε is the regularity parameter of U_0. Bałaban's Theorem 1 [8] is proved by induction over k. The reference at step k is the solution at step $k-1$, and passing from units of δ_{k-1} to units of δ_k multiplies its regularity parameter by a power of L. Without improvement the constants would grow geometrically in k. Bałaban's Section F [8] closes the induction with a local argument. On a cube around a given point the minimizer is compared with the solution of a variational problem relative to the trivial configuration. That problem is posed on nested domains: the configuration is frozen outside the cube, and the averaging becomes finer towards its boundary.

We stay on the torus. The observation is that Theorem 25.1 holds uniformly for *all* regular references. Near a point y we replace the induction reference by a configuration that is trivial (in a local gauge) near y and coincides with the minimizer far from y. Relative to this *spliced* reference, the first-order formula of Theorem 25.1(c) shows that the minimizer near y

is determined by the coarse field near y. The errors are exponentially small in the splicing radius, or quadratic in the a priori bound. The coarse field near y, in an axial gauge centred at y, is controlled by the curvature of V alone. This gives the improvement (Section 29), and the induction closes (Section 30).

This section provides the geometric input: gauges on large cubes (Lemma 28.3), a local gauge adapted to the coarse axial gauge (Lemma 28.4), and the splicing (Lemma 28.6).

Conventions for Part VI. We keep the conventions of Parts IV and V and assume in addition $M \geq 5L$. The torus has side $n \geq n_0$, where n_0 is fixed in Lemma 29.2. All boxes below have sides smaller than n, so they are identified with boxes in $\eta\mathbb{Z}^d$. Side lengths and shifts of boxes are tacitly rounded to multiples of η; this changes the constants below only by bounded factors. A *box* is a set $\{x \in \mathbb{T}_\eta : \ a_\mu \leq x_\mu \leq b_\mu\}$, and the bonds of a box are the bonds with both endpoints in it. We write $\Lambda_r(z) = \{x : |x - z|_\infty \leq r\}$, as in Section 6.

28.1 Regularity on a box

Definition 28.1 (Regularity on a box)**.** Let Λ be a box, and let A be a fine Lie field on the bonds of Λ. Put

$$\|A\|_{1,\alpha;\Lambda} = |A|_\Lambda + |\partial A|_\Lambda + \sup\{[\partial A]_{\alpha,\Lambda_3(z')} : \ \Lambda_3(z') \subset \Lambda\},$$

where $|\partial A|_\Lambda$ is the maximum of $|\partial_\nu A_\mu(x)|$ over all differences that involve only points of Λ. A real configuration U is *σ-regular on Λ* if there is a real gauge transformation u on Λ with $U^u = e^{\mathrm{i}\eta A}$ on the bonds of Λ and $\|A\|_{1,\alpha;\Lambda} \leq \sigma$.

This property is invariant under real gauge transformations of U, and it passes to subboxes. By Definition 18.1, an (M,ε)-regular configuration is 3ε-regular on every cube $\Lambda_{4M+2}(z)$. Conversely, if $0 < \sigma \leq 1$, a configuration that is σ-regular on every cube $\Lambda_{4M+2}(z)$ is (M,σ)-regular.

Two facts on gauge transformations of configurations of the form $e^{\mathrm{i}\eta A}$ are used repeatedly. Let λ be a real scalar field on a box with $|\lambda| \leq 1$. Then $(e^{\mathrm{i}\eta A})^{e^{\mathrm{i}\lambda}} = e^{\mathrm{i}\eta A^\lambda}$, where, by (24.1) with $U_0 = 1$,

$$A^\lambda(b) = e^{\mathrm{i}\,\mathrm{ad}\,\lambda(b_-)}A(b) - \mathfrak{d}(-\mathrm{ad}\,\lambda(b_-))\partial\lambda(b) + \eta\,\mathfrak{b}(-\lambda(b_-), A(b), -\partial\lambda(b)). \tag{28.1}$$

Consequently

$$\|A^\lambda\|_{1,\alpha;\Lambda} \leq C(\|A\|_{1,\alpha;\Lambda} + |\partial\lambda|_\Lambda + |\partial\partial\lambda|_\Lambda + \sup_{\Lambda_3(z')\subset\Lambda}[\partial\partial\lambda]_{\alpha,\Lambda_3(z')}), \tag{28.2}$$

provided the right-hand side is at most 1. Indeed, $A^\lambda(b)$ is an analytic function of $\lambda(b_-)$, $A(b)$ and $\partial\lambda(b)$, and its first differences and their Hölder quotients are controlled by the chain rule for lattice differences, as in the proof of Lemma 12.3. The term with the factor η is handled by (23.2).

28.2 Gauges on large cubes

Lemma 28.2 (Patching)**.** *There is $C_P \geq 1$, depending only on d, α and G, with the following property. Let Λ be a box, μ a direction, $a \in \mathbb{R}$ and $w \geq 18$. Put $\Lambda' = \{x \in \Lambda : x_\mu \leq a + w\}$ and $\Lambda'' = \{x \in \Lambda : x_\mu \geq a\}$, and suppose that all sides of Λ are at most $3w$. If U is σ-regular on Λ' and on Λ'', and $C_P w\sigma \leq 1$, then U is $C_P\sigma$-regular on Λ.*

Proof. Let u', u'' be the gauges, with $U^{u'} = e^{\mathrm{i}\eta A'}$ on Λ' and $U^{u''} = e^{\mathrm{i}\eta A''}$ on Λ''. On the slab $O = \Lambda' \cap \Lambda''$ the transformation $g = u'(u'')^{-1}$ satisfies

$$e^{\mathrm{i}\eta A'(b)} = g(b_-)\, e^{\mathrm{i}\eta A''(b)}\, g(b_+)^{-1} \qquad \text{for the bonds } b \text{ of } O.$$

Fix $x_0 \in O$. Replacing u'' by $g(x_0)u''$, which replaces A'' by $R(g(x_0))A''$ and does not change its norm, we may assume $g(x_0) = 1$.

Size of g. By the relation above, $g(b_+) = e^{-\mathrm{i}\eta A'(b)} g(b_-) e^{\mathrm{i}\eta A''(b)}$. Every point of O is joined to x_0 by a lattice path in O of length at most $3dw$. By the bi-invariance of the metric (Lemma 21.1(b)), $g = e^{\mathrm{i}\lambda}$ on O with λ real and $|\lambda| \le 6dw\sigma \le c_G$; here we take $C_P \ge 6d/c_G$.

Derivatives of λ. The relation gives $\lambda(b_+) = \ell\big(e^{-\mathrm{i}\eta A'(b)} e^{\mathrm{i}\lambda(b_-)} e^{\mathrm{i}\eta A''(b)}\big)$. Hence $\partial\lambda(b) = \mathfrak{p}\big(\lambda(b_-), \eta A'(b), \eta A''(b); A'(b), A''(b)\big)$, where $\mathfrak{p}$ is analytic, linear in the last two arguments, and bounded by $C(|A'(b)| + |A''(b)|)$. Taking one more difference and using the chain rule, $|\partial\lambda|_O + |\partial\partial\lambda|_O + [\partial\partial\lambda]_{\alpha,\Lambda_3(z')} \le C\sigma$ for every $\Lambda_3(z') \subset O$.

Gluing. Let χ_1 be a smooth function on $\mathbb{R}$ with $\chi_1 = 1$ on $(-\infty, \frac{1}{3}]$ and $\chi_1 = 0$ on $[\frac{2}{3}, \infty)$, and put $\chi(x) = \chi_1((x_\mu - a)/w)$. Define

$$u = u' \ \text{ on } \{x_\mu \le a + \tfrac{w}{3}\}, \qquad u = e^{\mathrm{i}\chi\lambda} u'' \ \text{ on } O, \qquad u = u'' \ \text{ on } \{x_\mu \ge a + \tfrac{2w}{3}\}.$$

The definitions agree on the overlaps, because $e^{\mathrm{i}\lambda} u'' = g u'' = u'$. Hence $U^u = e^{\mathrm{i}\eta A}$ on the bonds of Λ, with $A = A'$, $A = A''$, or $A = (A'')^{\chi\lambda}$ as in (28.1), on the three regions. Since $|\partial^\beta \chi| \le C w^{-|\beta|}$ and $|\lambda| \le 6dw\sigma$, the field $\chi\lambda$ satisfies $|\partial(\chi\lambda)| + |\partial\partial(\chi\lambda)| + [\partial\partial(\chi\lambda)]_{\alpha,\Lambda_3(z')} \le C\sigma$; the terms in which derivatives fall on χ carry factors $w^{-1} \cdot w\sigma$ or $w^{-2} \cdot w\sigma$. By (28.2), $\|(A'')^{\chi\lambda}\|_{1,\alpha;O} \le C\sigma$.

Finally, since $w \ge 18$, every cube $\Lambda_3(z') \subset \Lambda$ (of side $6 \le w/3$) and every difference lie entirely in one of the three regions; so $|\partial A|_\Lambda \le C\sigma$. For a pair of differences in a Hölder quotient whose points lie in a common region, use the bound on that region (on the left and right regions, the regularity on Λ' and Λ''). If no region contains the points of both differences, their base points are at distance at least $w/3 - 2\eta \ge 4$ in the direction μ, and

$$\frac{|\partial A(x) - \partial A(x'')|}{|x - x''|^\alpha} \le 2C\sigma\, 4^{-\alpha}.$$

Hence $\|A\|_{1,\alpha;\Lambda} \le C_P\sigma$, after increasing C_P. □

Lemma 28.3 (Large cubes)**.** *There are $C_E \ge 1$ and $p \ge 1$, depending only on d, α and G, with the following property. Let $18 \le r \le R$, and let U be σ-regular on every cube $\Lambda_r(z)$, $z \in \mathbb{T}_\eta$. If $C_E(R/r)^p R\sigma \le 1$, then U is $C_E(R/r)^p\sigma$-regular on every cube $\Lambda_R(z)$.*

Proof. We enlarge boxes in rounds. Let $T = 2\eta\lfloor R/\eta \rfloor$ be the side of the cubes $\Lambda_R(z)$; $T < n$ by the conventions of Part VI, under which the cubes $\Lambda_R(z)$ are boxes. At the beginning of a round all sides equal some ℓ. If $\ell < T$, put

$$\delta = \min\Big(T - \ell,\ \eta\Big\lfloor \frac{\ell}{2\eta} \Big\rfloor\Big),$$

and, for $\mu = 1, \dots, d$ in turn, pass to the boxes whose side in direction μ is $\ell + \delta$; each of them is the union of two boxes of the current shape, shifted against each other by δ in direction μ. The overlap has width $w = \ell - \delta \ge \frac{\ell}{2}$, and during the round all sides lie between ℓ and $\ell + \delta \le \frac{3}{2}\ell \le 3w$. Initially $\ell = \ell_0 = 2\eta\lfloor r/\eta \rfloor \ge 36$, since 18 is a multiple of η. Hence $w \ge 18$, all sides of the unions are at most $3w$, and all boxes have sides at most $T < n$, so none of them wraps around the torus. By Lemma 28.2 and induction, after j steps U is $C_P^j\sigma$-regular on every box of the current shape, as long as $C_P \cdot 3R \cdot C_P^j\sigma \le 1$. A round with $\delta < T - \ell$ gives the side $\ell + \delta \ge \frac{3}{2}\ell - \eta$, so after i such rounds the side is at least $(\frac{3}{2})^i(\ell_0 - 2\eta) + 2\eta \ge (\frac{3}{2})^i \cdot \frac{16}{9} r$,

since $\ell_0 - 2\eta \geq 2r - 4\eta \geq \frac{16}{9}r$. Since a round with $\delta < T - \ell$ ends with a side less than $T \leq 2R$, the number i of such rounds satisfies $(\frac{3}{2})^i < \frac{9}{8} \cdot \frac{R}{r}$; as $\frac{9}{8} < \frac{3}{2}$, this gives $i \leq \lceil \log_{3/2}(R/r) \rceil$. So after at most $\lceil \log_{3/2}(R/r) \rceil + 1$ rounds, the last one with $\delta = T - \ell$, all sides equal T, and every cube $\Lambda_R(z)$ is a box of the current shape. The number of steps is $m \leq d(\log_{3/2}(R/r) + 2)$, so $C_P^m \leq C_P^{2d}(R/r)^{d \log C_P / \log(3/2)}$. Hence the lemma holds with $p = d \log C_P / \log(3/2)$ and $C_E = 3C_P^{2d+1}$. The rounding to multiples of η costs at most one round, so it changes C_E but not p. □

The loss is only polynomial in R/r. This is what makes the splicing below possible: the polynomial loss is beaten by the exponential decay of the operators of Part III.

28.3 A gauge adapted to the coarse axial gauge

For $y, y' \in \mathbb{T}$ let $\Gamma^{\mathrm{ax}}_{y,y'}$ be the contour on $\mathbb{T}$ from y to y' that changes the coordinates in the order $d, d-1, \dots, 1$. Let $\phi_1 \in C^\infty(\mathbb{R})$ be supported in $[-1,1]$, with $\phi_1(0) = 1$ and $\sum_{m \in \mathbb{Z}} \phi_1(t - m) = 1$. For instance, $\phi_1(t) = \vartheta(1 - |t|)$, where $\vartheta : [0,1] \to [0,1]$ is smooth, vanishes with all its derivatives at 0, and satisfies $\vartheta(s) + \vartheta(1-s) = 1$. Put $\phi(x) = \prod_\mu \phi_1(x_\mu)$. Then $\phi(y' - y'') = \delta_{y'y''}$ for $y', y'' \in \mathbb{T}$, and $\sum_{y' \in \mathbb{T}} \phi(x - y') = 1$.

Lemma 28.4 (Adapted local gauge)**.** *There are $C_8, q \geq 1$, depending only on d, L, G, α and M, with the following property. Let $U \in \mathfrak{U}_0(\varepsilon_0)$ be real and (M, ε_0)-regular, with $\varepsilon_0 \leq c_1$. Suppose that $V = U^{(k)} \in \mathfrak{U}_k(\varepsilon_1)$ with $\varepsilon_1 \leq 2\varepsilon_0$. Let $R \geq 4M + 2$ be an integer and $y \in \mathbb{T}$, and assume $C_8 R^q \varepsilon_0 \leq 1$. Then there is a real gauge transformation u on $\Lambda_R(y)$ such that:*

(i) $U^u = e^{\mathrm{i}\eta A}$ on the bonds of $\Lambda_R(y)$, with $\|A\|_{1,\alpha;\Lambda_R(y)} \leq C_8 R^q \varepsilon_0$;

(ii) for every bond c of $\mathbb{T}$ with both endpoints in $\Lambda_R(y)$,

$$V^u(c) := u(c_-) V(c) u(c_+)^{-1} = e^{\mathrm{i}B_y(c)}, \qquad |B_y(c)| \leq d\,(|c_- - y|_\infty + 1)\,\varepsilon_1.$$

Proof. Step 1: a gauge on a large cube. By Lemma 28.3 with $r = 4M + 2 \geq 18$, $\Lambda_{R+3}(y)$ carries a real gauge u_1 with $U^{u_1} = e^{\mathrm{i}\eta A_1}$ and $\|A_1\|_{1,\alpha;\Lambda_{R+3}(y)} \leq \sigma := 3C_E(R+3)^p \varepsilon_0$.

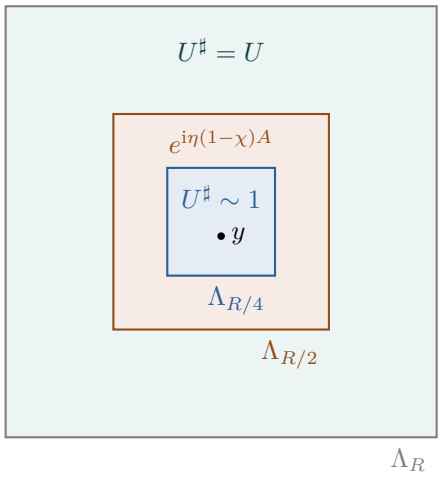
Step 2: the coarse axial gauge. Let c be a bond of $\mathbb{T}$ with both endpoints in $\Lambda_{R+1}(y)$. Then $B(c_-) \cup B(c_+) \subset \Lambda_{R+3}(y)$. By Lemma 21.8, applied to U^{u_1} with the union of blocks $B(c_-) \cup B(c_+)$ (the averages are local, so u_1 may be extended arbitrarily outside the cube), $V^{u_1}(c) = e^{\mathrm{i}B_1(c)}$ with $|B_1(c)| \leq \sigma_1 := \sigma + 4d\varepsilon_0$. For unit points $y' \in \Lambda_{R+1}(y)$ put $h(y') = V^{u_1}(\Gamma^{\mathrm{ax}}_{y,y'})$, so that $h(y) = 1$ and $|\lambda(y')| \leq d(R+1)\sigma_1$ for $\lambda(y') = \ell(h(y'))$. For a bond $c = (y', y'')$ as above,

$$(V^{u_1})^h(c) = h(y') V^{u_1}(c) h(y'')^{-1} = V^{u_1}\big(\Gamma^{\mathrm{ax}}_{y,y'} \cup c \cup (\Gamma^{\mathrm{ax}}_{y,y''})^{-1}\big)$$

is the holonomy of a closed contour based at y. As in Lemma 21.2(ii), the two coordinate-ordered contours coincide up to the segment in the direction of c, and after it they run parallel at distance 1. So the loop is spanned by at most $d(|y' - y|_\infty + 1)$ plaquettes of $\mathbb{T}$. Holonomies of closed contours are conjugated by gauge transformations, so Lemma 21.2 applied to V gives $|\ell((V^{u_1})^h(c))| \leq d(|y' - y|_\infty + 1)\varepsilon_1$. Moreover $h(y'') = (V^{u_1})^h(c)^{-1}\, h(y')\, V^{u_1}(c)$. By bi-invariance, the distance between $h(y'')$ and $h(y')$ is at most $|\ell((V^{u_1})^h(c))| + |B_1(c)|$. Near 1 the logarithm is Lipschitz with respect to this distance. Hence

$$|\lambda(y'') - \lambda(y')| \leq C(\sigma_1 + R\varepsilon_1) \leq C\sigma' \quad \text{for neighbours } y', y'', \qquad \sigma' = \sigma_1 + R\varepsilon_0.$$

Step 3: smooth extension. For fine points $x \in \Lambda_R(y)$ put $\tilde\lambda(x) = \sum_{y'} \phi(x - y')\lambda(y')$; only points y' with $|y' - x|_\infty < 1$ contribute. Then $\tilde\lambda = \lambda$ at the points of $\mathbb{T}$ and $|\tilde\lambda| \leq \max|\lambda|$. Since the lattice differences of $\sum_{y'} \phi(x - y') = 1$ vanish, every difference of order 1, 2 or 3 of $\tilde\lambda$



In the gauge u of Lemma 28.4:

- if $B(c_-) \cup B(c_+) \subset \Lambda_{R/4}$, the reference average is 1, the data are the coarse axial field, $|B^\sharp(c)| \le d(|c_- - y| + 1)\varepsilon_1$;
- in the transition region the bounds are $O(R^q\varepsilon_0)$;
- $B^\sharp(c) = 0$ if $B(c_-) \cup B(c_+)$ does not meet $\Lambda_{R/2}$;
- $J_{U^\sharp} = J_U$ on bonds with no endpoint in $\Lambda_{R/2+1}$.

Figure 3: The spliced reference configuration of Definition 28.5.

is $\sum_{y'} \partial^\beta\phi(x-y')(\lambda(y') - \lambda(y_x))$, where y_x is a unit point with $|x - y_x|_\infty < 1$. Hence these differences are bounded by $C\sigma'$. The same holds for the Hölder quotients of $\partial\partial\tilde\lambda$, which are bounded by the third differences on short distances and by twice the second differences on distances at least 1.

Let $u = e^{i\tilde\lambda}u_1$ on $\Lambda_R(y)$. At the unit points, $u = hu_1$, which gives (ii). By (28.2), $U^u = e^{i\eta A}$ with $\|A\|_{1,\alpha;\Lambda_R(y)} \le C(\sigma + \sigma') \le C_8R^q\varepsilon_0$, for suitable C_8 and $q = p + 1$. The smallness conditions used ($\sigma \le 1$, $|\tilde\lambda| \le c_G$, and the hypotheses of Lemmas 21.8 and 21.2) follow from $C_8R^q\varepsilon_0 \le 1$ if C_8 is large. □

28.4 Splicing

Fix a smooth function χ_0 on $\mathbb{R}^d$ with $\chi_0 = 1$ on $\{|x|_\infty \le \frac14\}$, $\chi_0 = 0$ on $\{|x|_\infty \ge \frac12\}$ and $0 \le \chi_0 \le 1$. For $R \ge 1$ and $y \in \mathbb{T}$ put $\chi(x) = \chi_0((x-y)/R)$, so that $|\partial^\beta\chi| \le C_\beta R^{-|\beta|}$. For a positively oriented bond b write $\chi(b) = \chi(b_-)$. Definition 28.5 below prescribes $U^\sharp$ on positively oriented bonds, and $U^\sharp(-b) = U^\sharp(b)^{-1}$.

Definition 28.5 (Spliced configuration)**.** Let U be real, $y \in \mathbb{T}$, and let u be a real gauge transformation on $\Lambda_R(y)$ with $U^u = e^{i\eta A}$ on the bonds of $\Lambda_R(y)$. The *spliced configuration* $U^\sharp$ is defined by

$$(U^\sharp)^u = e^{i\eta(1-\chi)A} \text{ on the bonds of } \Lambda_R(y), \qquad U^\sharp = U \text{ on all other bonds.}$$

This is consistent: a bond of $\Lambda_R(y)$ with no endpoint in $\Lambda_{R/2}(y)$ has $\chi(b) = 0$, so the two prescriptions agree on it, and every bond with an endpoint in $\Lambda_{R/2}(y)$ is a bond of $\Lambda_R(y)$. See Figure 3.

Lemma 28.6 (Splicing)**.** *There is $C_9 \ge 1$, depending only on d, L, G, α and M, with the following property. Let $R \ge 16M + 16$, and let U be real and (M, ε_0)-regular with $U \in \mathfrak{U}(\varepsilon_0)$. Let y, u and A be as in Definition 28.5, with $\|A\|_{1,\alpha;\Lambda_R(y)} \le s$ and $C_9(\varepsilon_0 + s) \le c_1$. Then:*

(a) $U^\sharp = U$ on all bonds with no endpoint in $\Lambda_{R/2}(y)$, and $(U^\sharp)^u = 1$ on the bonds of $\Lambda_{R/4}(y)$.

(b) $U^\sharp$ is $(M, C_9(\varepsilon_0+s))$-regular, $|F_{U^\sharp}| \le C_9(\varepsilon_0+s)$ and $|J_{U^\sharp}| \le \varepsilon_0 + C_9s$. Moreover $J_{U^\sharp} = J_U$ on bonds with no endpoint in $\Lambda_{R/2+1}(y)$, and $J_{U^\sharp} = 0$ on the bonds of $\Lambda_{R/4-1}(y)$.

(c) *Let* $W^\sharp = (U^\sharp)^{(k)}$ *and* $V = U^{(k)}$. *For every bond* c *of* $\mathbb{T}$, $|\ell(V(c)W^\sharp(c)^{-1})| \le C_9(\varepsilon_0 + s)$. *If* $B(c_-) \cup B(c_+)$ *does not meet* $\Lambda_{R/2}(y)$, *then* $W^\sharp(c) = V(c)$. *If* $B(c_-) \cup B(c_+) \subset \Lambda_{R/4}(y)$, *then* $u(c_-)W^\sharp(c)u(c_+)^{-1} = 1$.

Proof. (a) follows from the definition, since $\chi = 1$ on $\Lambda_{R/4}(y)$.

(b) *Regularity.* If $\Lambda_{4M+2}(z) \subset \Lambda_R(y)$, the gauge u gives $(U^\sharp)^u = e^{i\eta(1-\chi)A}$ there, and $\|(1-\chi)A\|_{1,\alpha} \le Cs$ by the Leibniz rule, since $R \ge 1$. Otherwise $|z-y|_\infty > R - 4M - 2$. Then every point of $\Lambda_{4M+2}(z)$ has distance more than $R - 8M - 4 \ge R/2$ from y, so $U^\sharp = U$ on $\Lambda_{4M+2}(z)$, and the gauges of U apply. The curvature is bounded in the same way, using $|F_{e^{i\eta B}}| \le C(|\partial B| + |B|^2)$.

Current. The current at a bond depends only on the configuration on the plaquettes containing it and on their neighbours. This gives the two identities. In general, compute in the gauge u, which does not change $|J|$. For a configuration $e^{i\eta B}$ and a plaquette p with bonds $b_1, \dots, b_4$ as in Definition 21.3,

$$F^s_{e^{i\eta B}}(p) = \eta^{-2}\mathfrak{S}(\eta B(b_1), \eta B(b_2), \eta B(b_3), \eta B(b_4)), \qquad \mathfrak{S}(z) = P_{\mathfrak{g}} \sin \ell(e^{iz_1}e^{iz_2}e^{-iz_3}e^{-iz_4}),$$

with F^s as in Lemma 23.2. The function $\mathfrak{S}$ is analytic near 0, and $\mathfrak{S}(z) = z_1 + z_2 - z_3 - z_4 + \mathfrak{S}_2(z)$ with $|\mathfrak{S}_2(z)| \le C|z|^2$ and $|\nabla\mathfrak{S}_2(z)| \le C|z|$. Let $t = 1 - \chi$ and $A^\sharp = tA$, and let x_p be the base point of p. Then

$$F^s_{e^{i\eta A^\sharp}}(p) = t(x_p)F^s_{e^{i\eta A}}(p) + E(p),$$

$$E(p) = \sum_{i=1}^{4} \pm\frac{t(b_i) - t(x_p)}{\eta} A(b_i) + \eta^{-2}\Big(\mathfrak{S}_2(\eta A^\sharp(b_1), \dots) - t(x_p)\,\mathfrak{S}_2(\eta A(b_1), \dots)\Big).$$

Since $|\partial^\beta t| \le CR^{-|\beta|}$ and $|A| + |\partial A| \le s$, we get $|E| \le Cs$. Between neighbouring plaquettes, the arguments $\eta A^\sharp(b_i)$, $\eta A(b_i)$ and $t(x_p)$ change by at most $C\eta^2 s$ and $C\eta/R$. With $|\nabla\mathfrak{S}_2(z)| \le C|z|$ and the product rule, the difference quotients of E are bounded by Cs. By Lemma 12.2, $D^T_{e^{i\eta B}} = \partial^* + T^T_B$ with T_B local and bounded by $C|B|$. Using $J_{e^{i\eta A}} = D^T_{e^{i\eta A}}F^s_{e^{i\eta A}} = R(u)J_U$ (Lemma 23.2(a)),

$$J_{e^{i\eta A^\sharp}} = t\,R(u)J_U + [\partial^*, t]F^s_{e^{i\eta A}} + (T^T_{A^\sharp}t - tT^T_A)F^s_{e^{i\eta A}} + D^T_{e^{i\eta A^\sharp}}E.$$

Each term after the first is bounded by Cs, because $|F^s_{e^{i\eta A}}| \le Cs$ and $|\partial t| \le C/R$. Hence $|J_{U^\sharp}| \le \varepsilon_0 + Cs$.

(c) The average $U^{(k)}(c)$ depends only on the bonds of $B(c_-) \cup B(c_+)$ (Lemma 21.5). This gives the first identity. The second follows from (a) and covariance, since $\mathfrak{a}^{(k)}(1) = 1$; the gauge u may be extended arbitrarily outside $\Lambda_R(y)$. In the remaining case $B(c_-) \cup B(c_+) \subset \Lambda_{R/2+2}(y) \subset \Lambda_R(y)$. Lemma 21.8, applied to U^u and to $(U^\sharp)^u$ on the union of blocks $B(c_-) \cup B(c_+)$, gives $V^u(c) = e^{iB'}$ and $(W^\sharp)^u(c) = e^{iB''}$ with $|B'| \le s + 4d\varepsilon_0$ and $|B''| \le s + 4dC'(\varepsilon_0 + s)$. Here C' is the constant obtained in (b) for the curvature of $U^\sharp$. So (c) holds with $C_9 \ge (1 + 4d)(1 + C')$; the same bound applies in the second case. By covariance and Lemma 21.1(b), $|\ell(VW^{\sharp -1})(c)| = |\ell(e^{iB'}e^{-iB''})| \le |B'| + |B''|$. □

29 Localized bounds and the local improvement

29.1 Localized bounds for the relative problem

Theorem 25.1 bounds the solution in sup norms over the whole torus. Its proof gives more. By the first-order formula and the Euler–Lagrange equation, the solution near a point is controlled by the data near that point, up to exponentially decaying contributions from far away and a quadratic error.

In the setting of Theorem 25.1, let A_* and ω be as there. For $y \in \mathbb{T}$ let c_y be the centre of $B(y)$, and let u_y be any real gauge transformation in which U_0 satisfies (B) on $\Lambda_4(c_y)$ with parameter $C\varepsilon$; write $\tilde{A}_{*,y} = R(u_y)A_*$. The bounds below hold uniformly in the choice of u_y. Put

$$\mathsf{b}(y) = \max\{|B(c)| : \ |c_- - y|_\infty \le 1\}, \qquad \mathsf{j}(y) = \max\{|J_{U_0}(b)| : \ b_- \in B(y)\},$$

$$\mathsf{N}(y) = |A_*|_{\Lambda_1(c_y)} + |\nabla A_*|_{\Lambda_1(c_y)} + |\Delta_{U_0} A_*|_{\Lambda_1(c_y)} + [\partial \tilde{A}_{*,y}]_{\alpha,\Lambda_1(c_y)} + \max\{|\omega(c)| : \ |c_- - y|_\infty \le 2\},$$

where c runs over the bonds of $\mathbb{T}$.

Proposition 29.1 (Localized bounds)**.** *There are $C_7, \gamma_7 > 0$, independent of k, n, η and U_0, such that, under the hypotheses of Theorem 25.1, for every $y \in \mathbb{T}$:*

(a) $\mathsf{N}(y) \le C_7 \sum_{y' \in \mathbb{T}} e^{-\gamma_7 |y-y'|}(\mathsf{b}(y') + \mathsf{j}(y')) + C_7(\varepsilon+\beta)^2$*;*

(b) the configuration $U_\flat = e^{i\eta A_*} U_0$, which is gauge equivalent to the minimizer U_*, satisfies

$$|F_{U_\flat}(p)| \le |F_{U_0}(p)| + C_7 \mathsf{N}(y), \qquad |J_{U_\flat}(b)| \le C_7 \mathsf{N}(y)$$

for all plaquettes p and bonds b with base point in $B(y)$.

Proof. Throughout, sums of products of exponentially decaying kernels are again exponentially decaying, with a smaller rate; γ_7 is the rate obtained at the end. We use the pointwise bounds of Parts II and III at U_0, which hold by Corollary 22.8, on cubes $\Lambda_1(z)$ instead of blocks (Remark 7.2).

Step 1: first derivatives. By Theorem 25.1(c), $A_* = \mathbf{H}B - \mathbf{C}J_{U_0} + \mathcal{R}$ with $|\mathcal{R}|_{C^1} \le C(\varepsilon+\beta)^2$. By the pointwise bounds of Theorem 19.9 (Theorem 16.1(b),(d)),

$$|\mathbf{H}B|_{\Lambda_1(c_y)} + |\nabla \mathbf{H}B|_{\Lambda_1(c_y)} \le C \sum_{y'} e^{-\gamma|y-y'|} \mathsf{b}(y'),$$

$$|\mathbf{C}J_{U_0}|_{\Lambda_1(c_y)} + |\nabla \mathbf{C}J_{U_0}|_{\Lambda_1(c_y)} \le C \sum_{y'} e^{-\gamma|y-y'|} \mathsf{j}(y').$$

Let $\mathsf{N}_1(y) = |A_*|_{\Lambda_1(c_y)} + |\nabla A_*|_{\Lambda_1(c_y)}$. Then $\mathsf{N}_1(y)$ is bounded by the right-hand side of (a).

Step 2: the multiplier. By the proof of Proposition 27.2, for every bond c of $\mathbb{T}$ and $\beta_0 \in \mathfrak{g}$ with $|\beta_0| = 1$,

$$\operatorname{tr}(\beta_0\, \omega(c)) = \langle \varphi, J_{U_0} + \nabla \mathcal{N}(A_*) \rangle + \langle D\varphi, DA_* \rangle,$$

with $|\varphi(x)| + |\nabla\varphi(x)| \le Ce^{-\gamma'|x-c_-|}$. Moreover $|\nabla \mathcal{N}(A_*)|_\infty \le C(\varepsilon + |A_*|_\infty)|A_*|_{C^1} \le C(\varepsilon+\beta)^2$ by Proposition 23.3(a). Hence

$$|\omega(c)| \le C \sum_{y'} e^{-\gamma'|c_- - y'|}(\mathsf{j}(y') + \mathsf{N}_1(y')) + C(\varepsilon+\beta)^2.$$

Step 3: second order. By Propositions 23.3 and 27.2,

$$D^T D A_* = d\mathcal{Q}(A_*)^T \omega - J_{U_0} - \nabla \mathcal{N}(A_*).$$

By Proposition 22.6(c), the kernel of $d\mathcal{Q}(A_*)^T = \mathbf{Q}^T + dC_k(A_*)^T$ connects a bond b only to bonds c with $b_- \in S_c$. Hence $|d\mathcal{Q}(A_*)^T \omega|_{\Lambda_1(c_y)} \le C \max\{|\omega(c)| : |c_- - y|_\infty \le 2\}$. Since $RD^T A_* = 0$, we have $\Delta_{U_0} A_* = D^T D A_* + D P_{\mathcal{V}} D^T A_*$. The second term is bounded on $\Lambda_1(c_y)$ by $C \sum_{y'} e^{-\gamma|y-y'|} |A_*|_{B(y')}$, by the pointwise bound on $\partial P_{\mathcal{V}} D^T$ in Proposition 14.2(e) (Theorem 18.2(g)) and $D = \partial + T$. This bounds $|\Delta_{U_0} A_*|_{\Lambda_1(c_y)}$. For the Hölder quotient, write

Δ_{U_0} in the gauge u_y in the normal form of Lemma 12.2. Then $-\Delta\tilde{A}_{*,y} = R(u_y)\Delta_{U_0}A_* - \sum_\mu \partial^*_\mu(a_\mu\tilde{A}_{*,y}) - b\tilde{A}_{*,y}$ on $\Lambda_3(c_y)$, with $|a_\mu| + [a_\mu]_\alpha + |b| \le C\varepsilon$. Theorem 6.3(d) gives

$$[\partial\tilde{A}_{*,y}]_{\alpha,\Lambda_1(c_y)} \le C\Big(|\Delta_{U_0}A_*|_{\Lambda_2(c_y)} + \max_{|y'-y|_\infty\le 3} \mathsf{N}_1(y')\Big).$$

The cube $\Lambda_2(c_y)$ is covered by the cubes $\Lambda_1(c_{y'})$ with $|y'-y|_\infty \le 1$; the values of ω that enter there are bounded by Step 2 for all bonds. Combining Steps 1–3 gives (a).

(b) By (23.1), $U_\flat(\partial p) = e^{\mathrm{i}\eta^2\zeta_p(A_*)}U_0(\partial p)$ with $|\zeta_p(A_*)| \le C(|A_*| + |\nabla A_*|)$ on the bonds of p. Lemma 21.1(b) gives the bound on the curvature. By Proposition 23.3(d) and Proposition 27.2, $J_{U_\flat} = \Psi(A_*)^{-T}d\mathcal{Q}(A_*)^T\omega$. As in Step 3, this is bounded at b by $C\max\{|\omega(c)| : |c_- - y|_\infty \le 2\}$. □

29.2 The local improvement

Lemma 29.2 (Local improvement)**.** *There are $\varepsilon_{\mathrm{imp}} > 0$, $C_{\mathrm{imp}} \ge 1$ and an integer n_0, depending only on d, L, G, a, α and M, with the following property. Let $k \ge 1$, $n \ge n_0$, and let $V \in \mathfrak{U}_k(\varepsilon_1)$ be a real configuration on $\mathbb{T}$. Let $U \in \mathfrak{B}(V)$ be a critical point of $S|_{\mathfrak{B}(V)}$. Suppose that U is (M,ε_0)-regular and $U \in \mathfrak{U}(\varepsilon_0)$, with $\varepsilon_0 \le \varepsilon_{\mathrm{imp}}$. Then U is (M,ε_0')-regular and $U \in \mathfrak{U}(\varepsilon_0')$, where*

$$\varepsilon_0' = C_{\mathrm{imp}}\varepsilon_1 + \tfrac{1}{2}\varepsilon_0.$$

If $\varepsilon_0' > 1$, the regularity conclusion is read as $(M,1)$-regularity; it gives the three bounds of Definition 18.1 with ε_0' in place of ε.

The point is that C_{imp} does not depend on ε_0. Iterating the lemma brings the regularity parameter down to $2C_{\mathrm{imp}}\varepsilon_1$, whatever the a priori bound, as long as the latter is at most $\varepsilon_{\mathrm{imp}}$.

Proof. If $\varepsilon_1 > 2\varepsilon_0$ there is nothing to prove, since then $\varepsilon_0' \ge \varepsilon_0$. So let $\varepsilon_1 \le 2\varepsilon_0$.

Choice of constants. Let C_7, γ_7 be as in Proposition 29.1, C_8, q as in Lemma 28.4, and C_9 as in Lemma 28.6. Put

$$K_R = 2C_9(1 + C_8R^q), \qquad S_1 = \sum_{w\in\mathbb{Z}^d} e^{-\gamma_7|w|}(1+|w|), \qquad T(t) = \sum_{w\in\mathbb{Z}^d,\,|w|_\infty>t} e^{-\gamma_7|w|},$$

and $K_1 = \max(5, C_7)$. Fix an integer $R \ge 64(M+1)$ with

$$K_1C_7K_R\,T(R/8) \le \tfrac{1}{4};$$

this is possible because K_R grows polynomially and T decays exponentially. Let $n_0 = 4R + 16$. Choose $\varepsilon_{\mathrm{imp}}$ so small that $K_1C_7K_R^2\varepsilon_{\mathrm{imp}} \le \frac{1}{4}$, $K_R\varepsilon_{\mathrm{imp}} \le c_6$, $\varepsilon_{\mathrm{imp}} \le \theta_0$, $C_8R^q\varepsilon_{\mathrm{imp}} \le 1$ and $C_9(1 + C_8R^q)\varepsilon_{\mathrm{imp}} \le c_1$.

Step 1: the adapted gauge. Fix $y \in \mathbb{T}$. By Lemma 28.4 there is a real gauge transformation u on $\Lambda_R(y)$ with $U^u = e^{\mathrm{i}\eta A}$, $\|A\|_{1,\alpha;\Lambda_R(y)} \le s := C_8R^q\varepsilon_0$, and $V^u(c) = e^{\mathrm{i}B_y(c)}$ with $|B_y(c)| \le d(|c_- - y|_\infty + 1)\varepsilon_1$ for the bonds of $\mathbb{T}$ in $\Lambda_R(y)$.

Step 2: the spliced reference. Let $U^\sharp$ be the spliced configuration of Definition 28.5, built with this u. By Lemma 28.6, $U^\sharp$ is $(M,\varepsilon^\sharp)$-regular with $|J_{U^\sharp}| \le \varepsilon^\sharp$, where $\varepsilon^\sharp = C_9(\varepsilon_0 + s)$. Moreover $V(c)W^\sharp(c)^{-1} = e^{\mathrm{i}B^\sharp(c)}$ with $|B^\sharp|_\infty \le \beta^\sharp := C_9(\varepsilon_0 + s)$, where $W^\sharp = (U^\sharp)^{(k)}$. Then $\varepsilon^\sharp + \beta^\sharp \le K_R\varepsilon_0 \le c_6$. So Theorem 25.1 applies with the reference $U^\sharp$ and the coarse configuration V.

Step 3: identification. By Theorem 25.1(a), $S|_{\mathfrak{B}(V)}$ has exactly one critical orbit in $\mathfrak{B}(V) \cap \mathfrak{U}(\theta_0)$. Since U is critical and lies in $\mathfrak{U}(\varepsilon_0) \subset \mathfrak{U}(\theta_0)$, it belongs to this orbit. Hence

$U = (e^{\mathrm{i}\eta A^\sharp_*} U^\sharp)^v$ for a real gauge transformation v, where $A^\sharp_*$ is the Landau representative relative to $U^\sharp$. Applying u, we get on the bonds of $\Lambda_{R/4}(y)$, where $(U^\sharp)^u = 1$,

$$U^{uv^{-1}} = e^{\mathrm{i}\eta \tilde{A}}, \qquad \tilde{A} = R(u) A^\sharp_*.$$

In this gauge $\nabla_{U^\sharp}$ is the ordinary difference, and u is a gauge in which $U^\sharp$ satisfies (B) near every block in $\Lambda_{R/4-4}(y)$.

Step 4: the data near y. Apply Proposition 29.1 with reference $U^\sharp$. Let $y' \in \mathbb{T}$ with $|y' - y|_\infty \le R/4 - 3$. Every bond c with $|c_- - y'|_\infty \le 1$ has $B(c_-) \cup B(c_+) \subset \Lambda_{R/4}(y)$. So, by Lemma 28.6(c), $(W^\sharp)^u(c) = 1$, and

$$|B^\sharp(c)| = |\ell(V^u(c))| = |B_y(c)| \le d(|y' - y|_\infty + 2)\varepsilon_1.$$

Moreover $\mathsf{j}(y') = 0$ by Lemma 28.6(b). For the remaining y' we only use $\mathsf{b}(y') + \mathsf{j}(y') \le K_R \varepsilon_0$. Let $y_1 \in \mathbb{T}$ with $|y_1 - y|_\infty \le 4M + 4$. For $|y' - y|_\infty > R/4 - 3$ we have $|y_1 - y'|_\infty > R/8$, because $R \ge 64(M+1)$. Therefore

$$\mathsf{N}(y_1) \le C_7 d S_1 (4M+7)\, \varepsilon_1 + C_7 K_R\, T(R/8)\, \varepsilon_0 + C_7 K_R^2 \varepsilon_0^2.$$

Step 5: conclusion. Let $z \in \mathbb{T}_\eta$, and choose $y \in \mathbb{T}$ with $|z - y|_\infty \le 1$. Then $\Lambda_{4M+2}(z) \subset \Lambda_{4M+3}(y) \subset \Lambda_{R/4-4}(y)$. By Step 3, $U^{uv^{-1}} = e^{\mathrm{i}\eta \tilde{A}}$ on $\Lambda_{4M+2}(z)$. On it, $|\tilde{A}|$, $|\partial \tilde{A}|$ and the Hölder quotients of $\partial \tilde{A}$ on cubes $\Lambda_1(c_{y_1})$ are bounded by $\mathsf{N}(y_1)$, with the gauge $u_{y_1} = u$. Put $H = \max_{|y_1 - y|_\infty \le 4M+4} \mathsf{N}(y_1)$. Let $x \ne x'$ be points of a cube $\Lambda_3(z') \subset \Lambda_{4M+2}(z)$. If $|x - x'|_\infty < \frac{1}{2}$, then in each coordinate x and x' differ by a multiple of $\eta = 1/N$ less than $\frac{1}{2}$, hence by at most $\frac{1}{2} - \frac{\eta}{2}$, and by at most $\frac{1}{2} - \eta$ if N is even. A difference involves points at most 2η beyond its base point in each coordinate (for $\partial_\mu \tilde{A}_\mu$, the point $x + 2\eta e_\mu$). So the points of the two differences span, in each coordinate, an interval of length at most $\frac{1}{2} + \frac{3\eta}{2} \le 1$ if $N \ge 3$ is odd, and at most $\frac{1}{2} + \eta \le 1$ if N is even. The centres c with $|p - c|_\infty \le 1$ for all these points p form, in each coordinate, a closed interval of length at least 1. The block centres form a grid of spacing 1, so one of them lies in it: these points lie in a common cube $\Lambda_1(c_{y_1})$, and then $|y_1 - y|_\infty \le 4M + 4$. If $|x - x'|_\infty \ge \frac{1}{2}$, the Hölder quotient is at most $4 \sup |\partial \tilde{A}|$. Hence $|\tilde{A}| \le H$, $|\partial \tilde{A}| \le H$, and the Hölder quotients of $\partial \tilde{A}$ on the cubes $\Lambda_3(z') \subset \Lambda_{4M+2}(z)$ are at most $4H$. These are the three bounds of Definition 18.1, and each is at most $K_1 H$, since $K_1 \ge 5$. By Proposition 29.1(b), with $F_{U^\sharp} = 0$ near y, the curvature and the current of U on the plaquettes and bonds based in $B(y)$ are bounded by $C_7 \mathsf{N}(y)$. All these quantities are gauge invariant. Every plaquette and bond is based in some block, and every z is close to some y. With the choice of R and $\varepsilon_{\mathrm{imp}}$, Step 4 gives the lemma with $C_{\mathrm{imp}} = K_1 C_7 d S_1 (4M+7)$:

$$K_1 \max \mathsf{N} \le C_{\mathrm{imp}} \varepsilon_1 + \tfrac{1}{4}\varepsilon_0 + \tfrac{1}{4}\varepsilon_0. \qquad \square$$

Remark 29.3 (Comparison with [8, Sect. F]). The structure of the argument is Bałaban's: near a point the minimizer equals, up to small errors, **H** applied to the coarse field in an axial gauge centred at the point; the latter grows only linearly with the distance, with slope given by the curvature of V, while **H** decays exponentially. Compare [8, (159)–(167)]. The difference is in how the problem is localized. Bałaban formulates a new variational problem on a nested sequence of cubes around the point. The configuration is frozen outside, and the problem is solved relative to the trivial configuration. Here the problem is not changed. Only the reference configuration is changed, and Theorem 25.1 is applied once more on the torus. This is possible because that theorem is uniform in the reference. The splicing is a purely geometric operation (Lemma 28.6). As in [8, (166)], the quadratic error forces the a priori bound to be small in terms of the localization radius.

30 The induction over the number of steps

We now prove Bałaban's variational theorem on the torus by induction over k. Each step has three parts:

- the solution of the $(k-1)$-step problem for the trivial extension of V is a reference with *exactly* the right averages [8, Sect. A];
- Theorem 25.1 solves the problem near this reference, with an a priori regularity bound that is worse by a power of L;
- Lemma 29.2, iterated, restores the regularity bound with a constant independent of k.

30.1 The trivial extension

Let $k \geq 1$. A bond b of $\mathbb{T}^{(k-1)}$ in direction μ is a *crossing bond* if its endpoints lie in different blocks, $b_- \in B_k(y)$ and $b_+ \in B_k(y+e_\mu)$; we then write $c(b) = (y, y+e_\mu)$. For a configuration V on $\mathbb{T}$ define V_0 on $\mathbb{T}^{(k-1)}$ by

$$V_0(b) = V(c(b)) \ \text{ for crossing bonds}, \qquad V_0(b) = 1 \ \text{ for bonds inside a block}$$

[8, (11)].

Lemma 30.1 (Trivial extension)**.** *(a)* $\mathfrak{a}_k(V_0) = V$.

(b) Let p be a plaquette of $\mathbb{T}^{(k-1)}$. If p has crossing bonds in both of its directions, then $V_0(\partial p) = V(\partial p')$, where p' is the plaquette of $\mathbb{T}$ spanned by the four blocks containing the corners of p. Otherwise $V_0(\partial p) = 1$. Hence $F_{V_0}(p) = L^2 F_V(p')$ or 0, and $V \in \mathfrak{U}_k(\varepsilon_1)$ implies $V_0 \in \mathfrak{U}_{k-1}(L^2\varepsilon_1)$.

Proof. (a) Let $c = (y, y+e_\mu)$ and $x \in B_k(y)$. The pieces $\Gamma^k_{y,x}$ and $\Gamma^k_{y+e_\mu,x(c)}$ of $\Gamma^k_{c,x}$ stay inside blocks. The straight piece $[x, x(c)]$ contains exactly one crossing bond, from $B_k(y)$ to $B_k(y+e_\mu)$. The same holds for $[c]$. Hence $V_0(\Gamma^k_{c,x}) = V(c) = V_0([c])$, all logarithms in Definition 21.4 vanish, and $\mathfrak{a}_k(V_0)(c) = V_0([c]) = V(c)$.

(b) The two bonds of p in direction μ are crossing or not simultaneously, and they connect the same pair of blocks if the bonds in direction ν do not cross. In that case $V_0(\partial p)$ is $V(c)V(c)^{-1} = 1$ or 1. If both pairs cross, the four bonds carry the values of V on the four bonds of $\partial p'$, in the order of Definition 21.3. Finally $F_{V_0}(p) = \delta_{k-1}^{-2}\ell(V_0(\partial p)) = L^2\ell(V(\partial p')) = L^2 F_V(p')$. □

In units of δ_{k-1}, curvatures are multiplied by $\delta_{k-1}^2 = L^{-2}$. So in its own units, V_0 satisfies the same curvature bound ε_1 as V.

30.2 Change of units

As in the proof of Theorem 24.2, the $(k-1)$-step structure on $(\mathbb{T}_\eta, \mathbb{T}^{(k-1)})$, in units of δ_{k-1}, is the $(k-1)$-step structure on a torus of side Ln with fine spacing $L\eta$. The Wilson action changes by a constant factor, so critical points and minimizers are the same.

Lemma 30.2 (Change of units)**.** *There is C_L, depending only on d, α, G, L and M, with the following property. Let U_0 be a real configuration on $\mathbb{T}_\eta$ which, in units of δ_{k-1}, lies in $\mathfrak{U}(\theta)$ and is (M,θ)-regular. If $C_L\theta \leq 1$, then in units of $\delta_k = 1$, $U_0 \in \mathfrak{U}(L^3\theta)$ and U_0 is $(M, C_L\theta)$-regular.*

Proof. Passing to units of δ_k divides lengths by L and multiplies Lie fields by L, their differences and curvatures by L^2, currents by L^3, and Hölder quotients of differences by $L^{2+\alpha}$. This gives the first claim. For the second, let $r' = (4M+2)/L$. Every cube $\Lambda_{r'}(z)$ carries a gauge with $|A| \le L\theta$ and $|\partial A| \le L^2\theta$, and with Hölder quotients of ∂A at most $L^{2+\alpha}\theta$ on cubes of radius $3/L$. Let $r = r' - 2$. Two points of $\Lambda_r(z)$ at distance at most $3/L$ lie in a common cube of radius $3/L$ contained in $\Lambda_{r'}(z)$. For points at larger distance the quotient is at most $2|\partial A|(L/3)^\alpha$. Hence U_0 is $3L^3\theta$-regular on every cube $\Lambda_r(z)$. Since $M \ge 5L$, $r \ge 18$. Lemma 28.3 with $R = 4M+2$ gives the claim, with $C_L = 3C_E(4M+2)^{p+1}L^3$. □

30.3 The theorem

Theorem 30.3 (Bałaban's variational theorem on the torus)**.** *Assume the conventions of Part VI, in particular $M \ge \max(M_0, 5L)$. Let $d \ge 3$, and let θ_0 be as in Theorem 25.1 and n_0 as in Lemma 29.2. There are $a_1, B_3 > 0$, depending only on d, L, G, a, α and M, such that the following holds for all $k \ge 0$, all $n \ge n_0$ and every real configuration V on $\mathbb{T}$ with $V \in \mathfrak{U}_k(\varepsilon_1)$, $\varepsilon_1 \le a_1$.*

(a) $S|_{\mathfrak{B}(V)}$ has exactly one critical orbit in $\mathfrak{B}(V) \cap \mathfrak{U}(\theta_0)$. This orbit is the set of minimizers of S on $\mathfrak{B}(V) \cap \mathfrak{U}(\theta_0)$.

(b) Every element $U_k(V)$ of this orbit lies in $\mathfrak{U}(B_3\varepsilon_1)$ and is $(M, B_3\varepsilon_1)$-regular.

The constants do not depend on k, n or η.

This is [8, Theorem 1] for a single torus with the small-field condition everywhere. Part (b) is the regularity statement [8, (8)–(10)] for cubes of the fixed size M. As in [8], the elements of the orbit are called *background fields* or *minimal configurations*.

Proof. Constants. Let C_{base} be the constant of the case $k = 0$ below, and put

$$B_3 = \max(4C_{\mathrm{imp}}, C_{\mathrm{base}}), \qquad K = (1 + C_6)C_L.$$

Choose a_1 so small that $C_L B_3 a_1 \le \min(c_6, 1)$, $K B_3 a_1 \le \varepsilon_{\mathrm{imp}}$, $B_3 a_1 \le \theta_0$, and $a_1 \le c_1$, $d(8M+6)a_1 \le c_G$. These choices do not depend on k.

The case $k = 0$. Then $\eta = 1$, $\mathbb{T}_\eta = \mathbb{T}$, $\mathfrak{B}(V) = \{V\}$ and $\mathfrak{G}_0 = \{1\}$. We first prove (b); then $V \in \mathfrak{U}(B_3\varepsilon_1) \subset \mathfrak{U}(\theta_0)$, and (a) is trivial. The current $J_V = D_V^T F_V^s$ is a combination of at most $4d$ values of F_V^s, so $|J_V| \le C\varepsilon_1$. For regularity, fix z and a point $z_0 \in \mathbb{T}$ with $|z - z_0|_\infty \le 1$, and let u be the axial gauge on $\Lambda_{4M+2}(z)$ with $V^u(\Gamma^{\mathrm{ax}}_{z_0,x}) = 1$. As in the proof of Lemma 28.4, $|\ell(V^u(b))| \le d(4M+4)\varepsilon_1$ for every bond b of $\Lambda_{4M+2}(z)$. On the unit lattice, for $A = \ell(V^u)$, $|\partial A| \le 2|A|$, and the Hölder quotients of ∂A are at most $2|\partial A| \le 4|A|$, since distinct points have distance at least 1. So V is $(M, C_{\mathrm{base}}\varepsilon_1)$-regular and lies in $\mathfrak{U}(C_{\mathrm{base}}\varepsilon_1)$, with C_{base} depending only on d, G and M.

From $k-1$ to k. Let $V \in \mathfrak{U}_k(\varepsilon_1)$ with $\varepsilon_1 \le a_1$, and let V_0 be its trivial extension. In units of δ_{k-1} the torus has side $Ln \ge n_0$, and V_0 has curvature at most ε_1 (Lemma 30.1). By the induction hypothesis there is a minimal configuration $U_0 = U_{k-1}(V_0)$. In units of δ_{k-1} it lies in $\mathfrak{U}(B_3\varepsilon_1)$ and is $(M, B_3\varepsilon_1)$-regular. By Lemma 30.2, in units of δ_k,

$$U_0 \in \mathfrak{U}(L^3 B_3 \varepsilon_1), \qquad U_0 \text{ is } (M, \varepsilon)\text{-regular}, \qquad \varepsilon := C_L B_3 \varepsilon_1 \le c_6.$$

Since $U_0^{(k-1)} = V_0$, Lemma 30.1(a) gives

$$W_k = U_0^{(k)} = \mathfrak{a}_k(V_0) = V,$$

so the hypotheses of Theorem 25.1 hold with $\beta = 0$; note $|J_{U_0}| \le L^3 B_3 \varepsilon_1 \le \varepsilon$. Parts (a) and (b) of that theorem give (a).

For (b), let U be an element of the orbit. By Theorem 25.1(e), U is $(M, \varepsilon_0^{(0)})$-regular and lies in $\mathfrak{U}(\varepsilon_0^{(0)})$, with $\varepsilon_0^{(0)} = (1 + C_6)\varepsilon = KB_3\varepsilon_1 \le \varepsilon_{\text{imp}}$. If $\varepsilon_0^{(0)} \le B_3\varepsilon_1$ there is nothing to prove, so let $\varepsilon_0^{(0)} > B_3\varepsilon_1 \ge 4C_{\text{imp}}\varepsilon_1$. Define $\varepsilon_0^{(m+1)} = C_{\text{imp}}\varepsilon_1 + \frac{1}{2}\varepsilon_0^{(m)}$. If $\varepsilon_0^{(m)} \ge 2C_{\text{imp}}\varepsilon_1$, then $\varepsilon_0^{(m+1)} \le \varepsilon_0^{(m)}$. By Lemma 29.2 and induction over m, U is $(M, \varepsilon_0^{(m)})$-regular and lies in $\mathfrak{U}(\varepsilon_0^{(m)})$ for all m. Since

$$\varepsilon_0^{(m)} - 2C_{\text{imp}}\varepsilon_1 = 2^{-m}(\varepsilon_0^{(0)} - 2C_{\text{imp}}\varepsilon_1) \le 2^{-m}KB_3\varepsilon_1,$$

we have $\varepsilon_0^{(m)} \le 2C_{\text{imp}}\varepsilon_1 + \frac{1}{2}B_3\varepsilon_1 \le B_3\varepsilon_1$ once $2^m \ge 2K$. This proves (b) with the same B_3, and closes the induction. □

The induction uses Theorem 25.1 with $\beta = 0$: the reference has exactly the prescribed averages. On the nested domains of [8] this is not the case, and β is of order ε_1 [8, (14)]. Theorem 25.1 allows $\beta > 0$, which is used in Lemma 29.2 and in Corollary 30.5 below.

30.4 Consequences

Corollary 30.4 (Local regularity from gauge-invariant conditions)**.** *There are $a_2, B_5 > 0$, depending only on d, L, G, a, α and M, such that for all $k \ge 0$ and $n \ge n_0$ every real configuration $U \in \mathfrak{U}(\theta)$ on $\mathbb{T}_\eta$ with $\theta \le a_2$ is $(M, B_5\theta)$-regular. More precisely, U is gauge equivalent, by a real gauge transformation, to $e^{i\eta A}U_k(U^{(k)})$. Here A lies in Balaban's Landau gauge relative to $U_k(U^{(k)})$, and A obeys the bounds of Theorem 24.2 with $\varepsilon + \theta + \beta = (2B_3 + 1)\theta$.*

Proof. Let $k = 0$. Then $\eta = 1$, $\mathbb{T}_\eta = \mathbb{T}$ and $U^{(0)} = U$. By the case $k = 0$ of the proof of Theorem 30.3, $\mathfrak{B}(U) = \{U\}$, so $U_0(U) = U$, and the representation holds with $A = 0$, for which the gauge condition and the bounds are trivial. The curvature of U is at most $\theta \le a_1$, so Theorem 30.3(b), with $V = U$ and $\varepsilon_1 = \theta$, shows that U is $(M, B_3\theta)$-regular.

Let $k \ge 1$ and $V = U^{(k)}$. By Proposition 21.7, $V \in \mathfrak{U}_k(2\theta)$. Let $2\theta \le a_1$ and $(2B_3+1)\theta \le c_4$. By Theorem 30.3, $U_0 = U_k(V)$ exists. It is $(M, 2B_3\theta)$-regular, satisfies $|J_{U_0}| \le 2B_3\theta$, and has $U_0^{(k)} = V$. By Lemma 22.2(c), the $\mathfrak{G}_0$-orbit of U contains a configuration U^{ax} with $U' = U^{\text{ax}}U_0^{-1} \in \text{Ax}_k$. Transformations in $\mathfrak{G}_0$ do not change the averages, so $U'^{[k]} = (U^{\text{ax}})^{(k)}V^{-1} = 1$. Theorem 24.2, with $\varepsilon = 2B_3\theta$ and $\beta = 0$, gives $\tilde{U} = e^{i\eta A}$ in the Landau gauge with $(\tilde{U}U_0)^{\mathsf{w}_k(\tilde{U})} = U^{\text{ax}}$ and with the stated bounds. Finally, $e^{i\eta A}U_0$ is $(M, 2B_3\theta + CB_1(2B_3 + 1)\theta)$-regular, by the argument in the proof of Proposition 27.4(c). That argument uses the Hölder bound on the unit cubes $\Lambda_1(z)$. For the field A of Theorem 24.2 this bound is given by the proof of Proposition 24.5 (Theorem 6.3(d)) and Remark 7.2. So the corollary holds with $B_5 = 2B_3 + CB_1(2B_3 + 1)$, which is at least the constant B_3 of the case $k = 0$. □

Corollary 30.4 is the torus version of [6, Sect. F, Prop. 6] and of the regularity conditions [7, (3.35)–(3.36)]. It removes the hypothesis of Part III for all configurations of Bałaban's small-field class: on tori of side $n \ge n_0$, regularity follows from the gauge-invariant bounds on curvature and current. The proof uses the background field as a reference: the local gauges are the Landau gauge relative to $U_k(U^{(k)})$, composed with the local gauges of the background field. The latter come out of the induction.

Corollary 30.5 (Analyticity around a background field [8, Sect. G])**.** *Let V be as in Theorem 30.3, and let U_0 be an element of its minimal orbit. For $k \ge 1$, Theorem 25.1 applies with this reference and $\varepsilon = B_3\varepsilon_1$. For real B on $\mathbb{T}$ with $|B|_\infty \le \beta$ and $B_3\varepsilon_1 + \beta \le c_6$, the critical orbit for $e^{iB}V$ in $\mathfrak{B}(e^{iB}V) \cap \mathfrak{U}(\theta_0)$ is unique, and its Landau representative $A_*(B)$ relative to U_0 satisfies $A_*(0) = 0$. The map $B \mapsto A_*(B)$ extends analytically to complex B, and $dA_*(B)\delta B$ decays exponentially away from the support of δB, as in Theorem 25.1(f).*

Proof. U_0 is $(M, B_3\varepsilon_1)$-regular with $|J_{U_0}| \le B_3\varepsilon_1$, and $W_k = V$. For $B = 0$, the configuration $U_0 = \mathsf{U}(0)$ is critical, and $A = 0$ lies in the Landau gauge with $\mathcal{Q}(1) = 1$; by uniqueness, $A_*(0) = 0$. For $k \ge 1$ the rest is Theorem 25.1.

Let $k = 0$. The operators of Parts II–V need $k \ge 1$ (Section 2), but the conclusions hold directly. Here $\eta = 1$, $\mathbb{T}_\eta = \mathbb{T}$, $\mathfrak{G}_0 = \{1\}$ and $U_0 = V$, by the case $k = 0$ of the proof of Theorem 30.3. The averaging, the correspondence transformation and $\mathcal{Q}$ are the identity, so $\mathfrak{B}(e^{\mathrm{i}B}V) = \{e^{\mathrm{i}B}V\}$, and the relative field of $e^{\mathrm{i}B}V$ is $A_*(B) = B$; the Landau gauge condition is void. By Lemma 21.1(b), the curvature of $e^{\mathrm{i}B}V$ is at most $\varepsilon_1 + 4|B|_\infty$, and its current, a combination of at most $4d$ values of F^s, is at most $4d$ times this. Since $\varepsilon_1 \le B_3\varepsilon_1$, both are at most $20d\, c_6 \le \min(\theta_0, c_1)$, by the choice of c_6 in the proof of Theorem 25.1. So the configuration $e^{\mathrm{i}B}V$ lies in $\mathfrak{U}(\theta_0) \cap \mathfrak{U}_0(c_1)$, and its orbit is the unique critical orbit. The map $B \mapsto B$ is entire, $A_*(0) = 0$, and $|A_*(B)|_{C^1} \le 3|B|_\infty$ by (23.2). Finally $dA_*(B)\delta B = \delta B$ is supported on the bond c, and its covariant gradient on the bonds within distance 1 of c. This is the decay of Theorem 25.1(f), with $C_6 \ge 3e^{\gamma'}$. □

Remark 30.6 (Cubes of size of order ε_1^{-1}). Bałaban also proves gauges on cubes of side $2M' \le c\,\varepsilon_1^{-1}$, with bounds $B_3M'\varepsilon_1$ linear in the size [8, (9), (10)]. Theorem 30.3 and Lemma 28.3 give gauges on cubes of side $2R$ with bounds $C(R/M)^p B_3\varepsilon_1$, polynomial in R, for $R^{p+1}\varepsilon_1$ small. Linear bounds should follow by running Lemma 29.2 with a splicing radius proportional to R, with R limited to a power of ε_1^{-1} by the quadratic error. We do not prove this. The regularity at the fixed scale M is all that Parts II–V require.

Part VII: One step of the renormalization group

31 The renormalization group transformation

This part carries out one step of Bałaban's renormalization group for pure Yang–Mills theory in the small-field region [9]. The plan and the target are described in the companion blueprint (`RG-blueprint`) [14], referred to below as "the blueprint". Here we derive an exact representation of one step as a Gaussian integral around the background field, and we prove the linear estimates on which the analysis of that integral rests:

- the coercivity and decay of the fluctuation covariance;
- the analyticity of the coarse effective action and the decay of its derivatives;
- the structure of the background-dependent normalization.

The polymer expansion of the interaction, its renormalization and the iteration are the subject of later parts.

Conventions for Part VII. We keep the conventions of Parts IV–VI. In addition $d = 3$ in the statements about couplings, although the constructions of this part work for $d \ge 3$, and G is semisimple. Semisimplicity is not used in this part; it is needed for the renormalization in the next one. We work in the *units of step* k: the unit lattice is $\mathbb{T} = \mathbb{T}^{(k)}$, the fine lattice is $\mathbb{T}_\eta$ with $\eta = L^{-k}$, and the next coarse lattice is $\mathbb{T}' = \mathbb{T}^{(k+1)}$, of spacing L. Its blocks are $B'(y) = B_{k+1}(y)$, with L^d points of $\mathbb{T}$, and the one-step averaging is $\mathfrak{a}' = \mathfrak{a}_{k+1}$. The torus $\mathbb{T}$ has side Ln, where n is the side of $\mathbb{T}'$ in its own units. We assume $n \ge n_{\delta=1}$, with n_δ from Proposition 31.5, and $n \ge 80M'_L$, with M'_L from Theorem 34.4. Fields on $\mathbb{T}$ carry the unweighted norm $\|Z\|^2 = \sum_b |Z(b)|^2$.

31.1 The model and the transformation

Let $\epsilon = L^{-N}$ be the finest lattice spacing, g the bare coupling, and $g_k^2 = g^2 L^k \epsilon$ the dimensionless coupling at step k. The initial density on the ϵ-lattice is $\rho_0(U) = \exp(-g_0^{-2} S_{\mathrm{lat}}(U) - E_0)$, where $S_{\mathrm{lat}}(U) = \sum_p (\mathsf{N} - \operatorname{Re}\operatorname{tr} U(\partial p))$. In the units of step k the ϵ-lattice has spacing $\eta = L^{-k}$, and $g_0^{-2} S_{\mathrm{lat}} = g_k^{-2} S$ with the action S of Section 23. Indeed $S = \eta^{d-4} S_{\mathrm{lat}}$, $d = 3$, and $g_0^2 = g_k^2 \eta$.

The averaging $\mathfrak{a}'$ uses logarithms, so it is defined only on part of the configuration space. Let $\operatorname{dom}\mathfrak{a}'$ be the set of real configurations V on $\mathbb{T}$ such that, for every positively oriented bond c of $\mathbb{T}'$ and every $x \in B'(c_-)$,

$$\left| V(\Gamma_{c,x}^{k+1})\, V([c])^{-1} - 1 \right| \le c_G .$$

For $V \in \operatorname{dom}\mathfrak{a}'$ the logarithms in Definition 21.4 are defined and lie in $\mathfrak{g}$, by the choice of c_G (Section 21), so $\mathfrak{a}'(V)$ is defined and real. Outside this set a logarithm may be undefined or may leave $\mathfrak{g}$. The set $\operatorname{dom}\mathfrak{a}'$ is closed, and $\mathfrak{a}'$ is continuous on it. It is invariant under real gauge transformations u, which conjugate the holonomies above by $u(c_-)$. It contains $\mathfrak{U}_k(b)$ whenever $dbL^2 \le c_G$: by Lemma 21.5(a), with $\delta_{k+1} = L$, the logarithms of these holonomies are bounded by dbL^2, and $|h-1| \le |\ell(h)|$ for $h \in G$. If $dbL^2 < c_G$, it contains $\mathfrak{U}_k(b)$ in its interior.

For an integrable density ρ on configurations V on $\mathbb{T}$ that vanishes outside $\operatorname{dom}\mathfrak{a}'$, let $T\rho$ be the image of the measure $\rho(V) \prod_b dV(b)$ under $\mathfrak{a}'$:

$$\int \varphi \, d(T\rho) = \int \prod_b dV(b)\ \varphi(\mathfrak{a}'(V))\, \rho(V)$$

for bounded Borel functions φ of configurations on $\mathbb{T}'$. Here b runs over the positively oriented bonds of $\mathbb{T}$, and $dV(b)$ is the normalized Haar measure. With $\varphi = 1$, the total mass of $T\rho$ is $\int \prod_b dV(b)\, \rho(V)$. We write

$$(T\rho)(V') = \int \prod_b dV(b)\ \prod_c \delta\big(\mathfrak{a}'(V)(c)\, V'(c)^{-1}\big)\ \rho(V), \tag{31.1}$$

where c runs over the positively oriented bonds of $\mathbb{T}'$ and δ is the δ-function of G at 1 with respect to the Haar measure. This formula is a notation for the image measure. If δ is replaced by a continuous function $\delta_n \ge 0$ on G with $\int \delta_n = 1$, supported in a neighbourhood of 1 that shrinks as $n \to \infty$, the right side is a continuous density, and the measures with these densities converge weakly to $T\rho$: for continuous φ this follows from Fubini's theorem, the invariance of the Haar measure and the uniform continuity of φ. The left side denotes a density of $T\rho$ with respect to $\prod_c dV'(c)$, and it is used only where such a density exists. A sufficient condition is that ρ vanishes outside an open subset of $\operatorname{dom}\mathfrak{a}'$ on which $\mathfrak{a}'$ is a submersion, i.e. its derivative has rank $\dim\mathfrak{g}$ times the number of positively oriented bonds of $\mathbb{T}'$. Then, near each point of this set, $\mathfrak{a}'$ is a coordinate projection in suitable analytic coordinates, and for almost every V' the density at V' is the integral of ρ over the fibre $\{\mathfrak{a}'(V) = V'\}$ with respect to the measure that the two Haar measures induce on it. Integrability of ρ alone does not make this fibre integral finite at every V'. If ρ is a bounded Borel function and vanishes outside a compact subset of the open set, the fibre integral converges absolutely at every V'; it is then taken as the density at every V', and the usual rules for δ-functions compute it. In that case $\int \prod_c dV'(c)\, (T\rho)(V') = \int \prod_b dV(b)\, \rho(V)$. The density is used only for the small-field step below. There the integrand is bounded, and the cutoff confines it to a compact subset of the region of Lemma 32.4, which gives the submersion, and Proposition 32.5 computes the density. Passing to the units of step $k+1$ relabels the configurations, without any Jacobian since they are group valued, and replaces S by $L^{4-d}S = LS$. Hence $g_k^{-2} S$ becomes $g_{k+1}^{-2} S$ with $g_{k+1}^2 = L g_k^2$.

If ρ vanishes outside $\operatorname{dom}\mathfrak{a}'$ and $\rho(V^u) = \rho(V)$ for all real gauge transformations u on $\mathbb{T}$, then $T\rho$ is gauge invariant on $\mathbb{T}'$: it is invariant under $V' \mapsto V'^{u'}$ for every real gauge transformation u' on $\mathbb{T}'$. This follows from $\mathfrak{a}'(V^u) = \mathfrak{a}'(V)^{u|_{\mathbb{T}'}}$ (Lemma 21.5), applied to an extension u of u', from the invariance of $\operatorname{dom}\mathfrak{a}'$, and from the invariance of the Haar measure under $V \mapsto V^u$. All densities below are gauge invariant.

The small-field step. In the small-field region the density at step k has, by assumption, the form

$$\rho_k(V) = \exp\Big(-\frac{1}{g_k^2} S(U_k(V)) + \mathfrak{E}_k(V) \Big) \qquad \text{for } V \in \mathfrak{U}_k(\tilde{h}), \tag{31.2}$$

with $U_k(V)$ the background field of Theorem 30.3, a gauge-invariant function $\mathfrak{E}_k$, measurable and bounded on $\mathfrak{U}_k(\tilde{h})$, and a threshold $\tilde{h} \le a_1$. The renormalization group studied here is $\rho \mapsto T(\chi^{\mathrm{w}}\rho)$, where χ^{w} restricts the fluctuation field around the background of the step to $|\cdot| \le w$ (Definition 32.3). The cutoff depends on V' through the base configuration of the step. In $T(\chi^{\mathrm{w}}\rho)$ it is taken at the base of $\mathfrak{a}'(V)$, and it is set to 0 unless $V \in \operatorname{dom}\mathfrak{a}'$ and $\mathfrak{a}'(V)$ satisfies the hypotheses of Proposition 32.5, under which the base is defined. That proposition shows that, for such V', the cutoff of V' vanishes outside the interior of $\operatorname{dom}\mathfrak{a}'$, so that the restriction to $\operatorname{dom}\mathfrak{a}'$ removes nothing, and that $\mathfrak{a}'$ is a submersion wherever the cutoff is 1. The cutoff is a Borel function of V. Indeed, near each such V', Corollary 30.5 gives an element U' of the minimal orbit that depends continuously on V'. For this choice the cutoff is the indicator function of a closed set composed with continuous maps: $\mathfrak{a}'$, the base $U'^{(k)}$, the axial representative of Lemma 22.2(a), and ℓ. The cutoff does not depend on the choice of U' (Section 32), and finitely many such neighbourhoods cover the compact set of these V'. This is the analogue of [15, (172)]; Bałaban's version is [9, (7)–(13)]. The large-field complement is not treated here. In Section 48 the small-field condition is also imposed on the white noise, $T_{\mathrm{sf}}(\rho) = T(\chi^{\mathrm{w}}\chi_\xi\rho)$ (Definition 48.1), as in [15, (206)]. The cutoff χ_ξ also depends on V', and it is treated in the same way. It does not depend on the choice of U' either (Section 48), and it is Borel by the same argument, since C_k depends continuously on the base (Corollary 34.5).

31.2 Curvature of background fields

Theorem 30.3 bounds the curvature of $U_k(V)$ by $B_3\varepsilon_1$, where B_3 depends on L. The small-field thresholds of consecutive steps are compatible only if the curvature of the background is bounded with a constant that does *not* grow with L. That is proved here.

The constants of the pointwise bounds of Parts II and III must therefore be independent of L and of M; note that $M \ge 5L$ in Part VI. We make this explicit. Bałaban's averaging pair enters only through one parameter.

Definition 31.1 (Parameter of an averaging pair)**.** Let U be a real $(1,\varepsilon)$-regular background and $(Q', \mathbf{Q})$ an averaging pair with properties (i), (iii) and (iv) of Remark 19.12. Let $\varepsilon_Q \ge 0$. The pair has *parameter* ε_Q *at* U if the following holds for every $y \in \mathbb{T}$ and every real gauge transformation u in which (B) holds with parameter ε on $\Lambda_4(c_y)$. Here (B) on $\Lambda_4(c_y)$ refers to the bonds and differences that involve only points of $\Lambda_4(c_y)$, and the Hölder quotient on a cube $\Lambda_3(z'') \subset \Lambda_4(c_y)$ is taken over the pairs of differences that involve only points of $\Lambda_3(z'')$. If U is (M,ε)-regular, every gauge of Definition 18.1 on a cube containing $\Lambda_4(c_y)$ satisfies (B) with parameter ε on $\Lambda_4(c_y)$ in this sense. Computed in the gauge u, on the blocks and tubes contained in $\Lambda_4(c_y)$:

(a) $Q'(U^u) - Q_0$ and $\mathbf{Q}(U^u) - \mathbf{Q}$ satisfy the bounds of Lemma 12.5(b) with $C\varepsilon$ replaced by ε_Q: block norms, the bounds from sup norms to values, and the same for the transposes;

(b) the defect E satisfies the bounds of Lemma 12.5(c) with $C\varepsilon$ replaced by ε_Q, and its transpose satisfies $|E^T\omega|_\infty \le \varepsilon_Q|\omega|_\infty$.

By Lemma 12.5, the model pair $(Q'(U), \mathbf{Q}(U))$ of Section 12 has parameter $C_{\mathrm{mod}}\varepsilon$, with C_{mod} depending only on d.

Lemma 31.2 (Monotonicity of regularity)**.** *If U is (M,ε)-regular and $1 \le M' \le M$, then U is (M',ε)-regular.*

Proof. The cube $\{x : |x-z|_\infty \le 4M'+2\}$ lies in $\{x : |x-z|_\infty \le 4M+2\}$. Restrict the gauge transformation of Definition 18.1 to it. The bonds with both endpoints in the smaller cube have both endpoints in the larger one, and every cube $\Lambda_3(z')$ in the smaller cube lies in the larger one. So the three bounds of Definition 18.1 persist. □

Lemma 31.3 (Uniformity of the constants of Parts II–III)**.** *Let $d \ge 3$ and fix $a > 0$, $\alpha \in (0,1)$. Let M_0 be the constant of Part III (Theorem 19.6). There are ε_{U}, θ_Q, C_H, $\gamma_H > 0$, depending only on d, a, α and $\dim \mathfrak{g}$, with the following property. Let $k \ge 1$ and let the side of $\mathbb{T}$ be at least $8M_0$. Let $M \ge M_0$, let U_0 be real and (M,ε)-regular with $\varepsilon \le \varepsilon_{\mathrm{U}}$, and let $(Q', \mathbf{Q})$ be an averaging pair with properties (i)–(iv) of Remark 19.12 and parameter $\varepsilon_Q \le \theta_Q$ at U_0. Then:*

(a) Theorem 18.2 (with $X = \mathbb{T}_\eta$), Theorems 19.6 and 19.7 (with $X = \mathbb{T}_\eta$), and Theorem 19.9 hold at U_0 with this pair, in the form stated there: existence, covariance, identities, coercivity, decay and the pointwise bounds in local gauges, with no comparison with the operators at $U = 1$. Their constants c, C, γ, c_* depend only on d, a, α, $\dim \mathfrak{g}$: neither on M, nor on $N = L^k$, n, k, nor on the pair beyond $\varepsilon_Q \le \theta_Q$.

(b) Let $\mathbf{H} = \mathbf{H}(U_0)$ be the minimizer map of Theorem 19.9 (with $W = 0$) and ∇ the covariant difference at U_0. For every unit-lattice field B,

$$|\mathbf{H}B|_{B(y)} + |\nabla \mathbf{H}B|_{B(y)} \le C_H \sum_{y'} e^{-\gamma_H |y-y'|} |B(y')|. \tag{31.3}$$

(c) For Bałaban's pair $(Q'_k(U_0), \mathbf{Q}_k(U_0))$ we have $E = 0$, and there are $c_Q, C_Q > 0$ such that the pair has parameter $\varepsilon_Q \le C_Q\varepsilon$ for $\varepsilon \le c_Q$. Here c_Q, C_Q may depend on d, L, G, a, α and M (the conventions of Part IV). Hence (a) and (b) hold for this pair whenever $\varepsilon \le \varepsilon_{\mathrm{B}} = \min(\varepsilon_{\mathrm{U}}, \theta_Q/C_Q, c_Q)$. In particular (31.3) holds with C_H, γ_H independent of L and M, for ε small depending on L and M.

If the side of $\mathbb{T}$ is less than $8M_0$, (a)–(c) hold as well, with the same constants.

Proof. Step 0: reduction to $M = M_0$. For $X = \mathbb{T}_\eta$ every j lies in $J(\mathbb{T}_\eta)$, so $m_{\mathbb{T}_\eta} = 0$. Hence the torus operators $\mathfrak{A}'_{\mathbb{T}_\eta}$, G', Γ', $P_{\mathcal{V}}$, R, S, Π, $\mathfrak{A}_\pi = \mathfrak{A}^{\mathbb{T}_\eta}$, $\mathbf{G}$, $\mathbf{\Gamma}_\pi$, $\mathbf{H}$, $\mathbf{C}$ are defined without reference to M. By Lemma 31.2, U_0 is (M_0,ε)-regular. Apply every result of Part III with $M = M_0$; their hypotheses are "$M \ge M_0$ and (M,ε)-regular". Their constants then depend on M_0, which depends only on $d, a, \alpha, \dim \mathfrak{g}$. Coercivity and decay are properties of the fixed torus operators, so proving them at scale M_0 proves them for U_0. This proves the independence of M. The scale M_0 also enters through the partition of unity (Lemma 17.3), which needs the side of $\mathbb{T}$ to be at least $8M_0$. If the side is smaller, the partition at scale M_0 is the trivial one of Section 17. The cube of Definition 18.1 at scale M_0 then covers $\mathbb{T}_\eta$, so one real gauge gives the bounds of that definition on all of $\mathbb{T}_\eta$. The proof of Theorem 19.6 treats this case, and the other results in (a) do not use the partition. In Steps 1–4 this gauge serves as the local gauge of every cube, and (a)–(c) follow with the same constants.

Step 1: Part I inputs. The model pair at $U = 1$ is $(Q_0, \mathbf{Q})$ on blocks of side 1 with N^d fine points. The following constants are used, and none depends on $N \ge 2$ (Section 2: L enters only through $N = L^k$; at $N = 1$ the trial field of Theorem 18.2(c) degenerates, which is why $k \ge 1$):

- the block Poincaré constants (Lemma 3.1, Corollary 3.3), depending on d;

- the gauge-field coercivity constant c_g (Theorem 9.6), depending on d and a: on a directly, on the constants C_P, C_W (Lemma 9.3) and C_B (Lemma 4.4), which depend only on d, and on the bound in Proposition 8.8(e);
- the decay of $P_{\mathcal{V}}(1)$ and $dP_{\mathcal{V}}(1)d^*$ (Proposition 8.8), depending on d and a;
- the Combes–Thomas constants (Lemmas 5.5, 5.6, 5.7), explicit in the coercivity and conjugation constants;
- the interior regularity constant C_{reg} (Theorem 6.3), depending on d and α, for all $N \geq 1$.

All these are statements on the fine lattice with unit blocks. Distances, decay rates and the norms $|\cdot|_{B(y)}$ are measured in unit-lattice units. The supports of the averaging kernels, $B(y)$ and $S_c = B(c_-) \cup B(c_+)$, consist of a fixed number of unit blocks, whatever L is.

Step 2: where the averaging pair enters. We list every use of the pair in the proofs of the results in (a). In each, the model bound $C\varepsilon$ is replaced by ε_Q, and the conclusion holds once $\varepsilon_Q \leq \theta_Q$, with θ_Q a threshold of the same kind as ε_{U}.

(A1) Theorem 18.2(a): in the local gauge, $|(Q'\lambda)(y) - (Q_0\tilde{\lambda})(y)| \leq \varepsilon_Q \|\lambda\|_{B(y)}$, by Definition 31.1(a) and the locality (iv). The error term becomes $C\varepsilon_Q^2\|\lambda\|_B^2$ and is absorbed.

(A2) Theorem 18.2(b): hypothesis (ii) of Lemma 5.7 uses that $Q'^T Q'$ is block-diagonal, by (iv), and bounded by $(1+\varepsilon_Q)^2$.

(A3) Theorem 18.2(c): the trial field φ built with the model transports in the local gauge satisfies $\|D\varphi\| \leq C\|g\|$, as before. For the general pair, $Q'\varphi = g + r$ with $\|r\| \leq C(\varepsilon+\varepsilon_Q)\|g\|$, so $\langle Q'^T g, \varphi\rangle \geq (1-C(\varepsilon+\varepsilon_Q))\|g\|^2$. The bound $M'_X \geq c^2$ uses $\|Q'^T g\| = \|g\|$, which is (i).

(A4) Theorem 18.2(d),(e): algebraic. They use (i) for the surjectivity of Q', and (iv) for the facts that Q' maps $\ell^2_0(X)$ to $\ell^2(X_1)$ and that $Q'^T Q'$ does not couple blocks.

(A5) Theorem 18.2(g) and Theorem 14.1(b),(d): $a(Q'^T Q' - Q_0^* Q_0)$ is pointwise admissible with constants $(C\varepsilon_Q, 0, 1)$ (Lemma 13.6(c) with $C\varepsilon$ replaced by ε_Q). The pointwise bound on H' uses $|Q'^T g|_{B(y)} \leq (1+\varepsilon_Q)|g(y)|$, from the transposed bound in Definition 31.1(a).

(A6) Lemmas 15.1, 15.3 and 19.2: (iii), with the bounds on E from Definition 31.1(b). The real configuration V enters only through D_V, whose transport factors are unitary.

(A7) Lemma 19.5, Step 2: $a(\mathbf{Q}^T\mathbf{Q} - \mathbf{Q}^*\mathbf{Q})$ and the terms of $\mathcal{K}_2$ are admissible perturbations of size $C(\varepsilon_Q + \varepsilon_Q^2)$. The comparison of the scalar ingredients uses (A1)–(A5).

(A8) Theorem 19.6 and Theorem 19.7(a): $\mathbf{Q}^T\mathbf{Q}$ is local and bounded by $(1+\varepsilon_Q)^2$, by (ii) and (iv). This is all that the double commutators (Lemma 17.5) and hypothesis (ii) of Lemma 5.7 use.

(A9) Theorem 19.7(b): in the local gauge, $a(\mathbf{Q}^T\mathbf{Q} - \mathbf{Q}^*\mathbf{Q})$ is pointwise admissible of type 1 with constants $(C\varepsilon_Q, 0, 1)$, and $a\mathbf{Q}^*\mathbf{Q}$ with $(0, C, 1)$. The terms $\mathbf{Q}^T ESu$, $DRG'(E^T\mathbf{Q}u)$ and $DRG'(E^T ESu)$ of $\mathcal{K}_2$ are of the form $\mathbf{Q}^T$(bounded) or DRG'(bounded) with a factor ε_Q; this uses the transposed bound on E of Definition 31.1(b). Only sup bounds of the averages enter. No Hölder bound on their kernels is needed, since they contribute to F_u only.

(A10) Lemma 19.8: the model trial fields in the local gauge, with the Neumann series around the diagonally dominant matrix T_U. The margin is $m_\perp(\sigma_0 - \sigma_+ - \sigma_-) \geq m_\perp/5$ for $N \geq 16$, and a fixed margin for $N < 16$. The general pair adds a local error of size $C(\varepsilon + \varepsilon_Q)$, which is absorbed for $\varepsilon \leq \varepsilon_{\mathrm{U}}$ and $\varepsilon_Q \leq \theta_Q$. The slice correction uses $\|ESL\| \leq C\varepsilon_Q$.

(A11) Theorem 19.9: $\mathbf{Q}_\pi \mathbf{G} \mathbf{Q}_\pi^T$ has decay, with constants controlled by (ii) and (iv). The pointwise bound on $\mathbf{H}B = \mathbf{G}J$, $J = \mathbf{Q}_\pi^T \mathbf{\Gamma}_\pi B$, uses three facts:

- $|\mathbf{Q}^T \omega|_{B(y)} \le (1 + C\varepsilon_Q) \max |\omega|$, the maximum over the bonds c with S_c meeting $B(y)$ (Lemma 2.6(c) for $\mathbf{Q}^*$, and the transposed bound of Definition 31.1(a));
- the identity $D^T \mathbf{Q}^T = Q'^T D_V^T + E^T$, from (iii), with the transposed bound on E of Definition 31.1(b);
- $|Q'^T g| \le (1 + \varepsilon_Q)|g|$.

The transposed sup bounds of the pair and of E enter here, and also in (A5) and (A9). The pointwise bound on $\mathbf{C}$ uses in addition the forward bounds from sup norms to values of $\mathbf{Q}$ and of E (Definition 31.1(a),(b)), through $\mathbf{Q}_\pi \mathbf{G} J = \mathbf{Q}\mathbf{G}J - ES\mathbf{G}J$. In each case it enters only as a factor $1 + C\varepsilon_Q$, so it affects the smallness condition, not the constants.

The explicit contour transports $\Gamma_{y,x}$ of the model pair enter only in (A3) and (A10), as stated in Remark 19.12. The reality in (i) is used in the Gårding inequality (Lemma 19.4) and in the variational characterizations of H' and $\mathbf{H}$. No other property of the pair is used.

Step 3: the constants. Choose θ_Q, and ε_{U} not larger than the threshold ε_0 of Parts II–III, below the thresholds of (A1)–(A11), of Lemma 13.2, and of Lemma 13.5 (that is, ε_*, depending on d, α, γ). Then:

- Theorem 18.2 gives c, C, γ depending on $d, a, \alpha, \dim \mathfrak{g}$, under $(1, \varepsilon)$-regularity.
- Lemma 19.5 and Theorem 19.6 give $\frac{c_g}{2}$ and c_* for $M \ge M_0$. In their proofs M_0 is fixed by $Ce^{-\gamma M_0} \le c_g/8$ and by the absorption of the terms CM_0^{-2}, with C, γ from the previous item; so M_0 is the constant of Part III and depends only on $d, a, \alpha, \dim \mathfrak{g}$. In Step 2 of the proof of Lemma 19.5 the pair at U^u is compared directly with the model pair $(Q_0, \mathbf{Q})$ at $U = 1$, by Definition 31.1(a) and (A7); so Step 3 concerns the model pair only. For Bałaban's pair the pair at $U = 1$ is the model pair anyway (Proposition 22.7(c)).
- Theorem 19.7(a) with $X = \mathbb{T}_\eta$ follows from Lemma 5.7, with c_* and a conjugation constant C_1 built from the scalar decay constants. The rate is $\gamma_0 = \min(\alpha_1, \frac{1}{3}, c_*/(2C_1))$. It does not involve M.
- Theorem 19.7(b) follows from Lemma 18.3, with $\gamma' = \frac{1}{2}\min(\gamma, \gamma_0)$ and a constant depending on $d, \alpha, \gamma, \gamma_0, C_\delta$. Here C_δ comes from the pointwise bounds of Theorem 18.2(g). The letter M in Lemma 18.3 is the L^2 amplitude of Lemma 13.5, not the regularity scale. That lemma is used with its invariant inputs: the small inputs are $|u| + |\nabla u|$ on the cubes $\Lambda_1(c_z)$, with weights $e^{-\gamma|y-z|}$ and the full covariant gradient ($|u| + |Du|$ for scalar fields). Only the small differential terms multiply them; the large smoothing terms use the local L^2 bounds from Combes–Thomas. The terms containing the pair or E in (A5), (A9) and (A11) use their forward and transposed value bounds, not lattice derivatives or Hölder quotients of the averaging kernels. Every composition uses a positive rate no larger than the inverse and perturbation rates (Lemma 13.3).
- Theorem 19.9: $\mathbf{\Gamma}_\pi$ has decay by Lemma 5.5, with the lower bound from (A10). The sum form of the pointwise bound of $\mathbf{G}$ (Theorem 15.4(c), at regular backgrounds) and (A11) give (b). Here γ_H is half the smallest of the rates above, and C_H is a product of the constants above and S_{γ_H} (Lemma 5.2).

Every constant in this chain depends only on $d, a, \alpha, \dim \mathfrak{g}$. This proves (a) and (b).

Step 4: (c). By Proposition 22.7(a), (b), (d), (f), Bałaban's pair has properties (i), (iii) (with $E = 0$ and $V = W_k$ real) and (iv). An (M, ε)-regular U_0 has curvature at most $C_{\mathrm{curv}}\varepsilon$ (proof of Corollary 22.8), so $U_0 \in \mathfrak{U}_0(C_{\mathrm{curv}}\varepsilon)$. Choose c_Q so that $\varepsilon \le c_Q$ implies

$\max(1, C_{\mathrm{curv}})\varepsilon \le c_3$ and the hypotheses of Lemma 21.8. Then, in each gauge of Definition 31.1, Proposition 22.7(e) applies on $\Lambda_4(c_y)$, with ε replaced by $\max(1, C_{\mathrm{curv}})\varepsilon$. It gives the kernel bounds with a constant that depends on L, by the conventions of Part IV. Call the resulting constant C_Q. The kernel bounds give the block norms by the Schur test, and the sup bounds of the transposes by column sums. Hence $\varepsilon_Q \le C_Q\varepsilon$, and $\varepsilon_Q \le \theta_Q$ for $\varepsilon \le \theta_Q/C_Q$. □

Remark 31.4 (Where L and M do enter). (a) Part IV provides only an L-dependent constant C_Q in Lemma 31.3(c). The corrections r_j of Lemma 22.5 are carried by the contours $\Gamma^j_{c,x}$, which share their initial bonds near the corner c_-. So the kernel of $\mathbf{Q}_k(U_0) - \mathbf{Q}$ may concentrate near the corners of the scale-j blocks, and its sup norm, which controls the transposes, is bounded only by $O(C_L\varepsilon)$, by the bookkeeping in the proof of Proposition 22.6(c). Whether this is sharp is not settled here. Lemma 31.3 uses the transposes in (A5), (A9) and (A11). This is why (31.3) is stated only for ε small depending on L. This only affects how small the regularity parameter must be, not C_H or γ_H.

(b) Two results of Part III have constants that depend on M. The random-walk expansion (Theorem 20.2(a)) gives decay at a rate of order $M^{-1}\log M$, and Theorem 19.10(b) has the factor e^{CM}. Neither is used in Lemma 31.3: there, decay comes from Combes–Thomas. Since $M \ge 5L$ in Part VI, no bound that must be independent of L may use them.

(c) The Dirichlet operators $\mathfrak{A}^{X,\kappa}$ for $X \ne \mathbb{T}_\eta$ depend on M through the mass profile m_X. Their constants are still independent of $M \ge M_0$ (Theorem 19.6), but Step 0 does not apply to them, because the operators themselves change with M.

Let C_H, γ_H be the constants of Lemma 31.3(b). By Lemma 31.3(c), (31.3) holds at every real (M,ε)-regular background U_0 with $M \ge M_0$ and $\varepsilon \le \varepsilon_{\mathrm{B}}$, with $\mathbf{H}(U_0)$ the minimizer map for Bałaban's pair $(Q'_k(U_0), \mathbf{Q}_k(U_0))$. Put

$$C_* = \max\Big(1,\ 2dC_H \sum_{w\in\mathbb{Z}^d} e^{-\gamma_H|w|}\big(|w|_\infty + 2\big)\Big).$$

Then C_* depends only on d, a, α and G.

Proposition 31.5 (Curvature of background fields)**.** *For every $\delta > 0$ there are $a_\delta \in (0, a_1]$ and an integer $n_\delta \ge n_0$, both depending on d, L, G, a, α, M and δ, such that for all $k \ge 0$, $n \ge n_\delta$ and $V \in \mathfrak{U}_k(\varepsilon_1)$ with $\varepsilon_1 \le a_\delta$, every element of the minimal orbit satisfies $U_k(V) \in \mathfrak{U}_0\big((C_* + \delta)\varepsilon_1\big)$.*

Proof. For $k = 0$, $U_0(V) = V$ and $C_* \ge 1$. Let $k \ge 1$ and $U = U_k(V)$. By Theorem 30.3, U is (M,ε_0)-regular and lies in $\mathfrak{U}(\varepsilon_0)$ with $\varepsilon_0 = B_3\varepsilon_1$, and U is critical. Fix $R \ge 64(M+1)$ as below and a point $y \in \mathbb{T}$. For ε_0 small enough, depending on R, Steps 1–3 of the proof of Lemma 29.2 apply with this R. This gives a real gauge transformation w and, on the bonds of $\Lambda_{R/4}(y)$, $U^w = e^{\mathrm{i}\eta\tilde{A}}$ with $\tilde{A} = R(u)A^\sharp_*$. Here $A^\sharp_*$ is the Landau representative relative to the spliced reference $U^\sharp$, for which $\varepsilon^\sharp + \beta^\sharp \le K_R\varepsilon_0$.

Let p be a plaquette of $\mathbb{T}_\eta$ with base point in $B(y)$. Its curvature satisfies $|F_U(p)| = |F_{e^{\mathrm{i}\eta\tilde{A}}}(p)| \le |(\partial\tilde{A})(p)| + C|\tilde{A}|^2$, where $\partial\tilde{A}(p) = \partial_\mu\tilde{A}_\nu - \partial_\nu\tilde{A}_\mu$; this follows from Lemma 21.1(c). By Theorem 25.1(c), $A^\sharp_* = \mathbf{H}^\sharp B^\sharp - \mathbf{C}^\sharp J_{U^\sharp} + \mathcal{R}$ with $|\mathcal{R}|_{C^1} \le C(K_R\varepsilon_0)^2$, where $\mathbf{H}^\sharp = \mathbf{H}(U^\sharp)$ and $\mathbf{C}^\sharp = \mathbf{C}(U^\sharp)$.

- In the gauge u, $\nabla_{U^\sharp}$ is the ordinary difference near y. The background $U^\sharp$ is real and $(M,\varepsilon^\sharp)$-regular with $\varepsilon^\sharp \le K_R\varepsilon_0$ (Lemma 28.6). We assume $K_RB_3a_\delta \le \varepsilon_{\mathrm{B}}$ of

Lemma 31.3(c); this is one more condition on a_δ, which already depends on L and M. By (31.3) applied at $U^\sharp$ (Lemma 31.3(c)),

$$|\partial(R(u)\mathbf{H}^\sharp B^\sharp)|_{B(y)} \le C_H \sum_{y'} e^{-\gamma_H|y-y'|}|B^\sharp(y')|.$$

- For $|y'-y|_\infty \le R/4-3$, Step 4 of that proof gives $|B^\sharp(y')| \le d(|y'-y|_\infty+2)\varepsilon_1$. These terms contribute at most $\frac12 C_*\varepsilon_1$ to $|\partial\tilde A_\mu|$.
- The other terms, the term $\mathbf{C}^\sharp J_{U^\sharp}$ and the remainder contribute at most $CK_R\,T(R/8)\varepsilon_0 + C(K_R\varepsilon_0)^2$, as in Step 4. Here T is defined as in that proof, with γ_7 replaced by the smaller of γ_H and the decay rate of $\mathbf{C}^\sharp$.

Altogether

$$|F_U(p)| \le C_*\varepsilon_1 + C(K_R T(R/8) + K_R^2\varepsilon_0)B_3\varepsilon_1 + C(K_R\varepsilon_0)^2.$$

Choose R with $CK_RT(R/8)B_3 \le \delta/2$, then $a_\delta \le a_1$ so that the remaining terms are at most $\frac\delta2\varepsilon_1$ for $\varepsilon_0 = B_3\varepsilon_1 \le B_3 a_\delta$; the bound $a_\delta \le a_1$ is needed for Theorem 30.3. Every plaquette has its base point in some block. The torus must contain the cubes $\Lambda_{R+3}(y)$ without wrapping, and $n \ge n_0$ is needed for Theorem 30.3; this fixes $n_\delta = \max(n_0, 4R+16)$. □

The proposition makes the small-field sets of consecutive steps compatible. If V' has curvature at most h' in the units of step $k+1$, then the curvature of $U_{k+1}(V')^{(k)}$, in the units of step k, is at most $L^{-2}(1+Ch')(C_*+\delta)h'$, by Proposition 21.7. With $h' = L^{1/2}h$ (the ratio g_{k+1}/g_k), $\delta \le C_*$ and $Ch' \le 1$, this is at most $\frac12 h$ as soon as $L^{3/2} \ge 8C_*$. Since C_* does not depend on L, this is a condition on L alone.

31.3 The background of a step

Lemma 31.6 (Composition of background fields)**.** *Let V' be a real configuration on $\mathbb{T}'$ whose curvature, in the units of step $k+1$, is at most $\varepsilon_1' \le \min\left(a_{\delta=1},\, a_1/(2C_*+2)\right)$, with a_δ from Proposition 31.5, and $(C_*+1)\varepsilon_1' \le c_1$. Let $U' = U_{k+1}(V')$ be an element of its minimal orbit, and $V^0 = U'^{(k)}$. Then:*

(a) $V^0 \in \mathfrak{U}_k(\varepsilon_1)$, in the units of step k, with $\varepsilon_1 = 2(C_*+1)L^{-2}\varepsilon_1' \le a_1$, and $\mathfrak{a}'(V^0) = V'$;

(b) U' belongs to the minimal orbit of the k-step problem for V^0: $U_k(V^0) = U'$;

(c) for every real V with $\mathfrak{a}'(V) = V'$ and $V \in \mathfrak{U}_k(a_1')$, $a_1' = \min(a_1, \theta_0 L^{-3}B_3^{-1})$, we have $S(U_k(V)) \ge S(U')$. Equality holds iff $V = (V^0)^u$ for a real u with $u|_{\mathbb{T}'} = 1$.

Proof. (a) By Proposition 31.5 (with $\delta = 1$), U' has curvature at most $(C_*+1)\varepsilon_1'$ in the units of step $k+1$. By Proposition 21.7 its k-fold average has curvature at most twice this, in the same units, which is the claim in the units of step k. Moreover $\mathfrak{a}'(V^0) = U'^{(k+1)} = V'$.

(b) Near U', the constraint manifold $\{U^{(k)} = V^0\}$ is a submanifold of $\{U^{(k+1)} = V'\}$, since $U^{(k)} = V^0$ implies $U^{(k+1)} = \mathfrak{a}'(V^0) = V'$. As sets, $\mathfrak{B}_k(V^0) \subset \mathfrak{B}_{k+1}(V')$ is false: the cap $U \in \mathfrak{U}_0(c_1)$ in the definition of $\mathfrak{B}$ is weaker in the units of step k. Only the local inclusion is used. The configuration U' is a critical point of S on the manifold $\{U^{(k+1)} = V'\}$ near U' and lies on the submanifold $\{U^{(k)} = V^0\}$. So it is critical on $\mathfrak{B}_k(V^0)$. In the units of step k its curvature and current are smaller than in the units of step $k+1$, so $U' \in \mathfrak{U}(\theta_0)$ in the units of step k. By Theorem 30.3(a) at step k, U' lies in the unique critical orbit, which is the minimal orbit.

(c) By Theorem 30.3(b), $U_k(V) \in \mathfrak{U}(B_3 a_1')$ in the units of step k. In the units of step $k+1$ this gives curvature and current at most $L^3 B_3 a_1' \le \theta_0$. Hence $U_k(V) \in \mathfrak{B}_{k+1}(V') \cap \mathfrak{U}(\theta_0)$, and $S(U_k(V)) \ge S(U')$. If equality holds, $U_k(V)$ is a minimizer of the $(k+1)$-step problem, so

$U_k(V) = U'^u$ with $u|_{\mathbb{T}'} = 1$. Then $V = U_k(V)^{(k)} = (V^0)^{u|_{\mathbb{T}}}$. The converse follows from gauge invariance. □

So V^0 is the minimizer of the coarse action $V \mapsto S(U_k(V))$ on the one-step constraint set, and the background of the $(k+1)$-step problem is the background of V^0. We call V^0 the *base configuration* of the step. It depends on the choice of U' in its orbit only through a gauge transformation on $\mathbb{T}$ that is trivial on $\mathbb{T}'$.

In the units of step k, U' is $(M, B_3\varepsilon_1')$-regular and lies in $\mathfrak{U}(B_3\varepsilon_1')$. Indeed, passing from the units of step $k+1$ to those of step k multiplies Lie fields by L^{-1}, curvatures by L^{-2} and currents by L^{-3}, and the cubes of Definition 18.1 become larger, which only helps (Lemma 31.2). Throughout this part we write

$$\varepsilon := B_3\varepsilon_1'$$

for the regularity parameter of U' as a reference configuration for the k-step problem.

32 Gauge fixing and fluctuation coordinates

Throughout this section V' is as in Lemma 31.6, $W := V^0$ is the base configuration and $W' = \mathfrak{a}'(W) = V'$. We use the notation of Section 22, with the reference U', so that $W_k = W$ and $W_{k+1} = W'$. Configurations on $\mathbb{T}$ are written $V = XW$, with the relative configuration X, and $X = e^{\mathrm{i}Z}$ with the *fluctuation field* Z; in the units of step k we have $\delta_k = 1$. The integrand of (31.1) is a function of X through $V = XW$. By invariance of the Haar measure, $dV(b) = dX(b)$.

32.1 The one-step axial gauge

Definition 32.1 (Tree, crossing bonds, eliminated bonds, free bonds)**.** We use the following sets of bonds of $\mathbb{T}$.

- The *tree* $\mathcal{T}$ is the set of bonds of $\mathbb{T}$ that belong to some contour $\Gamma^{k+1}_{y,x}$, $y \in \mathbb{T}'$, $x \in B'(y)$.
- A bond of $\mathbb{T}$ is *crossing* if its endpoints lie in different blocks $B'(\cdot)$.
- For a bond $c = (y, y + Le_\mu)$ of $\mathbb{T}'$, let $b_0(c)$ be the last bond of the straight contour $[c]$, from $y + (L-1)e_\mu$ to $y + Le_\mu$. It is a crossing bond.
- The bonds of $\mathbb{T}$ that are neither in $\mathcal{T}$ nor of the form $b_0(c)$ are the *free bonds*; their set is $\mathfrak{F}$.

The contour $\Gamma^{k+1}_{y,x}$ changes the coordinates in the order $d, \dots, 1$, so the path to x passes only through points whose own paths are initial segments of it. Hence $\mathcal{T} \cap B'(y)$ is a spanning tree of $B'(y)$ rooted at y, and every $x \in B'(y) \setminus \{y\}$ is the endpoint of exactly one tree bond b_x, the last bond of $\Gamma^{k+1}_{y,x}$. Since $X_W(\gamma b) = X_W(\gamma)\, R(W(\gamma))X(b)$, induction along the tree shows:

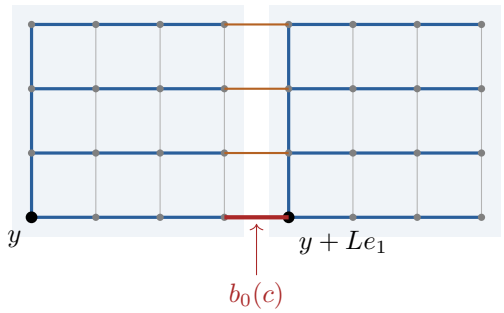
$$X \text{ is axial at scale } k+1 \iff X(b) = 1 \text{ for all } b \in \mathcal{T}. \tag{32.1}$$

See Figure 4.

Let $\mathcal{G}' = \{u : \mathbb{T} \to G :\ u|_{\mathbb{T}'} = 1\}$. By Lemma 22.2(a), with $h = 1$, every $\mathcal{G}'$-orbit of relative configurations at scale k contains exactly one axial configuration X^{ax}. No smallness is needed for this.

Lemma 32.2 (Faddeev–Popov identity for the axial gauge)**.** *Let f be an integrable function of configurations on $\mathbb{T}$ with $f(V^u) = f(V)$ for all $u \in \mathcal{G}'$. Then*

$$\int \prod_b dV(b)\, f(V) = \int \prod_{b \notin \mathcal{T}} dX(b)\ f(XW)\Big|_{X=1 \text{ on } \mathcal{T}}.$$



bold: tree $\mathcal{T}$ ($X = 1$, axial gauge)

orange: crossing bonds

red: $b_0(c)$, eliminated by the constraint

thin: further free bonds

Figure 4: Two blocks $B'(y)$, $B'(y + Le_1)$ of $\mathbb{T}$ for $d = 2$, $L = 4$. The tree runs first in direction 2, then in direction 1.

Proof. For a relative configuration X, let $g = g(X) \in \mathcal{G}'$ be the transformation with $X^{(g)}$ axial (Lemma 22.2(a)). Its value $g(x)$ depends only on X on the tree bonds of $\Gamma^{k+1}_{y,x}$. Hence, for fixed tree variables, the map $(X(b))_{b\notin\mathcal{T}} \mapsto (X^{(g)}(b))_{b\notin\mathcal{T}}$ is bondwise a left and right translation, and it preserves the product Haar measure. Since $f((XW)^g) = f(XW)$ and $X^{(g)} = 1$ on $\mathcal{T}$,

$$\int \prod_b dX(b)\, f(XW) = \int \prod_{b\in\mathcal{T}} dX(b) \int \prod_{b\notin\mathcal{T}} dX(b)\, f(X^{(g)}W) = \int \prod_{b\notin\mathcal{T}} dX(b)\, f(XW)\Big|_{X=1 \text{ on } \mathcal{T}}.$$

The tree integrals give 1, by normalization of the Haar measure. □

Definition 32.3 (Fluctuation cutoff)**.** For $w \le \frac{1}{2}c_G$ and a configuration V on $\mathbb{T}$ let X^{ax} be the axial representative of the $\mathcal{G}'$-orbit of VW^{-1}. Put $\chi^{\mathrm{w}}(V) = 1$ if $|\ell(X^{\mathrm{ax}}(b))| \le w$ for all bonds b, and $\chi^{\mathrm{w}}(V) = 0$ otherwise.

The function χ^{w} depends on V' through W. It is invariant under $\mathcal{G}'$ by construction. It does not depend on the choice of U' in its orbit. Indeed, replacing U' by U'^v with $v|_{\mathbb{T}'} = 1$ replaces W by W^v and the axial representative by its transform, whose relative field is $R(v(b_-))X^{\mathrm{ax}}(b)$; this has the same norm. Under global gauge transformations χ^{w} is covariant in the same way.

32.2 Elimination of the constraint

For axial $X = e^{\mathrm{i}Z}$, i.e. $Z = 0$ on $\mathcal{T}$, and a bond c of $\mathbb{T}'$ put

$$\ell_c(Z) = \ell(\overline{X}(c)), \qquad \overline{X} = \mathfrak{a}'(XW)\, W'^{-1}.$$

The constraint in (31.1) is $\ell_c(Z) = 0$ for all c. Let $l' = l_{k+1}$ be the linear one-step operator of Lemma 22.5 at the background W. Let $\mathsf{a} := 2(C_* + 1)\varepsilon'_1$. By Lemma 31.6 and Proposition 31.5, it bounds the curvature both of W and of the fine reference U', in the units of step $k+1$. The latter is what Lemma 22.5 needs.

Lemma 32.4 (Local elimination)**.** *There are $C, c > 0$, depending only on d and G, such that the following holds with $\rho_L = cL^{-d-1}$ and $r_L = c^2L^{-2d-1}$, provided $L^d\mathsf{a} \le c$. Moreover $d\mathsf{a} + 4dL^2 r_L \le \frac{1}{2}c_G$.*

(a) ℓ_c is analytic for complex Z with $|Z| \le cL^{-1}$ on the bonds with both endpoints in $B'(c_-) \cup B'(c_+)$, and depends only on these values. Among the bonds $b_0(c')$ it depends only on $Z(b_0(c))$. Moreover $\ell_c(Z) = L(l'Z)(c) + O(L^2|Z|^2)$.

(b) $\partial\ell_c/\partial Z(b_0(c)) = L^{1-d}R(W([c_-, b_0(c)_-]))\,(1 + \mathfrak{r}_c(Z))$ with $|\mathfrak{r}_c(Z)| \le C(L^d\mathsf{a} + L^{d+1}|Z|)$.

(c) *For complex Z on the free bonds with $|Z| \le r_L$ there is exactly one $\zeta_c(Z)$ with $|\zeta_c(Z)| \le \rho_L$ such that $\ell_c = 0$ when $Z(b_0(c)) = \zeta_c(Z)$. It is analytic, real for real Z, depends only on Z on the bonds of (a), and satisfies $|\zeta_c(Z)| \le CL^d \max|Z|$, the maximum running over those bonds.*

(d) *Let $\hat{Z}(Z)$ be the field equal to Z on $\mathfrak{F}$, to $\zeta_c(Z)$ on $b_0(c)$ and to 0 on $\mathcal{T}$, and write $\hat{Z} = \hat{Z}^1 Z + \hat{Z}^2(Z) + \hat{Z}^{\ge 3}(Z)$ (Taylor expansion). Then $\hat{Z}^1$ is a linear bijection from fields on $\mathfrak{F}$ onto*

$$\mathcal{Z} = \{B : \ B = 0 \text{ on } \mathcal{T}, \ l'B = 0\},$$

and $\|Z\| \le \|\hat{Z}^1 Z\| \le CL^d \|Z\|$. The maps $\hat{Z}^2$, $\hat{Z}^{\ge 3}$ are supported on the bonds $b_0(c)$ and satisfy $|\hat{Z}^2(Z)(b_0(c))| \le CL^{3d+1} \max|Z|^2$, the maximum over the bonds of (a).

Proof. (a), (b) For axial X we have $\tau_X = 1$, so $\overline{X} = \tilde{\mathfrak{a}}_{k+1}(X)$ (Definition 22.3). We apply Lemma 22.5 in the units of step $k+1$. There $\delta_k = L^{-1}$ and $X = e^{\mathrm{i}\delta_k B}$ with $B = LZ$, so

$$\ell_c(Z) = (l'B)(c) + c_{k+1}(B)(c) = L(l'Z)(c) + c_{k+1}(LZ)(c),$$

with the bounds of that lemma for $|B| \le c_2$. This gives (a) and the analyticity, with the bonds stated there.

Next consider the coefficient of $Z(b_0(c))$ in $(q_{k+1}Z)(c)$. The straight contour $[x, x(c)]$ contains $b_0(c)$ if and only if x is one of the points $c_- + me_\mu$, $0 \le m < L$. For these x the transport $R(W(\Gamma^{k+1}_{c_-,x}))R(W([x, b_0(c)_-]))$ equals $R(W([c_-, b_0(c)_-]))$, since both paths run along the same line. So the coefficient is $L \cdot L^{-d} L^{-1} R(W([c_-, b_0(c)_-]))$, and in $L(q_{k+1}Z)(c)$ it is $L^{1-d} R(W([c_-, b_0(c)_-]))$. The bound on r_{k+1} in Lemma 22.5(b) and the Lipschitz bound for c_{k+1} in (c), applied to $B = LZ$, change it by at most $CL\mathsf{a} + CL^2|Z|$, i.e. relatively by $C(L^d\mathsf{a} + L^{d+1}|Z|)$. The factor L^{d+1} is genuine: the coefficient L^{1-d} is what remains after the cancellation of terms of order one between $W([c])$ and the L^d loops, and the non-abelian corrections act on those terms. Here a bounds the curvature of the fine reference U' in the units of step $k+1$, as in Lemma 22.5. A bond $b_0(c')$ has both endpoints in $B'(c_-) \cup B'(c_+)$ only if $c' = c$.

(c) Put $M_c = L^{1-d} R(W([c_-, b_0(c)_-]))$ and consider $\Phi(z) = z - M_c^{-1} \ell_c(Z, z)$, where z is the value on $b_0(c)$. By (b), $|\Phi'(z)| \le \frac{1}{2}$ on $|z| \le \rho_L$ if c is small, since there $L^{d+1}|z| \le c$. By (a), $|\ell_c(Z, 0)| \le CL \max|Z|$, so $|\Phi(0)| \le C_0 L^d \max|Z|$, with C_0 depending only on d and G. For $|Z| \le r_L = c^2 L^{-2d-1}$ this is at most $C_0 c\, \rho_L \le \frac{1}{2}\rho_L$ for $c \le 1/(2C_0)$. (With the same constant c in ρ_L and r_L the contraction would not close.) So Φ is a contraction of $\{|z| \le \rho_L\}$ into itself, and its fixed point is the unique zero there. Analyticity follows as in Lemma 26.1. Reality follows from uniqueness, since $\overline{X} \in G$ for real Z. Finally, $\mathsf{a} \le L^{-d} c \le c$ and $L^2 r_L \le c^2$, so $d\mathsf{a} + 4dL^2 r_L \le dc + 4dc^2 \le \frac{1}{2} c_G$ for $c \le c_G/(10d)$.

(d) Differentiating $\ell_c(\hat{Z}(Z)) = 0$ at $Z = 0$ gives $L(l'\hat{Z}^1 Z)(c) = 0$ for all c, so $\hat{Z}^1 Z \in \mathcal{Z}$. Conversely, $B \in \mathcal{Z}$ is determined by its values on $\mathfrak{F}$: by (b) at $Z = 0$, the equation $(l'B)(c) = 0$ fixes $B(b_0(c))$. This gives the bijection. The bound on $\hat{Z}^1$ follows from $|\zeta^1_c(Z)|^2 \le CL^{2d} \max|Z|^2 \le CL^{2d} \sum |Z(b)|^2$, the sum over the bonds of (a), since each bond lies in a bounded number of these sets. The bound on $\hat{Z}^2$ follows from Cauchy's estimate, since $|\zeta_c| \le \rho_L$ on $|Z| \le r_L$. □

32.3 The exact formula

Let σ be the density of the normalized Haar measure in exponential coordinates: $d(e^{\mathrm{i}Z}) = \sigma(Z)\, dZ$ for $|Z| < c_G$, where dZ is the Lebesgue measure on $\mathfrak{g}$ given by the trace form. It is analytic, positive and Ad-invariant near 0, and $\sigma(Z)/\sigma(0) = \det\big(\sinh(\frac{\mathrm{i}}{2}\operatorname{ad} Z)/(\frac{\mathrm{i}}{2}\operatorname{ad} Z)\big) = 1 + O(|Z|^2)$. The δ-function of G is then $\delta(e^{\mathrm{i}w}) = \delta(w)/\sigma(w)$ near 1.

Proposition 32.5 (One step in fluctuation coordinates)**.** *Let V' be as in Lemma 31.6, with $L^d\mathsf{a} \le c$, and let $w \le r_L$ (so $w \le \rho_L$). Let ρ be gauge invariant, and measurable and bounded on the support of χ^{w}. Then χ^{w} vanishes outside the interior of $\operatorname{dom}\mathfrak{a}'$, $\mathfrak{a}'$ is a submersion at every point where $\chi^{\mathrm{w}} = 1$, and the density of $T(\chi^{\mathrm{w}}\rho)$ at V', defined in Section 31, is*

$$T(\chi^{\mathrm{w}}\rho)(V') = \int_{(\mathfrak{g})^{\mathfrak{F}}} \prod_{b\in\mathfrak{F}} \sigma(Z(b))\, dZ(b) \;\; \chi^{\mathrm{w}}(\hat{Z}(Z)) \prod_c \mathsf{J}_c(Z)\; \rho(e^{\mathrm{i}\hat{Z}(Z)}W),$$

where $\chi^{\mathrm{w}}(\hat{Z}) = \prod_b \mathbf{1}(|\hat{Z}(b)| \le w)$ and

$$\mathsf{J}_c(Z) = \frac{\sigma(\zeta_c(Z))}{\sigma(0)\,\det \partial\ell_c/\partial Z(b_0(c))}\,,$$

evaluated at $Z(b_0(c)) = \zeta_c(Z)$. For real Z this determinant is positive (see the proof), so it equals its absolute value, which is what the change of variables produces. The integrand is defined to be 0 if $|Z(b)| > w$ for some $b \in \mathfrak{F}$. On the remaining set $|Z|_\infty \le w \le r_L$, so $\zeta_c(Z)$, $\hat{Z}(Z)$, $\mathsf{J}_c(Z)$ and $\sigma(Z(b))$ are defined, and the factor $\chi^{\mathrm{w}}(\hat{Z}(Z))$ imposes the cutoff also on the bonds $b_0(c)$. The same convention is used for all integrals over fields on $\mathfrak{F}$ below. Each J_c is an analytic function of Z on the bonds of Lemma 32.4(a), and $\mathsf{J}_c(Z) = L^{(d-1)\dim\mathfrak{g}}(1+O(L^d\mathsf{a}+L^{2d+1}|Z|_\infty))$, with $|Z|_\infty$ the sup over the free bonds. The value on $b_0(c)$ is ζ_c, with $|\zeta_c| \le CL^d|Z|_\infty$.

Proof. Domain and submersion. Let $\chi^{\mathrm{w}}(V) = 1$, and let $X = e^{\mathrm{i}Z}$ be the axial representative of Definition 32.3. Then $|Z(b)| \le w$ on all bonds, and $V = (XW)^u$ for some $u \in \mathcal{G}'$. The holonomy $(XW)(\partial p)$ of a plaquette is the product of four unitary conjugates of the $e^{\pm\mathrm{i}Z(b)}$ and of $W(\partial p)$. By Lemma 21.1(b) and Lemma 31.6(a), XW, and hence V, has curvature at most $L^{-2}\mathsf{a} + 4w$ in the units of step k. Since $dL^2(L^{-2}\mathsf{a} + 4w) \le d\mathsf{a} + 4dL^2 r_L \le \frac{1}{2}c_G$ (Lemma 32.4), V lies in the interior of $\operatorname{dom}\mathfrak{a}'$ (Section 31). By Lemma 32.4(a),(b), the derivative of $Z \mapsto (\ell_c(Z))_c$ with respect to the variables $Z(b_0(c))$ is block diagonal, with the invertible blocks $L^{1-d}R(W([c_-, b_0(c)_-]))(1+\mathfrak{r}_c)$; here $|\mathfrak{r}_c| \le \frac{1}{2}$, as in the proof of Lemma 32.4(c), since $|Z| \le w \le \rho_L$. Since $\ell_c(Z) = \ell(\mathfrak{a}'(e^{\mathrm{i}Z}W)(c)\, V'(c)^{-1})$, and the exponential and the logarithm are local diffeomorphisms, $\mathfrak{a}'$ is a submersion at XW. It is one at V as well, since $\mathfrak{a}'(Y^u) = \mathfrak{a}'(Y)$ for $u \in \mathcal{G}'$.

The density. The density at V' is the integral of $\chi^{\mathrm{w}}\rho$ over the fibre $\{\mathfrak{a}'(V) = V'\}$, with respect to the measure induced by the two Haar measures (Section 31). We write its computation with δ-functions. The reduction to axial X below is Lemma 32.2 for this fibre integral, with the same proof. For fixed tree variables, the substitution $X \mapsto X^{(g)}$ of that proof is a bondwise translation of the other variables. It preserves their Haar measure, and it does not change $\mathfrak{a}'$, since $g \in \mathcal{G}'$. So it maps the part of the fibre with these tree variables onto the axial part of the fibre, with the induced measures, and the integral over the tree variables gives 1. The elimination of $Z(b_0(c))$ is then the change of variables $z \mapsto \ell_c(Z, z)$ at fixed free variables.

The integrand $\prod_c \delta(\mathfrak{a}'(V)(c)V'(c)^{-1})\, \chi^{\mathrm{w}}(V)\rho(V)$ is invariant under $\mathcal{G}'$: for ρ by assumption, for χ^{w} by construction, and for the constraint by Lemma 21.5(b). By Lemma 32.2 we may restrict to axial X. Then χ^{w} requires $|\ell(X(b))| \le w$ on all bonds, and on this set $X = e^{\mathrm{i}Z}$ with $dX(b) = \sigma(Z(b))dZ(b)$. By Lemma 32.4(a), the δ-function for c involves, among the variables $Z(b_0(\cdot))$, only $Z(b_0(c))$. So the integrals over these variables factorize. For fixed free variables, $Z(b_0(c)) \mapsto \ell_c$ has at most one zero in $|Z(b_0(c))| \le w$, by Lemma 32.4(c). It is $\zeta_c(Z)$ if $|\zeta_c(Z)| \le w$; otherwise the integrand vanishes, which is accounted for by the factor $\chi^{\mathrm{w}}(\hat{Z}(Z))$. Hence

$$\int \sigma(z)\, dz\; \delta(e^{\mathrm{i}\ell_c(Z,z)})\,(\cdots) = \int \sigma(z)\frac{\delta(\ell_c(Z,z))}{\sigma(\ell_c(Z,z))}(\cdots)\, dz = \frac{\sigma(\zeta_c)}{\sigma(0)\,|\det \partial_z\ell_c|}(\cdots)\Big|_{z=\zeta_c}.$$

Positivity of the determinant. By Lemma 32.4(b),

$$\det \partial_z \ell_c = L^{(1-d)\dim \mathfrak{g}} \det R(W([c_-, b_0(c)_-])) \det(1+\mathfrak{r}_c).$$

Here $\det R(\cdot) = 1$ on $\mathfrak{g}$, since G is connected (Section 12). In the proof of Lemma 32.4(c), $|\mathfrak{r}_c| \le \frac{1}{2}$ wherever ζ_c is defined, so $t \mapsto \det(1+t\mathfrak{r}_c)$ does not vanish on $[0,1]$, and $\det(1+\mathfrak{r}_c) > 0$ for real Z. Hence $|\det \partial_z \ell_c| = \det \partial_z \ell_c$ for real Z, and J_c is the analytic function of the statement. The bound on J_c follows from Lemma 32.4(b),(c), since $L^{d+1}|\zeta_c| \le CL^{2d+1}|Z|_\infty$ and $\sigma(\zeta_c)/\sigma(0) = 1 + O(|\zeta_c|^2)$. □

This is Bałaban's representation [9, (9)–(18)], with the elimination performed on one bond per coarse bond, so that the Jacobians are local [9, (17), (20)]. It is exact. No small-field approximation has been made beyond the insertion of χ^{w}.

33 The coarse effective action

The integrand of Proposition 32.5 depends on the fluctuation through the action of the background field, $S(U_k(e^{\mathrm{i}\hat{Z}}W))$. In this section we study this function of the coarse configuration. Its gradient is the Lagrange multiplier of the variational problem; its derivatives decay along trees; and its Hessian at the base configuration is comparable with the operator $\mathbf{\Gamma}_\pi - a$ of Part II.

Setting. In this section $k \ge 1$: the k-step operators of Parts II–V at U' need at least one averaging step (Section 2). The first step, $k = 0$, is treated in Remark 34.9. Let W be a real configuration on $\mathbb{T}$ in $\mathfrak{U}_k(\varepsilon_1)$, $\varepsilon_1 \le a_1$, and let U' be an element of its minimal orbit. Let ε be a number such that U' is (M,ε)-regular with $|J_{U'}| \le \varepsilon$; by Theorem 30.3 one may take $\varepsilon = B_3\varepsilon_1$, and in Section 32 we take $\varepsilon = B_3\varepsilon_1'$ (end of Section 31). We assume $\varepsilon \le c_6/2$, and we put $r_1 := c_6/4$. In Section 32, W is the base configuration V^0 and U' the $(k+1)$-step background. For complex unit-lattice fields B with $|B|_\infty < 2r_1$, so that $\varepsilon + |B|_\infty < c_6$, let $A_*(B)$ be the Landau representative relative to U' given by Theorem 25.1(f), for the coarse configuration $e^{\mathrm{i}B}W$, and put

$$\mathcal{S}(B) = \Phi(A_*(B)), \qquad \Phi(A) = S(e^{\mathrm{i}\eta A}U').$$

The bounds of Proposition 33.1(c) are stated on the smaller ball $|B|_\infty < r_1$. The margin r_1 between the two balls is used only for Cauchy's estimate in the proof of the third-derivative bound. For real B, $\mathcal{S}(B)$ is the minimum of S on $\mathfrak{B}(e^{\mathrm{i}B}W) \cap \mathfrak{U}(\theta_0)$: by Theorem 25.1(a),(b) the orbit of $\mathsf{U}(A_*(B))$ is the set of minimizers there, and $S(\mathsf{U}(A)) = \Phi(A)$. If moreover $e^{\mathrm{i}B}W \in \mathfrak{U}_k(a_1)$, this orbit is the minimal orbit of Theorem 30.3(a), and $\mathcal{S}(B) = S(U_k(e^{\mathrm{i}B}W))$; this is the case in all uses below. In particular $\mathcal{S}$ depends only on the configuration $e^{\mathrm{i}B}W$, and not on the choice of the reference U'.

For bonds $c_1, \dots, c_m$ of $\mathbb{T}$ let $\tau(c_1, \dots, c_m) = \min_z \sum_i |c_i - z|$; for $m = 2$ this is $|c_1 - c_2|$. It satisfies $\tau(c_1, \dots, c_m) \le |c_1 - x| + \tau(x, c_2, \dots, c_m)$.

33.1 Gradient and decay of derivatives

Proposition 33.1 (The coarse effective action)**.** *There are $C, \gamma_1 > 0$, independent of k, n, η and W, such that:*

(a) $\mathcal{S}$ is analytic on $\{|B|_\infty < 2r_1\}$ and real for real B.

(b) $\nabla\mathcal{S}(B) = \omega(B)$, where for real B, $\omega(B)$ is the multiplier of Proposition 27.2 for the coarse configuration $e^{\mathrm{i}B}W$; ω is analytic, and $|\omega(B)|_\infty \le C(\varepsilon + |B|_\infty)$ for $|B|_\infty < 2r_1$.

(c) For $|B|_\infty < r_1$ and all bonds c_1, c_2, c_3:

$$|\partial_{c_1}\partial_{c_2}\mathcal{S}(B)| \le Ce^{-\gamma_1|c_1-c_2|}, \qquad |\partial_{c_1}\partial_{c_2}\partial_{c_3}\mathcal{S}(B)| \le Ce^{-\gamma_1\tau(c_1,c_2,c_3)},$$

where ∂_c is the derivative with respect to $B(c)$, the components being suppressed.

(d) Put $\omega_0 = \omega(0)$ and $\Delta^0 = d^2\mathcal{S}(0)$. For $|B|_\infty \le r_1/2$,

$$\mathcal{S}(B) = \mathcal{S}(0) + \langle\omega_0, B\rangle + \tfrac{1}{2}\Delta^0(B,B) + \mathcal{S}_3(B), \qquad |\mathcal{S}_3(B)| \le C|B|_\infty\|B\|^2.$$

Proof. (a) A_* is analytic with $|A_*(B)|_{C^1} \le C_6(\varepsilon + |B|_\infty) \le c_5$, and Φ is analytic there (Proposition 23.3).

(b) For real B, Proposition 27.2 gives $\nabla\Phi(A_*(B)) = d\mathcal{Q}(A_*(B))^T\omega(B)$. Differentiating the constraint $\ell(\mathcal{Q}(e^{i\eta A_*(B)})) = B$ gives $d\mathcal{Q}(A_*)\,dA_*(B) = I$. Hence

$$d\mathcal{S}(B)h = \langle\nabla\Phi(A_*), dA_*h\rangle = \langle\omega, d\mathcal{Q}(A_*)dA_*h\rangle = \langle\omega, h\rangle.$$

Applying $\mathbf{H}^T$ to the Euler–Lagrange equation gives

$$\omega(B) = \big((I + dC_k(A_*)\mathbf{H})^{-1}\big)^T\mathbf{H}^T\nabla\Phi(A_*(B)).$$

This expression is analytic in complex B and defines the continuation of ω. The identities $\nabla\Phi(A_*) = d\mathcal{Q}(A_*)^T\omega$ and $\nabla\mathcal{S} = \omega$ hold for real B, hence for complex B by the identity theorem: the real ball is a totally real submanifold of maximal dimension. The bound for real B is Proposition 27.2; for complex B it follows from the formula, Lemma 24.1, Proposition 23.3(a), and the identity $\mathbf{H}^TD^TDA_* = (\mathbf{\Gamma}_\pi - a)\mathbf{Q}A_*$. This is the second identity in the proof of Lemma 26.2; it applies since $RD^TA_* = RD^TA'_* = 0$ also for complex B (Lemma 26.1, $A'_* \in \mathbf{H}B + \mathcal{X}$). The bound $|A_*|_{C^1} \le C(\varepsilon + |B|_\infty)$ alone would not bound D^TDA_* pointwise.

(c) *Second derivatives.* We prove the bound on the larger ball $|B|_\infty < 2r_1$. Let $|B|_\infty < 2r_1$. Differentiating $d\mathcal{S}(B)h = \langle\nabla\Phi(A_*), dA_*h\rangle$ once more,

$$d^2\mathcal{S}(B)[h_1,h_2] = d^2\Phi(A_*)[dA_*h_1, dA_*h_2] - \langle\omega, d^2C_k(A_*)[dA_*h_1, dA_*h_2]\rangle,$$

where we used $\langle\nabla\Phi, d^2A_*[h_1,h_2]\rangle = \langle\omega, d\mathcal{Q}\,d^2A_*[h_1,h_2]\rangle$ and the second derivative of the constraint, $d\mathcal{Q}\,d^2A_*[h_1,h_2] + d^2C_k(A_*)[dA_*h_1, dA_*h_2] = 0$. Now:

- $d^2\Phi(A) = D^TD + d^2\mathcal{N}(A)$, where $d^2\mathcal{N}(A)$ is a sum of plaquette-local terms with bounded coefficients (proof of Proposition 23.3(c));
- $d^2C_k(A)[u,v](c)$ is bounded by $C\|u\|_{S_c}\|v\|_{S_c}$, by polarization of Proposition 22.6(c);
- by Theorem 25.1(f), which holds for complex B with $\varepsilon + |B|_\infty < c_6$, and here $\varepsilon + |B|_\infty < 2r_1 + 2r_1 = c_6$: $|dA_*\delta_{c_i}| + |\nabla dA_*\delta_{c_i}| \le Ce^{-\gamma'|x-c_i|}$.

Hence $|\partial_{c_1}\partial_{c_2}\mathcal{S}(B)| \le C\sum_x e^{-\gamma'(|x-c_1|+|x-c_2|)}\eta^d \le Ce^{-\gamma'|c_1-c_2|/2}$ for $|B|_\infty < 2r_1$.

Third derivatives. Let $|B|_\infty < r_1$, and let e be a unit vector of $\mathfrak{g}$. For complex t with $|t| < 2r_1 - |B|_\infty$ we have $|B + te\,\delta_{c_3}|_\infty < 2r_1$. Since $2r_1 - |B|_\infty > r_1$, the function $t \mapsto \partial_{c_1}\partial_{c_2}\mathcal{S}(B + te\,\delta_{c_3})$ is analytic on a neighbourhood of the closed disc $|t| \le r_1$, and, by the bound just proved on the larger ball, it is bounded there by $Ce^{-\gamma'|c_1-c_2|/2}$. By Cauchy's estimate on the circle $|t| = r_1$,

$$|\partial_{c_1}\partial_{c_2}\partial_{c_3}\mathcal{S}(B)| \le Cr_1^{-1}e^{-\gamma'|c_1-c_2|/2}.$$

The left side is symmetric in c_1, c_2, c_3. So the bound holds with $|c_1 - c_2|$ replaced by the largest distance D between two of the three bonds. Taking $z = c_1$ in the definition of τ gives

$\tau(c_1,c_2,c_3) \le |c_2-c_1|+|c_3-c_1| \le 2D$. This proves (c) with $\gamma_1 = \gamma'/4$. No bound on the second or higher derivatives of A_* on the fine lattice is used.

(d) Fix B and a bond c, and let $R(s) = \omega(sB)(c)$ for complex s with $|s|\,|B|_\infty < r_1$. By (c), $|R'(s)| = |d\omega(sB)B(c)| \le C\sum_{c'} e^{-\gamma_1|c-c'|}|B(c')| =: K_c$ on this disc. By the Schwarz lemma applied to $R'(s)-R'(0)$, $|R'(s)-R'(0)| \le 4K_c|B|_\infty r_1^{-1}$ for $|s|\le 1$, when $|B|_\infty \le r_1/2$. By (b),

$$\mathcal{S}_3(B) = \int_0^1 \langle \omega(tB) - \omega_0 - t\,d\omega(0)B, B\rangle\, dt, \qquad |\mathcal{S}_3(B)| \le C|B|_\infty \sum_c K_c|B(c)| \le C|B|_\infty \|B\|^2,$$

by the Schur test. □

Part (b) is the discrete form of the statement that the derivative of the minimal action with respect to the constraint is the Lagrange multiplier. Part (c) is what makes the coarse action, and the Gaussian measure it defines, local.

33.2 The Hessian at the base configuration

Recall that $\mathbf{H} = \mathbf{H}(U')$, $\mathcal{X} = \{A : RD^TA = 0,\ \mathbf{Q}A = 0\}$ and $\mathbf{\Gamma}_\pi = \mathbf{\Gamma}_\pi(U')$ are taken at U' with the averaging pair $(Q'_k(U'), \mathbf{Q}_k(U'))$, and that

$$\mathbf{H}^T D^T D\mathbf{H} = \mathbf{\Gamma}_\pi - a. \tag{33.1}$$

This follows from the first identity in the proof of Lemma 26.2 and $S\mathbf{H} = G'RD^T\mathbf{H} = 0$. At $U' = 1$ the right-hand side is the operator $\mathbf{D}$ of Proposition 11.5.

Proposition 33.2 (The Hessian). *(a) For real B,*

$$\Delta^0(B,B) = \min\{q_2(A,A) :\ \mathbf{Q}A = B,\ RD^TA = 0\},$$
$$q_2(A,A) = \|DA\|^2 + d^2\mathcal{N}(0)[A,A] - \langle \omega_0, d^2C_k(0)[A,A]\rangle,$$

and the minimum is attained at $A = dA_*(0)B$.

(b) $|\Delta^0(B,B) - \langle B, (\mathbf{\Gamma}_\pi - a)B\rangle| \le C\varepsilon\|B\|^2$.

(c) Δ^0 is real symmetric with decay (C,γ_1).

(d) For every real gauge transformation u on $\mathbb{T}_\eta$, replacing (W,U') by $(W^{u|_{\mathbb{T}}}, U'^u)$ replaces Δ^0 by $R(u)\Delta^0R(u)^{-1}$ and ω_0 by $R(u)\omega_0$, where $R(u)$ acts on unit-lattice fields by $R(u(b_-))$.

Proof. (a) Let A^1 satisfy $\mathbf{Q}A^1 = B$ and $RD^TA^1 = 0$. Put $A(t) = tA^1 - \mathbf{H}\mathcal{D}(tA^1)$. By Lemma 26.1, $RD^TA(t) = 0$ and $\ell(\mathcal{Q}(e^{i\eta A(t)})) = tB$. Moreover $\mathcal{D}(tA^1) = t^2C_k^{(2)}(A^1) + O(t^3)$, where $C_k^{(2)}(A) = \frac12 d^2C_k(0)[A,A]$. For real t near 0 the configuration $\mathsf{U}(A(t))$ lies in $\mathfrak{B}(e^{itB}W)\cap \mathfrak{U}(\theta_0)$. By Theorem 25.1(b), $\mathcal{S}(tB) \le \Phi(A(t))$. Expand Φ with Proposition 23.3:

$$\Phi(A(t)) = \mathcal{S}(0) + t\langle J, A^1\rangle + t^2\Big(\tfrac12\|DA^1\|^2 + \tfrac12 d^2\mathcal{N}(0)[A^1,A^1] - \langle J, \mathbf{H}C_k^{(2)}(A^1)\rangle\Big) + O(t^3),$$

with $J = J_{U'} = \nabla\Phi(0) = \mathbf{Q}^T\omega_0$. Then $\langle J, A^1\rangle = \langle \omega_0, B\rangle$ and $\langle J, \mathbf{H}C_k^{(2)}(A^1)\rangle = \langle \omega_0, C_k^{(2)}(A^1)\rangle$, since $\mathbf{QH} = I$. The terms of first order agree with those of $\mathcal{S}(tB)$. So the inequality for $t\to 0$ gives $\Delta^0(B,B) \le q_2(A^1,A^1)$.

For equality, expand $\mathcal{S}(tB) = \Phi(A_*(tB))$ directly. We have $A_*(tB) = tA^1 + t^2A^2 + O(t^3)$ with $A^1 = dA_*(0)B$, both in the slice, and the constraint gives $\mathbf{Q}A^1 = B$, $\mathbf{Q}A^2 + C_k^{(2)}(A^1) = 0$. Hence the coefficient of t^2 is $\frac12\|DA^1\|^2 + \frac12 d^2\mathcal{N}(0)[A^1,A^1] + \langle J, A^2\rangle$, and $\langle J, A^2\rangle = \langle \omega_0, \mathbf{Q}A^2\rangle = -\langle \omega_0, C_k^{(2)}(A^1)\rangle$. So $\Delta^0(B,B) = q_2(A^1,A^1)$.

(b) Write $A = \mathbf{H}B + h$ with $h \in \mathcal{X}$. Then $\|DA\|^2 = \|D\mathbf{H}B\|^2 + \|Dh\|^2$, since $\langle Dh, D\mathbf{H}B\rangle = 0$ for $h\in\mathcal{X}$, and $\|D\mathbf{H}B\|^2 = \langle B, (\mathbf{\Gamma}_\pi - a)B\rangle$ by (33.1). By Proposition 23.3(c) at $A = 0$, $|d^2\mathcal{N}(0)[A,A]| \le C\varepsilon(\|A\|^2 + \|\nabla A\|^2)$. By Proposition 22.6(c) and $|\omega_0| \le C\varepsilon$, $|\langle \omega_0, d^2C_k(0)[A,A]\rangle| \le C\varepsilon\|A\|^2$. Moreover:

- $\|\mathbf{H}B\| + \|\nabla\mathbf{H}B\| \le C\|B\|$, by the decay of $\mathbf{H}$ and $D\mathbf{H}$, the identity $D^T\mathbf{H}B = P_{\mathcal{V}}D^T\mathbf{H}B$, and Lemma 26.4;
- $\|h\|^2 + \|\nabla h\|^2 \le c_X^{-1}\|Dh\|^2$, by Lemma 26.5.

Hence

$$\langle B, (\mathbf{\Gamma}_\pi - a)B\rangle - C\varepsilon\|B\|^2 + (1 - C\varepsilon)\|Dh\|^2 \le q_2(A, A).$$

Taking the minimum over h gives the lower bound; $h = 0$ gives the upper bound.

(c) is Proposition 33.1(c) at $B = 0$. (d) follows from the covariance of the variational problem, Theorem 25.1(g), and $\mathcal{S}$ depending only on $e^{\mathrm{i}B}W$: $(e^{\mathrm{i}B}W)^{u|_{\mathbb{T}}} = e^{\mathrm{i}R(u)B}W^{u|_{\mathbb{T}}}$. □

33.3 Dependence on the base configuration

Let $\hat\beta$ be a complex unit-lattice field with $|\hat\beta|_\infty \le r_1/4$, and $\hat W = e^{\mathrm{i}\hat\beta}W$ (bondwise product). For real $\hat\beta$ we have $\mathcal{S}_{\hat W}(B) = \mathcal{S}_W(B \star \hat\beta)$, with $B \star \hat\beta = \ell(e^{\mathrm{i}B}e^{\mathrm{i}\hat\beta})$ bondwise. We use this formula as the definition for complex $\hat\beta$. It defines $\mathcal{S}_{\hat W}$ analytically in $(\hat\beta, B)$ for $|B|_\infty \le r_1/4$, and in particular $\omega_0(\hat W)$ and $\Delta^0(\hat W)$.

Proposition 33.3 (Dependence on the base)**.** *$\omega_0(\hat W)$ and $\Delta^0(\hat W)$ are analytic in $\hat\beta$. Their derivatives with respect to $\hat\beta(c)$ satisfy*

$$|\partial_{\hat\beta(c)}\omega_0(\hat W)(c_1)| \le Ce^{-\gamma_1|c-c_1|}, \qquad |\partial_{\hat\beta(c)}\Delta^0(\hat W)(c_1, c_2)| \le Ce^{-\gamma_1\tau(c,c_1,c_2)}.$$

Proof. $\omega_0(\hat W) = \partial_B\mathcal{S}_W(B \star \hat\beta)|_{B=0}$ and $\Delta^0(\hat W) = \partial_B^2\mathcal{S}_W(B \star \hat\beta)|_{B=0}$. The map $(B, \hat\beta) \mapsto B \star \hat\beta$ is bondwise analytic. Its derivatives in B at $B = 0$ are bondwise analytic functions of $\hat\beta$. The chain rule expresses the derivatives in terms of $d^m\mathcal{S}_W(\hat\beta)$, $m \le 3$, contracted with bondwise factors. Proposition 33.1(c) at $B = \hat\beta$, with $|\hat\beta|_\infty \le r_1/4 < r_1$, gives the bounds. □

Proposition 33.3 controls the dependence on small changes of the base configuration. Changes that are not small, for example replacing the configuration far away by a different one, require a representation of the operators by terms that depend on the background only locally. This is the random-walk expansion of Section 20, which will be developed for the present operators in the next part.

34 The fluctuation covariance and the step

We return to the setting of Section 32: $W = V^0$ is the base configuration of the step, $U' = U_{k+1}(V')$, and $\varepsilon = B_3\varepsilon_1'$. The objects of Section 33 are taken at this W and U'.

34.1 The quadratic form of the step

Lemma 34.1 (Expansion in the free variables)**.** *Let $r_2 = \min(cL^{-2d-2}, cL^{-d}a_1')$ with $c > 0$ small, depending only on d, G, L and $r_1 = c_6/4$ of Proposition 33.1 (the proof needs $2cL^{-2d-2} \le \frac12 r_L$, $CL^{-d-2}c \le \frac14 r_1$ and $Cc \le \frac12$), a_1' as in Lemma 31.6(c), and assume $4(C_* + 1)L^{-2}\varepsilon_1' \le a_1'$. For complex fields Z on $\mathfrak{F}$ with $|Z|_\infty \le r_2$ put $F(Z) = \mathcal{S}(\hat Z(Z))$. Then F is analytic, $F(Z) = S(U_k(e^{\mathrm{i}\hat Z(Z)}W))$ for real Z, and*

$$F(Z) = S(U') + \tfrac12\Delta_k(Z, Z) + \mathcal{V}(Z), \qquad \Delta_k(Z, Z) = \Delta^0(\hat Z^1 Z, \hat Z^1 Z) + 2\langle\omega_0, \hat Z^2(Z)\rangle,$$

with $|\mathcal{V}(Z)| \le C_L|Z|_\infty\|Z\|^2$. In particular F has no linear term.

Proof. By Lemma 32.4, $\hat{Z}$ is analytic on $|Z|_\infty \le 2r_2 \le r_L$ with $|\hat{Z}(Z)|_\infty \le CL^d|Z|_\infty \le r_1/2$, so F is analytic on the disc of radius $2r_2$ by Proposition 33.1(a). For real Z the configuration $e^{\mathrm{i}\hat{Z}(Z)}W$ satisfies the one-step constraint, and its curvature is at most $2(C_*+1)L^{-2}\varepsilon_1' + CL^d r_2 \le a_1'$. Here $|\hat{Z}|_\infty \le CL^d|Z|_\infty$, so $CL^d r_2 \le a_1'/2$ by the choice of r_2, and we assume $4(C_*+1)L^{-2}\varepsilon_1' \le a_1'$. By Lemma 31.6(c), $F(Z) \ge F(0) = S(U')$, so $dF(0) = \langle \omega_0, \hat{Z}^1 \cdot \rangle = 0$. The second-order term follows from $\hat{Z}(tZ) = t\hat{Z}^1 Z + t^2\hat{Z}^2(Z) + O(t^3)$ and Proposition 33.1(d). For the remainder, $\nabla F(Z) = d\hat{Z}(Z)^T \omega(\hat{Z}(Z))$. The analyticity of $\hat{Z}$ and the bound $|\hat{Z}(Z)|_\infty \le r_1/2$ hold on the larger disc $|Z|_\infty \le 2cL^{-2d-2}$, since only the first term of the minimum defining r_2 enters them. By Proposition 33.1(c) and the locality of $\hat{Z}$ (range $2L$), d^2F has a kernel bounded by $C_L e^{-\gamma_1|b-b'|}$ on this larger disc. The argument of Proposition 33.1(d), applied to F on this disc with $dF(0) = 0$, gives the bound on $\mathcal{V}$. Its constant C_L depends on cL^{-2d-2}, but not on a_1'. □

This is the quadratic form [9, (53)–(54)]. The term $2\langle \omega_0, \hat{Z}^2 \rangle$ comes from the curvature of the constraint set and is Bałaban's term $-\langle H_1 \hat{D}^{(2)}(A), J \rangle$.

34.2 Coercivity

The lower bound for Δ_k is proved in two steps:

- at the trivial background, a Poincaré inequality on the one-step constraint set combined with Proposition 11.5;
- at a regular background, an IMS localization on the unit lattice, as in Section 19.

Let q be the one-step tube average at the trivial background, i.e. l' with $W = 1$.

Lemma 34.2 (Poincaré inequality for one step)**.** *There is C_{P}, depending only on d and L, such that every real field B on the bonds of $\mathbb{T}$ with $B = 0$ on $\mathcal{T}$ satisfies*

$$\|B\|^2 \le C_{\mathrm{P}}(\|d_1 B\|^2 + \|qB\|^2),$$

where $d_1 B$ is the unit-lattice curl.

Proof. All estimates are local in a pair of neighbouring blocks.

Bonds inside a block. Let $b = (x, x')$ be a bond of $B'(y)$ not in $\mathcal{T}$. By Lemma 21.2(ii) the loop $\Gamma^{k+1}_{y,x} \cup b \cup (\Gamma^{k+1}_{y,x'})^{-1}$ is spanned by at most dL plaquettes. Since $B = 0$ on the tree, the abelian Stokes formula gives $B(b) = \sum_i \pm d_1 B(p_i)$, so $|B(b)|^2 \le dL \sum_i |d_1 B(p_i)|^2$.

Crossing bonds. Let $c = (y, y')$, $y' = y + Le_\mu$, and for x in the face $\mathsf{F} = \{x \in B'(y) : x_\mu = y_\mu + L - 1\}$ let $b_x = (x, x + e_\mu)$. For $x, \tilde{x} \in \mathsf{F}$ the closed contour

$$\Gamma^{k+1}_{y,x}\, b_x\, (\Gamma^{k+1}_{y',x+e_\mu})^{-1}\, \Gamma^{k+1}_{y',\tilde{x}+e_\mu}\, b_{\tilde{x}}^{-1}\, (\Gamma^{k+1}_{y,\tilde{x}})^{-1}$$

lies in a box of side $2L$ and has at most $4dL + 2$ bonds. By Lemma 21.2(iii) it is spanned by at most $12d^2L^2$ plaquettes, so $B(b_x) - B(b_{\tilde{x}})$ is a sum of at most $12d^2L^2$ values $\pm d_1 B(p)$.

The average. $(qB)(c) = L^{-d} \sum_{x \in B'(y)} L^{-1} B([x, x(c)])$. Each straight contour $[x, x(c)]$ contains exactly one crossing bond $b_{\tilde{x}}$, with $\tilde{x} \in \mathsf{F}$ on the same line as x, and $L - 1$ bonds inside the two blocks. Each $\tilde{x}$ occurs for L values of x. Hence the mean m_c of $B(b_{\tilde{x}})$ over $\tilde{x} \in \mathsf{F}$ satisfies $|m_c| \le L|(qB)(c)| + (L-1)\max|B(b)|$, the maximum running over the inner bonds of the two blocks. Then $|B(b_x)| \le |m_c| + \max_{\tilde{x}} |B(b_x) - B(b_{\tilde{x}})|$.

Squaring and summing, each plaquette and each value $(qB)(c)$ is used a number of times bounded in terms of d and L. □

Lemma 34.3 (Local comparison)**.** *Let $M' \geq \max(M, L)$ be an integer and z a point of the torus (not necessarily of $\mathbb{T}$). Assume $n \geq 80M'$ and $\varepsilon \leq \varepsilon_{\mathrm{cmp}}(M')$, where $\varepsilon_{\mathrm{cmp}}(M') > 0$ depends only on M', d, L, G, a, α and M, and is fixed in the proof. There are a real gauge transformation u on $\Lambda_{40M'}(z)$ and $C_{M'}$ such that, for every real field B on the bonds of $\mathbb{T}$ starting in $\Lambda_{M'}(z)$ and vanishing on $\mathcal{T}$, with $B^u = R(u)B$:*

$$\big|\langle B, (\mathbf{\Gamma}_\pi - a)B\rangle - \langle B^u, \mathbf{D}B^u\rangle\big| \leq (C_{M'}\varepsilon + Ce^{-\gamma M'})\|B\|^2, \qquad \big|\|l'B\| - \|qB^u\|\big| \leq C_{M'}\varepsilon\|B\|.$$

Here $\mathbf{D} = \mathbf{\Gamma}(1) - a$ is the operator of Proposition 11.5, and C, γ do not depend on M'.

Proof. The configuration U' is 3ε-regular on every cube $\Lambda_{4M+2}(z'')$ (Section 28). Let $z_0 \in \mathbb{T}_\eta$ with $|z - z_0|_\infty \leq \eta$, so that $\Lambda_{40M'}(z) \subset \Lambda_{40M'+1}(z_0)$. Apply Lemma 28.3 with $r = 4M+2 \geq 18$, $R = 40M' + 1$ and $\sigma = 3\varepsilon$, on the cube $\Lambda_R(z_0)$. Its hypothesis $C_E(R/r)^p R\sigma \leq 1$ is the first condition defining $\varepsilon_{\mathrm{cmp}}(M')$. The boxes in its proof have sides at most $3R = 120M' + 3 < 80M'L \leq Ln$, the side of $\mathbb{T}$, so they do not wrap around the torus. This gives a real gauge u on $\Lambda_{40M'}(z)$ with $U'^u = e^{\mathrm{i}\eta A}$ and $\|A\|_{1,\alpha} \leq C(40M'/M)^p\varepsilon$. Extend u to a real gauge transformation $\tilde{u}$ of $\mathbb{T}_\eta$. Then $\mathbf{\Gamma}_\pi(U'^{\tilde{u}}) = R(\tilde{u})\mathbf{\Gamma}_\pi(U')R(\tilde{u})^{-1}$ and $l'(W^{\tilde{u}})R(\tilde{u}) = R(\tilde{u})l'(W)$, by Proposition 22.7(f), Proposition 16.3 and Lemma 21.5(b).

Let χ be a smooth cutoff equal to 1 on $\Lambda_{20M'}(z)$ and 0 outside $\Lambda_{30M'}(z)$, and put $U^\natural = e^{\mathrm{i}\eta\chi A}$. It satisfies (B) with parameter $C_{M'}\varepsilon$. The second condition defining $\varepsilon_{\mathrm{cmp}}(M')$ is that $C_{M'}\varepsilon$ lies below the thresholds of Parts II and III. The configuration $U^\natural$ coincides with $U'^{\tilde{u}}$ on $\Lambda_{20M'}(z)$. By Theorem 19.10(b), the kernels of $\mathbf{\Gamma}_\pi(U'^{\tilde{u}})$ and $\mathbf{\Gamma}_\pi(U^\natural)$ on $\Lambda_{M'}(z)$ differ by at most $C\sum_w e^{-\gamma(|y-w|+|w-y'|)}$, summed over w within $10M$ of the complement of $\Lambda_{20M'}(z)$. These w are at distance at least $9M'$ from $\Lambda_{M'}(z)$, which gives the error $Ce^{-\gamma M'}$ after a Schur bound.

By Part II at the background $U^\natural$, with the averaging pair $(Q'_k(U^\natural), \mathbf{Q}_k(U^\natural))$ (Remark 19.12), $\mathbf{\Gamma}_\pi(U^\natural) - \mathbf{\Gamma}(1)$ has decay $(C_{M'}\varepsilon, \gamma)$. This follows from Theorem 16.1(a) and Lemma 13.3.

Finally, by Lemma 21.8, whose hypotheses are the third condition defining $\varepsilon_{\mathrm{cmp}}(M')$, $W^{\tilde{u}} = e^{\mathrm{i}B_W}$ near $\Lambda_{M'}(z)$ with $|B_W| \leq C_{M'}\varepsilon$. So the transports in $l'(W^{\tilde{u}})$ are within $C_{M'}\varepsilon$ of 1 there. Together with Lemma 22.5(b), this gives $|l'(W^{\tilde{u}})B^u - qB^u| \leq C_{M'}\varepsilon\|B\|$. □

Theorem 34.4 (Coercivity on the constraint set)**.** *There are $c_L > 0$, depending only on d and L, and $M'_L \geq M$ and $\varepsilon_L > 0$, depending only on d, L, G, a, α and M, such that, if $\varepsilon \leq \varepsilon_L$ and $n \geq n_0$ with $n_0 \geq 80M'_L$,*

$$\langle B, (\mathbf{\Gamma}_\pi - a)B\rangle \geq c_L\|B\|^2 \qquad \text{for all real } B \in \mathcal{Z}.$$

Proof. Let $\{\chi_j\}$ be the partition of Lemma 17.3 at scale M', restricted to the unit lattice, acting on bond fields through b_-. The operator $K = \mathbf{\Gamma}_\pi - a$ is bounded with decay (C, γ) (Theorem 19.9). By Lemma 17.4 and the argument of Lemma 17.5 for kernels, $\sum_j(\chi_j(x) - \chi_j(x'))^2|K(x,x')| \leq CM'^{-2}|x - x'|^2|K(x,x')|$, so

$$\langle B, KB\rangle \geq \sum_j \langle \chi_j B, K\chi_j B\rangle - CM'^{-2}\|B\|^2.$$

Each $\chi_j B$ vanishes on $\mathcal{T}$ and is supported on bonds starting in a cube of side $2M'$. By Lemma 34.3, Proposition 11.5 and Lemma 34.2,

$$\begin{aligned}\langle \chi_j B, K\chi_j B\rangle &\geq c_D\|d_1(\chi_j B)^{u_j}\|^2 - \delta_1\|\chi_j B\|^2 \\ &\geq c_D C_{\mathrm{P}}^{-1}\|\chi_j B\|^2 - c_D\|q(\chi_j B)^{u_j}\|^2 - \delta_1\|\chi_j B\|^2,\end{aligned}$$

with $\delta_1 = C_{M'}\varepsilon + Ce^{-\gamma M'}$. Moreover $\|q(\chi_j B)^{u_j}\|^2 \leq 2\|l'\chi_j B\|^2 + 2C_{M'}^2\varepsilon^2\|\chi_j B\|^2$. The operator l' has range $2L$, and $|\chi_j(x) - \chi_j(x')| \leq CL/M'$ within this range. With the bounded overlap

of the χ_j, this gives $\sum_j \|l'\chi_j B\|^2 \le 2\|l'B\|^2 + C(L/M')^2\|B\|^2$. For $B \in \mathcal{Z}$, $l'B = 0$. Summing over j with $\sum_j \|\chi_j B\|^2 = \|B\|^2$,

$$\langle B, K\rangle B \ge \Big(c_D C_{\mathrm{P}}^{-1} - CM'^{-2} - Cc_D L^2 M'^{-2} - Ce^{-\gamma M'} - C_{M'}\varepsilon\Big)\|B\|^2.$$

Choose $M' = M'_L$ so large that the terms independent of ε are at most $\frac{1}{4}c_D C_{\mathrm{P}}^{-1}$. Then choose $\varepsilon_L \le \varepsilon_{\mathrm{cmp}}(M'_L)$, below the threshold of Corollary 22.8 for Theorem 19.9 (which gives the decay of K used in the IMS step), so small that $C_{M'_L}\varepsilon_L \le \frac{1}{4}c_D C_{\mathrm{P}}^{-1}$, and put $c_L = \frac{1}{2}c_D C_{\mathrm{P}}^{-1}$. Lemma 34.3 applies, since $n \ge 80M'_L$ and $\varepsilon \le \varepsilon_{\mathrm{cmp}}(M'_L)$. □

34.3 The fluctuation covariance

Corollary 34.5 (Fluctuation covariance)**.** *Let $\varepsilon \le \varepsilon_L$ and $CL^{3d+1}\varepsilon \le \frac{1}{4}c_L$. Then Δ_k, as an operator on real fields on $\mathfrak{F}$, is symmetric with*

$$\tfrac{1}{4}c_L\|Z\|^2 \le \Delta_k(Z,Z) \le C_L\|Z\|^2.$$

It has decay (C_L, γ_1), and $C_k = \Delta_k^{-1}$ has decay (C'_L, γ_C). For complex base configurations $\hat{W} = e^{\mathrm{i}\hat{\beta}}W$ with $|\hat{\beta}|_\infty \le c'_L$, where $c'_L \le r_1/4$ is a small constant, define

$$\Delta_k(\hat{W})(Z,Z) = \Delta^0(\hat{W})(\hat{Z}^1_{\hat{W}}Z, \hat{Z}^1_{\hat{W}}Z) + 2\langle\omega_0(\hat{W}), \hat{Z}^2_{\hat{W}}(Z)\rangle,$$

with the objects of Proposition 33.3 and the elimination of Lemma 32.4 at the base $\hat{W}$. Then:

- *$\Delta_k(\hat{W})$ and $C_k(\hat{W})$ are analytic in $\hat{\beta}$;*
- *$\operatorname{Re}\Delta_k(\hat{W})(Z,\bar{Z}) \ge \frac{1}{8}c_L\|Z\|^2$;*
- *the decay bounds persist, and $|\partial_{\hat{\beta}(c)}\Delta_k(\hat{W})(b,b')| + |\partial_{\hat{\beta}(c)}C_k(\hat{W})(b,b')| \le C_L e^{-\gamma_C\tau(c,b,b')}$.*

Proof. By Proposition 33.2(b) and Theorem 34.4, $\Delta^0(\hat{Z}^1Z, \hat{Z}^1Z) \ge (c_L - C\varepsilon)\|\hat{Z}^1Z\|^2 \ge \frac{1}{2}c_L\|Z\|^2$, since $\hat{Z}^1Z \in \mathcal{Z}$ and $\|\hat{Z}^1Z\| \ge \|Z\|$. By Lemma 32.4(d) and $|\omega_0| \le C\varepsilon$, $|2\langle\omega_0, \hat{Z}^2(Z)\rangle| \le CL^{3d+1}\varepsilon\|Z\|^2$. The upper bound and the decay follow from Proposition 33.2(c) and the locality of $\hat{Z}^1$, $\hat{Z}^2$. Lemma 5.5 gives the decay of C_k.

For complex bases, the elimination of Lemma 32.4 is analytic in $\hat{\beta}$. The function ℓ_c is an analytic function of the finitely many bond variables. The proof of Lemma 22.5 used unitarity of $R(W(\cdot))$ only to bound transports along contours of at most $3dL$ bonds, and these remain bounded by $1 + CL|\hat{\beta}|_\infty$ for complex bases, which is harmless for c'_L small. Together with Proposition 33.3, this shows that $\Delta_k(\hat{W}) - \Delta_k(W)$ has decay $(C_L|\hat{\beta}|_\infty, \gamma_1)$ and derivatives with tree decay. The lower bound follows for c'_L small. Lemma 5.5, in the complex form of Section 13, gives the decay of $C_k(\hat{W})$. Finally $\partial C_k = -C_k(\partial\Delta_k)C_k$, and $\tau(c,b_1,b_2) + |b - b_1| + |b_2 - b'| \ge \tau(c,b,b')$ gives the tree bound. □

Remark 34.6 (The constants do not depend on M). The constants c_L, C_L, C'_L, γ_1, γ_C and c'_L of Theorem 34.4 and Corollary 34.5, and the constant C_L of Theorem 34.8, do not depend on $M \ge M_0$:

- $c_L = \frac{1}{2}c_D C_{\mathrm{P}}^{-1}$, with c_D of Proposition 11.5 and C_{P} of Lemma 34.2; they depend only on d and L.
- The upper bound and the decay (C_L, γ_1) of Δ_k come from Proposition 33.2(c), that is, Proposition 33.1(c) at $B = 0$, and from the locality of $\hat{Z}^1$ and $\hat{Z}^2$. The constants of Proposition 33.1 come from Theorem 25.1, whose constants may be taken independent of $M \ge M_0$ (the paragraph after the theorem), from Propositions 23.3 and 22.6 (the

first assumes only $(1,\varepsilon)$-regularity; the proofs of Sections 21 and 22 do not use M, see Section 46.1, Setting; Lemma 32.4 depends only on d and G), and from $r_1 = c_6/4$. The comparison of Proposition 33.2(b) uses Lemma 26.5, whose constant comes from Theorem 19.6 and does not depend on $M \geq M_0$.

- C'_L and γ_C come from these by Lemma 5.5. The radius $c'_L \leq r_1/4$ and the bounds at complex bases, through Proposition 33.3, are of the same kind.
- The constant C_L of Theorem 34.8(d) comes from the proof of Lemma 34.1, on the disc $|Z|_\infty \leq 2cL^{-2d-2}$, where c depends only on d, G, L and c_6, none of which involves M.

The parameter M enters this part only through thresholds and radii: through $M'_L \geq M$; through the constant $C_{M'}$ and the factor e^{CM} of Theorem 19.10(b) in Lemma 34.3, which are absorbed by the choice of M'_L and of $\varepsilon_L \leq \varepsilon_{\mathrm{cmp}}(M'_L)$; and through a'_1, a_{step} and r_2, which involve the constants a_1, B_3 of Theorem 30.3 and a_δ of Proposition 31.5.

The fluctuation field lives on the unit lattice $\mathbb{T}$, where all operators are bounded. So the decay of the covariance follows from Combes–Thomas directly. In particular we do not need Bałaban's representation of the axial-gauge covariance through Landau-gauge operators on the fine lattice [7, (3.183)–(3.185)]. The fine-lattice Green functions of Parts I–III enter only through Δ^0, i.e. through the coarse effective action.

34.4 The determinant

The Gaussian integral over the free variables produces the factor $\det(\Delta_k)^{-1/2}$, which depends on the background. We write its logarithm as a sum of local densities.

Proposition 34.7 (Determinant densities)**.** *For real bases let*

$$\log \Delta_k = \int_0^\infty \Big(\frac{1}{1+t} - (\Delta_k + t)^{-1} \Big) dt, \qquad \ell_k(b) = \mathrm{tr}_{\mathfrak{g}} \left[(\log \Delta_k)(b,b) \right], \quad b \in \mathfrak{F}.$$

Then $\log\det \Delta_k = \sum_{b\in\mathfrak{F}} \ell_k(b)$*, and:*

(a) $\log \Delta_k$ *has decay* $(C_L, \gamma_C/2)$*; in particular* $|\ell_k(b)| \leq C_L$*;*

(b) $\ell_k(b)$ *is invariant under real gauge transformations of* (W, U')*;*

(c) for complex bases $\hat{W} = e^{\mathrm{i}\hat{\beta}} W$ *as in Corollary 34.5, the same integral defines* $\ell_k(b)$ *analytically, and* $|\partial_{\hat{\beta}(c)} \ell_k(b)| \leq C_L e^{-\gamma_C |b-c|/2}$*;*

(d) at the trivial base $W = 1$*,* $\ell_k(b)$ *depends only on the position of* b *relative to the blocks* $B'(\cdot)$*.*

Proof. (a) For $t \geq 0$, $\Delta_k + t$ has decay $(C_L + t, \gamma_1)$ and is bounded below by $\frac{1}{4}c_L + t$. By Lemma 5.5 its inverse has decay $(C(c_L/4+t)^{-1}, \gamma_C)$, with γ_C independent of t, since the ratio of the lower bound to the decay constant does not degenerate as $t \to \infty$. The integrand equals $(\Delta_k - 1)(1+t)^{-1}(\Delta_k + t)^{-1}$, which has decay $(C_L(1+t)^{-1}(c_L/4+t)^{-1}, \gamma_C/2)$ and is integrable. For a positive symmetric matrix the integral is the logarithm, and $\mathrm{Tr}\log = \log\det$.

(b) The fluctuation coordinates are covariant: a real gauge transformation u acts on fields on $\mathfrak{F}$ bondwise by $R(u(b_-))$, which is orthogonal. By Proposition 33.2(d) and the covariance of the elimination, $\Delta_k \mapsto R(u)\Delta_k R(u)^{-1}$. Hence $\log \Delta_k$ is conjugated in the same way, and its diagonal traces are invariant.

(c) For complex bases, $\mathrm{Re}(\Delta_k + t) \geq \frac{1}{8}c_L + t$, so the integral converges. It is analytic and agrees with the logarithm for real bases. The derivative is $\int_0^\infty (\Delta_k + t)^{-1} (\partial\Delta_k)(\Delta_k + t)^{-1} dt$. By the tree bound of Corollary 34.5 for $\partial\Delta_k$, its diagonal kernel at b is bounded by $C_L \sum_{b_1,b_2} e^{-\gamma_C(|b-b_1|+|b_2-b|)} e^{-\gamma_1 \tau(c,b_1,b_2)} (c_L/8+t)^{-2}$, which integrates to $C_L e^{-\gamma_C|b-c|/2}$.

(d) At $W = 1$ the operator Δ_k commutes with the translations by $L\mathbb{Z}^d$. □

Proposition 34.7 is the starting point of the analysis of Bałaban's normalization factor [9, (61)–(63)]. What remains is to show that $\ell_k(b;W) - \ell_k(b;1)$ is a gauge-invariant function of the background near b, up to exponentially small terms, and that it is small. This requires the comparison of Δ_k at bases that differ by a large amount far from b. It is part of the localization of the background dependence. This is carried out in Part VIII (Section 38) with Dirichlet approximants, and in Part IX with weakened random walks.

34.5 One step as a Gaussian integral

Theorem 34.8 (One small-field step)**.** *There are $a_{\mathrm{step}} > 0$ and $r_2 > 0$, depending only on d, L, G, a, α and M, and C_L, depending only on d, L, G, a and α, with the following property. Let $k \geq 1$. Let V' be real on $\mathbb{T}'$ with curvature at most $\varepsilon_1' \leq a_{\mathrm{step}}$ in the units of step $k+1$. Let $W = V^0$, U' and the fluctuation coordinates be as above, and let $w \leq r_2$. Let ρ be a gauge-invariant density of the form (31.2) on $\mathfrak{U}_k(\tilde{h})$, with $a_1 \geq \tilde{h} \geq 2(C_*+1)L^{-2}\varepsilon_1' + Cw$, and let $g = g_k$. Then*

$$T(\chi^{\mathrm{w}}\rho)(V') = \mathcal{N}_k \exp\Big(-\frac{1}{g^2}S(U') - \frac{1}{2}\sum_{b\in\mathfrak{F}}\ell_k(b) + \sum_c \log \mathsf{j}_c\Big) \\ \times \int d\mu_{C_k}(Y)\ \chi^{\mathrm{w}}(\hat{Z}(gY)) \exp\Big(-\frac{1}{g^2}\mathcal{V}(gY) + \mathfrak{J}(gY) + \mathfrak{E}\big(e^{\mathrm{i}\hat{Z}(gY)}W\big)\Big),$$

where the integrand vanishes unless $g|Y(b)| \leq w$ for all $b \in \mathfrak{F}$, as in Proposition 32.5, and:

(a) $\mathcal{N}_k = (2\pi g^2)^{|\mathfrak{F}|\dim\mathfrak{g}/2}\,\sigma(0)^{|\mathfrak{F}|}\,L^{(d-1)\dim\mathfrak{g}\,|\mathcal{B}(\mathbb{T}')|}$ does not depend on the background; $|\mathcal{B}(\mathbb{T}')|$ is the number of bonds of $\mathbb{T}'$;

(b) $\mathsf{j}_c = L^{-(d-1)\dim\mathfrak{g}}\mathsf{J}_c(0) = 1 + O(L^d\mathsf{a})$ is a local function of the base near c;

(c) $d\mu_{C_k}$ is the Gaussian measure with the covariance C_k of Corollary 34.5;

(d) $\mathcal{V}$ is analytic for complex $|Z|_\infty \leq r_2$, of third order, with $|g^{-2}\mathcal{V}(gY)| \leq C_L g|Y|_\infty\|Y\|^2$;

(e) $\mathfrak{J}(Z) = \sum_{b\in\mathfrak{F}}\log(\sigma(Z(b))/\sigma(0)) + \sum_c \log(\mathsf{J}_c(Z)/\mathsf{J}_c(0))$ is a sum of local analytic terms, bounded by $C|Z(b)|^2$ and by $CL^{2d+1}|Z|_\infty$ respectively, with $|Z|_\infty$ over the free bonds (equivalently by $CL^{d+1}w$ on the support of χ^{w}, where all $|\hat{Z}(b)| \leq w$);

(f) $\mathfrak{E}(e^{\mathrm{i}\hat{Z}(Z)}W) = \mathfrak{E}(e^{\mathrm{i}\eta A_*(\hat{Z}(Z))}U')$, where $\mathfrak{E}(U)$ denotes the gauge-invariant function $\mathfrak{E}_k$ expressed through the background field.

The term $g_k^{-2}S(U')$ in the units of step k is $g_{k+1}^{-2}S(U')$ in the units of step $k+1$.

Proof. Choose a_{step} so small that $4(C_*+1)L^{-2}a_{\mathrm{step}} \leq a_1'$, and that Lemma 31.6, Lemma 32.4 (with $L^d\mathsf{a} \leq c$), Theorem 34.4 and Corollary 34.5 apply with $\varepsilon = B_3\varepsilon_1'$. By Proposition 32.5, $T(\chi^{\mathrm{w}}\rho)(V')$ is an integral over Z on $\mathfrak{F}$. On the support of χ^{w}, which bounds all bonds including $b_0(c)$, the configuration $e^{\mathrm{i}\hat{Z}(Z)}W$ has curvature at most $2(C_*+1)L^{-2}\varepsilon_1' + Cw \leq \tilde{h}$. So (31.2) applies, and $\rho(e^{\mathrm{i}\hat{Z}}W) = \exp(-g^{-2}F(Z)+\mathfrak{E})$ with F as in Lemma 34.1. Since $\mathsf{j}_c = L^{-(d-1)\dim\mathfrak{g}}\mathsf{J}_c(0)$ by (b), write $\mathsf{J}_c(Z) = L^{(d-1)\dim\mathfrak{g}}\mathsf{j}_c \cdot \mathsf{J}_c(Z)/\mathsf{J}_c(0)$ and $\sigma(Z(b)) = \sigma(0)\,\sigma(Z(b))/\sigma(0)$. The Gaussian integral

$$\int \prod_{b\in\mathfrak{F}} dZ(b)\, e^{-\frac{1}{2g^2}\Delta_k(Z,Z)} f(Z) = (2\pi g^2)^{|\mathfrak{F}|\dim\mathfrak{g}/2}(\det\Delta_k)^{-1/2}\int d\mu_{C_k}(Y) f(gY)$$

and Proposition 34.7 give the formula. (d) follows from Lemma 34.1. (e) follows from $\sigma(Z)/\sigma(0) = 1 + O(|Z|^2)$ and Proposition 32.5. (f) is gauge invariance: by Theorem 25.1(c), $U_k(e^{\mathrm{i}\hat{Z}}W)$ and $e^{\mathrm{i}\eta A_*(\hat{Z})}U'$ differ by a real gauge transformation of $\mathbb{T}_\eta$, which need not lie in $\mathfrak{G}_0$. □

Remark 34.9 (The first step). Theorem 34.8 is stated for $k \geq 1$, because Sections 33 and 34 use the k-step operators of Parts II–V. For $k = 0$ the same representation holds, with its own constants $a_{\mathrm{step}}^{(0)}$, $r_2^{(0)}$ and C_L, depending on the same quantities. At $k = 0$, $\eta = 1$, $\mathbb{T} = \mathbb{T}_\eta$ and $U_0(V) = V$ (Theorem 30.3), so $W = U'^{(0)} = U'$, $A_*(B) = B$ and $\mathcal{S}(B) = S(e^{\mathrm{i}B}U')$. The proof of Theorem 34.8 applies with the following inputs.

- Lemma 31.6, Lemma 32.4 and Proposition 32.5 hold at $k = 0$, with $V^0 = U'$: the 0-fold average is the identity, and the other statements concern the one-step averaging $\mathfrak{a}'$ at the base W and Theorem 30.3 for $k = 0$ and $k = 1$. For Lemma 31.6(c) at $k = 0$: a real $V \in \mathfrak{U}_0(a_1')$ with $\mathfrak{a}'(V) = V'$ satisfies $U_0(V) = V \in \mathfrak{U}(B_3 a_1')$, by Theorem 30.3(b) at $k = 0$. In the units of step 1 its curvature and current are at most $L^3 B_3 a_1' \leq \theta_0$, so $V \in \mathfrak{B}_1(V') \cap \mathfrak{U}(\theta_0)$, and Theorem 30.3(a) at step 1 gives $S(V) \geq S(U')$, with equality only on the orbit of U', as in the proof of Lemma 31.6(c).
- Lemma 34.1 holds with this $\mathcal{S}$, with Δ^0 the Hessian of $B \mapsto S(e^{\mathrm{i}B}U')$ at 0 and $\omega_0 = J_{U'}$. The linear term vanishes by Lemma 31.6(c) at $k = 0$. The cubic bound $|\mathcal{V}(Z)| \leq C_L |Z|_\infty \|Z\|^2$ follows from Cauchy's estimate for the plaquette terms of S, each of which depends on Z, through $\hat{Z}$, on at most C_L bonds within distance $2L + 2$ of the plaquette.
- Theorem 34.4 is not used. The lower bound $\Delta_0 \geq \frac{1}{2} C_{\mathrm{P}}^{-1}$ of Proposition 50.2(b) replaces $\frac{1}{4} c_L$ in Corollary 34.5 and in the proof of Proposition 34.7. Its proof treats the local operator Δ^0 directly and does not use Theorem 34.8. The decay of Δ_0 and of $C_0 = \Delta_0^{-1}$ follows from the locality of Δ_0 and Lemma 5.5.
- The constant $a_{\mathrm{step}}^{(0)}$ is chosen so that Lemma 31.6, Lemma 32.4 (with $L^d \mathsf{a} \leq c$) and $4(C_* + 1) L^{-2} \varepsilon_1' \leq a_1'$ hold, together with the smallness conditions on $\varepsilon = B_3 \varepsilon_1'$ in the proof of Proposition 50.2(b), among them $\varepsilon \leq \varepsilon_{\mathrm{cmp}}(M')$. The radius $r_2^{(0)} \leq r_L$ is chosen as r_2 in Lemma 34.1, with r_1 replaced by 1.

The formula of Theorem 34.8 then holds at $k = 0$, with $g = g_0$, $\mathfrak{E} = \mathfrak{E}_0$, and $\mathcal{N}_0$, $\mathfrak{j}_c$ and $\mathfrak{J}$ as in (a), (b) and (e).

Remark 34.10 (What Parts VIII–X have to do). Theorem 34.8 is exact on the small-field set. The analysis of its Gaussian integral is the content of Parts VIII–X:

- a cluster expansion, in Part X, of the Gaussian integral of Theorem 34.8 (with the cutoff $\chi^{\mathrm{w}}(\hat{Z}(gY))$ and the exponent $-g^{-2}\mathcal{V}(gY) + \mathfrak{J}(gY) + \mathfrak{E}(e^{\mathrm{i}\hat{Z}(gY)}W)$), which requires a localized (random-walk) representation of C_k and of the coefficients of $\mathcal{V}$;
- the localization of the dependence on the background of the densities $\ell_k(b)$, of $\mathfrak{j}_c$ and of the new activities;
- the normalization at $U' = 1$, which removes the vacuum energy, and the proof that the remainder is of second order in the curvature of the background.

The inputs proved here are the coercivity and decay of C_k (Corollary 34.5), the tree decay of the derivatives of the coarse action (Proposition 33.1) and of its dependence on the base (Proposition 33.3), and the analyticity of all ingredients in a complex neighbourhood.

Part VIII: Localization of the background dependence

35 Dirichlet operators for the averaged problem

The objects of Part VII depend on the background field everywhere on the torus. The cluster expansion and the renormalization of the next stage need them as sums of terms that depend on the background only locally, with exponential decay in the size of the localization region. This part constructs such representations. The principle is simple:

- every object is approximated by a version computed with Dirichlet operators on a domain X, which depends only on the background in a fixed neighbourhood of X;
- the approximations on increasing domains are compared, and the differences are telescoped.

The comparison of domains is the content of Theorem 19.10(a), extended here to the averaged operators $\mathbf{\Gamma}$, $\mathbf{H}$, $\mathbf{C}$ (this section), to the coarse Hessian and the fluctuation covariance (Section 37), to the determinant (Section 38), and, with interior pointwise bounds proved in Section 36, to the nonlinear background map and the interaction (Section 39). No truncation of the background itself is needed, so no gauge has to be fixed on large regions. The Dirichlet versions are covariant by construction.

Conventions for Part VIII. We keep the setting of Part VII and work in the units of step k. The background U is a real configuration on $\mathbb{T}_\eta$ that is (M,ε)-regular with $|J_U| \le \varepsilon$, and all operators are taken with the averaging pair $(Q'_k(U), \mathbf{Q}_k(U))$. In applications $U = U'$ is the background of the step, but criticality is not used except where stated. Domains are unions of unit blocks, as in Definition 17.1. For a domain X and $r \ge 0$ let $X_{(r)}$ be the union of the blocks $B(y) \subset X$ with $\operatorname{dist}(y, \mathbb{T}_\eta \setminus X) > r$, and $X^{(r)}$ the union of the blocks within distance r of X. In these definitions, and in the definitions of the seams Σ below, distances are ∞-distances. Put

$$\mathfrak{B}(X) = \{c \text{ bond of } \mathbb{T} : \ B(c_-) \cup B(c_+) \subset X\}.$$

Potentials W are allowed to have range $r_W \le 3$: $(WA)(b)$ depends on A and U on bonds within distance r_W of b. They are real for real U: $W(U)$ maps real fields to real fields. With this change the L^2 statements of Section 19 (coercivity, decay, comparison of domains) hold verbatim, with $\hat{X}$ replaced by $X^{(r_W+1)}$. Their proofs use W only through its norm, its range in A, its decay, its covariance and its reality; its range in U enters only the locality, i.e. the dependence on U only on $X^{(r_W+1)}$. The pointwise statements additionally require Hölder coefficients and are not used for such W. The potential W_2 of Section 37 on the torus depends on U non-locally, through ω_0; this is harmless for torus statements, and on domains we use its Dirichlet version W_2^X, which depends only on U on $\hat{X}$. It has range ≤ 3 in A and is real symmetric at real U, but it depends on U on all of $\hat{X}$. So existence, coercivity and decay hold for it as above, while its comparison of domains is Proposition 35.1(d) with $W \ne W'$, together with Proposition 37.4(d). We write $\hat{X}$ for $X^{(4)}$.

35.1 Definitions

Let $\mathbf{G}^X = (\mathfrak{A}^{X,\kappa})^{-1}$ be the Dirichlet Green function of Theorem 19.7, built with a potential W supported in X. Let $\mathbf{Q}_X$ be $\mathbf{Q}$ restricted to $\ell^2_1(X)$ followed by the restriction to $\mathfrak{B}(X)$; for

$c \in \mathfrak{B}(X)$, $(\mathbf{Q}A)(c)$ depends only on A on the bonds of $B(c_-) \cup B(c_+)$. Define, on $\ell^2(\mathfrak{B}(X))$,

$$\mathbf{\Gamma}^X = (\mathbf{Q}_X \mathbf{G}^X \mathbf{Q}_X^T)^{-1}, \qquad \mathbf{H}^X = \mathbf{G}^X \mathbf{Q}_X^T \mathbf{\Gamma}^X, \qquad \mathbf{C}^X = \mathbf{G}^X - DG'_X R_X G'_X D_X^T - \mathbf{H}^X \mathbf{Q}_X \mathbf{G}^X.$$

When the potential matters we write $\mathbf{\Gamma}^X[W]$, and so on. For $X = \mathbb{T}_\eta$ these are the operators of Theorem 19.9, because $m_{\mathbb{T}_\eta} = 0$ and $\mathbf{Q}_\pi = \mathbf{Q}$ when $E = 0$ (Lemma 15.1(c)).

Since $E = 0$, the slice projection drops out on domains as well: $\mathbf{Q}_X D S_X = D_V Q' S_X = 0$ on $\mathfrak{B}(X)$, because $Q' G'_X R_X = 0$ (Theorem 18.2(d)). The mass term κm_X near $\mathbb{T}_\eta \setminus X$ and the exterior divergence term $\mathfrak{A}_X^{\mathrm{ext}}$ destroy some of the exact identities of Section 15, e.g. $\mathbf{G}^X D R_X = D G'_X R_X$. They are not needed here: the Dirichlet operators serve only as approximants.

35.2 Bounds

Proposition 35.1 (Dirichlet averaged operators)**.** *There are $c, C, \gamma > 0$, independent of X, k, n, η and of $M \geq M_0$, such that for every domain X with $\mathfrak{B}(X) \neq \emptyset$ and every potential W of range ≤ 3, real for real U, with $\|W\| \leq \varepsilon$ and supported in X:*

(a) $c \leq \mathbf{Q}_X \mathbf{G}^X \mathbf{Q}_X^T \leq C$ (for non-symmetric W the lower bound means $\mathrm{Re}\langle b, \mathbf{Q}_X \mathbf{G}^X \mathbf{Q}_X^T b\rangle \geq c\|b\|^2$, as in (e)); $\mathbf{\Gamma}^X$ has decay (C, γ).

(b) $\mathbf{Q}_X \mathbf{H}^X = I$. The operators $\mathbf{H}^X$, $D\mathbf{H}^X$, $\mathbf{C}^X$, $D\mathbf{C}^X$, $\mathbf{C}^X D^T$ and $D\mathbf{C}^X D^T$ have decay (C, γ).

(c) They depend only on U and W on the bonds of $\hat{X}$, and they are covariant under real gauge transformations on $\hat{X}$.

(d) (Comparison of domains.) *Let $X' \subset X$, let W and W' be potentials supported in X and X', and let Σ be the set of unit points within distance $10M$ of $X \setminus X'$. For $y, y' \in X'_1$ and each $F \in \{\mathbf{\Gamma}, \mathbf{H}, \mathbf{C}\}$ (for $\mathbf{\Gamma}$, and on the coarse side of $\mathbf{H}$, the indices are unit bonds starting at y, y' and lying in $\mathfrak{B}(X')$),*

$$\|\mathbf{1}_y (F^X[W] - F^{X'}[W']) \mathbf{1}_{y'}\| \leq C \sum_{w \in \Sigma} e^{-\gamma(|y-w|+|w-y'|)} + \frac{C}{\varepsilon} \sum_{w \in X'_1} e^{-\gamma(|y-w|+|w-y'|)} \|W - W'\|_w,$$

where $\|W - W'\|_w$ is the norm of $\mathbf{1}_w(W - W')$. The same bounds hold for DF, and for $\mathbf{C} D_X^T$ and $D\mathbf{C} D_X^T$ (derivatives on the left, and D_X^T on the right). They also hold for $\mathbf{G}$ itself and for $P\mathbf{G}Q$ with $P \in \{1, D, D^T, \nabla\}$ and $Q \in \{1, D^T, D\}$, as in Theorem 19.10(a), with the potential W on X and W' on X', for all unit points y, y'. For $\mathbf{H}$, $D\mathbf{H}$, $\mathbf{C}$ and their derivatives the bound holds on the fine side for all unit points as well, with the operators extended by zero.

(e) (Complex perturbations.) *Let U_δ be as in Proposition 19.11, with $\delta \leq \delta_0$ and $C\delta S_{\gamma/4} \leq \frac{c}{2}$, and let W extend holomorphically to U_δ as there. Then (a), (b) and (d) hold at U_δ, with the lower bound in (a) replaced by* Re*-coercivity with the constant $\frac{c}{2}$, and the operators are analytic in δA.*

Proof. (a) The upper bound follows from $\|\mathbf{G}^X\| \leq C$ and $\|\mathbf{Q}\| \leq C$. For the lower bound, let b be a field on $\mathfrak{B}(X)$. The trial fields of Lemma 19.8 for the bonds $c \in \mathfrak{B}(X)$ are supported in tubes inside $B(c_-) \cup B(c_+) \subset X$, so they lie in $\ell^2_1(X)$. The matrix T_U of that proof, restricted to $\mathfrak{B}(X)$, is still diagonally dominant. This gives $\Phi_b \in \ell^2_1(X)$ with $\mathbf{Q}_X \Phi_b = b$ and $\|\!|\Phi_b|\!\|_X \leq C\|b\|$. Put $u = \mathbf{G}^X \mathbf{Q}_X^T b$. By Theorem 19.6, $\mathrm{Re}\langle b, \mathbf{Q}_X \mathbf{G}^X \mathbf{Q}_X^T b\rangle = \mathrm{Re}\langle u, \mathfrak{A}^{X,\kappa} u\rangle \geq c_* \|\!|u|\!\|^2_{X,+}$. On

the other hand $\|b\|^2 = \langle \mathbf{Q}_X^T b, \Phi_b\rangle = \langle \mathfrak{A}^{X,\kappa} u, \Phi_b\rangle \le C\|\!|u|\!\|_{X,+}\|\!|\Phi_b|\!\|_X \le C\|\!|u|\!\|_{X,+}\|b\|$. Here $\langle \mathfrak{A}_X^{\mathrm{ext}} u, \Phi_b\rangle = \langle \mathbf{1}_{\mathbb{T}_\eta\setminus X} D^T u, \mathbf{1}_{\mathbb{T}_\eta\setminus X} D^T \Phi_b\rangle = 0$, since Φ_b lives on bonds with both endpoints in $B(c_-)\cup B(c_+) \subset X$. Hence $\mathrm{Re}\langle b, \mathbf{Q}_X \mathbf{G}^X \mathbf{Q}_X^T b\rangle \ge c\|b\|^2$, which is the lower bound in (a); the argument does not use the symmetry of W. Decay follows from Lemma 5.5.

(b) The identity is the definition. Decay follows from (a), Theorem 19.7(a), Theorem 18.2(b),(d) and Lemma 5.2.

(c) holds for the ingredients by Sections 18 and 19, and for $\mathbf{Q}_X$ by Proposition 22.7(d),(f).

(d) For $\mathbf{G}$ this is Theorem 19.10(a) when $W = W'$ on X'. A change of the potential is handled by the resolvent identity $\mathbf{G}^{X'}[W] - \mathbf{G}^{X'}[W'] = -\mathbf{G}^{X'}[W]\,\Pi_{X'}^T (W - W')\Pi_{X'}\,\mathbf{G}^{X'}[W']$ and the decay of both factors. For $\mathbf{Q}_X \mathbf{G}^X \mathbf{Q}_X^T$ the kernel bound follows, since $\mathbf{Q}$ is local. For $\mathbf{\Gamma}$ apply Lemma 17.7 on the unit lattice, with $\mathfrak{A} = \mathbf{Q}_X \mathbf{G}^X \mathbf{Q}_X^T$ and $\mathfrak{A}' = \mathbf{Q}_{X'} \mathbf{G}^{X'} \mathbf{Q}_{X'}^T$. The operator Δ of that lemma has two kinds of entries. For $c \in \mathfrak{B}(X')$ they are the entries of $\mathbf{Q}(\mathbf{G}^X - \mathbf{G}^{X'})\mathbf{Q}^T$, which have seam form. For $c \in \mathfrak{B}(X) \setminus \mathfrak{B}(X')$ they are entries $(\mathbf{Q}\mathbf{G}^X\mathbf{Q}^T)(c, c')$ with c within distance 2 of $X \setminus X'$, so again of seam form. For $\mathbf{H}$ and $\mathbf{C}$ use the telescoping argument of Theorem 18.2(e). With derivatives the argument is the same: Theorem 19.10(a) holds with derivatives, the factors $D\mathbf{G}^X$, $\mathbf{G}^X D_X^T$, $D\mathbf{G}^X D_X^T$, DG'_X, $G'_X D_X^T$ have decay, and $\mathbf{C}^X[W]^T = \mathbf{C}^X[W^T]$, where W^T has the properties of a potential that these bounds use (see below). Indeed $\mathfrak{A}^{X,\kappa}[W]^T = \mathfrak{A}^{X,\kappa}[W^T]$, since only $\mathfrak{B}_X$ contains W. So the bound for $\mathbf{C}^X D_X^T$ follows from that for $D\mathbf{C}^X$ at W^T, since kernel bounds pass to transposes; the bound for $D\mathbf{C}^X D_X^T$ follows by applying D on the left to the telescoping identity for $\mathbf{C}^X D_X^T$. The transpose W^T is real for real U, covariant, supported in X, of norm at most ε and of range ≤ 3 in A, and it extends holomorphically with W; the bounds of (a), (b) and (d) use the potential only through these properties (Conventions). Its range in U can be larger, up to $2r_W$, since an entry $W(b', b)$ depends on U near b'. This range is not used: $\mathbf{C}^X[W^T] = \mathbf{C}^X[W]^T$ depends only on U and W on $\hat{X}$, by (c) for W. For $W \ne W'$, $\|W^T - W'^T\|_w = \|(W - W')\mathbf{1}_w\|$ is bounded by the norms $\|W - W'\|_{w'}$ of the boundedly many blocks w' near w, since $W - W'$ has range ≤ 3 in A. A block $w' \notin X'_1$ among them contributes only if $B(w') \subset X \setminus X'$; then $w' \in \Sigma$ and $\|W - W'\|_{w'} = \|\mathbf{1}_{w'} W\| \le \varepsilon$, so its term joins the first sum. For $P\mathbf{G}Q$ with $W \ne W'$, apply P and Q to the resolvent identity. The factors $P\mathbf{G}^{X'}[W]\Pi_{X'}^T$ and $\Pi_{X'}\mathbf{G}^{X'}[W']Q$ have decay, by Theorem 19.7(a), which holds for these potentials (Conventions), and Theorem 18.2(d) for $S_{X'}$. This gives the bound for $\mathbf{G}$ stated at the end of (d). For $\mathbf{H}$, $\mathbf{C}$ and their derivatives on the fine side, the telescoping uses only Theorem 19.10(a), which holds for all unit points, and the decay of all factors at all blocks; so the bound holds at all fine unit points.

The constants. They do not depend on $M \ge M_0$. The proof uses Theorems 18.2, 19.7(a) and 19.10(a), Lemma 19.8 and the locality of $\mathbf{Q}$. The constants of these results do not depend on $M \ge M_0$: M enters the Dirichlet operators only through the mass profile m_X, whose coefficient κ and coercivity constant are those of Theorem 19.6 (Remark 31.4(c)), and the seam width $10M$ enters only through Σ.

(e) Let U_δ and W be as in (e). The averages at U_δ are those of Proposition 46.4, with the real reference U, and $\mathbf{Q}_X(U_\delta) - \mathbf{Q}_X(U)$ is a local operator of size $C\delta$ (Proposition 46.4(v)). By Proposition 19.11, $\mathbf{G}^X(U_\delta)$ exists, with the decay of Theorem 19.7(a), and $\mathbf{G}^X(U_\delta) - \mathbf{G}^X(U)$ has decay $(C\delta, \gamma/4)$ by Lemma 13.3, with γ the common rate γ_0 there. So $\mathbf{Q}_X \mathbf{G}^X \mathbf{Q}_X^T$ at U_δ is $K_0 + \delta K$, where $K_0 \ge c$ is its value at U, by (a), and δK has decay $(C\delta, \gamma/4)$. By Lemma 13.3 in its unit-lattice form, for $C\delta S_{\gamma/4} \le \frac{c}{2}$, the operator $K_0 + \delta K$ is invertible, with $\mathrm{Re}\langle b, (K_0 + \delta K)b\rangle \ge \frac{c}{2}\|b\|^2$, and its inverse $\mathbf{\Gamma}^X(U_\delta)$ has decay. Then (b) holds at U_δ by the same compositions, and (d) by the same proof, since Theorem 19.10 holds at U_δ (Proposition 19.11) and Lemma 17.7 is algebraic. So (a), (b) and (d) hold at U_δ, with the lower bound in (a) replaced by Re-coercivity with the constant $\frac{c}{2}$. The operators are analytic in δA, as compositions and inverses of analytic families. □

35.3 Interior pointwise bounds

Sections 37 and 38 use Proposition 35.1 only through kernel bounds in block L^2 norms. The nonlinear constructions of Section 39 also need pointwise bounds with derivatives at interior points of a domain. They are provided by the following theorem, which is proved in Section 36. For a field h and a block $B(y)$ put

$$|h|_{1,\alpha;y} = |h|_{B(y)} + |\nabla h|_{B(y)} + [\nabla \tilde{h}_y]_{\alpha,B(y)},$$

where ∇ is the covariant gradient and $\tilde{h}_y$ is h in a real gauge in which (B) holds on $\Lambda_4(c_y)$. For a coarse field, $|B(y')|$ is the maximum over the unit bonds starting at y'.

Theorem 35.2 (Interior pointwise bounds)**.** *Let $W \in \mathcal{W}_{\mathrm{loc}}(\varepsilon)$, and build the Dirichlet operators on a domain X with the potential $\mathbf{1}_X W \mathbf{1}_X$. There are $\varepsilon_{\mathrm{ID}}, C, \gamma > 0$, independent of X, k, n, η and of $M \geq M_0$, such that the following holds for $\varepsilon \leq \varepsilon_{\mathrm{ID}}$, every domain X and every $y \in X_{(4)}$; the sums run over unit points y', and $|f|_{B(y')}$, $|g|_{B(y')}$ are suprema over the bonds, respectively plaquettes, that have a point in $B(y')$.*

(a) $|\mathbf{C}^X f|_{1,\alpha;y} \leq C \sum_{y'} e^{-\gamma|y-y'|} |f|_{B(y')}$ *for fine fields $f \in \ell^2_1(X)$.*

(b) $|\mathbf{C}^X D_X^T g|_{1,\alpha;y} \leq C \sum_{y'} e^{-\gamma|y-y'|} (|g|_{B(y')} + [\tilde{g}_{y'}]_{\alpha,\Lambda_1(c_{y'})})$ *for 2-forms g, where $D_X^T = \mathbf{1}_X D^T$.*

(c) $|\mathbf{H}^X B|_{1,\alpha;y} \leq C \sum_{y'} e^{-\gamma|y-y'|} |B(y')|$ *for coarse fields B on $\mathfrak{B}(X)$.*

(d) $|D^T D \mathbf{H}^X Y|_{B(y)} \leq C \sum_{y'} e^{-\gamma|y-y'|} |Y(y')|$ *for coarse fields Y on $\mathfrak{B}(X)$.*

(e) (Comparison of domains.) *Let $X' \subset X$, let Σ be the set of unit points within distance $10M$ of $X \setminus X'$, and put $K_\Sigma(y,y') = \sum_{w\in\Sigma} e^{-\gamma(|y-w|+|w-y'|)}$. For $y \in X'_{(4)}$ and sources supported on bonds and plaquettes inside X', respectively on $\mathfrak{B}(X')$, the differences between the operators of (a)–(d) on X and on X' satisfy the bounds (a)–(d) with $e^{-\gamma|y-y'|}$ replaced by $K_\Sigma(y,y')$.*

Part (d), and (e) for it, use the bound $|J_U| \leq \varepsilon$ on the current; the other parts do not. The theorem is stated for the real background U; (d) is not claimed at the complex backgrounds of Proposition 19.11.

For $X = \mathbb{T}_\eta$ there is no mass term, and (c), (d) are contained in Theorem 19.9, Lemma 24.1 and the identity $D^T D\mathbf{H} = \mathbf{Q}^T(\mathbf{\Gamma}_\pi - a) + S^T \mathfrak{F}^T D\mathbf{H}$ of Lemma 26.2; (a) and (b) add Hölder bounds on the gradient. On a proper domain that identity fails. The mass term and the exterior divergence term break $\mathbf{G}^X D R_X = D G'_X R_X$, so $\mathbf{H}^X$ need not lie in the Dirichlet slice, and $S_X \mathbf{H}^X \neq 0$. Then $D^T D \mathbf{H}^X$ contains the term $D^T \mathfrak{F} S_X \mathbf{H}^X$, whose coefficient involves second derivatives of the background. Part (d) is a consequence of gauge invariance of the action, which replaces these second derivatives by the current (Lemma 36.6).

Remark 35.3 (Why Part III does not give Theorem 35.2 directly). The pointwise bounds of Parts II and III are proved by a *global* bootstrap (Lemmas 13.5 and 18.3). The local regularity estimate at a block y involves sup norms of the solution on neighbouring blocks with a small factor ε, and these are absorbed by taking a maximum over *all* blocks. On a domain X the maximum would include blocks adjacent to the Dirichlet cut, where no pointwise estimate is available. Comparison with torus operators (Lemma 17.7) does not help either: it produces sources at the seam that are controlled only in L^2.

Section 36 replaces the global maximum by an absorption inside one cube, at all scales. Two observations make this work:

- The small nonlocal terms of $\mathfrak{A}^{X,\kappa}$, of Riesz type, all come from the slice projection Π_X. They become local once two scalar potentials are treated as unknowns. The other nonlocal terms factor through the unit lattice and are bounded by local L^2 norms; for the averaging term this uses $E = 0$, through $\Pi_X^T \mathbf{Q}^T \mathbf{Q} \Pi_X = \mathbf{Q}^T \mathbf{Q}$.
- For local systems, lower-order terms carry positive powers of the radius, so only the principal perturbations must be small.

Two alternatives are not needed here: Bałaban's treatment of Dirichlet propagators with the random-walk expansion [3, 7], and a canonical truncation of the background in a Coulomb gauge on boxes.

36 Interior regularity for Dirichlet operators

This section proves Theorem 35.2. The pointwise bounds of Parts II and III come from a global bootstrap (Lemma 13.5), which does not work on a domain (Remark 35.3). We replace it by an absorption argument inside a single cube, at all scales. Two observations make this possible.

- The nonlocal terms of $\mathfrak{A}^{X,\kappa}$ are of two kinds. Terms that factor through the unit lattice, such as $D P_{\mathcal{V},X} D_X^T$ and $a \mathbf{Q}^T \mathbf{Q}$, are bounded pointwise by local L^2 norms. The remaining nonlocal terms are small, but of Riesz type; they all come from the slice projection Π_X. They disappear if two scalar potentials, $\sigma = S_X u$ and a dual potential τ, are treated as additional unknowns. The triple (u, σ, τ) satisfies a *local* elliptic system with L^2-controlled sources (Lemma 36.4).
- For local systems the loss of domain in Theorem 6.3 is harmless. We apply that theorem at a radius proportional to the distance between two nested cubes and absorb with an iteration lemma (Lemma 36.1). This is the absorption technique of elliptic regularity theory, as in the hole-filling arguments of [1, Ch. V]. Lower-order terms carry positive powers of the radius, so only the principal perturbations need to be small.

No estimate near the boundary of X is used.

Conventions. We keep the setting of Part VIII. We may assume $\eta \le \frac{1}{16}$. Otherwise values, differences and Hölder quotients are bounded by block L^2 norms, as in Step 0 of the proof of Theorem 6.3, and Theorem 35.2 follows from the decay in Proposition 35.1(b). For $z \in \mathbb{R}^d$ we write $\Lambda_r = \Lambda_r(z)$. We fix a real gauge in which $U = e^{\mathrm{i}\eta A}$ on $\Lambda_4(z)$, with A satisfying (B) there; it exists because U is (M, ε)-regular. All fields are written in this gauge, and the tilde is omitted. Then $D_\mu = \partial_\mu + T_\mu$, $\mathfrak{F}\lambda = c\lambda$ and the normal forms of Lemma 12.2 hold on Λ_3. For a set Q of lattice points, $|h|_Q$ is the supremum, $[h]_{\alpha,Q}$ the Hölder quotient and $\|h\|_Q$ the L^2 norm over Q. An equation holds *on* Λ_r if it holds at every point, respectively at every bond or plaquette starting at a point, of Λ_r. Bounds "on Λ_r" may involve values within distance 2η of Λ_r; we do not indicate this enlargement. A *source in local form* on Λ_r is a field $\mathcal{S} = F + \sum_\mu \partial_\mu^* g_\mu$, componentwise for gauge fields, and we put

$$\mathfrak{n}_r(\mathcal{S}) = |F|_{\Lambda_r} + |g|_{\Lambda_r} + [g]_{\alpha,\Lambda_r}, \qquad \mathfrak{n}_r^0(\mathcal{S}) = |F|_{\Lambda_r} + |g|_{\Lambda_r}.$$

36.1 Absorption at all scales

Lemma 36.1 (Iteration [1, Ch. V, Lemma 3.1])**.** *Let $0 < \rho_0 < R_0$, and let $\Psi \ge 0$ be bounded on $[\rho_0, R_0]$. Suppose there are $\theta \in [0, 1)$ and $A, B, \beta \ge 0$ such that*

$$\Psi(t) \le \theta\, \Psi(t') + A(t' - t)^{-\beta} + B \qquad \text{for all } \rho_0 \le t < t' \le R_0.$$

Then $\Psi(\rho_0) \le C(\theta,\beta)(A(R_0-\rho_0)^{-\beta}+B)$.

Proof. Choose $\lambda \in (0,1)$ with $\theta\lambda^{-\beta} < 1$, and put $t_i = \rho_0 + (1-\lambda^i)(R_0-\rho_0)$, so that $t_{i+1}-t_i = (1-\lambda)\lambda^i(R_0-\rho_0)$. By induction,

$$\Psi(\rho_0) \le \theta^k \Psi(t_k) + \sum_{i<k} \theta^i \Big(A(1-\lambda)^{-\beta}\lambda^{-i\beta}(R_0-\rho_0)^{-\beta} + B \Big).$$

Since Ψ is bounded, the first term tends to 0 as $k \to \infty$. Summing the geometric series gives the bound. □

Lemma 36.2 (Local regularity at radius s)**.** *Let $\eta \le s \le 1$, $x \in \eta\mathbb{Z}^d$, and write $\Lambda_r = \Lambda_r(x)$ in this lemma. If $-\Delta u = F + \sum_\mu \partial_\mu^* g_\mu$ on Λ_{2s}, then, with C as in Theorem 6.3,*

$$|u|_{\Lambda_s} + s|\partial u|_{\Lambda_s} + s^{1+\alpha}[\partial u]_{\alpha,\Lambda_s} \le C(s^2|F|_{\Lambda_{2s}} + s|g|_{\Lambda_{3s}} + s^{1+\alpha}[g]_{\alpha,\Lambda_{3s}} + s^{-d/2}\|u\|_{\Lambda_{2s}}),$$
$$|u|_{\Lambda_s} + s^{\alpha}[u]_{\alpha,\Lambda_s} \le C(s^2|F|_{\Lambda_{2s}} + s|g|_{\Lambda_{3s}} + s^{-d/2}\|u\|_{\Lambda_{2s}}).$$

Proof. The function $u_s(x') = u(x+sx')$ lives on the lattice $(\eta/s)\mathbb{Z}^d$, with spacing at most 1. It satisfies $-\Delta u_s = s^2 F_s + \sum_\mu \partial_\mu^*(s g_{\mu,s})$, with lattice derivatives at spacing η/s, where $F_s(x') = F(x+sx')$, and similarly for g. Moreover $\|u_s\|_{\Lambda_2(0)} = s^{-d/2}\|u\|_{\Lambda_{2s}}$, $\partial u_s = s(\partial u)_s$ and $[\partial u_s]_{\alpha,} = s^{1+\alpha}[\partial u]_{\alpha,}$. The proof of Theorem 6.3 uses only that the spacing is at most 1; for spacing larger than $\frac{1}{16}$ Step 0 applies. Apply (d) and (b) of that theorem to u_s. □

Proposition 36.3 (Interior estimates for scalar equations)**.** *There is $\varepsilon_S > 0$, depending only on d, α and $\dim \mathfrak{g}$, with the following property. For $0 < \rho < r \le 3$ there is C, depending in addition on $r-\rho$, such that for $\varepsilon \le \varepsilon_S$: if v is a scalar field with $D^T D v = \mathcal{S}$ on Λ_r, for a source $\mathcal{S}$ in local form, then*

$$|v|_{\Lambda_\rho} + |\partial v|_{\Lambda_\rho} + [\partial v]_{\alpha,\Lambda_\rho} \le C(\mathfrak{n}_r(\mathcal{S}) + \|v\|_{\Lambda_r}),$$
$$|v|_{\Lambda_\rho} + [v]_{\alpha,\Lambda_\rho} \le C(\mathfrak{n}_r^0(\mathcal{S}) + \|v\|_{\Lambda_r}).$$

Proof. If $r-\rho < 40\eta$, both bounds follow from $|v(x)| \le \eta^{-d/2}\|v\|_{\Lambda_r}$, $|\partial v| \le 2\eta^{-1}\sup|v|$, $[w]_{\alpha,} \le 2\eta^{-\alpha}\sup|w|$ and $\eta^{-1} \le 40(r-\rho)^{-1}$, so assume $r-\rho \ge 40\eta$. By Lemma 12.2, $-\Delta v = (F - bv) + \sum_\mu \partial_\mu^*(g_\mu - a_\mu v)$ on Λ_r.

First bound. Let $\Psi(t) = \sup_{x\in\Lambda_t}(|v(x)| + |\partial v(x)|)$. Let $\rho \le t < t' \le r$ and $s = (t'-t)/6$. Suppose first $s \ge 3\eta$, so that $\Lambda_{3s}(x)$, enlarged by 2η, lies in $\Lambda_{t'}$ for $x \in \Lambda_t$. Two points of a cube Q of side $\ell \le 6$ are joined by a monotone lattice path in Q, so $[v]_{\alpha,Q} \le C\ell^{1-\alpha}|\partial v|_Q$, and similarly $[a_\mu]_{\alpha,Q} \le C|\partial a_\mu| \le C\varepsilon$. Hence on $\Lambda_{3s}(x)$

$$|bv| + |a_\mu v| + s^\alpha [a_\mu v]_{\alpha,\Lambda_{3s}(x)} \le C\varepsilon\, \Psi(t').$$

The first inequality of Lemma 36.2 on $\Lambda_{3s}(x)$, divided by s, gives

$$|v(x)| + |\partial v(x)| \le C\,\mathfrak{n}_r(\mathcal{S}) + C_S\varepsilon\,\Psi(t') + Cs^{-1-d/2}\|v\|_{\Lambda_r}.$$

If $s < 3\eta$, then $|v(x)|+|\partial v(x)| \le 3\eta^{-1-d/2}\|v\|_{\Lambda_r} \le C(t'-t)^{-1-d/2}\|v\|_{\Lambda_r}$, since $\eta^{-1} \le 18(t'-t)^{-1}$. So $\Psi(t) \le C_S\varepsilon\Psi(t') + C(t'-t)^{-1-d/2}\|v\|_{\Lambda_r} + C\mathfrak{n}_r(\mathcal{S})$ for all $\rho \le t < t' \le r$. For $C_S\varepsilon \le \frac12$, Lemma 36.1 on $[\rho', r]$, $\rho' = \frac12(\rho+r)$, bounds $\Psi(\rho')$. For the Hölder quotient let $s_1 = (r-\rho)/8 \ge \eta$ and $x, x' \in \Lambda_\rho$. If $|x-x'|_\infty \le s_1$, both points lie in $\Lambda_{s_1}(x)$, and Lemma 36.2 on $\Lambda_{3s_1}(x) \subset \Lambda_{\rho'}$ bounds the quotient, with the bound on $\Psi(\rho')$ inserted. Otherwise the quotient is at most $2s_1^{-\alpha}\Psi(\rho)$.

Second bound. The same argument with $\Psi_0(t) = |v|_{\Lambda_t}$ and the second inequality of Lemma 36.2. Now $s^2|bv| + s|a_\mu v| \le C\varepsilon s\Psi_0(t')$, and for $s < 3\eta$ we use $|v(x)| \le \eta^{-d/2}\|v\|_{\Lambda_r}$. □

36.2 The local system

Lemma 36.4 (Local form of the Dirichlet equation)**.** *Let X be a domain, $W \in \mathcal{W}_{\mathrm{loc}}(\varepsilon)$, and let $u \in \ell^2_1(X)$ satisfy $\mathfrak{A}^{X,\kappa} u = \mathcal{S}$. Put*

$$\sigma = S_X u, \qquad \tau = G'_X D_X^T \mathfrak{B}_X \Pi_X u,$$
$$\omega_1 = (M'_X)^{-1} Q' G'_X D_X^T u, \qquad \omega_2 = (M'_X)^{-1} Q' G'_X \tau, \qquad \pi_i = G'_X Q'^T \omega_i.$$

Then $Q'\sigma = 0$, and at every point, bond and plaquette at ∞-distance at least 2η from $\mathbb{T}_\eta \setminus X$:

$$D^T D u + D D^T u - D^T \mathfrak{F} \sigma + W(u - D\sigma) - D\tau + \kappa m_X u = \mathcal{S} + D\pi_1 - D\pi_2 - a\mathbf{Q}^T \mathbf{Q} u, \tag{36.1}$$
$$D^T D \sigma = D^T u - \pi_1, \tag{36.2}$$
$$D^T D \tau = \mathfrak{F}^T (Du - \mathfrak{F}\sigma) + D^T \big(W(u - D\sigma)\big) - a Q'^T Q' \tau, \tag{36.3}$$
$$D^T D \pi_i = Q'^T \omega_i - a Q'^T Q' \pi_i. \tag{36.4}$$

Moreover there are C, γ, independent of X, k, n, η, such that for every y

$$\|\sigma\|_{B(y)} + \|\tau\|_{B(y)} + \sum_i \big(|\omega_i(y)| + \|\pi_i\|_{B(y)}\big) \le C \sum_{y'} e^{-\gamma|y-y'|} \big(\|u\|_{B(y')} + \|Du\|_{B(y')}\big),$$

where $\|Du\|_{B(y')}$ is taken over the plaquettes starting in $B(y')$.

Proof. Since $E = 0$, $\mathbf{Q}\Pi_X = \mathbf{Q} - D_V Q' S_X = \mathbf{Q}$, because $Q' S_X = Q' G'_X R_X D_X^T = 0$ by Theorem 18.2(d). So the averaging term of $\mathfrak{A}^X$ is $a\mathbf{Q}^T\mathbf{Q}$. Next, $\Pi_X^T = 1 - S_X^T D_X^T$ with $S_X^T = D R_X G'_X$. Hence

$$\Pi_X^T \mathfrak{B}_X \Pi_X u = \mathfrak{B}_X \Pi_X u - D R_X \tau, \qquad R_X \tau = \tau - P_{\mathcal{V},X} \tau = \tau - \pi_2,$$

and $D R_X D_X^T u = D D_X^T u - D\pi_1$. Away from $\mathbb{T}_\eta \setminus X$ the projections $\mathbf{1}_X$ act trivially. There $\mathfrak{B}_X \Pi_X u = D^T D(u - D\sigma) + W(u - D\sigma)$, and $D^T D D \sigma = D^T \mathfrak{F} \sigma$. This gives (36.1). By Lemma 19.2, $\sigma \in \operatorname{ran} S_X \subset \mathcal{N}_X$, so $Q'\sigma = 0$ and $\mathfrak{A}'_X \sigma = D^T D \sigma$ there. Since $\mathfrak{A}'_X \sigma = R_X D_X^T u = D_X^T u - \pi_1$, this gives (36.2). At a point of X, D^T involves only bonds with an endpoint at that point, so $\mathbf{1}_X D^T \mathbf{1}_X D^T = \mathbf{1}_X D^T D^T = \mathbf{1}_X \mathfrak{F}^T$ on 2-forms. Hence

$$D_X^T \mathfrak{B}_X \Pi_X u = \mathbf{1}_X \mathfrak{F}^T D \Pi_X u + D_X^T \mathbf{1}_X W \Pi_X u,$$

with $D\Pi_X u = Du - \mathfrak{F}\sigma$ away from $\mathbb{T}_\eta \setminus X$. Together with $\mathfrak{A}'_X \tau = D_X^T \mathfrak{B}_X \Pi_X u$ this gives (36.3), and (36.4) is $\mathfrak{A}'_X \pi_i = Q'^T \omega_i$. For the bounds, write $\tau = G'_X \mathbf{1}_X \mathfrak{F}^T (Du - \mathfrak{F}\sigma) + G'_X D_X^T \mathbf{1}_X W(u - D\sigma)$. The operators S_X, DS_X, G'_X, $G'_X D_X^T$, $(M'_X)^{-1}$ have decay (Theorem 18.2(b)–(d)). The operators $\mathfrak{F}$, W, Q' are local and bounded, and $|Q'\lambda(y)| \le C\|\lambda\|_{B(y)}$. Compositions of operators with decay have decay (Lemma 5.2). □

Proposition 36.5 (Interior estimates for the local system)**.** *There is $\varepsilon_L > 0$, depending only on d, α, $\dim \mathfrak{g}$ and κ, with the following property. For $0 < \rho < r \le 3$ there is C, depending in addition on $r - \rho$, such that for $\varepsilon \le \varepsilon_L$ the following holds. Let $W \in \mathcal{W}_{\mathrm{loc}}(\varepsilon)$, let m be a real function with $|m| \le 1$, and let u be a gauge field and σ, τ scalar fields with*

$$\begin{aligned} D^T D u + D D^T u - D^T \mathfrak{F} \sigma + W(u - D\sigma) - D\tau + \kappa m u &= \mathcal{S}_u, \\ D^T D \sigma - D^T u &= \mathcal{S}_\sigma, \\ D^T D \tau - \mathfrak{F}^T (Du - \mathfrak{F}\sigma) - D^T \big(W(u - D\sigma)\big) &= \mathcal{S}_\tau \end{aligned} \tag{36.5}$$

on Λ_r, for sources in local form. Put $\mathfrak{s} = \mathfrak{n}_r(\mathcal{S}_u) + \mathfrak{n}_r(\mathcal{S}_\sigma) + \mathfrak{n}_r(\mathcal{S}_\tau)$ and $\mathfrak{m} = \|u\|_{\Lambda_r} + \|\sigma\|_{\Lambda_r} + \|\tau\|_{\Lambda_r}$. Then

$$\sum_{v \in \{u,\sigma,\tau\}} \Big(|v|_{\Lambda_\rho} + |\partial v|_{\Lambda_\rho} + [\partial v]_{\alpha,\Lambda_\rho} \Big) \le C(\mathfrak{s} + \mathfrak{m}).$$

Proof. As in Proposition 36.3 we may assume $r-\rho \ge 40\eta$.

Step 1: normal form. Let a_μ, b be the coefficients of the normal forms of Lemma 12.2, for scalar fields and, componentwise, for D^TD+DD^T on gauge fields. On 2-forms, $(D^T\omega)_\nu = \sum_\mu(\partial_\mu^* + T_\mu^T)\omega_{\mu\nu}$. Then on Λ_r, componentwise,

$$-\Delta u = F_u + \textstyle\sum_\mu \partial_\mu^* g_{u,\mu}, \qquad -\Delta\sigma = F_\sigma + \sum_\mu \partial_\mu^* g_{\sigma,\mu}, \qquad -\Delta\tau = F_\tau + \sum_\mu \partial_\mu^* g_{\tau,\mu},$$

where, with $\mathcal{S}_\bullet = F_{\mathcal{S}_\bullet} + \sum \partial^* g_{\mathcal{S}_\bullet}$,

$$\begin{aligned}
F_u &= F_{\mathcal{S}_u} - bu + T^T(\mathfrak{F}\sigma) - W(u - D\sigma) + D\tau - \kappa m u, \qquad & g_{u,\mu} &= g_{\mathcal{S}_u,\mu} - a_\mu u + (\mathfrak{F}\sigma)_{\mu\cdot},\\
F_\sigma &= F_{\mathcal{S}_\sigma} + T^T u - b\sigma, \qquad & g_{\sigma,\mu} &= g_{\mathcal{S}_\sigma,\mu} + u_\mu - a_\mu\sigma,\\
F_\tau &= F_{\mathcal{S}_\tau} + \mathfrak{F}^T(Du - \mathfrak{F}\sigma) + T^T\big(W(u-D\sigma)\big) - b\tau, \qquad & g_{\tau,\mu} &= g_{\mathcal{S}_\tau,\mu} + \big(W(u-D\sigma)\big)_\mu - a_\mu\tau.
\end{aligned}$$

The coefficients a_μ, b, T, c and those of W are bounded by $C\varepsilon$, and so are their Hölder quotients; moreover $|\partial a_\mu| \le C\varepsilon$ (Lemmas 12.2, 12.3 and the definition of $\mathcal{W}_{\mathrm{loc}}$). Hence on every cube $Q \subset \Lambda_r$ of side at most 6, with sup norms over Q (enlarged by 2η):

$$\begin{aligned}
|F_u| &\le |F_{\mathcal{S}_u}| + C\varepsilon(|u| + |\sigma| + |\partial\sigma| + |\tau|) + |\partial\tau| + \kappa|u|,\\
|g_u| + [g_u]_{\alpha,Q} &\le |g_{\mathcal{S}_u}| + [g_{\mathcal{S}_u}]_{\alpha,Q} + C\varepsilon(|u| + |\sigma| + [u]_{\alpha,Q} + [\sigma]_{\alpha,Q}),\\
|F_\sigma| &\le |F_{\mathcal{S}_\sigma}| + C\varepsilon(|u| + |\sigma|),\\
|g_\sigma| + [g_\sigma]_{\alpha,Q} &\le |g_{\mathcal{S}_\sigma}| + [g_{\mathcal{S}_\sigma}]_{\alpha,Q} + |u| + [u]_{\alpha,Q} + C\varepsilon(|\sigma| + [\sigma]_{\alpha,Q}),\\
|F_\tau| &\le |F_{\mathcal{S}_\tau}| + C\varepsilon(|u| + |\partial u| + |\sigma| + |\partial\sigma| + |\tau|),\\
|g_\tau| + [g_\tau]_{\alpha,Q} &\le |g_{\mathcal{S}_\tau}| + [g_{\mathcal{S}_\tau}]_{\alpha,Q} + C\varepsilon(|u| + |\sigma| + |\partial\sigma| + |\tau|)\\
&\qquad + C\varepsilon([u]_{\alpha,Q} + [\sigma]_{\alpha,Q} + [\partial\sigma]_{\alpha,Q} + [\tau]_{\alpha,Q}).
\end{aligned}$$

On a cube Q of side $\ell \le 6$, $[v]_{\alpha,Q} \le C\ell^{1-\alpha}|\partial v|_Q$ for every v, by joining points with monotone lattice paths in Q.

Step 2: one scale. Suppose first $r \le r_*$, with $r_* \le 3$ fixed in Step 3. For $v \in \{u,\sigma,\tau\}$ let $A_v(t) = \sup_{x\in\Lambda_t}(|v(x)| + |\partial v(x)|)$. Let $\rho \le t < t' \le r$, $s = (t'-t)/12$, and suppose $s \ge 3\eta$. For $x \in \Lambda_t$ the cube $\Lambda_{9s}(x)$, enlarged by 2η, lies in $\Lambda_{t'}$. Apply the first inequality of Lemma 36.2 on $\Lambda_{3s}(x)$ to each equation, and for σ also at scale $3s$ on $\Lambda_{9s}(x)$; divide by s and use Step 1. This gives, with all A_v evaluated at t':

$$\begin{aligned}
&|\sigma(x)| + |\partial\sigma(x)| + s^\alpha[\partial\sigma]_{\alpha,\Lambda_{3s}(x)} \le C_4(A_u + \varepsilon A_\sigma) + C\big(\mathfrak{s} + s^{-1-d/2}\mathfrak{m}\big),\\
&|\tau(x)| + |\partial\tau(x)| \le C_2\varepsilon(A_u + A_\sigma + A_\tau) + C\big(\mathfrak{s} + s^{-1-d/2}\mathfrak{m}\big),\\
&|u(x)| + |\partial u(x)| \le C_3(\varepsilon + \kappa s)A_u + C_3\varepsilon A_\sigma + C_3 s A_\tau + C\big(\mathfrak{s} + s^{-1-d/2}\mathfrak{m}\big).
\end{aligned}$$

In the bound for τ, the Hölder quotient of g_τ contains $\varepsilon[\partial\sigma]_{\alpha,\Lambda_{3s}(x)}$, which is bounded by the first line. In the bound for u, the terms $D\tau$ and $\kappa m u$ belong to F_u, which enters with the factor s. The constants $C_2, C_3, C_4 \ge 1$ depend only on d, α, $\dim\mathfrak{g}$. If $s < 3\eta$, then $|v(x)| + |\partial v(x)| \le 3\eta^{-1-d/2}\|v\|_{\Lambda_r} \le C(t'-t)^{-1-d/2}\mathfrak{m}$ for each v.

Step 3: weights and absorption. Let $s_* = \big(16C_3(1+\kappa)\big)^{-1}$, $r_* = \min(3, 12s_*)$, $w_\sigma = (8C_4)^{-1}$, $w_\tau = 4C_3 s_*$, and

$$\Psi(t) = A_u(t) + w_\sigma A_\sigma(t) + w_\tau A_\tau(t).$$

For $r \le r_*$ we have $s \le s_*$, so $C_3\kappa s \le \frac{1}{16}$ and $C_3 s \le \frac{1}{4}w_\tau$. Adding the bounds of Step 2 with the weights,

$$\begin{aligned}
\Psi(t) \le{}& \Big(C_3\varepsilon + \tfrac{3}{16} + w_\tau C_2\varepsilon\Big)A_u + (C_3 + C_4 w_\sigma + C_2 w_\tau)\varepsilon A_\sigma + \Big(\tfrac14 + C_2\varepsilon\Big)w_\tau A_\tau\\
&+ C\big(\mathfrak{s} + (t'-t)^{-1-d/2}\mathfrak{m}\big).
\end{aligned}$$

For $\varepsilon \le \varepsilon_L$ small this is at most $\frac12\Psi(t') + C(\mathfrak{s} + (t'-t)^{-1-d/2}\mathfrak{m})$. The same holds if $s < 3\eta$, by Step 2, since $w_\sigma, w_\tau \le 1$. Lemma 36.1 on $[\rho', r]$, $\rho' = \frac12(\rho + r)$, gives $\Psi(\rho') \le C(\mathfrak{s} + \mathfrak{m})$. Since the weights are fixed positive numbers, $A_u + A_\sigma + A_\tau$ is bounded on $\Lambda_{\rho'}$.

Step 4: Hölder quotients. With these sup bounds, Step 1 bounds $F_\bullet$, $g_\bullet$ and the Hölder quotients of g_u, g_σ on $\Lambda_{\rho'}$. Lemma 36.2 at the fixed scale $s_1 = (r-\rho)/40 \ge \eta$ then bounds $[\partial\sigma]_\alpha$, on cubes $\Lambda_{s_1}(x)$, $x \in \Lambda_{(\rho+\rho')/2}$, then $[g_\tau]_\alpha$, and then $[\partial\tau]_\alpha$, and $[\partial u]_\alpha$, on cubes $\Lambda_{s_1}(x)$, $x \in \Lambda_\rho$. Pairs of points of Λ_ρ at distance larger than s_1 are handled by the sup bounds, as in Proposition 36.3.

Step 5: general r. If $r > r_*$, apply Steps 2–4 on the cubes $\Lambda_{r_1}(x)$, $x \in \Lambda_\rho$, $r_1 = \min(r_*, r-\rho)$, which lie in Λ_r. Combine the resulting bounds on $\Lambda_{r_1/2}(x)$ as in Step 4. □

The small cubes in Step 3 are needed only because of the mass term, which is not small: its contribution $\kappa s|u|$ is made small by the radius. The coupling of σ to u is not small either, but it is one-directional: σ feeds back into u and τ only with a factor ε, and this is what the weights express.

36.3 A consequence of gauge invariance

The bound on $D^T D\mathbf{H}^X$ needs pointwise control of $D^T\mathfrak{F}\lambda$. Its coefficient $\partial^* c$ involves second derivatives of the background, which are not bounded uniformly in η. Gauge invariance of the action replaces it by the current.

Lemma 36.6 (Divergence of the curvature term)**.** *Let U be real and $(1,\varepsilon)$-regular with $\varepsilon \le c_5$. Let $W_0 = \nabla^2\mathcal{N}(0)$, where $\mathcal{N}$ is the remainder of Proposition 23.3 with $U_0 = U$. Then $(W_0 h)(b)$ depends only on h on bonds within distance 2η of b, and $|W_0 h(b)| \le C\varepsilon \max_{b'} |h(b')|$ over these bonds. For every scalar field λ,*

$$D^T\mathfrak{F}\lambda = -W_0 D\lambda + \mathcal{J}_U(\lambda), \qquad \mathcal{J}_U(\lambda)(b) = \tfrac{\mathrm{i}}{2}\big[J_U(b),\, \lambda(b_-) + R(U(b))\lambda(b_+)\big].$$

In particular $|D^T\mathfrak{F}\lambda(b)| \le C\varepsilon \max_{b'} |D\lambda(b')| + |J_U(b)|\big(|\lambda(b_-)| + |\lambda(b_+)|\big)$.

Proof. The operator W_0. By Proposition 23.3, $\Phi(A) = S(e^{\mathrm{i}\eta A}U)$ satisfies $\nabla\Phi(0) = J_U$ and $\nabla^2\Phi(0) = D^T D + W_0$. Differentiate the formula for $\mathcal{N}$ in the proof of that proposition twice at $A = 0$. There $\rho(0) = 0$, $d\rho(0) = \rho^{(1)}$ and $d^2\rho(0) = d^2\rho^{(2)}(0)$. The cubic part of $\mathcal{R}$ does not contribute, and the quadratic part of $\mathcal{R}$ has coefficients $O(\varepsilon^2\eta^4)$. Hence W_0 is the sum of:

- the operator of the form $h \mapsto \langle d^2\rho^{(2)}(0)[h,\cdot], F^s\rangle$, bounded by $C\varepsilon$ since $|F^s| \le C\varepsilon$;
- $D^T\rho^{(1)} + \rho^{(1)T}D$ and $\rho^{(1)T}\rho^{(1)}$, where $\rho^{(1)}$ is local with coefficients bounded by $C\varepsilon\eta$ and D, D^T cost at most $C\eta^{-1}$;
- the Hessian of the quadratic part of $\mathcal{R}$, bounded by $C\varepsilon^2\eta^4 \cdot \eta^{-2}$.

All are local, with the stated range.

The identity. Let λ be real and $g_t = e^{\mathrm{i}t\lambda}$. For small t and A define A_t by $e^{\mathrm{i}\eta A_t}U = (e^{\mathrm{i}\eta A}U)^{g_t}$, i.e. bondwise

$$e^{\mathrm{i}\eta A_t(b)} = e^{tP}e^{\mathrm{i}\eta A(b)}e^{-tQ}, \qquad P = \mathrm{i}\lambda(b_-), \quad Q = \mathrm{i}R(U(b))\lambda(b_+).$$

By gauge invariance of the Wilson action, $\Phi(A_t) = \Phi(A)$. Let $X_\lambda(A) = \frac{d}{dt}A_t|_{t=0}$. The Baker–Campbell–Hausdorff formula gives, to first order in t and in $Y = \mathrm{i}\eta A(b)$,

$$\log\big(e^{tP}e^{Y}e^{-tQ}\big) = Y + t(P-Q) + \tfrac{t}{2}[P+Q, Y] + O(t^2) + O(t|Y|^2).$$

Hence $X_\lambda(0) = -D\lambda$ and $dX_\lambda(0)h(b) = \frac{\mathrm{i}}{2}[\lambda(b_-) + R(U(b))\lambda(b_+), h(b)]$. Differentiating $\langle\nabla\Phi(A), X_\lambda(A)\rangle = 0$ at $A = 0$ in the direction h gives

$$-\langle (D^T D + W_0)h, D\lambda\rangle + \langle J_U, dX_\lambda(0)h\rangle = 0 \qquad \text{for all real } h.$$

Since $\mathrm{tr}(J[\mu,h]) = \mathrm{tr}([J,\mu]h)$, this says $(D^TD + W_0)D\lambda = \mathcal{J}_U(\lambda)$. Finally $D^TDD\lambda = D^T\mathfrak{F}\lambda$. Both sides are complex linear in λ, so the identity holds for complex λ. The bound follows from $|[X,Y]| \le 2|X||Y|$ and the bound on W_0. □

In the continuum this is the identity $D^*[F,\lambda] = [D^*F,\lambda] + \dots$: the divergence of the curvature is the current.

36.4 Proof of Theorem 35.2

The constants come from Proposition 35.1, Lemma 36.4, Propositions 36.3, 36.5 and 22.7(d),(e), and, for (d), Lemma 36.6; none of them depends on $M \ge M_0$. Fix a domain X and $y \in X_{(4)}$, and let $z = c_y$. Since the ∞-distance from y to $\mathbb{T}_\eta \setminus X$ exceeds 4, the cube $\Lambda_3(z)$ lies in X at ∞-distance more than $\frac12 \ge 8\eta$ from $\mathbb{T}_\eta \setminus X$. So the equations of Lemma 36.4 hold on $\Lambda_3(z)$. We use concentric cubes $\Lambda_3 = \Lambda_{r_0} \supset \Lambda_{r_1} \supset \dots \supset \Lambda_{r_{12}} = \Lambda_1 \supset B(y)$, with $r_j = 3 - j/6$. Each application of Proposition 36.3 or 36.5 passes from one cube to the next, and we do not record the indices. We write

$$\mathfrak{E}(f) = \sum_{y'} e^{-\gamma|y-y'|}|f|_{B(y')},$$

and similarly for 2-forms and coarse fields. Here γ may decrease from line to line, but remains independent of X, k, n, η. Two facts are used repeatedly:

- $|h|_{\Lambda_3} \le C\sum_{|y'-y|_\infty\le 4}|h|_{B(y')}$ and $\|h\|_{\Lambda_3} \le C\sum_{|y'-y|_\infty\le 4}\|h\|_{B(y')}$. Every point of $\Lambda_3 \subset X$ lies in a block $B(y') \subset X$, and these blocks have $|y'-y|_\infty \le 4$;
- for Bałaban's averaging, $|\mathbf{Q}A(c)| \le C\|A\|_{B(c_-)\cup B(c_+)}$, $|Q'\lambda(y')| \le C\|\lambda\|_{B(y')}$, and $\mathbf{Q}^T$, Q'^T are bounded from sup norms to sup norms (Proposition 22.7(d),(e)).

The basic estimate. Let $u \in \ell^2_1(X)$ satisfy $\mathfrak{A}^{X,\kappa}u = \mathcal{S}$, with $\mathcal{S}$ in local form on Λ_3, and put $\mathfrak{L}(u) = \sum_{y'} e^{-\gamma|y-y'|}(\|u\|_{B(y')} + \|Du\|_{B(y')})$. By Lemma 36.4, the L^2 norms of σ,τ,π_i on Λ_3 and the values $|\omega_i|$ at the unit points near y are bounded by $C\mathfrak{L}(u)$. By (36.4) and Proposition 36.3, $|\pi_i| + |D\pi_i| \le C\mathfrak{L}(u)$ on the next cube, and $|a\mathbf{Q}^T\mathbf{Q}u| + |aQ'^TQ'\tau| \le C\mathfrak{L}(u)$ there. So (u,σ,τ) satisfies (36.5) with $m = m_X$ and

$$\mathcal{S}_u = \mathcal{S} + D\pi_1 - D\pi_2 - a\mathbf{Q}^T\mathbf{Q}u, \qquad \mathcal{S}_\sigma = -\pi_1, \qquad \mathcal{S}_\tau = -aQ'^TQ'\tau.$$

Proposition 36.5 gives

$$\sum_{v\in\{u,\sigma,\tau\}} \Big(|v| + |\partial v| + [\partial v]_{\alpha,}\Big)_{\Lambda_{r_j}} \le C(\mathfrak{n}_{r_{j-2}}(\mathcal{S}) + \mathfrak{L}(u)). \tag{36.6}$$

In the local gauge, $\nabla u = \partial u + Tu$ with $|T|$ and $[T]_\alpha$ bounded by $C\varepsilon$. So (36.6) bounds $|u|_{1,\alpha;y}$.

(c) Let $u = \mathbf{H}^X B$. Then $\mathfrak{A}^{X,\kappa}u = \mathbf{Q}_X^T\mathbf{\Gamma}^X B$, a source with $g = 0$ and $|F|_{\Lambda_3} \le C\mathfrak{E}(B)$ by Proposition 35.1(a). Also $\mathfrak{L}(u) \le C\mathfrak{E}(B)$ by the decay of $\mathbf{H}^X$ and $D\mathbf{H}^X$ (Proposition 35.1(b)). Apply (36.6).

(a) Let $u = \mathbf{C}^X f$, and put $h_0 = G'_X D_X^T f$ and $\sigma_2 = G'_X R_X h_0$. Then $\sigma_2 \in \mathcal{N}_X$ and $D_X^T D\sigma_2 = R_X h_0$ (Theorem 18.2(d)), so $\mathfrak{A}^X D\sigma_2 = DR_X h_0$ by the domain version of Lemma 15.3(a) (Lemma 19.2). With $\mathfrak{A}^{X,\kappa}\mathbf{G}^X = 1$ and $\mathfrak{A}^{X,\kappa}\mathbf{H}^X = \mathbf{Q}_X^T\mathbf{\Gamma}^X$ this gives

$$\mathfrak{A}^{X,\kappa}\mathbf{C}^X f = f - DR_X h_0 - \kappa m_X D\sigma_2 - \mathfrak{A}_X^{\mathrm{ext}} D\sigma_2 - \mathbf{Q}_X^T\mathbf{\Gamma}^X\mathbf{Q}_X\mathbf{G}^X f.$$

The term $\mathfrak{A}_X^{\mathrm{ext}} D\sigma_2$ lives on the bonds crossing ∂X, so it vanishes on Λ_3 and does not enter the estimate. We put the remaining terms in local form.

- $D^TDh_0 = D^Tf - aQ'^TQ'h_0$ in the interior, with $D^Tf = \partial^*f + T^Tf$. The second bound of Proposition 36.3 gives $|h_0| + [h_0]_\alpha, \le C\mathfrak{E}(f)$, using $\|h_0\|_{B(y')} \le C\sum e^{-\gamma|y'-y''|}|f|_{B(y'')}$ by the decay of $G'_X D_X^T$.

- $Dh_0 = \partial h_0 + Th_0$ and $\partial_\mu h_0 = -\partial_\mu^*(S_\mu h_0)$. So Dh_0 is a source in local form, with g bounded by $|h_0|$ and $[h_0]_{\alpha,}$.
- $R_X h_0 = h_0 - \pi_3$ with $\pi_3 = G'_X Q'^T (M'_X)^{-1} Q' G'_X h_0$. As for π_i, $|\pi_3| + |D\pi_3| \le C\mathfrak{E}(f)$.
- $D^T D\sigma_2 = h_0 - \pi_3$, since $Q'\sigma_2 = 0$. The first bound of Proposition 36.3 gives $|D\sigma_2| \le C\mathfrak{E}(f)$, using $\|\sigma_2\|_{B(y')} \le C \sum e^{-\gamma|y'-y''|} |f|_{B(y'')}$ by the decay of $G'_X R_X G'_X D_X^T$.
- $|\mathbf{Q}_X^T \mathbf{\Gamma}^X \mathbf{Q}_X \mathbf{G}^X f| \le C\mathfrak{E}(f)$, by the decay of $\mathbf{\Gamma}^X$ and $\mathbf{G}^X$.

Also $\mathfrak{L}(\mathbf{C}^X f) \le C\mathfrak{E}(f)$ by the decay of $\mathbf{C}^X$ and $D\mathbf{C}^X$. Apply (36.6).

(b) Let $u = \mathbf{C}^X D_X^T g$. In the same way,

$$\mathfrak{A}^{X,\kappa} u = D_X^T g - DR_X h_3 - \kappa m_X D\sigma_3 - \mathfrak{A}_X^{\text{ext}} D\sigma_3 - \mathbf{Q}_X^T \mathbf{\Gamma}^X \mathbf{Q}_X \mathbf{G}^X D_X^T g,$$

where again $\mathfrak{A}_X^{\text{ext}} D\sigma_3$ vanishes on Λ_3, with $h_3 = G'_X D_X^T D_X^T g = G'_X \mathbf{1}_X \mathfrak{F}^T g$ and $\sigma_3 = G'_X R_X h_3$. In the interior $D_X^T g = \partial^* g + T^T g$, a source in local form whose norm involves $[g]_{\alpha,\Lambda_3}$. Two points of Λ_3 at ∞-distance at most $\frac{1}{2}$ lie in a common cube $\Lambda_1(c_{y'})$, so $[g]_{\alpha,\Lambda_3} \le C \sum_{|y'-y|_\infty \le 4} (|g|_{B(y')} + [\tilde{g}_{y'}]_{\alpha,\Lambda_1(c_{y'})})$. Here the gauge of z and that of y' differ on $\Lambda_1(c_{y'})$ by a gauge transformation whose differences are bounded by $C\varepsilon$, which changes Hölder quotients by at most $C\varepsilon|g|$. Next, $D^T D h_3 = \mathfrak{F}^T g - aQ'^T Q' h_3$ is bounded, so $|h_3| + |Dh_3| \le C\mathfrak{E}(g)$ by Proposition 36.3, with the L^2 input from the decay of $G'_X \mathbf{1}_X \mathfrak{F}^T$ (and of $G'_X R_X$ for σ_3), and $DR_X h_3$ is a bounded source. The remaining terms are as in (a), with the decay of $\mathbf{G}^X D_X^T$, $\mathbf{C}^X D_X^T$ and $D\mathbf{C}^X D_X^T$.

(d) Let $u = \mathbf{H}^X Y$, and let $\sigma, \tau, \pi_i, \omega_i$ be as in Lemma 36.4. By (c) and (36.6), u, σ, τ and their first derivatives are bounded by $C\mathfrak{E}(Y)$ on Λ_{r_2}. Put $\theta = D_X^T u$, so that $DD^T u = D\theta$ in the interior. Apply D^T to (36.1). By $D^T D^T = \mathfrak{F}^T$, $D^T(D^T \mathfrak{F}\sigma) = \mathfrak{F}^T \mathfrak{F}\sigma$, and by (36.3),

$$\mathfrak{F}^T Du - \mathfrak{F}^T \mathfrak{F}\sigma + D^T(W(u - D\sigma)) - D^T D\tau = aQ'^T Q'\tau.$$

The curvature terms and the potential terms cancel exactly. With $E = 0$, $D^T \mathbf{Q}^T = (\mathbf{Q}D)^T = Q'^T D_V^T$. Together with (36.4) this gives

$$D^T D\theta = Q'^T D_V^T (\mathbf{\Gamma}^X Y - a\mathbf{Q}u) + Q'^T(\omega_1 - \omega_2) - aQ'^T Q'(\pi_1 - \pi_2 + \tau) - \kappa D^T(m_X u).$$

Here $D^T(m_X u) - m_X \theta$ consists of terms $(\partial^* m_X) \cdot$ (shifted u) and (differences of m_X) $\cdot$ T^T(shifted u), so it is bounded by $C(|u| + |\nabla u|)$, since $|\partial m_X| \le C$. So the right-hand side is bounded by $C\mathfrak{E}(Y)$ on Λ_{r_2}. Proposition 36.3, with $\|\theta\|_{\Lambda_{r_2}} \le C|\nabla u|_{\Lambda_{r_2}}$, gives $|D\theta| \le C\mathfrak{E}(Y)$ on Λ_{r_3}. Now solve (36.1) for $D^T Du$:

$$D^T Du = \mathbf{Q}^T \mathbf{\Gamma}^X Y + D\pi_1 - D\pi_2 - a\mathbf{Q}^T \mathbf{Q}u - D\theta + D^T \mathfrak{F}\sigma - W(u - D\sigma) + D\tau - \kappa m_X u.$$

Every term is bounded by $C\mathfrak{E}(Y)$ on Λ_{r_3}. For $D^T \mathfrak{F}\sigma$ this is Lemma 36.6, with $|J_U| \le \varepsilon$. This proves (d).

(e) Let $X' \subset X$, $y \in X'_{(4)}$ and $z = c_y$. Then $\Lambda_3(z)$ is interior to both domains. Consider one of the operators, say $u = \mathbf{C}^X f$ and $u' = \mathbf{C}^{X'} f$ for f supported in X', and let δ denote differences of corresponding quantities on X and X': $\delta u = u - u'$, $\delta\sigma$, $\delta\tau$, $\delta\pi_i$, $\delta\omega_i$, δh_0, $\delta\sigma_2$, and so on. Write K_Σ-sums for bounds of the form $C \sum_{y'} K_\Sigma(y, y') |f|_{B(y')}$.

L^2 differences. Every operator in the construction satisfies the comparison of domains in block L^2 norms, with kernels of seam form, also when composed with D on the left. This covers G', $G'D^T$, $DG'D^T$, S, R, $P_\mathcal{V}$, $(M')^{-1}$ (Theorem 18.2(e)), $\mathbf{G}$ (Theorem 19.10(a)), and $\mathbf{\Gamma}$, $\mathbf{H}$, $\mathbf{C}$ (Proposition 35.1(d)). The same holds with D on the left and D_X^T on the right. The local operators $\mathbf{1}_X$, D_X^T, $\mathfrak{B}_X$ differ from their versions on X' only within distance 1 of $X \setminus X'$. These extensions are recorded after Proposition 35.1. For blocks $B(y') \not\subset X'$ we have $\delta u = u$ there, and y' lies in Σ, so the decay of the operators on X alone gives seam form. By telescoping, $\mathfrak{L}(\delta u)$, the L^2 norms of all δ-quantities near y, and $|\delta\omega_i|$ near y are K_Σ-sums.

The difference system. On $\Lambda_3(z)$ the equations of Lemma 36.4 for X and for X' have the same local operators, except for the mass terms. Hence $(\delta u, \delta\sigma, \delta\tau)$ satisfies (36.5) with $m = m_X$. Its sources are the differences of the sources, together with $-\kappa(m_X - m_{X'})u'$. The scalar differences satisfy local equations: for instance $D^T D\delta h_0 = -aQ'^T Q'\delta h_0$, because f is the same, and $D^T D\delta\pi_i = Q'^T\delta\omega_i - aQ'^T Q'\delta\pi_i$. So Proposition 36.3 bounds them pointwise, with derivatives, by K_Σ-sums. The only new terms are the mass differences $\kappa(m_X - m_{X'})u'$ and, in (a), $\kappa(m_X - m_{X'})D\sigma_2'$. We have $m_X - m_{X'} = -\sum_{j\in J(X)\setminus J(X')} \chi_j^2$. For such j, $\Lambda_j \subset X$ but $\Lambda_j \not\subset X'$, so Λ_j meets $X \setminus X'$. Its ∞-diameter is at most $6M+2$, so $m_X - m_{X'}$ vanishes outside the $8M$-neighbourhood of $X \setminus X'$. At a point $x \in \Lambda_3(z)$ where it does not vanish, there is $w \in \Sigma$ with $|x - w| \le \sqrt{d}$, and $|u'(x)| + |D\sigma_2'(x)| \le C\mathfrak{E}(f)$ by the bounds already proved for X', applied on $\Lambda_{r_6}(z)$; the difference argument then runs on the cubes $\Lambda_{r_6} \supset \cdots \supset \Lambda_{r_{12}}$. Since $|y - x| \le 4\sqrt{d}$,

$$e^{-\gamma|y-y'|} \le e^{2\gamma(|y-x|+|x-w|)} e^{-\gamma(|y-w|+|w-y'|)},$$

so these terms are K_Σ-sums as well. Proposition 36.5, applied to the differences on the inner cubes, proves (e) for (a). The other operators are treated in the same way; for (d), apply D^T to the difference system as in the proof of (d), and use Lemma 36.6 for $D^T\mathfrak{F}\delta\sigma$. □

37 Localization of the Hessian and of the fluctuation covariance

Status. The localization of this section and of Section 39 is of radius type (Section 39, Summary). The polymer expansion of Part X needs volume type, which Part IX provides. The main line uses from this section Lemma 37.2, Definition 37.3, Proposition 37.4, Lemma 37.5 and Proposition 37.6. Theorem 37.7 and Corollary 37.8 (Section 37.5) are superseded by the weakened representations of Section 41 and are not used later.

37.1 The telescoping principle

For a finite set S of unit points and $m \ge 0$ let $\Lambda_m(\mathsf{S})$ be the union of the blocks $B(y)$ with $\operatorname{dist}(y, \mathsf{S}) \le m$, and let m_{S} be the least m with $\Lambda_m(\mathsf{S}) = \mathbb{T}_\eta$.

Lemma 37.1 (Telescoping)**.** *Let F^X be numbers attached to the domains $X = \Lambda_m(\mathsf{S})$, $0 \le m \le m_{\mathsf{S}}$, with $F^{\mathbb{T}_\eta} = F$ and $|F^{\Lambda_m(\mathsf{S})} - F^{\Lambda_{m-1}(\mathsf{S})}| \le Ke^{-\gamma m}$. Then*

$$F = \sum_{m=0}^{m_{\mathsf{S}}} F_m, \qquad F_0 = F^{\Lambda_0(\mathsf{S})}, \quad F_m = F^{\Lambda_m(\mathsf{S})} - F^{\Lambda_{m-1}(\mathsf{S})}, \qquad |F_m| \le Ke^{-\gamma m} \ (m \ge 1).$$

If each F^X is a function of the background that depends only on its restriction to a set $\mathcal{R}(X)$, is analytic, and is invariant (or covariant) under real gauge transformations, and if $\mathcal{R}(\Lambda_{m-1}(\mathsf{S})) \subset \mathcal{R}(\Lambda_m(\mathsf{S}))$ for $1 \le m \le m_{\mathsf{S}}$, then F_m has the same properties with $\mathcal{R}(\Lambda_m(\mathsf{S}))$.

The proof is the identity of the telescoping sum. By the inclusion, $F^{\Lambda_m(\mathsf{S})}$ and $F^{\Lambda_{m-1}(\mathsf{S})}$ both depend only on the background on $\mathcal{R}(\Lambda_m(\mathsf{S}))$, and so does F_m. Theorem 37.7, Proposition 38.1 and Theorem 39.4 are applications of this lemma; Corollary 39.5 uses the same telescoping identity directly. In these three applications $\mathcal{R}(X)$ is $\widehat{X^+}$ (Theorem 37.7, Proposition 38.1) or $\hat{X}$ together with the bonds of $\mathfrak{B}(X)$ for the coarse field (Theorem 39.4); these sets increase with X, so the inclusion holds. The estimates needed are comparison bounds for approximants on nested domains, whose differences have seam form.

37.2 The coarse Hessian as an operator of Part II

Let $\mathbf{H}_0 = \mathbf{H}[0]$ be the minimizer map without potential, and for a regular background U put

$$\omega_0 = \mathbf{H}_0^T J_U, \qquad \langle A, W_2 A\rangle = d^2\mathcal{N}(0)[A,A] - \langle \omega_0, d^2 C_k(0)[A,A]\rangle,$$

with $\mathcal{N}$ the nonlinear part of the action at U (Proposition 23.3) and C_k that of the gauge-adapted averaging (Proposition 22.6).

Lemma 37.2 (The Hessian is $\mathbf{\Gamma}_\pi[W_2] - a$)**.** *$W_2$ is a real symmetric bounded operator with $\|W_2\| \le C\varepsilon$ and $\|\mathbf{1}_y W_2 \mathbf{1}_{y'}\| \le C\varepsilon e^{-\gamma|y-y'|}$. If U is critical, i.e. $U = U'$ is the background of the k-step problem for $W = U^{(k)}$, then ω_0 is its multiplier and*

$$\Delta^0 = \mathbf{\Gamma}_\pi[W_2] - a,$$

where Δ^0 is the Hessian of Proposition 33.2 and $\mathbf{\Gamma}_\pi[W_2]$ is the operator of Theorem 19.9 with the potential W_2.

Proof. At $A = 0$ the second derivative $d^2\mathcal{N}(0)$ is a sum of plaquette-local bilinear forms. By the first part of Lemma 36.6, it is a local operator with $|d^2\mathcal{N}(0)[h,h']| \le C\varepsilon \|h\|\,\|h'\|$; its proof is the computation of the proof of Proposition 23.3(c) at $A = 0$, with the lattice trick (23.2). By Proposition 22.6(c), $|d^2 C_k(0)[h,h](c)| \le C\|h\|^2_{S_c}$. By Cauchy–Schwarz on blocks, as in the proof of Proposition 37.4(b), $|\omega_0(c)| \le \sum_{y'} \|\mathbf{1}_{y'}\mathbf{H}_0\mathbf{1}_c\|\,\|J_U\|_{B(y')} \le C\varepsilon$, by the block decay of $\mathbf{H}_0$ (Theorem 19.9) and $|J_U| \le \varepsilon$. Hence the bounds. The coefficients $\omega_0(c)$ depend on U through the decaying kernel of $\mathbf{H}_0^T$.

If U is critical, the Euler–Lagrange equation at $B = 0$ gives $J = \mathbf{Q}^T\omega$. So $\mathbf{H}_0^T J = (\mathbf{Q}\mathbf{H}_0)^T\omega = \omega$. By Proposition 33.2(a), $\Delta^0(B,B) = \min\{\|DA\|^2 + \langle A, W_2 A\rangle : \mathbf{Q}A = B,\ RD^TA = 0\}$. By Theorem 16.1(b) with the potential W_2, the minimizer is $\mathbf{H}[W_2]B$. On the constraint set $\langle A, \mathfrak{A}_\pi A\rangle = \|DA\|^2 + \langle A, W_2 A\rangle + a\|B\|^2$. The minimum of $\langle A, \mathfrak{A}_\pi A\rangle$ over $\{\mathbf{Q}A = B\}$ is $\langle B, \mathbf{\Gamma}_\pi B\rangle$ and is attained at $\mathbf{H}[W_2]B$. Only the L^2 properties of W_2 enter: the coercivity of Theorem 19.6 uses W only through $\|W\|$, and the variational characterization is algebraic. □

37.3 Dirichlet Hessians

Definition 37.3 (Dirichlet Hessian)**.** For a domain X put

$$\omega_0^X = (\mathbf{H}_0^X)^T \mathbf{1}_X J_U \quad \text{on } \mathfrak{B}(X),$$
$$\langle A, W_2^X A\rangle = d^2\mathcal{N}(0)[\mathbf{1}_X A, \mathbf{1}_X A] - \langle \omega_0^X, d^2 C_k(0)[A,A]\rangle_{\mathfrak{B}(X)},$$

for $A \in \ell^2_1(X)$, and

$$\Delta^{0,X} = \mathbf{\Gamma}^X[W_2^X] - a \qquad \text{on } \ell^2(\mathfrak{B}(X)).$$

Proposition 37.4 (Dirichlet Hessians)**.** *(a) For $X = \mathbb{T}_\eta$: $\omega_0^{\mathbb{T}_\eta} = \omega_0$ and $\Delta^{0,\mathbb{T}_\eta} = \mathbf{\Gamma}_\pi[W_2] - a$. This is Δ^0 if U is critical.*

(b) $|\omega_0^X| \le C\varepsilon$, $\|W_2^X\| \le C\varepsilon$, and $\Delta^{0,X}$ has decay (C,γ), uniformly in X.

(c) ω_0^X, W_2^X and $\Delta^{0,X}$ depend only on U on $\hat{X}$ and are covariant under real gauge transformations. They are analytic under complex perturbations of U as in Proposition 19.11. The potential W_2^X is real symmetric at real U and has range ≤ 3 in A, but it depends on U on all of $\hat{X}$. It satisfies the hypotheses of Proposition 19.11 for existence, Re*-coercivity and decay; its comparison of domains is (d). The analyticity and the changes under complex perturbations come from:*

- *$d^2\mathcal{N}(0)$ and the kernel of $d^2C_k(0)$ are local and analytic in the background (Propositions 23.3 and 22.6, and Proposition 46.4 at complex references);*
- *$\langle b, \omega_0^X\rangle = \langle F_U^s, D\mathbf{H}_0^X b\rangle$, with $D\mathbf{H}_0^X$ analytic by Proposition 19.11 at $W = 0$;*
- *the changes of size $C\delta$ follow from Cauchy's estimate on a ball of twice the radius.*

(d) For $X' \subset X$ and $c, c' \in \mathfrak{B}(X')$,

$$|\omega_0^X(c) - \omega_0^{X'}(c)| \le C\varepsilon \sum_{w\in\Sigma} e^{-\gamma|c-w|},$$
$$|\Delta^{0,X}(c,c') - \Delta^{0,X'}(c,c')| \le C \sum_{w\in\Sigma} e^{-\gamma'(|c-w|+|w-c'|)},$$

with Σ as in Proposition 35.1(d) and some $\gamma' > 0$.

Proof. (a)–(c) follow from Proposition 35.1 and Lemma 37.2. The bound $|\omega_0^X| \le C\varepsilon$ needs no pointwise estimate. By Cauchy–Schwarz on blocks, $|\omega_0^X(c)| \le \sum_{y'} \|\mathbf{1}_{y'}\mathbf{H}_0^X\mathbf{1}_c\|\,\|J_U\|_{B(y')} \le C\varepsilon\sum_{y'} e^{-\gamma|c-y'|}$, using only the block kernel bounds of $\mathbf{H}_0^X$ and $|J_U| \le \varepsilon$. (d) The bound for ω_0 follows from Proposition 35.1(d) for $\mathbf{H}_0$, since J_U is local and $|J_U| \le \varepsilon$. Then $W_2^X - W_2^{X'}$ on X' is bounded near w by $C\varepsilon\sum_{s\in\Sigma} e^{-\gamma|w-s|}$. The $d^2\mathcal{N}$ parts differ only on bonds within distance 1 of $X \setminus X'$. Insert this into Proposition 35.1(d) for $\mathbf{\Gamma}$. Since $\sum_w e^{-\gamma(|c-w|+|w-c'|)}e^{-\gamma|w-s|} \le Ce^{-\gamma'(|c-s|+|s-c'|)}$, the result has seam form.

Complex perturbations in (c). Let U_δ be as in Proposition 19.11, of size $\delta < 2\delta_*$, with δ_* fixed at the end of this paragraph. In a local real gauge of U, $U_\delta = e^{\mathrm{i}\eta A_0}$ with A_0 satisfying (B) with parameter $C(\varepsilon+\delta)$ (Section 46). This uses the bound on $\partial\delta A$: with a sup bound alone, $H_p - 1$ would only be $O(\eta\delta)$. Hence $|H_p(U_\delta) - 1| \le C\eta^2(\varepsilon+\delta)$, $\|R(U_\delta(b))^{\pm1}\| \le 2$ and $|F^s_{U_\delta}| \le C(\varepsilon+\delta)$ (step (v) of the proof of Proposition 46.7). The proof of the bound on W_0 in Lemma 36.6 uses only these three bounds, the transports being bounded by 2 instead of 1. So $d^2\mathcal{N}_{U_\delta}(0)$ is a local operator of L^2 norm at most $C(\varepsilon+\delta)$. Read ω_0^X through the identity $\langle b, \omega_0^X\rangle = \langle F^s, D\mathbf{H}_0^X b\rangle$ of (c), and use Proposition 46.4(iii),(vi) for $d^2C_k(0)$. Then $W_2^X(U_\delta)$ is holomorphic in δA, of range ≤ 3, with $\|W_2^X(U_\delta)\| \le C(\varepsilon+\delta)$, for every δA of size $< 2\delta_*$. The Schwarz lemma on the complex lines $t \mapsto U_{t\,\delta A}$, $|t| < 2\delta_*/\delta$, gives $\|W_2^X(U_\delta) - W_2^X(U)\| \le C(\varepsilon + 2\delta_*)\,\delta/\delta_*$. For $\delta \le \delta_*$ this is at most $C(\varepsilon+2\delta_*)$. So W_2^X satisfies the range-3 clause of Proposition 19.11 at these perturbations once $C'(\varepsilon+2\delta_*) \le c$, with c the stability threshold of Lemma 13.2. Here $\delta_* > 0$ is fixed once, depending on d, L, G and α, for all ε with $\varepsilon \le \frac12\min(\varepsilon_0, c_3')$, $C\varepsilon \le \frac12 c_5$ and $C'\varepsilon \le \frac12 c$; it does not depend on $M \ge M_0$, since the radii and constants involved are those of Parts II, III and VIII, which do not depend on $M \ge M_0$, and those of Propositions 46.4 and 23.3. It satisfies $2C\delta_* \le \frac12\min(\varepsilon_0, c_3')$, $2C\delta_* \le \frac12 c_5$, $2C'\delta_* \le \frac12 c$, and the radius conditions of Propositions 19.11 (with $W = 0$), 46.4 and 35.1(e) at size $2\delta_*$. With $C'\varepsilon \le \frac12 c$, a threshold of this section, the clause holds for all $\delta \le \delta_*$. Moreover $\|W_2^X(U_\delta) - W_2^X(U)\| \le C_W\delta$ with $C_W = C(1+2\delta_*)/\delta_*$, a fixed constant. The constant c_L' of Lemma 37.5 is taken with $C_{\mathrm{cmp}}c_L' \le \delta_*$, with $C_{\mathrm{cmp}} = 4$ of Section 46. □

37.4 The fluctuation operator on domains

We return to the step of Part VII. Its operator Δ_k acts on fields on the free bonds $\mathfrak{F}$ of $\mathbb{T}$ (Definition 32.1), via the elimination maps $\hat{Z}^1$, $\hat{Z}^2$ of Lemma 32.4. These depend only on $W = U^{(k)}$ within distance $2L$, and $\hat{Z}^1\delta_b$ is supported within distance $2L$ of b. For a domain X and free bonds b, b' for which $\hat{Z}^1\delta_b$, $\hat{Z}^1\delta_{b'}$ are supported in $\mathfrak{B}(X)$, define the *Dirichlet entries*

$$\Delta_k^{(X)}(b,b') = \Delta^{0,X}(\hat{Z}^1\delta_b, \hat{Z}^1\delta_{b'}) + (\text{polarization of } Z \mapsto 2\langle\omega_0^X, \hat{Z}^2(Z)\rangle)(b,b'),$$

and $\Delta_k^{(X)}(b,b') = 0$ otherwise. For $X = \mathbb{T}_\eta$ these are the entries of Δ_k (Lemma 34.1).

Fix a margin $R_0 \geq 2L+2$. For a domain Y let $Y^+ = Y^{(R_0)}$, let $\mathfrak{F}(Y)$ be the set of free bonds starting in Y, and define the *local fluctuation operator*

$$\Delta_Y = \mathbf{1}_{\mathfrak{F}(Y)} \Delta_k^{(Y^+)} \mathbf{1}_{\mathfrak{F}(Y)} \qquad \text{on } \ell^2(\mathfrak{F}(Y)).$$

For a regular background U that is not necessarily critical, we *define* $\Delta_k = \Delta_k^{(\mathbb{T}_\eta)}$ and $C_k = \Delta_k^{-1}$, with the elimination at the base $W = U^{(k)}$. This agrees with Part VII when U is critical (Proposition 37.4(a)).

Lemma 37.5 (Coercivity at general backgrounds)**.** *Under the smallness hypotheses of Corollary 34.5, but for any real regular U with $|J_U| \leq \varepsilon$ (critical or not), $\frac{1}{4}c_L \leq \Delta_k \leq C_L$ on $\ell^2(\mathfrak{F})$. For complex perturbations of size at most c_L'/C_{AB} in the local gauges, if moreover $\varepsilon \leq \varepsilon_{\mathrm{cc}}$ and ε is below the thresholds of the proof of Proposition 37.4(c), with c_L' decreased so that $C_{\mathrm{cmp}} c_L' \leq \delta_*$ (proof of Proposition 37.4(c)) and so that the changes in the proof are at most $\frac{1}{8}c_L$, $\operatorname{Re}\Delta_k \geq \frac{1}{8}c_L$.*

Proof. Theorem 34.4 and Lemma 34.3, at the scale M_L' and for $\varepsilon \leq \varepsilon_{\mathrm{cmp}}(M_L')$, which is part of the smallness hypotheses of Corollary 34.5, do not use criticality: they give $\mathbf{\Gamma}_\pi[0] - a \geq c_L$ on $\mathcal{Z}$. By Lemma 37.2, $\|W_2\| \leq C\varepsilon$. Lemma 13.3, applied to $\mathbf{QG}[W]\mathbf{Q}^T$, gives $\|\mathbf{\Gamma}_\pi[W_2] - \mathbf{\Gamma}_\pi[0]\| \leq C\varepsilon$. Hence $\Delta^{0,\mathbb{T}_\eta} \geq (c_L - C\varepsilon)$ on $\mathcal{Z}$. The term $2\langle \omega_0, \hat{Z}^2\rangle$ is bounded by $CL^{3d+1}\varepsilon\|Z\|^2$, since $|\omega_0| \leq C\varepsilon$. At a complex perturbation of size at most c_L'/C_{AB} the base is $e^{\mathrm{i}\beta}W$ with $|\beta|_\infty \leq c_L'$ (Proposition 46.4(i)), so Corollary 34.5(c) applies there, and all operators change by $O(c_L')$ in norm (Proposition 19.11, and for W_2 the proof of Proposition 37.4(c)); the constant contains C_W. We decrease c_L' so that these changes are at most $\frac{1}{8}c_L$. Decreasing c_L' keeps Corollary 34.5(c) true, and δ_* does not depend on M, so the constants of Remark 34.6 remain free of M. □

Proposition 37.6 (Local fluctuation operators)**.** *There is $R_0(L,M) \leq C_L M$ such that for $R_0 \geq R_0(L,M)$, under the hypotheses of Lemma 37.5:*

(a) $\Delta_{\mathbb{T}_\eta} = \Delta_k$, *and* $\|\Delta_Y - \mathbf{1}_{\mathfrak{F}(Y)}\Delta_k \mathbf{1}_{\mathfrak{F}(Y)}\| \leq C_L e^{-\gamma'(R_0 - 2L - 10M)/2}$, *with γ' of Proposition 37.4(d), taken valid also at the complex perturbations of Lemma 37.5. Hence $\Delta_Y \geq \frac{1}{8}c_L$, and $C_Y = \Delta_Y^{-1}$ has decay (C_L, γ_C) uniformly in Y.*

(b) Δ_Y *and* C_Y *depend only on U on $\widehat{Y^+}$. They are covariant, and analytic under the complex perturbations of Lemma 37.5, with* $\operatorname{Re}\Delta_Y \geq \frac{1}{16}c_L$. *At complex perturbations the current J_U is not bounded uniformly in η. Wherever ω_0 enters, write $\langle b, \omega_0^X\rangle = \langle F_U^s, D\mathbf{H}_0^X b\rangle$ and use the decay of $D\mathbf{H}_0^X$, as in Section 41, level (L4); the proofs of Proposition 37.4(b),(c),(d) and Lemma 37.5 are to be read this way. In the proof of Proposition 37.4(d) this gives $\langle b, \omega_0^X - \omega_0^{X'}\rangle = \langle F_U^s, D(\mathbf{H}_0^X - \mathbf{H}_0^{X'})b\rangle$, which is of seam form, at all fine unit points, by Proposition 35.1(d),(e) for $D\mathbf{H}_0$ at the complex perturbation.*

(c) For $Y \subset Y'$ and $b, b' \in \mathfrak{F}(Y)$,

$$|(C_{Y'} - C_Y)(b,b')| + |(\Delta_{Y'} - \Delta_Y)(b,b')| \leq C_L \sum_{w \in \Sigma_Y} e^{-\gamma''(|b-w| + |w-b'|)},$$

where Σ_Y is the set of unit points within distance $R_0 + 10M + 2L$ of $Y' \setminus Y$.

(d) (Uniformity in $r \geq 0$.) $\operatorname{Re}(\Delta_Y + r) \geq \frac{1}{16}c_L + r$, *and $(\Delta_Y + r)^{-1}$ has decay $(C_L(\frac{1}{16}c_L + r)^{-1}, \gamma_C)$ uniformly in r and Y. Also (c) holds for $(\Delta_Y + r)^{-1}$, with an extra factor $(\frac{1}{16}c_L + r)^{-1}$ on the right. Indeed, r cancels from the operator of Lemma 17.7, and conjugation does not see it, since $(K + r)_\alpha - (K + r) = K_\alpha - K$. This is the form used at level (L5) of Section 41 and in Proposition 38.1.*

Proof. (a) For $b, b' \in \mathfrak{F}(Y)$ the supports of $\hat{Z}^1\delta_b$, $\hat{Z}^1\delta_{b'}$ lie at distance at least $R_0 - 2L$ from $\mathbb{T}_\eta \setminus Y^+$. By Proposition 37.4(d) with $X = \mathbb{T}_\eta$ and $X' = Y^+$, together with the locality of $\hat{Z}^1$, $\hat{Z}^2$,

$$|\Delta_k(b,b') - \Delta_k^{(Y^+)}(b,b')| \le C_L e^{-\gamma'(R_0-2L-10M)/2} e^{-\gamma'|b-b'|/2}.$$

Here the sum over the points w of the seam is taken with half of the rate in the distance to the seam, which absorbs the number of these points. A Schur bound gives the norm estimate, with a constant C_{seam}. By the same proof, read as in (b), the bound holds also at the complex perturbations of Lemma 37.5; we take C_{seam} and γ' valid at real U and at these perturbations. Put $R_0(L,M) := 2L + 10M + \frac{2}{\gamma'}\log^+ \frac{16 C_{\mathrm{seam}}}{c_L}$. For $R_0 \ge R_0(L,M)$ the norm in (a) is at most $\frac{1}{16}c_L$, at real U and at these complex perturbations. The constants C_{seam}, c_L and γ' do not depend on M, so $R_0(L,M) \le C_L M$ for $M \ge 1$; and $R_0(L,M) \ge 2L+2$. Then $\mathbf{1}_{\mathfrak{F}(Y)}\Delta_k \mathbf{1}_{\mathfrak{F}(Y)} \ge \frac{1}{4}c_L$, as a compression of Δ_k (Lemma 37.5). This gives the lower bound, since $\frac{1}{4} - \frac{1}{16} \ge \frac{1}{8}$, and Lemma 5.5 gives the decay.

(b) follows from Proposition 37.4(c), the locality of the elimination, and (a) for complex perturbations: there $\operatorname{Re}\mathbf{1}_{\mathfrak{F}(Y)}\Delta_k\mathbf{1}_{\mathfrak{F}(Y)} \ge \frac{1}{8}c_L$ by Lemma 37.5, so $\operatorname{Re}\Delta_Y \ge \frac{1}{8}c_L - \frac{1}{16}c_L$.

(c) Apply Lemma 17.7 with $\mathfrak{A} = \Delta_{Y'}$ and $\mathfrak{A}' = \Delta_Y$. The operator $\Delta = \Delta_{Y'}\mathbf{1}_{\mathfrak{F}(Y)} - \mathbf{1}_{\mathfrak{F}(Y)}\Delta_Y\mathbf{1}_{\mathfrak{F}(Y)}$ has two kinds of entries. For $b \in \mathfrak{F}(Y)$ they are $\Delta_k^{(Y'^+)}(b,b') - \Delta_k^{(Y^+)}(b,b')$, of seam form by Proposition 37.4(d), since $Y'^+ \setminus Y^+$ lies within distance R_0 of $Y' \setminus Y$. For $b \in \mathfrak{F}(Y') \setminus \mathfrak{F}(Y)$ they are entries $\Delta_k^{(Y'^+)}(b,b')$ with b starting in $Y' \setminus Y$, bounded by $C_L e^{-\gamma_1|b-b'|}$. Both have seam form, and the decay of $C_{Y'}$ and C_Y gives the bound for C. The bound for Δ is the first kind of entry. □

37.5 Local expansion of the covariance

This subsection is not on the main line (see the status paragraph at the beginning of the section).

Theorem 37.7 (Localization of the fluctuation covariance)**.** *For free bonds b, b' let $\mathsf{S} = \{b_-, b'_-\}$ and $\Lambda_m = \Lambda_m(\mathsf{S})$. Then*

$$C_k(b,b') = \sum_{m\ge 0} C_{k,m}(b,b'), \qquad \Delta_k(b,b') = \sum_{m\ge 0} \Delta_{k,m}(b,b'),$$

with $C_{k,0} = C_{\Lambda_0}(b,b')$ and $C_{k,m} = C_{\Lambda_m}(b,b') - C_{\Lambda_{m-1}}(b,b')$ for $m \ge 1$, and similarly for Δ_k. Each $C_{k,m}(b,b')$ and $\Delta_{k,m}(b,b')$ depends only on U on $\widehat{\Lambda_m^+}$. They are covariant, analytic under complex perturbations, and bounded by

$$|C_{k,m}(b,b')| + |\Delta_{k,m}(b,b')| \le C'_L e^{-\gamma_2(m+|b-b'|)}.$$

Proof. Lemma 37.1 with Proposition 37.6(c) for $Y = \Lambda_{m-1} \subset Y' = \Lambda_m$. The points $w \in \Sigma_Y$ have distance at least $m - 1 - R_0 - 10M - 2L$ from S. So

$$\sum_{w\in\Sigma_Y} e^{-\gamma''(|b-w|+|w-b'|)} \le C_L e^{-\gamma''(m-R'_0)/2} e^{-\gamma''|b-b'|/2},$$

with $R'_0 = R_0 + 10M + 2L + 1$. The factor $e^{\gamma'' R'_0/2}$ is absorbed into C'_L, and $\gamma_2 = \gamma''/2$. □

Corollary 37.8 (Locality under large changes of the background)**.** *Let U_1, U_2 be two backgrounds satisfying the hypotheses above, which coincide on the bonds of $\hat{Y}$ for a domain Y. Then for free bonds b, b' starting in Y, with $\delta = \operatorname{dist}(\{b_-, b'_-\}, \mathbb{T}_\eta \setminus Y)$,*

$$|C_k(U_1)(b,b') - C_k(U_2)(b,b')| + |\Delta_k(U_1)(b,b') - \Delta_k(U_2)(b,b')| \le C_L e^{-\gamma_2(\delta - R'_0)}.$$

Proof. The terms with $\widehat{\Lambda_m^+} \subset \hat{Y}$, i.e. $m \le \delta - R_0'$, coincide. The others sum to at most $C_L e^{-\gamma_2(\delta - R_0')}$. □

Corollary 37.8 compares two arbitrary regular backgrounds that agree on a region. They may differ by large amounts elsewhere, which Proposition 33.3 does not cover. For backgrounds of two renormalization group steps, $C_k(U_i)$ is the true fluctuation covariance of Part VII.

Remark 37.9 (Use in the cluster expansion). *Note added.* The radius-type localization of this section is not sufficient for the polymer expansion (Section 39). The main line now uses the weakened random walks of Part IX, with the operators of this section as local inverses; Proposition 37.6 is used there at level (L5). The original remark follows. Theorem 37.7 is the input for a cluster expansion with an interpolated covariance, e.g. the Brydges–Kennedy forest formula. That formula needs only the decay of the kernel and the positivity of the interpolated covariance. The entries of C_k then appear in the activities, and expanding each entry by the theorem makes the dependence on the background local. Bałaban's random-walk expansion [7] is an alternative. It can be built from the local inverses C_Y, with the parametrix of Section 20 and the local entries $\Delta_{k,m}$ in the remainder, but it is not needed here.

38 Localization and normalization of the determinant

Status. The main line uses from this section the local densities $\ell_Y(b)$ defined below (through Corollary 41.6), the background activities (Definition 38.2), the graded domains (Definition 38.3) with (38.1), Lemmas 38.4 and 38.5, and Remark 38.7. Proposition 38.1, the telescoping of the densities, and Corollary 38.8 are of radius type and are not used later; they are superseded by Proposition 41.7, which gives the one-loop source in polymer form at the complex points of the activity domain.

38.1 Local densities

For a domain Y and $b \in \mathfrak{F}(Y)$ let

$$\ell_Y(b) = \operatorname{tr}_{\mathfrak{g}} \left[(\log \Delta_Y)(b,b)\right], \qquad \log \Delta_Y = \int_0^\infty \Big(\frac{1}{1+t} - (\Delta_Y + t)^{-1} \Big) dt,$$

with Δ_Y the local fluctuation operator of Proposition 37.6. For $Y = \mathbb{T}_\eta$ this is the density $\ell_k(b)$ of Proposition 34.7.

Proposition 38.1 (Localization of the determinant)**.** *For $b \in \mathfrak{F}$ let $\Lambda_m = \Lambda_m(\{b_-\})$. Then*

$$\ell_k(b) = \sum_{m \ge 0} \ell_{k,m}(b), \qquad \ell_{k,0}(b) = \ell_{\Lambda_0}(b), \quad \ell_{k,m}(b) = \ell_{\Lambda_m}(b) - \ell_{\Lambda_{m-1}}(b),$$

where $\ell_{k,m}(b)$ depends only on U on $\widehat{\Lambda_m^+}$. It is invariant under real gauge transformations and analytic under complex perturbations of U, and $|\ell_{k,m}(b)| \le C_L e^{-\gamma_3 m}$.

Proof. For $Y \subset Y'$ and $t \ge 0$, Lemma 17.7 applied to $\Delta_{Y'} + t$ and $\Delta_Y + t$ gives, as in Proposition 37.6(c),

$$|((\Delta_{Y'} + t)^{-1} - (\Delta_Y + t)^{-1})(b,b)| \le C_L (c_L/16 + t)^{-2} \sum_{w \in \Sigma_Y} e^{-\gamma'' |b-w|}.$$

The lower bound and the decay constants of $\Delta_Y + t$ are uniform in t, as in Proposition 34.7(a). Integrating in t bounds $|\ell_{Y'}(b) - \ell_Y(b)|$ by the same sum without the factor $(c_L/16 + t)^{-2}$. Lemma 37.1 gives the expansion and the bound. The trace of the diagonal kernel of an operator that is covariant under $R(u)$ is invariant. Analyticity follows from Proposition 37.6(b). □

At the trivial background $U = 1$ the same construction gives numbers $\ell_{k,m}(b;1)$. They satisfy $\sum_{b,m} \ell_{k,m}(b;1) = \log\det \Delta_k(1)$, which is all that is used. (The individual pieces need not depend only on the position of b relative to the blocks $B'(\cdot)$. The mass profiles of the Dirichlet operators are built on the grid $M'\mathbb{Z}^d$, which is not invariant under $L\mathbb{Z}^d$.)

38.2 Normalized background activities

Definition 38.2 (Background activities)**.** Let P be a domain and $\varepsilon_a, \delta_a > 0$. A *background activity at P* is a function $F(U)$ with the following properties:

- it is defined for the complex configurations $U = e^{\mathrm{i}\eta\,\delta A} U_r$ such that U_r is real, (M, ε_a)-regular and satisfies $|J_{U_r}| \le \varepsilon_a$, and $|\delta A|, |\partial\delta A|, [\partial\delta A]_\alpha \le \delta_a$ in the local gauges of U_r;
- it depends only on U on the bonds of P;
- it is analytic and invariant under real gauge transformations.

It is *normalized* if $F(1) = 0$. We write $\|F\| = \sup|F|$ over the domain of definition.

The activities of the renormalization group are evaluated on a smaller, graded domain (blueprint, Appendix F, domain D3‴). It allows a real part with small curvature and current, and a complex part that is small in sup norm and smaller still in its derivatives. Only a neighbourhood of the domain P matters.

For a domain P let $P^{(3)}$ be the set of points at unit-lattice ℓ^∞-distance at most 3 from P, with the bonds joining them; so $\Lambda_3(x) \subset P^{(3)}$ for $x \in P$. For a real configuration U_r and a $\mathfrak{g}^c$-valued field A on the bonds of a set Y, let $|\nabla A|_Y$ be the maximum of $|D_{U_r,\nu} A_\mu(x)|$ over the differences that involve only points of Y. Below, $\nabla A(x)$ denotes the components at x with this property. Let

$$[\nabla A]_{\alpha,Y} = \sup\Big\{ \frac{|\nabla A(x) - R(U_r(\Gamma_{x,x'}))\nabla A(x')|}{|x-x'|^\alpha} : \; x \ne x' \in \Lambda_3(z') \subset Y \Big\},$$

where $\Gamma_{x,x'}$ is the lattice path from x to x' that changes the coordinates in the order $1, \dots, d$. Both quantities are invariant under real gauge transformations ($U_r \mapsto U_r^u$, $A(b) \mapsto R(u(b_-))A(b)$). In a real gauge on a box Λ' in which $U_r = e^{\mathrm{i}\eta A_r}$ with $\|A_r\|_{1,\alpha;\Lambda'} \le \sigma \le 3$, the field $\tilde A$ that represents A satisfies, on Λ',

$$|\nabla A - \partial\tilde A| \le C\sigma|A|, \qquad \big|[\nabla A]_\alpha - [\partial\tilde A]_\alpha\big| \le C\sigma\big(|A| + |\partial\tilde A|\big). \tag{38.1}$$

Indeed $D_{U_r,\nu}\tilde A_\mu - \partial_\nu \tilde A_\mu = \eta^{-1}\big(R(e^{\mathrm{i}\eta A_{r,\nu}}) - 1\big)\tilde A_\mu(\cdot + \eta e_\nu)$, whose Hölder quotients are bounded by $C\sigma(|A| + |\partial\tilde A|)$, since $|A_r| + |\partial A_r| \le \sigma$ and $|x-x'|^{1-\alpha} \le C$ on the cubes. Moreover $\|R(U_r(\Gamma_{x,x'})) - 1\| \le C|x-x'|\sigma$ for x, x' in a cube $\Lambda_3(z')$.

Definition 38.3 (Graded domains and activities)**.** Let P be a domain and $\theta_r, \delta_1, \delta_0 > 0$. The set $\mathcal{U}(P) = \mathcal{U}(P;\theta_r,\delta_1,\delta_0)$ consists of the complex configurations U on $\mathbb{T}_\eta$ for which there are

- a real configuration U_r on $\mathbb{T}_\eta$ with $|F_{U_r}| < \theta_r$ and $|J_{U_r}| < \theta_r$ everywhere, and
- a $\mathfrak{g}^c$-valued field A on the bonds of $P^{(3)}$ with $U = e^{\mathrm{i}\eta A} U_r$ there, $|A| < \delta_0$, and $|\nabla A|_{P^{(3)}} + [\nabla A]_{\alpha,P^{(3)}} < \delta_1$, with ∇ taken with respect to U_r.

The set is invariant under real gauge transformations, and it contains 1. A *graded activity at P* is a function F of $U|_P$ that is defined and holomorphic on an open set containing $\{U|_P : U \in \mathcal{U}(P)\}$ and invariant under real gauge transformations on $\mathcal{U}(P)$. It is *normalized* if $F(1) = 0$. We write $\|F\| = \sup_{\mathcal{U}(P)}|F|$.

The domain D3‴ of the blueprint is $\mathcal{U}(X; h^{1-\epsilon/2}, h^{1-\epsilon}, h^{1-2\epsilon})$, with $0 < \epsilon < \frac14$.

Lemma 38.4 (Complex gauge invariance)**.** *Let F be a graded activity at P, let $U \in \mathcal{U}(P)$, and let λ_1 be a $\mathfrak{g}^c$-valued scalar field on $\mathbb{T}_\eta$ such that $U^{e^{\mathrm{i}t\lambda_1}} \in \mathcal{U}(P)$ for all $t \in [0,1]$. Then $F(U^{e^{\mathrm{i}\lambda_1}}) = F(U)$.*

Proof. Let $\mathcal{O}$ be the open set on which F is holomorphic, and W the set of $\mathfrak{g}^c$-valued scalar fields λ with $U^{e^{\mathrm{i}\lambda}}|_P \in \mathcal{O}$. Since $\lambda \mapsto U^{e^{\mathrm{i}\lambda}}|_P$ is holomorphic, W is open and $\Phi(\lambda) = F(U^{e^{\mathrm{i}\lambda}})$ is holomorphic on W. Let W_0 be the connected component of W that contains the segment $\{t\lambda_1 : t \in [0,1]\}$. Since $\mathcal{U}(P)$ is invariant under real gauge transformations, W_0 contains a ball of real fields around 0, and Φ is constant on it. This ball is a nonempty open subset of the real points, which form a totally real subspace of maximal dimension. By the identity theorem Φ is constant on W_0, so $\Phi(\lambda_1) = \Phi(0)$. □

Lemma 38.5 (Quadratic bound for normalized activities)**.** *There are $C_0 \geq 1$ and $c_0 > 0$, depending only on d, α and G (C_0 is fixed first, then $c_0 \leq c(C_0)$), with the following property. Let G be semisimple, $n \geq n_0$, $0 < \theta_r \leq a_2$ (a_2 from Corollary 30.4) and $0 < \delta_1 \leq \delta_0$. Let P be a connected domain, and let F be a normalized graded activity at P. Let $U = e^{\mathrm{i}\eta A'} U_r$ on $P^{(3)}$ be a representation as in Definition 38.3, with $|F_{U_r}|, |J_{U_r}| \leq \theta < \theta_r$. Let $\Lambda = \Lambda_R(z) \supset P^{(3)}$ with $2R < n$, and put*

$$B_P = 3C_E B_5 \max\Big(1, \frac{R}{4M+2}\Big)^p, \qquad \mathfrak{m} = |\nabla A'|_{P^{(3)}} + [\nabla A']_{\alpha, P^{(3)}},$$

with C_E, p from Lemma 28.3 and B_5 from Corollary 30.4. Let D_N be the diameter of the set of bonds of $P^{(3)}$ for the following distance between parallel bonds: the least unit-lattice length of a chain of parallel bonds of $P^{(3)}$ in which each bond is the translate of the previous one by one fine lattice step, in any direction. Put

$$c_* = \delta_0 + (1 + D_N)(|A'| + B_P\theta),$$

and assume $RB_P\theta \leq 1$, $c_* \leq c_0$ and $|A'| < \frac{1}{2}\delta_0$. Put

$$\mathfrak{m}_1 = (1 + C_0 c_*)\,\mathfrak{m} + C_0|A'|(|A'| + (1+D_N)B_P\theta), \qquad \mathfrak{s} = (1 + C_0 c_*)\Big(1 + \frac{D_N\delta_1}{\delta_0}\Big)\frac{\mathfrak{m}_1 + B_P\theta}{\delta_1}.$$

If $\mathfrak{s} \leq \frac{1}{2}$, then

$$|F(U)| \leq 2\,\mathfrak{s}^2\,\|F\|.$$

Proof. The proof uses two cancellations. At the base point $x_0 \in P$, an affine complex gauge removes the value of the complex field (Step 2). What remains are differences of the field. They are summed along chains of parallel bonds and bounded by the covariant quantity $\mathfrak{m}$, through the derivative estimate (38.2). At the configuration 1, normalization and real gauge invariance remove the terms of order 0 and 1 of an interpolation (Step 3).

Constants. Let $c_{\mathrm{an}} \in (0,1]$, depending only on d and G, be so small that the following expansions hold whenever $c_* \leq c_{\mathrm{an}}$, with constants depending only on d and G: (24.1) for complex λ, with its bound on $\mathfrak{b}$, and the power series of $e^{-\mathrm{i}\,\mathrm{ad}\,X}$ and $\mathfrak{d}(X)$, for the fields of Step 2; the Baker–Campbell–Hausdorff formula, for the fields of Step 3. Such a c_{an} exists, since these fields are bounded by Cc_* under the hypotheses of the lemma, with C depending only on d and G (Steps 2 and 3). We assume $c_0 \leq c_{\mathrm{an}}$; in particular $c_* \leq 1$. Below, C denotes a constant that depends only on d, α and G, through (38.1), (23.2) and these expansions. Each such C is valid for every $c_* \leq c_{\mathrm{an}}$, and none depends on C_0 or on c_0. By the definition of c_*,

$$B_P\theta \leq c_*, \qquad D_N B_P\theta \leq c_*, \qquad (1+D_N)|A'| \leq c_*, \qquad \delta_0 \leq c_*.$$

Moreover $D_N \geq 1$. Indeed, $P^{(3)}$ contains every bond at a point of P together with its translates by one unit, and a chain from a bond to such a translate has length at least 1.

Step 1: a real gauge on Λ. By Corollary 30.4, U_r is $(M, B_5\theta)$-regular, hence $3B_5\theta$-regular on every cube $\Lambda_{4M+2}(z')$. If $R \geq 4M+2$, Lemma 28.3 with $r = 4M+2$ applies, since $RB_P\theta \leq 1$. If $R < 4M+2$, Λ lies in a cube $\Lambda_{4M+2}(z)$. In both cases there is a real gauge u on Λ, extended arbitrarily to $\mathbb{T}_\eta$, with $U_r^u = e^{\mathrm{i}\eta A_r}$ on the bonds of Λ and $\|A_r\|_{1,\alpha;\Lambda} \leq B_P\theta$. Let $\tilde{A}'(b) = R(u(b_-))A'(b)$ represent A' in this gauge. Then $U^u = e^{\mathrm{i}\eta\tilde{A}'}U_r^u$ on $P^{(3)}$, and $|\tilde{A}'| = |A'|$. For a field B on the bonds of $P^{(3)}$, let $|\partial B|$ and $[\partial B]_\alpha$ be the quantities $|\nabla B|_{P^{(3)}}$ and $[\nabla B]_{\alpha,P^{(3)}}$ for the real configuration 1. They are built from the flat differences $\partial_\mu B_\nu(y) = \eta^{-1}(B_\nu(y+\eta e_\mu) - B_\nu(y))$ that involve only points of $P^{(3)}$. By (38.1), with $\sigma = B_P\theta \leq 1$, we have the derivative estimate

$$|\partial\tilde{A}'| + [\partial\tilde{A}']_\alpha \leq (1 + CB_P\theta)\,\mathfrak{m} + CB_P\theta|A'|. \tag{38.2}$$

Real gauge invariance gives $F(U) = F(U^u)$.

Step 2: an affine complex gauge. Fix $x_0 \in P$, use the coordinates of Λ, and put

$$\lambda_1(x) = -\sum_\mu (x - x_0)_\mu\, \tilde{A}'_\mu(x_0) \quad (x \in \Lambda), \qquad \lambda_1 = 0 \text{ off } \Lambda.$$

Here $\tilde{A}'_\mu(x_0)$ is the value of $\tilde{A}'$ on the bond $(x_0, x_0 + \eta e_\mu)$, which lies in $P^{(3)}$ since $x_0 \in P$. Then $\partial_\nu\lambda_1 = -\tilde{A}'_\nu(x_0)$ and $\partial\partial\lambda_1 = 0$ on Λ. On $P^{(3)}$, $|\lambda_1| \leq (1 + D_N)|A'| \leq c_*$, since $|x - x_0|_1$ is at most the length of a chain as in the definition of D_N, plus 1. Moreover $D_{U_r^u}\lambda_1 - \partial\lambda_1 = O(|A_r||\lambda_1|)$, and its derivatives are $O((|\partial A_r| + [\partial A_r]_\alpha)|\lambda_1| + |A_r||\partial\lambda_1|)$. In particular $|D_{U_r^u}\lambda_1| \leq |A'| + CB_P\theta(1 + D_N)|A'| \leq C|A'|$ on $P^{(3)}$. By (24.1), which holds for complex λ by analytic continuation, $(U^u)^{e^{-\mathrm{i}t\lambda_1}} = e^{\mathrm{i}\eta B_t}U_r^u$ on $P^{(3)}$ for $t \in [0,1]$, with

$$B_t(b) = e^{-\mathrm{i}t\,\mathrm{ad}\,\lambda_1(b_-)}\tilde{A}'(b) + t\,\mathfrak{d}(t\,\mathrm{ad}\,\lambda_1(b_-))D_{U_r^u}\lambda_1(b) + \eta\,\mathfrak{b}(t\lambda_1(b_-), \tilde{A}'(b), tD_{U_r^u}\lambda_1(b)).$$

The cancellation at the base point. Let b be a bond of $P^{(3)}$ in direction ν at x. Split $B_t(b) = M_t(b) + E_t(b)$. The main part is

$$M_t(b) = \tilde{A}'(b) + t\,\partial_\nu\lambda_1(x) = \tilde{A}'_\nu(x) - t\tilde{A}'_\nu(x_0) = (1-t)\,\tilde{A}'_\nu(x) + t(\tilde{A}'_\nu(x) - \tilde{A}'_\nu(x_0)),$$

and the rest is

$$\begin{aligned} E_t(b) = {} & (e^{-\mathrm{i}t\,\mathrm{ad}\,\lambda_1(x)} - 1)\tilde{A}'(b) + t(\mathfrak{d}(t\,\mathrm{ad}\,\lambda_1(x)) - 1)D_{U_r^u}\lambda_1(b) \\ & + t(D_{U_r^u}\lambda_1 - \partial\lambda_1)(b) + \eta\,\mathfrak{b}(t\lambda_1(x), \tilde{A}'(b), tD_{U_r^u}\lambda_1(b)). \end{aligned}$$

The gauge is chosen so that $t\,\partial_\nu\lambda_1(x_0) = -t\tilde{A}'_\nu(x_0)$ cancels the value of the field at the base point. At $t = 1$ the main part is the difference $\tilde{A}'_\nu(x) - \tilde{A}'_\nu(x_0)$, which vanishes at $x = x_0$. So the sup norm $|A'|$ enters B_1 only through E_1. By the bounds on λ_1 and $D_{U_r^u}\lambda_1$ above and the bound on $\mathfrak{b}$ in (24.1), the four terms of E_t are at most $C(1 + D_N)|A'|^2$, $C(1 + D_N)|A'|^2$, $C(1 + D_N)B_P\theta|A'|$ and $C|A'|^2$. Hence

$$|E_t(b)| \leq C(1 + D_N)|A'|(|A'| + B_P\theta).$$

The summation along a chain. Let b^0 be the bond in direction ν at x_0. It lies in $P^{(3)}$. By the definition of D_N there is a chain $b^0, b^1, \ldots, b^m = b$ of bonds of $P^{(3)}$ in direction ν, with $b^{i+1} = b^i \pm \eta e_{\mu_i}$ and $m\eta \leq D_N$. Then

$$\tilde{A}'_\nu(x) - \tilde{A}'_\nu(x_0) = \sum_{i=0}^{m-1} (\tilde{A}'(b^{i+1}) - \tilde{A}'(b^i)),$$

and each term is $\pm\eta\,\partial_{\mu_i}\tilde{A}'_\nu(y_i)$. Here y_i is the initial point of that one of b^i, b^{i+1} whose translate by ηe_{μ_i} is the other. This difference involves only the endpoints of b^i and b^{i+1}, which lie in $P^{(3)}$. It is a covariant difference up to a transport. Indeed, in the gauge u,

$$\eta\,\partial_\mu\tilde{A}'_\nu(y) = \eta\,R(u(y))(D_{U_r,\mu}A'_\nu)(y) - \big(R(e^{\mathrm{i}\eta A_{r,\mu}(y)})-1\big)\tilde{A}'_\nu(y+\eta e_\mu),$$

and the last term is at most $C\eta B_P\theta|A'|$. This is the first inequality of (38.1), which enters (38.2). Summing over the chain,

$$|\tilde{A}'_\nu(x)-\tilde{A}'_\nu(x_0)| \le m\eta\,|\partial\tilde{A}'| \le D_N|\partial\tilde{A}'|,$$

and (38.2) bounds $|\partial\tilde{A}'|$ by $\mathfrak{m}$ and $B_P\theta|A'|$. Since $|\tilde{A}'_\nu(x)|\le|A'|$, we get $|M_t(b)|\le(1-t)|A'|+tD_N|\partial\tilde{A}'|$. With the bound on E_t, for $t\in[0,1]$,

$$\begin{aligned}|B_t| &\le |A'| + D_N|\partial\tilde{A}'| + C(1+D_N)|A'|(|A'|+B_P\theta),\\ |B_1| &\le D_N|\partial\tilde{A}'| + C(1+D_N)|A'|(|A'|+B_P\theta).\end{aligned} \tag{38.3}$$

Derivatives. Since $\partial\partial\lambda_1=0$, the differences of the main part are those of $\tilde{A}'$: the gauge does not change them. Differentiating E_t with the chain rule in all four terms (for the term with η, $\eta|\partial f|\le|\partial f|$ and $\eta[\partial f]_\alpha\le 2\eta^{1-\alpha}|\partial f|\le 2|\partial f|$, with $|\partial\mathfrak{b}|$ given by the chain rule),

$$|\partial B_t|+[\partial B_t]_\alpha \le |\partial\tilde{A}'|+[\partial\tilde{A}']_\alpha + C\big(|A'|^2+D_N|A'|(|\partial\tilde{A}'|+[\partial\tilde{A}']_\alpha)+(1+D_N)B_P\theta|A'|\big). \tag{38.4}$$

The constant C_0. The field B_t is given in the gauge u. By (38.1) with $\sigma=B_P\theta$,

$$|\nabla B_t|_{P^{(3)}} + [\nabla B_t]_{\alpha,P^{(3)}} \le (1+CB_P\theta)\big(|\partial B_t|+[\partial B_t]_\alpha\big) + CB_P\theta|B_t|,$$

with ∇ taken with respect to U_r^u. Insert (38.4) and (38.3), and then (38.2). Use $1+D_N\le 2D_N$, the bounds of $B_P\theta$, $D_NB_P\theta$ and $D_N|A'|$ by c_*, and $(1+Cc_*)(1+C'c_*)\le 1+(C+C'+CC')c_*$, which holds for $c_*\le 1$. For instance, the coefficient of $|\partial\tilde{A}'|+[\partial\tilde{A}']_\alpha$ is at most $(1+CB_P\theta)(1+CD_N|A'|)+CD_NB_P\theta\le 1+C'c_*$. This gives a constant C_2, depending only on d, α and G, such that for all $t\in[0,1]$

$$|\nabla B_t|_{P^{(3)}} + [\nabla B_t]_{\alpha,P^{(3)}} \le \mathfrak{m}_2, \qquad |B_t|\le|A'|+D_N\mathfrak{m}_2, \qquad |B_1|\le D_N\mathfrak{m}_2, \tag{38.5}$$

where $\mathfrak{m}_2 = (1+C_2c_*)\,\mathfrak{m} + C_2|A'|(|A'|+(1+D_N)B_P\theta)$. We take $C_0\ge C_2$. Then $\mathfrak{m}_2\le\mathfrak{m}_1$, and (38.5) gives, for all $t\in[0,1]$,

$$|\nabla B_t|_{P^{(3)}} + [\nabla B_t]_{\alpha,P^{(3)}} \le \mathfrak{m}_1, \qquad |B_t|\le|A'|+D_N\mathfrak{m}_1, \qquad |B_1|\le D_N\mathfrak{m}_1.$$

Since $1+C_0c_*\ge 1$ and $1+D_N\delta_1/\delta_0\ge\max(1,D_N\delta_1/\delta_0)$, we have $\mathfrak{s}\ge\mathfrak{m}_1/\delta_1$ and $\mathfrak{s}\ge D_N\mathfrak{m}_1/\delta_0$. So $\mathfrak{s}\le\frac12$ gives $\mathfrak{m}_1\le\frac12\delta_1$ and $D_N\mathfrak{m}_1\le\frac12\delta_0$. Together with $|A'|<\frac12\delta_0$, this gives $|B_t|<\delta_0$, and $\mathfrak{m}_1\le\frac12\delta_1$ gives $|\nabla B_t|_{P^{(3)}}+[\nabla B_t]_{\alpha,P^{(3)}}<\delta_1$. Hence $(U^u)^{e^{-\mathrm{i}t\lambda_1}}\in\mathcal{U}(P)$ for $t\in[0,1]$. Its real part is U_r^u, whose curvature and current have the same size as those of U_r, hence are below θ_r. By Lemma 38.4, $F(U^u)=F\big((U^u)^{e^{-\mathrm{i}\lambda_1}}\big)$.

Step 3: a perturbation of 1. On the bonds of $P^{(3)}$ write $e^{\mathrm{i}\eta B_1}e^{\mathrm{i}\eta A_r}=e^{\mathrm{i}\eta A_{\mathrm{tot}}}$. By the Baker–Campbell–Hausdorff formula, $A_{\mathrm{tot}}=B_1+A_r+\eta\,\mathfrak{c}$ with $|\mathfrak{c}|\le C|B_1||A_r|$. With (23.2) and (38.1), there is a constant C_3, depending only on d, α and G, such that

$$|A_{\mathrm{tot}}|\le(1+C_3c_*)(D_N\mathfrak{m}_1+B_P\theta), \qquad |\partial A_{\mathrm{tot}}|+[\partial A_{\mathrm{tot}}]_\alpha\le(1+C_3c_*)(\mathfrak{m}_1+B_P\theta).$$

Here the corrections are bounded by $|B_1||A_r|+B_P\theta|B_1|\le c_*\mathfrak{m}_1$ and by $|B_1|B_P\theta\le\delta_0B_P\theta$. From Step 2 these bounds use only $|\nabla B_1|_{P^{(3)}}+[\nabla B_1]_{\alpha,P^{(3)}}\le\mathfrak{m}_1$ and $|B_1|\le D_N\mathfrak{m}_1\le\frac12\delta_0$, and they are linear in $\mathfrak{m}_1$. So C_3 does not depend on C_0. We take $C_0\ge C_3$. For complex s let

$V_s = e^{\mathrm{i}\eta s A_{\mathrm{tot}}}$ on the bonds of $P^{(3)}$ and $V_s = 1$ elsewhere. With real part 1 and the field sA_{tot}, for which $\nabla = \partial$, the configuration V_s lies in $\mathcal{U}(P)$ for $|s| < s_0$, where

$$s_0^{-1} = \max\Big(\frac{|A_{\mathrm{tot}}|}{\delta_0},\ \frac{|\partial A_{\mathrm{tot}}| + [\partial A_{\mathrm{tot}}]_\alpha}{\delta_1}\Big) \le \mathfrak{s}.$$

Indeed, $C_3 \le C_0$ and $\delta_1 \le \delta_0$ give $|A_{\mathrm{tot}}|/\delta_0 \le (1 + C_0 c_*)((D_N \delta_1/\delta_0)\,\mathfrak{m}_1/\delta_1 + B_P\theta/\delta_1) \le \mathfrak{s}$, and the second quotient is at most $(1 + C_0 c_*)(\mathfrak{m}_1 + B_P\theta)/\delta_1 \le \mathfrak{s}$. So $f(s) = F(V_s)$ is holomorphic for $|s| < s_0$, with $s_0 \ge 2$, and bounded there by $\|F\|$. Moreover:

- $f(1) = F((U^u)^{e^{-\mathrm{i}\lambda_1}}) = F(U)$, since the two configurations agree on P;
- $f(0) = F(1) = 0$;
- $f'(0) = 0$. Indeed, for a constant real gauge transformation v and small complex fields B on P, real gauge invariance gives $F(e^{\mathrm{i}\eta R(v)B}) = F(e^{\mathrm{i}\eta B})$. Differentiating at $B = 0$, the gradient $\operatorname{grad} F(1)(b) \in \mathfrak{g}^c$ is $\mathrm{Ad}(G)$-invariant for every bond b. For semisimple G the only such vector is 0 [9, (31)–(32)].

Cauchy's estimate gives $|f(1)| \le \|F\| \sum_{j\ge 2} s_0^{-j} \le 2 s_0^{-2}\|F\| \le 2\mathfrak{s}^2\|F\|$.

The order of the constants, and the slack. The proof needs $C_0 \ge \max(1, C_2, C_3)$. The constants C_2 and C_3 come from (38.1), (23.2) and the expansions fixed with c_{an}. They depend only on d, α and G, and they hold for every $c_* \le c_{\mathrm{an}}$. Neither depends on C_0 or on c_0. So $C_0 = \max(1, C_2, C_3)$ is fixed first. Then c_0 is any number in $(0, c_{\mathrm{an}}]$. No condition links c_0 to C_0, so $c(C_0) = c_{\mathrm{an}}$ in the statement, and decreasing c_0 does not change C_0. The hypothesis $c_* \le c_0$ is used only to apply the expansions and, through $c_* \le 1$, to turn products of small factors into terms linear in c_*. The slack lies in the hypotheses. The factors $1 + C_0 c_*$ in $\mathfrak{m}_1$ and $\mathfrak{s}$ absorb the factors $1 + C_2 c_*$ and $1 + C_3 c_*$. The condition $\mathfrak{s} \le \frac{1}{2}$ gives $D_N \mathfrak{m}_1 \le \frac{1}{2}\delta_0$, and $|A'| < \frac{1}{2}\delta_0$ takes the other half of δ_0; this keeps the path of Step 2 in $\mathcal{U}(P)$. The same condition gives $s_0 \ge 2$, so the Cauchy sum is at most twice its first term. □

Remark 38.6 (The bound on D3‴). (a) With $\theta_r = h^{1-\epsilon/2}$, $\delta_1 = h^{1-\epsilon}$, $\delta_0 = h^{1-2\epsilon}$, the factor $1 + D_N\delta_1/\delta_0$ is $1 + D_N h^\epsilon$, and $B_P\theta/\delta_1 \le B_P h^{\epsilon/2}$. The constant B_5, which depends on L and M, and the ratio $R/(4M+2)$ enter only through $B_P\theta$. For polymers with $D_N \le h^{-\epsilon/2}$ and $B_P \le h^{-\epsilon/4}$ we have $c_* = o(1)$ as $h \to 0$, uniformly. So the bound is $2(1+o(1))(\mathfrak{m}/\delta_1)^2\|F\|$ plus terms that vanish as $h \to 0$. The leading coefficient is $1+o(1)$, with no universal factor. This is the form used in the blueprint, Appendix F. There is no splitting into real and imaginary parts and no current of a complex configuration: the interpolation runs through perturbations of 1.

(b) The term $C_0|A'|^2$ in $\mathfrak{m}_1$ cannot be dropped. Take $U_r = 1$ and A' constant with $[A'_\mu, A'_\nu] \ne 0$. Then $\mathfrak{m} = 0$, but U has curvature of order $|A'|^2$ and is not complex-gauge equivalent to 1.

(c) The lemma needs a cube $\Lambda \supset P^{(3)}$ with $2R < n$, so it does not apply to polymers that wrap around the torus. These need a separate treatment. On the main line they are handled by Proposition 49.4 (large polymers), Proposition 41.7(d) and Lemma 50.1.

Remark 38.7 (Background activities restrict to graded activities). Let F be a background activity at P in the sense of Definition 38.2. Suppose $\theta_r \le a_2$, $\max(1, B_5)\theta_r \le \varepsilon_a$ and $C(\delta_0 + \delta_1) < \delta_a$. Then F defines a graded activity at P with $\|F\|$ not larger. Indeed, for $U \in \mathcal{U}(P)$ let χ be a cutoff at unit scale, equal to 1 at unit distance at most 1 from P and to 0 at distance at least 2, and put $\tilde U = e^{\mathrm{i}\eta\chi A} U_r$, with $\tilde U = U_r$ off $P^{(3)}$. By Corollary 30.4, U_r is $(M, B_5\theta_r)$-regular, and $|J_{U_r}| \le \theta_r$. By (38.1), χA obeys the bounds of Definition 38.2 with $C(\delta_0 + \delta_1)$. Here (38.1) is applied with $\sigma \le 3\varepsilon_a$. Since F depends only on the bonds of P, and

$\tilde U = U$ there, $F(\tilde U)$ is a function of $U|_P$. It is holomorphic near every point $U_0|_P$, $U_0 \in \mathcal{U}(P)$. Indeed, fix (U_r, A) for U_0. For W near $U_0|_P$, let $\tilde U_W$ equal W on P and $\tilde U_0$ elsewhere. Since $C(\delta_0 + \delta_1) < \delta_a$ is strict, $\tilde U_W$ stays in the domain of F for W near $U_0|_P$. So $G(W) = F(\tilde U_W)$ is holomorphic, and it equals the restricted activity.

The lemma converts the size of the activity into a bound that is quadratic in the gauge-invariant curvature and current, and in the derivatives of the complex part. The polynomial factors in the diameter are absorbed by the exponential decay of the activities.

Corollary 38.8 (The one-loop vacuum polarization)**.** *Define* $\mathfrak{e}_m(b; U) = -\frac{1}{2}(\ell_{k,m}(b; U) - \ell_{k,m}(b; 1))$. *Then*

$$-\tfrac{1}{2} \log \det \Delta_k(U) = -\tfrac{1}{2} \log \det \Delta_k(1) + \sum_{b \in \mathfrak{F}} \sum_{m \geq 0} \mathfrak{e}_m(b; U),$$

and each $\mathfrak{e}_m(b; \cdot)$ *is a normalized background activity at* $\widehat{\Lambda_m^+}$*, with* $\|\mathfrak{e}_m(b)\| \leq C_L e^{-\gamma_3 m}$*, for constants* ε_a, δ_a *depending on* L*. For real* $U \in \mathfrak{U}(\theta)$ *with* θ *small,*

$$\sum_{m \geq 0} |\mathfrak{e}_m(b; U)| \leq C_L(\theta^2 + e^{-cn}).$$

Proof. The identity follows from Proposition 38.1, at U and at 1, and Proposition 34.7. Normalization holds by construction. The properties of a background activity follow from Proposition 38.1. The domain of definition consists of the regular, not necessarily critical, backgrounds and their complex perturbations. On it, Δ_Y is coercive with uniform constants by Lemma 37.5 and Proposition 37.6; here $\varepsilon_a \leq \varepsilon_{\text{cc}}$, ε_a is below the thresholds of the proof of Proposition 37.4(c), and $C_{\text{cmp}}\delta_a \leq c'_L/C_{\text{AB}}$, so that every configuration of Definition 38.2 is a complex perturbation as in Lemma 37.5. Hence the sup bound $\|\mathfrak{e}_m(b)\| \leq C_L e^{-\gamma_3 m}$ holds on the whole domain. For the bound, take $\theta_r = \delta_1 = \delta_0 = c_{\text{ol}}$ in Remark 38.7, with $c_{\text{ol}} > 0$ so small that $c_{\text{ol}} \leq \min(\frac{1}{2}c_0, a_2)$, $\max(1, B_5)c_{\text{ol}} \leq \varepsilon_a$ and $2Cc_{\text{ol}} < \delta_a$, with C of that remark, so that $\mathfrak{e}_m(b; \cdot)$ is a normalized graded activity at $\widehat{\Lambda_m^+}$ with sup at most $C_L e^{-\gamma_3 m}$. Apply Lemma 38.5 with $U_r = U$, $A' = 0$ and a cube $\Lambda \supset (\widehat{\Lambda_m^+})^{(3)}$ of side at most $C(1 + m + R_0)$. Then $\mathfrak{m} = 0$, $D_N \leq C(1+m+R_0)$ and $B_P \leq C_L(1+m+R_0)^p$, so $\mathfrak{s} \leq C_L(1+m+R_0)^{p+1}\theta$. When this is at most a small constant, all hypotheses hold, and the lemma gives $C_L(1 + m + R_0)^{2p+2}\theta^2 e^{-\gamma_3 m}$. This requires $C(1 + m + R_0) \leq n$ as well. For larger m use $|\mathfrak{e}_m| \leq C_L e^{-\gamma_3 m}$, together with $e^{-\gamma_3 m/2} \leq C\theta^2$ when $m \geq c'\theta^{-1/(p+1)}$, and $e^{-\gamma_3 m} \leq e^{-cn}$ when $C(1 + m + R_0) > n$. Summing over m gives the bound.

The term e^{-cn} cannot be dropped. A flat background with non-trivial holonomy around the torus has $\theta = 0$ but is not gauge equivalent to 1, and $\ell_k(b; U) - \ell_k(b; 1)$ is then generically nonzero, of order e^{-cn}. It reflects polymers that wrap around the torus. In the renormalization group these sources are controlled by the size of the final torus (Lemma 50.1). At the last steps the torus side is fixed, so e^{-cn} is not small compared with h^2 unless the final torus side is at least of order $\log g_*^{-1}$, as in the first alternative of (T3). □

This proves the input of Proposition A.5 of the blueprint at real backgrounds, under a global small-field bound. It does not supersede that proposition: the pieces here are of radius type ($e^{-\gamma_3 m}$) rather than polymer type, and neither the scaling L^{-1} nor bounds at complex points of the activity domain are derived here (blueprint, Appendix F). Since Lemma 38.5 allows complex points, and Remark 38.7 makes $\mathfrak{e}_m(b; \cdot)$ a graded activity, the argument above also bounds $\mathfrak{e}_m(b; U)$ at complex U, provided U has the form of Definition 38.3 on $(\widehat{\Lambda_m^+})^{(3)}$. Turning this into the polymer bounds of D3‴, including the factor L^{-1} per coarse block, is not done here. The vacuum part $-\frac{1}{2} \log \det \Delta_k(1)$ is independent of the background and enters the vacuum energy. The normalized activities are of second order in the curvature and current. The coefficient C_L does not contain g_k: this is a one-loop effect.

39 Localization of the background map and of the interaction

Status. This section is not on the main line. Its localization is of radius type (Summary below). It is superseded by the weakened background map of Section 45 and the step quantities of Section 47. Only the scaling of the plaquette terms in the proof of Theorem 39.4 is used later, in Proposition 47.5.

The interaction $\mathcal{V}$ of Theorem 34.8, and the old activities composed with the background map $Z \mapsto e^{i\eta A_*(\hat{Z}(Z))}U$, depend on the fluctuation and on the background through the nonlinear map $B \mapsto A_*(B)$ of Part V. That map is the fixed point of a contraction. We approximate it by fixed points of maps built with Dirichlet operators and with cut-off sources, so that every nonlinear expression is evaluated in the interior of a domain. The contraction needs pointwise bounds with derivatives for the Dirichlet operators at interior points. These are provided by Theorem 35.2, proved in Section 36.

39.1 Comparison of fixed points

For fields h write $|h|_1(x) = |h(x)| + |\nabla h(x)|$, the gradient involving only neighbours of x, and for a unit point y put $a_y(h) = \sup_{x \in B(y)} |h|_1(x)$. The comparison is made in these block quantities. Theorem 35.2 gives bounds from block suprema of the data to block quantities of the output; it gives no bound on a fine-lattice kernel, and none is used.

Lemma 39.1 (Comparison of fixed points)**.** *Let Ω be a domain, let χ be a function with $0 \le \chi \le 1$, $\chi = 1$ on $\Omega_{(2)}$, $\chi = 0$ on $\mathbb{T}_\eta \setminus \Omega_{(1)}$, $|\nabla\chi| \le 1$, and let $\Sigma \supset \Omega \setminus \Omega_{(2)}$. Let $\mathcal{T}$ be a map of fields on $\mathbb{T}_\eta$ with fixed point A, and $\mathcal{T}_\Omega$ a map of fields on Ω with fixed point A^Ω. Assume, for the unit points y with $B(y) \subset \Omega$:*

(i) for fields A_1, A_2 in a ball containing A and $\tilde{A} = \chi A^\Omega + (1-\chi)A$,

$$a_y(\mathcal{T}(A_1) - \mathcal{T}(A_2)) \le \sum_{y'} K(y,y')\, a_{y'}(A_1 - A_2), \qquad \sup_y \sum_{y'} K(y,y') e^{\gamma'|y-y'|} \le \tfrac{1}{4};$$

(ii) $a_y(\mathcal{T}(\tilde{A}) - \mathcal{T}_\Omega(A^\Omega)) \le \sigma e^{-\gamma' \operatorname{dist}(y,\Sigma)}$.

Then $a_y(A - A^\Omega) \le 2\sigma e^{-\gamma' \operatorname{dist}(y,\Sigma)}$ for $B(y) \subset \Omega$.

Proof. Let $\delta = A - A^\Omega$ on Ω, and $\|\delta\|_w = \sup_y a_y(\delta) e^{\gamma' \operatorname{dist}(y,\Sigma)}$, the supremum over the y with $B(y) \subset \Omega$; it is finite. Then

$$\delta = (\mathcal{T}(A) - \mathcal{T}(\tilde{A})) + (\mathcal{T}(\tilde{A}) - \mathcal{T}_\Omega(A^\Omega)), \qquad A - \tilde{A} = \chi\delta.$$

Since χ vanishes on the outer layer $\Omega \setminus \Omega_{(1)}$, the field $\chi\delta$ vanishes off $\Omega_{(1)}$, its gradient involves δ only on Ω, and $|\nabla(\chi\delta)| \le |\nabla\delta| + |\nabla\chi||\delta|$. So $a_{y'}(\chi\delta) \le 2a_{y'}(\delta)$ if $B(y') \subset \Omega$, and $a_{y'}(\chi\delta) = 0$ otherwise. Since $e^{-\gamma' \operatorname{dist}(y',\Sigma)} \le e^{\gamma'|y-y'|} e^{-\gamma' \operatorname{dist}(y,\Sigma)}$, assumption (i) bounds a_y of the first term by $\frac{1}{2}\|\delta\|_w e^{-\gamma' \operatorname{dist}(y,\Sigma)}$. Assumption (ii) bounds the second. Hence $\|\delta\|_w \le \frac{1}{2}\|\delta\|_w + \sigma$. □

The hybrid field $\tilde{A}$ is what allows a comparison of the torus map with a map defined only on Ω, without boundary estimates for the latter.

39.2 Dirichlet background maps

Let U be a background as in Section 35, and let B be a complex unit-lattice field with $|B|_\infty \le r_1$. Recall from Section 26 that $A_*(B) = A' - \mathbf{H}\mathcal{D}(A')$ with $A' = A_1 + \mathbf{H}B$, where A_1 is the fixed point of

$$\mathcal{T}(A_1) = -\mathbf{C}(J + \nabla\mathcal{V}(A_1 + \mathbf{H}B)).$$

The gradient $\nabla\mathcal{V}$ is the sum of the terms listed in the proof of Lemma 26.2(a). Write each of them with the derivatives outside, e.g. $\mathbf{H}^T D^T DA' = (D\mathbf{H})^T(DA')$ and $D^T D\mathbf{H}\mathcal{D} = D^T(D\mathbf{H}\mathcal{D})$. Then it is a composition of:

- the operators $\mathbf{H}$, $\mathbf{H}^T$, $D\mathbf{H}$, $(D\mathbf{H})^T$;
- the derivatives $d\mathcal{D}$, $d\mathcal{D}^T$;
- local functions of A' and $\nabla A'$ (from $\nabla\mathcal{N}$);
- one outer D^T, absorbed by $\mathbf{C}D^T$.

Definition 39.2 (Dirichlet background map)**.** Fix $R_1 \geq 10$. For a domain X let $\Omega_X = X_{(4)}$. Let ζ be a smooth function on $\mathbb{T}_\eta$ with $\zeta = 1$ on $X_{(R_1)}$, $\zeta = 0$ off $X_{(R_1-2)}$, and bounded derivatives (up to second order and Hölder), and let $\hat{\zeta}$ be the indicator of the unit bonds in $\mathfrak{B}(X_{(R_1)})$. Fine fields on Ω_X are extended by 0. Every pairing over X that involves the field A', such as $(D\mathbf{H}^X)^T(DA')$, is evaluated with $\zeta A'$ in place of A', so that the extension by zero creates no jump. Define:

- $\mathcal{D}^X(A')$ as the fixed point of $Y \mapsto \hat{\zeta} C_k(A' - \mathbf{H}^X Y)$;
- $\nabla\mathcal{V}^X$ by replacing, in the expression for $\nabla\mathcal{V}$:
  - every operator by its Dirichlet version on X (Proposition 35.1) with $W = 0$;
  - $\mathcal{D}$ by $\mathcal{D}^X$;
  - J and $\nabla\mathcal{N}$ by ζJ and $\zeta\nabla\mathcal{N}$;

  and multiplying the result by ζ;
- $\mathcal{T}^X(A_1) = -\mathbf{C}^X\zeta(J + \nabla\mathcal{V}^X(A_1 + \mathbf{H}^X\hat{\zeta}B))$, restricted to Ω_X;
- $A^X_*(B) = A'^X - \mathbf{H}^X\hat{\zeta}\mathcal{D}^X(A'^X)$ on Ω_X, where $A'^X = A^X_1 + \mathbf{H}^X\hat{\zeta}B$ and A^X_1 is the fixed point of $\mathcal{T}^X$.

Every nonlinear expression is evaluated on $X_{(R_1-2)} \subset \Omega_X$, where Theorem 35.2 gives pointwise bounds with derivatives. By construction:

- $A^X_*(B)$ depends only on U on $\hat{X}$ and on B on $\mathfrak{B}(X)$;
- it is covariant under real gauge transformations on $\hat{X}$;
- it is analytic in (B, U);
- for $X = \mathbb{T}_\eta$ it equals $A_*(B)$.

Proposition 39.3 (Comparison of background maps)**.** *There are $r_4, C, \gamma_4 > 0$ such that for $\varepsilon + |B|_\infty \leq r_4$:*

(a) $\mathcal{T}^X$ is a contraction on $\{|A_1|_1 \leq \rho_2 \text{ on } \Omega_X\}$, uniformly in X, and $|A^X_*(B)|_1 \leq C(\varepsilon + |B|_\infty)$ on Ω_X;

(b) the Hölder quotients $[\nabla A^X_*(B)]_{\alpha,B(y)}$, for blocks in Ω_X, are bounded by $C(\varepsilon + |B|_\infty)$, by the Hölder parts of Theorem 35.2;

(c) for domains $X' \subset X$ and $x \in \Omega_{X'}$,

$$|A^X_*(B) - A^{X'}_*(B)|_1(x) \leq C(\varepsilon + |B|_\infty)\, e^{-\gamma_4 \operatorname{dist}(x,\Sigma')},$$

where Σ' is the set of points within distance $R_1 + 10M + 4$ of $(\mathbb{T}_\eta \setminus X') \cup (X \setminus X')$. The Hölder quotients $[\nabla(A^X_*(B) - A^{X'}_*(B))^\sim_y]_{\alpha,B(y)}$, for blocks $B(y) \subset \Omega_{X'}$, obey the same bound with $x = y$.

Proof. (a) The proof of Proposition 26.3 applies on Ω_X, with Lemma 24.1 replaced by Theorem 35.2. The sources are cut off to $X_{(R_1)}$, so they are evaluated where the interior bounds hold. The map $\mathcal{D}^X$ is defined by the contraction of Lemma 26.1: $C_k(A)(c)$ for $c \in \mathfrak{B}(X_{(R_1)})$ depends on A on $S_c \subset \Omega_X$, and only the sup norm of $\mathbf{H}^X Y$ there enters.

(b) follows from the Hölder parts of Theorem 35.2 as in (a). (c) Apply Lemma 39.1 with $\mathcal{T} = \mathcal{T}^X$ (regarded as a map of fields on Ω_X), $\mathcal{T}_\Omega = \mathcal{T}^{X'}$, $\Omega = \Omega_{X'}$ and $\Sigma = \Sigma' \cap \Omega_{X'}$. Choose r_4 so that the bound of (a) gives $|A_1^X|_1, |A_1^{X'}|_1 \le \frac{1}{6}\rho_2$. Since $|\nabla\chi| \le 1$, $|\chi a + (1-\chi)b|_1 \le 3\max(|a|_1, |b|_1)$, so the hybrid field $\tilde{A}$ also lies in $\{|A_1|_1 \le \frac{1}{2}\rho_2\}$, where (i) is proved. The lemma gives $a_y(A_1^X - A_1^{X'}) \le 2\sigma e^{-\gamma' \operatorname{dist}(y,\Sigma)}$. For $x \in B(y)$, $\operatorname{dist}(x,\Sigma') \ge \operatorname{dist}(y,\Sigma') - 1$, which gives the pointwise form of (c).

Hypothesis (i). The derivative of $\mathcal{T}^X$ in A_1 is $-\mathbf{C}^X \zeta\, d\nabla\mathcal{V}^X(A'^X)$. On the torus, (27.2) in the proof of Proposition 27.5 bounds $d\nabla\mathcal{V}$ in block form, with the factor $C(\varepsilon + \rho_2)$. Its proof uses the block bounds of $\mathbf{H}$, $D\mathbf{H}$, their transposes and $D^T D\mathbf{H}$, the kernels of $d\mathcal{D}$ and $d\mathcal{D}^T$, the local bound for $d\nabla\mathcal{N}$, and the mixed bound (26.1) of Lemma 26.1. The same proof applies to $\nabla\mathcal{V}^X$, with Theorem 35.2 in place of Lemma 24.1. Indeed $\hat{\zeta}$ restricts $d^2 C_k$ to the coarse bonds of $X_{(R_1)}$, whose sets S_c are interior, and ζ restricts the sources and the output to $X_{(R_1-2)}$. So $\mathbf{H}^X$, $D\mathbf{H}^X$, their transposes and $D^T D\mathbf{H}^X$ are needed only at interior blocks, where Theorem 35.2(c),(d) gives block bounds with exponential decay. In particular

$$|(d^2\mathcal{D}^X[h,\cdot])^T\omega|_{B(y)} \le C|\omega|_\infty \sum_{y'} e^{-\gamma|y-y'|} a_{y'}(h),$$
$$|d\nabla\mathcal{V}^X(A')h|_{B(y)} \le C(\varepsilon+\rho_2) \sum_{y'} e^{-\gamma|y-y'|} a_{y'}(h),$$

with $|f|_{B(y)}$ the supremum of $|f|$ on $B(y)$. By Theorem 35.2(a),(b), $a_y(\mathbf{C}^X\zeta f) \le C\sum_{y'} e^{-\gamma|y-y'|}|f|_{B(y')}$. Composing, (i) holds with $K(y,y') = C_{\mathrm{map}}(\varepsilon+\rho_2)e^{-\gamma_{\mathrm{map}}|y-y'|/2}$. Then $\sum_{y'} K(y,y')e^{\gamma'|y-y'|} \le \frac{1}{4}$ for $\gamma' \le \gamma_{\mathrm{map}}/4$, by the choice of θ_0 in Section 27: $\varepsilon + |B|_\infty \le r_4 \le \theta_0$, where we take $r_4 \le \theta_0$, and $\rho_2 \le (8B_1 + C_H + K)\theta_0$, with B_1, C_H and K as there (this K is the constant $4C_C(1+C_H)$ of Section 26, not the kernel above). No bound on a fine-lattice kernel of $d\mathcal{T}^X$ is used.

Hypothesis (ii). $\mathcal{T}^X(\tilde{A}) - \mathcal{T}^{X'}(A^{X'})$ is a sum of three kinds of terms:

- differences of Dirichlet operators on X and X', which have seam form by the pointwise comparison of Theorem 35.2(e) and are applied to fields bounded by $C(\varepsilon + |B|_\infty)$;
- differences of the cutoffs $\zeta_X - \zeta_{X'}$ and $\hat{\zeta}_X - \hat{\zeta}_{X'}$, supported within distance R_1 of $\mathbb{T}_\eta \setminus X'$;
- terms in which $\tilde{A}$ and $A^{X'}$ differ, i.e. outside $\Omega_{X'(2)}$, which lie in Σ'.

Each kind contributes at most $C(\varepsilon + |B|_\infty)e^{-\gamma \operatorname{dist}(y,\Sigma')}$ to a_y, after composition with the decaying block kernels. Since M may be much larger than R_1, the mass terms of $\mathfrak{A}^{X,\kappa}$ and $\mathfrak{A}^{X',\kappa}$ are active on parts of $X_{(R_1)}$; their difference is part of the comparison of domains and is included in Σ'. The same comparison applies to $\mathcal{D}^X$ and to $A_*^X = A'^X - \mathbf{H}^X\hat{\zeta}\mathcal{D}^X(A'^X)$.

Hölder quotients in (c). Write $\delta = A_1^X - A_1^{X'}$ on $\Omega_{X'}$ as $(\mathcal{T}^X(A_1^X) - \mathcal{T}^X(\tilde{A})) + (\mathcal{T}^X(\tilde{A}) - \mathcal{T}^{X'}(A_1^{X'}))$. The derivative of $\mathcal{T}^X$ is $\mathbf{C}^X\zeta$ applied to $d\nabla\mathcal{V}^X$, and the block bound for $d\nabla\mathcal{V}^X$ in the proof of hypothesis (i) maps the block quantities $a_{y'}$ to block suprema, with the same exponential weights. Theorem 35.2(a),(c) therefore bound the Hölder quotients of the gradient of the first term by C times the weighted bound on $a_{y'}(\chi\delta)$ just proved. The second term is bounded by the Hölder parts of Theorem 35.2(e), as in the proof of hypothesis (ii). The same applies to $\mathbf{H}^X\hat{\zeta}\mathcal{D}^X$, which gives the Hölder part of (c). □

39.3 Localization of the interaction

Write $\Phi(A) - \Phi(0) = \sum_p s_p(A)$, where, with $H = U(\partial p)$ and $H_A = (e^{i\eta A}U)(\partial p)$,

$$s_p(A) = \tfrac{1}{2}\eta^{d-4} \operatorname{tr}\left(H + H^{-1} - H_A - H_A^{-1}\right)$$

is the difference of the plaquette terms of the analytic Wilson action of Section 23. It depends only on A on the bonds of p and is holomorphic in complex A; for real A it equals $\eta^{d-4} \operatorname{Re}\operatorname{tr}(H - H_A)$. The real part itself is not holomorphic, and it is not used at complex fields. Then $|s_p(A) - s_p(A')| \le C\eta^d(\varepsilon + |A|_1 + |A'|_1)\,|A - A'|_1$ near p, by the expansion of Proposition 23.3.

Theorem 39.4 (Localization of the interaction)**.** *Let $r_5 = \min\left(r_2, \tfrac{1}{2}r_L, r_4/(1 + 2CL^d)\right)$, with r_2 of Lemma 34.1, r_L and C of Lemma 32.4(c), and r_4 of Proposition 39.3; and let $\gamma_5 = \gamma_4$ of Proposition 39.3. In the setting of Theorem 34.8, with $U = U'$ and $\varepsilon \le r_5$, there are pieces $\mathcal{V}_{p,m}$, indexed by plaquettes p of $\mathbb{T}_\eta$ and $m \ge 0$, such that for complex Z with $|Z|_\infty \le r_5$:*

$$\mathcal{V}(Z) = \sum_p \sum_{m \ge 0} \mathcal{V}_{p,m}(Z),$$

where:

- *$\mathcal{V}_{p,m}$ depends only on Z and U in $\widehat{\Lambda_m(\{p\})}$ (enlarged by $2L$ for Z);*
- *it is analytic and invariant under real gauge transformations;*
- *it is of third order in Z, with $|\mathcal{V}_{p,m}(Z)| \le C_L |Z|^3_{\Lambda_m(\{p\})} e^{-\gamma_5 m}\eta^d$.*

Here $|Z|_\Lambda$ is the supremum of Z on the free bonds within distance $2L$ of Λ.

Proof. For a domain X containing p in $X_{(R_1+1)}$, put $F_p^X(B) = s_p(A_*^X(B)) - s_p(A_*^X(0))$. For $X = \mathbb{T}_\eta$ this is $s_p(A_*(B)) - s_p(A_*(0))$, and $A_*(0) = 0$ since U' is critical. By Proposition 39.3(c), for $X = \Lambda_m(\{p\})$ and $X' = \Lambda_{m-1}(\{p\})$, $m \ge m_1 = R_1 + 10M + 5$,

$$|F_p^{\Lambda_m}(B) - F_p^{\Lambda_{m-1}}(B)| \le C(\varepsilon + |B|_\infty)^2 e^{-\gamma_4(m-m_1)}\eta^{d-4}\eta^4,$$

since s_p carries the factor η^{d-4} and is of order η^4 in the fields. Lemma 37.1, started at m_1 (the pieces with $m < m_1$ are 0, and $F_{p,m_1} = F_p^{\Lambda_{m_1}}$), gives $\mathcal{S}(B) - \mathcal{S}(0) = \sum_p \sum_m F_{p,m}(B)$, with pieces depending only on B and U in $\widehat{\Lambda_m(\{p\})}$, and $|F_{p,m}(B)| \le Ce^{\gamma_4 m_1}(\varepsilon + |B|_\infty)^2 e^{-\gamma_5 m}\eta^d$. For $|Z|_\infty \le 2r_5$ we have $|\hat{Z}(Z)|_\infty \le CL^d|Z|_\infty$ (Lemma 32.4(c)), so $\varepsilon + |\hat{Z}(Z)|_\infty \le r_4$, and Proposition 39.3 applies at $B = \hat{Z}(Z)$. Put $G_{p,m}(Z) = F_{p,m}(\hat{Z}(Z))$. These are analytic, vanish at $Z = 0$, and by Lemma 32.4 depend on Z only within distance $2L$ of $\Lambda_m(\{p\})$.

Let T_2G denote the Taylor polynomial of degree 2 at $Z = 0$. Since the sums converge absolutely,

$$\sum_{p,m} T_2 G_{p,m} = T_2(\mathcal{S}(\hat{Z}) - \mathcal{S}(0)) = \tfrac{1}{2}\Delta_k(Z, Z)$$

by Lemma 34.1; the linear term vanishes. Hence $\mathcal{V}_{p,m} = G_{p,m} - T_2G_{p,m}$ sum to $\mathcal{V}$. For the bound fix Z with $0 < |Z|_\infty \le r_5$, and put $g(t) = \mathcal{V}_{p,m}(tZ)$. The comparison bounds hold on the complex ball of radius $2r_5$, where $\sup|G_{p,m}| \le C(\varepsilon + 2r_5)^2 e^{-\gamma_5 m}\eta^d$. So g is analytic for $|t| < 2r_5/|Z|_\infty$. By Cauchy's estimate each Taylor term of $t \mapsto G_{p,m}(tZ)$ of degree at most 2 obeys the same bound there, so $|g(t)| \le 4C(\varepsilon + 2r_5)^2 e^{-\gamma_5 m}\eta^d$. Since g vanishes to third order at $t = 0$, the maximum principle for $g(t)/t^3$ on the discs $|t| \le \varrho$, $\varrho < 2r_5/|Z|_\infty$, and the limit $\varrho \to 2r_5/|Z|_\infty$ give

$$|\mathcal{V}_{p,m}(Z)| \le 4C(\varepsilon + 2r_5)^2(2r_5)^{-3}|Z|^3_\infty e^{-\gamma_5 m}\eta^d \le \tfrac{9}{2}Cr_5^{-1}|Z|^3_\infty e^{-\gamma_5 m}\eta^d,$$

using $\varepsilon \le r_5$. The factor r_5^{-1} is part of C_L. Since $G_{p,m}$ depends on Z only within distance $2L$ of $\Lambda_m(\{p\})$, $|Z|_\infty$ may be replaced by $|Z|_{\Lambda_m(\{p\})}$. Gauge invariance follows from the covariance of A_*^X and the invariance of s_p. □

The same argument localizes the old activities composed with the background map. An activity has its own domain, so the radius depends on it. Let P be a domain, let m_1 be the least m with $P^{(2)} \subset \Omega_{\Lambda_m(P)}$, and let χ_P be a smooth function, equal to 1 on $P^{(1)}$ and to 0 off $P^{(2)}$, with bounded derivatives. For $m \ge m_1$ the field $\chi_P A_*^{\Lambda_m(P)}(\hat{Z}(Z))$ is defined on $\mathbb{T}_\eta$; the cutoff, not an extension by zero, keeps its derivatives bounded. By Proposition 39.3, Lemma 32.4(c) and (38.1), there is C_{a}, depending only on d, α, G, L, M and R_1, such that the following holds in the local gauges of U', for complex Z with $|Z|_\infty \le r_5$. The sup norm, the gradient and the Hölder quotient of $\chi_P A_*^{\Lambda_m(P)}(\hat{Z}(Z))$ are at most $C_{\mathrm{a}}(\varepsilon + L^d|Z|_\infty)$. For $m > m_1$, those of $\chi_P(A_*^{\Lambda_m(P)} - A_*^{\Lambda_{m-1}(P)})(\hat{Z}(Z))$ are at most $C_{\mathrm{a}}(\varepsilon + L^d|Z|_\infty)e^{-\gamma_4 m}$. For the latter, the points of $P^{(2)}$ are at distance at least $m - R_1 - 10M - 8$ from the set Σ' of Proposition 39.3(c), and the factor $e^{\gamma_4(R_1+10M+8)}$ is part of C_{a}.

Corollary 39.5 (Composed activities)**.** *Let F be a background activity at a domain P, with parameters ε_a, δ_a (Definition 38.2). Assume $\varepsilon \le \varepsilon_a$ and $C_{\mathrm{a}}\varepsilon \le \frac{1}{4}\delta_a$, and let $0 < r_F \le r_5$ with $C_{\mathrm{a}}L^d r_F \le \frac{1}{4}\delta_a$. Then, for complex Z with $|Z|_\infty \le r_F$,*

$$F\big(e^{\mathrm{i}\eta A_*(\hat{Z}(Z))}U'\big) = \sum_{m \ge m_1} F_m(Z),$$

where F_m depends only on U' in $\widehat{\Lambda_m(P)}$ and on Z within distance $2L$ of $\Lambda_m(P)$, is analytic and gauge invariant, and satisfies $|F_m| \le 2e^{\gamma_4 m_1}\|F\|e^{-\gamma_4 m}$.

Proof. For $m \ge m_1$ let $A_m = \chi_P A_*^{\Lambda_m(P)}(\hat{Z}(Z))$ and $V_m = e^{\mathrm{i}\eta A_m}U'$. On $P^{(1)}$, V_m equals $e^{\mathrm{i}\eta A_*^{\Lambda_m(P)}(\hat{Z}(Z))}U'$, so by the locality of F the value $F(V_m)$ does not depend on the choice of χ_P. If $\Lambda_m(P) = \mathbb{T}_\eta$, then $A_*^{\Lambda_m(P)} = A_*$ and $F(V_m)$ is the left side. Put $F_{m_1} = F(V_{m_1})$ and $F_m = F(V_m) - F(V_{m-1})$ for $m > m_1$; the sum is finite. The reference U' is real, (M, ε_a)-regular and $|J_{U'}| \le \varepsilon_a$, since $\varepsilon \le \varepsilon_a$. By the hypotheses and the choice of C_{a}, A_{m-1} has sup norm, gradient and Hölder quotient at most $\frac{1}{2}\delta_a$, and $A_m - A_{m-1}$ at most $\frac{1}{2}\delta_a e^{-\gamma_4 m}$. So $f(t) = F\big(e^{\mathrm{i}\eta(A_{m-1}+t(A_m-A_{m-1}))}U'\big)$ is analytic and bounded by $\|F\|$ for $|t| < e^{\gamma_4 m}$, and the Schwarz lemma for $f - f(0)$ gives $|F_m| = |f(1) - f(0)| \le 2\|F\|e^{-\gamma_4 m}$. Also $|F_{m_1}| \le \|F\|$. The dependence on U' and Z follows from that of A_*^X (on U on $\hat{X}$ and on B on $\mathfrak{B}(X)$) and from the range $2L$ of $\hat{Z}$. Analyticity follows from that of A_*^X, $\hat{Z}$ and F, and gauge invariance from the covariance of A_*^X and $\hat{Z}$ and the invariance of F; χ_P is scalar. □

39.4 Summary

The background-dependent ingredients of one renormalization group step have the following representations as sums of local pieces with exponential decay:

- the fluctuation covariance and the quadratic form, entrywise (Theorem 37.7);
- the determinant, as a background-independent constant plus normalized activities of second order in the curvature and current (Corollary 38.8);
- the interaction, as pieces of third order in the fluctuation (Theorem 39.4), and the old activities composed with the background map (Corollary 39.5).

The first two use only kernel bounds. The last two also use the interior pointwise bounds of Theorem 35.2.

All four are of *radius type*. A piece that depends on the fields and on the background in a ball Λ_m is bounded by $e^{-\gamma m}$. The polymer expansion of the next stage needs *volume type*, i.e. a fixed small factor for every cube of the dependence region, as in [15, (116)–(122)]. A ball of radius m contains of order m^d cubes, so in $d \geq 2$ the bound $e^{-\gamma m}$ does not suffice. Bałaban and Dimock obtain volume-type localization from random-walk expansions with weakening parameters. For the operators of Parts II and III these require the nested walks of Section 41. The Dirichlet operators of Part III and Theorem 35.2 are the local propagators of such walks. The results of this part are therefore inputs to that construction, but they do not replace it. The local Jacobians $\mathfrak{J}$ and the factors j_c of Theorem 34.8 are local already. These are the inputs I8b, I8c and I9b of the blueprint, in radius-type form. The plan at this point was that the next part would:

- construct volume-type localizations by random walks with weakening, with the Dirichlet operators of Parts III and VIII as local propagators;
- perform the cluster expansion of the Gaussian integral with these activities;
- normalize the resulting activities at the trivial background (Lemma 38.5);
- control the change of units in the norm of the blueprint, Appendix A.

This was carried out in Parts IX and X, with volume-type inputs from Part IX in place of the radius-type ones above, and the activity norm of Section 49 in place of that of the blueprint, Appendix A.

Part IX: Weakened random walks

40 Weakenable families

The localization of Part VIII is of radius type: a piece that depends on the data in a ball of radius m is bounded by $e^{-\gamma m}$. The polymer expansion of the next stage needs volume type, a fixed small factor for every cube of the dependence region, as in [15, (116)–(122)]. Bałaban and Dimock obtain this from random walks with weakening parameters $s_\square$, with analyticity for $|s_\square| \leq e^{\kappa}$. This part constructs such weakened walks for all the operators of the step, in L^2 norms. Section 42 explains what is missing for pointwise norms.

Setting. We keep the conventions of Part VIII. The background U is real and (M, ε)-regular, or a complex perturbation of such a background as in Proposition 19.11. Here ρ is a fixed number, chosen in Section 41, independent of M. The torus side satisfies $n \geq C(\rho + C_v)M$, so that the sets below do not wrap around. We use the grid points z_j, the functions χ_j, $h_j = \chi_j^2$ and the sets K_j, Λ_j of Section 17, and the enlarged domains

$$\Lambda_j^v = \text{the union of the blocks meeting } \{x : |x - z_j|_\infty \leq \rho M\}.$$

For $\rho = 9$ these are the domains Λ_j^w of Section 20. We have $|h_j(x) - h_j(x')| \leq CM^{-1}|x - x'|$, and every point lies in at most 3^d supports K_j.

Boxes and polymers. Fix a *product paving* $\mathcal{P} = \{I_{a_1}^1 \times \cdots \times I_{a_d}^d\}$ of $\mathbb{T}_\eta$. Here each $\{I_a^\mu\}_a$ is a partition of the circle $\mathbb{R}/n\mathbb{Z}$ into intervals of integer length between $M/4$ and M. Its elements are called *boxes*; they are rectangular unions of unit blocks. A *polymer* is a $*$-connected union X of boxes: any two boxes of X are joined by a chain of boxes of X, consecutive ones sharing at least a corner. Because the paving is a product, two boxes that do not touch are separated

in some direction by a whole interval, so they are at distance at least $M/4$. At most 2^d boxes lie within distance $r < M/8$ of a point, and each box touches at most 3^d boxes. Write $|X|$ for the number of boxes of X, and $X^{(r)}$ for the r-neighbourhood in lattice units. For each j let $X(j)$ be the union of the boxes that meet $(\Lambda_j^v)^{(R_v)}$. Here $R_v \le C_v M$ is a fixed margin, chosen at each level to contain the dependence region of the local propagators. The local operators on a domain Y depend on the background only on $\hat{Y}$, and those of the fluctuation operator only on $\widehat{Y^+}$, with $Y^+ = Y^{(R_0)}$ and the margin $R_0 \ge R_0(L,M)$ of Proposition 37.6(b). We take $R_v \ge R_0 + 4 + 2r_{\mathrm{loc}}$, where r_{loc} is the largest range of the local operators of the step; then $(\Lambda_j^v)^{(R_v)}$ contains these sets for $Y = \Lambda_j^v$, together with their $2r_{\mathrm{loc}}$-neighbourhoods. Then $X(j)$ is a polymer, and $|X(j)| \le \mathsf{N}_{\mathrm{b}}$, with N_{b} depending only on d, ρ and C_v. (The subscript distinguishes this number of boxes from the matrix size N of $G \subset U(\mathsf{N})$.)

Monomials. For $s = (s_\square)_{\square\in\mathcal{P}} \in \mathbb{C}^{\mathcal{P}}$ and a function $n : \mathcal{P} \to \{0,1,2,\dots\}$ with finite support, put $s^n = \prod_\square s_\square^{n(\square)}$, with $\deg s^n = \sum_\square n(\square)$ and $\operatorname{supp} s^n = \{\square : n(\square) \ge 1\}$. For a union of boxes X, s^X denotes the monomial with $n = \mathbf{1}_X$. For $\kappa > 0$ let $\mathbb{D}_\kappa = \{s : |s_\square| \le e^\kappa \text{ for all } \square\}$. Repeated factors in a monomial are harmless; only the support and the degree enter.

40.1 Decay norms

The field spaces are those of Parts II–VII: scalar fields, gauge fields and 2-forms on $\mathbb{T}_\eta$, and unit-lattice fields on $\mathbb{T}$, including fields on the free bonds $\mathfrak{F}$. All of them decompose over unit blocks, with projections $\mathbf{1}_y$, $y \in \mathbb{T}$. For an operator F between two such spaces and $\gamma \ge 0$ put

$$|F|_\gamma = \max\Big\{ \sup_y \sum_{y'} e^{\gamma|y-y'|} \|\mathbf{1}_y F \mathbf{1}_{y'}\|,\ \sup_{y'} \sum_y e^{\gamma|y-y'|} \|\mathbf{1}_y F \mathbf{1}_{y'}\| \Big\}.$$

Lemma 40.1 (Decay norms). *(a) $\|F\| \le |F|_0 \le |F|_\gamma$, and $|FG|_\gamma \le |F|_\gamma |G|_\gamma$.*

(b) If F has decay (C,γ') with $\gamma' > \gamma$, then $|F|_\gamma \le C S_{\gamma'-\gamma}$, with $S_\beta = \sum_y e^{-\beta|y|}$.

(c) $|[F,h_j]|_{\gamma/2} \le C(\gamma M)^{-1} |F|_\gamma$ for $0 < \gamma \le 1$, with C depending only on d.

Proof. (a) The first inequality is the Schur test. For the second, use

$$\|\mathbf{1}_y FG \mathbf{1}_{y''}\| \le \sum_{y'} \|\mathbf{1}_y F \mathbf{1}_{y'}\|\, \|\mathbf{1}_{y'} G \mathbf{1}_{y''}\|, \qquad e^{\gamma|y-y''|} \le e^{\gamma|y-y'|} e^{\gamma|y'-y''|}.$$

(b) is immediate. (c) Write

$$\mathbf{1}_y [F,h_j] \mathbf{1}_{y'} = \mathbf{1}_y F \mathbf{1}_{y'} (h_j - h_j(c_y)) - (h_j - h_j(c_y)) \mathbf{1}_y F \mathbf{1}_{y'}.$$

On $B(y)$ we have $|h_j - h_j(c_y)| \le CM^{-1}$, and on $B(y')$ we have $|h_j - h_j(c_y)| \le CM^{-1}(1+|y-y'|)$. Hence $\|\mathbf{1}_y [F,h_j] \mathbf{1}_{y'}\| \le CM^{-1}(1+|y-y'|)\|\mathbf{1}_y F \mathbf{1}_{y'}\|$. Finally $(1+t)e^{-\gamma t/2} \le C\gamma^{-1}$. □

On the fine lattice D is not bounded uniformly in η. As in Parts II and III, operators are therefore estimated together with the versions carrying derivatives. For an operator F on fine fields put

$$|F|_\gamma^D = |F|_\gamma + |DF|_\gamma + |FD^T|_\gamma + |DFD^T|_\gamma,$$

where D stands for the covariant derivative appropriate to the spaces involved. For scalar operators followed by D on gauge fields, and so on, the obvious variants are meant.

40.2 Weakenable families

Definition 40.2 (Weakenable family)**.** Let $r \le M/16$. A *weakenable family with margin* r is a collection $(X_\omega, n_\omega, O_\omega)_{\omega\in\Omega}$ with the following properties:

- each X_ω is a polymer;
- n_ω is a function on $\mathcal{P}$ with $\operatorname{supp} s^{n_\omega}$ equal to the set of boxes of X_ω;
- each O_ω is an operator whose kernel is supported in $X_\omega^{(r)} \times X_\omega^{(r)}$, and which depends on the background, and on any other data, only on $X_\omega^{(r)}$.

We write $O(s) = \sum_\omega s^{n_\omega} O_\omega$, $\deg\omega = \deg s^{n_\omega}$, and

$$\|O\|_{\kappa,\gamma} = \sup_{\square\in\mathcal{P}} \sum_{\omega:\,\square\subset X_\omega} e^{\kappa\deg\omega}|O_\omega|_\gamma, \qquad \|O\|^D_{\kappa,\gamma} = \text{the same with } |O_\omega|^D_\gamma.$$

We say that $O(s)$ *represents* an operator O if $O(1) = O$, the series converging absolutely.

The data dependence is part of the definition. It is what makes the weakening also localize the dependence on the background. A family with margin r can always be converted to one with margin 0, on the level of boxes. Replace X_ω by $X_\omega^\sharp$, the union of the boxes meeting $X_\omega^{(r)}$, and multiply the monomial by $s^{X_\omega^\sharp\setminus X_\omega}$. Degrees grow by at most a factor 3^d, and every term then depends on the data, and acts on fields, only inside $X_\omega^\sharp$. We use this conversion whenever dependence inside a union of boxes, not a neighbourhood of it, is needed. This is the case for the factorization over disjoint polymers in the cluster expansion.

Additive local terms are the terms of a family that come neither from a walk (Theorem 41.4), nor from a product of families (Lemma 40.4), nor from a local operator applied to a family (Lemma 40.5). In Section 41 they are the identities in $R_X = \mathbf{1}_X - P_{\mathcal{V},X}$ and $\Pi_X = I - DS_X$ (level (L2)), the local part $d^2\mathcal{N}(0)|_X$ of W_2^X (level (L4)), and the constant $-a$ in $\Delta^{0,X} = \mathbf{\Gamma}^X[W_2^X] - a$ (the repetition of level (L4)). The constant r of level (L5) belongs to the local part of the operator there and is not a term of a family. Such a term T, a multiple of the identity or a local operator of range at most r, is split canonically, box by box. A bond, a plaquette or a unit bond is based at its initial point, and a point at itself. For a box $\square$ let $\mathbf{1}_\square$ be the projection onto the field components at the sites based in $\square$. Then

$$T = \sum_{\square\in\mathcal{P}} T\mathbf{1}_\square,$$

and $T\mathbf{1}_\square$ is a term with polymer $\square$ and monomial $s_\square$; it has margin at most r, a bound that does not depend on M. Since the boxes partition $\mathbb{T}_\eta$, this is a decomposition of T. It uses only the paving and the base points, so it involves no choice; in particular it commutes with the translations that preserve the paving. The split is used in the L^2 norms of this part. A split by indicators does not preserve Hölder tiers, so no tier bound is claimed for an additive term standing alone (Section 44). The constant $\frac{1}{1+t}$ in the determinant densities is split by the partition of unity h_j (proof of Corollary 41.6), which is canonical as well.

Lemma 40.3 (Basic properties)**.** *Let O be a weakenable family with $\|O\|_{\kappa,\gamma} < \infty$.*

(a) For $s \in \mathbb{D}_\kappa$ the series $O(s)$ converges absolutely, $|O(s)|_\gamma \le 2^d\|O\|_{\kappa,\gamma}$, and $s \mapsto O(s)$ is analytic in the interior of $\mathbb{D}_\kappa$. The same holds with $|\cdot|^D$.

(b) (Localization.) *Let Z be a union of boxes, and let $s_\square = 0$ for all $\square \not\subset Z$. Then only the terms with $X_\omega \subset Z$ contribute. $O(s)$ is block diagonal with respect to the $*$-components Z_i of Z, i.e. $\mathbf{1}_{Z_i^{(r)}} O(s) \mathbf{1}_{Z_{i'}^{(r)}} = 0$ for $i \ne i'$, and $O(s)$ vanishes off $Z^{(r)} \times Z^{(r)}$. The block on Z_i depends on the data only on $Z_i^{(r)}$, and on the weakening parameters only through $s_\square$, $\square \subset Z_i$.*

(c) (Covariance.) *If every O_ω is covariant under real gauge transformations, so is $O(s)$.*

Proof. (a) A term contributes to $\mathbf{1}_y O(s)\mathbf{1}_{y'}$ only if $y \in X_\omega^{(r)}$. Then X_ω contains one of the at most 2^d boxes within distance r of y. Hence

$$\sum_{y'} e^{\gamma|y-y'|}\|\mathbf{1}_y O(s)\mathbf{1}_{y'}\| \le \sum_{\square \text{ near } y} \sum_{\omega:\square\subset X_\omega} |s^{n_\omega}|\,|O_\omega|_\gamma \le 2^d\|O\|_{\kappa,\gamma},$$

and similarly for the column sums. The convergence is uniform on $\mathbb{D}_\kappa$, so the sum is analytic. (b) If $X_\omega \not\subset Z$, then $s^{n_\omega} = 0$. A polymer $X_\omega \subset Z$ lies in one component Z_i, and its monomial involves only the boxes of X_ω. Distinct components are at distance at least $M/4 > 2r$, so $X_\omega^{(r)}$ meets no other $Z_{i'}^{(r)}$. (c) is clear. □

Lemma 40.4 (Products)**.** *Let $F_1, \dots, F_n$ be weakenable families with margins at most r, and let $0 < \delta \le 1$. Index the product $F_1 \cdots F_n$ by the tuples $(\omega_1, \dots, \omega_n)$ for which $X_{\omega_i}^{(r)}$ and $X_{\omega_{i+1}}^{(r)}$ meet for every i. All other products vanish, and this index set is determined by the geometry alone, not by the background. Give each tuple the domain $\bigcup_i X_{\omega_i}$ and the monomial $\prod_i s^{n_{\omega_i}}$. This is a weakenable family with margin r, and*

$$\|F_1 \cdots F_n\|_{\kappa,\gamma} \le n \prod_{i=1}^{n} (C_d\delta^{-2}\|F_i\|_{\kappa+\delta,\gamma}).$$

Proof. If the product of the terms does not vanish, consecutive kernel supports meet. So $X_{\omega_i}^{(r)} \cap X_{\omega_{i+1}}^{(r)} \neq \emptyset$, and the boxes involved are within distance $2r < M/4$. Hence they touch, and the union is a polymer. The monomial has the right support, and the product of the terms has kernel in the union of the supports and depends only on the data there. By Lemma 40.1(a), $|\prod_i F_{i,\omega_i}|_\gamma \le \prod_i |F_{i,\omega_i}|_\gamma$.

Fix a box $\square$ and let i be the first index with $\square \subset X_{\omega_i}$. Given ω_i, sum over ω_{i+1}: it must meet $X_{\omega_i}^{(2r)}$, which is covered by at most $3^d|X_{\omega_i}|$ boxes. So the sum is at most $3^d|X_{\omega_i}|\,\|F_{i+1}\|_{\kappa,\gamma}$. Continue to the right, and in the same way to the left of i. Each factor receives at most two factors $|X_{\omega_l}| \le \deg\omega_l$, from its neighbours. Since $\sum_\omega e^{\kappa\deg\omega}(\deg\omega)^2|F_\omega| \le C\delta^{-2}\sum_\omega e^{(\kappa+\delta)\deg\omega}|F_\omega|$, each factor contributes at most $C_d\delta^{-2}\|F_l\|_{\kappa+\delta,\gamma}$. Summing over i gives the factor n. □

Lemma 40.5 (Local factors and commutators)**.** *Let O be a weakenable family with margin r.*

(a) Let $\mathcal{L}$ be a bounded operator of range r_0 that depends on the data only within distance r_0 of its support. Examples are $\mathbf{Q}$, $\mathbf{Q}^T$, Q', Q'^T, D_V, $\mathfrak{F}$, $W \in \mathcal{W}_{\mathrm{loc}}$, multiplication by h_j or m_X, and the elimination maps $\hat{Z}^1$, $\hat{Z}^2$. Then $\mathcal{L}O$ and $O\mathcal{L}$, with the same domains and monomials, are weakenable with margin $r + 2r_0$, and $\|\mathcal{L}O\|_{\kappa,\gamma} \le |\mathcal{L}|_\gamma\,\|O\|_{\kappa,\gamma}$, with $|\mathcal{L}|_\gamma \le e^{\gamma r_0'}|\mathcal{L}|_0$, where r_0' is the range in block units. The block Schur norm $|\mathcal{L}|_0$ is at most $C(d, r_0')\max_{y,y'}\|\mathbf{1}_y\mathcal{L}\mathbf{1}_{y'}\|$. It can exceed the operator norm $\|\mathcal{L}\|$, so the latter cannot be used here.

(b) $[O, h_j]$, with the same domains and monomials, is weakenable with $\|[O,h_j]\|_{\kappa,\gamma/2} \le C(\gamma M)^{-1}\|O\|_{\kappa,\gamma}$. The same holds with $|\cdot|^D$, up to an additional term $CM^{-1}\|O\|_{\kappa,\gamma}^D$ coming from $[D, h_j]$.

Proof. (a) Apply Lemma 40.1(a) term by term. The supports grow by the range of $\mathcal{L}$. (b) Apply Lemma 40.1(c) term by term. For the derivative versions use $D[O_\omega, h_j] = [DO_\omega, h_j] - [D, h_j]O_\omega$, where $[D, h_j]$ is a multiplication by differences of h_j, of size CM^{-1}. □

41 Weakened inverses by nested walks

In this section domains are arbitrary unions of blocks, together with the empty set. The family of domains is closed under intersection with the Λ_j^v. For domains $X' \subset X$ and a width $w > 0$, let $\Sigma_w(X, X')$ be the set of blocks within distance w of $X \setminus X'$, the *seam*. The width depends on the level of the construction: $w = 10M$ at the levels (L1)–(L4) below, and $w = R_0 + 10M + 2L$ at the level (L5), with $R_0 = R_0(L, M)$ from Proposition 37.6. We fix ρ so large that

$$(\rho - 1)M - w \geq \tfrac{1}{4}(\rho - 1)M \qquad \text{at every level,} \tag{41.1}$$

e.g. $\rho \geq 2R_0/M + 30$. This is possible because $R_0(L, M) \leq C_L M$. The number N_b of Section 40 then depends on L, but not on M.

41.1 Seam norms

For an operator F, a set of blocks Σ and $\gamma \geq 0$, let $|F|_{\gamma,\Sigma}$ be the decay norm of Section 40, with the weight $e^{\gamma|y-y'|}$ replaced by $e^{\gamma(|y-y'|+d_\Sigma(y)+d_\Sigma(y'))}$, where $d_\Sigma = \operatorname{dist}(\cdot, \Sigma)$. A kernel of seam form,

$$\|\mathbf{1}_y F \mathbf{1}_{y'}\| \leq C \sum_{w \in \Sigma} e^{-\gamma'(|y-w|+|w-y'|)},$$

has $|F|_{\gamma'/4,\Sigma} \leq C S^2_{\gamma'/2}$. Indeed $|y-y'| \leq |y-w|+|w-y'|$, $d_\Sigma(y) \leq |y-w|$ and $d_\Sigma(y') \leq |w-y'|$. Conversely, $|F|_{\gamma,\Sigma} < \infty$ gives $\|\mathbf{1}_A F \mathbf{1}_{A'}\| \leq |F|_{\gamma,\Sigma} e^{-\gamma(d_\Sigma(A)+d_\Sigma(A'))}$. If F is supported on $\Sigma \times \mathbb{T}_\eta$ or $\mathbb{T}_\eta \times \Sigma$, then $|F|_{\gamma,\Sigma} \leq |F|_{2\gamma}$. Since $d_\Sigma(y) \leq |y - y_1| + d_\Sigma(y_1)$,

$$|AFB|_{\gamma,\Sigma} \leq |A|_{2\gamma}\, |F|_{\gamma,\Sigma}\, |B|_{2\gamma}. \tag{41.2}$$

For weakenable families, $\|O\|_{\kappa,\gamma,\Sigma}$ is defined with $|O_\omega|_{\gamma,\Sigma}$. By (41.2), Lemma 40.4 holds with one factor measured in the seam norm and the others in $|\cdot|_{2\gamma}$.

Factorized families. On the fine lattice the non-local parts of the operators contain derivatives, and are unbounded uniformly in η. As in Parts II and III, they are written as finite sums of products with derivatives outside. Here we write them as $\sum_i X_i^T M_i Y_i$ with $X_i^T \in \{1, D^T\}$ and $Y_i \in \mathfrak{D} \cup \{1\}$. A left gradient D, where present, is made part of the middle factor M_i. A *factorized weakenable family* is such a sum in which each M_i is a weakenable family. Its norm is $\sum_i \|M_i\|_{\kappa,\gamma}$: the middle factors are measured without further derivatives. The derivatives X_i^T, Y_i are kept outside the terms. A left D^T is absorbed by the factor to its left in a product, and Y_i by the local propagator to its right, whose derivative norms are controlled by (A2) below.

Derivative norms of the middle factors are neither used nor available in general. At level (L3) the middle factor $M = WDG'R$ has DM containing the derivatives of the coefficients of W, which are only Hölder continuous: for $W = w_q I$ with $w_q(x) = c\varepsilon q^{-\alpha} \sin(qx_1)$ and c small, a member of $\mathcal{W}_{\mathrm{loc}}(\varepsilon)$, $|DM|$ is at least of order $\varepsilon q^{1-\alpha}$ for $1 \ll q \lesssim \eta^{-1}$. In the same way, derivatives of the curvature factor $\mathfrak{F}$ are not controlled at complex references.

Definition 41.1 (Compatible representations)**.** Factorized weakenable representations $O^X(s) = \sum_i X_i^T M_i^X(s) Y_i$ of operators O^X on $\ell^2(X)$, one for each domain X, are *compatible*, with constants (C_O, γ) and seam width w, if:

- the index sets do not depend on X, matched terms carry the same domain and monomial, and a term vanishes on X if one of its local domains does not meet X;
- $\|M_i^X\|_{\kappa,2\gamma} \leq C_O$ for all X and i;
- for $X' \subset X$, the differences $\mathbf{1}_{X'} M_i^X(s) \mathbf{1}_{X'} - M_i^{X'}(s)$ satisfy $\|\cdot\|_{\kappa,\gamma,\Sigma_w(X,X')} \leq C_O$.

The last condition is the weakened form of the comparison of domains (Theorem 18.2(e), Theorem 19.10(a), Proposition 35.1(d), Proposition 37.6(c)). On 2-forms, $\mathbf{1}_X$ denotes the projection onto the fields vanishing on the plaquettes with no corner in X.

41.2 The walk

Let $\mathfrak{A}^X$ be operators on $\ell^2(X)$, one for each domain X, with $\mathfrak{A} = \mathfrak{A}^{\mathbb{T}_\eta}$, satisfying:

(A1) *Structure.* $\mathfrak{A}^X = \mathfrak{A}^X_{\mathrm{loc}} + \kappa_m m_X + \mathfrak{N}^X$, where:

- $\mathfrak{A}^X_{\mathrm{loc}}$ is a local operator of range r_0, possibly of second order and containing a block-local averaging term, which depends on X only through cut terms at ∂X. For example, $\mathbf{1}_X D D^T \mathbf{1}_X$ differs from $D\mathbf{1}_X D^T$ by a cut term.
- $\kappa_m m_X$ is the mass term of Section 19 in the gauge case, and 0 otherwise.
- $\mathfrak{N}^X$ is the non-local part.

(A2) *Local propagators.* For every j and every domain X, $\mathfrak{A}^{\Lambda_j^v \cap X}$ is invertible on $\ell^2(\Lambda_j^v \cap X)$. Its inverse G_j^X, extended by zero, depends on the data only on $(\Lambda_j^v \cap X)^\wedge$, satisfies $|G_j^X|^D_{\gamma_0} \le C_G$, and obeys the comparison of domains in seam form with width w, with derivatives, and constants (C_G, γ_0). Here, and in the derivative norms of the outputs below, D stands for every first-order derivative that occurs as a factor Y_i, X_i^T or as an external derivative. In the gauge case these are D and the divergence $D^T_{X_j}$ of the local domain $X_j = \Lambda_j^v \cap X$. The bounds for $D^T_{X_j} G_j^X$ and $D^T_{X_j} G_j^X D^T$ come from the norm $\|\!| \cdot |\!\|_{X_j,+}$ (Theorem 19.7(a)). In the gauge case (A2) also includes the full gradient ∇G_j^X, $\nabla G_j^X D^T$, and the comparison of domains for these and for the torus divergence (Theorem 19.10(a)). With the Dirichlet conditions of Section 19, even the torus divergence $D^T G_j^X$ is bounded. The regrouping in the proof of Lemma 41.3 is kept, since it needs only the divergence of the local domain.

(A3) *Non-local part.* $\mathfrak{N}^X = \sum_i X_i^T \mathfrak{N}_i^X Y_i$ has compatible factorized representations $\mathfrak{N}^X(s)$ with constants $(C_{\mathfrak{N}}, \gamma_0)$ at $\kappa + 2\delta$. Lemma 41.3 at κ' uses $\mathfrak{N}$ at $\kappa' + \delta$, and Theorem 41.4 at κ uses Lemma 41.3 at $\kappa + \delta$. So each walk costs 2δ in the weakening rate.

Lemma 41.2 (Parametrix on domains)**.** *Under (A1), (A2), with $K^X = \sum_j h_j G_j^X$ on $\ell^2(X)$,*

$$\mathfrak{A}^X K^X = I - \sum_j \mathcal{R}_j^X,$$
$$\mathcal{R}_j^X = -\Big([\mathfrak{A}^X_{\mathrm{loc}}, h_j] + [\mathfrak{N}^X, h_j] + h_j (\mathfrak{N}^X \mathbf{1}_{\Lambda_j^v \cap X} - \mathbf{1}_{\Lambda_j^v \cap X} \mathfrak{N}^{\Lambda_j^v \cap X} \mathbf{1}_{\Lambda_j^v \cap X}) \Big) G_j^X.$$

Proof. As in Lemma 20.1, $\mathfrak{A}^X h_j G_j^X = h_j \mathfrak{A}^X G_j^X + [\mathfrak{A}^X, h_j] G_j^X$, and m_X commutes with h_j.

- *Local parts.* On $\operatorname{supp} h_j$ the local parts of $\mathfrak{A}^X$ and $\mathfrak{A}^{\Lambda_j^v \cap X}$ agree on fields supported in $\Lambda_j^v \cap X$. The only possible difference is a cut term at $\partial \Lambda_j^v$, which is at distance at least $(\rho - 1)M > r_0$ from $\operatorname{supp} h_j \subset K_j$.
- *Mass profiles.* These agree on K_j. If $K_i \cap K_j \neq \emptyset$, then $\Lambda_i \subset \Lambda_j^v$ for $\rho \ge 6$. So $\Lambda_i \subset X$ iff $\Lambda_i \subset \Lambda_j^v \cap X$, i.e. $i \in J(X)$ iff $i \in J(\Lambda_j^v \cap X)$, and only such i contribute to m on K_j.

Hence $h_j \mathfrak{A}^X G_j^X = h_j + h_j (\mathfrak{N}^X \mathbf{1}_{\Lambda_j^v \cap X} - \mathbf{1} \mathfrak{N}^{\Lambda_j^v \cap X} \mathbf{1}) G_j^X$. Sum over j, using $\sum_j h_j = 1$. □

Since $h_j \mathbf{1}_{X \setminus (\Lambda_j^v \cap X)} = 0$, the seam term equals $h_j \mathbf{1}_{X'} (\mathfrak{N}^X - \mathfrak{N}^{X'}) \mathbf{1}_{X'} G_j^X$ with $X' = \Lambda_j^v \cap X$.

The weakened step. Insert the representations of (A3), and give each step the monomial $s^{X(j)}$. Each term of $\mathcal{R}_j^X$ is then of the form $D^T A + B$, with the left derivative coming from X_i^T, and

$$\begin{aligned}\mathcal{R}^X(s) &= -\sum_j s^{X(j)}\Big([\mathfrak{A}_{\mathrm{loc}}^X, h_j] + [\mathfrak{N}^X(s), h_j] + h_j \mathbf{1}_{X'}(\mathfrak{N}^X(s) - \mathfrak{N}^{X'}(s))\mathbf{1}_{X'}\Big) G_j^X \\ &= D^T \mathcal{A}^X(s) + \mathcal{B}^X(s),\end{aligned}$$

with weakenable families $\mathcal{A}^X$ and $\mathcal{B}^X$ ($\mathcal{A}^X = 0$ at the scalar levels). Each term has domain $X(j) \cup X_\omega$. This is a polymer, since X_ω meets $X(j)$ for every non-vanishing term. Put

$$G^X(s) = \sum_{n\geq 0} \Big(\sum_j s^{X(j)} h_j G_j^X\Big) \mathcal{R}^X(s)^n.$$

In a product, every D^T in front of $\mathcal{A}$ is absorbed by the factor on its left, which is one of $h_j G_j^X D^T$, $\mathcal{A}^X D^T$, $\mathcal{B}^X D^T$. So the products are products of the bounded factors

$$\mathcal{F}_0 = \sum_j s^{X(j)} h_j G_j^X, \quad \mathcal{F}_0 D^T, \quad \mathcal{A}^X, \quad \mathcal{A}^X D^T, \quad \mathcal{B}^X, \quad \mathcal{B}^X D^T.$$

This replaces the use of the identity of Lemma 36.6. That identity needs a real background, while the weakened walks are needed for complex perturbations as well.

Lemma 41.3 (Step operators)**.** *Assume (A1)–(A3) and*

$$|[\mathfrak{A}_{\mathrm{loc}}^X, h_j] G_j^X|_{\gamma_0} + |[\mathfrak{A}_{\mathrm{loc}}^X, h_j] G_j^X D^T|_{\gamma_0} \leq C_G C_{\mathrm{loc}} M^{-1}.$$

Then, with $\gamma = \gamma_0/8$ and $\gamma_0 \leq 1$,

$$\begin{aligned}&\|\mathcal{A}^X\|_{\kappa,\gamma} + \|\mathcal{A}^X D^T\|_{\kappa,\gamma} + \|\mathcal{B}^X\|_{\kappa,\gamma} + \|\mathcal{B}^X D^T\|_{\kappa,\gamma} \leq \theta, \\ &\theta := C\, e^{\kappa \mathsf{N}_{\mathrm{b}}}\Big(C_G C_{\mathrm{loc}} M^{-1} + C\delta^{-1}(\gamma_0 M)^{-1} C_{\mathfrak{N}} C_G + C\delta^{-1} C_{\mathfrak{N}} C_G e^{-\gamma_0(\rho-1)M/16}\Big).\end{aligned}$$

The constant C includes the number of j with $\square \subset X(j)$, which is bounded independently of M.

Proof. The local commutator. This is the hypothesis, times $e^{\kappa \mathsf{N}_{\mathrm{b}}}$ from the monomial.

Regrouping. Let $X' = \Lambda_j^v \cap X$. Consider a term $X_i^T M_i^X Y_i^X$ of $\mathfrak{N}^X$ with $Y_i^X = D_X^T$; the matching term of $\mathfrak{N}^{X'}$ has $Y_i^{X'} = D_{X'}^T$. Put $\sigma_j = (D_X^T - D_{X'}^T) G_j$. It is supported at points of $X \setminus X'$, so $h_j \sigma_j = 0$, and $h_j D_X^T G_j = h_j D_{X'}^T G_j$. Then the non-local commutator and the seam term of this term are

$$\begin{aligned}[X_i^T M_i^X D_X^T, h_j] G_j &= [X_i^T M_i^X, h_j] D_{X'}^T G_j + X_i^T M_i^X [D_X^T, h_j] G_j - h_j X_i^T M_i^X \sigma_j, \\ h_j (X_i^T M_i^X D_X^T - X_i^T M_i^{X'} D_{X'}^T) G_j &= h_j X_i^T (M_i^X - M_i^{X'}) D_{X'}^T G_j + h_j X_i^T M_i^X \sigma_j,\end{aligned}$$

and the terms with σ_j cancel exactly. We use the steps in this regrouped form. Then only $D_{X'}^T G_j$ and $D_{X'}^T G_j D^T$ enter, which (A2) bounds, and $[D_X^T, h_j]$ is a multiplication by differences of h_j.

The non-local commutator. It is bounded by Lemma 40.5(b), applied to the factors M_i, together with the commutators $[X_i^T, h_j]$, $[Y_i, h_j]$, which are of size CM^{-1}. The derivatives Y_i are absorbed by G_j^X, using $|G_j^X|^D$. In the terms $X_i^T [M_i, h_j] Y_i G_j$ and $X_i^T M_i [Y_i, h_j] G_j$ a left D^T stays in front and gives the form $D^T \mathcal{A}$; then $\mathcal{A} D^T$ contains $Y_i G_j D^T$, again controlled by (A2). So only the norms $\|M_i\|$ enter, never derivatives of M_i.

The seam term. By Definition 41.1, the differences $\mathbf{1}_{X'}(\mathfrak{N}_i^X - \mathfrak{N}_i^{X'})\mathbf{1}_{X'}$ are of seam form for $\Sigma_w(X, X')$. By (41.1), every block of this seam lies at distance at least $\frac{1}{4}(\rho-1)M$ from K_j. The factor h_j restricts the output to K_j, so the seam norm and (41.2) give the factor $e^{-\gamma_0(\rho-1)M/16}$.

Counting. The terms of the non-local commutator and of the seam term built from a term ω of $\mathfrak{N}$ carry the monomial $s^{X(j)}s^{n_\omega}$, of degree $|X(j)| + \deg\omega \le \mathsf{N}_{\mathrm{b}} + \deg\omega$. So the factor $e^{\kappa\mathsf{N}_{\mathrm{b}}}$ multiplies them as well. For a fixed term ω of $\mathfrak{N}$, the number of j for which the commutator or the seam term is non-zero is at most $C|X_\omega| \le C\deg\omega$. This is absorbed by evaluating $\|\mathfrak{N}\|$ at $\kappa+\delta$, at the cost of δ^{-1}. The versions with D^T on the right follow from (A2) and (A3) in the same way.

The power of C_G. Each term of $\mathcal{A}^X$, $\mathcal{B}^X$, $\mathcal{A}^X D^T$ and $\mathcal{B}^X D^T$ contains exactly one factor of the form $PG_j^X P'$ with $P, P' \in \{1, D, D^T\}$, which (A2) bounds by C_G; so θ is linear in C_G. This matters at level (L5), where $C_G = C_L(\frac{1}{16}c_L + r)^{-1} < 1$ for large r. □

Theorem 41.4 (Weakened inverses)**.** *Assume (A1)–(A3) and the hypothesis of Lemma 41.3. For (b) assume also the comparison of domains of the local commutators: for domains $X' \subset X$ the zero extensions of $[\mathfrak{A}_{\mathrm{loc}}^X, h_j]G_j^X - [\mathfrak{A}_{\mathrm{loc}}^{X'}, h_j]G_j^{X'}$ and of $[\mathfrak{A}_{\mathrm{loc}}^X, h_j]G_j^X D^T - [\mathfrak{A}_{\mathrm{loc}}^{X'}, h_j]G_j^{X'} D^T$ are of seam form for $\Sigma_w(X, X')$, with constants $(C_G C_{\mathrm{loc}}, \gamma_0)$. Let $\delta \in (0, 1]$. If $M \ge M_1(\kappa, \delta, C_G, C_{\mathrm{loc}}, C_{\mathfrak{N}}, \gamma_0, \rho, C_v)$, then for every domain X:*

(a) $G^X(s)$ is a weakenable family with margin at most $r + 2r_0$ that represents $(\mathfrak{A}^X)^{-1}$, with

$$\|G^X\|^D_{\kappa,\gamma_0/8} \le C(\kappa,\delta)e^{(\kappa+\delta)\mathsf{N}_{\mathrm{b}}}C_G;$$

(b) the representations of $G^X(s)$, $DG^X(s)$, $G^X(s)D^T$ and $DG^X(s)D^T$ are compatible with constants $(C_{\mathrm{cpt}}, \gamma_0/32)$ and seam width w. Here C_{cpt} depends only on κ, δ, γ_0, C_G, C_{loc}, $C_{\mathfrak{N}}$ and on ρ and C_v (through N_{b}); it may be taken independent of $M \ge M_1$, of X and of k, n, η;

(c) each term is covariant under real gauge transformations if the G_j^X and the terms of $\mathfrak{N}^X(s)$ are.

Proof. (a) After the regrouping of the derivatives, the n-th term is a sum of 2^n products of $n+1$ bounded factors from the list above. By Lemma 40.4 and Lemma 41.3 at $\kappa+\delta$,

$$\left\|\mathcal{F}_0\mathcal{R}^X(s)^n\right\|_{\kappa,\gamma} \le (n+1)\big(C\delta^{-2}e^{(\kappa+\delta)\mathsf{N}_{\mathrm{b}}}C_G\big)\big(2C\delta^{-2}\theta\big)^n.$$

The series converges if $2C\delta^{-2}\theta \le \frac{1}{2}$, which holds for M large, by the form of θ. With derivatives, D falls on the first factor, which is controlled by $|DG_j^X|$ and $[D, h_j]$. D^T falls on the last factor, controlled by $\|\mathcal{A}^X D^T\|$, $\|\mathcal{B}^X D^T\|$ or $|G_j^X D^T|$. At $s = 1$ the series is $K^X \sum_n (\sum_j \mathcal{R}_j^X)^n$, which by Lemma 41.2 is a right inverse of $\mathfrak{A}^X$. The space $\ell^2(X)$ is finite-dimensional, since the torus is finite. So this right inverse of $\mathfrak{A}^X$ is its inverse. No coercivity or positivity of any operator at $s \ne 1$ is used: the series is the definition of $G^X(s)$. Data locality and the domains hold term by term.

(b) Use the same index set on X and on $X' \subset X$. On X' a term vanishes if one of its local domains $\Lambda_j^v \cap X'$ is empty. The sets $X(j)$ and the monomials do not depend on the domain, so matched terms have the same domain and monomial. On each field space let $P = \mathbf{1}_{X'}$ be the orthogonal projection onto the fields of X' (Definition 17.1; on 2-forms, the fields vanishing on the plaquettes with no corner in X'), and $Q = \mathbf{1}_X - P$. On gauge fields Q is not $\mathbf{1}_{X\setminus X'}$: a bond from X' to $X \setminus X'$ is a field of both domains. The fields in the range of Q lie within one lattice step of $X \setminus X'$. After the regrouping, a term on X is a product of bounded pieces: the h_j; the local propagators G_j^X with the derivatives of (A2), among them the external D

of the first factor and the external D^T of the last; the local commutators $[\mathfrak{A}^X_{\mathrm{loc}}, h_j]G^X_j$, also with D^T on the right; the middle factors of the non-local parts, also commuted with h_j; and the multiplications $[X_i^T, h_j]$, $[Y_i, h_j]$ and, with an external D, $[D, h_j]$. Every factor of a word ends with a local propagator, possibly with a derivative of (A2) on its left, so the left D^T of the next step is absorbed into it, as in $G^X_j D^T$ or $[\mathfrak{A}^X_{\mathrm{loc}}, h_j]G^X_j D^T$. Insert $P+Q$ between the pieces of a term on X, and telescope, using $P\Phi' P = \Phi'$ for the pieces Φ' on X'. The difference $P(\text{term on } X)P - (\text{term on } X')$ is a sum of products, each containing one of the following factors:

- a difference of local propagators $P(G^X_j - G^{X'}_j)P$, with the derivatives applied before the comparison, of seam form by (A2), since $\Lambda^v_j \cap X' \subset \Lambda^v_j \cap X$ and $(\Lambda^v_j \cap X)\setminus(\Lambda^v_j \cap X') \subset X \setminus X'$;
- a difference of non-local parts, on X and X' or on $\Lambda^v_j \cap X$ and $\Lambda^v_j \cap X'$, of seam form by (A3), also after commutation with h_j, which commutes with P;
- a difference of local commutators $P[\mathfrak{A}^X_{\mathrm{loc}}, h_j]G^X_j P - [\mathfrak{A}^{X'}_{\mathrm{loc}}, h_j]G^{X'}_j$, also with D^T on the right, of seam form by the hypothesis of (b);
- a cross term $P\Phi Q$ of a piece Φ, a difference $P\Phi P - \Phi'$ of the multiplications $[X_i^T, h_j]$, $[Y_i, h_j]$, $[D, h_j]$, or a piece containing G^X_j whose domain $\Lambda^v_j \cap X'$ is empty. These are bounded and supported on the seam, where $|\cdot|_{\gamma,\Sigma} \le |\cdot|_{2\gamma}$.

By (41.2) and Lemma 40.4, each such product is bounded in the seam norm. No bare derivative stands next to P or Q: every derivative is part of a bounded piece. A local commutator $[\mathfrak{A}_{\mathrm{loc}}, h_j]$ never stands without its propagator: at (L1) and (L3) it is a differential operator, not bounded uniformly in η. A term with n steps has at most $C(n+1)$ pieces, so the sum over the position of the distinguished factor costs a factor $C(n+1)$, which is absorbed. The steps are taken in the regrouped form of the proof of Lemma 41.3 on both domains. On X' the divergences of the local domains agree at the points of X', so their differences are again of seam form: $P\mathbf{1}_{\Lambda^v_j \cap X} D^T G^X_j P - \mathbf{1}_{\Lambda^v_j \cap X'} D^T G^{X'}_j = \mathbf{1}_{\Lambda^v_j \cap X'} D^T (G^X_j - G^{X'}_j)P$, which (A2) bounds through the torus divergence. The constant C_{cpt} collects the bound of (a), which is proportional to C_G, and the seam constants of (A2), (A3) and of the local commutators. It may be taken as its value at $M = M_1$, since the bounds of the series decrease in M. It cannot be taken independent of C_G: for $\mathfrak{A}^X = tI_X$ on unit-lattice fields and $\mathfrak{N} = 0$ the family is $t^{-1}\sum_j s^{X(j)} h_j$, whose norm is at least $2^{-d}t^{-1}$ by Lemma 40.3(a).

(c) is clear. □

41.3 The levels

Theorem 41.4 is applied at five levels, and (L3), (L4) are applied twice, without and with the potential W_2. At each level the hypotheses follow from the previous levels and from Parts III, VII and VIII. There is no circularity: the order is

$$\text{(L1)} \to \text{(L2)} \to \text{(L3), (L4) with } W = 0 \to \omega_0,\ W_2 \to \text{(L3), (L4) with } W_2 \to \text{(L5)}.$$

Every use of Lemma 40.4 costs δ in κ, every use of Theorem 41.4 costs 2δ, each costs a fixed factor in γ, and there are boundedly many of them. The parameters are chosen in the order L, then κ (see below), then ρ by (41.1), then M large depending on all of these, and finally ε small depending on M. The constants C_G, C_{loc}, $C_{\mathfrak{N}}$ and the rates γ_0 of all levels do not depend on M: for $M \ge M_0$, and, at a level whose non-local parts come from Theorem 41.4(b) at earlier levels, for M at least the thresholds M_1 of those levels, with C_{cpt} taken at these thresholds. So the thresholds are fixed level by level, in the order above, and M_1 denotes the largest of them. At (L1)–(L3) this holds for the torus operators by Lemma 31.2 and Step 0 of the proof of Lemma 31.3, for the Dirichlet operators by Theorem 19.6 (Remark 31.4(c)),

and for the non-local parts by Theorem 41.4(b) at the earlier levels. At (L4) and (L5) the inputs are Propositions 35.1, 37.4 and 37.6. They are built from these Dirichlet operators, from the averaging operators, whose proofs do not use M (Section 46.1, Setting), and from the constants c_L, C_L, C'_L, γ_1 and γ_C of Part VII, which do not depend on $M \geq M_0$ (Remark 34.6). Theorem 19.10(b), the one result of Part III whose constant contains e^{CM}, does not enter the constants of (A2): (A2) uses Theorem 19.10(a), and (b) enters only through the coercivity of Part VII, in Lemma 34.3, where it is absorbed by the choice of M'_L and ε_L. The margin $R_0 \leq C_L M$ enters only ρ and the seam width. So each threshold M_1 is a number, fixed before M. The smallness of ε is another matter. The thresholds of Parts II and III do not depend on M, and ε_{B} of Lemma 31.3(c) may be taken at $M = M_0$ (Lemma 31.2). The thresholds of Parts VI–VIII depend on M, for instance a_1 of Theorem 30.3 (whose constant B_3 also depends on M), a_δ of Proposition 31.5 and $\varepsilon_L \leq \varepsilon_{\mathrm{cmp}}(M'_L)$ with $M'_L \geq M$ (Theorem 34.4); this is why ε is chosen last.

(L1) *Scalar operators.*

- $\mathfrak{A}'^X = \mathbf{1}_X(D^TD + aQ'^TQ')\mathbf{1}_X$, with $\mathfrak{N} = 0$, since Q'^TQ' is block-local and is counted in $\mathfrak{A}_{\mathrm{loc}}$.
- (A2) is Theorem 18.2(b),(e), on the domains $\Lambda_j^v \cap X$.
- The hypothesis of Lemma 41.3 is $|[\mathfrak{A}', h_j]G'_j|_{\gamma_0} \leq CM^{-1}$, in the decay norm, with D^T on the right, from Theorem 20.2(a), whose proof applies verbatim on the domains $\Lambda_j^v \cap X$. The comparison of the local commutators in Theorem 41.4(b) holds as well. Since h_j commutes with $\mathbf{1}_X$ and $\mathbf{1}_X\mathbf{1}_{X'} = \mathbf{1}_{X'}$, the difference is $\mathbf{1}_X[\mathfrak{A}', h_j](G'_{\Lambda_j^v \cap X} - G'_{\Lambda_j^v \cap X'}) + (\mathbf{1}_X - \mathbf{1}_{X'})[\mathfrak{A}', h_j]G'_{\Lambda_j^v \cap X'}$, with $\mathfrak{A}' = D^TD + aQ'^TQ'$. By the discrete Leibniz rule, $[\mathfrak{A}', h_j]u$ is a sum of multiplications of size CM^{-1} applied to u and to Du, up to shifts by η, and of $a[Q'^TQ', h_j]u$. So the first term is of seam form by Theorem 18.2(e) for G' and DG', also with D^T on the right. The second is bounded by the same expansion and Theorem 18.2(b), and supported at the points of $X \setminus X'$.

This gives weakened $G'^X(s)$, together with $DG'^X(s)$, $G'^X(s)D^T$ and $DG'^X(s)D^T$. The walk has only local steps, as in [15, Lemma 6].

(L2) *Scalar averaged operators.*

- $Q'G'^XQ'^T$ and $M'^X = Q'(G'^X)^2Q'^T$ on $\ell^2(X_1)$, with $\mathfrak{A}_{\mathrm{loc}} = 0$.
- Their weakened representations come from (L1), Lemma 40.4 and Lemma 40.5(a). Compatibility follows from (L1).
- (A2) holds for the Dirichlet versions on $\Lambda_j^v \cap X$, by Theorem 18.2(c),(e).

Theorem 41.4 gives $\Gamma'^X(s)$ and $(M'^X)^{-1}(s)$. Products then give H'^X, $P_{\mathcal{V}}^X$, R_X, S_X, DS_X and Π_X, with derivatives.

(L3) *Gauge-field operators.*

- $\mathfrak{A}^{X,\kappa_m}$, with $\mathfrak{A}^X_{\mathrm{loc}} = \mathbf{1}_X(D^TD + DD^T + W + a\mathbf{Q}^T\mathbf{Q})\mathbf{1}_X$ on $\ell^2_1(X)$, the Dirichlet restriction of the torus operator, which includes the exterior divergence term $\mathfrak{A}^{\mathrm{ext}}_X$, and mass term $\kappa_m m_X$.
- Using $E = 0$ as in Lemma 36.4,

$$\mathfrak{N}^X = -DP_{\mathcal{V},X}D^T_X - \mathfrak{B}_X DS_X - (\mathfrak{B}_X DS_X)^T + S^T_X D^T_X \mathfrak{B}_X DS_X,$$

with $\mathfrak{B}_X D = D^T \mathfrak{F} + W D$. This is a factorized family with $X_i^T \in \{1, D^T\}$. Its terms are listed in the proof of Lemma 44.9. For example, $D^T \mathfrak{F} S_X$ has $X_i^T = D^T$, $M_i = \mathfrak{F} G'_X R_X$ and $Y_i = D_X^T$; $-D P_{\mathcal{V},X} D_X^T$ has $X_i^T = 1$, $M_i = D P_{\mathcal{V},X}$ and $Y_i = D_X^T$. Every middle factor is bounded in the plain norm by (L1), (L2) with derivatives, by the sup bound on $\mathfrak{F}$, and by the norm of W: the sup bound of its coefficients for $W \in \mathcal{W}_{\mathrm{loc}}$, the family norm for W_2. A left gradient always falls on a scalar propagator, never on the coefficients of $\mathfrak{F}$ or W, so no derivative of these coefficients is needed. The compatibility of the middle factors follows from Theorem 41.4(b) at (L1), (L2), which includes the derivatives of G'^X and $P_{\mathcal{V}}^X$, and from the locality of $\mathfrak{F}$ and W.

- (A2) is Theorem 19.7(a) and Theorem 19.10(a) on the domains $\Lambda_j^v \cap X$.
- The hypothesis of Lemma 41.3 holds on every domain. The local commutator is $[\mathfrak{A}_{\mathrm{loc}}^X, h_j] = \mathbf{1}_X [D^T D + D D^T + W + a \mathbf{Q}^T \mathbf{Q}, h_j] \mathbf{1}_X$, since h_j commutes with $\mathbf{1}_X$. The terms $D^T [D, h_j] G_j$ and $D [D^T, h_j] G_j$ need the full gradient of G_j on the support of ∂h_j, also where it meets ∂X. This is bounded by Lemma 19.3 and Theorem 19.7(a). The same holds with D^T on the right, and at complex perturbations. On the torus one may also use the localized Hodge identity at interior blocks, as in the proof of Lemma 44.4. The comparison of the local commutators in Theorem 41.4(b) follows from the same expansion. With $\mathfrak{A}_{\mathrm{loc}}^{\mathbb{T}_\eta} = D^T D + D D^T + W + a \mathbf{Q}^T \mathbf{Q}$, the difference is $\mathbf{1}_X [\mathfrak{A}_{\mathrm{loc}}^{\mathbb{T}_\eta}, h_j] (G_j^X - G_j^{X'}) + (\mathbf{1}_X - \mathbf{1}_{X'}) [\mathfrak{A}_{\mathrm{loc}}^{\mathbb{T}_\eta}, h_j] G_j^{X'}$. In the first term $[\mathfrak{A}_{\mathrm{loc}}^{\mathbb{T}_\eta}, h_j]$ acts through D, the torus divergence D^T, ∇, $\mathbf{Q}$ and W, with coefficients and local maps of size $C M^{-1}$; Theorem 19.10(a) compares these derivatives of the local propagators, also with D^T on the right, in seam form. In the second term $\mathbf{1}_X - \mathbf{1}_{X'}$ projects onto the bonds within one lattice step of $X \setminus X'$, and $[\mathfrak{A}_{\mathrm{loc}}^{\mathbb{T}_\eta}, h_j] G_j^{X'}$ is bounded as above.

The result is $\mathbf{G}^X(s)$, together with $D \mathbf{G}^X(s)$, $\mathbf{G}^X(s) D^T$ and $D \mathbf{G}^X(s) D^T$.

(L4) *Averaged gauge operators and the Hessian.*

- $\mathbf{Q}_X \mathbf{G}^X \mathbf{Q}_X^T$ on $\ell^2(\mathfrak{B}(X))$, with $\mathfrak{A}_{\mathrm{loc}} = 0$.
- (A2) is Proposition 35.1(a),(d), and (e) at complex perturbations of the background.

This gives $\mathbf{\Gamma}^X(s)$, hence $\mathbf{H}^X(s)$, $\mathbf{C}^X(s)$ and $\omega_0^X(s) = \mathbf{H}_0^X(s)^T \mathbf{1}_X J_U$. Then

$$W_2^X(s) = d^2 \mathcal{N}(0)\big|_X - \omega_0^X(s) \cdot d^2 C_k(0)$$

is a weakenable family. Its first term is local, and the second is $\omega_0^X(s)$ times the bounded local kernel of $d^2 C_k(0)$ (Proposition 22.6(c)).

For complex perturbations of the background, J_U is not bounded uniformly in η. There one writes $\langle b, \omega_0 \rangle = \langle J_U, \mathbf{H}_0 b \rangle$ with $J_U = D^T F_U^s$ (Lemma 23.2), and moves the derivative onto $\mathbf{H}_0 b$, using $|D \mathbf{H}_0|$. The same remark applies to Proposition 37.4(c).

In the repetition of (L3) the potential $W_2^X(s)$ is put into the non-local part $\mathfrak{N}$, with $X_i = Y_i = 1$, not into $\mathfrak{A}_{\mathrm{loc}}$. It depends on X globally, through ω_0^X, so it does not satisfy the requirement of (A1) on local parts. Its plain norm and its compatibility follow from (L3), (L4) with $W = 0$ and Theorem 41.4(b). In this repetition the local propagators are the inverses of the Dirichlet operators with the potential $W_2^{X_j}$, $X_j = \Lambda_j^v \cap X$, which depends on U on all of $\hat{X}_j$. So (A2) is not Theorems 19.7(a) and 19.10(a) literally. Existence, Re-coercivity and decay come from Proposition 19.11, whose hypotheses $W_2^{X_j}$ satisfies (Proposition 37.4(b),(c)); the comparison of domains

comes from Propositions 35.1(d), which allows different potentials on the two domains, and 37.4(d), as stated in Section 19. These comparisons include $P \in \{D, D^T, \nabla\}$, so the comparison of the local commutators in Theorem 41.4(b) follows as at the first pass of (L3); the local part is the same. Repeating (L3) and (L4) with the potential W_2^X gives $\mathbf{\Gamma}^X[W_2^X](s)$ and the Dirichlet Hessians $\Delta^{0,X}(s)$. Through the local maps $\hat{Z}^1$, $\hat{Z}^2$ it gives the operators $\Delta_k^{(X)}(s)$ and $\Delta_Y(s)$ of Section 37.

(L5) *The fluctuation operator.*

- $\mathfrak{A}^X = \Delta_X + r$ on $\ell^2(\mathfrak{F}(X))$, for $r \geq 0$, with $\mathfrak{A}_{\mathrm{loc}} = r$ and $\mathfrak{N}^X = \Delta_X$ from (L4). Since Δ_X is built from $\Delta_k^{(X^+)}$, the representations of (L4) are used on the domains $X^+ = X^{(R_0)}$.
- (A2) is Proposition 37.6(c),(d), with the seam width $w = R_0 + 10M + 2L$. The local propagators on $Y = \Lambda_j^v \cap X$ depend on the background on $\widehat{Y^+}$ (Proposition 37.6(b)), not only on $\hat{Y}$; this set lies in $(\Lambda_j^v)^{(R_v)}$ by the choice of R_v in Section 40, which is the form of the data dependence of (A2) used at this level.
- The local operators satisfy $\mathrm{Re}(\Delta_{\Lambda_j^v \cap X} + r) \geq \frac{1}{16} c_L + r$. So the local propagators $(\Delta_{\Lambda_j^v \cap X} + r)^{-1}$ have norm at most $(\frac{1}{16} c_L + r)^{-1}$ and decay $(C_L(\frac{1}{16} c_L + r)^{-1}, \gamma_C)$, uniformly in r (Proposition 37.6(d)).
- The margin R_v of $X(j)$ is at least $R_0 + 4 + 2r_{\mathrm{loc}}$ (Section 40).

The constant $C_G = C_L(\frac{1}{16} c_L + r)^{-1}$ of (A2) is at most $16 C_L / c_L$. Since $[r, h_j] = 0$, $C_{\mathrm{loc}} = 0$, and $C_{\mathfrak{N}}$ does not depend on r. So M_1 and the constants of Theorem 41.4 are uniform in r, and the bound of (a) carries the factor C_G. It gives $R_r(s)$, a representation of $(\Delta_k + r)^{-1}$, with $\|R_r\|_{\kappa,\gamma} \leq C(c_L + r)^{-1}$.

Remark 41.5 (Boundary conditions of the gauge-field operators). The verification of 31 October 2026 found that an earlier version of the gauge-field Dirichlet operators did not support the walks on proper domains.

- *The old operators.* They contained only the interior divergence D_X^T, so the normal components on the bonds crossing ∂X were free. These are relative boundary conditions, and the normal trace of $\mathbf{G}^X g$ was then of order 1. The local commutator $[D\mathbf{1}_X D^T, h_j]$ produced, on a crossing bond at the inner point x, the term $-\eta^{-1}([D^T, h_j]u)(x)$, whose main part is $\eta^{-1}(\partial_n h_j) u_n$. In L^2 this is of order $\eta^{-1/2} M^{-1}$. So the hypothesis of Lemma 41.3 failed uniformly in η on every proper domain.
- *The repair.* The operators now contain the exterior divergence term $\mathfrak{A}_X^{\mathrm{ext}}$ (Section 19). Their local part is the Dirichlet restriction of the torus operator, so all components satisfy Dirichlet conditions. Coercivity holds in $\|\!| \cdot |\!\|_{X,+}$, and the full gradient is controlled up to ∂X (Lemma 19.3). The local commutators are then bounded, and the cutoff proof of the comparison of domains (Theorem 19.10(a)) leaves no boundary terms.
- *What did not change.* The torus operators, since $\mathfrak{A}_{\mathbb{T}_\eta}^{\mathrm{ext}} = 0$, and hence Parts IV–VII. The scalar operators and the slice objects S_X, R_X, $P_{\mathcal{V},X}$. Enlarging the scalar domain is not needed, and would not restore exact identities on domains, since gradients of scalars supported at exterior points do not lie in $\ell_1^2(X)$. The mass term is kept, because the exterior divergence alone gives no L^2 control near the boundary.

Corollary 41.6 (Weakened covariance)**.** *For $M \geq M_1(\kappa, L)$, the formula*

$$C_k^{1/2}(s) = \frac{1}{\pi} \int_0^\infty \frac{dr}{\sqrt{r}} R_r(s)$$

defines a weakenable family representing $C_k^{1/2}$. Its norm $\|C_k^{1/2}\|_{\kappa',\gamma'}$ is finite for $\kappa' = \kappa - C\delta$, and some $\gamma' > 0$ depending on L. Here $C \le 20$. The walks of (L1)–(L5) and the repetitions of (L3), (L4) with W_2, seven in all, cost at most 2δ each. The products between them, at most six, cost δ each. Its terms are covariant and depend on the background only on their domains. Its margin is that of the families $R_r(s)$, independent of r and of M; it is at most $M/16$ for M large, as the definition of weakenable families requires (Section 40). The norm is bounded uniformly in the background, real or complex as in Section 40; the terms are analytic in the complex perturbation of the background, and real for real backgrounds and real s. In the same way the diagonal entries

$$\ell_k(b) = \sum_j h_j(b)\, \ell_{\Lambda_j^v}(b) - \int_0^\infty \operatorname{tr}_{\mathfrak{g}} \left[R_t^{(\ge 1)}(b,b)\right] dt$$

are represented by weakenable families of numbers $(X_\omega, n_\omega, O_\omega)_{\omega\in\Omega_b}$. Here $R_t^{(\ge 1)}$ collects the walk terms of $(\Delta_k + t)^{-1}$ with at least one step, and $\ell_{\Lambda_j^v}$ are the densities of Section 38. Moreover $O_\omega = 0$ unless $b \in X_\omega^{(r)}$, and

$$\|\ell_k(b)\|_{\kappa'} := \sup_{\square} \sum_{\omega\in\Omega_b:\, \square\subset X_\omega} e^{\kappa' \deg \omega} |O_\omega| \le C,$$

uniformly in b and in the background, real or complex as in Section 40.

Proof. By (L5), $\|R_r\|_{\kappa',\gamma'} \le C(c_L + r)^{-1}$, which is integrable against $r^{-1/2}dr$. Interchanging the sum over terms with the integral gives a representation whose terms are r-integrals of terms of R_r, with the same domains. At $s = 1$ it is $C_k^{1/2}$, by $\frac{1}{\pi}\int_0^\infty r^{-1/2}(\lambda + r)^{-1}dr = \lambda^{-1/2}$, which also holds for operators with positive real part [15, (192)–(193)]. The bound of (L5) is uniform in the background and in r, and the terms of $R_r(s)$ are analytic in the background and real at real backgrounds and real s, since they are built from real operators by sums, products and resolvents at real r; the r-integral preserves these properties.

For the determinant, the walk terms of $(\Delta_k + t)^{-1}$ with no steps are $\sum_j h_j(\Delta_{\Lambda_j^v} + t)^{-1}$. Since $\sum_j h_j(b) = 1$, the constant $\frac{1}{1+t}$ in the formula of Proposition 34.7 splits accordingly. So these terms contribute $\sum_j h_j(b)\ell_{\Lambda_j^v}(b)$. The terms with $n \ge 1$ steps contain $n+1$ resolvents and are $O((c_L + t)^{-2})$, hence integrable in t. More precisely, by (L5), $\|R_t^{(\ge 1)}\|_{\kappa',\gamma'} \le C(c_L + t)^{-2}$, uniformly in t and in the background. Since $|\operatorname{tr}_{\mathfrak{g}} O(b,b)| \le C|O|_{\gamma'}$, integration in t gives the norm bound for the terms with steps. The terms without steps are bounded by $|\ell_{\Lambda_j^v}(b)| \le C_L$, and their monomials $s^{X(j)}$ have degree at most $C(\rho)$. The index set does not depend on t, so the t-integral commutes with the sum over terms. □

By Lemma 40.3(b), at $s_\square = 0$ outside a union of boxes Z, all these unit-lattice objects split into independent blocks, one for each $*$-component of Z. After the conversion to margin 0 of Section 40, each block acts on fields and depends on the background only inside its component.

- *Scope.* This is the volume-type localization of the fluctuation covariance, of the coarse Hessian and of the determinant. Dimock's localization step [15, (225)–(233)] uses in addition sup bounds on the field $W_k(s)$ [15, Lemma 16, (100)]. Its Yang–Mills analogue needs the fine-lattice field $\mathbf{H}(s)Z$ pointwise, which is the subject of Section 42.

- *The first cube.* Unlike [15, (102)], the first cube of a walk is included in its monomial. So $G^X(0) = 0$, and the background dependence is cut together with the field; the price is a factor $e^{\kappa \mathsf{N}_{\mathrm{b}}}$ per step, which is harmless.

- *The order of the parameters.* In [15] the analyticity radius is $M^{1/2}$, i.e. $\kappa_1 = \frac{1}{2}\log M$. Here κ is fixed first. The rate $\kappa'_w = \kappa - C\delta$ must satisfy the conditions of Proposition 49.6

and Proposition 48.5. These involve κ_n, κ_0, $c_\sharp$, $c'_\sharp$, $c_\natural$, $c_\flat$ and C_d, not L. Choosing κ after L and before M is consistent with this.

41.4 The one-loop source in polymer form

Corollary 38.8 bounds the one-loop term by pieces of radius type. The polymer expansion needs pieces of volume type, at the complex points of the activity domain D3‴ (blueprint, Appendices B and F). Both follow from Corollary 41.6 and Lemma 38.5.

For $b \in \mathfrak{F}$ let $(X_\omega, n_\omega, O_\omega)_{\omega\in\Omega_b}$ be the weakenable family of numbers of Corollary 41.6 that represents $\ell_k(b)$, converted to margin 0, and write $X^\sharp_\omega$ for its polymers; $\deg\omega$ denotes the degree before the conversion. The index set, the polymers and the monomials are determined by the geometry of the walks (Lemma 40.4), not by the background; only the numbers $O_\omega = O_\omega(U)$ depend on it. For a polymer X put

$$\mathfrak{e}(X,U) = -\frac{1}{2}\sum_{b\in\mathfrak{F}} \sum_{\omega\in\Omega_b:\, X^\sharp_\omega = X} \big(O_\omega(U) - O_\omega(1)\big).$$

We write κ' for the rate of Corollary 41.6, $\kappa_2 = 3^{-d}\kappa'$, and C for constants that may depend on L, M, κ and ρ.

Proposition 41.7 (One-loop source in polymer form)**.** *Let G be semisimple and $q = 2p+6$, with p from Lemma 28.3. There is $c_a > 0$ such that the following holds.*

(a) $-\frac{1}{2}\log\det\Delta_k(U) = -\frac{1}{2}\log\det\Delta_k(1) + \sum_X \mathfrak{e}(X,U)$*, with absolute convergence, for real U in the setting of Section 40, and for complex U in the ball $\mathcal{B}$ of the proof of Proposition 46.7.*

(b) Each $\mathfrak{e}(X,\cdot)$ depends only on U on the bonds of X, and is a normalized graded activity at X on $\mathcal{U}(X;c_a,c_a,c_a)$ (Definition 38.3). Moreover

$$\sup_{\mathcal{U}(X;c_a,c_a,c_a)} |\mathfrak{e}(X,\cdot)| \le C|X|^2 e^{-\kappa_2|X|}.$$

(c) Let X have diameter at most $n/2-8$, and let $U = e^{\mathrm{i}\eta A'}U_r \in \mathcal{U}(X;c_a,c_a,c_a)$ with $|F_{U_r}|, |J_{U_r}| \le \theta$. Put

$$\mathfrak{t} = |\nabla A'|_{X^{(3)}} + [\nabla A']_{\alpha,X^{(3)}} + |A'|^2 + \theta.$$

If $C(1+|X|)^q(\mathfrak{t}+|A'|) \le 1$, then

$$|\mathfrak{e}(X,U)| \le C(1+|X|)^q\, \mathfrak{t}^2\, e^{-\kappa_2|X|}.$$

(d) Let $0 < \epsilon < \frac{1}{4}$ and $h \le h_0$, with h_0 depending on L, M, κ, ρ and ϵ. For every polymer X of diameter at most $n/2-8$, and every $U \in \mathcal{U}(X; h^{1-\epsilon/2}, h^{1-\epsilon}, h^{1-2\epsilon})$, i.e. every point of D3‴ at X,

$$|\mathfrak{e}(X,U)| \le C\, h^{2-2\epsilon}\, e^{-(\kappa_2-1)|X|}.$$

For polymers of larger diameter, $|\mathfrak{e}(X,U)| \le C|X|^2 e^{-\kappa_2|X|} \le Ce^{-(\kappa_2-1)|X|}e^{-cn/M}$.

Proof. (a) For real U: by Proposition 34.7, $\log\det\Delta_k = \sum_{b\in\mathfrak{F}}\ell_k(b)$, at U and at 1. By Corollary 41.6, $\ell_k(b) = \sum_{\omega\in\Omega_b} O_\omega$, with absolute convergence. Group the terms by $X^\sharp_\omega$. For complex U, both sides are holomorphic in the perturbation on the convex ball $\mathcal{B}$, and they agree at its real points; the identity theorem applies as in step (iii) of the proof of Proposition 46.7.

(b) Each O_ω depends on the background only on $X^{(r)}_\omega \subset X^\sharp_\omega$. It is invariant under real gauge transformations, since it is the diagonal trace of a covariant kernel or a term

$h_j(b)\ell_{\Lambda_j^v}(b)$. It is analytic under the complex perturbations of Section 40. These are those of Proposition 19.11. The averaging and the averaged backgrounds at complex references are given by Proposition 46.4, and the remaining inputs by Proposition 46.7. A configuration of Definition 38.2 with $\varepsilon_a \le \varepsilon_{\max}$ is a complex reference of size at most $C_{\mathrm{cmp}}\delta_a$, with $C_{\mathrm{cmp}} = 4$ of the first item of Section 46; so this needs $\varepsilon_a \le \varepsilon_{\max}$ and $C_{\mathrm{cmp}}\delta_a < \delta_P$. So $\mathfrak{e}(X,\cdot)$ is a normalized background activity at X in the sense of Definition 38.2, for some $\varepsilon_a, \delta_a > 0$. Remark 38.7 makes it a graded activity on $\mathcal{U}(X; c_a, c_a, c_a)$, if $c_a \le a_2$, $\max(1, B_5)c_a \le \varepsilon_a$ and $2Cc_a < \delta_a$. For the bound, let ω satisfy $X_\omega^\sharp = X$. Every box of X touches a box of X_ω, so $\deg\omega \ge |X_\omega| \ge 3^{-d}|X|$. Hence

$$\sum_{\omega\in\Omega_b:\, X_\omega^\sharp = X} |O_\omega| \le \sum_{\square\subset X} \sum_{\omega:\,\square\subset X_\omega} e^{\kappa' \deg\omega}|O_\omega|\, e^{-\kappa' 3^{-d}|X|} \le C|X| e^{-\kappa_2|X|},$$

by the norm bound of Corollary 41.6, uniformly in b and in the background. At most $CM^d|X|$ bonds $b \in \mathfrak{F}$ contribute.

(c) Apply Lemma 38.5 with $P = X$, $\theta_r = \delta_1 = \delta_0 = c_a \le \frac{1}{2}c_0$, and a cube $\Lambda \supset X^{(3)}$ of side $2R = 2\,\mathrm{diam}(X) + 8 < n$. Since $\mathrm{diam}(X) \le M|X|$, $R/(4M+2) \le C(1+|X|)$, so $B_P \le C(1+|X|)^p$. Also $D_N \le CM(1+|X|)$, since X is a $*$-connected union of boxes of side at most M. Hence

$$\mathfrak{m}_1 \le C(\mathfrak{m} + |A'|^2 + (1+|X|)^{p+1}|A'|\theta), \qquad \mathfrak{s} \le C(1+|X|)^{p+2}\mathfrak{t},$$

and $c_* \le \frac{1}{2}c_0 + C(1+|X|)^{p+1}(|A'|+\theta)$, $RB_P\theta \le C(1+|X|)^{p+1}\theta$. For a suitable C, the condition $C(1+|X|)^q(\mathfrak{t}+|A'|) \le 1$ implies all hypotheses of the lemma and $\mathfrak{s} \le \frac{1}{2}$. The lemma and (b) give $|\mathfrak{e}(X,U)| \le 2\mathfrak{s}^2 C|X|^2 e^{-\kappa_2|X|} \le C(1+|X|)^{2p+6}\mathfrak{t}^2 e^{-\kappa_2|X|}$.

(d) The domain $\mathcal{U}(X; h^{1-\epsilon/2}, h^{1-\epsilon}, h^{1-2\epsilon})$ lies in $\mathcal{U}(X; c_a, c_a, c_a)$ for h small. At its points, $\mathfrak{t} \le h^{1-\epsilon} + h^{2-4\epsilon} + h^{1-\epsilon/2} \le 3h^{1-\epsilon}$, since $\epsilon < \frac{1}{3}$, and $|A'| < h^{1-2\epsilon}$. If $C(1+|X|)^q \cdot 4h^{1-2\epsilon} \le 1$, (c) gives $|\mathfrak{e}(X,U)| \le 9C(1+|X|)^q h^{2-2\epsilon} e^{-\kappa_2|X|}$, and $(1+|X|)^q e^{-|X|} \le C$. Otherwise $|X| \ge ch^{-(1-2\epsilon)/q}$, and (b) gives $|\mathfrak{e}(X,U)| \le C|X|^2 e^{-\kappa_2|X|} \le h^{2-2\epsilon} e^{-(\kappa_2-1)|X|}$ for $h \le h_0$. For polymers of diameter larger than $n/2 - 8$, $|X| \ge cn/M$. □

Remark 41.8 (Use in the renormalization group). (a) *Why the affine gauge matters.* Without the complex gauge step of Lemma 38.5, the sup norm $|A'| < h^{1-2\epsilon}$ would enter $\mathfrak{s}$ linearly, and (d) would give only $h^{2-4\epsilon}$. With it, $|A'|$ enters only through $|A'|^2$ and the smallness condition.

(b) *The next step.* In the units of step k, the activity domain of step $k+1$ lies in that of step k, and $h_k \le h_{k+1}$. So (d) gives $|\mathfrak{e}(X,U)| \le Ch_{k+1}^{2-2\epsilon} e^{-(\kappa_2-1)|X|}$ there. These are the one-loop sources of size $O(h_{k+1}^{2-2\epsilon})$ required by the recursion. The constant depends on L and M, which is allowed, since g_* is chosen last. The scaling of the next domain (blueprint, Appendix F) gives an extra factor L^{-4} per polymer of step k, which the recursion does not need.

(c) *What is achieved.* All polymers, wrapping ones included, are normalized by construction: $\mathfrak{e}(X,1) = 0$. Each $\mathfrak{e}(X,\cdot)$ depends on the background only inside X. Joint analyticity in the weakening parameters is not needed, since the determinant lies outside the fluctuation integral (Theorem 34.8) and $\mathfrak{e}$ is extracted at $s = 1$.

(d) *The domain.* Part (d) uses D3‴ in the form of Definition 38.3, with the complex part controlled on $X^{(3)}$. In the blueprint's notation this means a domain margin of at least 3, not 0. (In Section 48, r_E denotes the margin of dependence of an activity; the domain margin is then $r_E + 3$.)

(e) *Inputs supplied in Part X.* Four further points are needed to use (d) in the flow; they are proved in Part X.

- *The decay rate.* $\kappa_2 - 1 > 0$ needs $\kappa' > 3^d$. Matching the rate of the activity norm needs $3^{-d}\kappa' - 1$ above that rate, a factor 3^d from the conversion of margins on top of the blueprint's condition $\kappa_1 \geq \kappa + 3\kappa_0 + 4$. The rates are fixed in Proposition 49.6 and Theorem 49.8(3).
- *The polymer scale.* The polymers here are unions of boxes of the paving, with sides between $M/4$ and M, while the activity norm uses $d_{\tilde{M}}$. The paving is matched to the polymer grid of Section 49.
- *Reblocking and the background.* The polymers are reblocked into polymers of step $k+1$ by Lemma 49.2 and Remark 49.3. The background U is related to the next coarse configuration through the base $W = U'^{(k)}$ (Section 47, Setting).
- *Wrap-around.* The polymers whose 3-neighbourhood lies in no cube of side less than n give the sources $Ce^{-cn/M}$. They are controlled by the size of the final torus (Lemma 50.1).

42 Pointwise norms: what is missing

The covariance, the coarse Hessian and the determinant enter the step through L^2 bounds, and for them Section 41 is complete. The background field is different. To localize the interaction and the composed activities, the weakened background map $A_*(s)$ must be built with weakened operators $\mathbf{H}(s)$, $\mathbf{C}(s)$, analytic on $\mathbb{D}_\kappa$. Its construction is a contraction in sup norms with derivatives (Proposition 26.3), or in $C^{1,\alpha}$ (blueprint, Appendix C). So, for every $s \in \mathbb{D}_\kappa$, it needs bounds of the form

$$|\mathbf{C}(s)f|_{1,\alpha;y} \leq C \sum_{y'} e^{-\gamma|y-y'|} |f|_{B(y')}, \tag{42.1}$$

together with the same bounds for $\mathbf{H}(s)$ and $\mathbf{C}(s)D^T$.

Where the difficulty is. The simplest way to obtain (42.1) is to make the step operators $\mathcal{R}(s)$ bounded from sup norms to sup norms. Then every walk term is bounded term by term, as in [15, (97)–(99)]. At level (L3) this fails without further input, because of the seam term $h_j \mathbf{1}_{X'}(\mathfrak{N}(s) - \mathfrak{N}^{X'}(s))\mathbf{1}_{X'} G_j$.

- It is small only through the cancellation between its two parts. Weakening only one of them destroys the cancellation: for $|s_\square| = e^\kappa$, the terms without steps are multiplied by $e^{\kappa \mathsf{N}_\flat}$. So both parts are weakened, compatibly, and the terms of $\mathfrak{N}^{X'}(s)$ act on $u = G_j f$ throughout Λ_j^v.
- The interior estimate (Proposition 36.5) bounds u pointwise at distance at least 4 from $\partial\Lambda_j^v$; near $\partial\Lambda_j^v$ only its L^2 norm is controlled.
- The terms of $\mathfrak{N}^{X'}(s)$ that start near the boundary are products whose factors need pointwise information on their inputs. Interior regularity with L^2 sources controls values in $d = 3$, but not gradients. The commutators $[\mathfrak{A}_{\mathrm{loc}}, h_{j'}]$ are of first order.

Note, however, that absolute convergence does not force all factors to be sup-to-sup. It suffices that $\mathcal{R}(s)$ is a contraction in some Banach space adapted to the geometry. This leaves two routes.

Route 1: boundary regularity. Prove regularity up to the boundary for the Dirichlet operators on the boxes of the construction:

> *(BR)* For a box X, the operators G'_X, $G'_X D_X^T$, $\mathbf{G}^X$, $\mathbf{G}^X D_X^T$, $\mathbf{H}^X$ and $\mathbf{C}^X$ satisfy the bounds of Theorem 35.2 and Proposition 36.5 at all blocks of X, not only at interior ones.

Then all factors are sup-to-sup, and the construction of Section 41 carries over to (42.1). Bałaban proves estimates of this kind for his Dirichlet propagators [2, 3, 7].

For the scalar Dirichlet Laplacian on a lattice box, odd reflection across the plane of the exterior layer gives a problem on the full lattice. At edges and corners, the reflections across the faces commute. The absorption argument of Section 36 can then be repeated on half-cubes and quarter-cubes at the boundary:

- the small perturbations become sources for the reflected equation. Their reflected coefficients may jump across the reflection plane, which is harmless only because u vanishes on the exterior layer;
- the averaging terms are block-local and the boxes are unions of blocks, so they remain L^2-controlled sources;
- the mass term is of order zero.

The gauge case is not covered by this sketch. On $\ell^2_1(X)$ the bonds crossing ∂X are kept, and since the exterior divergence is part of the operator, all components satisfy Dirichlet conditions for $D^T D + DD^T$ (Section 19). The reflection then has to respect the lattice structure of the crossing bonds. L^2 regularity of second order on boxes is classical in the continuum, but the pointwise statement on the lattice, with these operators, has not been carried out.

Route 2: a mixed norm, without boundary estimates. At the top level $X = \mathbb{T}_\eta$, every pointwise output is needed only on the supports K_j. Every Dirichlet cut involved lies at distance at least $\frac{1}{4}(\rho - 1)M$ from these supports, by (41.1). Two observations suggest that a norm mixing L^2 and pointwise control suffices:

- pieces of a kernel that jump a distance of at least 2 need only L^2 input, by interior regularity on a cube where the source vanishes;
- a chain of walk factors from a cut to K_j passes through many interior steps. Each step of the form commutator times propagator gains one derivative, and each step of order zero times propagator gains two. So in $d = 3$ a few interior steps take block-L^2 data to $C^{1,\alpha}$.

The norm would be block-L^2 within a few units of the cuts and pointwise elsewhere, with exponential weights. Its analytic input is a set of interior estimates with L^p data, variants of Proposition 36.5, together with the off-diagonal smoothing just described. These estimates are proved in Section 43 (Corollary 43.8). No boundary regularity is needed. The step bounds are carried out in Section 44, in block L^6 and Hölder norms rather than a mixed norm. They give (42.1) for $\mathbf{G}(s)$, $\mathbf{H}(s)$, $\mathbf{C}(s)$ and $\mathbf{C}(s)D^T$ on the torus (Theorems 44.10 and 44.12). A trace inequality on boxes (Lemma 44.5) handles the Dirichlet cuts. $\mathbf{C}(s)$ is represented through the torus identity $\mathbf{C} = \mathbf{G}\Pi^T - \mathbf{HQG}$.

Summary. Sections 40 and 41 give volume-type localization of everything that enters the step through L^2 bounds: the fluctuation covariance and its square root, the coarse Hessian, and the determinant. The background map, and with it the interaction and the composed activities, need in addition pointwise control of the weakened operators, (42.1). Either route above would provide it. Route 2 stays closer to the interior methods of this note. Route 1 is Bałaban's.

43 Interior estimates with L^p data

Route 2 of Section 42 needs interior estimates for the local propagators whose sources are controlled only in L^p norms. Sections 6 and 36 assume bounded sources. This section supplies the missing estimates:

(I1) sources in L^p, $p > d$, give $C^{1,\beta}$ solutions, $\beta \le 1 - d/p$ (Theorem 43.2(a), Propositions 43.4(a) and 43.7(a));

(I2) sources $F + \partial^* g$ with $F \in L^p$, $\frac{d}{2} < p < d$, and $g \in L^{p^*}$ give bounded C^β solutions with gradients in L^{p^*}, $\beta \le 2 - d/p$ (Theorem 43.2(b),(c), Propositions 43.4(b) and 43.7(b)). In $d = 3$ with $p = 2$ this is $C^{1/2}$ with gradients in L^6;

(I3) the Calderón–Zygmund bound behind the gradient part of (I2) (Lemma 43.1(d));

(I4) the $C^{2,\beta}$ Schauder estimate for scalar equations without averaging term (Proposition 43.5).

The proofs follow Sections 6 and 36. The only new analytic inputs are the Hardy–Littlewood–Sobolev and the Calderón–Zygmund inequalities on $\eta\mathbb{Z}^d$, both uniform in η by scaling. The input (I4) of the blueprint was formulated for the scalar local propagators. In that form it is false, because of the averaging term (Remark 43.6). The consequences for the weakened operator $\mathbf{C}(s)$ are discussed at the end of the section.

Notation. For a set Q of lattice points and $1 \le p < \infty$ let $\|h\|_{p,Q} = (\eta^d \sum_{x \in Q} |h(x)|^p)^{1/p}$, so that $\|h\|_{2,Q} = \|h\|_Q$; for gauge fields and 2-forms the sum also runs over the components. We write p' for the dual exponent and, for $1 < p < d$, $p^* = dp/(d-p)$, so that $1/p - 1/p^* = 1/d$. For $0 < \beta < 1$, $[h]_{\beta,Q}$ is the Hölder quotient with exponent β; thus $[h]_{\alpha,Q} = [h]_{\alpha,Q}$. Where Hölder quotients of the coefficients enter, namely in part (a) of Propositions 43.4 and 43.7, in Proposition 43.5 and in Corollary 43.8(a), we assume $\beta \le \alpha$, the exponent of hypothesis (B). On a cube Q of side at most 6, $[h]_{\beta,Q} \le C[h]_{\alpha,Q}$. Hence the coefficients of Lemmas 12.2 and 12.3, and those of $W \in \mathcal{W}_{\mathrm{loc}}(\varepsilon)$, have β-Hölder quotients bounded by $C\varepsilon$ on such cubes. Constants depend on d, $\dim \mathfrak{g}$, α, β and the exponents p, q, but not on η, k, n.

43.1 Two inequalities on the fine lattice

Lemma 43.1 (Kernel bounds)**.** *Let $d \ge 3$, $1 < p < \infty$, $x \in \eta\mathbb{Z}^d$, and let h be a field on $\eta\mathbb{Z}^d$.*

(a) If $0 \le s < d/p'$ and $R \ge \eta$: $\sum_{|x'-x| \le R} \eta^d (|x - x'| + \eta)^{-s} |h(x')| \le C R^{d/p' - s} \|h\|_p$.

(b) If $s > d/p'$ and $r \ge \sqrt{d}\,\eta$: $\sum_{|x'-x| > r} \eta^d |x - x'|^{-s} |h(x')| \le C r^{d/p' - s} \|h\|_p$.

(c) (Hardy–Littlewood–Sobolev.) *Let $0 < s < d$ and $1 < p < q < \infty$ with $1/q = 1/p - (d - s)/d$, and $k_s(x) = (|x| + \eta)^{-s}$. Then $\|k_s \star h\|_q \le C \|h\|_p$.*

(d) (Calderón–Zygmund.) *Let $K = \partial_\nu \partial_\mu^* g_\eta$. Then $\|K \star h\|_p \le C \|h\|_p$ for finitely supported h.*

Here $\|\cdot\|_p$ is the norm on all of $\eta\mathbb{Z}^d$, and C depends only on d, s, p, q.

Proof. (a) and (b) follow from Hölder's inequality and Lemma 6.2(a), respectively (c), with the exponent sp', which is smaller, respectively larger, than d.

(c) For $x' \in \eta\mathbb{Z}^d$ let $W(x') = x' + [-\eta/2, \eta/2)^d$ and $H = \sum_{x'} h(x') \mathbf{1}_{W(x')}$, so that $\|H\|_{L^p(\mathbb{R}^d)} = \|h\|_p$. For $w \in W(x)$ and $z \in W(x')$ we have $|w - z| \le |x - x'| + \sqrt{d}\eta \le (1 + \sqrt{d})(|x - x'| + \eta)$, so $k_s(x - x') \le C|w - z|^{-s}$. With the Riesz potential $I_s F(w) = \int_{\mathbb{R}^d} |w - z|^{-s} F(z)\, dz$ we get, for every $w \in W(x)$,

$$(k_s \star |h|)(x) = \sum_{x'} \int_{W(x')} k_s(x - x') |H(z)|\, dz \le C\, (I_s |H|)(w).$$

Averaging the q-th power over $w \in W(x)$ and summing over x gives $\|k_s \star h\|_q \le C\||I_s|H|\|_{L^q(\mathbb{R}^d)}$. The Hardy–Littlewood–Sobolev theorem [12, Ch. V, §1.2, Theorem 1] bounds the right-hand side by $C\|H\|_{L^p}$.

(d) By the proof of Proposition 6.1, $\eta^d K(x-x') = K_1(x/\eta - x'/\eta)$ with $K_1 = \partial_\nu \partial_\mu^* g_1$ on $\mathbb{Z}^d$, and both sides of the inequality scale by $\eta^{d/p}$. So we may take $\eta = 1$. On $\ell^2(\mathbb{Z}^d)$, convolution with K_1 is the Fourier multiplier

$$\frac{(e^{\mathrm{i}\xi_\nu} - 1)(e^{-\mathrm{i}\xi_\mu} - 1)}{\sum_\lambda 4\sin^2(\xi_\lambda/2)}, \qquad \xi \in [-\pi, \pi]^d,$$

of modulus at most 1, since $|e^{\mathrm{i}t} - 1| = 2|\sin(t/2)|$ and $4|\sin(\xi_\nu/2)\sin(\xi_\mu/2)| \le 2\sin^2(\xi_\nu/2) + 2\sin^2(\xi_\mu/2)$. So it is bounded on ℓ^2 with norm at most 1. By Proposition 6.1, $|K_1(z)| \le C(1+|z|)^{-d}$, and the first differences of K_1 are bounded by $C(1+|z|)^{-d-1}$. Let $y \ne y'$ and $|x-y| \ge 2|y-y'|$. A monotone lattice path from y to y' has at most $\sqrt{d}|y-y'|$ steps, and its points w satisfy $|x-w| \ge \frac{1}{2}|x-y|$. Hence $|K_1(x-y) - K_1(x-y')| \le C|y-y'||x-y|^{-d-1}$. Lemma 6.2(c), together with a bounded number of points with $|x-y| < \sqrt{d}$, gives Hörmander's condition

$$\sup_{y \ne y'} \sum_{|x-y| \ge 2|y-y'|} |K_1(x-y) - K_1(x-y')| \le C.$$

The same holds for $K_1(-\cdot)$. With counting measure and Euclidean distance, $\mathbb{Z}^d$ is a space of homogeneous type. The Calderón–Zygmund theorem on spaces of homogeneous type [13, Ch. I] gives weak type $(1,1)$. Marcinkiewicz interpolation gives the bound for $1 < p \le 2$, and duality, applied to the kernel $K_1(-\cdot)$, for $2 \le p < \infty$. □

43.2 The flat estimates

Theorem 43.2 (Interior regularity with L^p data)**.** *Let $d \ge 3$, let $\eta \le 1$ be any lattice spacing, with Λ_r as in Theorem 6.3, and let $u, F : \Lambda_3 \to \mathbb{R}$ and $g : \Lambda_3 \to \mathbb{R}^d$ satisfy $-\Delta u = F + \sum_\mu \partial_\mu^* g_\mu$ at every point of Λ_2.*

(a) (I1) *If $p > d$ and $0 < \beta \le 1 - d/p$, then $|u|_{\Lambda_1} + |\partial u|_{\Lambda_1} + [\partial u]_{\beta,\Lambda_1} \le C(\|F\|_{p,\Lambda_2} + |g|_{\Lambda_3} + [g]_{\beta,\Lambda_3} + \|u\|_{\Lambda_2})$.*

(b) (I2), values. *If $\frac{d}{2} < p < d$, $q > d$ and $0 < \beta \le \min(2 - d/p, 1 - d/q)$, then $|u|_{\Lambda_1} + [u]_{\beta,\Lambda_1} \le C(\|F\|_{p,\Lambda_2} + \|g\|_{q,\Lambda_3} + \|u\|_{\Lambda_2})$.*

(c) (I2), gradients. *If $1 < p < d$, then $\|\partial u\|_{p^*,\Lambda_1} \le C(\|F\|_{p,\Lambda_2} + \|g\|_{p^*,\Lambda_3} + \|u\|_{\Lambda_2})$.*

For $d = 3$, (b) and (c) with $p = 2$ and $q = p^* = 6$ give, for sources $F + \partial^* g$ with $F \in L^2$ and $g \in L^6$, solutions in $C^{1/2}$ with gradients in L^6.

Proof. We use the proof of Theorem 6.3. Step 0 is unchanged, since $\|h\|_{p,\Lambda_1} \le C|h|_{\Lambda_1}$. So assume $\eta \le \frac{1}{16}$. The representation (6.2) holds:

$$u = \mathcal{G}\Phi_1 + \mathcal{G}\Phi_2 + \sum_\mu (\partial_\mu^* g_\eta) \star (\chi g_\mu) + \sum_\mu (\partial_\mu^* g_\eta) \star m_\mu \qquad \text{on } \Lambda_{1+2\eta},$$

with $\Phi_1 = \chi F - \sum_\mu (\partial_\mu^* \chi) S_\mu^{-1} g_\mu$. Step 2 does not involve the sources. The terms with Φ_2 and m_μ have two derivatives bounded by $C\|u\|_{\Lambda_2}$ on Λ_1, and they are Lipschitz together with their first derivatives. So they satisfy all three bounds. Moreover $\|\Phi_1\|_{p,\Lambda_2} \le \|F\|_{p,\Lambda_2} + C\|g\|_{p,\Lambda_3}$. Since Λ_3 has bounded volume, $\|g\|_{p,\Lambda_3}$ is bounded by $C|g|_{\Lambda_3}$ in (a), by $C\|g\|_{q,\Lambda_3}$ in (b), where $q > p$, and by $C\|g\|_{p^*,\Lambda_3}$ in (c). Recall $|g_\eta(y)| \le C(|y| + \eta)^{2-d}$, and that k-th differences of g_η are bounded by $C(|y| + \eta)^{2-d-k}$, $k \le 3$ (Proposition 6.1). All points below lie in $\Lambda_{1+2\eta}$, and sums run over the support of Φ_1, respectively of χg, which lies in Λ_2.

(a) The term $\sum_\mu (\partial_\mu^* g_\eta) \star (\chi g_\mu)$ is treated as in Step 5 of the proof of Theorem 6.3, with β in place of α. For $\mathcal{G}\Phi_1$, Lemma 43.1(a) with $s = d-2$ and $s = d-1$ gives $|\mathcal{G}\Phi_1| + |\partial\mathcal{G}\Phi_1| \le C\|\Phi_1\|_p$; here $s < d/p'$ because $p > d$. For the Hölder quotient of $k \star \Phi_1$, $k = \partial_\mu g_\eta$, let $x, x' \in \Lambda_1$ and $\eta \le \delta = |x - x'| \le 1$, and split the sum at $|x - x''| = 4\sqrt{d}\delta$, as in Step 3 of that proof. On the near part, Lemma 43.1(a) with $s = d-1$ and $R = (4\sqrt{d}+1)\delta$ gives the bound $C\delta^{d/p'-d+1}\|\Phi_1\|_p = C\delta^{1-d/p}\|\Phi_1\|_p$. On the far part, $|k(x-x'') - k(x'-x'')| \le C\delta|x-x''|^{-d}$, by the path argument of Step 3. Lemma 43.1(b) with $s = d > d/p'$ and $r = 4\sqrt{d}\delta$ gives $C\delta \cdot \delta^{d/p'-d}\|\Phi_1\|_p = C\delta^{1-d/p}\|\Phi_1\|_p$. Since $\delta \le 1$ and $\beta \le 1 - d/p$, this is at most $C\delta^\beta\|\Phi_1\|_p$. Pairs with $\delta > 1$ are covered by the sup bound.

(b) By Lemma 43.1(a), $|\mathcal{G}\Phi_1(x)| \le C\|\Phi_1\|_p$, since $d - 2 < d/p'$ for $p > \frac{d}{2}$, and $|(\partial_\mu^* g_\eta) \star (\chi g_\mu)(x)| \le C\|g\|_q$, since $d - 1 < d/q'$ for $q > d$. For the Hölder quotient of $\mathcal{G}\Phi_1$ split as in (a). The near part is bounded by $C\delta^{d/p'-d+2}\|\Phi_1\|_p = C\delta^{2-d/p}\|\Phi_1\|_p$. On the far part the differences of g_η are bounded by $C\delta|x-x''|^{1-d}$, and Lemma 43.1(b) with $s = d-1 > d/p'$ (since $p < d$) gives $C\delta \cdot \delta^{d/p'-d+1}\|\Phi_1\|_p = C\delta^{2-d/p}\|\Phi_1\|_p$. The Hölder quotient of $(\partial_\mu^* g_\eta) \star (\chi g_\mu)$ is bounded as in (a), with $(q, \chi g_\mu)$ in place of (p, Φ_1). This gives $C\delta^{1-d/q}\|g\|_q$.

(c) We have $\partial\mathcal{G}\Phi_1 = (\partial g_\eta) \star \Phi_1$ with $|\partial g_\eta| \le C k_{d-1}$. Lemma 43.1(c) with $s = d-1$, for which $1/q = 1/p - 1/d = 1/p^*$, gives $\|\partial\mathcal{G}\Phi_1\|_{p^*} \le C\|\Phi_1\|_p$. Next, $\partial_\nu((\partial_\mu^* g_\eta) \star (\chi g_\mu)) = (\partial_\nu \partial_\mu^* g_\eta) \star (\chi g_\mu)$, and Lemma 43.1(d) bounds its L^{p^*} norm by $C\|g\|_{p^*,\Lambda_2}$. □

Lemma 43.3 (Rescaled estimates)**.** *Let $\eta \le s \le 1$, $x \in \eta\mathbb{Z}^d$, $\Lambda_r = \Lambda_r(x)$, and $-\Delta u = F + \sum_\mu \partial_\mu^* g_\mu$ on Λ_{2s}. Under the conditions of Theorem 43.2(a), (b), (c), respectively,*

$$|u|_{\Lambda_s} + s|\partial u|_{\Lambda_s} + s^{1+\beta}[\partial u]_{\beta,\Lambda_s} \le C\big(s^{2-d/p}\|F\|_{p,\Lambda_{2s}} + s|g|_{\Lambda_{3s}} + s^{1+\beta}[g]_{\beta,\Lambda_{3s}} + s^{-d/2}\|u\|_{\Lambda_{2s}}\big),$$
$$|u|_{\Lambda_s} + s^\beta[u]_{\beta,\Lambda_s} \le C\big(s^{2-d/p}\|F\|_{p,\Lambda_{2s}} + s^{1-d/q}\|g\|_{q,\Lambda_{3s}} + s^{-d/2}\|u\|_{\Lambda_{2s}}\big),$$
$$\|\partial u\|_{p^*,\Lambda_s} \le C\big(\|F\|_{p,\Lambda_{2s}} + \|g\|_{p^*,\Lambda_{3s}} + s^{-1-d/2+d/p^*}\|u\|_{\Lambda_{2s}}\big).$$

Proof. As for Lemma 36.2. On the lattice $(\eta/s)\mathbb{Z}^d$ we have $\|h_s\|_{p,\Lambda_r(0)} = s^{-d/p}\|h\|_{p,\Lambda_{rs}}$ for $h_s(x') = h(x + sx')$, and $\partial u_s = s(\partial u)_s$. For the third bound, $\|\partial u_s\|_{p^*,\Lambda_1(0)} = s^{1-d/p^*}\|\partial u\|_{p^*,\Lambda_s}$, while $\|s^2 F_s\|_{p,\Lambda_2(0)} = s^{2-d/p}\|F\|_{p,\Lambda_{2s}}$ and $\|s g_s\|_{p^*,\Lambda_3(0)} = s^{1-d/p^*}\|g\|_{p^*,\Lambda_{3s}}$. Since $d/p - d/p^* = 1$, the two data terms carry the same power s^{1-d/p^*} as the left-hand side. □

43.3 Scalar equations

We keep the conventions of Section 36: the local gauge on $\Lambda_4(z)$, sources in local form $\mathcal{S} = F + \sum_\mu \partial_\mu^* g_\mu$, and bounds on Λ_r that may involve values within 2η of Λ_r. For a given representation of $\mathcal{S}$ put

$$\mathfrak{n}_r^{p,\beta}(\mathcal{S}) = \|F\|_{p,\Lambda_r} + |g|_{\Lambda_r} + [g]_{\beta,\Lambda_r}, \qquad \mathfrak{m}_r^{p,q}(\mathcal{S}) = \|F\|_{p,\Lambda_r} + \|g\|_{q,\Lambda_r}.$$

Proposition 43.4 (Scalar interior estimates with L^p data)**.** *There is $\varepsilon_S' > 0$ with the following property. For $0 < \rho < r \le 3$ there is C, depending in addition on $r - \rho$, such that for $\varepsilon \le \varepsilon_S'$: if v is a scalar field with $D^T D v = \mathcal{S}$ on Λ_r, then:*

(a) if $p > d$ and $\beta \le 1 - d/p$: $|v|_{\Lambda_\rho} + |\partial v|_{\Lambda_\rho} + [\partial v]_{\beta,\Lambda_\rho} \le C(\mathfrak{n}_r^{p,\beta}(\mathcal{S}) + \|v\|_{\Lambda_r})$;

(b) if $\frac{d}{2} < p < d$ and $\beta \le 2 - d/p$: $|v|_{\Lambda_\rho} + [v]_{\beta,\Lambda_\rho} + \|\partial v\|_{p^*,\Lambda_\rho} \le C(\mathfrak{m}_r^{p,p^*}(\mathcal{S}) + \|v\|_{\Lambda_r})$.

Proof. As in Proposition 36.3, we may assume $r - \rho \ge 40\eta$, and we write $-\Delta v = (F - bv) + \sum_\mu \partial_\mu^*(g_\mu - a_\mu v)$ on Λ_r.

(a) Repeat the proof of the first bound of Proposition 36.3, with exponent β and with the first inequality of Lemma 43.3 in place of that of Lemma 36.2. The perturbation bv is part of F, and $\|bv\|_{p,\Lambda_{2s}(x)} \le C\varepsilon s^{d/p}|v|_{\Lambda_{t'}}$. So it enters, as before, with the factor s^2. After division by s the source enters as $s^{1-d/p}\|F\|_{p,\Lambda_r} \le \|F\|_{p,\Lambda_r}$. The rest is unchanged.

(b) Let $A(t) = |v|_{\Lambda_t} + \|\partial v\|_{p^*,\Lambda_t}$ and $\beta_0 = 2 - d/p$; note $1 - d/p^* = \beta_0$. Let $\rho \le t < t' \le r$, $s = (t'-t)/6$, and suppose $s \ge 3\eta$. For $x \in \Lambda_t$, the second inequality of Lemma 43.3 with $q = p^*$ gives

$$|v(x)| \le C\big(s^{\beta_0}\|F - bv\|_{p,\Lambda_{2s}(x)} + s^{\beta_0}\|g - av\|_{p^*,\Lambda_{3s}(x)} + s^{-d/2}\|v\|_{\Lambda_r}\big).$$

Since $s \le 1$, $\|bv\|_{p,\Lambda_{2s}(x)} + \|av\|_{p^*,\Lambda_{3s}(x)} \le C\varepsilon|v|_{\Lambda_{t'}}$, and this is at most $C\mathfrak{m}_r^{p,p^*}(\mathcal{S}) + C\varepsilon A(t') + Cs^{-d/2}\|v\|_{\Lambda_r}$. For the gradient, cover Λ_t by the cubes $\Lambda_s(x_i)$, with x_i the points of Λ_{t+s} on a grid of lattice points of spacing $h = \eta\lfloor s/\eta\rfloor$. Since $s \ge 3\eta$, $\frac{1}{2}s \le h \le s$. Every point of Λ_t is within $\frac{1}{2}h$ of a grid point, in each coordinate, so these cubes cover Λ_t, and Lemma 43.3 applies at each $x_i \in \eta\mathbb{Z}^d$. The cubes $\Lambda_{3s}(x_i)$ lie in $\Lambda_{t+4s} \subset \Lambda_{t'}$ and overlap at most 13^d times. Apply the third inequality of Lemma 43.3 on each cube, raise it to the power p^* and sum over i. For the data use $\sum_i a_i^{p^*} \le (\sum_i a_i^p)^{p^*/p}$ and $\sum_i a_i^{p^*} \le (\sum_i a_i^2)^{p^*/2}$, valid since $p^* > p$ and $p^* > d > 2$. This gives

$$\|\partial v\|_{p^*,\Lambda_t} \le C\big(\mathfrak{m}_r^{p,p^*}(\mathcal{S}) + \|bv\|_{p,\Lambda_{t'}} + \|av\|_{p^*,\Lambda_{t'}} + s^{-\theta}\|v\|_{\Lambda_r}\big), \qquad \theta = 1 + \tfrac{d}{2} - \tfrac{d}{p^*},$$

and the perturbations are bounded by $C\varepsilon|v|_{\Lambda_{t'}}$. If $s < 3\eta$, then $A(t) \le C\eta^{-1-d/2}\|v\|_{\Lambda_r} \le C(t'-t)^{-1-d/2}\|v\|_{\Lambda_r}$. So $A(t) \le C'_S\varepsilon A(t') + C(t'-t)^{-1-d/2}\|v\|_{\Lambda_r} + C\mathfrak{m}_r^{p,p^*}(\mathcal{S})$, and for $C'_S\varepsilon \le \frac{1}{2}$ Lemma 36.1 on $[\rho', r]$, $\rho' = \frac{1}{2}(\rho + r)$, bounds $A(\rho')$. The Hölder quotient on Λ_ρ follows from the second inequality of Lemma 43.3 at the scale $s_1 = (r-\rho)/8$, with the sup bound inserted, as in Proposition 36.3. □

Proposition 43.5 (Schauder estimate, (I4))**.** *There is $\varepsilon''_S > 0$ with the following property. For $0 < \rho < r \le 3$ there is C, depending in addition on $r - \rho$, such that for $\varepsilon \le \varepsilon''_S$: if v is a scalar field with $D^T Dv = F$ on Λ_r, then*

$$\max_{\mu,\nu}\big(|\partial_\mu\partial_\nu v|_{\Lambda_\rho} + [\partial_\mu\partial_\nu v]_{\beta,\Lambda_\rho}\big) \le C\big(|F|_{\Lambda_r} + [F]_{\beta,\Lambda_r} + \|v\|_{\Lambda_r}\big).$$

The same holds for the covariant second derivatives $D_\mu D_\nu v$.

Proof. We may assume $r - \rho \ge 40\eta$. Otherwise $\eta^{-1} < 40(r-\rho)^{-1}$, and $|v(x)| \le \eta^{-d/2}\|v\|_{\Lambda_r}$ gives the bounds: second differences, flat or covariant (the transports are unitary), are at most $4\eta^{-2-d/2}\|v\|_{\Lambda_r}$, and their β-Hölder quotients at most $8\eta^{-2-\beta-d/2}\|v\|_{\Lambda_r}$, since distinct lattice points are at distance at least η. So the proposition holds with $C = 12\big(40/(r-\rho)\big)^{2+\beta+d/2}$ in this case. Let $\rho_1 = \frac{1}{2}(\rho + r)$ and $\mathfrak{N} = |F|_{\Lambda_r} + [F]_{\beta,\Lambda_r} + \|v\|_{\Lambda_r}$. The first bound of Proposition 36.3, with exponent β and $g = 0$, gives $|v| + |\partial v| + [\partial v]_\beta \le C\mathfrak{N}$ on Λ_{ρ_1}. Write $-\Delta v = F - bv - \sum_\mu \partial^*_\mu(a_\mu v)$, fix ν and put $w = \partial_\nu v$. Since $\partial_\nu h = -\partial^*_\nu(S_\nu h)$ and ∂_ν commutes with ∂^*_μ,

$$-\Delta w = \sum_\mu \partial^*_\mu G_\mu, \qquad G_\mu = -\delta_{\mu\nu}S_\nu(F - bv) - \partial_\nu(a_\mu v) \qquad \text{on } \Lambda_{\rho_1}.$$

Here a_μ is a multiplication operator composed with a shift by at most η, so $\partial_\nu(a_\mu v)$ is a sum of terms (differences of the coefficients) $\cdot$ (shifted v) and (coefficients) $\cdot$ (shifted ∂v). By Lemma 12.2, $|a_\mu| + |\partial a_\mu| + |b| \le C\varepsilon$, and the β-Hölder quotients of ∂a_μ and b are bounded by $C\varepsilon$. Hence $|G| + [G]_\beta \le |F| + [F]_\beta + C\varepsilon(|v| + |\partial v| + [\partial v]_\beta) \le C\mathfrak{N}$ on Λ_{ρ_1}. Let $s_1 = (\rho_1 - \rho)/4 \ge \eta$. For $x \in \Lambda_\rho$, the first inequality of Lemma 36.2 on $\Lambda_{3s_1}(x) \subset \Lambda_{\rho_1}$, with exponent β, $F = 0$ and $\|w\|_{\Lambda_{2s_1}(x)} \le C|\partial v|_{\Lambda_{\rho_1}}$, bounds $|\partial w| + [\partial w]_\beta$ on $\Lambda_{s_1}(x)$ by $C\mathfrak{N}$. Pairs of points at distance larger than s_1 are handled by the sup bound. Finally, $D_\mu D_\nu v - \partial_\mu\partial_\nu v$ is a sum of terms (coefficient) $\cdot$ (shifted v or ∂v), with coefficients bounded by $C\varepsilon$ whose β-Hölder quotients are bounded by $C\varepsilon$. They include $\partial_\mu\tau_\nu$, which is only C^α by Lemma 12.2, and $\beta \le \alpha$ suffices. □

Remark 43.6 (The averaging term and (I4)). Proposition 43.5 needs a Hölder source. For the scalar Dirichlet propagator, $v = G'_X f$ satisfies $D^T D v = f - aQ'^T Q' v$ in the interior. The function $Q'^T Q' v$ jumps across the faces of the blocks, by a times the difference of the averages of v on adjacent blocks, up to parallel transport. This jump is not small. A jump across a single face already destroys the Hölder bound on second derivatives: for $-\Delta v = \mathbf{1}_{\{x_1>0\}}$ in the continuum, $\partial_1^2 v$ is discontinuous, and on the lattice $[\partial_1 \partial_1 v]_\beta$ is of order $\eta^{-\beta}$. Jumps across the faces meeting at an edge destroy even the sup bound. If $-\Delta v = \sum_{i=1}^4 c_i \mathbf{1}_{Q_i}$ for the four quadrants Q_i around an edge, then $\partial_1 \partial_2 v$ grows like $(c_1 - c_2 + c_3 - c_4)\log(1/|x|)$ at the edge. On the lattice this becomes $\log(1/\eta)$. Here the coefficient is a mixed second difference of the block averages, which is generically non-zero. So (I4) does not hold for the scalar local propagators G'_Y themselves. Proposition 43.5 applies when the averaging term vanishes, i.e. to fields σ with $Q'\sigma = 0$. For instance, $\sigma = G'_X R_X h$ satisfies $D^T D\sigma = R_X h = h - P_{\mathcal{V},X} h$ in the interior (Theorem 18.2(d)). Here $P_{\mathcal{V},X} h = G'_X Q'^T (M'_X)^{-1} Q' G'_X h$ is $C^{1,\beta}$ there, by Proposition 36.3. Hence $|\partial\partial\sigma| + [\partial\partial\sigma]_\beta$ is bounded at $y \in X_{(4)}$ by $|h| + [h]_\beta$ nearby plus weighted L^2 norms.

43.4 The local system

Proposition 43.7 (System interior estimates with L^p data)**.** *There is $\varepsilon'_L > 0$, depending in addition on the mass parameter κ_m of Section 19, with the following property. For $0 < \rho < r \le 3$ there is C, depending in addition on $r - \rho$, such that for $\varepsilon \le \varepsilon'_L$ the following holds. Let W, m, (u, σ, τ) and the sources $\mathcal{S}_u, \mathcal{S}_\sigma, \mathcal{S}_\tau$ be as in Proposition 36.5, with (36.5) on Λ_r, and let $\mathfrak{m} = \|u\|_{\Lambda_r} + \|\sigma\|_{\Lambda_r} + \|\tau\|_{\Lambda_r}$.*

(a) If $p > d$ and $\beta \le 1 - d/p$:

$$\sum_{v \in \{u,\sigma,\tau\}} \big(|v| + |\partial v| + [\partial v]_\beta\big)_{\Lambda_\rho} \le C\Big(\sum_\bullet \mathfrak{n}_r^{p,\beta}(\mathcal{S}_\bullet) + \mathfrak{m}\Big).$$

(b) If $\frac{d}{2} < p < d$ and $\beta \le 2 - d/p$:

$$\sum_{v \in \{u,\sigma,\tau\}} \big(|v|_{\Lambda_\rho} + [v]_{\beta,\Lambda_\rho} + \|\partial v\|_{p^*,\Lambda_\rho}\big) \le C\Big(\sum_\bullet \mathfrak{m}_r^{p,p^*}(\mathcal{S}_\bullet) + \mathfrak{m}\Big).$$

Proof. (a) Repeat the proof of Proposition 36.5 with exponent β and with Lemma 43.3 in place of Lemma 36.2. The perturbations in F_u, F_σ, F_τ are bounded pointwise, as in Step 1 there. On $\Lambda_{2s}(x)$ their L^p norms are at most $Cs^{d/p}$ times their sup norms, so they enter with the same powers of s as before. The sources enter as $s^{1-d/p}\|F_{\mathcal{S}_\bullet}\|_p \le \|F_{\mathcal{S}_\bullet}\|_p$. In Step 2 the Hölder quotient of $\partial\sigma$ at scale $3s$ is bounded by the first inequality of Lemma 43.3, which gives $s^\beta [\partial\sigma]_\beta \le \cdots + s^{1-d/p}\|F_\sigma\|_p$. The coefficients of W have β-Hölder quotients bounded by $C\varepsilon$, so Step 1 holds with β in place of α. Steps 3–5 are unchanged.

(b) Replace $A_v(t)$ in Steps 2 and 3 of the proof of Proposition 36.5 by $A_v(t) = |v|_{\Lambda_t} + \|\partial v\|_{p^*,\Lambda_t}$. Values are bounded at each $x \in \Lambda_t$ by the second inequality of Lemma 43.3, with $q = p^*$. The L^{p^*} norms of gradients are bounded by the third inequality on the cubes $\Lambda_s(x_i)$ of the proof of Proposition 43.4(b), summed in ℓ^{p^*}. The perturbations of Step 1 there are estimated cube by cube:

- terms bounded pointwise by $|v|$, in $F_\bullet$: their L^p norm on a cube of side $4s$ is at most $Cs^{d/p}$ times the sup, and the ℓ^{p^*} sum over the Cs^{-d} cubes gives the factor $s^{d/p - d/p^*} = s$; for values the factor is $s^{\beta_0 + d/p} = s^2$;
- gradient terms in $F_\bullet$ ($D\tau$ in F_u; $\varepsilon\partial u$ and $\varepsilon\partial\sigma$ in F_τ and F_u): by Hölder on a cube of side $4s$, $\|\partial v\|_{p,\Lambda_{2s}(x_i)} \le Cs\|\partial v\|_{p^*,\Lambda_{2s}(x_i)}$, which gives the factor s (and $s^{1+\beta_0}$ for values);

- terms in $g_\bullet$: their L^{p^*} norms are bounded by $C\varepsilon$ times $|v|$ or $\|\partial v\|_{p^*}$, except for u_μ in g_σ, which has coefficient 1. The potential W is local with bounded coefficients, so $\|Wh\|_{p^*,Q} \le C\varepsilon\|h\|_{p^*,Q'}$ for a slightly larger cube Q'.

Hölder quotients of the coefficients are not needed. So Step 2 gives inequalities of the same form as there:

$$A_\sigma(t) \le C_4(A_u + \varepsilon A_\sigma) + R, \qquad A_\tau(t) \le C_2\varepsilon(A_u + A_\sigma + A_\tau) + R,$$
$$A_u(t) \le C_3(\varepsilon + \kappa_m s)A_u + C_3\varepsilon A_\sigma + C_3 s A_\tau + R,$$

with the A_v on the right evaluated at t' and $R = C\big(\sum_\bullet \mathfrak{m}_r^{p,p^*}(\mathcal{S}_\bullet) + (t'-t)^{-\theta'}\mathfrak{m}\big)$ for some $\theta' > 0$. Step 3, with the same weights and small cubes, and Lemma 36.1 bound $A_u + A_\sigma + A_\tau$ on $\Lambda_{\rho'}$. The Hölder quotients of the values follow at a fixed scale from the second inequality of Lemma 43.3. In Step 5, for $r > r_*$, cover Λ_ρ by the cubes $\Lambda_{r_1/2}(x_i)$, with x_i on a grid of spacing $r_1/2$. There are at most $C(r_1)$ of them. Sum the p^*-th powers of the bounds on these cubes. □

In Proposition 36.5 the bound for τ needed the Hölder quotient of $\partial\sigma$, through the term $WD\sigma$ in g_τ. In (b) only $\|\partial\sigma\|_{p^*}$ enters, so this step disappears.

43.5 Consequences for the Dirichlet operators

For a block y and fields on $\eta\mathbb{Z}^d$ write $\mathfrak{E}^p_y(F) = \sum_{y'} e^{-\gamma|y-y'|}\|F\|_{p,B(y')}$. For a 2-form or a gauge field g write $\mathfrak{E}^\beta_y(g) = \sum_{y'} e^{-\gamma|y-y'|}(|g|_{B(y')} + [\tilde g_{y'}]_{\beta,\Lambda_1(c_{y'})})$, as in Theorem 35.2(b).

Corollary 43.8 (Dirichlet propagators with L^p data)**.** *In the setting of Theorem 35.2 there are $\varepsilon', C, \gamma > 0$, independent of X, k, n, η, such that the following holds for $\varepsilon \le \varepsilon'$, every domain X and every $y \in X_{(4)}$. Let O be one of the operators G'_X, $\mathbf{G}^X$, and let $f = F + D^T_X g$, with F a scalar, respectively gauge, field and g a gauge field, respectively 2-form, supported in X.*

(a) If $p > d$ and $\beta \le 1 - d/p$: $|Of|_{1,\beta;y} \le C\big(\mathfrak{E}^p_y(F) + \mathfrak{E}^\beta_y(g)\big)$.

(b) If $d = 3$: $|Of|_{B(y)} + [\widetilde{Of}_y]_{1/2,\Lambda_1(c_y)} + \|\nabla Of\|_{6,B(y)} \le C\big(\mathfrak{E}^2_y(F) + \mathfrak{E}^6_y(g)\big)$.

Here $|\cdot|_{1,\beta;y}$ is $|\cdot|_{1,\alpha;y}$ of Section 35 with exponent β. In particular $\nabla G'_X D^T_X$ and $\nabla\mathbf{G}^X D^T_X$ map L^6 to L^6 locally, with exponential decay. This is the Calderón–Zygmund input (S5) of the blueprint for $p = 6$, which is the case used there. The section gives gradient bounds in L^q only for $q = p^* \in (d, \infty)$.

Proof. For $O = \mathbf{G}^X$ let $u = \mathbf{G}^X f$, so that $\mathfrak{A}^{X,\kappa_m}u = f$. Proceed as in the basic estimate of the proof of Theorem 35.2. By Lemma 36.4, (u, σ, τ) satisfies (36.5) on $\Lambda_3(c_y)$, with

$$\mathcal{S}_u = f + D\pi_1 - D\pi_2 - a\mathbf{Q}^T\mathbf{Q}u, \qquad \mathcal{S}_\sigma = -\pi_1, \qquad \mathcal{S}_\tau = -aQ'^T Q'\tau.$$

In the local gauge, $D^T_X g = \partial^* g + T^T g$ is in local form, with $T^T g$ part of F. The terms $D\pi_i$, $a\mathbf{Q}^T\mathbf{Q}u$ and $aQ'^T Q'\tau$ are bounded pointwise by $C\mathfrak{L}(u)$, as in the proof of Theorem 35.2. By the L^2 decay of $\mathbf{G}^X$, $D\mathbf{G}^X$, $\mathbf{G}^X D^T_X$ and $D\mathbf{G}^X D^T_X$ (Theorem 19.7(a)), together with $\|h\|_{B(y')} \le \|h\|_{p,B(y')}$ for $p \ge 2$, we have $\mathfrak{L}(u) \le C(\mathfrak{E}^p_y(F) + \mathfrak{E}^q_y(g))$ for $p, q \ge 2$. Here $\mathfrak{E}^q_y(g)$ is defined like $\mathfrak{E}^p_y(F)$, and $\mathfrak{E}^q_y(g) \le \mathfrak{E}^\beta_y(g)$. The Hölder quotient of g is converted to the local gauge as in the proof of Theorem 35.2(b). Proposition 43.7(a), respectively (b) with $p = 2$, gives the claim. For $O = G'_X$, $v = G'_X f$ satisfies $D^T Dv = f - aQ'^T Q'v$ in the interior. Here $|Q'v(y')| \le C\|v\|_{B(y')}$, and G'_X, DG'_X, $G'_X D^T_X$, $DG'_X D^T_X$ have L^2 decay (Theorem 18.2(b)). Apply Proposition 43.4. □

Consequences for route 2. This paragraph records the state before Section 44, which carries out route 2 and supersedes this paragraph in part. With Corollary 43.8, the interior smoothing mechanisms (S2), (S3) and (S5) of the blueprint hold for the Dirichlet propagators G'_Y and $\mathbf{G}^Y$ with potentials $W \in \mathcal{W}_{\mathrm{loc}}(\varepsilon)$. The levels (L1)–(L3) are compositions of these with bounded block-local operators, and (S1), (S4) were already available. The levels with the potential W_2^X (range up to 3, non-local through ω_0) are not covered: the cube bound $\|Wh\|_{p^*,Q} \le C\varepsilon\|h\|_{p^*,Q'}$ of Proposition 43.7(b) fails at scales $s \ll 3$. This is harmless as long as pointwise bounds are needed only for the operators of the background map, which are built with $W \in \mathcal{W}_{\mathrm{loc}}$. For $C^{1,\beta}$ outputs from $\mathscr{V}$ the exponent restriction is $\beta \le \min(\alpha, \frac{1}{2})$ in $d = 3$; with $\alpha = \frac{1}{2}$, $\beta = \frac{1}{2}$ is allowed. Two remarks on the blueprint follow.

- *(I4) for* $\mathbf{C}(s)$. By Remark 43.6, second derivatives are available only for fields annihilated by Q'. In a weakened product $DG'(s)R(s)G'(s)D^T$ the identity $Q'G'R = 0$ does not hold term by term. So the term-by-term argument proposed for $\mathbf{C}(s)$ does not work. Applying the unweakened operator to $\mathbf{C}(s)f$ and treating the weakening errors as sources does not help either, for the same reason (Remark 44.11). The fix is the torus identity $\mathbf{C} = \mathbf{G}\Pi^T - \mathbf{HQG}$ of Section 44, which needs no second derivatives of scalar fields.
- *Mixed norms.* The step bounds of route 2 are carried out in Section 44, in block L^6 and Hölder norms. Two interior factors at the output suffice there.

44 Regularity tiers and step bounds

This section carries out the main step of route 2 of Section 42. It proves the pointwise bounds (42.1) for the weakened operators $\mathbf{G}(s)$, $\mathbf{H}(s)$, $\mathbf{C}(s)$ and $\mathbf{C}(s)D^T$ on the torus, uniformly on $\mathbb{D}_\kappa$. The only analytic inputs are the interior estimates of Section 43 and the L^2 theory of Section 41. No regularity at the Dirichlet cuts is used. The construction rests on two observations.

- *Outputs are interior.* Every term of a walk ends, on the left, with a first factor h_jG_j, evaluated on K_j. Every local step $[\mathfrak{A}_{\mathrm{loc}}, h_j]G_j$ is evaluated on $\operatorname{supp}\nabla h_j \subset K_j$. Both sets lie at ∞-distance at least $(\rho-1)M - 1$ from the cut $\partial\Lambda_j^v$. Near the cuts, only L^2 information enters, and it enters only as an input.
- *One or two interior factors lift* L^2 *to* L^6. By (I2), one interior propagator takes L^2 data to $C^{1/2}$ values with L^6 gradients. At the scalar levels the steps carry no derivative in front, so the first factor alone does this. At the gauge level, by the regrouping of the derivatives in Section 41, a factor carrying a D^T on the right always receives its input from a factor of type $\mathcal{A}$, which has already lifted it. So the last one or two factors of a term suffice, and the rest of the term can be estimated in L^2.

The intermediate norms are block L^6 norms and block Hölder norms, which compose exactly like the L^2 decay norms. The source space $L^6 + D^T(C^\beta)$ of the blueprint is not needed as such. Its divergence part is carried by the factors $\mathcal{A}$ of the step and by the regrouping rule.

For $\mathbf{C}(s)$ the formula $\mathbf{G} - DG'RG'D^T - \mathbf{HQG}$, weakened term by term, does not work: the weakened $DG'(s)R(s)G'(s)D^T$ has no uniform C^1 bound (Remark 44.11). We use instead the torus identity $\mathbf{C} = \mathbf{G}\Pi^T - \mathbf{HQG}$, which involves only first derivatives of scalar fields.

Setting. We keep the setting of Section 41 at the levels (L1)–(L4), with $W \in \mathcal{W}_{\mathrm{loc}}(\varepsilon)$, and of Section 43. We take $d = 3$ and $\beta = \min(\alpha, \frac{1}{2})$. The level with W_2 is not treated. The background is real: the interior estimates of Sections 36 and 43 are proved for real backgrounds, and the complex perturbations of the background used for analyticity in Sections 40 and 41 are not covered here. The weakening parameters s are complex. $X'_{(r)}$ denotes the blocks of X'

at distance more than r from $\mathbb{T}_\eta \setminus X'$, as in Part VIII. As there, distances between blocks and sets are ∞-distances: in $X'_{(r)}$, $K_j^{(r)}$ and Y_j, in the seams Σ_w and the seam distance d_Σ, and in the radii of the near parts of Definition 44.1. Only the exponential weights $e^{\gamma|y-y'|}$ use the Euclidean distance. The inequalities $d_\Sigma(y) \le |y-w|$ for $w \in \Sigma$ and $d_\Sigma(y) \le |y-y_1| + d_\Sigma(y_1)$ of Section 41 hold with the ∞-distance, since $|\cdot|_\infty \le |\cdot|$.

44.1 Tiers and tier norms

For a block y and a fine field g (scalar, gauge field or 2-form) put

$$\mathfrak{n}_y^0(g) = \|g\|_{B(y)}, \qquad \mathfrak{n}_y^1(g) = \|g\|_{6,B(y)}, \qquad \mathfrak{n}_y^2(g) = |g|_{B(y)} + [\tilde g_y]_{\beta,\Lambda_1(c_y)},$$
$$\mathfrak{n}_y^3(g) = |g|_{B(y)} + |\nabla g|_{B(y)} + [\nabla \tilde g_y]_{\beta,\Lambda_1(c_y)},$$

with $\tilde g_y$ the field in a real gauge in which (B) holds on $\Lambda_4(c_y)$, as in Theorem 35.2. In $\mathfrak{n}_y^3$ the Hölder quotient is taken on $\Lambda_1(c_y)$, which is larger than $B(y)$, as in $\mathfrak{n}_y^2$. The interior estimates give this directly: Propositions 36.5, 43.4 and 43.7 hold on $\Lambda_\rho(c_y)$ with $\rho = 1$. Tier 3 is only ever an output tier. In Theorems 44.10 and 44.12, $|\cdot|_{1,\beta;y}$ is understood in this sense. For coarse fields all four are $|\omega(y)|$. Since blocks have volume 1, $\mathfrak{n}_y^0 \le \mathfrak{n}_y^1 \le C\mathfrak{n}_y^2$. Moreover $\mathfrak{n}_y^2(g) \le C \max_{|y'-y|_\infty \le 1} \mathfrak{n}_{y'}^3(g)$, which costs 1 in the radius of the bounds below; we do not record this. For the derivative of a field h, $\mathfrak{n}_y^2(\nabla h) \le C\mathfrak{n}_y^3(h)$. The index $t \in \{0,1,2,3\}$ is the *tier*. In words: L^2, L^6, C^β and $C^{1,\beta}$.

Definition 44.1 (Tier bounds)**.** Let $0 \le a, b \le 3$, $\gamma \ge 0$, $R \in [0,\infty]$, and let Y be a set of blocks. An operator F satisfies a *tier bound* $a \to b$ *on* Y *with radius* R and constant c if there are kernels $k, k' \ge 0$ such that for all g and all $y \in Y$

$$\mathfrak{n}_y^b(Fg) \le \sum_{|y'-y|_\infty \le R} k(y,y')\, \mathfrak{n}_{y'}^a(g) + \sum_{y'} k'(y,y')\, \mathfrak{n}_{y'}^0(g), \qquad \sup_{y\in Y} \sum_{y'} e^{\gamma|y-y'|}(k+k')(y,y') \le c.$$

The infimum of such c is $|F|^{a\to b}_{\gamma;R,Y}$. For Y the set of all blocks and $R = \infty$ we write $|F|^{a\to b}_\gamma$, the *global* tier norm; then the second sum can be absorbed in the first.

The first sum is the *near part*: regularity of the input is required only near y. The second sum, the *far part*, uses only L^2. Interior estimates give bounds of this split form directly. Global tier norms are needed for walks of unbounded length. Split forms with finite radius are used only for boundedly many factors.

Lemma 44.2 (Tier calculus)**.** *(a)* $|F|^{0\to 0}_\gamma \le |F|_\gamma$*, the decay norm of Section 40.*

(b) (Global composition.) $|FG|^{a\to c}_\gamma \le |F|^{b\to c}_\gamma |G|^{a\to b}_\gamma$.

(c) (Local composition.) *Suppose* $|F|^{b\to c}_{\gamma;R_F,Y_F} \le c_F$ *and* $|G|^{a\to b}_{\gamma;R_G,Y_G} \le c_G$*. Suppose also that every block within distance* R_F *of* Y_F *lies in* Y_G*, and that* $|G|_\gamma < \infty$*. Then*

$$|FG|^{a\to c}_{\gamma;R_F+R_G,Y_F} \le c_F\,(c_G + |G|_\gamma).$$

(d) (Commutators.) $|[F,h_j]|^{a\to b}_{\gamma/2;R,Y} \le C(\gamma M)^{-1}|F|^{a\to b}_{\gamma;R,Y}$ *for* $0 < \gamma \le 1$.

(e) (Seams.) *With the seam weights of Section 41, define* $|F|^{a\to b}_{\gamma,\Sigma}$ *like* $|F|^{a\to b}_\gamma$*, with* $e^{\gamma|y-y'|}$ *replaced by* $e^{\gamma(|y-y'|+d_\Sigma(y)+d_\Sigma(y'))}$*, and similarly for split forms. Then* $|AFB|^{a\to d}_{\gamma,\Sigma} \le |A|^{c\to d}_{2\gamma}|F|^{b\to c}_{\gamma,\Sigma}|B|^{a\to b}_{2\gamma}$*, and the analogue of (c) holds.*

Proof. (a) $\|\mathbf{1}_y F g\| \le \sum_{y'} \|\mathbf{1}_y F \mathbf{1}_{y'}\| \, \|\mathbf{1}_{y'} g\|$.

(b) Insert the bound for G into that for F. The kernels compose as matrices. Weighted row sums are submultiplicative, by $e^{\gamma|y-y''|} \le e^{\gamma|y-y'|} e^{\gamma|y'-y''|}$.

(c) In the near part of F, y' lies within R_F of $y \in Y_F$, hence in Y_G. So the bound for G applies at y', and its near part has radius R_G. In the far part of F use $\mathfrak{n}^0_{y'}(Gg) \le \sum_{y''} \|\mathbf{1}_{y'} G \mathbf{1}_{y''}\| \mathfrak{n}^0_{y''}(g)$. Then compose as in (b).

(d) As in Lemma 40.1(c), write $[F, h_j]g = F(\varphi g) - \varphi F g$ with $\varphi = h_j - h_j(c_y)$. By Lemma 17.3, $|\partial^k h_j| \le C M^{-k}$. Hence multiplication by φ changes $\mathfrak{n}^t_{y'}$ by at most a factor $CM^{-1}(1 + |y - y'|)$, for every t; for $t = 3$ use the discrete Leibniz rule. It needs $|h|$ and $|\nabla h|$ on all of $\Lambda_1(c_y)$, while $\mathfrak{n}^3_y$ contains them only on $B(y)$; they follow from the values on $B(y)$ and the Hölder quotient on $\Lambda_1(c_y)$, up to the local-gauge correction $C\varepsilon|h|$. Finally use $(1 + t)e^{-\gamma t/2} \le C\gamma^{-1}$.

(e) As (41.2), using $d_\Sigma(y) \le |y - y_1| + d_\Sigma(y_1)$. □

Weakenable families. For a weakenable family put

$$\|O\|^{a\to b}_{\kappa,\gamma} = \sup_{\square} \sum_{\omega:\square\subset X_\omega} e^{\kappa \deg \omega} |O_\omega|^{a\to b}_\gamma,$$

and similarly for split forms and seam norms. Lemma 40.3(a) and Lemma 40.4 hold in these norms, with margins increased by 1: Hölder quotients at y are taken on $\Lambda_1(c_y)$. Their proofs use only subadditivity of the seminorms, the submultiplicativity of Lemma 44.2(b), and the supports. In products, the tiers must match: each factor is measured from the tier that its right neighbour delivers.

44.2 Interior factors

Lemma 44.3 (Local Sobolev inequality)**.** *For v on Λ_2, $\|v\|_{6,\Lambda_1} \le C(\|\partial v\|_{\Lambda_2} + \|v\|_{\Lambda_2})$.*

Proof. Let χ be the cutoff of Step 1 of the proof of Theorem 6.3, rescaled to Λ_2, and $w = \chi v$, extended by zero. Then $w = \mathcal{G}(-\Delta w) = \sum_\mu (\partial^*_\mu g_\eta) \star (\partial_\mu w)$. The kernel is bounded by Ck_2, so Lemma 43.1(c) with $s = 2$, $p = 2$, $q = 6$ gives $\|w\|_6 \le C\|\partial w\|_2$. Finally, $\|\partial w\|_2 \le C(\|\partial v\|_{\Lambda_2} + \|v\|_{\Lambda_2})$. □

Lemma 44.4 (Interior factors)**.** *Let G_j be a local propagator of level (L1) or (L3) on a domain $Z_j = \Lambda^v_j \cap X$, i.e. G'_{Z_j} or $\mathbf{G}^{Z_j}$. Let φ be a function supported in $K^{(1)}_j$ with $|\partial^k \varphi| \le C$ for $k \le 2$, and let $Y_j = \{y :$ all blocks within distance 8 of y lie in $(Z_j)_{(4)}\}$. Then the following tier bounds hold on Y_j, with radius 4, rate $\gamma_1 = \gamma_0/2$ and a constant C_T, independent of j, X, M, k, n, η:*

$$\begin{array}{lllll} \varphi G_j: & 0 \to 2 \text{ (values)}, & 1 \to 3, & \\ \varphi G_j D^T: & 0 \to 1 \text{ (values)}, & 1 \to 2, & 2 \to 3, \\ \varphi D G_j: & 0 \to 1, & 1 \to 2, & \\ \varphi D G_j D^T: & 1 \to 1, & 2 \to 2. & \end{array}$$

In the gauge case, $\varphi \mathbf{G}^{Z_j} D$ satisfies $2 \to 3$ on scalar data supported in Z_j, at distance at least 2η from $\mathbb{T}_\eta \setminus Z_j$. All these operators also have L^2 decay (C, γ_0). Outside $K^{(2)}_j$ they vanish.

Proof. At $y \in Y_j$ the cube $\Lambda_3(c_y)$ is interior to Z_j. The bounds follow from the proof of Corollary 43.8, from Propositions 43.4 and 43.7, and from Proposition 36.5 for the Hölder data. In that proof the local data enter only through their norms on $\Lambda_3(c_y)$, which gives the near part with radius 4. The rest enters through $\mathfrak{L}(u)$, which gives the far part, by the L^2 decay of the propagators with derivatives. Weighted row sums with rate $\gamma_1 < \gamma_0$ are bounded by $CS_{\gamma_0-\gamma_1}$ (Lemma 40.1(b)). The entry $\varphi G_j D^T : 0 \to 1$ is Lemma 44.3. It needs

the full gradient of $v = G_j D^T g$ in L^2 near y. In the scalar case this is $\|Dv\|$. In the gauge case, for a cutoff χ supported in the interior, summation by parts on the full lattice gives $\|\partial(\chi v)\|^2 = \|d(\chi v)\|^2 + \|d^*(\chi v)\|^2$ in the local gauge, up to $C\varepsilon |\!|\!| \chi v |\!|\!|^2$. So the full gradient is controlled by $\|Dv\|$, $\|D_{Z_j}^T v\|$ and $\|v\|$ near y. The L^2 inputs used here and in the far parts are the decay of G_j, DG_j, $G_j D^T$, $DG_j D^T$ (Theorems 18.2(b), 19.7(a)), and, in the gauge case, of $D_{Z_j}^T \mathbf{G}^{Z_j}$ and $D_{Z_j}^T \mathbf{G}^{Z_j} D^T$. The latter follow as in Theorem 19.7(a), since $|\!|\!| \cdot |\!|\!|_X$ contains $\|D_X^T A\|$. The bound $1 \to 1$ for $\varphi D G_j D^T$ is the gradient part of Proposition 43.7(b) with $p = 2$, $p^* = 6$, i.e. the Calderón–Zygmund input. For $\varphi \mathbf{G}^{Z_j} D$, note that $D\lambda = \partial\lambda + T\lambda$ and $\partial_\mu \lambda = -\partial_\mu^*(S_\mu \lambda)$, so $D\lambda$ is a source in local form with Hölder data $S_\mu \lambda$. For λ supported in Z_j away from $\mathbb{T}_\eta \setminus Z_j$, the restriction of $D\lambda$ to $\ell_1^2(Z_j)$ is $D\mathbf{1}_{Z_j}\lambda$, and the L^2 decay of $\mathbf{G}^{Z_j} D \mathbf{1}_{Z_j}$ is the transpose of that of $D_{Z_j}^T \mathbf{G}^{Z_j}$. Scalar data near the cut are treated in Lemma 44.6. Multiplication by φ preserves all tiers, by the Leibniz rule. □

For $X = \mathbb{T}_\eta$ the blocks of $K_j^{(2)}$ lie in Y_j, because K_j is at ∞-distance at least $(\rho - 1)M - 1$ from $\partial\Lambda_j^v$. So for the top-level walks these bounds hold at every block where the factors do not vanish. In particular they are global tier bounds.

Scalar gradients at the cut. The space $\ell_1^2(Z)$ contains the bonds crossing ∂Z. So for a scalar field λ on $\mathbb{T}_\eta$, the restriction of $D\lambda$ to $\ell_1^2(Z)$ is $D(\mathbf{1}_Z \lambda) + c_Z(\lambda)$, where $c_Z(\lambda)$ is supported on the crossing bonds and has size $\eta^{-1}|\lambda|$ at the exterior endpoint. This surface term is not bounded in L^2 uniformly in η. By duality, $\langle w, \mathbf{G}^Z c_Z(\lambda)\rangle = \mathrm{b}(w^*, \mathbf{G}^Z c_Z(\lambda))$ is bounded by $|\lambda|$ times the L^1 trace $\sum_b \eta^{d-1}|v(b)|$ of $v = \mathbf{G}^Z w^*$ on the crossing bonds (proof of Lemma 44.6). We need the following discrete trace inequality. A *box* is a domain of the form $Z = \{x : \ x_\mu \in I_\mu, \ \mu = 1, \dots, d\}$, where each I_μ is either the whole circle or a lattice arc whose complement has at least two points. The local domains Λ_j^v are boxes, at every $\eta \le 1$. Indeed, $(\Lambda_j^v)^{(R_v)} \subset X(j)$ with $R_v \ge 1$, and $X(j)$ does not wrap around the torus, by the torus side of Section 40. If in some direction the complement of the interval of Λ_j^v had fewer than $2R_v$ points, then $(\Lambda_j^v)^{(R_v)}$, and with it $X(j)$, would wrap around the torus in that direction. So each such complement has at least $2R_v \ge 2$ points. What the proof uses is the product structure, which rules out re-entrant corners. Write $\mathfrak{B}_{\mathrm{cr}}(y')$ for the bonds that have one endpoint in $Z \cap B(y')$ and the other endpoint outside Z.

Lemma 44.5 (Trace inequality on boxes)**.** *Let Z be a box, i.e. a product of intervals that is a union of unit blocks, so that each side that does not wrap around the torus has length at least 1; as in the definition of a box, in each direction in which Z does not wrap around, the complement of its interval has at least two points. For a union of unit blocks this holds automatically if $\eta \le \frac{1}{2}$. Let U be a real $(1, \varepsilon)$-regular background with $\varepsilon \le 1$, or a complex perturbation of such a background of size δ_c small (Proposition 19.11). Then for every $A \in \ell_1^2(Z)$ and every block y' meeting ∂Z,*

$$\sum_{b \in \mathfrak{B}_{\mathrm{cr}}(y')} \eta^{d-1} |A(b)| \le C \big(\|A\| + \|DA\| + \|D_Z^T A\| \big)_{B(y')^{(2)}},$$

where the norms are those of $|\!|\!| \cdot |\!|\!|_Z$, localized to $B(y')^{(2)}$. Thus $\|DA\|$ runs over all plaquettes with a bond in $\mathfrak{B}(Z)$, including those crossing ∂Z, and $\|D_Z^T A\|$ over the points of Z. The crossing plaquettes carry the boundary condition. With interior plaquettes only, the lemma would be false: the lattice gradient of a harmonic function has normal trace only in $H^{-1/2}$. The constant depends only on d and $\dim \mathfrak{g}$.

The side length matters. For a slab of one fine layer, at $U = 1$, put $A_1 = a$ on the normal bonds crossing both faces and $A = 0$ elsewhere. Then $dA = 0$ and $d_Z^* A = 0$, while the localized

norm is $O(\sqrt{\eta})|a|$ and the trace is of order $|a|$. The local domains Λ_j^v, and the boxes to which the lemma is applied, are unions of unit blocks.

Proof. Step 1: an exact identity at $U = 1$. Let $\mathfrak{B}(Z)$ be the set of bonds with an endpoint in Z, so that $A \in \ell_1^2(Z)$ vanishes off $\mathfrak{B}(Z)$. By Lemma 2.2 on the torus,

$$\|dA\|^2 + \|d^*A\|^2 = \|\partial A\|^2 = \sum_{\mu,\nu} \|\partial_\mu A_\nu\|^2,$$

a sum over all pairs of parallel bonds at distance η. Split $d^*A = \mathbf{1}_Z d^*A + \mathbf{1}_{Z^c} d^*A$. Let $x \notin Z$. A bond at x lies in $\mathfrak{B}(Z)$ only if its other endpoint lies in Z. Since Z is a product of intervals, x has at most one neighbour in Z. Indeed, x must differ from Z in exactly one coordinate ν, and by exactly η. So $(d^*A)(x) = \pm\eta^{-1}A(b_x)$, where b_x is the unique bond from x into Z, or $(d^*A)(x) = 0$ if there is none. Let b'_x be the bond parallel to b_x that continues outward from x. It has no endpoint in Z, so $A(b'_x) = 0$, and the pair (b_x, b'_x) contributes exactly $\eta^d\eta^{-2}|A(b_x)|^2$ to $\|\partial A\|^2$. The pairs (b_x, b'_x) are distinct for distinct x. Hence

$$\|dA\|^2 + \|d_Z^*A\|^2 = \|\partial A\|^2 - \|\mathbf{1}_{Z^c} d^*A\|^2 \;\geq\; \|\partial A\|^2_{\mathfrak{B}(Z)},$$

where $\|\partial A\|^2_{\mathfrak{B}(Z)}$ is the sum over the pairs of parallel bonds that *both* lie in $\mathfrak{B}(Z)$. All omitted terms are non-negative. Convexity of the box enters through the uniqueness of b_x.

Step 2: a one-dimensional trace. Let $b \in \mathfrak{B}_{\mathrm{cr}}(y')$ point in the direction $\pm e_\nu$ normal to the face. The parallel bonds $b_m = b \mp m\eta e_\nu$, $0 \leq m < \eta^{-1}$, lie in $\mathfrak{B}(Z)$, since Z is a union of blocks and has side at least 1. Consecutive ones form pairs of $\|\partial A\|_{\mathfrak{B}(Z)}$. Averaging $|A(b)| \leq |A(b_m)| + \sum_{j<m} |A(b_j) - A(b_{j+1})|$ over m and summing over the crossing bonds of $B(y')$ with weight η^{d-1} gives

$$\sum_{b \in \mathfrak{B}_{\mathrm{cr}}(y')} \eta^{d-1}|A(b)| \leq C\big(\|A\| + \|\partial A\|_{\mathfrak{B}(Z)}\big)_{B(y')^{(1)}},$$

by Cauchy–Schwarz on a region of bounded volume.

Step 3: localization. Let χ be a smooth cutoff at unit scale, equal to 1 on $B(y')^{(1)}$ and supported in $B(y')^{(2)}$, and put $(\chi A)(b) = \chi(b_-)A(b)$. Then $\chi A \in \ell_1^2(Z)$. Moreover $d(\chi A) - \chi\, dA$ and $d_Z^*(\chi A) - \chi\, d_Z^* A$ are sums of differences of χ times shifted values of A, bounded by $C\|A\|_{B(y')^{(2)}}$. Apply Steps 1 and 2 to χA.

Step 4: the background. All quantities in the lemma are invariant under real gauge transformations. By regularity, on a cube containing $B(y')^{(3)}$ (the regularity cubes have half-side $4M+2 \geq 6$) there is a real gauge with $U = e^{\mathrm{i}\eta A_0}$ satisfying (B). There, applying Lemma 12.2 in this local gauge, $D = \partial + T$ and $D^T = \partial^* + T^T$, with T local and $|T| \leq C\varepsilon$. So $\|d(\chi A)\| \leq \|D(\chi A)\| + C\varepsilon\|\chi A\|$, and similarly for d_Z^*. Finally, $\|D(\chi A)\| \leq \|DA\|_{B(y')^{(2)}} + C\|A\|_{B(y')^{(2)}}$ by the Leibniz rule for D, and similarly for D_Z^T. For complex perturbations, $D_{U_\delta} = D_U + T_\delta$ with T_δ local of size $C\delta_c$ (Proposition 19.11). □

The continuum analogue is the Gaffney–Friedrichs inequality for fields with vanishing tangential trace on a convex domain. On a box the lattice version is an exact identity. The exterior divergence cancels exactly against the outward normal jumps. The remaining boundary terms, the tangential jumps, are non-negative and are dropped. At re-entrant corners the uniqueness of b_x, and with it the inequality, would fail, as in the continuum.

Lemma 44.6 (Cut term)**.** *Let Z be a box in the sense of Lemma 44.5. Then, for scalar λ on $\mathbb{T}_\eta$,*

$$\|\mathbf{G}^Z c_Z(\lambda)\|_{B(y)} + \|D\mathbf{G}^Z c_Z(\lambda)\|_{B(y)} + \|D_Z^T \mathbf{G}^Z c_Z(\lambda)\|_{B(y)} \leq C \sum_{y'} e^{-\gamma_1|y-y'|}|\lambda|_{B(y')^{(1)}},$$

the sum running over the blocks y' of Z meeting ∂Z. The neighbourhood $B(y')^{(1)}$ is needed because $c_Z(\lambda)$ takes λ at the exterior endpoint of the crossing bonds. Consequently, $\mathbf{1}_{\ell^2_1(Z)}D$ followed by $\varphi\mathbf{G}^Z$ satisfies the bound $2\to 3$ of Lemma 44.4 at blocks of Y_j, with a far part in $\mathfrak{n}^2$ instead of $\mathfrak{n}^0$ at the blocks meeting ∂Z.

Proof. L^2 input. Theorem 19.7(a), proved with Combes–Thomas in the norm $|\!|\!| \cdot |\!|\!|_{X,+}$, gives decay (C,γ_0) for all operators $P\,\mathbf{G}^X Q$ with $P \in \{1, D, D_X^T\}$ and $Q \in \{1, D^T, D\mathbf{1}_X\}$. For $Q = D\mathbf{1}_X$ this uses $\langle u, D\mathbf{1}_X g\rangle = \langle D_X^T u, g\rangle \le |\!|\!| u |\!|\!|_X \|g\|$.

Proof of the bound. For a field u, $\|u\|_{B(y)}$ is the supremum of $|\langle w, u\rangle| = |\mathrm{b}(w^*, u)|$ over the fields w supported in $B(y)$ with $\|w\| \le 1$. Here w^* is the pointwise adjoint, which has the same support and norm. The operator $\mathbf{G}^Z$ is symmetric for the bilinear form b. At real references this is its symmetry, since b-transposes of real operators are their adjoints on real fields. At complex references the kernel of $\mathbf{G}^Z$ is holomorphic in the perturbation, and the identity $(\mathbf{G}^Z)^T = \mathbf{G}^Z$ extends from the real points by the identity theorem, as in step (iii) of the proof of Proposition 46.7. Hence $\mathrm{b}(w^*, \mathbf{G}^Z c_Z(\lambda)) = \mathrm{b}(\mathbf{G}^Z w^*, c_Z(\lambda))$, and similarly with $D^T w^*$ and $D\mathbf{1}_Z w^*$ for the derivatives. Moreover $|\mathrm{b}(v, c_Z(\lambda))| \le C\sum_b \eta^{d-1}|v(b)|\,|\lambda|$, the sum running over the crossing bonds, with λ taken at their exterior endpoints. Apply Lemma 44.5 to $v = \mathbf{G}^Z w^*$, respectively $\mathbf{G}^Z D^T w^*$ and $\mathbf{G}^Z D\mathbf{1}_Z w^*$. This needs $|\!|\!| v |\!|\!|_Z$ near the crossing blocks, which the L^2 input just stated provides, with exponential decay.

The consequence. Write $\lambda = \lambda_{\mathrm{in}} + \lambda_{\mathrm{cut}}$ with a smooth cutoff, so that λ_{in} vanishes within distance 2 of $\mathbb{T}_\eta \setminus Z$ and λ_{cut} is supported within distance 3 of it. The source $D(\mathbf{1}_Z\lambda_{\mathrm{in}})$ is covered by Lemma 44.4. At $y \in Y_j$, the source $D(\mathbf{1}_Z\lambda_{\mathrm{cut}}) + c_Z(\lambda)$ is far. It enters only through the L^2 decay of $\{1, D, D_Z^T\}\mathbf{G}^Z D\mathbf{1}_Z$ and the bound just proved, i.e. as a far part. That far part is of tier 0 for λ_{cut}, and is bounded by $|\lambda|$ for c_Z. □

A far part in $\mathfrak{n}^2$ is harmless in global tier norms. It is used only in Theorem 44.10(b), where the input is globally of tier 2.

44.3 The scalar levels on domains: lifting

Lemma 44.7 (Lift lemma)**.** *Let Z be a domain, and let $G'^Z(s)$, $P_{\mathcal{V}}^Z(s)$, $R_Z(s)$ be the weakened families of (L1), (L2) on Z. Let $Y = Z_{(16)}$. With radius 16 and constants $C(\kappa,\delta)$ at κ (the families being evaluated at $\kappa+\delta$), the following tier bounds hold on Y:*

(Z1) $G'^Z(s)\colon 0 \to 2$ (values), and $DG'^Z(s)\colon 0 \to 1$;

(Z2) $G'^Z(s)D^T\colon 0 \to 1$ (values), and $DG'^Z(s)D^T\colon 1 \to 1$;

(Z3) $P_{\mathcal{V}}^Z(s)$, $P_{\mathcal{V}}^Z(s)D^T\colon 0 \to 2$ (values), and $DP_{\mathcal{V}}^Z(s)$, $DP_{\mathcal{V}}^Z(s)D^T\colon 0 \to 1$;

(Z4) $R_Z(s)\colon t \to t$ for $t \le 2$. Here, inside products, $R_Z(s) = \mathbf{1}_Z - P_{\mathcal{V}}^Z(s)$ with the literal identity, which needs no monomial. No tier bound is claimed for $R_Z(s)$ standing alone. It is used alone only in L^2, with its identity split box by box (Section 40), and that split does not preserve tiers.

The differences between $Z = \mathbb{T}_\eta$ and $Z = X'$ (restricted to X') satisfy the same bounds in seam form on $Y = X'_{(16)}$, with seam width w.

Proof. Structure of the terms. The walk of (L1) has only local steps, and at the scalar levels $\mathcal{A} = 0$ (Section 41). Indeed $[D^T D, h_j]G'_j = D^T((\partial h_j)\cdot SG'_j) + (\partial h_j)\cdot DG'_j$, up to shifts, and the derivative in front can be expanded onto G'_j, since DG'_j has decay. So a term of $G'^Z(s)$ with n steps is a product $\Phi_0\Phi_1\cdots\Phi_n$ with:

- $\Phi_0 = \mathcal{F}_0$;
- $\Phi_i = \mathcal{B}$ for $i \ge 1$, a sum of terms $\varphi DG'_j$, $\varphi G'_j$ and $[aQ'^T Q', h_j]G'_j$;

- an external D^T, as in $G'(s)D^T$, carried by Φ_n.

Every factor has L^2 decay norm bounded by C, and by θ for $i \geq 1$ (Lemma 41.3). Every factor vanishes outside $K_j^{(2)}$ for its j. At blocks of Y every factor is interior in the sense of Lemma 44.4.

(Z1). Estimate $\Phi_1 \cdots \Phi_n$ in L^2, i.e. $0 \to 0$ by Lemma 44.2(a). Lemma 44.4 gives $0 \to 2$ for the values of Φ_0 and $0 \to 1$ for $D\Phi_0$. Lemma 44.2(c) composes, and every block within distance 4 of Y is interior.

(Z2). For $n = 0$ the term is $\mathcal{F}_0 D^T$ acting on the external input. This gives $0 \to 1$ for the values, and $1 \to 1$ for $D\mathcal{F}_0 D^T$, by Lemma 44.4. For $n \geq 1$ the external D^T sits on Φ_n, which is estimated in L^2, and (Z1) applies.

(Z3). $Q'G'^Z(s)$ and $Q'G'^Z(s)D^T$ map L^2 to coarse fields, by the L^2 bounds. Coarse fields are bounded by L^2 norms, and $(M'^Z)^{-1}(s)$ is a unit-lattice family. Finally Q'^T maps coarse fields to bounded fields, so its output has tier 1. Apply (Z1) to $G'^Z(s)Q'^T$.

(Z4). $R_Z(s) = \mathbf{1}_Z - P_{\mathcal{V}}^Z(s)$. The identity preserves tiers, and (Z3) bounds the second term.

Sums and seams. Summing over the terms is Theorem 41.4(a), with the norms of at most three factors replaced by their tier norms. For the differences, telescope as in Theorem 41.4(b). If the distinguished difference factor is among $\Phi_2, \dots, \Phi_n$, it is estimated in the L^2 seam norm, and Lemma 44.2(e) composes. If it is Φ_0 or Φ_1, it is a difference $\varphi(G'_{\Lambda_j^v \cap \mathbb{T}_\eta} - G'_{\Lambda_j^v \cap X'})$ of interior propagators, at blocks interior to both domains. Its argument solves the same equation, so the difference solves $D^T D\, \delta v = -aQ'^T Q' \delta v$ there. Proposition 36.3, with this source bounded by L^2 norms, bounds every tier of δv at y by $\|\delta v\|$ near y. The cross terms $\mathbf{1}_{X'} F \mathbf{1}_{X \setminus X'}$ at Φ_0 or Φ_1 vanish within distance 16 of $y \in X'_{(16)}$, so they enter only through L^2. The latter is of seam form by Theorem 18.2(e). □

Remark 44.8. The near part in (Z2) requires tier 1 of the input only near y, and only for the term with $n = 0$. All other terms need only L^2 input. This is why no tier bounds near ∂Z are needed. Long chains are estimated in L^2, where the Dirichlet cuts are harmless. Regularity is recovered at the end, at an interior output block.

44.4 Step bounds for the top-level walks

We now take $X = \mathbb{T}_\eta$ at the levels (L1) and (L3). Write G_j for the local propagators on Λ_j^v and $u = G_j g$.

Rates. At each of these levels let γ_0 be at most:

- the rate of (A2), from Theorems 18.2 and 19.7;
- the rate γ of Corollary 43.8, which enters the far parts of Lemma 44.4;
- at (L3), $\frac{1}{32}$ of the smaller of the output and compatibility rates of the walks of (L1) and (L2) (Theorem 41.4).

Then Lemma 44.4 holds with rate $\gamma_0/2$, and Lemmas 44.7 and 44.9 hold with rate $2\gamma_0$, and γ_0 in seam form, as (A3) requires. The rates decrease by fixed factors at each level. There are boundedly many levels, so the final rates are positive constants.

Lemma 44.9 (The non-local part in tier norms)**.** *Let $Z \in \{\mathbb{T}_\eta\} \cup \{\Lambda_j^v\}$, and let $X_i^T M_i^Z(s) Y_i$ be the terms of $\mathfrak{N}^Z(s)$ at level (L3), with the factorization of Section 41. Then, on $Z_{(32)}$ and with constants $C'_{\mathfrak{N}}(\kappa, \delta)$:*

- *if $X_i^T = 1$, then $M_i^Z(s)$ maps $0 \to 1$, except for the term $-DR_Z(s)G'^Z(s)D^T W$, which maps $1 \to 1$ with near radius 16;*

- *if $X_i^T = D^T$, then $M_i^Z(s)$ maps $0 \to 2$;*
- *the differences between $Z = \mathbb{T}_\eta$ and $Z = \Lambda_j^v$ satisfy the same bounds in seam form.*

Proof. By Section 41, (L3), and $(\mathfrak{B}DS)^T = S^T(\mathfrak{F}^T D + D^T W)$ with $S^T = DRG'$, the terms of $\mathfrak{N}$ are (all on Z, at s). As in Section 41, a left gradient D is part of the middle factor. Only sup norms and Hölder quotients of the coefficients of $\mathfrak{F}$ and W enter below, never their derivatives.

(i) $-DP_{\mathcal{V}}D^T$: $X^T = 1$, $M = DP_{\mathcal{V}}$, $Y = D^T$. By (Z3), $DP_{\mathcal{V}}\colon 0 \to 1$.

(ii) $-D^T\mathfrak{F}S$ and $-WDS$, with $S = G'RD^T$: here $X^T = D^T$, respectively $X^T = 1$, $Y = D^T$, and $M = \mathfrak{F}G'R$, respectively $M = WDG'R$. By (Z4), (Z1), $G'R\colon 0 \to 2$ (values) and $DG'R\colon 0 \to 1$. The operators $\mathfrak{F}$ and W are local of order 0, with Hölder coefficients.

(iii) $-DRG'\mathfrak{F}^T D$ and $-DRG'D^T W$: $X^T = 1$, $M = DRG'\mathfrak{F}^T$, respectively $M = DRG'D^T W$, and $Y = D$, respectively $Y = 1$. By (Z1), (Z3), $DRG'\mathfrak{F}^T\colon 0 \to 1$. By (Z2), (Z3), $DRG'D^T\colon 1 \to 1$.

(iv) $S^T D^T \mathfrak{B}DS = DRG'(\mathfrak{F}^T\mathfrak{F} + D^T WD)G'RD^T$: $X^T = 1$, $M = DRG'(\mathfrak{F}^T\mathfrak{F} + D^T WD)G'R$, $Y = D^T$. The inner $G'R$ and $DG'R$ map $0 \to 2$ (values) and $0 \to 1$, by (Z1), (Z4). So $\mathfrak{F}^T\mathfrak{F}G'R$ maps $0 \to 1$, and $WDG'R$ maps $0 \to 1$. The outer DRG' and $DRG'D^T$ map $1 \to 1$, by (Z1)–(Z4).

Compose with Lemma 44.2(c). At most two compositions of split bounds of radius 16 occur, so the radius is at most 32 and $Y = Z_{(32)}$ suffices. The seam statement follows from the seam part of Lemma 44.7 and Lemma 44.2(e). □

The constants $C'_{\mathfrak{N}}(\kappa, \delta)$ do not depend on M, k, n, η or the background.

Theorem 44.10 (Weakened walks in tier norms)**.** *At the levels (L1) and (L3) on $\mathbb{T}_\eta$, let $\mathcal{F}_0, \mathcal{A}, \mathcal{B}$ be the factors of Section 41. Then*

$$\|\mathcal{A}\|_{\kappa,\gamma}^{1\to 2} + \|\mathcal{A}D^T\|_{\kappa,\gamma}^{1\to 2} + \|\mathcal{B}\|_{\kappa,\gamma}^{1\to 1} + \|\mathcal{B}D^T\|_{\kappa,\gamma}^{1\to 1} \le \theta',$$

with $\gamma = \gamma_0/8$. Here $\theta' = \theta'(\kappa)$ has the form of θ in Lemma 41.3, with C_G replaced by $\max(C_T, C_G)$ and $C_{\mathfrak{N}}$ by $C'_{\mathfrak{N}}(\kappa + \delta, \delta)$ of Lemma 44.9:

$$\theta'(\kappa) = Ce^{\kappa \mathsf{N}_{\mathrm{b}}}\Big(\max(C_T, C_G) C_{\mathrm{loc}} M^{-1} \\ + C\delta^{-1}\big((\gamma_0 M)^{-1} + e^{-\gamma_0(\rho-1)M/16}\big) C'_{\mathfrak{N}}(\kappa+\delta, \delta) \max(C_T, C_G)\Big).$$

The factor $\max(C_T, C_G)$ comes from Lemma 44.2(c). The walk at κ uses the step factors at $\kappa + \delta$. It converges if $2C_d\delta^{-2}\theta'(\kappa+\delta) \le \frac{1}{2}$, which holds for $M \ge M_1'$. We choose M_1' with θ' replaced by $2\theta'$; the coercivity constants are the fixed ones of Proposition 46.7, and every decay bound that enters is taken at the fixed rate of the differences there, a quarter of the minimum of the perturbation rate and the two chosen inverse decay rates in Lemma 13.3; neither depends on δ_c, so the same numbers serve at real and at complex references (Proposition 46.7, the step "The constants" of the proof). At complex references of size $\delta_c < \delta_P$, since $C_{\mathrm{cx}}\delta_P \le 1$, the inputs of θ' change by at most a factor $1 + \sigma_{\mathrm{cx}}$, $(1 + \sigma_{\mathrm{cx}})^3 \le 2$ ($\sigma_{\mathrm{cx}} = 2^{1/3} - 1$), and θ' is a sum of products of at most three of them, so $2\theta'$ bounds θ' there (Proposition 46.7, the step "The constants" of the proof). So M_1' does not depend on δ_P, and it serves at these references as well. Then $\|G(s)\|_{\kappa,\gamma}^{1\to 3} \le 4C_d\delta^{-2}C_j e^{(\kappa+\delta)\mathsf{N}_{\mathrm{b}}} \max(C_T, C_G)$, with C_j the number of j with $\square \subset X(j)$. For $M \ge M_1'$ the walks $G(s) = \mathcal{F}_0 \sum_n \mathcal{R}(s)^n$ satisfy, with constants $C(\kappa, \delta)$:

(a) $G(s)\colon 1 \to 3$; $G(s)D^T\colon 1 \to 2$ (values) and $2 \to 3$; $DG(s)\colon 1 \to 2$; $DG(s)D^T\colon 1 \to 1$;

(b) in the gauge case, $\mathbf{G}(s)D\colon 2 \to 3$ *on scalar data.*

All of these are global tier norms of weakenable families. In particular

$$|\mathbf{G}(s)f|_{1,\beta;y} \le C\sum_{y'} e^{-\gamma|y-y'|}|f|_{B(y')}, \qquad s \in \mathbb{D}_\kappa,$$

and the same holds for $G'(s)$*. For* $\mathbf{G}(s)D^T$ *and* $DG'(s)D^T$ *the analogous statements are the tier bounds of (a). For* $DG'(s)D^T$ *these are* $1 \to 1$ *and* $2 \to 2$ *only: a bound from sup norms to sup norms fails, as the Calderón–Zygmund estimate does at* $p = \infty$*.*

Proof. The input of a step. A step acts on $u = G_j g$ with g of tier 1. By Corollary 43.8 on Λ_j^v, u is of tier 3, and Du, $D^T u$ are of tier 2, at every block of $(\Lambda_j^v)_{(16)}$. Near the cut only their L^2 norms are controlled. We check that each part of the step requires regularity of u only at blocks of $(\Lambda_j^v)_{(16)}$:

- *local commutators:* they live on $K_j^{(1)}$;
- *seam term:* its output is restricted to K_j by h_j. The near parts of the bounds of Lemma 44.9 have radius at most 32, much smaller than $(\rho-1)M$;
- *non-local commutator:* $[F, h_j]u = F(\varphi u) - \varphi F u$ with $\varphi = h_j - h_j(c_y)$. If $y \notin K_j^{(34)}$, then $\varphi = h_j$ vanishes on the 32-neighbourhood of y. So neither term requires regularity of u near y. If $y \in K_j^{(34)}$, the near parts involve blocks of $(\Lambda_j^v)_{(16)}$.

Step factors. The local commutator $[\mathfrak{A}_{\mathrm{loc}}, h_j]G_j$ consists of:

- $D^T(\varphi G_j)$ and $\varphi \nabla G_j$, with $|\partial^k \varphi| \le CM^{-1}$; in the gauge case ∇G_j stands for DG_j and $D_{Z_j}^T G_j$, from $[D^T D + DD^T, h_j]$; by the discrete Leibniz rule, $D[D^T, h_j]G_j$ is a sum of terms $\varphi' G_j$ with $|\varphi'| \le CM^{-2}$ and of terms $\varphi \nabla G_j$ with ∇ the full covariant gradient, up to shifts by at most η, and the latter are bounded like $\varphi D G_j$, since $\mathfrak{n}_y^2(\nabla h) \le C\mathfrak{n}_y^3(h)$ and (A2) includes ∇G_j;
- φG_j, with $|\varphi| \le CM^{-2}$;
- $[aQ'^T Q', h_j]G_j$, respectively $[a\mathbf{Q}^T\mathbf{Q}, h_j]G_j$, of size CM^{-1}, whose output is bounded on each block by L^2 norms (tier 1);
- $[W, h_j]G_j$, of size $C\eta M^{-1}$.

By Lemma 44.4, the $\mathcal{A}$-parts φG_j map $1 \to 3$, and $\varphi G_j D^T$ maps $1 \to 2$. The $\mathcal{B}$-parts map $1 \to 1$, also with D^T on the right. All carry the factor CM^{-1}. The non-local parts are $X_i^T[M_i, h_j]Y_i G_j$, $[X_i^T, h_j]M_i Y_i G_j$, $X_i^T M_i [Y_i, h_j]G_j$ and the seam term:

- The steps are taken in the regrouped form of the proof of Lemma 41.3, so that $Y_i = D^T$ acts on G_j as the divergence $D_{\Lambda_j^v}^T$ of the local domain. By Lemma 44.9, Lemma 44.2(c),(d), and the L^2 bounds of $Y_i G_j$ and $Y_i G_j D^T$ from (A2), the commutator terms map $1 \to 1$ (and $1 \to 2$ for the $\mathcal{A}$-part) with the factor $C(\gamma_0 M)^{-1}$. Here (c) is applied with the case split of the first paragraph: for $y \notin K_j^{(34)}$ the near part vanishes, and for $y \in K_j^{(34)}$ the inclusion of (c) holds with $Y_F = K_j^{(34)}$.
- The seam term carries $e^{-\gamma_0(\rho-1)M/16}$, by the seam part of Lemma 44.9 and Lemma 44.2(e).
- The term $-DRG'D^T W$ needs tier 1 of Wu near the output. This is available by the first paragraph.

The counting is that of Lemma 41.3, which gives θ'.

The walk. A term with n steps is a product $\Phi_0 \cdots \Phi_n$ with the regrouping rule of the proof of Lemma 44.7. Measure:

- Φ_i, $i \geq 2$, in $1 \to 1$;
- Φ_1 in $1 \to 2$ if $\Phi_0 = \mathcal{F}_0 D^T$, which forces Φ_1 to be of type $\mathcal{A}$, and in $1 \to 1$ otherwise;
- Φ_0 in $2 \to 3$, respectively $1 \to 3$, by Lemma 44.4.

For $X = \mathbb{T}_\eta$ these are global tier norms, by the remark after Lemma 44.4. Lemma 40.4 in tier norms and $2C_d\delta^{-2}\theta'(\kappa+\delta) \leq \frac{1}{2}$ give (a) for $G(s)$. The other statements follow in the same way:

- an external D^T is carried by Φ_n, which maps $1 \to 1$ or $1 \to 2$, and $2 \to 2$ by Lemma 44.4;
- an external D (in (b)) is handled, for $n = 0$, by the entry $\varphi \mathbf{G}^{Z_j} D \colon 2 \to 3$ and by Lemma 44.6, which uses Lemma 44.5. For $n \geq 1$ the last factor $\Phi_n D$ must map $2 \to 1$, or $2 \to 2$ if it is of type $\mathcal{A}$. Its parts are $\varphi \nabla G_j D$, $\varphi G_j D$, $[a\mathbf{Q}^T\mathbf{Q}, h_j]G_j D$, $[W, h_j]G_j D$, $[M_i, h_j]Y_i G_j D$, $[X_i^T, h_j]M_i Y_i G_j D$ and $X_i^T M_i [Y_i, h_j] G_j D$ with $Y_i \in \{D^T_{\Lambda_j^v}, D, 1\}$, and the seam piece. Each is bounded by Lemma 44.4 for the interior part and by Lemma 44.6, whose L^2 input covers $P \in \{1, D, D_Z^T\}$, for the rest;
- a left D uses $D\mathcal{F}_0 \colon 1 \to 2$ and $D\mathcal{F}_0 D^T \colon 2 \to 2$.

The pointwise bound is (a) with $\mathfrak{n}^1_{y'}(f) \leq |f|_{B(y')}$. □

44.5 The constrained operators

Remark 44.11 (Why not the defining formula for $\mathbf{C}$). This remark is heuristic; it explains the choice below and is not used. Weaken $\mathbf{C} = \mathbf{G} - DG'RG'D^T - \mathbf{HQG}$ term by term, and put $\sigma_2(s) = G'(s)R(s)G'(s)D^T f$. Then $Q'\sigma_2(s) \neq 0$ for $s \neq 1$, in general. Each first factor $h_j G_j' k_j$ of the walk satisfies an equation with the averaging term, so $D^T D\sigma_2(s)$ is a sum of Hölder terms and of the jump terms $-a\sum_j s^{X(j)} h_j Q'^T Q' G_j' k_j$. These jump across block faces. At $s = 1$ the jumps cancel, since $D^T D\sigma_2 = Rh_0$ is Hölder there. For $s \neq 1$ they do not cancel in general. By Remark 43.6, $\partial\partial\sigma_2(s)$, and hence $\nabla D\sigma_2(s)$, is in general not bounded uniformly in η near block edges. The other two terms of $\mathbf{C}(s)$ are $C^{1,\beta}$ by Theorem 44.10, so the sum is not uniformly C^1. The same obstruction defeats the alternative of applying the unweakened operator to $\mathbf{C}(s)f$. In the equation for $u(s) = \mathbf{C}(s)f$, the term $DRD^T u(s)$ produces the source D applied to these jump terms. This is a divergence of bounded but discontinuous data, and it gives only $u(s) \in C^\beta$.

On the torus, Lemma 15.3(a) gives $\mathbf{G}DR = DG'R$. Since $E = 0$ (Proposition 22.7), $\mathbf{Q}\Pi = \mathbf{Q}$ (Lemma 15.1(c)), so $\mathbf{Q}_\pi = \mathbf{Q}$, $\mathbf{Q}_\pi^T = \mathbf{Q}^T$ and $\mathbf{\Gamma}_\pi = \mathbf{\Gamma}$. Hence

$$\begin{aligned} DG'RG'D^T &= \mathbf{G}\, DRG'D^T, & \mathbf{C} &= \mathbf{G}\,\Pi^T - \mathbf{HQG}, \\ \Pi^T &= 1 - DRG'D^T, & \mathbf{H} &= \mathbf{GQ}^T\mathbf{\Gamma}. \end{aligned} \tag{44.1}$$

Here $\Pi^T = 1 - S^T D^T$ with $S^T = DRG'$. We define the weakened operators on $\mathbb{T}_\eta$ by

$$\begin{aligned} \mathbf{H}(s) &= \mathbf{G}(s)\mathbf{Q}^T\mathbf{\Gamma}(s), \\ \mathbf{C}(s) &= \mathbf{G}(s) - \mathbf{G}(s)DG'(s)D^T + \mathbf{G}(s)DP_{\mathcal{V}}(s)G'(s)D^T - \mathbf{H}(s)\mathbf{QG}(s), \end{aligned}$$

with $G'(s)$, $R(s)$ from (L1), (L2), $\mathbf{G}(s)$ from (L3) and $\mathbf{\Gamma}(s)$ from (L4). This is $\mathbf{G}(s)\Pi^T(s) - \mathbf{H}(s)\mathbf{QG}(s)$ with $R = 1 - P_{\mathcal{V}}$ expanded, so that no identity is weakened on its own. By (44.1) these represent $\mathbf{H}$ and $\mathbf{C}$. They are products of weakenable families, so Lemma 40.4 gives their L^2 bounds, data locality and covariance, as in Section 41. On the torus they replace the representation of $\mathbf{C}$ in (L4).

Theorem 44.12 (Pointwise bounds for $\mathbf{H}(s)$ and $\mathbf{C}(s)$)**.** *For $M \geq M_1'$ and $s \in \mathbb{D}_\kappa$:*

$$|\mathbf{H}(s)B|_{1,\beta;y} \leq C \sum_{y'} e^{-\gamma|y-y'|}|B(y')|, \qquad |\mathbf{C}(s)f|_{1,\beta;y} \leq C \sum_{y'} e^{-\gamma|y-y'|}|f|_{B(y')},$$
$$|\mathbf{C}(s)D^T g|_{1,\beta;y} \leq C \sum_{y'} e^{-\gamma|y-y'|}\big(|g|_{B(y')} + [\tilde{g}_{y'}]_{\beta,\Lambda_1(c_{y'})}\big),$$

with $C = C(\kappa, \delta)$. These are (42.1) and Theorem 35.2(a)–(c) for the weakened operators on the torus, with β in place of α.

Proof. All tier norms below are global. Theorem 44.10 at level (L1) on $\mathbb{T}_\eta$ gives $G'(s)\colon 1 \to 3$ and $G'(s)D^T\colon 1 \to 2$ (values). By (Z3) on $\mathbb{T}_\eta$, $P_{\mathcal{V}}(s)\colon 0 \to 2$ (values). Hence $R(s) = 1 - P_{\mathcal{V}}(s)$, inside the product, maps $2 \to 2$.

- $\mathbf{H}(s)$. $\mathbf{\Gamma}(s)$ is a unit-lattice family. $\mathbf{Q}^T$ maps coarse fields to bounded fields, i.e. to tier 1. Apply $\mathbf{G}(s)\colon 1 \to 3$.
- $\mathbf{G}(s)f$. Tier $1 \to 3$, with $\mathfrak{n}^1_{y'}(f) \leq |f|_{B(y')}$.
- $\mathbf{G}(s)DR(s)G'(s)D^T f$. The scalar field $k = R(s)G'(s)D^T f$ is of tier 2. Theorem 44.10(b) gives $\mathbf{G}(s)Dk$ in tier 3.
- $\mathbf{H}(s)\mathbf{Q}\mathbf{G}(s)f$. $\mathbf{Q}\mathbf{G}(s)f$ is a coarse field, bounded by L^2 norms. Apply the bound for $\mathbf{H}(s)$.
- $\mathbf{C}(s)D^T g$. Use $\mathbf{G}(s)D^T\colon 2 \to 3$, and $R(s)G'(s)D^T D^T g = R(s)G'(s)\mathfrak{F}^T g$, where $\mathfrak{F}^T g$ is bounded, so this is of tier ≥ 2. The coarse term is as before.

The Hölder quotient of g is converted between local gauges as in the proof of Theorem 35.2(b). □

44.6 What this gives, and what remains

- *Given.* The linear inputs of the weakened background map on the torus: $\mathbf{G}(s)$, $\mathbf{H}(s)$, $\mathbf{C}(s)$, $\mathbf{C}(s)D^T$ in $C^{1,\beta}$, with exponential weights, uniformly on $\mathbb{D}_\kappa$. The exponent is $\beta = \min(\alpha, \frac{1}{2})$, so α should be fixed in $(0, \frac{1}{2}]$.
- *The trace inequality.* The scalar-gradient input $\mathbf{G}(s)Dk$, i.e. the gauge part $\mathbf{G}\,DRG'D^T$ of $\mathbf{C}$, needs Lemma 44.5. It holds because the local domains are boxes.
- *Not needed: $D^T D\mathbf{H}(s)$.* Section 39 bounds $\nabla\mathcal{V}$ with the pointwise bound of Theorem 35.2(d) on $D^T D\mathbf{H}^X$, which has no term-by-term analogue. For the weakened background map it is not needed. In Lemma 26.2, $D^T D\mathbf{H}$ occurs as $D^T(D\mathbf{H}\mathcal{D})$, which is absorbed by $\mathbf{C}(s)D^T$ (tier $2 \to 3$), and in $\mathbf{H}^T D^T DA' = (D\mathbf{H})^T(DA')$ and $\mathbf{H}^T D^T D\mathbf{H}\mathcal{D}$, which are coarse fields bounded by L^2 norms. The remainder $\nabla\mathcal{N}$ is bounded by C^1 norms (Proposition 23.3(a)). So Theorems 44.10 and 44.12 give all linear inputs of the contraction for $A_*(s)$.
- *Not given: the level with W_2.* Lemma 44.4 needs $W \in \mathcal{W}_{\mathrm{loc}}$, as noted after Corollary 43.8. The background map uses only the operators with $W \in \mathcal{W}_{\mathrm{loc}}$, so this is not needed for $A_*(s)$.
- *Domains.* Theorems 44.10 and 44.12 are stated on the torus. On a domain X the same bounds hold on $X_{(32)}$, by Lemma 44.7, for the scalar levels. For (L3) on domains one would need the lift argument for the top walk as well; this is not needed for $A_*(s)$.

- *Comparison with the blueprint.* The mixed L^2/pointwise norm near the cuts and the transport of regularity along chains of Appendix D are replaced by two simpler facts: tier norms compose exactly, and two interior factors at the output suffice. The remark on (I4) there and the equation route at the end of Section 43 are superseded by (44.1).

45 The weakened background map

This section constructs the weakened version of the background map $B \mapsto A_*(B)$ of Part V. It is the fixed point of a contraction built with the weakened operators of Section 44. The localization of the step needs three properties of it:

- analyticity in the weakening parameters $s \in \mathbb{D}_\kappa$;
- factorization when $s_\square = 0$ outside a union of boxes;
- a bound in $C^{1,\beta}$ that is linear in B and uniform in s.

The plan, and its review, are in the blueprint (Appendix E). Two points of the construction are not obvious.

- *The current is removed from the equation.* At a critical background $\mathbf{C}J = 0$. The weakened $\mathbf{C}(s)J$ does not vanish, and it would produce a field of size $C(\kappa)|J|$ even at $B = 0$. Without the source the fixed point vanishes at $B = 0$ for every s.
- *No second derivatives are needed.* The gradient of the interaction is written with derivatives outside, so that every term is either bounded or the divergence of a Hölder field. These are exactly the inputs of $\mathbf{C}(s)$ and $\mathbf{C}(s)D^T$ in Theorem 44.12. In particular no bound on $D^T D\mathbf{H}(s)$ is used.

Setting. We keep the setting of Section 44, with $d = 3$ and $W = 0$. We fix $\alpha \le \frac{1}{2}$, so that $\beta = \min(\alpha, \frac{1}{2}) = \alpha$; the activity domain of the next stage uses the exponent α.

- *The reference.* U_0 is a real (M, ε)-regular background with $|J_{U_0}| \le \varepsilon$, in the units of step k. It is $(1, \varepsilon)$-regular, so Proposition 23.3 applies. The bound on the current is used only in Theorem 45.5(c). Elsewhere only the regularity of U_0 enters, through $|\mathsf{F}| \le C\varepsilon$, where $\mathsf{F} = F^s_{U_0}$ is the symmetrized curvature of Lemma 23.2. We write F to avoid a clash with the weakening parameter s. In the renormalization group step, $U_0 = U_{k+1}(V_{k+1})$. That background is critical for the k-step problem (Lemma 31.6), but criticality is used only in Theorem 45.5(c).
- *The operators.* $\mathbf{G}(s)$, $\mathbf{H}(s)$, $\mathbf{C}(s)$ are the weakened operators on $\mathbb{T}_\eta$ of Section 44, at U_0, evaluated at $s \in \mathbb{D}_\kappa$.
- *Bilinear forms.* For complex fields, $\langle \cdot, \cdot \rangle$ denotes the bilinear form b, and gradients and transposes are taken with respect to it. On real fields this is the inner product of Part V.
- *Norms.* We write $|A|_* = \sup_y |A|_{1,\beta;y}$ for fine fields, and $|\omega|_\infty$ for coarse fields. For 2-forms,
$$|g|_{(2)} = \sup_y \big(|g|_{B(y)} + [\tilde{g}_y]_{\beta,\Lambda_1(c_y)} \big).$$
- *Constants.* C_P denotes constants that may depend on $\kappa, \delta, L, M, \rho, \alpha$, but not on k, n, η or $s \in \mathbb{D}_\kappa$. We assume $M \ge M_1'$, as in Theorem 44.12, and $M \ge 64$. Then the margin $r \le M/16$ of the weakened families satisfies $M/4 - 2r - 2 > 2$.
- *Norms and the activity domain.* The Hölder quotients in $|\cdot|_{1,\beta;y}$ are taken on $\Lambda_1(c_y)$, as in the definition of the tiers in Section 44. Pairs of points at distance at most 1 lie

in a common cube $\Lambda_1(c_y)$. Between the local gauges of Corollary 30.4 and those of the blocks, the change of gauge alters Hölder quotients by at most $C\varepsilon$ times the C^1 norm, as in the proof of Theorem 35.2(b). So $|A|_*$ bounds the quantities that define the graded activity domains (Definition 38.3: covariant derivatives, Hölder quotients on the cubes Λ_3 and the sup norm of the complex perturbation) up to a constant C_{conv}, not $1 + C\varepsilon$: pairs at distance more than 1 cost a factor 2 (Lemma 47.7).

45.1 The current at critical backgrounds

Lemma 45.1 (Current of critical backgrounds)**.** *Let U' be a real background field of the k-step problem, i.e. an element of the minimal orbit of Theorem 30.3, with $\varepsilon_1 \le a(L, M)$ small. Let $\mathbf{H}$ be its minimizer map. Then*

$$J_{U'} = \mathbf{Q}^T \omega_0, \qquad \omega_0 = \mathbf{H}^T J_{U'} = (D\mathbf{H})^T F^s_{U'},$$

and $|J_{U'}|_\infty \le C_J |F_{U'}|_\infty$. The constant C_J depends only on d, G, α and the parameter a of the averaging term. In particular it does not depend on L or M.

Proof. By Corollary 30.5, $A_*(0) = 0$ relative to $U_0 = U'$. The Euler–Lagrange equation (Theorem 25.1(d)) at $A_* = 0$ reads $\nabla\Phi(0) = d\mathcal{Q}(0)^T \omega_0$, with $\nabla\Phi(0) = J_{U'}$ (Proposition 23.3) and $d\mathcal{Q}(0) = \mathbf{Q}$ (Proposition 22.6). This is also the identity used in the proof of Proposition 33.1. Since $\mathbf{Q}\mathbf{H} = I$, $\mathbf{H}^T\mathbf{Q}^T = I$, so $\omega_0 = \mathbf{H}^T J_{U'}$. By Lemma 23.2(a), $J_{U'} = D^T F^s_{U'}$, so $\mathbf{H}^T J_{U'} = (D\mathbf{H})^T F^s_{U'}$. For a coarse bond c,

$$|\omega_0(c)| \le \sum_y \|\mathbf{1}_c (D\mathbf{H})^T \mathbf{1}_y\| \, \|F^s\|_{B(y)} \le C \, |F^s|_\infty,$$

by the L^2 decay of $D\mathbf{H}$ (Theorem 16.1(b), at regular backgrounds Theorem 19.9). Moreover $|F^s| \le 2|F|$ by Lemma 23.2(c), and $|\mathbf{Q}^T\omega|_\infty \le 2|\omega|_\infty$ once ε_1 is small depending on L. This follows from Proposition 22.7(e): $\mathbf{Q}^T$ differs from the transpose of the flat tube average by an operator of norm at most $C_L\varepsilon$ from sup norms to sup norms. The flat transpose has sup norm at most 1, since its kernel is nonnegative with column sums $\sum_y w_\mu(y, x) = 1$ (Lemma 2.6(a),(c)).

Constants. The constants of Parts II and III depend on the averaging pair only through the properties listed in Remark 19.12, by Lemma 31.3(a). So they do not depend on L, provided the background is regular with ε small depending on L and M, as there. For $M \ge M_0$ they do not depend on M either. □

With Proposition 31.5, the curvature and the current of $U_k(V)$ are both bounded by $(1 + C_J)(C_* + \delta)\varepsilon_1$. These constants do not grow with L or M. In the units of step k this gives $\theta(U_{k+1}) \le (1 + C_J)(C_* + \delta)L^{-3/2}h_k$. With the activity domain D3‴ of the blueprint (Appendix F), whose real budget is $h_k^{1-\epsilon/2}$, this puts the reference of the step into the domain for g_* small, without a condition on L.

45.2 The weakened equation

The linearizing map. For a fine field A', let $\mathcal{D}_s(A')$ be the coarse field X that solves

$$X = C_k(A' - \mathbf{H}(s)X), \tag{45.1}$$

with C_k the nonlinear part of the averaging (Proposition 22.6). At $s = 1$ this is the equation of Lemma 26.1.

The functional and its gradient. Put

$$\begin{aligned}\mathcal{V}_s(A') = &-\langle \mathbf{H}(s)\mathcal{D}_s(A'), J\rangle - \langle DA', D\mathbf{H}(s)\mathcal{D}_s(A')\rangle \\ &+ \tfrac{1}{2}\langle D\mathbf{H}(s)\mathcal{D}_s(A'), D\mathbf{H}(s)\mathcal{D}_s(A')\rangle + \mathcal{N}(A' - \mathbf{H}(s)\mathcal{D}_s(A')),\end{aligned}$$

with $J = J_{U_0}$ and $\mathcal{N}$ as in Proposition 23.3. At $s = 1$ this is the functional $\mathcal{V}$ of Section 26. Write $A = A' - \mathbf{H}(s)\mathcal{D}_s(A')$ and $d\mathcal{D}_s = d\mathcal{D}_s(A')$. Using $J = D^T\mathsf{F}$ (Lemma 23.2(a)), the gradient is

$$\nabla\mathcal{V}_s(A') = n_0(A') + D^T n_2(A'), \tag{45.2}$$

where

$$\begin{aligned}n_2(A') &= -D\mathbf{H}(s)\mathcal{D}_s(A'),\\ n_0(A') &= -d\mathcal{D}_s^T(D\mathbf{H}(s))^T(DA' - D\mathbf{H}(s)\mathcal{D}_s(A') + \mathsf{F}) + \nabla\mathcal{N}(A) - d\mathcal{D}_s^T\mathbf{H}(s)^T\nabla\mathcal{N}(A).\end{aligned}$$

This is the gradient formula of Lemma 26.2, term by term. There are three differences. $D^T D\mathbf{H}\mathcal{D}$ is kept as $D^T(D\mathbf{H}\mathcal{D})$; $\mathbf{H}^T D^T$ is written as $(D\mathbf{H})^T$; and $\mathbf{H}^T J$ is written as $(D\mathbf{H})^T\mathsf{F}$.

The map. For a coarse field B, put

$$\mathcal{T}_s(A_1) = -\mathbf{C}(s)\,\nabla\mathcal{V}_s(A_1 + \mathbf{H}(s)B), \tag{45.3}$$

on fine fields A_1 on $\mathbb{T}_\eta$, without the constraint $A_1 \in \mathcal{X}$. Given a fixed point A_1, put

$$A'_s = A_1 + \mathbf{H}(s)B, \qquad A_*(s;B) = A'_s - \mathbf{H}(s)\mathcal{D}_s(A'_s).$$

Compared with (26.3), the source J is missing. At a critical reference and $s = 1$ this changes nothing, because $\mathbf{C}J = 0$ there (Theorem 45.5(c)).

45.3 Bounds

Lemma 45.2 (Linear inputs)**.** *For $s \in \mathbb{D}_\kappa$:*

(a) $|\mathbf{C}(s)f|_* \le C_P|f|_\infty$ *and* $|\mathbf{C}(s)D^T g|_* \le C_P|g|_{(2)}$*;*

(b) $|\mathbf{H}(s)\omega|_* \le C_P|\omega|_\infty$ *and* $|D\mathbf{H}(s)\omega|_{(2)} \le C_P|\omega|_\infty$*;*

(c) $|\mathbf{H}(s)^T f|_\infty \le C_P|f|_\infty$ *and* $|(D\mathbf{H}(s))^T g|_\infty \le C_P|g|_\infty$ *for fine fields f and 2-forms g.*

Proof. (a) and the first part of (b) are Theorem 44.12, summed over y'. Its kernel $e^{-\gamma|y-y'|}$ has $\sum_{y'} e^{-\gamma|y-y'|} = S_\gamma$. For the second part of (b), write $D\mathbf{H}(s)\omega = D\mathbf{G}(s)\mathbf{Q}^T\mathbf{\Gamma}(s)\omega$. Here $\mathbf{Q}^T\mathbf{\Gamma}(s)$ maps coarse fields to bounded fine fields, i.e. to tier 1. Theorem 44.10(a) gives $D\mathbf{G}(s)\colon 1 \to 2$ in a global tier norm, which is the bound in $|\cdot|_{(2)}$. For (c), the values of a coarse field on the bonds starting at y are bounded by its block norm there. Then

$$\|\mathbf{1}_y\mathbf{H}(s)^T f\| \le \sum_{y'} \|\mathbf{1}_y\mathbf{H}(s)^T\mathbf{1}_{y'}\|\,\|f\|_{B(y')} \le C|\mathbf{H}(s)|_\gamma\,|f|_\infty,$$

because blocks have volume 1 and fields have at most C components and the decay norm $|\cdot|_\gamma$ is invariant under transposition. Finally $|\mathbf{H}(s)|_\gamma + |D\mathbf{H}(s)|_\gamma \le 2^d\|\mathbf{H}\|^D_{\kappa,\gamma}$ (Lemma 40.3(a)), and $\|\mathbf{H}\|^D_{\kappa,\gamma}$ is finite, since $\mathbf{H}(s) = \mathbf{G}(s)\mathbf{Q}^T\mathbf{\Gamma}(s)$ is a product of the output of Theorem 41.4(a) at level (L3), the local factor $\mathbf{Q}^T$ and the output at level (L4) (Lemmas 40.5(a) and 40.4). □

Lemma 45.3 (The weakened linearizing map)**.** *There is $r_P > 0$, depending on the constants C_P, with the following property. For $s \in \mathbb{D}_\kappa$ and complex A' with $|A'|_\infty \le r \le r_P$, equation (45.1) has a unique solution $\mathcal{D}_s(A')$ in $\{|X|_\infty \le r_P\}$. It is analytic in (s, A') for s in the interior of $\mathbb{D}_\kappa$, and*

$$|\mathcal{D}_s(A')|_\infty \le C_P r^2, \qquad |d\mathcal{D}_s(A')^T\omega|_\infty \le C_P r\,|\omega|_\infty.$$

Proof. Let K_H be the constant of Lemma 45.2(b), so that $|\mathbf{H}(s)X|_* \le K_H|X|_\infty$. For $|X|_\infty, |X'|_\infty \le r_P$, Proposition 22.6(b) gives, with $|A' - \mathbf{H}(s)X|_\infty \le (1+K_H)r_P$,

$$|C_k(A' - \mathbf{H}(s)X)|_\infty \le C(1+K_H)^2 r_P^2,$$
$$|C_k(A' - \mathbf{H}(s)X) - C_k(A' - \mathbf{H}(s)X')|_\infty \le 2C(1+K_H)r_P K_H |X - X'|_\infty.$$

For r_P small, the map is a contraction of the ball into itself, and $(1+K_H)r_P \le c_3$. The fixed point is analytic, since the map is analytic and the contraction is uniform. Inserting $|A'|_\infty \le r$ into the first bound gives $|X| \le C(r + K_H|X|)^2$, hence $|X| \le C_P r^2$. Differentiating (45.1) gives $d\mathcal{D}_s = (I + dC_k(A)\mathbf{H}(s))^{-1} dC_k(A)$, hence

$$d\mathcal{D}_s^T = dC_k(A)^T \big(I + \mathbf{H}(s)^T dC_k(A)^T\big)^{-1}.$$

By Proposition 22.6(c), $|dC_k(A)^T\omega|_\infty \le C|A|_\infty|\omega|_\infty$. By Lemma 45.2(c), the operator norm of $\mathbf{H}(s)^T dC_k(A)^T$ on coarse sup norms is at most $C_P C(1+K_H)r_P$, which is $\le \frac{1}{2}$ for r_P small. The Neumann series gives the bound. □

Lemma 45.4 (The gradient)**.** *There are $r'_P > 0$ and C_P such that for $s \in \mathbb{D}_\kappa$ and complex A' with $|A'|_{C^1} \le r \le 2r'_P$,*

$$|n_0(A')|_\infty \le C_P(\varepsilon + r)r, \qquad |n_2(A')|_{(2)} \le C_P r^2.$$

For $|A'|_{C^1}, |\tilde{A}'|_{C^1} \le r \le r'_P$,

$$|n_0(A') - n_0(\tilde{A}')|_\infty \le C_P(\varepsilon + r)|A' - \tilde{A}'|_{C^1}, \qquad |n_2(A') - n_2(\tilde{A}')|_{(2)} \le C_P r |A' - \tilde{A}'|_{C^1}.$$

Proof. Choose $r'_P \le r_P/2$ with $4r'_P \le c_5$ and $2K_H C_P r'_P \le 1$. By Lemma 45.3, $|\mathcal{D}_s| \le C_P r^2$. So $|A|_{C^1} \le r + K_H C_P r^2 \le 2r$, by Lemma 45.2(b) and $|\cdot|_{C^1} \le |\cdot|_*$. By Lemma 45.2(b), $|n_2|_{(2)} \le C_P|\mathcal{D}_s|_\infty \le C_P r^2$. The terms of n_0 are bounded as follows:

- $|DA'|_\infty \le C|A'|_{C^1}$, $|D\mathbf{H}(s)\mathcal{D}_s|_\infty \le C_P r^2$, and $|\mathsf{F}|_\infty \le C\varepsilon$ by regularity of U_0. Lemma 45.2(c) and Lemma 45.3 then bound the first term by $C_P r(r+\varepsilon)$.
- $|\nabla\mathcal{N}(A)|_\infty \le C(\varepsilon + 2r)2r$, by Proposition 23.3(a), for $2r \le c_5$.
- The last term is bounded by $C_P r \cdot C_P(\varepsilon + 2r)2r$.

These bounds hold on the complex ball $|A'|_{C^1} \le 2r'_P$, and n_0, n_2 are analytic there. The Lipschitz bounds follow from the Cauchy estimate in the direction $A' - \tilde{A}'$, as in the proof of Lemma 26.2(a). □

Only $|A'|_{C^1}$ enters, not the Hölder quotients of $\nabla A'$. The sources of $\mathbf{C}(s)$ are bounded, and those of $\mathbf{C}(s)D^T$ are 2-forms of the form $D\mathbf{H}(s)\omega$. The regularity of the output comes entirely from Lemma 45.2(a).

45.4 The theorem

Theorem 45.5 (Weakened background map)**.** *There are $\varepsilon_P, \varrho_P > 0$ and C_P, depending on $\kappa, \delta, L, M, \rho, \alpha$, such that, with $R_P = \varrho_P/K_H$, the following holds for $\varepsilon \le \varepsilon_P$, every reference U_0 as in the setting, and every complex coarse field B with $|B|_\infty \le R_P$.*

(a) (Existence and bounds.) *For every $s \in \mathbb{D}_\kappa$, $\mathcal{T}_s$ has a unique fixed point $A_1(s;B)$ in $\{|A_1|_* \le \varrho_P\}$. With $A_*(s;B)$ as above,*

$$|A_1(s;B)|_* \le C_P(\varepsilon + |B|_\infty)|B|_\infty, \qquad |A_*(s;B)|_* \le C_P|B|_\infty, \qquad A_*(s;0) = 0.$$

(b) (Analyticity.) *$A_*(s;B)$ is analytic in (s,B) for s in the interior of $\mathbb{D}_\kappa$ and $|B|_\infty < R_P$, continuous up to the boundary, and real for real s, B.*

(c) (Identification.) *If U_0 is critical, i.e. $U_0 = U_k(V)$ as in Corollary 30.5, then $A_*(1;B) = A_*(B)$ for $|B|_\infty \le \min(R_P, \beta_V)$, real or complex. Here β_V is the radius of Corollary 30.5 and Proposition 26.3, i.e. $\varepsilon + \beta_V \le \min(c_6, c_7)$.*

(d) (Factorization.) *Let Z be a union of boxes with $*$-components Z_i, and let $s_\square = 0$ for all $\square \not\subset Z$. Then $A_*(s;B)$ vanishes outside $Z^{(r)}$. On $Z_i^{(r)}$ it depends only on B on the coarse bonds of $Z_i^{(r)}$, on U_0 on $Z_i^{(r+1)}$, and on the weakening parameters only through $s_\square$, $\square \subset Z_i$. Here $r \le M/16$ is the largest margin of the weakened families $\mathbf{C}$, $\mathbf{H}$, $D\mathbf{H}$ and their transposes, including the range added by local factors (Lemma 40.5(a)).*

Proof. (a) Let $b = |B|_\infty \le R_P$ and $|A_1|_* \le \varrho_P$. Then $r := |A'|_* \le |A_1|_* + K_H b \le 2\varrho_P$. By (45.2), Lemma 45.2(a) and Lemma 45.4, with $2\varrho_P \le r'_P$,

$$|\mathcal{T}_s(A_1)|_* \le C_P|n_0|_\infty + C_P|n_2|_{(2)} \le C_P(\varepsilon + r)r, \qquad |\mathcal{T}_s(A_1) - \mathcal{T}_s(\tilde{A}_1)|_* \le C_P(\varepsilon + 2\varrho_P)|A_1 - \tilde{A}_1|_*.$$

Choose ε_P, ϱ_P with $4C_P(\varepsilon_P + 2\varrho_P) \le 1$ and $C(\varepsilon_P + R_P) \le K_H$, with C the constant of the estimate in (c). The threshold ε_P also contains the smallness conditions on ε of the inputs: $\varepsilon \le c_5$ (Proposition 23.3), $a \le c_3$ (Proposition 22.6), the condition "ε small depending on M" of Section 41, and the thresholds of Sections 36 and 43. Then $\mathcal{T}_s$ maps the ball $\{|A_1|_* \le \varrho_P\}$ into itself and is a contraction with constant $\frac{1}{4}$. At the fixed point, $|A_1|_* \le \frac{1}{4}(|A_1|_* + K_H b)$, so $|A_1|_* \le K_H b$ and $r \le 2K_H b$. Inserting this once more gives $|A_1|_* \le C_P(\varepsilon + 2K_H b)2K_H b \le C_P(\varepsilon + b)b$. Next, $|A_*|_* \le |A'_s|_* + K_H|\mathcal{D}_s(A'_s)|_\infty \le C_P b$. For $B = 0$:

- $\mathcal{D}_s(0) = 0$, since $X = 0$ solves (45.1) and the solution is unique;
- $\nabla\mathcal{N}(0) = 0$;
- every term of $n_0(0)$ and $n_2(0)$ contains $\mathcal{D}_s(0)$, $d\mathcal{D}_s(0) = dC_k(0)\cdots = 0$ or $\nabla\mathcal{N}(0)$.

So $\mathcal{T}_s(0) = 0$, and the fixed point is 0.

(b) $\mathcal{T}_s(A_1)$ is analytic in (s, A_1, B). The weakened operators are analytic in s in the interior of $\mathbb{D}_\kappa$, and bounded uniformly on $\mathbb{D}_\kappa$ (Lemma 40.3(a)). $\mathcal{D}_s$ is analytic by Lemma 45.3. The functions $\nabla\mathcal{N}$ and C_k are analytic. The iterates $\mathcal{T}_s^m(0)$ are therefore analytic, and they converge uniformly. For real s and B every term of the weakened families is a real operator, so the iterates are real.

(c) At $s = 1$, the weakened families represent the operators of Part V: $\mathbf{C}(1) = \mathbf{C}$ by (44.1), and $\mathbf{H}(1) = \mathbf{H}$. Hence $\mathcal{D}_1 = \mathcal{D}$ and $\nabla\mathcal{V}_1 = \nabla\mathcal{V}$. By Lemma 45.3 with the ball $|X| \le r_P$, $\mathcal{D}_1 = \mathcal{D}$. Since U_0 is critical, $A_*(0) = 0$. At $B = 0$ this forces the fixed point $A_{1,*}$ of Proposition 26.3 to vanish: $A_{1,*} = \mathbf{H}\mathcal{D}(A_{1,*})$ and $\mathbf{Q}A_{1,*} = 0$ give $\mathcal{D}(A_{1,*}) = 0$. Hence $0 = \mathcal{T}(0) = -\mathbf{C}(J + \nabla\mathcal{V}(0)) = -\mathbf{C}J$. So $\mathcal{T}_1$ coincides with the map $\mathcal{T}$ of (26.3). Let B be real with $|B|_\infty \le \min(R_P, \beta_V)$, and let $A_{1,*}$ be the fixed point of Proposition 26.3. The estimate in its proof, now without the term $C_C\varepsilon$ that came from $\mathbf{C}J$, gives $|A_{1,*}|_{C^1} \le C(\varepsilon + b)b \le K_H b$. By Lemma 45.4, only $|A'|_{C^1}$ enters the bound for $\mathcal{T}_1$. So $|A_{1,*}|_* = |\mathcal{T}_1(A_{1,*})|_* \le C_P(\varepsilon + 2K_H b)2K_H b \le \varrho_P$. Thus $A_{1,*}$ lies in the ball of (a), and by uniqueness $A_{1,*} = A_1(1;B)$. Then $A_*(1;B) = A_{1,*} + \mathbf{H}B - \mathbf{H}\mathcal{D}(A_{1,*} + \mathbf{H}B) = A_*(B)$, by the definition of A_* after Proposition 26.3. Both sides are analytic in B (Corollary 30.5 and (b)), so the identity extends to complex B.

(d) By Lemma 40.3(b), applied to the weakened families $\mathbf{C}$, $\mathbf{H}$ and their transposes and derivatives, the following hold at such s:

- each operator vanishes off $Z^{(r)} \times Z^{(r)}$;
- it is block diagonal over the $Z_i^{(r)}$;

- its block on Z_i depends on U_0 only on $Z_i^{(r)}$.

For $\mathbf{H}(s)$ and $\mathbf{H}(s)^T$ the coarse side consists of the bonds in $Z^{(r)}$.

Locality of the nonlinear maps.

- $C_k(A)(c)$ depends only on A on the bonds of $S_c = B(c_-) \cup B(c_+)$ (Proposition 22.6(a)), and on U_0 there (Proposition 22.7(d) and Lemma 21.5). It vanishes if $A = 0$ there, and so does $dC_k(A)(c)$.
- $\mathcal{N} = \sum_p n_p$ is a sum of plaquette terms, each a function of A and U_0 on the bonds of p, vanishing to second order at $A|_p = 0$. So $\nabla\mathcal{N}(A)(b)$ depends only on A on the plaquettes containing b, and vanishes if $A = 0$ there.
- F is local in U_0.

The class preserved by the iteration. Let $\mathcal{K}$ be the class of pairs (A_1, X) of fine and coarse fields with the following property. A_1 vanishes outside $Z^{(r)}$ and X outside $Z^{(r+1)}$. Here the coarse bonds of a set are the bonds starting in it. $X = \mathcal{D}_s(A')$ can be non-zero on coarse bonds that end in $Z^{(r)}$ but start outside it, since $C_k(\cdot)(c)$ depends on S_c; but $\mathbf{H}(s)$ sees only the coarse bonds starting in $Z^{(r)}$, by Lemma 40.3(b). The restrictions to $Z_i^{(r)}$, respectively $Z_i^{(r+1)}$, depend only on B on the coarse bonds of $Z_i^{(r)}$, on U_0 on $Z_i^{(r+1)}$, and on $s_\square$ for $\square \subset Z_i$. For B this holds because B enters the iterations only through $\mathbf{H}(s)B$ in $A'_s = A_1 + \mathbf{H}(s)B$, and the block of $\mathbf{H}(s)$ on Z_i reads only the coarse bonds starting in $Z_i^{(r)}$ (Lemma 40.3(b)). The blocks of the weakened operators have the last property by Lemma 40.3(b).

- *The inner iteration.* By the locality just described, the iteration of (45.1), started at 0, stays in $\mathcal{K}$. The components are at distance at least $M/4 - 2r - 2 > 2$, so the local maps do not connect them.
- *The outer iteration.* The same holds for n_0 and n_2, including the Neumann series for $d\mathcal{D}_s^T$, whose terms are products of operators of the class. Finally $\mathcal{T}_s$ maps into the range of $\mathbf{C}(s)$. So $\mathcal{T}_s$ preserves the class.

The fixed point is the limit of the iterates from 0, so it lies in the class, and so does $A_*(s; B)$. □

45.5 Remarks

Remark 45.6 (Why the current is removed). Keep J in (45.3), as in (26.3). At a critical reference $\mathbf{C}J = 0$, but for $s \neq 1$ the term $-\mathbf{C}(s)J = -\mathbf{C}(s)\mathbf{Q}^T\omega_0$ does not vanish, so $A_*(s; 0) \neq 0$. Then the composed activities of Proposition 47.8 do not vanish at $Z = 0$. The mixed pieces of the cluster expansion would have a part independent of the fluctuation, which destroys both the bound by τ and their vanishing at $Z = 0$. Nor is the term small in general. At complex references J is not bounded uniformly in η (Proposition 46.7(c)), and the source would have to be read as $-\mathbf{C}(s)D^T F^s$, with the operator $\mathbf{C}(s)D^T$ of Theorem 44.12. It is then of order C_P times the current of the real part and the derivatives of the complex part, and C_P grows with M, which is chosen after L. Removing J is the only change to the equation, and it is invisible at $s = 1$.

Remark 45.7 (References that are not critical). Parts (a), (b), (d) hold for every real regular reference. At a non-critical reference the fixed point at $s = 1$ is the critical point of $\mathfrak{F} - \langle \cdot, J \rangle$, not of $\mathfrak{F}$. This is the intended definition. The activities of the next step must be analytic on a domain that contains non-critical configurations. There the step formulas are definitions, and they must contain no source term. There is no linear term in the fluctuation. In the background map, the source $-\mathbf{C}(s)J$ is removed. The current still enters through $(D\mathbf{H}(s))^T\mathsf{F}$ in n_0, but only multiplied by $d\mathcal{D}_s^T = O(|A'|)$. At real critical backgrounds they agree with the renormalization group, which is where they are evaluated (blueprint, Appendix E, "Activities away from

critical backgrounds"). Complex references, i.e. complex perturbations of U_0, need the complex versions of the interior estimates. They are treated in Section 46 (Proposition 46.7(d)).

Remark 45.8 (Use in the step). In the step, $B = \hat{Z}(Z)$ with Z the free-bond fluctuation (Lemma 32.4). With the white-noise representation, $Z = g_k C_k^{1/2}(s)\xi$, $|\xi| \le p_{0,k}$.

- *Two radii.* The large radius R_P, which does not depend on g_k, is used for the Cauchy bounds on the Taylor remainders of order three in Z. The small one, $|B|_\infty \le C_P L^d p_{0,k} g_k$, gives $|A_*(s;B)|_* \le C_P^2 L^d p_{0,k} g_k$. This is below the budgets $\frac{1}{2}h_k^{1-\epsilon}$ and $\frac{1}{2}h_k^{1-2\epsilon}$ of the activity domain D3‴ (blueprint, Appendix F) once g_* is small, using the norm conversion of the setting.
- *Factorization.* The local elimination $\hat{Z}$ has range $2L$ in block units. So, after composition with $\hat{Z}$, the factorization of Theorem 45.5(d) requires $M \ge CL$.

These are the inputs of the localization lemma (Lemma 47.2) for the interaction pieces and the composed activities (Section 47).

Part X: The small-field flow

46 Complex references

The activities of the renormalization group must be analytic in the background on a complex domain (blueprint, Appendix F, domain D3‴). So the step evaluates the operators of Part IX, the interior estimates and the weakened background map at *complex* references. Sections 36–45 were written for real backgrounds. This section records where reality was used, and shows that every statement needed later extends to small complex perturbations. The multiscale averaged backgrounds of Part IV need a separate argument, given in Section 46.1 (Proposition 46.4). The results of Part IX are extended in Section 46.2.

Complex references. Let U_r be a real (M,ε)-regular background. A *complex reference* of size δ_c is $U = e^{\mathrm{i}\eta A'} U_r$, with A' a $\mathfrak{g}^c$-valued field such that, in the local real gauges of Definition 18.1 for U_r,

$$|A'| + |\nabla A'| + \sup_y [\nabla \tilde{A}'_y]_{\alpha,\Lambda_1(c_y)} < \delta_c.$$

- *Relation to Proposition 19.11.* That proposition writes $U\, e^{\mathrm{i}\eta\,\delta A}$ and uses ordinary derivatives. The two forms are equivalent, with $\delta A = R(U_r)^{-1}A'$. The two sizes are comparable: there is $\varepsilon_{\mathrm{cc}} > 0$, depending only on d, α and G, such that for $\varepsilon \le \varepsilon_{\mathrm{cc}}$ the δ of Proposition 19.11 and the size of A' above bound each other up to the factor $C_{\mathrm{cmp}} = 4$. Indeed, in the gauge of Definition 18.1 on a cube $\hat{\Lambda}$, where $U_r = e^{\mathrm{i}\eta A_r}$, the two fields differ by the factor $R(e^{\mathrm{i}\eta A_r})$, which is within $4\eta\varepsilon$ of 1, with difference quotients and Hölder quotients of size $C\eta\varepsilon$. Covariant and flat gradients differ by $C\varepsilon|A'|$, by (38.1), and the change between the gauges of the blocks and of the cubes costs at most $C\varepsilon$ times the C^1 norm, as in the proof of Theorem 35.2(b). Finally, two points at ℓ^∞-distance less than 1 lie in a common cube $\Lambda_1(c_y)$, and pairs at distance at least 1 have Hölder quotients at most twice the sup of the gradient.
- *The local gauge.* In a local gauge of U_r we have $U = e^{\mathrm{i}\eta A_0}$, with $A_0 = A' \oplus A_r$ combined bondwise by the Baker–Campbell–Hausdorff formula. Then A_0 is complex and satisfies (B) with parameter $C(\varepsilon + \delta_c)$.
- *Potentials.* Local potentials at a complex reference are taken in $\mathcal{W}^{\mathrm{an}}_{\mathrm{loc}}$ (Section 19), and satisfy the bounds of $\mathcal{W}_{\mathrm{loc}}(\varepsilon + C\delta_c)$ in the real gauges of U_r. The non-local potential

W_2^X satisfies the hypotheses of Proposition 19.11 for existence, Re-coercivity and decay; its comparison of domains is Proposition 37.4(d). Only its L^2 statements are used (Section 37). In Section 45, $W = 0$.

- *The domain D3‴.* It only needs complex perturbations near a polymer. These are extended by U_r outside a neighbourhood, with a smooth cutoff at unit scale, which gives a complex reference with $\delta_c \le Ch^{1-2\epsilon}$. In D3‴ the real part has only curvature and current at most $h^{1-\epsilon/2}$. Its (M,ε)-regularity comes from Corollary 30.4, with constants depending on L and M. Its current is at most ε for g_* small; Section 47 adds $|J_{U_r}| \le \varepsilon$ to the definition of a complex reference, for the coercivity of the Hessian.

46.1 Averaged backgrounds at complex references

Proposition 22.6 and Proposition 22.7 assume a real U_0, with real averaged backgrounds $W_j \in \mathfrak{U}_j(2a)$ (Proposition 21.7). Lemma 22.5 uses this through:

- the logarithms of Lemma 21.5(a);
- the curvature bound on H_x, E;
- Lemma 21.8.

The underlying lemmas are proved only for real configurations: Lemma 21.1(b), with the bi-invariant metric on G; Lemma 21.2; and Lemma 21.6, which uses a real gauge and gauge invariance of $|F|$. These results feed the bounds $|\mathbf{Q}A(c)| \le C\|A\|$ and $|Q'\lambda| \le C\|\lambda\|$, and the bounded transposes, used in Theorem 35.2 and in Part IX. A naive per-scale bound on complex transports would compound to 2^j over the scales.

We therefore work relative to the real reference U_r. Two observations make this work.

- The averaged backgrounds at U are gauge transforms of $X_j W_j(U_r)$, with $X_j = \mathcal{Q}_j(U')$, by explicit complex gauge transformations h_j (Lemma 46.2). This is proved directly from the one-step definitions. The axial gauge, Lemma 22.2(b),(c) and Theorem 22.4(a),(b),(e),(f) are not used at complex configurations.
- At the points of $\mathbb{T}^{(j)}$, h_j has size $C\delta_j|A'|$, not $C|A'|$. It is a product of the contour variables τ_{X_i}, $i < j$. Each τ_{X_i} is the exponential of an average of logarithms of relative holonomies along contours of length at most $d\delta_{i+1}$, and $\sum_{i<j}\delta_{i+1} \le 2\delta_j$. The scale-$j$ gradient of h_j carries the factor δ_j^{-1}, so it is bounded by the sup norm. Hence the relative fields are $O(|A'|)$ uniformly in j, and the losses per scale are summable.

Only the sup norm of A' is used, and the smallness of δ_c depends on d, L and the group $G \subset U(\mathsf{N})$, including the radius c_G of its logarithm chart, but not on M.

Setting. Let U_r be real and (M,ε)-regular with $\varepsilon \le \varepsilon_0$. By the proof of Corollary 22.8, $U_r \in \mathfrak{U}_0(a)$ with $a \le C\varepsilon$, where C does not depend on M (Lemma 12.3). The proofs of Sections 21 and 22 do not use M. So c_3 and the constants C, c of this subsection depend only on d, L and G. Let $c_3' \le c_3$ be such a constant, decreased below where needed, and assume $a \le c_3'$. Put $W_j^r = W_j(U_r) = U_r^{(j)}$. These are real and lie in $\mathfrak{U}_j(2a)$ by Proposition 21.7.

Let $U = U'U_r$ (bondwise), with $U' = e^{i\eta A'}$ and A' a $\mathfrak{g}^c$-valued field. Under a real gauge transformation u, $U_r \mapsto U_r^u$ and $A'(b) \mapsto R(u(b_-))A'(b)$. So $|A'|_\infty$ is gauge invariant, and the condition $|A'| < \delta_c$ in the local gauges of U_r is the condition $|A'|_\infty < \delta_c$. The derivative bounds on A' in the definition of complex references are not used in this subsection. Let $W_j(U) = U^{(j)} = \mathfrak{a}^{(j)}(U)$, whenever defined.

We use the gauge-adapted averaging of Section 22 relative to the *real* reference U_r, so there $W_j = W_j^r$. Applied to the relative configuration U' it gives

$$X_j = \mathcal{Q}_j(U') = e^{\mathrm{i}\delta_j B_j}, \qquad \tau_i = \tau_{X_i} \ \ (\text{on } \mathbb{T}^{(i+1)}), \qquad h_j(y) = \tau_0(y)\cdots\tau_{j-1}(y) \ \ (y \in \mathbb{T}^{(j)}),$$

with $h_0 = 1$. By Theorem 22.4(c), which holds for complex relative configurations (remark after that theorem), $h_j = \mathsf{w}_j(U')^{-1}$ at the points of $\mathbb{T}^{(j)}$. As in Definition 22.3 we use $\tilde{\mathfrak{a}}_j(X) = \overline{X}^{(\tau_X^{-1})}$. By Lemma 46.1 below, this coincides with $\overline{X^{(gx)}}$ on the domain of Lemma 22.5.

Norms of linear maps of $\mathfrak{g}^c$, and of matrices acting on $\mathbb{C}^{\mathsf{N}}$, are operator norms, written $\|\cdot\|$. The constants $C_{\mathrm{AB}} = 14d+3$, C_{loop} and $\kappa_{\mathrm{AB}} = 12dC_{\mathrm{AB}}$ depend only on d and N. Recall $\delta_j = L^{j-k}$, so that

$$\sum_{i\le k}\delta_i \le 2, \qquad \sum_{i<j}\delta_{i+1} = \delta_j\sum_{m<j}L^{-m} \le 2\delta_j. \tag{46.1}$$

Lemma 46.1 (Covariance under complex gauge transformations)**.** *Let $j \ge 1$ and let Y be a configuration on $\mathbb{T}^{(j-1)}$ with $|Y(\Gamma^j_{c,x})Y([c])^{-1} - 1| \le \frac{1}{2}c_G$ for all bonds c of $\mathbb{T}^{(j)}$ and all $x \in B_j(c_-)$. Let $g : \mathbb{T}^{(j-1)} \to G^c$ with $\|g(z)\|\|g(z)^{-1}\| \le 2$ for all z. Then $\mathfrak{a}_j(Y)$ and $\mathfrak{a}_j(Y^g)$ are defined, and $\mathfrak{a}_j(Y^g) = \mathfrak{a}_j(Y)^{g|_{\mathbb{T}^{(j)}}}$.*

Consequently, let X be a relative configuration at scale $j-1$ such that XW_{j-1} satisfies the hypothesis on Y, for some background W_{j-1} with $W_j = \mathfrak{a}_j(W_{j-1})$. Then $\overline{X^{(g)}} = \overline{X}^{(g|_{\mathbb{T}^{(j)}})}$.

Proof. The loops of Y^g are $g(c_-)hg(c_-)^{-1}$, with h the corresponding loop of Y, and $|ghg^{-1} - 1| \le \|g\|\|g^{-1}\||h-1| \le c_G$. Since $(ghg^{-1}-1)^n = g(h-1)^n g^{-1}$, the power series give $\ell(ghg^{-1}) = g\,\ell(h)\,g^{-1}$, and $\exp(\mathrm{i}gZg^{-1}) = g\,e^{\mathrm{i}Z}g^{-1}$. Insert this and $Y^g([c]) = g(c_-)Y([c])g(c_+)^{-1}$ into Definition 21.4. For relative configurations use $X^{(g)}W_{j-1} = (XW_{j-1})^g$. □

This makes the remark after Lemma 21.5 quantitative, without analytic continuation.

Lemma 46.2 (Representation of the averaged backgrounds)**.** *There is $\delta_{AB} > 0$, depending on d, L, G, such that for $a \le c_3'$ and $|A'|_\infty \le \delta_{AB}$:*

(a) $|\ell(\tau_i)| \le 3d\,\delta_{i+1}|A'|_\infty$ on $\mathbb{T}^{(i+1)}$ for $i < k$. The function $\lambda_j = \ell(h_j)$ on $\mathbb{T}^{(j)}$ is defined and satisfies $|\lambda_j| \le 7d\,\delta_j|A'|_\infty$ for $j \le k$.

(b) For every $j \le k$, $W_j(U)$ is defined and

$$W_j(U) = (X_jW_j^r)^{h_j}, \qquad \textit{i.e.} \quad W_j(U)(b) = h_j(b_-)\,X_j(b)\,W_j^r(b)\,h_j(b_+)^{-1}. \tag{46.2}$$

Equivalently, $U'^{[j]} = \mathcal{Q}_j(U')^{(h_j)} = \mathcal{Q}_j(U')^{(\mathsf{w}_j(U')^{-1})}$.

Proof. (a) Proposition 22.6(a) applies to the real reference U_r and the complex field A', and gives $|B_i|_\infty \le 2|A'|_\infty$. By Lemma 22.5, $\ell(\tau_i) = \delta_{i+1}\big(\gamma_{i+1}B_i + c^\tau_{i+1}(B_i)\big)$ with $|\gamma_{i+1}B_i| \le 2d|A'|$ and $|c^\tau_{i+1}(B_i)| \le 4C\delta_{i+1}|A'|^2$. This gives the first bound for $|A'| \le d/(4C)$. By (46.1), $s = \sum_{i<j}|\ell(\tau_i)| \le 6d\delta_j|A'|$. Lemma 21.1(c) gives $|\lambda_j| \le s + Cs^2 \le 7d\delta_j|A'|$ for $|A'|$ small.

(b) We use induction on j. For $j = 0$ we have $W_0(U) = U = U'U_r = X_0W_0^r$ and $h_0 = 1$. Let $j \ge 1$, assume (46.2) at $j-1$, and put $Y = X_{j-1}W^r_{j-1}$. With $h^r_x = W^r_{j-1}(\Gamma^j_{c,x})W^r_{j-1}([c])^{-1}$,

$$Y(\Gamma^j_{c,x})Y([c])^{-1} = (X_{j-1})_{W^r}(\Gamma^j_{c,x})\;h^r_x\;(X_{j-1})_{W^r}([c])^{-1}.$$

The relative transports are products of at most $(2d+2)L$ factors $e^{\mathrm{i}\delta_{j-1}R(W^r(\cdot))B_{j-1}(\cdot)}$, and $R(W^r)$ is unitary. Moreover $|\ell(h^r_x)| \le 2da\delta_j^2$, by Lemma 21.5(a) and Proposition 21.7. By Lemma 21.1(c) the loop has logarithm at most $2\big(4(d+1)\delta_j|A'| + 2da\delta_j^2\big)$. So it is within $\frac{1}{4}c_G$ of 1, after decreasing c_3' and δ_{AB}. By (a), $\|h_{j-1}\|\|h_{j-1}^{-1}\| \le e^{2|\lambda_{j-1}|} \le 2$. So the loops of

$W_{j-1}(U) = Y^{h_{j-1}}$, which are conjugates of those of Y by $h_{j-1}(c_-)$, are within $\frac{1}{2}c_G$ of 1; this is used in Lemma 46.3(b). By the induction hypothesis $W_{j-1}(U) = Y^{h_{j-1}}$, so Lemma 46.1 shows that $W_j(U) = \mathfrak{a}_j(W_{j-1}(U))$ is defined and

$$W_j(U) = \mathfrak{a}_j(Y)^{h_{j-1}|_{\mathbb{T}^{(j)}}}.$$

By the definitions of the relative average and of $\tilde{\mathfrak{a}}_j$,

$$\mathfrak{a}_j(Y) = \overline{X_{j-1}}\, W_j^r, \qquad \overline{X_{j-1}} = \tilde{\mathfrak{a}}_j(X_{j-1})^{(\tau_{j-1})} = X_j^{(\tau_{j-1})},$$

i.e. $\mathfrak{a}_j(Y) = (X_j W_j^r)^{\tau_{j-1}}$. Since $(Z^g)^{g'} = Z^{g'g}$ and $h_{j-1}\tau_{j-1} = h_j$ on $\mathbb{T}^{(j)}$, (46.2) follows at j. The last statement is (46.2) rewritten with the relative gauge action, together with $h_j = \mathsf{w}_j(U')^{-1}$ on $\mathbb{T}^{(j)}$. □

Lemma 46.3 (Averaged backgrounds at complex references)**.** *Possibly after decreasing δ_{AB} and c_3' (depending on d, L, G), the following holds for $a \le c_3'$, $|A'|_\infty \le \delta_c \le \delta_{AB}$ and $0 \le j \le k$.*

(a) $W_j(U) = e^{\mathrm{i}\delta_j \tilde{B}_j}\, W_j^r$ bondwise, where

$$\tilde{B}_j = B_j - D_{W_j^r}\lambda_j + \beta_j, \qquad |\beta_j|_\infty \le C\delta_j |A'|_\infty^2, \qquad |\tilde{B}_j|_\infty \le C_{\mathrm{AB}}|A'|_\infty. \tag{46.3}$$

Moreover $\big|\tilde{B}_j - \mathbf{Q}_j(U_r)A' + D_{W_j^r}\mathbf{\Gamma}_j A'\big|_\infty \le C|A'|_\infty^2$, where $\mathbf{\Gamma}_j A' = \sum_{i<j}\delta_{i+1}\gamma_{i+1}\mathbf{Q}_i(U_r)A'$ is evaluated at the points of $\mathbb{T}^{(j)}$.

(b) $\tilde{B}_j$ is holomorphic in A' on $\{|A'|_\infty < \delta_{AB}\}$, and real for real A'. Under a real gauge transformation u of $\mathbb{T}_\eta$ ($U_r \mapsto U_r^u$, $A'(b) \mapsto R(u(b_-))A'(b)$) we have $\tilde{B}_j(b) \mapsto R(u(b_-))\tilde{B}_j(b)$. In particular (a) holds in every real gauge, e.g. in the local gauges of U_r.

(c) Let γ be a contour of n bonds of $\mathbb{T}^{(j)}$ with $s = C_{\mathrm{AB}} n\delta_j|A'|_\infty \le s_0$, where s_0 depends only on N. Then $W_j(U)(\gamma) = e^{\mathrm{i}Z_\gamma}W_j^r(\gamma)$ with $|Z_\gamma| \le 2s$. (Here Z_γ is the Baker–Campbell–Hausdorff series of $\mathfrak{g}^c$-elements of Lemma 21.1(c), which lies in $\mathfrak{g}^c$; so no radius c_G enters.) Hence

$$\|R(W_j(U)(\gamma))^{\pm 1}\| \le e^{4s}, \qquad \|R(W_j(U)(\gamma)) - R(W_j^r(\gamma))\| \le e^{4s} - 1.$$

Applied at index $j-1$ to the contours of at most $3dL$ bonds of $\mathbb{T}^{(j-1)}$ used in the step $j-1 \to j$, where $4s \le 12dLC_{\mathrm{AB}}\delta_{j-1}\delta_c = \kappa_{\mathrm{AB}}\delta_j\delta_c$, this gives these bounds with e^{4s} replaced by $e^{\kappa_{\mathrm{AB}}\delta_j\delta_c}$.

(d) For $j \ge 1$, a bond c of $\mathbb{T}^{(j)}$ and $x \in B_j(c_-)$, let $E(U) = L^{-d}\sum_x H_x(U)$ and

$$H_x(U) = \ell\big(W_{j-1}(U)(\Gamma_{c,x}^j)\, W_{j-1}(U)([c])^{-1}\big).$$

Then

$$|H_x(U) - H_x(U_r)|, \quad |E(U) - E(U_r)| \le C_{\mathrm{loop}}\delta_j|A'|_\infty,$$
$$|H_x(U)| \le 2da\delta_j^2 + C_{\mathrm{loop}}\delta_j|A'|_\infty, \qquad |E(U)| \le 2da\delta_j^2 + C_{\mathrm{loop}}\delta_j|A'|_\infty.$$

Proof. (a) By Lemma 46.2, $V_j = W_j(U)(W_j^r)^{-1} = X_j^{(h_j)}$, i.e.

$$V_j(b) = e^{\mathrm{i}\lambda_j(b_-)}\, e^{\mathrm{i}\delta_j B_j(b)}\, e^{-\mathrm{i}R(W_j^r(b))\lambda_j(b_+)}.$$

The three exponents have total size $s \le (14d+2)\delta_j|A'| \le \frac{1}{4}$, by Lemma 46.2(a) and Proposition 22.6(a); here we used that $R(W_j^r)$ is unitary. By Lemma 21.1(c), $\ell(V_j(b)) =$

$\lambda_j(b_-)+\delta_j B_j(b)-R(W_j^r(b))\lambda_j(b_+)+\delta_j\beta_j(b)$ with $\delta_j|\beta_j|\le Cs^2$. This is (46.3), with $D_{W_j^r}\lambda(b)=\delta_j^{-1}(R(W_j^r(b))\lambda(b_+)-\lambda(b_-))$ as in Section 22. By Lemma 46.2(a), $|D_{W_j^r}\lambda_j|\le\delta_j^{-1}(|\lambda_j(b_+)|+|\lambda_j(b_-)|)\le 14d|A'|$. Hence $|\tilde B_j|\le(2+14d+C\delta_j|A'|)|A'|\le C_{\mathrm{AB}}|A'|$.

For the linear part, use $B_j=\mathbf{Q}_jA'+C_j(A')$ with $|C_j(A')|\le C|A'|^2$ (Proposition 22.6(b)). Further, $\lambda_j=\sum_{i<j}\ell(\tau_i)+O(\delta_j^2|A'|^2)$ and $\ell(\tau_i)=\delta_{i+1}\gamma_{i+1}\mathbf{Q}_iA'+O(\delta_{i+1}|A'|^2)$. By (46.1), $|\lambda_j-\mathbf{\Gamma}_jA'|\le C\delta_j|A'|^2$, and $D_{W_j^r}$ costs at most $2\delta_j^{-1}$.

(b) B_j and τ_i are holomorphic in A' by Proposition 22.6 and Lemma 22.5. Then λ_j and $\tilde B_j$ are holomorphic, since they are obtained from these by ℓ, products and Lemma 21.1(c). For real A' all configurations involved are real. For covariance, $U^u=e^{\mathrm{i}\eta R(u_-)A'}U_r^u$. Lemma 46.1 with real g, applied at each scale (its hypothesis holds by the proof of Lemma 46.2(b)), gives $W_j(U^u)=W_j(U)^{u|_{\mathbb{T}^{(j)}}}$, and likewise $W_j(U_r^u)=(W_j^r)^{u|_{\mathbb{T}^{(j)}}}$. Hence $V_j(b)\mapsto u(b_-)V_j(b)u(b_-)^{-1}$.

(c) $W_j(U)(\gamma)=(V_j)_{W_j^r}(\gamma)\,W_j^r(\gamma)$, and $(V_j)_{W_j^r}(\gamma)=\prod_i R(W_j^r(\gamma_{<i}))(V_j(b_i))$. For negatively oriented bonds, $V_j(-b)=R(W_j^r(b))^{-1}(V_j(b)^{-1})$. Since $R(W_j^r)$ is unitary, every factor is $e^{\mathrm{i}\delta_jZ_i}$ with $|Z_i|\le C_{\mathrm{AB}}|A'|$. Lemma 21.1(c) gives $|Z_\gamma|\le s+Cs^2\le 2s$ for $s\le s_0=\min(\frac14,C^{-1})$. By (12.1), $\|R(e^{\pm\mathrm{i}Z})\|\le e^{2|Z|}$ and $\|R(e^{\mathrm{i}Z})-1\|\le e^{2|Z|}-1$, and $R(W_j^r(\gamma))$ is unitary. The last statement is the case $n\le 3dL$ at index $j-1$, with $\delta_{j-1}=\delta_j/L$; it needs $3dC_{\mathrm{AB}}\delta_j\delta_c\le s_0$, which holds after decreasing δ_{AB}.

(d) We have

$$W_{j-1}(U)(\Gamma^j_{c,x})\,W_{j-1}(U)([c])^{-1}=e^{\mathrm{i}Z_1}\,e^{\mathrm{i}H}\,e^{-\mathrm{i}Z_2},$$

with $H=H_x(U_r)$, $e^{\mathrm{i}Z_1}=(V_{j-1})_{W^r}(\Gamma^j_{c,x})$ and $e^{\mathrm{i}Z_2}=(V_{j-1})_{W^r}([c])$. By (c), $|Z_1|+|Z_2|\le 4(d+1)C_{\mathrm{AB}}\delta_j|A'|$. By Lemma 21.5(a) and Proposition 21.7, $|H|\le 2da\delta_j^2\le 2dc_3'$. Let $\varphi(t)=\ell(e^{\mathrm{i}tZ_1}e^{\mathrm{i}H}e^{-\mathrm{i}tZ_2})-H$. For $|t|\le r_0/(|Z_1|+|Z_2|)$, with r_0 small depending on N (the logarithm is again a Baker–Campbell–Hausdorff series of $\mathfrak{g}^c$-elements, so c_G does not enter), it is holomorphic and bounded by a constant $K_{\mathrm{S}}(\mathsf{N})$, and $\varphi(0)=0$; this uses $2dc_3'\le r_0$. The Schwarz lemma gives $|\varphi(1)|\le(K_{\mathrm{S}}/r_0)(|Z_1|+|Z_2|)$. Average over x for E. □

Proposition 46.4 (Averaged backgrounds at complex references, (AB))**.** *There are $\delta_{AB}>0$ and $c_3'\le c_3$, depending only on d,L,G (through the constants of Part IV, including c_G), such that the following holds for $a\le c_3'$, every real $U_r\in\mathfrak{U}_0(a)$, and every complex reference $U=e^{\mathrm{i}\eta A'}U_r$ with $|A'|_\infty\le\delta_c<\delta_{AB}$. All bounds are uniform in k, in η and in $j\le k$. Below, relative configurations, τ_X, $\tilde{\mathfrak{a}}_j$, $\mathcal{Q}_j$, l_j, q_j, r_j, γ_j, q_j', $\mathbf{Q}_k$, Q_k' at the reference U are defined as in Section 22, with the complex backgrounds $W_j(U)$ in place of W_j.*

(i) (Backgrounds.) *$W_j(U)$ is defined for $j\le k$, and Lemmas 46.2 and 46.3 hold. In particular, relative to $W_j(U_r)$, $W_j(U)$ is a complex perturbation of size at most $C_{\mathrm{AB}}\delta_c$, in every real gauge.*

(ii) (One step.) *Lemma 22.5 holds at $W=W_{j-1}(U)$, with c_2 replaced by $\frac12c_2$. Parts (a), (c), (d) are unchanged and (e) is dropped. Part (b) is replaced by*

$$|(q_jB)(c)|^2\le e^{2\kappa_{\mathrm{AB}}\delta_j\delta_c}L^{-d}\sum_x L^{-1}\sum_{b\in[x,x(c)]}|B(b)|^2,$$
$$|(r_jB)(c)|\le C(a\delta_j^2+\delta_j\delta_c)\max|B|,\qquad|\gamma_jB|\le de^{\kappa_{\mathrm{AB}}\delta_j\delta_c}\max|B|.$$

(iii) (Gauge-adapted averaging.) *Proposition 22.6 holds at the reference U, for complex A with $|A|_\infty\le c_3'$. Parts (a), (b) and (d) hold in the same form, and "real for real A" is dropped. In (c) the kernel bounds become*

$$|K(c,b)|\le(1+Ca)\,e^{C_L\delta_c},\qquad|K_A(c,b)|\le Ce^{C_L\delta_c}|A|_\infty,$$
$$|d^2C_k(A)[h,h](c)|\le Ce^{C_L\delta_c}\|h\|^2_{S_c},$$

and the consequences of (c) (pointwise bounds by $\|h\|_{S_c}$, sup-norm bounds of the transposes) hold with their constants multiplied by $e^{C_L\delta_c}$.

(iv) (Linearized averages.) *$Q'_k(U)Q'_k(U)^T = I$, with the transpose taken with respect to* b. *Proposition 22.7(d), and (f) for real u, hold as stated. The tree transports satisfy $\|T_U(y,x)\| \le e^{\kappa_{\rm AB}\delta_c}$ and $\|T_U(y,x) - T_{U_r}(y,x)\| \le e^{\kappa_{\rm AB}\delta_c} - 1$. Hence $|Q'_k(U)\lambda(y)| \le e^{\kappa_{\rm AB}\delta_c}\|\lambda\|_{B(y)}$, and $Q'_k(U)^T$ is bounded on sup norms by $e^{\kappa_{\rm AB}\delta_c}$. Proposition 22.7(e) holds with ε replaced by $\varepsilon + \delta_c$. More precisely, suppose that on a cube containing $B(c_-) \cup B(c_+)$ we have $U_r = e^{i\eta A_r}$ in a real gauge, with $|A_r| \le \varepsilon$ and $a \le \varepsilon \le c'_3$. Then, in this gauge, $|K_{\mathbf{Q}_k(U)}(c,b) - K_{\mathbf{Q}}(c,b)| + |K_{Q'_k(U)}(y,x) - K_{Q_0}(y,x)| \le C_L(\varepsilon + \delta_c)$, and the same holds for the transposes.*

(v) (Closeness to the real reference.) *With the supports of Proposition 22.7(d),*

$$|K_{\mathbf{Q}_k(U)}(c,b) - K_{\mathbf{Q}_k(U_r)}(c,b)| \le C_L\delta_c, \qquad |K_{Q'_k(U)}(y,x) - K_{Q'_k(U_r)}(y,x)| \le e^{\kappa_{\rm AB}\delta_c} - 1,$$
$$\|R(W_k(U)(b)) - R(W_k(U_r)(b))\| \le e^{2C_{\rm AB}\delta_c} - 1.$$

So $\mathbf{Q}_k(U) - \mathbf{Q}_k(U_r)$, $Q'_k(U) - Q'_k(U_r)$, $D_{W_k(U)} - D_{W_k(U_r)}$ and their transposes are local operators of size $C_L\delta_c$.

(vi) (Analyticity.) *$W_j(U)$, l_j, $\mathbf{Q}_k(U)$, $Q'_k(U)$ and $D_{W_k(U)}$ are holomorphic in A' on $\{|A'|_\infty < \delta_{AB}\}$. The nonlinear part $C_k(A)$ at the reference U is holomorphic in (A', A) jointly.*

Proof. (i) is Lemmas 46.2 and 46.3.

(ii) The proof of Lemma 22.5 uses the reality of W in four places; everything else is algebraic, or uses only analyticity and Cauchy estimates.

(1) *Analyticity.* Relative transports are products of at most $3dL$ factors $e^{i\delta R(W(\gamma))B(b)}$, with γ a contour of at most $3dL$ bonds of $\mathbb{T}^{(j-1)}$. By Lemma 46.3(c), $\|R(W(\gamma))\| \le e^{\kappa_{\rm AB}\delta_j\delta_c} \le 2$ instead of 1. This is absorbed by $c_2 \to \frac{1}{2}c_2$. By Lemma 46.3(d), the arguments of all logarithms are within $C(c_2 + a\delta_j^2 + \delta_j\delta_c)$ of 1. The Cauchy estimates for c_j, c_j^τ are then unchanged.

(2) *The loops.* In the linear part, $|H_x|, |E| \le 2da\delta_j^2$ is replaced by $2da\delta_j^2 + C_{\rm loop}\delta_j\delta_c$, by Lemma 46.3(d). The bounds $|\mathfrak{f}(\operatorname{ad} H_x) - 1| \le C|H_x|$, $|\mathfrak{e}(\operatorname{ad} E) - 1| \le C|E|$ and $|R(e^{iE}) - R(h_x)| \le C(|E| + |H_x|)$ hold for complex H_x, E of size at most 1. Indeed $\mathfrak{e}$ is entire, $\mathfrak{f}$ is analytic for $\|\operatorname{ad}\| < 2\pi$ (Lemma 22.5), and $\|\operatorname{ad} H\| \le 2|H| \le 2$. Moreover $|R(W(\Gamma^j_{c,x})) - R(W_j(c))| = \|(R(h_x e^{-iE}) - 1)R(W_j(c))\| \le C(|E| + |H_x|)$, since $\|R(W_j(c))\| \le 2$.

(3) *The tube average.* In $q_jB(c)$ every value $B(b)$ is multiplied by a single transport along a contour of at most $(d+1)L$ bonds of $\mathbb{T}^{(j-1)}$, which has norm at most $e^{\kappa_{\rm AB}\delta_j\delta_c}$. The Cauchy–Schwarz inequality gives the bound on q_j; the bound on γ_j is the same.

(4) *The remainder r_j.* The algebra of the proof is unchanged: with all factors $\mathfrak{e}, \mathfrak{f}, R(h_x), R(e^{iE})$ equal to 1 and $R(W(\Gamma^j_{c,x})) = R(W_j(c))$, only q_jB remains. So r_jB is a sum of terms, each containing one of the differences in (2), multiplied by a contour sum bounded by $3dLe^{\kappa_{\rm AB}\delta_j\delta_c}\max|B|$, and by the prefactor L^{-1}. Hence $|r_jB| \le C(a\delta_j^2 + \delta_j\delta_c)\max|B|$.

(iii) *(a), (b).* By (ii), $|l_jB|_\infty \le (1 + C(a\delta_j^2 + \delta_j\delta_c))|B|_\infty$ and $|c_j(B)|_\infty \le C\delta_j|B|_\infty^2$. The induction of the proof of Proposition 22.6 gives, by (46.1) and $\sum_i \delta_i^2 \le \frac{4}{3}$,

$$|B_j|_\infty \le |A|_\infty \prod_{i\le j}(1 + C(a\delta_i^2 + \delta_i\delta_c) + 2C\delta_i|A|_\infty) \le |A|_\infty e^{C(2a+2\delta_c+4|A|_\infty)} \le 2|A|_\infty$$

for $a, |A|_\infty \le c'_3$ and $\delta_c < \delta_{AB}$. The bounds on C_j and the locality follow as there.

(c). We run the proof of Proposition 22.6(c) with nonnegative majorants. For a scale-$(m-1)$ field v write $|v|$ for the function $b \mapsto |v(b)|$. In $q_m v(c')$ each value $v(b)$ is multiplied by one transport of norm at most $e^{\kappa_{\mathrm{AB}}\delta_m\delta_c}$, by (ii)(3), and the scalar weights are those at $W=1$. So $|q_m v| \le e^{\kappa_{\mathrm{AB}}\delta_m\delta_c} q_m^{(1)}|v|$ pointwise, where $q_m^{(1)}$ is the tube average at $W=1$. By (ii)(4), $|r_m v| \le r_m^{\mathrm{maj}}|v|$, where r_m^{maj} is a nonnegative kernel with the support of the real case and at most C_L entries per row, each bounded by $C(a\delta_m^2+\delta_m\delta_c)$; it already accounts for the complex transports inside r_m. Hence

$$|l_j \cdots l_{i+1} v| \le e^{\kappa_{\mathrm{AB}}\delta_c \sum_m \delta_m} \big(q_j^{(1)} + r_j^{\mathrm{maj}}\big) \cdots \big(q_{i+1}^{(1)} + r_{i+1}^{\mathrm{maj}}\big)|v|.$$

The prefactor is at most $e^{2\kappa_{\mathrm{AB}}\delta_c}$, by (46.1). The product of majorants is the real case with unitary transports, except that the coefficients of the insertions are $C(a\delta_m^2+\delta_m\delta_c)$ instead of $Ca\delta_m^2$. The steps "Transports" and "Insertions" of that proof therefore apply verbatim, with $a\delta_m^2$ replaced by $a\delta_m^2+\delta_m\delta_c$. They show that the product is of type

$$\prod_m (1 + C_L(a\delta_m^2+\delta_m\delta_c)) \le \prod_m (1+C_L a\delta_m^2) \prod_m (1+C_L\delta_m\delta_c) \le (1+Ca)\, e^{C_L\delta_c}.$$

In the step "First derivatives", $dc_m(B)$ plays the role of r_m, with coefficients $C\delta_m|A|_\infty$ by the Cauchy estimates of (ii), and the same majorant argument applies. In the step "Second derivatives", the type bounds just proved replace those of the real case. So both steps hold with constants multiplied by $e^{C_L\delta_c}$. The consequences follow from the kernel bounds as before.

(d). The values of w_k at the points of $\mathbb{T}$ are given by Theorem 22.4(c), which is the recursion of Definition 22.3, and by (ii); this gives the same bounds. At fine points, the extensions $\mathcal{E}^{(i)}$ transport along the trees, with norm at most $e^{\kappa_{\mathrm{AB}}\delta_c}$ instead of 1.

(iv) For every $Y \in G^c$, $R(Y)^T = R(Y^{-1})$ with respect to b, since $\mathrm{tr}(YXY^{-1}Z) = \mathrm{tr}(XY^{-1}ZY)$. Hence $q_j'^T g(x) = R(W_{j-1}(U)(\Gamma^j_{y,x}))^{-1} g(y)$ and $q_j' q_j'^T = I$, as in the proof of Proposition 22.7(a). That proof uses only the volume factors, not unitarity. The Hermitian identity $Q_k' Q_k'^* = I$ need not hold for complex U; the b-transpose identity always holds. Locality (d) holds because Lemma 21.5(c) holds verbatim for complex configurations, and Lemma 22.5(a) holds by (ii). Covariance (f) follows from Lemma 46.1 with real g.

The tree transport $T_U(y,x)$ is a product over $m<k$ of $R(W_m(U)(\gamma_m))$, with $\gamma_m = \Gamma^{m+1}_{x_{m+1},x_m}$ a contour of at most $d(L-1)$ bonds of $\mathbb{T}^{(m)}$. By Lemma 46.3(c), $W_m(U)(\gamma_m) = e^{\mathrm{i}Z_m} W_m(U_r)(\gamma_m)$ with $|Z_m| \le 2dC_{\mathrm{AB}}\delta_{m+1}\delta_c$. By (12.1) and (46.1), $\|T_U\| \le e^{8dC_{\mathrm{AB}}\delta_c} \le e^{\kappa_{\mathrm{AB}}\delta_c}$. Telescoping the product gives $\|T_U - T_{U_r}\| \le e^{\kappa_{\mathrm{AB}}\delta_c} - 1$.

For (e), compute in the given gauge, extended arbitrarily to $\mathbb{T}_\eta$; this is allowed by (f). By locality, $\tilde B_j$ on the relevant bonds depends only on U_r, A' in the cube. By Lemma 21.8, $W_j(U_r) = e^{\mathrm{i}\delta_j A_{r,j}}$ with $|A_{r,j}| \le \varepsilon + 4da \le C\varepsilon$. By Lemma 46.3(a),(b) and Lemma 21.1(c), $W_j(U) = e^{\mathrm{i}\delta_j A_{0,j}}$ with $|A_{0,j}| \le C(\varepsilon+\delta_c)$. All transports along contours of at most $3dL$ bonds of $\mathbb{T}^{(j-1)}$ are then within $C(\varepsilon+\delta_c)\delta_j$ of 1. With (ii), $l_j - q_j(1)$ is a combination of at most C_L values with coefficients $C(\varepsilon+\delta_c)\delta_j$, using $a\delta_j^2 \le \varepsilon\delta_j$. The telescoping of the proof of Proposition 22.7(e), with the type bounds of (iii), gives the kernel bound $\sum_j C_L(\varepsilon+\delta_c)\delta_j e^{C_L\delta_c} \le C_L(\varepsilon+\delta_c)$. The same argument applies to Q_k'.

(v) The bound for Q_k' is the tree bound in (iv). For the unit-lattice derivative, $W_k(U)(b) = e^{\mathrm{i}\tilde B_k(b)} W_k(U_r)(b)$ with $|\tilde B_k| \le C_{\mathrm{AB}}\delta_c$, and (12.1) applies. For $\mathbf{Q}_k$ telescope:

$$\mathbf{Q}_k(U) - \mathbf{Q}_k(U_r) = \sum_{j\le k} l_k(U)\cdots l_{j+1}(U)\,\big(l_j(U) - l_j(U_r)\big)\, l_{j-1}(U_r)\cdots l_1(U_r).$$

By the formula for l_j in the proof of Lemma 22.5, $l_j B(c)$ is a sum of at most C_L terms. Each is a product of a bounded number of factors from $\{R(W(\gamma)), R(e^{\mathrm{i}E}), R(h_x), \mathfrak{e}(\mathrm{ad}\,E), \mathfrak{f}(\mathrm{ad}\,H_x)\}$ applied to one value $B(b)$, with γ a contour of at most $3dL$ bonds. Each factor is bounded by 2.

By Lemma 46.3(c),(d), each factor changes by at most $C\delta_j\delta_c$ from U_r to U. So $l_j(U) - l_j(U_r)$ is a combination of at most C_L values with coefficients $C_L\delta_j\delta_c$. With the type bounds of (iii), at U and at U_r, the kernel of the sum is bounded by $\sum_j C_L\delta_j\delta_c(1+Ca)e^{C_L\delta_c} \le C_L\delta_c$. The supports are those of Proposition 22.7(d). Transposes follow by summing the kernel bounds.

(vi) By Lemma 46.3(b), $W_j(U)$ is holomorphic in A'. On the domain of (ii), $\delta_j^{-1}\ell(\tilde{\mathfrak{a}}_j(e^{\mathrm{i}\delta_{j-1}B})(c))$ is jointly holomorphic in B and in the finitely many bond variables of $W_{j-1}(U)$ on which it depends. Its derivative at $B=0$, which is l_j, is therefore holomorphic in A'. The rest follows by composition, uniformly in k by the bounds of (iii)–(v). □

Remark 46.5 (Why the naive bound fails). The linear part in Lemma 46.3(a) explains why a per-scale bound fails. The derivative at $A'=0$ of the plain iterated average is $\mathbf{Q}_j - D_{W_j^T}\mathbf{\Gamma}_j$: the gauge-adapted linearization plus a coarse pure-gauge term. The plain one-step relative average has a sup-norm loss of up to a factor $2d+1$, which comes entirely from that pure-gauge term. The term does not accumulate, since $\mathbf{\Gamma}_j$ is a sum of δ_{i+1}-weighted contour averages.

Remark 46.6 (Scope of Proposition 46.4). (1) *Not proved, and not needed below.*

- A curvature bound for $W_j(U)$ of order $a+\delta_c$ in field-strength units for $j<k$, i.e. a complex analogue of Lemma 21.6 and Proposition 21.7. We only have $|H_x(U)| \le 2da\delta_j^2 + C_{\mathrm{loop}}\delta_j\delta_c$. This is weaker by δ_j^{-1} in the δ_c-term, but summable over scales. At the unit scale the two agree: $|F_{W_k(U)}| \le 2a + C\delta_c$, from Lemma 46.3(c) with $n=4$ and the Schwarz-lemma argument of (d).
- Derivative or Hölder bounds of $\tilde{B}_j$ for $0<j<k$.
- Theorem 22.4(a),(b),(e),(f) and Lemma 22.2(b),(c) at complex references. They are stated for real relative configurations; only Theorem 22.4(c),(d) and Lemma 22.2(a) hold for complex ones.

Lemma 22.5, Proposition 22.6 and Proposition 22.7 use the backgrounds only through transports along contours of bounded length. If a later argument needs any of the items above, it must be supplied separately.

(2) *Level of detail.* Proposition 46.4(ii)–(v) is a line-by-line modification of the proofs in Section 22. It inherits their level of detail, in particular the type bookkeeping of Proposition 22.6(c). The new inputs, Lemmas 46.1–46.3, are self-contained given Proposition 22.6 at real references.

(3) *Downstream.*

- The bounds $|\mathbf{Q}A(c)| \le C\|A\|_{S_c}$ and $|Q'\lambda(y)| \le C\|\lambda\|_{B(y)}$, and the sup-norm bounds of $\mathbf{Q}^T, Q'^T$, used in Theorem 35.2 and Part IX, follow from (iii) and (iv) with constants $e^{C_L\delta_c} \le 2$.
- The exact identities ($E=0$, Proposition 22.7(b), and those listed in step (iii) of the proof of Proposition 46.7) are not extended on all of $\{|A'|_\infty < \delta_{AB}\}$: there only the sup norm of A' is controlled, so the real points need not be regular backgrounds. They hold on the ball $\mathcal{B}$ of that proof, whose real points are regular, by (vi) and the identity theorem.

46.2 Extension of the results of Part IX

Proposition 46.7 (Extension to complex references)**.** *Let $\varepsilon_{\max}$ be half the smallest of the real-reference thresholds on ε that are used in the proof below, each computed with the multiplicative constants doubled. There is $\delta_P > 0$, depending on $L, M, \kappa, \rho, \alpha$, on the loss parameter δ of Part IX, on $\varepsilon_{\max}$ and on G, such that the following holds for $\delta_c < \delta_P$, at complex references of*

size δ_c whose real part is (M,ε)-regular with $\varepsilon \le \varepsilon_{\max}$. The statements hold with ε replaced by $\varepsilon + C\delta_c$. The Re*-coercivity constants and the decay rates do not depend on δ_c. The coercivity constants are those of Proposition 19.11, i.e. $\frac{c}{2}$ with c that of the real reference (Lemma 13.2), and at level (L5) $\frac{1}{8}c_L$ and $\frac{1}{16}c_L$ (Lemma 37.5, Proposition 37.6). The decay rates are those that the real-reference proofs give when every input decay bound is taken at the fixed rate of the differences, a quarter of the minimum of the perturbation rate and the two chosen inverse decay rates in Lemma 13.3 (the step "The constants" of the proof). The other constants, computed at the real reference with these coercivity constants and rates, change by at most a factor $1 + C_{\mathrm{cx}}\delta_c \le 2$, where C_{cx} depends on $L, M, \kappa, \delta, \rho, \alpha$ and G, but not on δ_c, and $C_{\mathrm{cx}}\delta_P \le 1$:*

(a) Theorem 35.2(a)–(c), Corollary 43.8, and Lemmas 44.4, 44.7, 44.9;

(b) Lemma 44.5, Lemma 44.6, and Theorems 44.10 and 44.12;

(c) Proposition 23.3(a)–(c). Here F^s and $J = D^T F^s$ are the holomorphic continuations, with $P_{\mathfrak{g}}$ replaced by its complex-linear extension $P_{\mathfrak{g}^c}$ and the summand continued without taking real parts. $|F^s| \le C(\varepsilon + \delta_c)$, while J need not be bounded uniformly in η;

(d) Theorem 45.5(a), (d), and (b) without the realness statement. The fixed point $A_*(s;B)$ is analytic in A', jointly with (s,B).

Not extended, and not needed at complex references: Theorem 35.2(d), the identity of Lemma 36.6 (its bound on W_0 extends; proof of Proposition 37.4(c)), Lemma 45.1 and Theorem 45.5(c). The "real for real arguments" statements of Propositions 22.6 and 23.3 and of Lemma 22.5(e) are dropped at complex references as well.

Proof. We go through the places where reality enters.

(i) *The L^2 theory.*

- Proposition 19.11 gives coercivity (as Re-coercivity), decay and comparison of domains for the Dirichlet operators of Parts III and VIII. It was proved for the averaging of Section 12; for the gauge-adapted averaging of Part IV it needs property (v) of Remark 19.12, i.e. Proposition 46.4(iv)–(vi) and step (iii) below. The averaged operators $\mathbf{\Gamma}^X$, $\mathbf{H}^X$, $\mathbf{C}^X$ follow by compositions and perturbed inverses.
- Part IX allows complex perturbations from the outset (Section 40, Setting).

(ii) *The local gauge in the interior estimates.* Instead of a real gauge in which U satisfies (B), use the real gauge of U_r. There $U = e^{\mathrm{i}\eta A_0}$ with complex A_0 satisfying (B) with parameter $C(\varepsilon + \delta_c)$.

- Lemmas 12.2 and 12.3 allow $\mathfrak{g}^c$-valued A.
- The absorption, iteration and L^p arguments use only sup norms and quotients, no maximum principle.
- The flat estimates (Theorems 6.3, 43.2, Lemmas 36.2, 43.1) apply to complex-valued fields componentwise.

(iii) *Algebraic and analytic identities.* Fix U_r and consider A' in the open ball $\mathcal{B} = \{\mathrm{size}(A') < \delta_P\}$. The size is a norm on the complex fields A', so $\mathcal{B}$ is convex. Real A' give real configurations $U = e^{\mathrm{i}\eta A'}U_r$ that are $(M, \varepsilon + C\delta_P)$-regular, with curvature at most $C(\varepsilon + \delta_P)$; this needs $\varepsilon_{\max} + C\delta_P \le \min(\varepsilon_0, c_3')$. At these real points the operators below are those of Parts IV–IX at the real background U: Proposition 46.4 defines all objects at the reference U through $W_j(U) = U^{(j)}$ by the formulas of Section 22, and inverses are unique. So every identity proved in the note for real regular backgrounds of small curvature holds at the real points of $\mathcal{B}$. These form a totally real subspace of maximal

dimension of the convex set $\mathcal{B}$. So an identity between two holomorphic functions on $\mathcal{B}$ that holds at the real points holds on all of $\mathcal{B}$, by the identity theorem. This is used in two stages.

(iii-a) Identities between objects of Proposition 46.4 and of the action. These are holomorphic on $\mathcal{B}$ by Proposition 46.4(vi), and since the action is a polynomial in the entries of U and U^{-1}. They are the exact covariance $E = 0$ (Proposition 22.7(b)); $J = D^T F^s$ (Lemma 23.2(a)); and $\langle \rho^{(1)}(A), F^s \rangle = 0$ in the proof of Proposition 23.3. After this stage the pair $(Q'_k, \mathbf{Q}_k)$ has property (v) of Remark 19.12 on $\mathcal{B}$, and step (i) applies. Stage (iii-a) holds with the same proof on every such ball whose real points satisfy $\varepsilon + C \cdot (\text{radius}) \le \min(\varepsilon_0, c'_3)$; in particular on the ball of radius $2\delta_*$ of the proof of Proposition 37.4(c).

(iii-b) Identities that involve inverses. After step (i), the operators of Parts III, VIII and IX are holomorphic on $\mathcal{B}$, as compositions and inverses of holomorphic families with uniformly bounded inverses, and uniformly convergent series of such. The identities are those of Lemma 19.2 used in Lemma 36.4; $\mathbf{G}DR = DG'R$ (Lemma 15.3(a)) and (44.1); the b-symmetry $(\mathbf{G}^Z)^T = \mathbf{G}^Z$ used in Lemma 44.6; and the identity of Proposition 41.7(a). For the last, $\log\det\Delta_k$ means $\sum_b \ell_k(b)$, with the integral of Proposition 34.7(c). On $\mathcal{B}$, $\operatorname{Re}\Delta_k \ge \frac{1}{8}c_L$ (Lemma 37.5; it needs $|J_{U_r}| \le \varepsilon$, which holds at every reference where Proposition 41.7 is applied, Section 47, Setting; its conditions $\varepsilon \le \varepsilon_{\text{cc}}$ and those of the proof of Proposition 37.4(c) hold since $\varepsilon \le \varepsilon_{\max}$), so the spectrum of Δ_k lies in $\{\operatorname{Re} z > 0\}$. There the integral is the principal logarithm: it is holomorphic on $\mathcal{B}$, real at the real points, and its exponential is $\det\Delta_k$.

No connectedness of the set of all complex references is needed. That set need not be connected: if G has a $U(1)$ factor, real regular backgrounds with different fluxes exist.

(iv) *Unitarity of transports and the averaging.* This is Proposition 46.4.

(v) *The action.* The proof of Proposition 23.3 uses a gauge with $|A_0|, |\partial A_0| \le \varepsilon$, $|R(H_p) - 1| \le C\varepsilon\eta^2$ and $|F^s| \le C\varepsilon$. These hold with $C(\varepsilon + \delta_c)$, using the holomorphic summand and $P_{\mathfrak{g}^c}$. The current J is not bounded uniformly in η. In the weakened background map it enters only as $(D\mathbf{H}(s))^T\mathsf{F}$ (Section 45).

(vi) *The trace inequality.* Lemma 44.5 is stated for complex perturbations.

(vii) *Analyticity in A'.* Every input is holomorphic in $A' \in \mathcal{B}$, by (i)–(v), and the contraction of Theorem 45.5 is uniform in A'. The factorization of Theorem 45.5(d), including its statement on the dependence on s, follows from the same proof: Lemma 40.3(b) allows complex backgrounds, the averaging operators are local at complex references (Proposition 46.4(iii),(iv)), and the nonlinear part is local by (c).

(viii) *The constants.* The Re-coercivity constants and the decay rates at complex references do not depend on δ_c. *The fixed rates.* For the operators of Parts II, III and VIII, Lemma 13.2 gives the hypotheses of Lemma 5.7 with the constants $c/2$, $C_1 + 1$ and $\min(\alpha_1, \gamma/2, 1)$, where c, C_1, α_1 are those at the real reference and γ is the rate of the perturbation. So these operators decay at the rate $\gamma_{\text{c}} = \min(\alpha_1, \gamma/2, \frac{1}{3}, c/(4(C_1+1)))$. By Lemma 13.3, their differences from the operators at U_r have decay $(C\delta, \gamma_{\text{c}}/4)$, with $\delta \le C_{\text{cmp}}\delta_c$. At level (L5) the lower bounds are $\frac{1}{8}c_L$ and $\frac{1}{16}c_L$, and the differences again decay at a quarter of the minimum of the perturbation rate and the two chosen inverse decay rates. Every input decay bound below, at real and at complex references, is taken at these quarter rates. At the rate γ_{c} itself the constants at a complex reference are only bounded by about twice the real ones.

The other constants. Order the levels of Sections 41 and 44, and the constants of Section 45, as in Section 41: $\ell = 1, \dots, N$. Each multiplicative constant K of level ℓ is $F_K(x_\ell)$, where x_ℓ collects the primitive inputs and the constants of the earlier levels, and F_K is a nonnegative combination of products of nonnegative, nondecreasing, positively 1-homogeneous functions of x_ℓ. These include sums, products, maxima, Neumann and walk series with nonnegative coefficients, and integrals of such, like the resolvent integral of Corollary 41.6. So $t \mapsto F_K((1+t)x_\ell)$ is convex. Put $\sigma_{\mathrm{cx}} = 2^{1/3} - 1$, so that $(1+\sigma_{\mathrm{cx}})^3 \le 2$. For $x' \le (1+t)x$ and $0 \le t \le \sigma_{\mathrm{cx}}$, convexity gives

$$F_K(x') \le F_K(x)\Big(1 + \frac{t}{\sigma_{\mathrm{cx}}}\Big(\frac{F_K((1+\sigma_{\mathrm{cx}})x)}{F_K(x)} - 1\Big)\Big).$$

At a complex reference the primitive inputs change by at most a factor $1 + C_{\mathrm{in}}\delta_c$: the averaging kernels by $e^{C_L\delta_c} \le 1 + 2C_L\delta_c$ (Proposition 46.4(iii)–(v)); the operators of Parts II, III and VIII by the differences above; and W_2^X by $C_{\mathrm{cmp}}C_W\delta_c$ (proof of Proposition 37.4(c)). Each input bound K is taken free of ε (with $\varepsilon \le 1$; e.g. $\|W_2^X\| \le C$), so an additive change $C\delta_c$ is a factor $1 + (C/K)\delta_c$; for inverses, Lemma 13.3 gives the change relative to the bound, also at level (L5), where $C_G < 1$. Put $C^{(\ell)} = (C_{\mathrm{in}}^{(\ell)}/\sigma_{\mathrm{cx}}) \max\big(1, \max_K(F_K((1+\sigma_{\mathrm{cx}})x_\ell)/F_K(x_\ell) - 1)\big)$, the maximum over the constants of level ℓ, with $C_{\mathrm{in}}^{(\ell)} = \max(C_{\mathrm{in}}, C^{(1)}, \dots, C^{(\ell-1)})$. Let $C_{\mathrm{cx}} = \max_\ell C^{(\ell)}$. For $\delta_c \le 1/C_{\mathrm{cx}}$ one has $C_{\mathrm{in}}^{(\ell)}\delta_c \le \sigma_{\mathrm{cx}}$ for every ℓ. So, by induction over ℓ, every input of every level changes by at most a factor $1 + \sigma_{\mathrm{cx}}$, and every constant by at most $1 + C_{\mathrm{cx}}\delta_c \le 2$. The values $F_K((1+\sigma_{\mathrm{cx}})x_\ell)$ are finite. Indeed, the step constants θ of Lemma 41.3 and θ' of Theorem 44.10 are sums of products of at most three inputs of their level, so their values at $(1+\sigma_{\mathrm{cx}})x_\ell$ are at most twice those at x_ℓ. Moreover M_1 is computed with the inputs doubled, and M_1' with θ' replaced by $2\theta'$. So for $M \ge M_1, M_1'$ the walks converge at $(1+\sigma_{\mathrm{cx}})x_\ell$, with ratio at most $\frac{1}{2}$. For the same reason $2\theta'$ bounds θ' at complex references, and the inputs of M_1 at most double there. C_{cx} is built from real-reference constants that hold for all ε below thresholds fixed before δ_P, such as ε_0 and c_5. So it depends on L, M, κ, δ, ρ, α and G, but not on δ_P. The thresholds M_1 and M_1', fixed in this way before M, serve at every complex reference with $\delta_c < \delta_P$, and M is chosen before δ_P without circularity. The proofs of Theorems 44.10 and 44.12 then apply verbatim.

(ix) *Not extended.*

- The identity of Lemma 36.6 differentiates real gauge transformations and needs $|J_U| \le \varepsilon$. It enters only Theorem 35.2(d), which is not used after Section 45.
- Lemma 45.1 and Theorem 45.5(c) concern real critical backgrounds, where they are used.

Smallness. δ_P must satisfy the following:

- $C_{\mathrm{AB}}\delta_P \le \min(c_L', cL^{-d-1})$, for the elimination of Lemma 32.4 at a complex base (Corollary 34.5, with $|\mathfrak{r}_c| \le \frac{1}{2}$) and for the coercivity of Lemma 37.5, used in Section 47; and $\delta_P \le c_L'$. Since $C_{\mathrm{cmp}}c_L' \le \delta_*$, with $C_{\mathrm{cmp}} = 4$ (first item of Section 46) and δ_* of the proof of Proposition 37.4(c), the perturbations of size $< C_{\mathrm{cmp}}\delta_P$, and the ball of twice that size used there for the Cauchy estimate, lie within the radii fixed with δ_*. These are the radii of Proposition 19.11 for $W = 0$ and for W_2, W_2^X, of Proposition 46.4, of Proposition 23.3, of Proposition 35.1(e), and of $E = 0$ (step (iii));
- $\varepsilon_{\max} + C\delta_P \le \min(\varepsilon_0, c_3', \varepsilon_L, \varepsilon_{\mathrm{ID}}, \varepsilon_{\mathrm{cc}}, \frac{1}{20})$, with ε_L the threshold of Proposition 36.5, $\varepsilon_{\mathrm{ID}}$ that of Theorem 35.2 and $\varepsilon_{\mathrm{cc}}$ that of the first item of Section 46; moreover $\varepsilon_{\max} + C\delta_P$ is below the other thresholds of Sections 36 and 43;

- $C(\varepsilon_{\max}+\delta_P)\le c_5$ of Proposition 23.3;
- $C_P(\varepsilon_{\max}+C\delta_P)$ small in Theorem 45.5, with C_P computed with the multiplicative constants doubled;
- at complex references the condition $(1+K_H)r_P\le c_3$ in the proof of Lemma 45.3 becomes $(1+2K_H)r_P\le c_3'$, with r_P computed with the multiplicative constants doubled;
- $C_{\mathrm{cx}}\delta_P\le 1$, for the factor $1+C_{\mathrm{cx}}\delta_c\le 2$.

At complex references the constant K_H is at most $2K_H$, by the step "The constants" of the proof. Later sections use $2K_H$ in place of K_H, and the radii R_P, r_P and the constants C_P, ε_P, ϱ_P of Theorem 45.5 computed with the multiplicative constants doubled. All of these are conditions on δ_P after $L, M, \kappa, \rho, \alpha$ and $\varepsilon_{\max}$ are fixed. □

Remark 46.8 (Use in the renormalization group). In the activity domain D3‴ the complex perturbations have size at most $Ch_k^{1-2\epsilon}$, which tends to 0 as $g_*\to 0$. So, once g_* is small, $\delta_c<\delta_P$, and the real parts are (M,ε)-regular with $\varepsilon\le\varepsilon_{\max}$: by Corollary 30.4, $\varepsilon\le B_5 h_k^{1-\epsilon/2}$, with B_5 fixed before g_*. This is consistent with the order of the parameters in Theorem 49.8. In the units of step k the budgets of the next step shrink further, which only helps. By Propositions 46.4 and 46.7, the new activities produced by the step are analytic functions of the complex background, as the norm requires.

47 The weakened step quantities

This section turns the ingredients of one step (Theorem 34.8) into the input of the cluster expansion. That input consists of functions of the weakening parameters $s\in\mathbb{D}_\kappa$, the white-noise field and the background, with three properties. They are analytic with uniform bounds; they factorize when $s_\square=0$ outside a union of boxes; and they are invariant under real gauge transformations. They are defined by source-free formulas for every background in the domain, not only at critical backgrounds. The plan and its review are in the blueprint (Appendix H).

The determinant is treated in Proposition 41.7, and the Gaussian measure through $C_k^{1/2}(s)$ (Corollary 41.6). This section treats the interaction (Section 47.2), the Jacobians (Section 47.3), the old activities composed with the background map (Section 47.4) and the Hessian away from critical backgrounds (Section 47.5), which Section 37 already defines. Section 47.1 gives the localization lemma that turns these functions of s into pieces of volume type.

Setting.

- *The reference.* U' is a reference in the sense of Section 45: real and (M,ε)-regular with $|J_{U'}|\le\varepsilon$, or a complex reference $U'=e^{i\eta A'}U_r'$ of size $\delta_c<\delta_P$ in the sense of Section 46, whose real part is (M,ε)-regular with $\varepsilon\le\varepsilon_{\max}$ (Proposition 46.7) and satisfies $|J_{U_r'}|\le\varepsilon$. Theorem 45.5(a), (b), (d) hold at it (Proposition 46.7(d)). The constant K_H and the radii R_P, r_P of Part IX are those computed with the multiplicative constants doubled (Proposition 46.7, end of the proof); they serve at every complex reference with $\delta_c<\delta_P$ and do not depend on δ_P. The thresholds M_1 and M_1' are those of Theorem 49.8(4) and do not depend on δ_P. At a complex reference, $|\cdot|_*$ is taken in the real gauges of U_r', and covariant derivatives with respect to U_r'; the other reading, with respect to U', differs by a local term of size $C\delta_c|A|$. Within this setting, where regularity is assumed separately, the current bound on U_r' is used at complex references only for coercivity (Proposition 47.9(d)); at real references it is also used in Theorem 45.5(c). On D3‴ it is used in addition for the (M,ε)-regularity of U_r' (Corollary 30.4), for the

background-activity domain (Proposition 41.7(b)) and in Lemma 38.5. It holds on D3‴, where $|J_{U'_r}| < h^{1-\epsilon/2}$.

- *Extension.* A point of the activity domain of step $k+1$ at a polymer Y' of step $k+1$ fixes U' only near Y'. It is extended to a reference by the cutoff of Remark 38.7, with the same real part U'_r. A piece indexed by a polymer Z is used only when Y' contains the LM-cubes meeting $Z^{(r')}$ (Remark 49.7). By the factorization below, such a piece does not depend on the extension.

- *The base.* For every reference put

$$W := U'^{(k)}, \qquad V' := U'^{(k+1)}.$$

  At critical backgrounds these are V^0 and V' of Lemma 31.6. They are local in U' and covariant, and analytic at complex references by Proposition 46.4(i). The elimination of Lemma 32.4 is taken at the base W. In that lemma the parameter a is defined through the curvature ε'_1 of V'; at a general reference it is replaced by a bound on the curvatures of the real parts U'_r and $(U'_r)^{(k)}$, which is all the proof uses. At complex bases the elimination is analytic in the reference, by the argument in the proof of Corollary 34.5 together with Proposition 46.4(i),(vi). This needs $C_{\mathrm{AB}}\delta_c \le c'_L$, which is part of the smallness of δ_P (Proposition 46.7). It gives ζ_c, $\hat{Z}$, $\hat{Z}^1$, $\hat{Z}^2$, with $|\hat{Z}(Z)|_\infty \le CL^d|Z|_\infty$.

- *The fields.* We write ξ for the white-noise field on $\mathfrak{F}$ (Dimock's W), since in Sections 47–50 W is the base, and $Z(s) = gC_k^{1/2}(s)\xi$. $A_*(s;B)$ is the weakened background map of Theorem 45.5 at the reference U'.

- *Margins.* Let r be the largest margin of the weakened families used: $\mathbf{C}(s)$, $\mathbf{H}(s)$, $D\mathbf{H}(s)$ and their transposes, and $C_k^{1/2}(s)$ (Corollary 41.6). The definition of weakenable families (Section 40) requires $r \le M/16$. The margins are sums of a bounded number of ranges of local operators (Lemma 40.5(a), Theorem 41.4(a), and the box split of the additive local terms, Section 40), independent of M; so $r \le M/16$ is a condition on M, listed in Theorem 49.8(4). A family with margin r is also a family with every larger margin up to $M/16$, since its kernel supports and its data dependence lie in every larger neighbourhood; so we may and do assume $r \ge 3$. Put $r' = r + 2L + 2$. We assume $M > 32(L+1)$, so that $2r' < M/4$. Two polymers that do not touch are then at distance at least $M/4 > 2r'$, so their r'-neighbourhoods are disjoint.

- *Constants.* C_P denotes constants that may depend on $\kappa, \delta, L, M, \rho, \alpha$, but not on k, n, η or s.

Remark 47.1 (Notation in Sections 47–50). Some letters are used here with a meaning that differs from earlier parts of the note. We keep them, and list them once.

- C_T is the bound on $C_k^{1/2}(s)$ of Section 47.2, not the tier constant of Section 44. C_{tor} is the torus constant of (T2).

- C_d, A, ϵ_* are the constants of Theorem 48.6; they are not the constants of Lemma 40.4, the contraction factor A of Theorem 49.8, or the bootstrap threshold of the earlier parts.

- κ and $\kappa' = \kappa - C\delta$ are the weakening rates of Part IX; Section 49 writes κ'_w for κ', since there κ' is also the reblocked rate of Lemma 49.2. κ_n is the rate of the activity norm. The rate κ in Theorem 48.6 is generic. Theorem 45.5 is stated on $\mathbb{D}_\kappa$ with κ the rate of its inputs $\mathbf{H}(s)$, $\mathbf{C}(s)$ at level (L4); that rate is at least κ', and the pieces below are formed on $\mathbb{D}_{\kappa'}$.

- $\ell_k(b;1)$ is the density at the reference $U' = 1$, not at $s = 1$.

- U' is the reference; in Section 46 it was also the relative factor $e^{\mathrm{i}\eta A'}$. W is the base; earlier it also denotes potentials and configurations.
- r is the margin; the resolvent parameter of Corollary 41.6 is unrelated. r_E is the margin of dependence of the old activities (Section 48).
- δ is the loss parameter of Part IX, δ_c the size of a complex reference, and δ_1, δ_0 (with primes) the budgets of a graded domain; δ_0 of Proposition 19.11 enters only through δ_P.
- B is the source constant of Proposition 49.6, unrelated to B_P, B_3, B_5 and to the coarse field B; C_{P} (roman) is the Poincaré constant of Proposition 50.2, unrelated to C_P. In Part IX, N_{b} is the box count of Section 40, and N, in Proposition 46.4 and earlier, is the matrix size of G. h_j is a partition of unity, unrelated to $h_k = p_k g_k$. a is the curvature parameter of Lemma 32.4, not the averaging parameter a.
- $\hat{Z}$ is the elimination map acting on fields; applied to a polymer (Remarks 47.3 and 49.3) it denotes the union of the LM-cubes meeting $Z^{(r')}$. The polymer Z and the field $Z(s)$ are distinguished by the argument.
- p is a plaquette in s_p and v_p, and p, $p_0 = p_{0,k}$ bound the white noise; $p_k = p(g_k)$ is the small-field scale. ζ is the bound on $\hat{Z}$, unrelated to ζ_c. τ is the composed-activity ratio of Proposition 47.8. F (Section 45) and F^s denote the same symmetrized curvature.
- ϵ is the exponent of D3‴ and, in Section 50, also the finest spacing; ε_k is the vacuum energy per volume, unrelated to the thresholds ε_0, ε_1 and to ε^0 of N_χ. $n = n_k$ is the side of $\mathbb{T}^{(k)}$ (see (T0)), and h_0, n_0 are thresholds, not values at step 0.

47.1 The localization lemma

For a polymer Y, a polymer $Z \supset Y$ and $t \in [0,1]^{Z\setminus Y}$, let $s(Z,t)$ be the point of $\mathbb{D}_\kappa$ with $s_\square = 1$ for $\square \subset Y$, $s_\square = t_\square$ for $\square \subset Z \setminus Y$, and $s_\square = 0$ otherwise.

Lemma 47.2 (Localization)**.** *Let $\kappa \geq 1$, let Y be a polymer, and let f be a function on $\mathbb{D}_\kappa$, analytic in the interior and continuous, with $|f| \leq F$. Suppose that, for every union of boxes $Z' \supset Y$ with $*$-component $Z'_0 \supset Y$, and every s with $s_\square = 0$ for $\square \not\subset Z'$, $f(s)$ depends on s only through $s_\square$, $\square \subset Z'_0$, and on the data only on $(Z'_0)^{(r')}$. Then*

$$f(1) = \sum_{Z \supset Y} f_{\mathrm{loc}}(Y,Z), \qquad f_{\mathrm{loc}}(Y,Z) = \int_{[0,1]^{Z\setminus Y}} \partial_{s_{Z\setminus Y}} f(s(Z,t))\, dt,$$

the sum running over the polymers $Z \supset Y$, where:

(a) $f_{\mathrm{loc}}(Y,Z)$ depends on the data only on $Z^{(r')}$;

(b) $|f_{\mathrm{loc}}(Y,Z)| \leq F\, e^{-(\kappa-1)|Z\setminus Y|}$;

(c) if f is invariant under real gauge transformations of the data, or analytic in further parameters with $|f| \leq F$ uniformly, then so is $f_{\mathrm{loc}}(Y,Z)$.

Proof. Write $f(1)$ as a function of the variables $s_\square$, $\square \not\subset Y$, with $s_\square = 1$ on Y. Apply $g(1) = g(0) + \int_0^1 g'(t)\, dt$ in each of these variables. This gives the sum over sets S of boxes outside Y of $\int_{[0,1]^S} \partial_{s_S} f\, dt$, evaluated at $s_\square = 1$ on Y, $s_\square = t_\square$ on S, and $s_\square = 0$ elsewhere. Let $Z' = Y \cup S$, with $*$-component $Z'_0 \supset Y$. By hypothesis, at these points f does not depend on $s_\square$ for $\square \subset S \setminus Z'_0$. So the term vanishes unless $S \subset Z'_0$, i.e. unless $Z = Y \cup S$ is a polymer. This gives the formula. Part (a) follows from the hypothesis at $Z' = Z$.

For (b), every point $s(Z,t)$ has $|s_\square| \leq 1$. In each differentiated variable, Cauchy's formula on the circle of radius $e^\kappa - 1$ around $t_\square$ stays in $\mathbb{D}_\kappa$, by continuity up to the boundary. Each

variable is differentiated once, so $|\partial_{s_S} f| \le F(e^{\kappa}-1)^{-|S|} \le F e^{-(\kappa-1)|S|}$ for $\kappa \ge 1$. Part (c) holds because differentiation in s and integration over t preserve both properties. □

Remark 47.3 (Margins). The pieces are indexed by polymers Z and depend on the data on $Z^{(r')}$, not inside Z. In the cluster expansion, polymers that do not touch have disjoint r'-neighbourhoods, so "touching" is the hard-core condition. Margins should not be converted to 0 for the composed activities. Replacing Z by $Z^\sharp$ (Section 40) turns the decay $e^{-(\kappa-1)|Z\setminus Y|}$ into $e^{-3^{-d}(\kappa-1)|Z^\sharp|+(\kappa-1)|Y|}$. For Y a single box, the factor $e^{(\kappa-1)|Y|}$ is a constant. For the composed activities, Y is the old polymer, and the factor would swamp the old decay. The next-step polymer of a piece is the union $\hat{Z}$ of the LM-cubes meeting $Z^{(r')}$. In the reblocking this costs at most $C|\hat{Z}|_{LM}$, counted in LM-cubes, with C independent of L and M, since $r' < M/8$.

47.2 The interaction pieces

Write $S(U) = \sum_p s_p(U)$, where s_p is the term of the plaquette p of $\mathbb{T}_\eta$ in the action (Section 23). Put $r_I = \min\big(r_L, c_I L^{-d} R_P\big)$, with r_L from Lemma 32.4 and R_P from Theorem 45.5 (computed with the multiplicative constants doubled, see the Setting), and c_I so small that $|\hat{Z}(Z)|_\infty < R_P$ for $|Z|_\infty < r_I$.

Definition 47.4 (Interaction pieces)**.** For a plaquette p, $s \in \mathbb{D}_\kappa$ and complex Z on $\mathfrak{F}$ with $|Z|_\infty < r_I$, put

$$v_p(s;Z) = s_p\big(e^{\mathrm{i}\eta A_*(s;\hat{Z}(Z))}U'\big) - T_2\big[s_p\big(e^{\mathrm{i}\eta A_*(s;\hat{Z}(\cdot))}U'\big)\big](Z),$$

where T_2 is the Taylor polynomial of degree 2 at $Z=0$, for fixed s and U'.

Proposition 47.5 (Interaction pieces)**.** *(a) $v_p(s;Z)$ is analytic in (s,Z) for s in the interior of $\mathbb{D}_\kappa$ and $|Z|_\infty < r_I$, continuous up to the boundary in s, and analytic in the reference.*

(b) $|v_p(s;Z)| \le C_P\,\eta^d\,(|Z|_\infty/r_I)^3$ for $|Z|_\infty \le \frac{1}{2}r_I$, uniformly in $s \in \mathbb{D}_\kappa$.

(c) (Identification.) *If U' is real and critical, then $\sum_p v_p(1;Z) = \mathcal{V}(Z)$ for $|Z|_\infty \le \min(\frac{1}{2}r_I, r_2)$ with $|\hat{Z}(Z)|_\infty \le \beta_V$.*

(d) (Factorization.) *Let $s_\square = 0$ for $\square \not\subset Z'$, a union of boxes with $*$-components Z'_i. Then $v_p = 0$ unless p has a point in $Z'^{(r+1)}$. If p has a point in $(Z'_i)^{(r+1)}$, then v_p depends on s only through $s_\square$, $\square \subset Z'_i$, on Z only on $(Z'_i)^{(r+2L)}$, and on U' only on $(Z'_i)^{(r')}$.*

(e) (Invariance.) *v_p is invariant under real gauge transformations u, acting by $U' \mapsto U'^u$ and $Z(b) \mapsto R(u(b_-))Z(b)$.*

Proof. (a) follows from Theorem 45.5(b), the analyticity of $\hat{Z}$ and of s_p, and Proposition 46.7(d). The Taylor coefficients at $Z=0$ are analytic in s and in the reference.

(b) Fix s and Z, and put $g(t) = v_p(s;tZ)$. It is analytic for $|t| < r_I/|Z|_\infty$ and vanishes to third order at $t=0$. Let $h(t) = s_p(e^{\mathrm{i}\eta A_*(s;\hat{Z}(tZ))}U') - s_p(U')$. Write $s_p(U) = \eta^{d-4}(\mathsf{N} - \operatorname{tr}\cos X_p)$ with $X_p = \ell(U(\partial p))$, as in the proof of Theorem 39.4. Then $|s_p(U)| \le C\eta^{d-4}|X_p|^2$ for $|X_p| \le 1$. By Theorem 45.5(a), $|A_*(s;B)|_* \le C_P|B|_\infty \le C_P R_P = C_P\varrho_P/K_H \le \frac{1}{8}$, since $4C_P(\varepsilon_P + 2\varrho_P) \le 1$ in its proof (with $K_H \ge 1$). So, in the gauge of U'_r, the four bond fields of p at $e^{\mathrm{i}\eta A_*}U'$ have sup norm at most $\frac{1}{8} + \varepsilon + C\delta_c \le \frac{1}{5}$, since $\varepsilon + C\delta_c \le \frac{1}{20}$ (Proposition 46.7, smallness). For $k \ge 1$, $\eta = L^{-k} \le \frac{1}{2}$, so $|U(\partial p) - 1| \le e^{4\eta/5} - 1 \le e^{2/5} - 1 \le \frac{1}{2}$. At $k=0$, where $\eta = 1$ and $A_* = B$ (Proposition 50.2), $|B|_\infty \le \rho_L$ (Lemma 32.4(c)), so, with $\varepsilon + C\delta_c \le \frac{1}{20}$ and $\rho_L \le \frac{1}{20}$ (Proposition 50.2), the bond fields have sup norm at most $\frac{1}{10}$ and $|U(\partial p) - 1| \le e^{2/5} - 1 \le \frac{1}{2}$. In both cases $|X_p| \le 1$ (Lemma 21.1(a)). By the expansion of

Lemma 12.3, $X_p = \eta^2(F_{U'}(p) + O(|A_*|_*))$, and $|F_{U'}| \le C(\varepsilon + \delta_c)$. So $|s_p| \le C\eta^{d-4}\eta^4 = C\eta^d$ at $e^{i\eta A_*}U'$ and at U', and $|h(t)| \le C_P\eta^d$ for $|t| < r_I/|Z|_\infty$. No expansion in A_* is made, so the current, which is not bounded uniformly in η at complex references, does not enter. By Cauchy's estimate each of the three Taylor terms of h of degree at most 2 is bounded by $C_P\eta^d$ there. So $g = h - T_2h$ is bounded by $4C_P\eta^d$ there, and vanishes to order three. The maximum principle for $g(t)/t^3$ on the discs $|t| \le \varrho$, $\varrho < r_I/|Z|_\infty$, and the limit $\varrho \to r_I/|Z|_\infty$ give $|g(1)| \le 4C_P\eta^d(|Z|_\infty/r_I)^3$.

(c) By Theorem 45.5(c), $A_*(1;B) = A_*(B)$ for $|B|_\infty \le \min(R_P, \beta_V)$. So, summing over the finitely many plaquettes,

$$\sum_p s_p(e^{i\eta A_*(1;\hat Z(Z))}U') = \mathcal{S}(\hat Z(Z)).$$

By Lemma 34.1, $\mathcal{S}(\hat Z(Z)) = S(U') + \frac12\Delta_k(Z,Z) + \mathcal{V}(Z)$, where $\mathcal{V}$ vanishes to third order. Hence $\sum_p T_2[\cdots] = S(U') + \frac12\Delta_k(Z,Z)$, and $\sum_p v_p(1;Z) = \mathcal{V}(Z)$.

(d) By Theorem 45.5(d), $A_*(s;B)$ vanishes outside $Z'^{(r)}$. On $(Z'_i)^{(r)}$ it depends on B only on the coarse bonds of $(Z'_i)^{(r)}$, on U' only on $(Z'_i)^{(r+1)}$, and on s only in Z'_i. If p has no point in $Z'^{(r+1)}$, then $A_* = 0$ on the bonds of p. So $s_p(e^{i\eta A_*}U') = s_p(U')$ for all Z, and $v_p = 0$. Otherwise, all bonds of p lie in $(Z'_i)^{(r+2)}$ for exactly one i, since the components are at distance at least $M/4$. The map $\hat Z$ has range $2L$ in block units, and the base $W = U'^{(k)}$ is local. This gives the dependence stated.

(e) $A_*(s)$ is covariant (Lemma 40.3(c) and the construction in Section 45), $\hat Z$ is covariant, and s_p is invariant. The Taylor coefficients of an invariant function are invariant. □

Grouping by boxes, put $V(\square;s) = g^{-2}\sum_p v_p(s;Z(s))$, the sum running over the plaquettes whose base point lies in $\square$. The weakened families of Part IX live on $\mathbb{D}_{\kappa'}$, $\kappa' = \kappa - C\delta$, and $|Z(s)|_\infty \le gC_T|\xi|_\infty$ there, with $C_T = 2^d\sqrt{d}\,\|C_k^{1/2}\|_{\kappa',\gamma'}$. Indeed, by Lemma 40.3(a) and Corollary 41.6, $|C_k^{1/2}(s)|_{\gamma'} \le 2^d\|C_k^{1/2}\|_{\kappa',\gamma'}$. The row sums of the block norms bound the operator on sup norms up to the factor $\sqrt{d}$ from $\|\xi\|_{B(y')} \le \sqrt{d}|\xi|_\infty$. By Corollary 41.6, C_T is uniform in the reference. Let $|\xi|_\infty \le p$ with $gC_Tp \le \frac12 r_I$. Then $V(\square;s)$ is defined and analytic on the interior of $\mathbb{D}_{\kappa'}$, continuous up to the boundary. By Corollary 41.6, Lemma 40.3(b) and Proposition 47.5(d), Lemma 47.2 applies to $f(s) = V(\square;s)$ on $\mathbb{D}_{\kappa'}$ with $Y = \square$, and $F = \sup_{\mathbb{D}_{\kappa'}}|V(\square;\cdot)| \le C_Pg^{-2}M^d(gC_Tp/r_I)^3$. With $Z(s) = gC_k^{1/2}(s)\xi$, this is of order $g\,|\xi|^3$. The control of ξ belongs to the cluster expansion.

47.3 The Jacobians

Proposition 47.6 (Jacobians). *(a) $\mathfrak{J}(Z) = \sum_{b\in\mathfrak{F}}\log(\sigma(Z(b))/\sigma(0)) + \sum_c\log(\mathsf{J}_c(Z)/\mathsf{J}_c(0))$ is a sum of terms, each analytic for $|Z|_\infty \le r_L$, depending on Z and on the base only within range $2L$ of one bond, vanishing at $Z = 0$ and bounded as in Theorem 34.8(e). The terms are invariant under real gauge transformations. Since Proposition 32.5 is written with* det*, which is positive at real references, they are analytic in Z and in the reference jointly, on the complex references with $\delta_c < \delta_P$, and real at real references and real Z.*

(b) By Proposition 32.5, $\mathfrak{j}_c = \det(1+\mathfrak{r}_c(0))^{-1}$, with $\mathfrak{r}_c$ from Lemma 32.4(b). It is an analytic function of the base near c, invariant under real gauge transformations, and $\mathfrak{j}_c = 1$ at the trivial base.

(c) Let G be semisimple. As a function of U' (in the units of step k) through $W = U'^{(k)}$, $\log\mathfrak{j}_c$ is a normalized graded activity at the blocks $B'(c_-)\cup B'(c_+)$ on $\mathcal{U}(\cdot;c_a,c_a,c_a)$ for $c_a \le c_JL^{-d-2}$, with sup at most C. Lemma 38.5 applies to it. On $D\mathfrak{Z}'''$ it gives $|\log\mathfrak{j}_c| \le C_Ph^{2-2\epsilon}$. (For abelian G, $\mathfrak{r}_c \equiv 0$ and $\mathfrak{j}_c \equiv 1$.)

Proof. (a) is Proposition 32.5 and Theorem 34.8(e), with Lemma 32.4(a),(c) for the locality. For invariance, a real gauge transformation u acts on the relative variables by $Z(b) \mapsto R(u(b_-))Z(b)$ and on ℓ_c by $R(u(c_-))$. The function σ is Ad-invariant, and $\det \partial \ell_c / \partial Z(b_0(c))$ changes by $\det R(u(c_-)) \det R(u(b_0(c)_-))^{-1} = 1$. The terms are analytic functions of Z and of the base, through ℓ_c and ζ_c, and the base and the elimination are analytic in the reference (Setting). At real references with $L^d \mathsf{a} \le c$ and $|Z|_\infty \le r_L$, $\partial \ell_c / \partial Z(b_0(c))$ is real, and its determinant is positive (Proposition 32.5). So the analytic terms are those of Theorem 34.8(e) there.

(b) At $Z = 0$ we have $\zeta_c(0) = 0$, since ℓ_c vanishes at $Z = 0$. By Lemma 32.4(b) and $\det R = 1$ on $\mathfrak{g}$, $\det \partial \ell_c / \partial Z(b_0(c)) = L^{(1-d)\dim \mathfrak{g}} \det(1 + \mathfrak{r}_c(0))$. This gives the formula. At the trivial base, $l' = q$, so $\partial \ell_c / \partial Z(b_0(c)) = L^{1-d}$ exactly and $\mathfrak{r}_c(0) = 0$. Invariance is (a) at $Z = 0$. For complex bases, $\det R(g) = 1$ for $g \in G^c$, since G is connected (Section 12). So the formula is the analytic continuation.

(c) On $\mathcal{U}(P; c_a, c_a, c_a)$, the base $W = U'^{(k)}$ is a complex perturbation of size Cc_a of a real base of curvature at most Cc_a, in the units of step k (Proposition 21.7, Proposition 46.4(i)). In Lemma 32.4, a is a curvature in the units of step $k+1$ (Setting), which is L^2 times the curvature in the units of step k. So $L^d \mathsf{a} \le C L^{d+2} c_a$ for the real part. The complex part of the base contributes to $\mathfrak{r}_c(0)$ through the remainder r_j of Proposition 46.4(ii) for the step from scale k to scale $k+1$. Proposition 46.4 is stated for $j \le k$; we apply it with $k+1$ in place of k, in the units of step $k+1$, where the curvature becomes $L^2 C c_a$ and the size of the complex part LCc_a. Its hypotheses hold for c_J small: $L^2 C c_a \le c_3'$ and $LCc_a < \delta_{AB}$. The remainder is relatively $CL^{d+1}c_a$. Hence $|\mathfrak{r}_c(0)| \le CL^{d+2}c_a \le \frac{1}{2}$ for c_J small. Then $\log \mathfrak{j}_c = -\log\det(1 + \mathfrak{r}_c(0))$ is analytic and bounded by C. It is normalized by (b), invariant by (b), and depends on U' only on the bonds of $B'(c_-) \cup B'(c_+)$. Lemma 38.5 with $\delta_1 = \delta_0 = c_a$ gives the bound, as in the proof of Proposition 41.7(d). □

The terms of $\mathfrak{J}(Z(s))$ are local in Z, but $Z(s)$ is not local in ξ. So $\mathfrak{J}(Z(s))$, grouped by boxes, goes through Lemma 47.2 like $V(\square; s)$.

47.4 The composed activities

The norm $|\cdot|_*$ of Theorem 45.5 and the quantities of Definition 38.3 are compared as follows.

Lemma 47.7 (Norm conversion)**.** *There is C_{conv}, depending only on d, α and G (through Theorem 35.2(b)), with the following property. Let U_r' be real and (M, ε)-regular with $\varepsilon \le \varepsilon_0$, and let A be a $\mathfrak{g}^c$-valued field on the bonds of $\mathbb{T}_\eta$, with $|A|_*$ taken in the real gauges of U_r' (Section 45, with $\beta = \alpha$). Then for every set Y,*

$$|A|_Y + |\nabla A|_Y + [\nabla A]_{\alpha, Y} \le C_{\mathrm{conv}} |A|_*,$$

with $|A|_Y$ the sup norm on the bonds of Y, and ∇ and the Hölder quotient of Definition 38.3 taken with respect to U_r'.

Proof. Every bond lies in a block $B(y)$, and $\mathfrak{n}_y^3(A)$ bounds $|A|_{B(y)}$ and $|\nabla A|_{B(y)}$. If the gradient in $\mathfrak{n}_y^3$ is read in the gauge of y as a flat derivative, (38.1) with $\sigma = C\varepsilon$ converts it, with an error $C\varepsilon|A|$. Consider a quotient in $[\nabla A]_{\alpha,Y}$, with $x \ne x'$ in a cube $\Lambda_3(z') \subset Y$. If $|x - x'| \ge 1$, it is at most $|\nabla A(x)| + |\nabla A(x')| \le 2|\nabla A|_Y$, since $R(U_r'(\Gamma_{x,x'}))$ is orthogonal. If $|x - x'| < 1$, then x, x' lie in a common cube $\Lambda_1(c_y)$ (Section 45, setting). In the real gauge of y, $U_r' = e^{i\eta A_r}$ with $\|A_r\|_{1,\alpha} \le C\varepsilon$ on $\Lambda_4(c_y)$ (Theorem 35.2(b)). There (38.1) and $\|R(U_r'(\Gamma_{x,x'})) - 1\| \le C|x - x'|\varepsilon$ bound the quotient by $[\partial \tilde{A}_y]_{\alpha, \Lambda_1(c_y)} + C\varepsilon(|A| + |\partial \tilde{A}_y|) \le C\mathfrak{n}_y^3(A)$. Taking the supremum over y gives the bound. □

The constant is not $1 + C\varepsilon$: pairs at distance at least 1 cost a factor 2.

Proposition 47.8 (Composed activities)**.** *Let X be a connected domain of step k (an old polymer), let $r_E \geq 0$ with $r_E + 3 \leq r$, and put $P = X^{(r_E)}$. Let $\mathcal{E}(X, \cdot)$ be a graded activity at P on $\mathcal{U}(P; \theta_r, \delta_1, \delta_0)$ (Definition 38.3), with sup $\|\mathcal{E}(X)\|$; so it depends on the configuration on P, and its domain is controlled on $P^{(3)} = X^{(r_E+3)}$. Let $U' = e^{\mathrm{i}\eta A'} U'_r$ be a reference as in the Setting, and suppose that on $P^{(3)}$ this is a representation as in Definition 38.3, with budgets $\theta_r, \delta'_1, \delta'_0$, where $\delta'_1 < \delta_1$ and $\delta'_0 < \delta_0$. Thus the real part of the representation is the U'_r in whose gauges $|\cdot|_*$ is measured. Let $0 < \zeta \leq \frac{1}{2} R_P$, and put*

$$\tau = C_1 C_{\mathrm{conv}} C_P \, \zeta \, \Big(\min(\delta_1 - \delta'_1, \ \delta_0 - \delta'_0) \Big)^{-1}.$$

Then, for $\tau < \frac{1}{2}$, $s \in \mathbb{D}_\kappa$ and $|\hat{Z}(Z)|_\infty \leq \zeta$:

(a) $e^{\mathrm{i}\eta A_*(s; \hat{Z}(Z))} U' \in \mathcal{U}(P; \theta_r, \delta_1, \delta_0)$, and

$$f(X; s, Z) = \mathcal{E}\big(X, e^{\mathrm{i}\eta A_*(s; \hat{Z}(Z))} U'\big) - \mathcal{E}(X, U')$$

is analytic in (s, Z) and in the reference jointly, continuous up to the boundary in s, with $|f(X; s, Z)| \leq 4\tau \|\mathcal{E}(X)\|$;

(b) if $s_\square = 0$ outside a union of boxes Z' that contains the boxes meeting X, then $f(X; s, Z)$ depends on s only through the boxes of the $*$-component Z'_0 of Z' containing them, and on Z and U' only on $(Z'_0)^{(r')}$;

(c) $f(X; s, Z)$ is invariant under real gauge transformations.

Proof. (a) For $|t| < \frac{1}{2\tau}$ consider $e^{\mathrm{i}\eta t A_*} U' = e^{\mathrm{i}\eta (t A_* \oplus A')} U'_r$ on $P^{(3)}$, with $A_* = A_*(s; \hat{Z}(Z))$, combined bondwise by the Baker–Campbell–Hausdorff formula. The real part U'_r is unchanged, so no current of A_* is needed. By Theorem 45.5(a), $|A_*|_* \leq C_P \zeta$. By Lemma 47.7, the sup norm, the covariant gradient and the Hölder quotient of A_* on $P^{(3)}$, with respect to U'_r, are at most $C_{\mathrm{conv}} C_P \zeta$. By the Baker–Campbell–Hausdorff formula and (23.2), the corresponding quantities of $tA_* \oplus A'$ exceed those of A' by at most $C_1 C_{\mathrm{conv}} C_P |t| \zeta < \frac{1}{2} \min(\delta_1 - \delta'_1, \delta_0 - \delta'_0)$, for a suitable C_1. So $tA_* \oplus A'$ satisfies the bounds of $\mathcal{U}(P; \theta_r, \delta_1, \delta_0)$ on $P^{(3)}$, strictly. Hence $t \mapsto \mathcal{E}(X, e^{\mathrm{i}\eta t A_*} U') - \mathcal{E}(X, U')$ is analytic for $|t| < \frac{1}{2\tau}$, bounded by $2\|\mathcal{E}(X)\|$, and vanishes at $t = 0$. Since $\tau < \frac{1}{2}$, the point $t = 1$ lies in the disc, and the Schwarz lemma gives $|f| \leq 2\|\mathcal{E}(X)\| \cdot 2\tau$ there.

Analyticity. $A_*(s; B)$ is analytic in (s, B) and in A' jointly (Theorem 45.5(b), Proposition 46.7(d)). $\hat{Z}$ is analytic in Z and in the base, and the base is analytic in A' (Proposition 46.4(vi), Setting). $\mathcal{E}(X, \cdot)$ is holomorphic on an open set containing $\mathcal{U}(P)|_P$, and the bounds above hold strictly. So f is analytic in (s, Z, A') jointly, as a composition. By (b), it depends on A' only on $(Z'_0)^{(r')}$, so it is analytic in the reference near that set. Continuity up to the boundary in s follows from Theorem 45.5(b).

(b) $\mathcal{E}(X, \cdot)$ depends on the configuration on $P = X^{(r_E)} \subset (Z'_0)^{(r)}$, and A_* on $(Z'_0)^{(r)}$ is determined by the data on $(Z'_0)^{(r')}$ and by s in Z'_0 (Theorem 45.5(d), and Lemma 32.4 for $\hat{Z}$). (c) follows from the covariance of A_* and $\hat{Z}$ and the invariance of $\mathcal{E}$. □

In the step, $\mathcal{E} = \mathcal{E}_k$ on the domain D3‴ of step k, with $r_E = 0$ and $P = X$, and U' is the extension (Setting) of a point of the domain of step $k+1$, in the units of step k. On $X^{(3)}$ the extension is not felt, since the pieces used have $Z^{(r')}$ inside the polymer of step $k+1$ (Setting), and $X^{(3)} \subset Z^{(r')}$. By the inclusion of domains in the proof of Proposition 49.4, (49.2), the representation on $X^{(3)}$ has the same real part and budgets $\delta'_1 = L^{-3/2-\epsilon/2} h_k^{1-\epsilon}$, $\delta'_0 = L^{-1/2-\epsilon} h_k^{1-2\epsilon}$. So $\min(\delta_1 - \delta'_1, \delta_0 - \delta'_0) \geq \frac{1}{2} h_k^{1-\epsilon}$ for $L^{1/2+\epsilon} \geq 2$. On the support of the small-field cutoff, ζ is of order $L^d g_k p_{0,k}$ ($\zeta = w$, Section 48). So $\tau \leq C_P L^d g_k p_{0,k} h_k^{-(1-\epsilon)} = o(1)$ as $g_* \to 0$. For $|\xi|_\infty \leq p$ with $C L^d g C_T p \leq \zeta$, the function $s \mapsto f(X; s, Z(s))$ is analytic on

the interior of $\mathbb{D}_{\kappa'}$ and bounded by $4\tau\|\mathcal{E}(X)\|$. Lemma 47.2 applies to it on $\mathbb{D}_{\kappa'}$, without conversion of margins (Remark 47.3). Here Y is the set of boxes of X when X is a union of boxes, as for the polymer grid; in general it is the union of the boxes meeting X.

47.5 The Hessian away from critical backgrounds

Section 37 already defines $\Delta_k = \Delta_k^{(\mathbb{T}_\eta)}$ and $C_k = \Delta_k^{-1}$ at regular backgrounds that need not be critical, with the elimination at the base $U^{(k)}$ (Definition 37.3 and the definition following it). We use this definition at every reference. At complex references, J_U is not bounded uniformly in η. So $\omega_0 = \mathbf{H}_0^T J_U$ is continued in the form

$$\omega_0 = (D\mathbf{H}_0)^T F^s_{U'},$$

which agrees with $\mathbf{H}_0^T J_U$ at real references, since $J_U = D^T F^s_U$ (Lemma 23.2). Here $\mathbf{H}_0$ is $\mathbf{H}$ with potential 0, and $F^s_{U'}$ is continued holomorphically. This is the form used at level (L4) of Section 41.

Proposition 47.9 (Hessian at a general reference)**.** *(a) At real critical references, Δ_k agrees with Lemma 34.1 and Corollary 34.5.*

(b) $|\omega_0|_\infty \le C(\varepsilon+\delta_c)$. The operator W_2 is real symmetric at real references, with $\|W_2\| \le C(\varepsilon+\delta_c)$ and L^2 decay. Only the L^2 theory of Part II is used for it, not the pointwise class $\mathcal{W}(\varepsilon,\gamma)$.

(c) Δ_k has decay (C_L,γ) for some $\gamma > 0$.

(d) For $\varepsilon+\delta_c$ small depending on L, $\operatorname{Re}\Delta_k(Z,\bar{Z}) \ge \frac{1}{8}c_L\|Z\|^2$.

(e) Δ_k is analytic in the reference and covariant under real gauge transformations. At $s=1$ it is represented by the weakened family $\Delta_k(s)$ of level (L4).

Proof. (a) is Lemma 37.2 and Proposition 37.4(a).

(b) At real references this is Proposition 37.4(b) and Lemma 37.2. At complex references use the block L^2 decay of $D\mathbf{H}_0$ and $|F^s_{U'}| \le C(\varepsilon+\delta_c)$ (Proposition 46.7(c)). The bound $\|W_2\| \le C(\varepsilon+\delta_c)$ and the L^2 decay at complex references come from Proposition 37.4(c). There $d^2\mathcal{N}(0)$ is bounded through its plaquette form, with the bounds $|R(H_p)-1| \le C\eta^2(\varepsilon+\delta_c)$ and $|F^s| \le C(\varepsilon+\delta_c)$ of step (v) of the proof of Proposition 46.7, which use the derivative part of δ_c. The L^2 operators at complex references come from Proposition 19.11, through step (i) of the proof of Proposition 46.7.

(c) is Proposition 37.4(b) for $X = \mathbb{T}_\eta$, together with the locality of $\hat{Z}^1$, $\hat{Z}^2$. At complex references the coefficient operators (the averages and the covariant derivatives) change by local operators of size $C_L\delta_c$ (Proposition 19.11, Proposition 46.4(v)), and by Lemma 13.3 the inverses in Δ_k change by $C_L\delta_c$ in the decay norm, which preserves the decay.

(d) This is Lemma 37.5, which does not use criticality, applied at the real part U'_r; it needs $|J_{U'_r}| \le \varepsilon$ (Setting); its conditions $\varepsilon \le \varepsilon_{\mathrm{cc}}$ and those of the proof of Proposition 37.4(c) hold by the smallness of ε (Proposition 46.7, smallness). The lemma covers complex perturbations of size at most c'_L/C_{AB}, and a complex reference with $\delta_c < \delta_P$ is one, since $C_{\mathrm{AB}}\delta_P \le c'_L$ (Proposition 46.7, smallness). As required by Proposition 37.6(b), ω_0 is read there in the form $(D\mathbf{H}_0)^T F^s_{U'}$.

(e) Analyticity and covariance are Proposition 37.4(c). Away from criticality, covariance follows directly from the covariance of $\mathbf{\Gamma}_\pi[W_2]$, $\mathbf{H}_0$, F^s and $\hat{Z}$. The representation is the construction of level (L4) with W_2. □

Remark 47.10 (Source-free step). For a general reference, the step quantities are those defined in this section. The exponent is $-g^{-2}\sum_p v_p(s; Z(s))$, with no terms of order 0, 1 or 2 in Z. The

Gaussian integral is taken through the white-noise representation $Z = gC_k^{1/2}\xi$, with $C_k = \Delta_k^{-1}$ as in Section 47.5; at complex references this representation, not a Gaussian measure, defines it. The determinant densities are those of Proposition 41.7, the Jacobians those of Proposition 47.6, and the composed activities those of Proposition 47.8. At real critical references they reproduce Theorem 34.8, by Propositions 47.5(c) and 47.9(a), and Theorem 45.5(c) for the composed activities. At a non-critical reference the polynomial $\sum_p T_2[\cdots]$ is not $S(U') + \frac{1}{2}\Delta_k$. This is harmless, since the exponent is a remainder by definition.

Remark 47.11 (What the cluster expansion receives). (a) *Pieces.* Lemma 47.2 applied to $V(\square; s)$, to the boxes of $\mathfrak{J}(Z(s))$, and to $f(X; s, Z(s))$ gives pieces indexed by polymers Z with volume-type decay $e^{-(\kappa'-1)|Z\setminus Y|}$. They depend on ξ and U' only on $Z^{(r')}$, are analytic and invariant, and are bounded in terms of $|\xi|$ on the region where $|Z(s)|_\infty \le \frac{1}{2}r_I$ and $|\hat{Z}(Z(s))|_\infty \le \zeta$.

(b) *Constants.* The vacuum terms $\mathcal{N}_k$ and $-\frac{1}{2}\sum_b \ell_k(b;1)$, and the new main term $g^{-2}S(U')$, are passed on unchanged.

(c) *Parameters.* The conditions on the rates are those of Proposition 48.5 and of Theorem 49.8(3), with $\kappa' = \kappa - C\delta$ and the rate κ_n of the activity norm. They supersede the provisional conditions of the blueprint (Appendix H). Further, $M > 32(L+1)$, $c_a \le c_J L^{-d-2}$, and $|\xi|_\infty \le p$ with $gC_T p \le \frac{1}{2}r_I$ and $C_w L^d g C_T p \le \zeta$; these give the two bounds in (a), since $|Z(s)|_\infty \le gC_T|\xi|_\infty$ (Section 47.2) and $|\hat{Z}(Z)|_\infty \le C_w L^d |Z|_\infty$ (Lemma 32.4(c),(d)). In Section 48, $p = p_0$ and $\zeta = w = C_w L^d C_T g p_0$; the first condition is part of (Q1), and the second holds with equality.

(d) *Not done here.* The small-field cutoff and its removal, the Mayer expansion, the bounds in $|\xi|$, the large-field boundary and the reblocking to step $k+1$ belong to P5–P7.

48 The cluster expansion

This section evaluates the fluctuation integral of one small-field step as a sum of polymer activities. It follows Dimock's small-field scheme [15, Sects. 4.3–4.6, App. B]:

- the small-field condition is imposed on the white noise;
- the integrand is localized in the white noise by the weakening parameters;
- the result is integrated by a Mayer and polymer expansion.

The inputs are those of Section 47 and Part IX. The plan and its review are in the blueprint (Appendix I). Notation is as in Section 47: ξ is the white-noise field on the free bonds $\mathfrak{F}$, W the base, U' the reference, and $g = g_k$.

Parameters. Let $p_0 = p_{0,k} \ge 1$. Put

$$w := C_w L^d C_T g p_0, \qquad \zeta := w,$$

with C_T from Section 47.2 and C_w the constant in $|\hat{Z}(Z)|_\infty \le C_w L^d |Z|_\infty$ (Lemma 32.4). We assume:

(Q1) $gC_T p_0 \le \frac{1}{2}r_I$, and $w \le \min(r_2, r_L, \frac{1}{2}c_G, \beta_V, \frac{1}{2}R_P)$;

(Q2) $2(C_*+1)L^{-2}\varepsilon_1' + Cw \le \tilde{h}$, where $\tilde{h}$ is the threshold of the density, together with the other hypotheses of Theorem 34.8 on the thresholds: $\tilde{h} \le a_1$ and $\varepsilon_1' \le a_{\text{step}}$;

(Q3) $\tau < \frac{1}{2}$ in Proposition 47.8, for the domains of the old activities.

In the renormalization group $\tilde{h} = p_k g_k$, $\varepsilon_1' = p_{k+1} g_{k+1}$ and $h = h_k$. Then (Q2) holds if $L^{3/2} \geq 4(C_* + 1) p_{k+1}/p_k$ and $C L^d C_T p_0 \leq \frac{1}{2} p_k$, and $h_k \leq a_1$, $h_{k+1} \leq a_{\text{step}}$ for g_* small. (Q1) holds for g_* small, since $g_k p_{0,k} \to 0$ (Section 49, small-field parameters). (Q3) holds for g_* small, since $\tau \leq C_P L^d g_k p_{0,k} h_k^{-(1-\epsilon)} = C_P L^d g_k^{\epsilon} p_{0,k} p_k^{-(1-\epsilon)} \to 0$ (Section 47.4).

48.1 The small-field step

Definition 48.1 (White-noise cutoff and small-field step)**.** For a configuration V on $\mathbb{T}$ in the support of χ^{w}, let Z_V be the free-bond values of the relative field of the axial representative of Definition 32.3, and $\xi_V = g^{-1} C_k^{-1/2} Z_V$, with C_k at the reference U'. Put

$$\chi_\xi(V) = \prod_{b \in \mathfrak{F}} \mathbf{1}(|\xi_V(b)| \leq p_0),$$

and define the small-field step by $T_{\text{sf}}(\rho) = T(\chi^{\text{w}} \chi_\xi \rho)$.

The function χ_ξ is invariant under the gauge transformations of Definition 32.3, by construction, and it is covariant under global ones. Indeed C_k is covariant (Proposition 47.9(e)), so ξ_V transforms bondwise like Z_V, and $|\xi_V(b)|$ is invariant. So T_{sf} is a renormalization group map on gauge-invariant densities, with a kernel that depends on V' through U'. It is defined for V' small field at step $k+1$, since C_k depends on U' everywhere. This is the small-field step of [15, (206)]; the difference

$$\mathcal{R}_{\text{LF}} = T(\chi^{\text{w}} \rho) - T_{\text{sf}}(\rho) = T(\chi^{\text{w}}(1 - \chi_\xi)\rho)$$

belongs to the large-field stage, which is not treated here. Because C_k is global in U', that stage will need a version of χ_ξ that depends on regions.

Lemma 48.2 (White noise)**.** *Assume (Q1). For $|\xi|_\infty \leq p_0$ and $s \in \mathbb{D}_{\kappa'}$, put $Z(s) = g C_k^{1/2}(s) \xi$. Then $|Z(s)|_\infty \leq g C_T p_0$ and $|\hat{Z}(Z(s))|_\infty \leq w$. In particular, on the support of χ_ξ:*

- *$\chi^{\text{w}} = 1$;*
- *all the step quantities of Section 47 are defined at $Z(s)$, for every $s \in \mathbb{D}_{\kappa'}$;*
- *the composed configurations lie in the domains of the old activities, by (Q3).*

Proof. $|C_k^{1/2}(s)\xi|_\infty \leq C_T |\xi|_\infty$ (Section 47.2), and $C_k^{1/2}(1) = C_k^{1/2}$ (Corollary 41.6), and $|\hat{Z}(Z)|_\infty \leq C_w L^d |Z|_\infty$ (Lemma 32.4(c),(d)). At $s = 1$, $\hat{Z}(Z(1))$ is the axial relative field, so $\chi^{\text{w}} = 1$. The other statements follow from (Q1), (Q3) and the definitions of r_I and ζ. □

Proposition 48.3 (The small-field step)**.** *Assume (Q1)–(Q3), and let ρ be as in Theorem 34.8. Then*

$$T_{\text{sf}}(\rho)(V') = \mathcal{N}_k N_\chi \exp\Big(-\frac{1}{g^2} S(U') - \frac{1}{2} \sum_{b \in \mathfrak{F}} \ell_k(b) + \sum_c \log \mathfrak{j}_c + \mathfrak{E}(U') \Big) \int d\mu^*(\xi)\, e^{\delta \mathcal{E}(\xi)},$$

where $d\mu^* = N_\chi^{-1} \chi_\xi \, d\mu_I$ is the product over $b \in \mathfrak{F}$ of standard Gaussian measures on $\mathfrak{g}$ restricted to $|\xi(b)| \leq p_0$. Further, $N_\chi = \exp(-\varepsilon^0 |\mathfrak{F}|)$ with $\varepsilon^0 = O(p_0^{\dim \mathfrak{g} - 2} e^{-p_0^2/2})$, and

$$\delta \mathcal{E}(\xi) = -\frac{1}{g^2} \mathcal{V}(Z) + \mathfrak{J}(Z) + (\mathfrak{E}(e^{\mathrm{i}\hat{Z}(Z)} W) - \mathfrak{E}(U')), \qquad Z = g C_k^{1/2} \xi.$$

Proof. By (Q2), the hypothesis of Theorem 34.8 holds with the cutoff w. This is the analogue of [15, Lemma 15]: on the support of χ^{w} the configuration $e^{\mathrm{i}\hat{Z}} W$ has curvature at most $2(C_* + 1) L^{-2} \varepsilon_1' + C w \leq \tilde{h}$, as in the proof of Theorem 34.8. The factor χ_ξ is a function of Z

on $\mathfrak{F}$. So the proof of Theorem 34.8 applies to $\chi^{\mathrm{w}}\chi_\xi\rho$, with χ_ξ carried along in the integrand. Substitute $Y = C_k^{1/2}\xi$, so that $d\mu_{C_k}(Y)$ becomes $d\mu_I(\xi)$. By Lemma 48.2, $\chi^{\mathrm{w}}(\hat{Z}(gY)) = 1$ on the support of χ_ξ. Pull out $\mathfrak{E}(U')$ [15, (211)–(212)]. The value of N_χ is the Gaussian mass of $\{|\xi(b)| \le p_0\}$ per bond, for $\mathfrak{g}$-valued $\xi(b)$. □

At references that are not real and critical, or outside the small-field region, the right-hand side is the definition of the step (Remark 47.10).

48.2 Localization in the white noise

Write the integrand with the weakening parameters:

$$\delta\mathcal{E}(\xi; s) = -\sum_{\square} V(\square; s) + \sum_{\square} \mathfrak{J}_\square(Z(s)) + \sum_X f(X; s, Z(s)),$$

with $V(\square; s)$ as in Section 47.2, $\mathfrak{J}_\square$ the terms of $\mathfrak{J}$ grouped by the box of their bond, and f as in Proposition 47.8. At $s = 1$ this is $\delta\mathcal{E}(\xi)$, by Corollary 41.6 ($C_k^{1/2}(1) = C_k^{1/2}$), Proposition 47.5(c) and Theorem 45.5(c).

Definition 48.4 (Decay functional)**.** A *decay functional* is a function $\mathsf{d} \ge 0$ on polymers with constants $c_\flat \in (0, 1]$, $c_\sharp \ge 1$, $c_\natural \ge 1$ such that

$$c_\flat|Z| - 1 \le \mathsf{d}(Z) \le c_\sharp(|Z| - 1), \qquad \mathsf{d}(Z_1 \cup Z_2) \le \mathsf{d}(Z_1) + \mathsf{d}(Z_2) + c_\natural$$

for touching polymers Z_1, Z_2. Examples are $|Z| - 1$ (with $c_\flat = c_\sharp = c_\natural = 1$), and Dimock's tree length d_M [15, (119)], with $c_\flat = 3^{-d}$ by [15, (122)], $c_\sharp = \sqrt{d}$ and $c_\natural = 2\sqrt{d}$. Adding the boxes of $Z \setminus X$ one at a time, each touching the previous union, gives $\mathsf{d}(Z) \le \mathsf{d}(X) + c_\natural|Z \setminus X|$ for $X \subset Z$.

The old activities are indexed by connected unions X of boxes. Each $\mathcal{E}_k(X, \cdot)$ depends on the configuration on $X^{(r_E)}$, where r_E is the margin of dependence, and is a graded activity at $X^{(r_E)}$ in the sense of Proposition 47.8, with its domain controlled on $X^{(r_E+3)}$. Part (b) of Proposition 47.8 needs $r_E \le r$. We assume $r_E + 3 \le r$ (for $r_E = 0$ this is $r \ge 3$, Section 47, Setting), which keeps $X^{(r_E+3)}$ inside $Z^{(r')}$ for $X \subset Z$, where the extension of the reference is not felt; in Section 49, $r_E = 0$. The paving of Section 40 is taken to be the polymer grid, which needs $M \mid n$. They are assumed to satisfy $\sup|\mathcal{E}_k(X, \cdot)| \le \|\mathcal{E}_k\| e^{-\kappa_n \mathsf{d}(X)}$, the supremum over the graded domain at $X^{(r_E)}$ ($\mathcal{U}_k(X)$ for $r_E = 0$), for a decay functional d and the rate κ_n of the activity norm. The norm itself is fixed in P6.

Proposition 48.5 (Localized pieces)**.** *Assume (Q1)–(Q3), $\kappa' - 1 \ge c_\natural\kappa_n + \log(1/c_\flat)$, and the decay of the old activities stated above. On the support of χ_ξ,*

$$\delta\mathcal{E}(\xi) = \sum_Z E_{\mathrm{p}}(Z) + \sum_Z E_{\mathrm{m}}(Z),$$

the sums running over polymers. Each piece depends on ξ and U' only on the bonds of $Z^{(r')}$, and on the old activities only through $\mathcal{E}_k(X, \cdot)$ with $X \subset Z$. Each piece is invariant under real gauge transformations, and real at real references for real ξ. The pure pieces are analytic in the reference on all complex references of Section 46 with $\delta_c < \delta_P$ whose real part is (M, ε)-regular with $\varepsilon \le \varepsilon_{\max}$. The mixed pieces $E_{\mathrm{m}}(Z)$ are analytic in the reference on the set of references at which the hypothesis of Proposition 47.8 on U' holds for every $X \subset Z$. The pieces satisfy, uniformly for $|\xi|_\infty \le p_0$,

$$|E_{\mathrm{p}}(Z)| \le \epsilon_{\mathrm{p}}\, e^{-(\kappa'-2)|Z|}, \qquad |E_{\mathrm{m}}(Z)| \le \epsilon_{\mathrm{m}}\, e^{-(\kappa_n-1)\mathsf{d}(Z)},$$

with $\epsilon_{\mathrm{p}} = C_P g p_0^3$ and $\epsilon_{\mathrm{m}} = C_P \tau \|\mathcal{E}_k\|$.

Proof. Apply Lemma 47.2 on $\mathbb{D}_{\kappa'}$ to each term. Its hypotheses hold by Lemma 48.2, Propositions 47.5(a),(d), 47.6(a) and 47.8(a),(b), and Lemma 40.3(b) and Corollary 41.6 for $C_k^{1/2}(s)$. These give analyticity in the reference on the domains stated, jointly with s.

- *Interaction.* With $Y = \square$, the bound $F \le C_P g^{-2} M^d (g C_T p_0 / r_I)^3 = C_P g p_0^3$ holds (Section 47.2).
- *Jacobians.* $F \le C_P(L^{2d+1} g p_0 + g^2 p_0^2) \le C_P g p_0^3$ (Theorem 34.8(e)), since $p_0 \ge 1$ and C_P may depend on L.
- *Pure pieces.* Put $E_{\mathrm{p}}(Z) = \sum_{\square \subset Z} (f_{\mathrm{loc}}(\square, Z)$ of these terms). Then $|E_{\mathrm{p}}(Z)| \le |Z| F e^{\kappa'-1} e^{-(\kappa'-1)|Z|}$, and $|Z| e^{-|Z|} \le 1$ gives the bound after adjusting C_P by $e^{\kappa'}$.
- *Mixed pieces.* Here $Y = X$. Indeed, $\mathcal{E}_k(X, \cdot)$ depends on the configuration on $X^{(r_E)}$ with $r_E \le r$, and by Theorem 45.5(d) and Lemma 40.3(b), A_* on $(Z_0')^{(r)}$ depends on s only through the boxes of Z_0'. So the hypothesis of Lemma 47.2 holds with Y the set of boxes of X. Proposition 47.8 gives $F \le 4\tau \|\mathcal{E}_k\| e^{-\kappa_n \mathsf{d}(X)}$, so

$$|f_{\mathrm{loc}}(X, Z)| \le 4\tau \|\mathcal{E}_k\| e^{-\kappa_n \mathsf{d}(X) - (\kappa'-1)|Z \setminus X|}.$$

  By Definition 48.4, $\mathsf{d}(Z) \le \mathsf{d}(X) + c_\natural |Z \setminus X|$. So

$$|f_{\mathrm{loc}}(X, Z)| \le 4\tau \|\mathcal{E}_k\| e^{-\kappa_n \mathsf{d}(Z)} e^{-(\kappa'-1-c_\natural \kappa_n)|Z \setminus X|}.$$

  Summing over $X \subset Z$, which over-counts the connected X harmlessly,

$$\sum_{X \subset Z} e^{-(\kappa'-1-c_\natural \kappa_n)|Z \setminus X|} \le \big(1 + e^{-(\kappa'-1-c_\natural \kappa_n)}\big)^{|Z|} \le e^{c_\flat |Z|} \le e^{\mathsf{d}(Z)+1}.$$

  This defines $E_{\mathrm{m}}(Z) = \sum_{X \subset Z} f_{\mathrm{loc}}(X, Z)$, with the bound stated.

Data dependence, analyticity and invariance are Lemma 47.2(a),(c). Reality: at a real reference and for real ξ and real s, all the functions localized are real (Corollary 41.6 for $C_k^{1/2}(s)$, and the realness of s_p, $\hat{Z}$, the Jacobians and the old activities). The localization takes real derivatives and integrals over $t \in [0,1]^{Z \setminus Y}$, so the pieces are real. Margins are not converted (Remark 47.3). □

48.3 The Mayer and polymer expansion

Two polymers are *incompatible* if they touch, i.e. if a box of one shares a point with a box of the other. Non-touching polymers are at distance at least $M/4 > 2r'$ (Section 47, Setting). So pieces of non-touching polymers depend on disjoint sets of free bonds, and are independent under the product measure $d\mu^*$.

We use the cluster expansion of Kotecký and Preiss [18], in the following form. Let $\mathcal{K}$ be a set of objects γ ("gas polymers"), each with a support $\operatorname{supp}\gamma$, a polymer, and a complex activity $k(\gamma)$. Call two objects compatible if their supports do not touch. Then

$$\sum_{\{\gamma_1, \dots, \gamma_n\} \text{ compatible}} \prod_i k(\gamma_i) = \exp\Big(\sum_{\mathcal{C}} \phi^T(\mathcal{C}) \prod_{\gamma \in \mathcal{C}} k(\gamma) \Big),$$

the sum running over the clusters $\mathcal{C}$, i.e. the finite multisets of objects whose incompatibility graph is connected, with the Ursell coefficients ϕ^T. This holds with absolute convergence if there are functions $a, b \ge 0$ on $\mathcal{K}$ with

$$\sum_{\gamma' \not\sim \gamma} |k(\gamma')| e^{a(\gamma') + b(\gamma')} \le a(\gamma) \qquad \text{for all } \gamma. \tag{48.1}$$

In that case $\sum_{\mathcal{C} \not\sim \gamma} |\phi^T(\mathcal{C}) \prod_{\gamma' \in \mathcal{C}} k(\gamma')| e^{\sum_{\gamma' \in \mathcal{C}} b(\gamma')} \le a(\gamma)$, the sum running over the clusters containing an object incompatible with γ [18].

Theorem 48.6 (Polymer expansion of the fluctuation integral)**.** *Let* d *be a decay functional. There are* C_d*,* A *and* $\epsilon_* > 0$*, depending only on* d *and the constants of* d*, with the following property. Let* $E(Z)$*, indexed by polymers, be complex functions of* ξ *and of further parameters. Suppose each* $E(Z)$ *depends on* ξ *only on the free bonds of* $Z^{(r')}$*, and on the other spatial data, such as the reference, only on* $Z^{(r')}$*; global parameters, such as the coupling or an interpolation parameter, are allowed. Suppose also*

$$|E(Z)| \le \epsilon\, e^{-\kappa(\mathsf{d}(Z)+c_\natural)} \qquad \text{for } |\xi|_\infty \le p_0,$$

with $\kappa \ge C_d$ *and* $\epsilon \le \epsilon_*$*. Then*

$$\int \exp\Big(\sum_Z E(Z)\Big) d\mu^* = \exp\Big(\sum_Y E^\#(Y)\Big),$$

where Y *runs over polymers, and:*

(a) $E^\#(Y)$ *depends only on the functions* $E(Z)$ *with* $Z \subset Y$*, and on the data only on* $Y^{(r')}$*;*

(b) $\sum_{Y \supset \square} |E^\#(Y)| e^{(\kappa - C_d)(\mathsf{d}(Y)+c_\natural)} \le A\epsilon$ *for every box* $\square$*;*

(c) if the $E(Z)$ *are analytic in a parameter, uniformly, so is* $E^\#(Y)$*; if they are invariant under real gauge transformations, so is* $E^\#(Y)$*, since* $d\mu^*$ *is invariant under the bondwise orthogonal action* $\xi(b) \mapsto R(u(b_-))\xi(b)$*;*

(d) if the $E(Z)$ *are even under a joint transformation of a parameter and* $\xi \mapsto -\xi$*, then* $E^\#(Y)$ *is even in the parameter;*

(e) if the $E(Z)$ *are real for real* ξ *at some value of the parameters, so is* $E^\#(Y)$ *at that value.*

Proof. Mayer expansion. Put $K(Z) = e^{E(Z)} - 1$, so $|K(Z)| \le 2|E(Z)|$ for $\epsilon \le 1$. Expand $\prod_Z (1 + K(Z))$ as a sum over finite sets of distinct polymers. Group each set into its connected components for touching, the clusters of polymers Γ, with supports $\operatorname{supp}\Gamma = \bigcup_{Z\in\Gamma} Z$, which are polymers. Distinct components do not touch, so the integral factorizes:

$$\int \prod_Z (1 + K(Z))\, d\mu^* = \sum_{\{\Gamma_1,\dots,\Gamma_n\} \text{ compatible}} \prod_i k(\Gamma_i), \qquad k(\Gamma) = \int \prod_{Z\in\Gamma} K(Z)\, d\mu^*.$$

Then $|k(\Gamma)| \le \prod_{Z\in\Gamma} 2\epsilon e^{-\kappa(\mathsf{d}(Z)+c_\natural)}$.

A tree count. For weights $u(Z) \ge 0$ let $S(\square)$ be the sum of $\prod_{Z\in\Gamma} u(Z)$ over the clusters Γ with a polymer containing $\square$, and $S_* = \sup_\square S(\square)$. Remove from such a Γ a polymer $Z_0 \ni \square$. The rest splits into touching-connected families, each containing a polymer that contains a box $\square'$ touching Z_0. Two such families cannot contain the same $\square'$, since they would touch. There are at most $3^d|Z_0|$ such boxes. Hence $S(\square) \le \sum_{Z_0 \ni \square} u(Z_0)(1 + S_*)^{3^d|Z_0|}$. Restrict first to polymers of size at most N, so that all sums are finite, and let $N \to \infty$. Put $\eta := \sup_\square \sum_{Z\ni\square} u(Z) e^{3^d|Z|}$, and scale all weights by $\lambda \in [0,1]$. Then S_* is a continuous function of λ that vanishes at $\lambda = 0$. While $S_* \le 1$, the inequality gives $S_* \le \lambda\eta$, since $(1 + S_*)^{3^d|Z_0|} \le e^{3^d|Z_0|}$. So S_* never exceeds $\lambda\eta$, and $S_* \le \eta$ whenever $\eta \le 1$.

Polymer expansion. Apply the Kotecký–Preiss theorem to the objects Γ, with $a(\Gamma) = A\epsilon|\operatorname{supp}\Gamma|$ and $b(\Gamma) = \beta(\mathsf{d}(\operatorname{supp}\Gamma) + c_\natural)$, $\beta = \kappa - C_d$. Along a spanning tree of the touching graph of Γ, the subadditivity of d gives

$$\mathsf{d}(\operatorname{supp}\Gamma) + c_\natural \le \sum_{Z\in\Gamma} (\mathsf{d}(Z) + c_\natural), \qquad |\operatorname{supp}\Gamma| \le \sum_{Z\in\Gamma} |Z| \le c_\flat^{-1} \sum_{Z\in\Gamma} (\mathsf{d}(Z) + c_\natural).$$

Objects Γ' incompatible with Γ have a polymer containing a box that touches $\operatorname{supp}\Gamma$, and there are at most $3^d|\operatorname{supp}\Gamma|$ such boxes. So the left side of (48.1) is at most $3^d|\operatorname{supp}\Gamma|\, S_*$, with

$$u(Z) = 2\epsilon\, e^{-(\kappa - \beta - A\epsilon c_\flat^{-1})(\mathsf{d}(Z)+c_\natural)} \le 2\epsilon\, e^{-(C_d - c_\flat^{-1})c_\flat|Z|},$$

for $A\epsilon \le 1$, using $\mathsf{d}(Z) + c_\natural \ge c_\flat |Z|$. The number of polymers $Z \ni \square$ with $|Z| = m$ is at most C_0^m, with C_0 depending on d. Choose $C_d = c_\flat^{-1}(\log C_0 + 3^d + 2)$. Then $\eta \le C_1\epsilon$, so $S_* \le C_1\epsilon$, and (48.1) holds with $A = 3^d C_1$ for $\epsilon \le \epsilon_*$.

The output. Group the clusters $\mathcal{C}$ of objects by $Y = \bigcup_{\Gamma\in\mathcal{C}} \operatorname{supp}\Gamma$, a polymer since $\mathcal{C}$ is connected. Put $E^{\#}(Y) = \sum_{\mathcal{C}:\,\cup\operatorname{supp}=Y} \phi^T(\mathcal{C}) \prod_{\Gamma\in\mathcal{C}} k(\Gamma)$. Then (a) holds, since each $k(\Gamma)$ involves only the $E(Z)$ with $Z \in \Gamma$, and $\bigcup_{Z\in\Gamma} Z^{(r')} \subset Y^{(r')}$. For (b), subadditivity gives $e^{\beta(\mathsf{d}(Y)+c_\natural)} \le \prod_{\Gamma\in\mathcal{C}} e^{b(\Gamma)}$. The clusters with $Y \supset \square$ are incompatible with the single-box object $\{\square\}$, a legitimate object since a box is a polymer. So the Kotecký–Preiss bound gives $\sum_{Y\supset\square} |E^{\#}(Y)| e^{\beta(\mathsf{d}(Y)+c_\natural)} \le a(\{\square\}) = A\epsilon$. For (c), all sums converge uniformly, and $d\mu^*$ is invariant under the bondwise orthogonal action with the symmetric cutoff. For (d), $d\mu^*$ is even, so each $k(\Gamma)$ is even in the parameter, and the combinatorics do not depend on ξ. For (e), $d\mu^*$ is a real measure, so each $k(\Gamma)$ is real, and the Ursell coefficients ϕ^T are real; the series converges absolutely. □

Corollary 48.7 (Two decay classes)**.** *In the situation of Proposition 48.5, put* $c'_\sharp = \max(c_\sharp, c_\natural)$. *Let* $\kappa_n - 1 \ge C_d$, $(\kappa'-2)/c'_\sharp \ge \kappa_n - 1$, $\epsilon_{\mathrm{p}} \le \frac12\epsilon_*$ *and* $\epsilon_{\mathrm{m}} e^{c_\natural(\kappa_n-1)} \le \frac14\epsilon_*$. *Then* $\int e^{\delta\mathcal{E}} d\mu^* = \exp(\sum_Y (E^{\#}_{\mathrm{p}}(Y) + E^{\#}_{\mathrm{m}}(Y)))$, *where:*

- $E^{\#}_{\mathrm{p}}$ *is the output of Theorem 48.6 for the pure pieces alone. It satisfies*

$$\sum_{Y\supset\square} |E^{\#}_{\mathrm{p}}(Y)| e^{((\kappa'-2)/c'_\sharp - C_d)(\mathsf{d}(Y)+c_\natural)} \le A\epsilon_{\mathrm{p}}.$$

- $E^{\#}_{\mathrm{m}} = E^{\#} - E^{\#}_{\mathrm{p}}$ *satisfies*

$$\sum_{Y\supset\square} |E^{\#}_{\mathrm{m}}(Y)| e^{(\kappa_n - 1 - C_d)(\mathsf{d}(Y)+c_\natural)} \le 4A\,\epsilon_{\mathrm{m}} e^{c_\natural(\kappa_n-1)}.$$

Proof. Since $\mathsf{d}(Z) + c_\natural \le c'_\sharp |Z|$, the pure pieces satisfy $|E_{\mathrm{p}}(Z)| \le \epsilon_{\mathrm{p}} e^{-((\kappa'-2)/c'_\sharp)(\mathsf{d}(Z)+c_\natural)}$. Theorem 48.6 gives the first bound. The mixed pieces satisfy $|E_{\mathrm{m}}(Z)| \le \epsilon_{\mathrm{m}} e^{c_\natural(\kappa_n-1)} e^{-(\kappa_n-1)(\mathsf{d}(Z)+c_\natural)}$. For the full system at the rate $\kappa_n - 1$, replace E_{m} by λE_{m} with complex $|\lambda| \le R = \epsilon_*/(2\epsilon_{\mathrm{m}} e^{c_\natural(\kappa_n-1)}) \ge 2$. Then $\epsilon = \epsilon_{\mathrm{p}} + |\lambda|\epsilon_{\mathrm{m}} e^{c_\natural(\kappa_n-1)} \le \epsilon_*$, and Theorem 48.6 applies uniformly in λ; $E^{\#}(Y;\lambda)$ is analytic in λ, and $E^{\#}(Y;0) = E^{\#}_{\mathrm{p}}(Y)$. Cauchy's formula on $|\lambda| = R \ge 2$ gives

$$|E^{\#}(Y;1) - E^{\#}(Y;0)| \le \frac{2}{R} \cdot \frac{1}{2\pi} \oint |E^{\#}(Y;\lambda)| \frac{|d\lambda|}{|\lambda|}.$$

Summing over Y under the integral, with $A\epsilon_*$ from Theorem 48.6(b), gives the second bound. □

The rate of the mixed class is the rate κ_n of the activity norm minus $1 + C_d$. It is recovered in P6 by the reblocking gain $d_M(X) \ge L\, d_{LM}(\bar X) - C|\bar X|$ of Dimock's tree length (blueprint, Appendix F). This needs $\kappa_n \ge C(C_d + \kappa_0 + 1)$.

48.4 The order g and the size of the pure sources

Replace g by a complex parameter t in the fluctuation only: $Z = tC_k^{1/2}(s)\xi$, and t^{-2} in front of the interaction pieces, as in the blueprint, Appendix G. The prefactor of Proposition 48.3 and $\mathfrak{E}(U')$ do not depend on t.

Proposition 48.8 (Pure sources)**.** *Let* $r_t = \min\left(c\, r_I^3 (C_P M^d C_T^3 p_0^3)^{-1},\ r_I (2C_T p_0)^{-1}\right)$, *with* c *small, the constant* C_P *including the factors* $e^{\kappa'}$. *For* $|t| \le r_t$ *the pure pieces, with* t *in place of*

g, satisfy the hypotheses of Theorem 48.6 with $\epsilon \le \frac{1}{2}\epsilon_*$, as in the proof of Corollary 48.7. The output $E^{\#}_{\mathrm{p}}(Y;t)$ is analytic and even in t, and $E^{\#}_{\mathrm{p}}(Y;0) = 0$. Hence, for $2g \le r_t$,

$$\sum_{Y \supset \square} |E^{\#}_{\mathrm{p}}(Y;g)| e^{((\kappa'-2)/c'_\sharp - C_d)(\mathrm{d}(Y)+c_\natural)} \le C_P g^2 p_0^6.$$

Proof. For $|t| \le r_t$: $|t| C_T p_0 \le \frac{1}{2} r_I$, by the second entry of the minimum. The pure pieces only need $|Z|_\infty \le \frac{1}{2} r_I$: Propositions 47.5(a),(b) and 47.6(a). The parts of (Q1) that involve w, Lemma 48.2 and the identification of Proposition 47.5(c) are used only at $t = g$, where they hold by (Q1). So the bounds of Proposition 48.5 for the pure pieces hold with g replaced by $|t|$. That is, $\epsilon_{\mathrm{p}}(t) \le C_P |t| p_0^3$, plus the Jacobian term $C_P |t| p_0$. This is at most $\frac{1}{2}\epsilon_*$ for c small. Analyticity in t follows from Theorem 48.6(c). Every pure piece depends on (t,ξ) only through t^{-2} and $t C_k^{1/2}(s)\xi$: the interaction pieces are Taylor remainders of order three in Z for every s (Definition 47.4), and $\mathfrak{J}$ depends on Z. Since $t^{-2} v_p(s;tY)$ is analytic at $t = 0$, being a remainder of order three, the pieces are analytic and even under $(t,\xi) \mapsto (-t,-\xi)$. The localization takes s-derivatives at fixed ξ. So Theorem 48.6(d) applies, and $E^{\#}_{\mathrm{p}}$ is even in t. At $t = 0$ every pure piece vanishes, so $E^{\#}_{\mathrm{p}}(Y;0) = 0$. So $E^{\#}_{\mathrm{p}}(Y;t)/t^2$ is analytic on $|t| \le r_t$. Cauchy's formula on $|t| = r_t$ gives, for $2g \le r_t$, $|E^{\#}_{\mathrm{p}}(Y;g)| \le 2(g/r_t)^2 \frac{1}{2\pi} \oint |E^{\#}_{\mathrm{p}}(Y;t)|\, |dt|/r_t$. Summing over Y under the integral, with Theorem 48.6(b) and $r_t^{-2} \le C_P p_0^6$ (both entries of the minimum, $p_0 \ge 1$), gives the bound. In particular there is no term of order g. □

48.5 The step in polymer form

Theorem 48.9 (Small-field step in polymer form)**.** *Let G be semisimple, and let ρ be as in Theorem 34.8. Assume (Q1)–(Q3), $2g \le r_t$, the decay of the old activities, and the hypotheses of Proposition 48.5 and Corollary 48.7. Then, at real critical references with V' small field,*

$$T_{\mathrm{sf}}(\rho)(V') = \mathcal{N}_k N_\chi \exp\Big(-\frac{1}{g^2} S(U') - \frac{1}{2}\sum_b \ell_k(b;1) + \sum_X \mathfrak{e}(X,U') + \sum_c \log \mathfrak{j}_c + \mathfrak{E}(U') + \sum_Y \big(E^{\#}_{\mathrm{p}}(Y,U') + E^{\#}_{\mathrm{m}}(Y,U')\big)\Big),$$

with $\mathfrak{e}$ from Proposition 41.7 and $\ell_k(b;1)$ the density of Proposition 34.7 at $U' = 1$. At other references the right-hand side is the definition of the step (Remark 47.10). Every term depends on U' only near its polymer, is invariant under real gauge transformations, and is real at real references (Theorem 48.6(e)). The terms $\mathfrak{e}(X)$, $\log \mathfrak{j}_c$ and $E^{\#}_{\mathrm{p}}$ are analytic in the reference on the complex references of Section 46 with $\delta_c < \delta_P$ whose real part is (M,ε)-regular with $\varepsilon \le \varepsilon_{\max}$ ($\mathfrak{e}$ and $\log \mathfrak{j}_c$ on the graded domains of Propositions 41.7 and 47.6). The terms $E^{\#}_{\mathrm{m}}(Y)$ and $\mathfrak{E}$ are analytic on the references at which Proposition 47.8 applies to the old polymers inside Y, respectively to all old polymers; Remark 49.7 shows that this suffices for the flow. Moreover:

- *$E^{\#}_{\mathrm{p}}$ is $O(g^2 p_0^6)$, with decay rate $(\kappa'-2)/c'_\sharp - C_d$ (Proposition 48.8);*
- *$E^{\#}_{\mathrm{m}}$ is $O(\tau \|\mathcal{E}_k\| e^{c_\natural \kappa_n})$, with decay rate $\kappa_n - 1 - C_d$ (Corollary 48.7);*
- *$\mathfrak{e}(X)$ and $\log \mathfrak{j}_c$ are $O(h^{2-2\epsilon})$ on $D\mathfrak{F}''$ (Propositions 41.7, 47.6), for polymers of diameter at most $n/2 - 8$. Larger polymers give $e^{-cn/M}$.*

Proof. Combine Proposition 48.3, Proposition 48.5, Corollary 48.7, Proposition 48.8, and Proposition 41.7 for the determinant. The locality in U' follows from the data dependence on $Y^{(r')}$ in Theorem 48.6(a). □

Remark 48.10 (Normalization and what remains). (a) *Normalization.* The new activities are normalized by subtracting their values at $U' = 1$. The constant $\sum_Y E^{\#}(Y, 1)$ goes into the vacuum energy, and $\log \mathcal{N}_k$, $\log N_\chi$ and $-\frac{1}{2}\sum_b \ell_k(b; 1)$ into the explicit normalization $\mathcal{Z}_{k+1}$ (Section 50.3). Only bounds per grid cube are needed for the vacuum energy; no periodicity and no Gaussian reference are used.

(b) *The old activities* at U', $\mathfrak{E}(U') = \sum_X \mathcal{E}_k(X, U')$, depend on U' on the margins of their own domains. Together with the reblocking and rescaling, they form the linear part of the flow (P6).

(c) *Not treated here.*

- The reblocking to polymers of step $k+1$, and the comparison of the rates with the activity norm. This includes the recovery of the loss C_d for the mixed class (P6, P7).
- The large-field remainder $\mathcal{R}_{\mathrm{LF}}$.
- The decay assumed for the old activities, which is part of the induction (P6, P7).
- The wrap-around polymers.

49 The activity norm and one step of the flow

This section defines the activity norm on the domain D3‴ and proves the bound for one step of the flow. The linear part follows from Lemma 38.5, and the sources from Sections 47 and 48. The plan and its review are in the blueprint (Appendix J). The iteration over the steps, the vacuum energy, the last steps and the target theorem are in Section 50.

Small-field parameters. We take $d = 3$. Let $g_{k+1}^2 = Lg_k^2$, and let $p_k = p(g_k)$, $p_{0,k} = p_0(g_k)$ for functions $p, p_0 \geq 1$ with the following properties: p is decreasing, so $p_{k+1} \leq p_k$; $p(g)/p(L^{1/2}g) \to 1$ as $g \to 0$, for fixed L, so $p_k/p_{k+1} \to 1$ uniformly; $p_0(g)/p(g) \to 0$; and $g^\beta p(g) \to 0$ as $g \to 0$ for every $\beta > 0$, so also $g^\beta p_0(g) \to 0$. For instance powers of $1 + \log g^{-1}$, as in [15]. Put $h_k = p_k g_k$, and fix $0 < \epsilon < \frac{1}{4}$. All statements are for $g_k \leq g_*$, with g_* small depending on all other parameters.

49.1 Polymers, domains and the norm

At step k, in the units of step k, the *polymer grid* is the paving of $\mathbb{T}_\eta$ by cubes of side M (the paving of Section 40 with all intervals of length M, which needs $M \mid n_k$). A *polymer* is a $*$-connected union of grid cubes. For a polymer X, let $d_M(X)$ be the length, in units of M, of the shortest tree in the continuum that meets every cube of X [15, (119)], and $|X|_M$ the number of its cubes. Then [15, (122), (123)]

$$d_M(X) \leq \sqrt{d}(|X|_M - 1), \qquad |X|_M \leq 3^d(1 + d_M(X)), \qquad \sum_{X \supset \square} e^{-\kappa_0 d_M(X)} \leq K_0, \tag{49.1}$$

with $\kappa_0 \geq 1$, K_0 depending only on d. (The first inequality replaces $d_M \leq |X|_M$ of [15], which is stated there for connected polymers.) There, polymers are connected through common faces [15, Sect. 3.1], and the second and the third inequality hold for $*$-connected polymers as well. For the second, let T be a tree that meets every cube of X, a finite union of segments, of length ℓ in units of M, and lift it to $\mathbb{R}^d$. Every cube of X contains a point t of T, so, in units of M, it is the image of a subset of $t + [-1, 1]^d$, and the union of the cubes lies in the image of $T + [-1, 1]^d$. A tree is obtained from a point by adding segments $[p, p + su]$, with p a point of the tree already built, $s > 0$ and $|u| = 1$. Each such segment enlarges $T + [-1, 1]^d$ by at

most s times the $(d-1)$-volume $2^{d-1}|u|_1 \le 2^{d-1}\sqrt{d}$ of the projection of $[-1,1]^d$ along u. So $T+[-1,1]^d$ has volume at most $2^d+2^{d-1}\sqrt{d}\,\ell$, and the projection to the torus does not increase volume. The cubes have volume 1 and disjoint interiors, so $|X|_M \le 2^d + 2^{d-1}\sqrt{d}\,\ell \le 3^d(1+\ell)$; the infimum over the trees gives the second inequality. In the proof of the third [15, Lemma 25], the polymers $X \supset \square$ with $|X|_M = n$ are counted by paths of length $2(n-1)$ along a spanning tree of the adjacency graph. Under $*$-adjacency a cube has 3^d-1 neighbours, so there are at most $(3^d-1)^{2(n-1)}$ such paths. The proof then uses the second inequality, in the form $d_M(X) \ge 3^{-d}|X|_M - 1$. So the third inequality holds for $*$-connected polymers, with κ_0 and K_0 depending only on d. So d_M is a decay functional (Definition 48.4), with the constants given there.

Definition 49.1 (Activities and norm)**.** The activity domain of step k at a polymer X is $\mathcal{U}_k(X) = \mathcal{U}(X; h_k^{1-\epsilon/2}, h_k^{1-\epsilon}, h_k^{1-2\epsilon})$ (Definition 38.3). An *activity of step k* is a family $\mathcal{E} = \{\mathcal{E}(X,\cdot)\}$, indexed by polymers, of graded activities at X on $\mathcal{U}_k(X)$. It is *normalized* if $\mathcal{E}(X,1) = 0$ for all X, wrapping ones included. For $\kappa > 0$ put

$$\|\mathcal{E}(X)\|_k = \sup_{\mathcal{U}_k(X)} |\mathcal{E}(X,\cdot)|, \qquad \|\mathcal{E}\|_{k,\kappa} = \sup_X \|\mathcal{E}(X)\|_k e^{\kappa d_M(X)}.$$

The domain is monotone, $\mathcal{U}_k(Y) \subset \mathcal{U}_k(X)$ for $X \subset Y$. Each $\mathcal{E}(X,\cdot)$ depends on the configuration only on X: the margin of dependence is 0, while the domain has the margin 3 of Definition 38.3.

Hypotheses on the activities of step k. With the rate κ_n of the norm:

(H1) $\mathcal{E}_k$ is an activity of step k in the sense of Definition 49.1;

(H2) each $\mathcal{E}_k(X,\cdot)$ depends on the configuration only on X;

(H3) $\|\mathcal{E}_k\|_{k,\kappa_n} < \infty$;

(H4) $\mathcal{E}_k$ is normalized, and invariant under real gauge transformations;

(H5) its polymers are polymers of the grid of step k;

(H6) $\mathcal{E}_k(X,U)$ is real for real U.

Torus hypotheses. Let n_k be the side, in the units of step k, of the torus of step k. The configurations of step k live on its unit lattice $\mathbb{T}^{(k)}$, and the background fields on its fine lattice $\mathbb{T}_\eta$, which has the same side in these units but spacing $\eta < 1$; the two lattices are different sets. This is the n of Parts VI, VIII and IX. In Part VII, n denotes the side of $\mathbb{T}'$, which is $n_{k+1} = n_k/L$. The torus side n_k satisfies:

(T0) $n_k \ge n_*(L, M, \rho)$, the largest of the torus conditions of the inputs. These are:

- n_0 of Lemma 38.5 and Corollary 30.4, for n_k; Lemma 38.5 needs only that its cube Λ does not wrap around $\mathbb{T}^{(k)}$, i.e. $2R < n_k$;
- n_δ of Proposition 31.5, for n_{k+1};
- $80M'$ with $M' \ge M'_L$ as in Theorem 34.4, also large enough for Proposition 50.2(b), for n_{k+1};
- $8M_0$ of Lemma 31.3, for n_{k+1};
- $C(\rho + C_v)M$ of Section 40, for n_k.

The conditions for $n_{k+1} = n_k/L$ are included in n_* with the factor L;

(T1) $LM \mid n_k$;

(T2) $n_k \geq C_{\mathrm{tor}} LM$, with C_{tor} depending only on d. It is chosen so that every polymer with $d_M(X) < L$ has $X^{(3)}$ in a cube of side less than $\frac{1}{2}n_k$, and diameter at most $\frac{1}{2}n_k - 8$. So such polymers are not wrapping in either sense: that of Lemma 38.5, and that of Proposition 41.7;

(T3) either $n_k \geq CM \log h_k^{-1}$, or the wrap-around one-loop pieces $Ce^{-cn_k/M}$ are carried as a separate source.

Wrapping polymers are those with diameter larger than $\frac{1}{2}n_k - 8$.

49.2 Reblocking

For a polymer X of step k let $\bar{X}$ be the union of the LM-cubes that contain a cube of X; equivalently, that meet X, since the grids are aligned. For an activity $\mathcal{E}$ of step k put $(\mathcal{B}\mathcal{E})(Y,\cdot) = \sum_{X\colon \bar{X}=Y} \mathcal{E}(X,\cdot)$, and let $\mathcal{B}\mathcal{E}$ act on the configurations in the units of step $k+1$ by rescaling. The LM-cubes are the grid cubes of step $k+1$, by (T1), and $d_{LM}(Y)$ in the units of step k is $d_M(Y)$ in those of step $k+1$.

Lemma 49.2 (Reblocking; [15, Lemma 10])**.** *Let $\kappa \geq \kappa_0 + 1$ and $\kappa' = L(\kappa - \kappa_0 - 1)$. For nonnegative numbers $a(X)$ with $a(X) \leq Ae^{-\kappa d_M(X)}$,*

$$\sum_{X\colon \bar{X}=Y} a(X) \leq 3^d K_0 L^d A e^{-\kappa' d_{LM}(Y)}.$$

If $a(X) = 0$ unless $d_M(X) \geq L$, then the same holds with the factor $3^d K_0 L^d e^{-L}$ and the rate $L(\kappa - \kappa_0 - 2)$, for $\kappa \geq \kappa_0 + 2$.

Proof. A minimal tree for X is a tree on the LM-cubes of $\bar{X}$, so $d_M(X) \geq L d_{LM}(\bar{X})$. Then $e^{-\kappa d_M(X)} \leq e^{-(\kappa-\kappa_0)Ld_{LM}(Y)} e^{-\kappa_0 d_M(X)}$. Every X with $\bar{X} = Y$ contains a grid cube of Y, so (49.1) gives $\sum_{\bar{X}=Y} e^{-\kappa_0 d_M(X)} \leq K_0 L^d |Y|_{LM} \leq 3^d K_0 L^d e^{d_{LM}(Y)}$. (Dimock writes 9 in place of 3^d in [15, (132)], which is inconsistent with his (122).) For the last statement use $e^{-(\kappa_0+1)d_M} \leq e^{-L} e^{-\kappa_0 d_M}$, with $\kappa - 1$ in place of κ. This is [15, (130)–(132), (139)]. □

Remark 49.3 (Pieces with margins). The pieces of Sections 47 and 48 are indexed by polymers Z and depend on the data on $Z^{(r')}$, with $r' < M/8 < LM$. For them use the enlarged closure $\hat{Z}$, the union of the LM-cubes meeting $Z^{(r')}$. Then $Ld_{LM}(\hat{Z}) \leq d_M(Z) + C|\hat{Z}|_{LM}$, with C depending only on d, since $r'/(LM) < 1/(8L)$. Indeed a minimal tree of Z meets every cube of Z, hence every LM-cube of $\bar{Z}$. Every other LM-cube of $\hat{Z}$ contains a point within r' of a cube of Z, so a branch of Euclidean length at most $\sqrt{d}(M + r') \leq 2\sqrt{d}M$ from the tree reaches it, since r' is an ℓ^∞-distance and $r' < M$. One branch per such cube costs at most $2\sqrt{d}/L$ in units of LM. The loss is counted in LM-cubes. With $|\hat{Z}|_{LM} \leq 3^d(1 + d_{LM}(\hat{Z}))$ it is absorbed by the L-gain of Lemma 49.2 once $L \geq C3^d$, at the price of a constant $e^{C\kappa 3^d}$.

49.3 The linear part

Proposition 49.4 (Linear part)**.** *Let $d = 3$ and G semisimple, and let $\mathcal{E}$ satisfy (H1)–(H5) at step k. Assume (T0)–(T2), $\kappa_n \geq 2(\kappa_0 + 1)$, $L \geq 4$ and $3^d K_0 L^d e^{-L} \leq L^{-1}$. For a polymer Y' of step $k+1$ and $U \in \mathcal{U}_{k+1}(Y')$ put*

$$(\mathcal{L}\mathcal{E})(Y', U) = \sum_{X\colon \bar{X}=Y'} \mathcal{E}(X, U),$$

with U read in the units of step k. Then $\mathcal{L}\mathcal{E}$ is a normalized activity of step $k+1$ satisfying (H1)–(H5), and

$$\|\mathcal{L}\mathcal{E}\|_{k+1,\kappa_n} \leq \left(2 \cdot 3^d K_0 L^{-\epsilon}(1 + o(1)) + L^{-1}\right) \|\mathcal{E}\|_{k,\kappa_n},$$

where $o(1) \to 0$ as $g_* \to 0$, for fixed L and M, uniformly in k.

Proof. Domains. Read a point $U \in \mathcal{U}_{k+1}(Y')$ in the units of step k. Fields scale by L^{-1}, gradients by L^{-2}, unit-scale Hölder quotients by at most $L^{-2-\alpha}$, curvatures by L^{-2} and currents by L^{-3}. The margin-3 neighbourhood of Y' contains that of every $X \subset Y'$ in the units of step k. With $h_{k+1} \le L^{1/2} h_k$, $U = e^{i\eta A'} U_r$ on $X^{(3)}$ with

$$\mathfrak{m} = |\nabla A'| + [\nabla A']_\alpha \le L^{-2} h_{k+1}^{1-\epsilon}, \qquad \theta \le L^{-2} h_{k+1}^{1-\epsilon/2}, \qquad |A'| \le L^{-1/2-\epsilon} h_k^{1-2\epsilon}.$$

So $U \in \mathcal{U}_k(X)$ with slack: with the same real part U_r,

$$\delta_1 - \mathfrak{m} \ge (1 - L^{-3/2-\epsilon/2}) h_k^{1-\epsilon}, \qquad \delta_0 - |A'| \ge (1 - L^{-1/2-\epsilon}) h_k^{1-2\epsilon}, \tag{49.2}$$

where $\delta_1 = h_k^{1-\epsilon}$, $\delta_0 = h_k^{1-2\epsilon}$. Both are at least $\frac{1}{2} h_k^{1-\epsilon}$ for $L^{1/2+\epsilon} \ge 2$ and $h_k \le 1$. The curvature and the current of U_r are below $\theta_r = h_k^{1-\epsilon/2}$ by the factors $L^{-2+(1-\epsilon/2)/2}$ and $L^{-3+(1-\epsilon/2)/2}$. Thus $\mathcal{LE}(Y', \cdot)$ is defined and analytic on an open neighbourhood of $\mathcal{U}_{k+1}(Y')$. It depends on U only on Y', is normalized and invariant, and Y' is a grid polymer of step $k+1$.

Small polymers, $d_M(X) < L$. By (T2), $X^{(3)}$ lies in a cube Λ of side $2R < \frac{1}{2} n_k + 6 < n_k$. Hence $B_P \le C(L, M)$ and $D_N \le CLM$. Apply Lemma 38.5 with $\theta_r = h_k^{1-\epsilon/2}$, $\delta_1 = h_k^{1-\epsilon}$, $\delta_0 = h_k^{1-2\epsilon}$. Its hypotheses hold for g_* small: $\theta < \theta_r$; $|A'| < \frac{1}{2}\delta_0$ for $L \ge 4$; $R B_P \theta \le 1$ and $c_* \le c_0$, since θ, $|A'|$ and δ_0 are positive powers of h_k times constants depending on L and M. Then

- $\mathfrak{m}/\delta_1 \le L^{-2} (h_{k+1}/h_k)^{1-\epsilon} \le L^{-3/2-\epsilon/2}$;
- $C_0 |A'| (|A'| + (1 + D_N) B_P \theta)/\delta_1 \le C(L, M) h_k^{1-3\epsilon}$;
- $B_P \theta / \delta_1 \le C(L, M) h_k^{\epsilon/2}$;
- c_* and $D_N \delta_1 / \delta_0$ are at most $C(L, M) h_k^{\epsilon}$.

So $\mathfrak{s} \le L^{-3/2-\epsilon/2}(1 + o(1))$, and $|\mathcal{E}(X, U)| \le 2 L^{-3-\epsilon}(1 + o(1)) \|\mathcal{E}\|_{k,\kappa_n} e^{-\kappa_n d_M(X)}$. Lemma 49.2 gives the contribution $2 \cdot 3^d K_0 L^{-\epsilon}(1 + o(1)) \|\mathcal{E}\|_{k,\kappa_n} e^{-\kappa' d_{LM}(Y')}$, with $\kappa' = L(\kappa_n - \kappa_0 - 1) \ge \kappa_n$.

Large polymers, $d_M(X) \ge L$, which include all wrapping ones. Here $|\mathcal{E}(X, U)| \le \|\mathcal{E}(X)\|_k$, since $U \in \mathcal{U}_k(X)$. Lemma 49.2 with the improved factor gives at most $3^d K_0 L^d e^{-L} \|\mathcal{E}\|_{k,\kappa_n} e^{-\kappa'' d_{LM}(Y')}$, with $\kappa'' = L(\kappa_n - \kappa_0 - 2) \ge \kappa_n$. This is at most $L^{-1} \|\mathcal{E}\|_{k,\kappa_n} e^{-\kappa_n d_{LM}(Y')}$, by hypothesis. This is [15, (139)–(140)]. □

Remark 49.5 (What the linear part uses). The constants $C(L, M)$ of Corollary 30.4 and Lemma 28.3 enter only multiplied by positive powers of h_k. The marginal factor $2 \cdot 3^d K_0 L^{-\epsilon}$ contains no constant chosen after L. The quadratic gain comes from complex gauge invariance (Lemma 38.4) and the vanishing of the linear term for semisimple G. The threshold $d_M(X) < L$ is Dimock's; it replaces the thresholds in D_N and B_P of the blueprint (Appendix F), since for small polymers all constants are fixed.

49.4 The sources

Proposition 49.6 (Sources)**.** *Assume the hypotheses of Theorem 48.9 and of Proposition 49.4, with $\|\mathcal{E}_k\|_{k,\kappa_n} \le K_{\mathcal{E}} h_k^{2-2\epsilon}$. Assume also (T3), $\kappa_n \ge 2(1 + C_d + \kappa_0)$, and $L \ge C 3^d + 3$ with the constant C of Remark 49.3. Assume finally that the rates satisfy $(3^{-d}\kappa'_w - 1)/c_\sharp \ge \kappa_n + \kappa_0 + 2$ and $(\kappa'_w - 2)/c'_\sharp - C_d \ge \kappa_n + \kappa_0 + 2$, where κ'_w is the weakening rate κ' of Part IX. Let the following terms of Theorem 48.9 be normalized by subtracting their values at $U' = 1$, then reblocked and rescaled:*

- *the one-loop pieces $\mathfrak{e}(X)$ of all polymers, wrapping ones included;*

- $\log \mathfrak{j}_c$;
- *the pure class* $E^{\#}_{\mathrm{p}}$;
- *the mixed class* $E^{\#}_{\mathrm{m}}$.

Call the resulting activities $\mathcal{E}^*_k$ *(the first three) and* $\mathcal{E}^{\mathrm{mix}}_k$*. They are activities of step* $k+1$ *with (H1)–(H5), and*

$$\|\mathcal{E}^*_k\|_{k+1,\kappa_n} \le B h_{k+1}^{2-2\epsilon}, \qquad \|\mathcal{E}^{\mathrm{mix}}_k\|_{k+1,\kappa_n} \le o(1)\,\|\mathcal{E}_k\|_{k,\kappa_n},$$

with B *depending on* L, M *and the other fixed parameters, but not on* k *or on the number of steps, under the first alternative of (T3). Under the second alternative of (T3), the same holds with* $\mathcal{E}^*_k$ *replaced by its part without the wrapping one-loop pieces. The wrapping pieces then form a separate activity, whose norm* $\|\cdot\|_{k+1,\kappa_n}$ *is at most* $Ce^{-cn_k/M}$.

Proof. One-loop. By Proposition 41.7(d), $|\mathfrak{e}(X,U)| \le C h_k^{2-2\epsilon} e^{-(\kappa_2-1)|X|}$ on $\mathcal{U}_k(X)$, with $\kappa_2 = 3^{-d}\kappa'_w$. This holds on $\mathcal{U}_{k+1}(Y')$ by the inclusion in the proof of Proposition 49.4. Since $|X| \ge d_M(X)/c_\sharp$, the rate in d_M is $(\kappa_2-1)/c_\sharp$. Remark 49.3 and Lemma 49.2 give $C(L,M)h_k^{2-2\epsilon} \le C(L,M)h_{k+1}^{2-2\epsilon}$ in the norm of step $k+1$. For wrapping polymers, Proposition 41.7(d) gives $|\mathfrak{e}(X)| \le Ce^{-(\kappa_2-1)|X|}e^{-cn_k/M}$. Under the first alternative of (T3) this is at most $Ch_k^{2-2\epsilon}e^{-(\kappa_2-1)|X|}$, and it is treated in the same way. Under the second it is the separate source. The $\mathfrak{e}(X)$ are already normalized.

Jacobians. By Proposition 47.6(c), $|\log \mathfrak{j}_c| \le C_P h^{2-2\epsilon}$ on D3‴, with $\log \mathfrak{j}_c = 0$ at $U' = 1$. Each term lives on one polymer of bounded size, and there are at most $C(L,M)$ of them per grid cube.

Pure class. By Proposition 48.8, the sum over $Y \supset \square$ of $|E^{\#}_{\mathrm{p}}(Y)|e^{\beta_{\mathrm{p}}(d_M(Y)+c_\natural)}$ is at most $C_P g^2 p_0^6$, with $\beta_{\mathrm{p}} = (\kappa'_w - 2)/c'_\sharp - C_d$. Subtracting the value at $U' = 1$ at most doubles this. After reblocking it is $C(L,M)g_k^2 p_{0,k}^6 = o(h_{k+1}^{2-2\epsilon})$, since $g_k^2 p_{0,k}^6 / h_{k+1}^{2-2\epsilon} \le g_k^{2\epsilon} p_{0,k}^6 \to 0$.

Mixed class. By Corollary 48.7, the sum over $Y \supset \square$ of $|E^{\#}_{\mathrm{m}}(Y)|e^{\beta(d_M(Y)+c_\natural)}$ is at most $C\epsilon_{\mathrm{m}} e^{c_\natural \kappa_n}$, with $\beta = \kappa_n - 1 - C_d$ and $\epsilon_{\mathrm{m}} = C_P \tau \|\mathcal{E}_k\|_{k,\kappa_n}$. Here $\tau \le C_P L^d g_k p_{0,k} h_k^{-(1-\epsilon)} \to 0$. Keep $e^{-\kappa_0 d_M(Y)}$ for the counting, and use the rest for the L-gain. By Remark 49.3,

$$e^{-(\beta-\kappa_0)d_M(Y)} \le e^{C(\beta-\kappa_0)3^d} e^{-(\beta-\kappa_0)(L-C3^d)d_{LM}(\hat{Y})}.$$

The counting of the Y with given $\hat{Y}$ is that of the proof of Lemma 49.2. For $L \ge C3^d + 3$ and $\kappa_n \ge 2(1 + C_d + \kappa_0)$ the resulting rate $(\beta - \kappa_0)(L - C3^d) - 1$ is at least κ_n. The bound is $C(L,M)e^{C\kappa_n 3^d} e^{c_\natural \kappa_n} \tau \|\mathcal{E}_k\| = o(1)\|\mathcal{E}_k\|$, since κ_n is fixed before g_*.

Properties. (H1) holds by Theorem 48.9 and the inclusion of domains, with the restriction of Remark 49.7. (H2) and (H5) hold since the reblocked polymer is $\hat{Y}$. (H3) is the bound. (H4) holds by the subtraction at $U' = 1$ and by Theorem 48.9. (H6) holds since all pieces are real for real s, real ξ and real references (Proposition 48.5, Corollary 41.6), and the Mayer expansion of real pieces against a real measure is real (Theorem 48.6(e)). □

Remark 49.7 (Extended references). A point of $\mathcal{U}_{k+1}(Y')$ fixes U' only near Y'. It is extended by the cutoff of Remark 38.7. In the cutoff band, $|\nabla(\chi A)| \sim |A|$, which can exceed $h_k^{1-\epsilon}$. So composed configurations of old polymers meeting the band need not lie in $\mathcal{U}_k(X)$. Therefore, before the Mayer expansion, all old activities and all pieces whose polymer is not contained in Y' (in the units of step k) are set to zero. By Theorem 48.6(a), the output $E^{\#}(Y)$ with $\hat{Y} = Y'$ is unchanged, since it depends only on the pieces of polymers inside Y. So the new activity at Y' is defined at every point of $\mathcal{U}_{k+1}(Y')$ and does not depend on the extension. Pieces with $Z \subset Y'$ near $\partial Y'$ may depend on the extension, but they do not enter $E^{\#}(Y)$ with $\hat{Y} = Y'$. At a real critical U', take $U_r = U'$ and $A = 0$, so that no extension is needed. The restricted

and the full systems then share every $E(Z)$ with $Z \subset Y$. By Theorem 48.6(a), applied also to $E^{\#}_{\mathrm{p}}$ and to the λ-family of Corollary 48.7, their outputs agree. Since $\sum_{Y'} \sum_{\hat{Y}=Y'} = \sum_{Y}$, the identity of Theorem 48.9 is not affected.

The collar inclusions. Let Y contribute to the output at Y', i.e. $\hat{Y} = Y'$, with $\hat{Y}$ the enlarged closure of Remark 49.3. Then $Y^{(r')} \subset Y'$, and the following holds.

- By Theorem 48.6(a), the output at Y uses only pieces $E(Z)$ with $Z \subset Y$.
- Each such piece depends on the reference only on $Z^{(r')} \subset Y^{(r')} \subset Y'$.
- A mixed piece uses old activities $\mathcal{E}_k(X)$ with $X \subset Z$. Their domain budget is needed on $X^{(3)} \subset Z^{(r')} \subset Y'$, since $r' = r + 2L + 2 \geq 3$.

So on every set that matters the extension is not felt. The real part there is that of U', and the slack (49.2) gives the hypotheses of Proposition 47.8. The same inclusions hold for the pure system and for the λ-family of Corollary 48.7.

To avoid an implicit definition, the restricted system for an output polymer Y' is defined on the input pieces with $Z \subset Y'$ only. Equivalently, it may be defined on a cutoff extension that equals U' on the 3-neighbourhood of Y', which is the margin of the next domain. Pieces near $\partial Y'$ whose collars cross it may depend on the extension, but they do not enter the outputs with $\hat{Y} = Y'$. At real critical references the restricted and the unrestricted systems share all inputs that enter, so their outputs coincide.

49.5 One step

Theorem 49.8 (One step of the flow)**.** *Let G be semisimple. Let the parameters be chosen in the following order:*

(1) $\epsilon \in (0, \frac{1}{4})$, *and* $\kappa_n \geq 2(1 + C_d + \kappa_0)$;

(2) $L \geq \max(10 \cdot 3^d K_0, (8C_*)^{2/3}, C3^d + 3, 4)$, *with* $L^{1/2+\epsilon} \geq 3$ *and* $3^d K_0 L^d e^{-L} \leq L^{-1}$. *Since* $C_* \geq 1$, $L^{3/2} \geq 8C_*$ *implies* $L^{3/2} \geq 4(C_* + 1)$;

(3) the weakening rate κ and the loss δ of Part IX, with $\kappa'_w = \kappa - C\delta$, the conditions of Proposition 49.6 and $\kappa'_w - 1 \geq c_\natural \kappa_n + \log(1/c_\flat)$ (Proposition 48.5);

(4) ρ, then M, with $M > 32(L+1)$, M a power of L, and

$$M \geq \max\left(M_0,\, 5L,\, M_1,\, M_1^{(0)},\, M_1',\, 64\right).$$

Here M_0 and $5L$ are the conditions of Theorem 30.3; $M_1 = M_1(\kappa, \delta, C_G, C_{\mathrm{loc}}, C_{\mathfrak{N}}, \gamma_0, \rho, C_v)$ is the largest of the thresholds of Theorem 41.4 at the levels of Section 41, fixed level by level (the paragraph before the levels), and $M_1^{(0)}$ its value for the operators of the first step (Proposition 50.2(c)); M_1' and 64 are the conditions of Section 45. M_1 is computed with the multiplicative constants of Part IX doubled, and M_1' with θ' replaced by $2\theta'$ (Theorem 44.10); for both, the coercivity constants are the fixed ones of Proposition 46.7, and every decay bound that enters is taken at the fixed rate of the differences there, a quarter of the minimum of the perturbation rate and the two chosen inverse decay rates in Lemma 13.3; neither depends on δ_c, so the same numbers serve at real and at complex references (Proposition 46.7, the step "The constants" of the proof). At complex references of size $\delta_c < \delta_P$ the inputs of M_1 change by at most a factor 2, and those of θ' by at most $1 + \sigma_{\mathrm{cx}}$ with $(1 + \sigma_{\mathrm{cx}})^3 \leq 2$. So M_1 and M_1' serve at complex references and do not depend on δ_P, and M is chosen before δ_P without circularity. The threshold $M_1^{(0)}$ is computed with the bounds of Proposition 50.2(b) and its proof at complex references ($\operatorname{Re}\Delta_0 \geq \frac{1}{2}C_{\mathrm{P}}^{-1}$, a norm change of at most $\frac{1}{8}C_{\mathrm{P}}^{-1}$, range $2L + 2$),

which do not depend on $\delta_{P,0}$. Further, $M \geq 16r$, where r is the largest margin of the weakened families used (Section 47, Setting); r does not depend on M, and $M \geq 16r$ is required by the definition of weakenable families (Section 40). Then the scales $M' \geq M'_L$ of Theorem 34.4 and of Proposition 50.2(b);

(5) the constants of the complex references: the threshold $\varepsilon_{\max}$, then δ_P (Proposition 46.7) and $\delta_{P,0}$ (Proposition 50.2), then $c_a \leq cL^{-d-2}$;

(6) g_*, small enough that, in particular, the real parts of the complex references of the step are (M,ε)-regular with $\varepsilon \leq \varepsilon_{\max}$.

Assume (T0)–(T3) at step k, and let $\mathcal{E}_k$ satisfy (H1)–(H6) with $\|\mathcal{E}_k\|_{k,\kappa_n} \leq K_{\mathcal{E}} h_k^{2-2\epsilon}$, where $K_{\mathcal{E}} = 2B$, B from Proposition 49.6. Let $\rho_k(V) = \mathcal{Z}_k \exp\big(-g_k^{-2} S(U_k(V)) + \mathcal{E}_k(U_k(V))\big)$ on $\mathfrak{U}_k(p_k g_k)$, with $\mathcal{E}_k(U) = \sum_X \mathcal{E}_k(X,U)$. Then, for V' small field at step $k+1$, in the units of step $k+1$,

$$T_{\mathrm{sf}}(\rho_k)(V') = \mathcal{Z}'_{k+1} \exp\Big(-\frac{1}{g_{k+1}^2} S(U') + \mathcal{E}_{k+1}(U')\Big), \qquad U' = U_{k+1}(V'),$$

where $\mathcal{Z}'_{k+1} > 0$ is a constant (the vacuum part; positive by (H6)), and $\mathcal{E}_{k+1} = \mathcal{L}\mathcal{E}_k + \mathcal{E}_k^{\mathrm{mix}} + \mathcal{E}_k^*$. The activity $\mathcal{E}_{k+1}$ satisfies (H1)–(H6) at step $k+1$, and

$$\|\mathcal{E}_{k+1}\|_{k+1,\kappa_n} \leq A\|\mathcal{E}_k\|_{k,\kappa_n} + Bh_{k+1}^{2-2\epsilon} \leq K_{\mathcal{E}} h_{k+1}^{2-2\epsilon}, \qquad A = 2\cdot 3^d K_0 L^{-\epsilon}(1+o(1)) + L^{-1} + o(1).$$

This holds under the first alternative of (T3). If the second alternative of (T3) is taken instead, then $\mathcal{E}_{k+1} = \mathcal{L}\mathcal{E}_k + \mathcal{E}_k^{\mathrm{mix}} + \mathcal{E}_k^* + \mathcal{E}_k^{\mathrm{wr}}$, where $\mathcal{E}_k^*$ is without the wrapping one-loop pieces and $\mathcal{E}_k^{\mathrm{wr}}$ is the separate activity of the wrapping pieces of Proposition 49.6. Then $\mathcal{E}_{k+1}$ satisfies (H1)–(H6), and

$$\|\mathcal{E}_{k+1}\|_{k+1,\kappa_n} \leq A\|\mathcal{E}_k\|_{k,\kappa_n} + Bh_{k+1}^{2-2\epsilon} + Ce^{-cn_k/M}.$$

The bound by $K_{\mathcal{E}} h_{k+1}^{2-2\epsilon}$ is not claimed then: a fixed exponential in the torus size is not absorbed by $h_{k+1}^{2-2\epsilon}$ as $g_k \to 0$ at fixed n_k.

Proof. By Theorem 48.9, with Proposition 41.7 for the determinant. The exponent consists of $-g_k^{-2}S(U')$, which is $-g_{k+1}^{-2}S(U')$ in the units of step $k+1$; the vacuum constants; $\mathfrak{E}(U') = \mathcal{E}_k(U')$; and the normalized pieces. The constants $\log \mathcal{N}_k$, $\log N_\chi$, $-\frac{1}{2}\sum_b \ell_k(b;1)$ and $\sum_Y E^{\#}(Y,1)$ are collected in $\mathcal{Z}'_{k+1}$. They are split into explicit normalizations and a vacuum energy in Section 50.3. The vacuum part $\sum_Y E^{\#}(Y,1)$ has a bound per grid cube, by Theorem 48.6(b) and Proposition 48.8. The normalizations are explicit; $\log \mathcal{N}_k$ contains $\log(2\pi g_k^2)$ per free bond, so they are not bounded per volume uniformly in k. Reblocking $\mathcal{E}_k(U')$ gives $\mathcal{L}\mathcal{E}_k$ (Proposition 49.4); the other terms give $\mathcal{E}_k^*$ and $\mathcal{E}_k^{\mathrm{mix}}$ (Proposition 49.6, Remark 49.7). The hypotheses of Theorem 48.9 hold:

- (Q1)–(Q3), for g_* small, since $p_{0,k}/p_k \to 0$, $g_k p_{0,k} \to 0$, $L^{3/2} \geq 4(C_*+1)$ and $p_{k+1} \leq p_k$ (Section 48); and $2g \leq r_t$ of Proposition 48.8, since $r_t \geq c p_{0,k}^{-3}$ with c depending on the fixed parameters, and $g_k p_{0,k}^3 \to 0$;
- $\epsilon_{\mathrm{p}} \leq \frac{1}{2}\epsilon_*$ and $\epsilon_{\mathrm{m}} e^{c_\natural(\kappa_n - 1)} \leq \frac{1}{4}\epsilon_*$, since $\epsilon_{\mathrm{p}} = C_P g_k p_{0,k}^3 \to 0$, $\epsilon_{\mathrm{m}} \leq C_P \tau K_{\mathcal{E}} h_k^{2-2\epsilon} \to 0$, and κ_n is fixed before g_*;
- the decay of the old activities, which is (H3).

For the recursion, $h_{k+1}^{2-2\epsilon} = L^{1-\epsilon}(p_{k+1}/p_k)^{2-2\epsilon} h_k^{2-2\epsilon}$. So $AK_{\mathcal{E}} h_k^{2-2\epsilon} + Bh_{k+1}^{2-2\epsilon} \leq K_{\mathcal{E}} h_{k+1}^{2-2\epsilon}$ holds if $AL^{-(1-\epsilon)}(p_k/p_{k+1})^{2-2\epsilon} \leq \frac{1}{2}$. This follows from $A \leq \frac{1}{4}L^{1-\epsilon}$, true for $L \geq 10\cdot 3^d K_0$ and g_* small, and $(p_k/p_{k+1})^{2-2\epsilon} \leq 2$, true for g_* small depending on L. □

Remark 49.9 (From the step to the target). (a) *The vacuum energy.* The constants are split into explicit normalizations $\mathcal{Z}_k$ and a vacuum energy ε_k per volume in Section 50.3. The flow $\varepsilon_{k+1} = L^3\varepsilon_k + \mathcal{L}_1\mathcal{E}_k + \varepsilon_k^*$ needs only the per-volume bounds of the proof. The mixed class at $U' = 1$ is part of $\mathcal{L}_1\mathcal{E}_k$.

(b) *The iteration* over $k \le K$ and the first step are in Section 50. Steps with $n_k < \max(C_{\mathrm{tor}}LM, n_*)$ are not covered by Theorem 49.8, so the final torus is chosen large enough (Lemma 50.1). That section also makes the choice between the alternatives of (T3), imposes the divisibility $LM \mid n_k$ for all k (with M a power of L), and proves the corollary on the partition function.

(c) *Not treated:* the large-field stage, which starts from $\mathcal{R}_{\mathrm{LF}}$ of Section 48.

50 The iteration and the small-field flow

This section iterates Theorem 49.8 over the steps. It treats the first step separately, defines the vacuum energy and its counterterm, and states the small-field flow and its consequence for the partition function. The plan and its review are in the blueprint (Appendix K). The model and the map T are those of Section 31. The finest spacing is $\epsilon = L^{-N}$. Here N is the total number of scales, unrelated to $N = L^k$ of Parts I–VII and to the matrix size N. The dimensionless couplings are $g_k^2 = g^2L^k\epsilon$, so $g_{k+1}^2 = Lg_k^2$, and $\rho_0(V) = \exp(-g_0^{-2}S_{\mathrm{lat}}(V) - E_0)$ on the ϵ-lattice $\mathbb{T}^{(0)}$. The parameters are chosen in the order of Theorem 49.8.

50.1 Scales and the torus

The assumptions on p and p_0 of Section 49 (small-field parameters) stay in force: $p_k = p(g_k)$, $p_{0,k} = p_0(g_k)$, p decreasing with $p(g)/p(L^{1/2}g) \to 1$, $p_0/p \to 0$, and $g^\beta p(g) \to 0$ for every $\beta > 0$. In addition $g \mapsto p(g)g$ is increasing. Powers of $1 + \log g^{-1}$ satisfy all of these. The lattice is assumed fine enough that $g_1 \le g_*$, i.e. $\epsilon \le g_*^2/(Lg^2)$. Let the torus have physical side ℓ. In the units of step k the lattice $\mathbb{T}^{(k)}$ has side $n_k = \ell L^{N-k}$. The iteration runs for $k = 0, \dots, K$, where K is the largest integer with $g_K \le g_*$. Then $L^{-1/2}g_* < g_K \le g_*$ and $n_K = \ell g^2/g_K^2 \ge \ell g^2/g_*^2$. Put

$$n_{\min} = \max\left(C_{\mathrm{tor}}M,\ n_*(L, M, \rho),\ CM(\log h_*^{-1} + \log L)\right), \qquad h_* = p(L^{-1/2}g_*)\, L^{-1/2}g_*,$$

with n_*, C_{tor} from (T0), (T2), and $h_* \le h_K$ the value of $h = pg$ at the smallest possible final coupling. We assume that the physical volume is not too small, $\ell g^2/g_*^2 \ge n_{\min}$, so that $n_K \ge n_{\min}$. We also assume that ℓL^N and M are powers of L, so that $M \mid n_K$. All bounds below are uniform in $g_k \le g_*$; this mirrors [15], whose final torus has side $L^{M+\Delta}$.

Lemma 50.1 (Torus hypotheses)**.** *For every $k < K$, (T0)–(T2) and the first alternative of (T3) hold.*

Proof. $n_k = L^{K-k}n_K \ge Ln_K$ gives (T0) and (T2), and $M \mid n_K$ gives (T1). For (T3): since $p_K \le p_k$ and $g_k^2 = L^{-(K-k)}g_K^2$, $\log h_k^{-1} \le \log h_K^{-1} + \frac{1}{2}(K-k)\log L$, and $\log h_K^{-1} \le \log h_*^{-1}$. Since $L^{K-k} \ge 1+(K-k)$, $n_k \ge n_K(1+(K-k)) \ge CM\log h_*^{-1} + CM(K-k)\log L \ge CM\log h_k^{-1}$. □

By Lemma 50.1 and Proposition 41.7(d), the wrapping one-loop pieces are bounded by $Ch_k^{2-2\epsilon}$, and no separate wrap-around source is needed.

50.2 The first step

At $k = 0$ the configurations V live on the ϵ-lattice, $U_0(V) = V$, and the background of the step is $U' = U_1(V')$, which is the case $k = 1$ of Theorem 30.3. The base is $W = U'^{(0)} = U'$. The general analysis of Sections 47–49 uses the k-step operators at U' with the pair $(Q'_k, \mathbf{Q}_k)$. At $k = 0$ these are the degenerate case of a single fine point per block: Lemma 31.3 excludes it, and Theorem 25.1 reduces to $\mathfrak{B}(V) = \{V\}$. The first step is simpler, and is treated directly.

Proposition 50.2 (The first step)**.** *Let $k = 0$, $\mathcal{E}_0 = 0$, and let V' be small field at step 1. Then the conclusions of Theorem 49.8 hold at $k = 0$, with $\mathcal{L}\mathcal{E}_0 = \mathcal{E}_0^{\mathrm{mix}} = 0$. The step quantities simplify as follows.*

(a) No background map. $\mathcal{S}(B) = S(e^{\mathrm{i}B}U')$, *and A_* is not used. The interaction pieces are $v_p(Z) = s_p(e^{\mathrm{i}\hat{Z}(Z)}U') - T_2[\cdots](Z)$, local in Z within range $2L + 2$.*

(b) The Hessian. $\Delta_0(Z, Z) = \Delta^0(\hat{Z}^1 Z, \hat{Z}^1 Z) + 2\langle \omega_0, \hat{Z}^2(Z)\rangle$, *where Δ^0 is the Hessian of $B \mapsto S(e^{\mathrm{i}B}U')$ at 0 and ω_0 is the gradient there. Δ_0 is local of range at most $2L + 2$, and $\operatorname{Re}\Delta_0 \geq \frac{1}{2}C_{\mathrm{P}}^{-1}$ on $\ell^2(\mathfrak{F})$, uniformly on the complex references.*

(c) The walks. $C_0^{1/2}(s)$ *and the determinant densities are given by Theorem 41.4 and Corollary 41.6, with $\mathfrak{A} = \Delta_0 + r$, $\mathfrak{N} = 0$ and no mass term.*

Proof. (a) For $k = 0$ the k-step variational problem is trivial: $U_0(e^{\mathrm{i}B}W) = e^{\mathrm{i}B}W$, so $\mathcal{S}(B) = S(e^{\mathrm{i}B}U')$. Lemma 32.4 and Proposition 32.5 are one-step statements for the averaging $\mathfrak{a}'$ from $\mathbb{T}$ to $\mathbb{T}'$ at the base $W = U'$, and they hold at $k = 0$. The interaction pieces are those of Definition 47.4 with $A_*(s; B)$ replaced by B. They are local, since s_p and $\hat{Z}$ are.

(b) At $k = 0$, Δ^0 is the second variation of the Wilson action on $\mathbb{T}$ at U'. It is a local operator of range 1, equal to $\|d_{U'}B\|^2$ up to terms bounded by $C|F_{U'}|\|B\|^2$, with $d_{U'}$ the covariant curl. Apply the proof of Theorem 34.4 with Δ^0 in place of $\mathbf{\Gamma}_\pi - a$:

- *IMS.* Localize at scale M'; since Δ^0 has range 1, the error is $CM'^{-2}\|B\|^2$.
- *Local gauges.* On each cube of side $2M'$ use a real gauge u_j in which $U' = e^{\mathrm{i}A_j}$ with $|A_j| + |\partial A_j| \leq C_{M'}\varepsilon$ (Definition 18.1, and Lemma 28.3, which needs $\varepsilon \leq \varepsilon_{\mathrm{cmp}}(M')$). By Proposition 23.3 at $\eta = 1$ (for $\varepsilon \leq c_5$), $\Delta^0(B, B) = \|D_{U'}B\|^2 + d^2\mathcal{N}(0)[B, B]$ with $|d^2\mathcal{N}(0)[B, B]| \leq C\varepsilon\|B\|^2$, since $\|\nabla B\|^2 \leq 4d\|B\|^2$ at $\eta = 1$. With Lemma 12.2 in the gauge u_j, which applies to the covariant curl, this gives $\Delta^0(\chi_j B, \chi_j B) \geq \|d_1(\chi_j B)^{u_j}\|^2 - C_{M'}\varepsilon\|\chi_j B\|^2$. This replaces Lemma 34.3.
- *Poincaré.* Lemma 34.2, together with $l'B = 0$ for $B = \hat{Z}^1 Z$ (Lemma 32.4(d)). In the gauge u_j, $|l'(\chi_j B) - R(u_j)^{-1}q((\chi_j B)^{u_j})| \leq C_{M'}\varepsilon\|\chi_j B\|$, by Lemma 22.5(b) and $|A_j| \leq C_{M'}\varepsilon$, as in the last paragraph of the proof of Lemma 34.3. The rest is as in the proof of Theorem 34.4.

This gives $\Delta^0(\hat{Z}^1 Z, \hat{Z}^1 Z) \geq (C_{\mathrm{P}}^{-1} - CM'^{-2} - CL^2M'^{-2} - C_{M'}\varepsilon)\|\hat{Z}^1 Z\|^2 \geq \frac{3}{4}C_{\mathrm{P}}^{-1}\|Z\|^2$, for M' large and then ε small, since $\|\hat{Z}^1 Z\| \geq \|Z\|$. At $k = 0$, $\eta = 1$ and $\omega_0 = \nabla_B S(e^{\mathrm{i}B}U')|_{B=0} = J_{U'}$ exactly. It is the multiplier of the trivial 0-step problem, not of the one-step constrained minimum. That $\langle \omega_0, \hat{Z}^1 Z\rangle = 0$, i.e. the absence of a linear term, follows from Lemma 31.6(c) at $k = 0$. The term $2\langle \omega_0, \hat{Z}^2(Z)\rangle$ is at most $CL^{3d+1}|J_{U'}|\|Z\|^2$, by Lemma 32.4(d), and $|J_{U'}|$ is small by Theorem 30.3. Away from criticality $\omega_0 := J_{U'}$, with the same bound. Δ_0 is an explicit local analytic function of U': the Hessian of the Wilson action, composed with the elimination of Lemma 32.4 at a complex base. Complex references change it by $O(\delta_c)$, so the lower bound persists for $\operatorname{Re}\Delta_0$. The range follows from that of $\hat{Z}^1$ and $\hat{Z}^2$. The IMS scale M' is chosen after L and before g_*, and $n_1 \geq 80M'$ is included in n_*.

(c) The hypotheses of Section 41 hold with $\mathfrak{A}^X = \mathbf{1}_X(\Delta_0 + r)\mathbf{1}_X$ on $\ell^2(\mathfrak{F}(X))$:

- (A1): $\mathfrak{A}^X$ is local, and $\mathfrak{N} = 0$;
- (A2): the compressions to $\Lambda_j^v \cap X$ have $\mathrm{Re} \geq \frac{1}{2}C_{\mathrm{P}}^{-1} + r$, hence decay by Lemma 5.5; the comparison of domains is in seam form, by the resolvent identity and the locality of Δ_0, as in Proposition 37.6(c);
- (A3) is empty.

The commutator hypothesis of Lemma 41.3 holds, since Δ_0 has range $2L+2$ and $|\nabla h_j| \leq C/M$. So does the comparison of the local commutators in Theorem 41.4(b): the difference is $\mathbf{1}_X[\Delta_0, h_j](G_j^X - G_j^{X'}) + (\mathbf{1}_X - \mathbf{1}_{X'})[\Delta_0, h_j]G_j^{X'}$, of seam form by (A2) and supported next to $X \setminus X'$, respectively. Then Theorem 41.4 and Corollary 41.6 apply. In the latter, the local densities $\ell_{\Lambda_j^v}$ are now the diagonal traces of $\log(\mathbf{1}_\Lambda \Delta_0 \mathbf{1}_\Lambda)$, $\Lambda = \Lambda_j^v$, by the formula of Proposition 34.7. Proposition 41.7 applies with one change. Its constants ε_a, δ_a at $k = 0$ satisfy $\varepsilon_a \leq \varepsilon_{\mathrm{cc}}$ and $C_{\mathrm{cmp}}\delta_a \leq \delta_{P,0}$, so that every configuration of Definition 38.2 is a complex reference of size at most $\delta_{P,0}$ (first item of Section 46). Parts (a), (b) obtained analyticity at complex references through Propositions 19.11, 46.4 and 46.7, which concern k-step operators. At $k = 0$ this is replaced by the direct statement in (b): Δ_0 is an explicit local analytic function of the reference, coercive under complex perturbations.

The rest of Sections 47–49 applies with these replacements:

- Proposition 47.5(a)–(c) and (e) with $A_* = B$, so that $|A_*|_* \leq C_P|B|$ holds trivially, and Proposition 47.6; in place of Proposition 47.5(d), the factorization of the composed pieces proved below;
- $r_I = r_L$, since Theorem 45.5 and its radius R_P are not used;
- in (Q1), the conditions $w \leq \beta_V$ and $w \leq \frac{1}{2}R_P$ are dropped: β_V enters only the identification through $A_*(1; B) = A_*(B)$, which is trivial for $A_* = B$, and R_P only through A_*;
- complex references at $k = 0$ are those of size $\delta_c \leq \delta_{P,0}$, where $\delta_{P,0}$ is such that the $O(\delta_c)$ change of Δ_0 in (b) is at most $\frac{1}{8}C_{\mathrm{P}}^{-1}$ in norm, and $\delta_{P,0} \leq \min(c'_L, cL^{-d-1})$ for the elimination at the complex base $W = U'$ (with $|\mathfrak{r}_c| \leq \frac{1}{2}$); moreover $\varepsilon + C\delta_{P,0} \leq \frac{1}{20}$ and $\rho_L \leq \frac{1}{20}$, for Proposition 47.5(b) at $k = 0$. Since the term $2\langle \omega_0, \hat{Z}^2(Z)\rangle$ is at most $\frac{1}{8}C_{\mathrm{P}}^{-1}\|Z\|^2$ for g_* small, this gives $\mathrm{Re}\,\Delta_0 \geq (\frac{3}{4} - \frac{1}{8} - \frac{1}{8})C_{\mathrm{P}}^{-1} = \frac{1}{2}C_{\mathrm{P}}^{-1}$;
- Section 48 without composed activities, and with Remark 34.9 in place of Theorem 34.8.

Factorization at $k = 0$. The first sentence of Proposition 47.5(d), a statement for a fixed field Z, does not hold at $k = 0$: there $v_p(Z)$ does not depend on s, and it need not vanish when all $s_\square$ do. For instance, at $U' = 1$, let Z be supported on one free bond b of a plaquette p that contains no bond $b_0(c)$. Then $v_p(Z) = f(Z(b)) - T_2[f](Z(b))$ with $f(X) = \mathsf{N} - \mathrm{tr}\cos X$, which is of order $|Z(b)|^4$ and not 0 for small real $Z(b) \neq 0$. What Lemma 47.2 needs is the factorization of the composed pieces $v_p(Z(s))$, $Z(s) = gC_0^{1/2}(s)\xi$, which we now prove. For a plaquette p let $B'(p)$ be the union of the blocks $B'(y)$, $y \in \mathbb{T}'$, that meet p. Its ∞-diameter is less than $2L$, since in each coordinate direction the points of p lie in at most two consecutive blocks, and it contains $B'(c_-) \cup B'(c_+)$ for every bond $b_0(c)$ of p. By Lemma 32.4(a),(c),(d) and Lemma 21.5(c), $\hat{Z}(Z)$ on the bonds of p depends on Z and on the base $W = U'$ only on the bonds of $B'(p)$, and so does $v_p(Z)$. If $Z = 0$ on the bonds of $\mathfrak{F}$ in $B'(p)$, then $\hat{Z}(Z) = 0$ on the bonds of p, since $\zeta_c(0) = 0$ by the uniqueness in Lemma 32.4(c); so $t \mapsto s_p(e^{\mathrm{i}\hat{Z}(tZ)}U')$ is constant, and $v_p(Z) = 0$.

Now let $s_\square = 0$ for $\square \not\subset Z'$, a union of boxes with $*$-components Z'_i. By (c), Corollary 41.6 and Lemma 40.3(b), $Z(s)$ vanishes off $Z'^{(r)}$; on $(Z'_i)^{(r)}$ it depends on ξ and on U' only on $(Z'_i)^{(r)}$, and on s only through $s_\square$, $\square \subset Z'_i$. Every term of $C_0^{1/2}(s)$ carries a monomial of

positive degree (Definition 40.2), so $Z(0)=0$. Hence $v_p(Z(s))=0$ unless $B'(p)$ meets $Z'^{(r)}$; in particular $v_p(Z(0))=0$. If $B'(p)$ meets $Z'^{(r)}$, then it meets $(Z'_i)^{(r)}$ for exactly one i and lies in $(Z'_i)^{(r+2L)}$, since its diameter is less than $2L$ and $2r'<M/4$. Then $v_p(Z(s))$ depends on s only through $s_\square$, $\square\subset Z'_i$, on ξ only on $(Z'_i)^{(r)}$, and on U' only on $(Z'_i)^{(r+2L)}\subset(Z'_i)^{(r')}$. For a box $\square\subset Z'_0$, every plaquette based in $\square$ has a point of $B'(p)$ in Z'_0, so this holds with $i=0$. So $V(\square;s)=g^{-2}\sum_p v_p(Z(s))$, the sum over the plaquettes based in $\square$, vanishes at $s=0$, depends on s only through the boxes of Z'_0, and depends on ξ and U' only on $(Z'_0)^{(r')}$. This is the hypothesis of Lemma 47.2.

Theorem 34.8 is stated for $k\geq 1$. Throughout Sections 47–49, in particular in Proposition 48.3 and Theorem 48.9, it is replaced at $k=0$ by Remark 34.9, with the constants listed there. Since $\mathcal{E}_0=0$, there is no linear part and no mixed class. The torus hypotheses at $k=0$ hold by Lemma 50.1. The source constant of the first step has its own values of the constants: $\frac12 C_{\mathrm P}^{-1}$ in place of $\frac18 c_L$, and its own C_T, r_I, r_t, κ' and $M_1^{(0)}$ (Theorem 49.8(4)). The constant B of Theorem 49.8 is taken as the maximum over the first step and the steps $k\geq 1$. The rate conditions of Proposition 49.6 hold at $k=0$ with the same choice of the weakening rate. □

50.3 The vacuum energy

Write the density of step k as

$$\rho_k(V)=\mathcal{Z}_k\exp\Big(-\frac{1}{g_k^2}S(U_k(V))-\varepsilon_k|\mathbb{T}^{(k)}|+\sum_X\mathcal{E}_k(X,U_k(V))\Big),\qquad V\in\mathfrak{U}_k(p_kg_k),\quad(50.1)$$

with $\mathcal{Z}_0=1$ and $\varepsilon_0=E_0/|\mathbb{T}^{(0)}|$. In the step $k\to k+1$, the constants of Theorem 48.9 are split as follows.

- *Explicit normalizations*: $\mathcal{Z}_{k+1}=\mathcal{Z}_k\,\mathcal{N}_kN_\chi\exp\big(-\frac12\sum_{b\in\mathfrak{F}}\ell_k(b;1)\big)$. These do not depend on the activities.
- *Vacuum energy*: $\varepsilon_{k+1}|\mathbb{T}^{(k+1)}|=\varepsilon_k|\mathbb{T}^{(k)}|-\sum_Y\big(E^{\#}_{\mathrm p}(Y,1)+E^{\#}_{\mathrm m}(Y,1)\big)$. The one-loop pieces and $\log \mathrm{j}_c$ vanish at $U'=1$.

Since $|\mathbb{T}^{(k)}|=L^3|\mathbb{T}^{(k+1)}|$,

$$\varepsilon_{k+1}=L^3\varepsilon_k+\mathcal{L}_1\mathcal{E}_k+\varepsilon_k^*,\qquad \mathcal{L}_1\mathcal{E}_k=-\frac{\sum_Y E^{\#}_{\mathrm m}(Y,1)}{|\mathbb{T}^{(k+1)}|},\quad \varepsilon_k^*=-\frac{\sum_Y E^{\#}_{\mathrm p}(Y,1)}{|\mathbb{T}^{(k+1)}|}.\qquad(50.2)$$

The notation $\mathcal{L}_1$ follows [15]; here $\mathcal{L}_1\mathcal{E}_k$ is the mixed class at $U'=1$, which is not linear in $\mathcal{E}_k$.

Proposition 50.3 (Vacuum energy)**.** *Under the hypotheses of Theorem 49.8 at step k,*

$$|\mathcal{L}_1\mathcal{E}_k|\leq o(1)\|\mathcal{E}_k\|_{k,\kappa_n},\qquad |\varepsilon_k^*|\leq C(L,M)g_k^2p_{0,k}^6.$$

Let $0<\beta'<2-2\epsilon$. There is a unique E_0 with $\varepsilon_K=0$; it is determined by the $\mathcal{L}_1\mathcal{E}_j+\varepsilon_j^*$, which do not depend on E_0. It satisfies $|E_0|/|\mathbb{T}^{(0)}|\leq Cg_0^{\beta'}$, and $|\varepsilon_k|\leq Cg_k^{\beta'}$ for all $k\leq K$, uniformly in K.

Proof. The number of grid cubes of step k is $|\mathbb{T}^{(k)}|M^{-3}=L^3M^{-3}|\mathbb{T}^{(k+1)}|$. Corollary 48.7 and Proposition 48.8 bound $\sum_{Y\supset\square}|E^{\#}(Y,1)|$ for each cube $\square$. This gives the two bounds. With $\|\mathcal{E}_k\|\leq K_{\mathcal{E}}h_k^{2-2\epsilon}$, and $g^{\beta''}p(g)\to0$, $g^{\beta''}p_0(g)\to0$ for every $\beta''>0$, $\delta_k:=\mathcal{L}_1\mathcal{E}_k+\varepsilon_k^*$ satisfies $|\delta_k|\leq Cg_k^{\beta'}$. The activities do not depend on the constants ε_k, so (50.2) is a linear recursion with given δ_k. Setting $\varepsilon_K=0$ gives $\varepsilon_k=-\sum_{j=k}^{K-1}L^{-3(j-k+1)}\delta_j$, hence

$$|\varepsilon_k|\leq C\sum_{j\geq k}L^{-3(j-k+1)}g_j^{\beta'}=Cg_k^{\beta'}\sum_{m\geq0}L^{-3(m+1)}L^{\beta'm/2}\leq Cg_k^{\beta'},$$

and $E_0=\varepsilon_0|\mathbb{T}^{(0)}|$. This is [15, Theorem 24, (284)–(288)]. □

50.4 The small-field flow

Theorem 50.4 (Small-field flow)**.** *Let G be semisimple, let the parameters be chosen as in Theorem 49.8, and let E_0 be as in Proposition 50.3. Let $\rho_{k+1} := \mathbf{1}_{\mathfrak{U}_{k+1}(p_{k+1}g_{k+1})}\, T_{\mathrm{sf}}(\rho_k)$ be the iterated small-field step (Definition 48.1), restricted at each step to the small-field set, where T_{sf} is defined. Then for every $k \le K$ and $V \in \mathfrak{U}_k(p_k g_k)$, ρ_k has the form (50.1), where:*

(i) $\mathcal{E}_k$ satisfies (H1)–(H6) of Section 49: its polymers are grid polymers; $\mathcal{E}_k(X,\cdot)$ depends on the configuration only on X, is analytic on the domain $\mathcal{U}_k(X)$ (D3′′′), is invariant under real gauge transformations, normalized, and real for real configurations;

(ii) $\|\mathcal{E}_k\|_{k,\kappa_n} \le K_{\mathcal{E}} h_k^{2-2\epsilon}$. In particular $\|\mathcal{E}_k\|_{k,\kappa_n} \le g_k^{\beta}$ for every $\beta < 2-2\epsilon$ and g_* small;

(iii) if $k < K$, then $\mathcal{E}_{k+1} = \mathcal{L}\mathcal{E}_k + \mathcal{E}_k^{\mathrm{mix}} + \mathcal{E}_k^{*}$. Here $\mathcal{L}$ is linear with $\|\mathcal{L}\mathcal{E}\| \le (2\cdot 3^d K_0 L^{-\epsilon}(1 + o(1)) + L^{-1})\|\mathcal{E}\|$; $\mathcal{E}_k^{\mathrm{mix}}$ is non-linear in $\mathcal{E}_k$, with $\|\mathcal{E}_k^{\mathrm{mix}}\| \le o(1)\|\mathcal{E}_k\|$; and $\|\mathcal{E}_k^{*}\| \le B h_{k+1}^{2-2\epsilon}$;

(iv) $|\varepsilon_k| \le C g_k^{\beta'}$, and $\mathcal{Z}_k$ is the explicit product of the normalizations of the steps; if $k < K$, then ε_{k+1} is given by (50.2).

The activities are evaluated at $U_k(V)$. This is legitimate: $U_k(V) \in \mathcal{U}_k(X)$ for every X, with real part $U_k(V)$ and complex part 0, since its curvature and current are at most $Ch_k \le h_k^{1-\epsilon/2}$ (Theorem 30.3, Proposition 31.5).

Proof. Induction on k, for arbitrary E_0; E_0 is fixed afterwards by Proposition 50.3, which uses only (i)–(iii), so there is no circularity. For $k = 0$, (i)–(iv) hold with $\mathcal{E}_0 = 0$. The step $0 \to 1$ is Proposition 50.2. For $1 \le k < K$, the step $k \to k+1$ is Theorem 49.8: the torus hypotheses hold by Lemma 50.1, and its other hypotheses by the induction hypothesis. The factor $\mathcal{Z}_k e^{-\varepsilon_k |\mathbb{T}^{(k)}|}$ is a constant and is absorbed into the constant of Theorem 49.8, since (31.2) carries none. The density at step k is of the form (31.2) required by Theorem 34.8, with $\mathfrak{E}_k = \mathcal{E}_k \circ U_k$. Item (iv) is (50.2) and Proposition 50.3. The bound $K_{\mathcal{E}} h_k^{2-2\epsilon} \le g_k^{\beta}$ holds since $h_k = p_k g_k$ and $g^{\beta''} p(g) \to 0$ for every $\beta'' > 0$. □

Corollary 50.5 (Small-field stability)**.** *The partition function of the modified model, $\mathcal{Z}^{\mathrm{sf}} = \int \rho_K(V)\, dV$, satisfies $\mathcal{Z}^{\mathrm{sf}} = \mathcal{Z}_K I_K$, with $\mathcal{Z}_K$ the explicit product of normalizations and*

$$e^{-c|\mathbb{T}^{(K)}|} \le I_K \le e^{c|\mathbb{T}^{(K)}|},$$

where c depends on g_* and the fixed parameters, but not on N or K. Since $|\mathbb{T}^{(K)}| = n_K^3$ with $\ell g^2/g_*^2 \le n_K < L\ell g^2/g_*^2$, the volume is bounded above and below in terms of the physical volume, and this is uniform in the lattice spacing.

Proof. By Theorem 50.4 with $\varepsilon_K = 0$,

$$I_K = \int_{\mathfrak{U}_K(p_K g_K)} \exp\Big(-\frac{1}{g_K^2} S(U_K(V)) + \sum_X \mathcal{E}_K(X, U_K(V))\Big) dV.$$

- *Activities.* $|\sum_X \mathcal{E}_K(X,U)| \le \sum_{\square} \sum_{X \supset \square} \|\mathcal{E}_K(X)\|_K \le M^{-3} |\mathbb{T}^{(K)}| K_0 \|\mathcal{E}_K\|_{K,\kappa_n}$, by (49.1) and $\kappa_n \ge \kappa_0$. This is $|\mathbb{T}^{(K)}|\, o(1)$.
- *Action.* $S \ge 0$. The curvature of $U_K(V)$ is at most $C p_K g_K$ on $\mathfrak{U}_K(p_K g_K)$, so $g_K^{-2} S(U_K(V)) \le C|\mathbb{T}^{(K)}| p_K^2$.
- *Volume.* The Haar measure of $\mathfrak{U}_K(p_K g_K)$ is at most 1. It contains the configurations $V = e^{iB}$ with $|B|_\infty \le c h_K$, whose Haar measure is at least $(c' h_K)^{\dim \mathfrak{g} \cdot |\mathcal{B}(\mathbb{T}^{(K)})|}$.

Together, $|\log I_K| \le C|\mathbb{T}^{(K)}|(p_K^2 + \log h_K^{-1} + 1)$. This is $c|\mathbb{T}^{(K)}|$, with c depending on g_* and L through $p_K \le p(L^{-1/2} g_*)$ and $h_K \ge h_*$. □

Remark 50.6 (On the corollary and on a Gaussian reference). (a) A comparison of the action $S(U_K(V))$ with a quadratic form on the small-field set is not available, and it is not used. Small-field configurations on a torus include non-trivial holonomy sectors, and nearly commuting constant modes make the difference of order $|\mathbb{T}|p_K^2$. The crude bounds suffice, because the final torus is fixed.

(b) A Gaussian reference in Lie-algebra variables has non-compact zero modes: the gauge directions of the tree bonds, the residual coarse gauge, and the harmonic modes. And $\mathcal{N}_k$ contains the Haar density $\sigma(0)^{|\mathfrak{F}|}$. A reference can be defined as the renormalization group of the quadratic density with the same conventions:

- the measure $\sigma(0)\,dA$;
- axial fields;
- the linearized constraint;
- χ_ξ with $C_k(1)$;
- a compact final domain.

Its normalizations are expected to agree with $\mathcal{Z}_k$ term by term; this is not checked here, and it is not used. Relating it to the free Gaussian would need $\sum_k |\mathfrak{F}_k| e^{-p_{0,k}^2/2} \le C|\mathbb{T}^{(K)}|$, a condition on the growth of $p_{0,k}$. The explicit form $\mathcal{Z}_K I_K$ avoids this.

(c) *The modified model.* Since T preserves integrals (Section 31), $\mathcal{Z}^{\mathrm{sf}}$ is the integral of ρ_0 times the product of the cutoffs of all steps, pulled back to the configurations of step 0. This defines the modified model. For $k \ge 1$ the density ρ_k vanishes outside $\mathfrak{U}_k(p_k g_k)$, which lies in $\operatorname{dom}\mathfrak{a}'$ as soon as $dL^2 p_k g_k \le c_G$ (Section 31). This holds for g_* small: $g \mapsto p(g)g$ is increasing and $g_k \le g_*$, so $p_k g_k \le p(g_*)g_*$, which tends to 0 as $g_* \to 0$, by $g^\beta p(g) \to 0$ with $\beta = 1$. So what is discarded at step k is $T((1-\chi^{\mathrm{w}}\chi_\xi)\rho_k)$, together with the region of V' outside the small-field set. At $k = 0$ the density ρ_0 is positive on all configurations, also outside $\operatorname{dom}\mathfrak{a}'$, where T is not defined; $T_{\mathrm{sf}}(\rho_0)$ is defined, since χ^{w} vanishes outside $\operatorname{dom}\mathfrak{a}'$ (Proposition 32.5, which holds at $k = 0$ by Remark 34.9). What is discarded at the first step is the measure $(1-\chi^{\mathrm{w}}\chi_\xi)\rho_0\,dV$ on the configurations of step 0, together with the region of V' outside the small-field set. In both cases this is more than $\mathcal{R}_{\mathrm{LF}}$.

(d) *Not treated.* The large-field stage, which starts from the discarded parts of every step, and the removal of the small-field modification.

Conclusion

51 Outlook

What has been done.

- *Part I* treats the trivial background $U = 1$ on the torus.
- *Part II* extends the results to backgrounds close to 1 in a global gauge, by perturbation.
- *Part III* replaces the global hypothesis by local regularity: on every cube of size of order M there is a gauge in which the background is close to 1. It provides:
  - Dirichlet operators on unions of blocks;

  - a local proof of coercivity by IMS localization;
  - locality of all operators in the domain and in the background;
  - Bałaban's random-walk expansion for the scalar Green function; for the gauge-field Green function it is completed in factorized form in Part IX (Section 41).

- *Part IV* constructs Bałaban's nonlinear averaging, the axial gauge and the gauge-adapted averaging, and proves the axial-to-Landau chart. The linearized averaging is exactly covariant ($E = 0$), and Parts II and III apply to it: the real statements under properties (i)–(iv) of Remark 19.12, the analytic ones with property (v) in addition.

- *Part V* solves the variational problem relative to a regular reference configuration. It gives existence, uniqueness and minimality of the background field, second-order and Hölder bounds, the Euler–Lagrange equation, analyticity in the coarse field, and exponential decay of the derivative.

- *Part VI* carries out the induction over the number of steps. This proves Bałaban's variational theorem on the torus with constants independent of k (Theorem 30.3). As a consequence, every real configuration with small curvature and current is regular (Corollary 30.4). So the hypothesis of Part III is a consequence of gauge-invariant conditions.

- *Part VII* writes one small-field renormalization group step exactly as a Gaussian integral around the background field (Theorem 34.8). It proves the linear input: coercivity and decay of the fluctuation covariance, analyticity and tree decay of the coarse effective action, and local determinant densities.

- *Part VIII* localizes the dependence on the background, including large changes, by Dirichlet approximants and telescoping:

  - for the fluctuation covariance and the coarse Hessian (Theorem 37.7, Corollary 37.8);
  - for the determinant, which becomes a constant plus normalized one-loop activities of second order in the curvature (Corollary 38.8);
  - for the interaction and the composed activities (Theorem 39.4, Corollary 39.5). This uses interior pointwise bounds with derivatives for the Dirichlet operators (Theorem 35.2). They are proved by absorption at all scales inside a single cube, for a local elliptic system in which two scalar potentials of the slice projection are additional unknowns (Section 36).

  This localization is of radius type. On the main line it is superseded by Parts IX and X. From Part VIII the main line keeps the interior bounds, the Dirichlet Hessians and local fluctuation operators, the local determinant densities, the graded domains and the quadratic bound for normalized activities (Lemma 38.5).

- *Part IX* gives the volume-type localization by random walks with weakening parameters: for the covariance, its square root, the coarse Hessian and the determinant in L^2 norms (Section 41); pointwise $C^{1,\beta}$ bounds for the weakened propagators, through interior L^p estimates and tier norms (Sections 43, 44); and the weakened background map (Theorem 45.5).

- *Part X* carries out the small-field renormalization group in Dimock's scheme:

  - complex references, including Bałaban's averaged backgrounds (Proposition 46.4, Proposition 46.7);

– the weakened step quantities and the localization lemma (Section 47);

– the white-noise cutoff and the Mayer and polymer expansion; by parity there is no term of order g (Section 48);

– the activity norm on the graded domain $\mathrm{D3}'''$, the linear part and one step of the flow (Theorem 49.8);

– the first step, the vacuum energy and the small-field flow (Theorem 50.4, Corollary 50.5; Theorem K).

How the induction closes. The reference at step k is the solution of the $(k-1)$-step problem for the trivial extension of V, and it has exactly the prescribed averages. Theorem 25.1 then solves the problem at step k. Its regularity bound is not good enough to continue, because the change of units from δ_{k-1} to δ_k costs a power of L. The local improvement (Lemma 29.2) uses only Theorem 25.1 itself, applied with a different reference. This reference is spliced to the trivial configuration near a given point and coincides with the solution far away. Near the point, the solution is then **H** applied to the coarse field in an axial gauge, up to an error that is exponentially small in the splicing radius times the a priori bound, plus a quadratic error. The coarse field in the axial gauge is controlled by the curvature of V. Two features make this work on the torus:

- the uniformity of Theorem 25.1 in the reference;
- the polynomial loss in the patching of local gauges (Lemma 28.3), which is beaten by the exponential decay of **H** and **C**.

Why the global hypothesis was not enough. Applied up to a real gauge transformation, hypothesis (B) is a condition on the real gauge orbit, so it does not break gauge invariance. But it is not local, and it is restrictive. By the isoperimetric argument of Section 12, every Wilson loop obeys a perimeter bound. Field strengths of size ε that are coherent over many blocks, and backgrounds with non-zero flux through a 2-cycle, are excluded. Moreover the renormalization group needs small-field conditions that factorize over local regions, and operators whose dependence on distant changes of the background is weak. Part III provides both: regularity is a local condition, and Theorem 19.10 gives the locality. Part VI shows that the local condition is implied by Bałaban's gauge-invariant small-field conditions.

A naive comparison of $G(U)$ and $G(U')$ through the resolvent identity would produce the difference $D_U - D_{U'}$, whose norm can be of order η^{-1} where the backgrounds differ. The proof of Theorem 19.10 avoids this. It compares each background with a Dirichlet operator on the region where they agree. Changing the domain at a fixed background produces cut couplings and differences of domain-dependent ingredients, including mass profiles and non-local terms, which are supported near the cut or exponentially small away from it. Derivative factors may occur at one or both ends of these terms; the estimates use the corresponding bounded undifferentiated or differentiated propagators.

Relation to Bałaban's construction. Bałaban works with Dirichlet operators on domains [7] and with local regularity on cubes [7, (3.35)–(3.36)], and builds the global operators by random walks. His positivity comes from [7, Theorem 3.11] and local axial-gauge estimates [4, Lemma 2.4]. Our route differs in several technical points:

- coercivity comes from IMS localization together with the coercivity of Part I;
- the local gauge-field propagators carry a mass term near the boundary of their domain;

- the normalization of the gauge transformation from the axial to the Landau gauge is built into a recursive correspondence map (Theorem 22.4). Bałaban's nonlinear averages of gauge transformations then appear only through the identity for pure gauges;
- the second-order regularity of the minimizer is derived from the Euler–Lagrange equation, which follows from gauge invariance and a transversality argument. We do not use the alternative equation [8, (143)];
- the local improvement of the regularity constants is obtained on the torus, by changing the reference configuration, instead of on nested domains (Lemma 29.2). The local regularity of general small-field configurations is derived from that of the background fields, through the chart (Corollary 30.4).

The random-walk expansion is used only for the explicit localized representation, not for the estimates.

Next steps.

(1) *Verification.* The small-field flow is written out, and its parts have been reviewed by independent AI agents, but the note has not been checked by a human expert. The companion documents (the blueprint, an audit and a verification report) record the plan, the reviews and the points to check first.

(2) *Large fields and nested domains,* where Dirichlet operators on irregular domains are essential. This is the analogue of [16, 17]. It must control what the small-field model of Theorem K discards at every step, and so remove the modification. Since the covariance C_k depends on the background everywhere, the white-noise cutoff of Section 48 will need a version that depends on regions. Part III treats arbitrary unions of blocks, so this is the natural setting for them. The splicing of Part VI is a torus device. On nested domains the reference no longer has exactly the prescribed averages, and the analysis on domains of Part III must be extended to the nonlinear problem.

(3) *Gauges on cubes of size up to* $c\,\varepsilon_1^{-1}$ with bounds linear in the size (Remark 30.6).

(4) *Gauge groups with abelian factors.* The irrelevance of normalized activities uses semisimplicity (Lemma 38.5). For abelian factors, gauge-invariant terms linear in the curvature can occur, and they must be treated separately.

A Decay of the lattice Green function and its derivatives

In this appendix $d \geq 3$ and g_1 is the Green function of $-\Delta$ on $\mathbb{Z}^d$, as in Section 6. For $\mu = 1, \dots, d$ let $D_\mu f(x) = f(x + e_\mu) - f(x)$ and $D_\mu^* f(x) = f(x - e_\mu) - f(x)$. A *word* ∇^β of length m is a composition of m operators from $\{D_\mu, D_\mu^*\}$.

Theorem A.1. *For every word ∇^β of length $m \leq 3$ there is C, depending only on d and β, such that*

$$|\nabla^\beta g_1(x)| \leq C(1 + |x|)^{2-d-m}, \qquad x \in \mathbb{Z}^d.$$

Write $\mathbb{T}^d = \mathbb{R}^d / 2\pi\mathbb{Z}^d$, identified with $[-\pi, \pi]^d$. We have

$$\nabla^\beta g_1(x) = \frac{1}{(2\pi)^d} \int_{\mathbb{T}^d} e^{\mathrm{i}p\cdot x}\, \psi(p)\, dp, \qquad \psi(p) = \frac{\sigma_\beta(p)}{\hat{\Delta}_1(p)}, \qquad \hat{\Delta}_1(p) = \sum_\mu 4 \sin^2 \frac{p_\mu}{2},$$

where σ_β is the product of the symbols $e^{\mathrm{i}p_\mu} - 1$ (for D_μ) and $e^{-\mathrm{i}p_\mu} - 1$ (for D_μ^*) of the letters of β. The theorem follows from the next two lemmas, applied with $s = m - 2$ and $K = d + m$.

Lemma A.2 (Symbol estimates)**.** *For every multi-index γ there is C_γ such that $|\partial_p^\gamma \psi(p)| \le C_\gamma |p|^{m-2-|\gamma|}$ for $p \in [-\pi,\pi]^d \setminus \{0\}$.*

Proof. Let $f = \hat{\Delta}_1$. By Lemma 4.1(a) with $\eta = 1$, $f(p) \ge \frac{4}{\pi^2}|p|^2$. Moreover $|\partial_\mu f| = 2|\sin p_\mu| \le 2|p|$, and all higher derivatives of f are bounded. Since $|p| \le \pi\sqrt{d}$, this gives $|\partial^\gamma f| \le C|p|^{2-|\gamma|}$ for all γ. By the Faà di Bruno formula, $\partial^\gamma(1/f)$ is a linear combination of terms $f^{-1-j}\prod_{i=1}^{j}\partial^{\gamma_i} f$ with $\gamma_1 + \cdots + \gamma_j = \gamma$. Each such term is bounded by $C|p|^{-2-2j}\prod_i |p|^{2-|\gamma_i|} = C|p|^{-2-|\gamma|}$. Each letter of σ_β is bounded by $|p_\mu| \le |p|$, and its derivatives are bounded. So by the Leibniz rule $|\partial^\gamma \sigma_\beta| \le C|p|^{\max(m-|\gamma|,0)} \le C'|p|^{m-|\gamma|}$. One more application of the Leibniz rule concludes the proof. □

Lemma A.3 (Decay of Fourier coefficients)**.** *Let ψ be smooth on $\mathbb{T}^d \setminus \{0\}$ and satisfy $|\partial^\gamma \psi(p)| \le C|p|^{s-|\gamma|}$ on $[-\pi,\pi]^d \setminus \{0\}$ for all $|\gamma| \le K$, where $s > -d$ and $K > d + s$. Then*

$$\Big| \int_{\mathbb{T}^d} e^{\mathrm{i}p\cdot x}\psi(p)\,dp \Big| \le C'(1+|x|)^{-d-s}, \qquad x \in \mathbb{Z}^d.$$

Proof. For $|x| \le 1$ the integral is bounded by $\int|\psi| < \infty$, because $s > -d$. Let $R = |x| > 1$, and choose j with $|x_j| \ge R/\sqrt{d}$. Fix $\zeta \in C_c^\infty(\mathbb{R}^d)$ with $\zeta = 1$ on $\{|p| \le 1\}$ and $\zeta = 0$ on $\{|p| \ge 2\}$. Since $R > 1$, $\zeta(R\,\cdot)$ is supported in $\{|p| \le 2/R\} \subset (-\pi,\pi)^d$, so it defines a smooth function on the torus. Split $\psi = \psi_1 + \psi_2$ with $\psi_1(p) = \zeta(Rp)\psi(p)$. Then

$$\Big| \int e^{\mathrm{i}p\cdot x}\psi_1 \Big| \le C \int_{|p|\le 2/R} |p|^s\,dp = C'' R^{-d-s}.$$

The function ψ_2 vanishes near 0, so it is smooth on the torus. Integrating by parts K times in p_j, without boundary terms by periodicity, gives

$$\int e^{\mathrm{i}p\cdot x}\psi_2 = \Big(\frac{\mathrm{i}}{x_j}\Big)^K \int e^{\mathrm{i}p\cdot x}\,\partial_j^K \psi_2.$$

By the Leibniz rule, $\partial_j^K \psi_2$ is a sum of terms $\partial_j^{K-i}\psi \cdot \partial_j^i[1-\zeta(R\,\cdot)]$. For $i = 0$ the term is bounded by $C|p|^{s-K}$ and vanishes for $|p| < 1/R$. For $i \ge 1$ it vanishes unless $1/R \le |p| \le 2/R$, where $R^i \le 2^i|p|^{-i}$; so it is again bounded by $C|p|^{s-K}$. Hence

$$\Big| \int e^{\mathrm{i}p\cdot x}\psi_2 \Big| \le C|x_j|^{-K} \int_{1/R\le|p|\le\pi\sqrt{d}} |p|^{s-K}dp \le C|x_j|^{-K}R^{K-s-d} \le C'R^{-d-s},$$

since $K - s > d$. □

Proof of Theorem A.1. By Lemma A.2 the hypotheses of Lemma A.3 hold with $s = m-2 > -d$ and every K. Take $K = d + m$. □

References

[1] M. Giaquinta, *Multiple Integrals in the Calculus of Variations and Nonlinear Elliptic Systems*, Annals of Mathematics Studies **105**, Princeton University Press, 1983.

[2] T. Bałaban, *Regularity and decay of lattice Green's functions*, Commun. Math. Phys. **89** (1983), 571–597.

[3] T. Bałaban, *Propagators and renormalization transformations for lattice gauge theories. I*, Commun. Math. Phys. **95** (1984), 17–40.

[4] T. Bałaban, *Propagators and renormalization transformations for lattice gauge theories. II*, Commun. Math. Phys. **96** (1984), 223–250.

[5] T. Bałaban, *Averaging operations for lattice gauge theories*, Commun. Math. Phys. **98** (1985), 17–51.

[6] T. Bałaban, *Spaces of regular gauge field configurations on a lattice and gauge fixing conditions*, Commun. Math. Phys. **99** (1985), 75–102.

[7] T. Bałaban, *Propagators for lattice gauge theories in a background field*, Commun. Math. Phys. **99** (1985), 389–434.

[8] T. Bałaban, *The variational problem and background fields in renormalization group method for lattice gauge theories*, Commun. Math. Phys. **102** (1985), 277–309.

[9] T. Bałaban, *Ultraviolet stability of three-dimensional lattice pure gauge field theories*, Commun. Math. Phys. **102** (1985), 255–275.

[10] T. Bałaban, A. Jaffe, *Constructive gauge theory*, in: Fundamental Problems of Gauge Field Theory (Erice 1985), G. Velo, A. S. Wightman (eds.), Plenum, New York, 1986, 207–263.

[11] J. M. Combes, L. Thomas, *Asymptotic behaviour of eigenfunctions for multiparticle Schrödinger operators*, Commun. Math. Phys. **34** (1973), 251–270.

[12] E. M. Stein, *Singular Integrals and Differentiability Properties of Functions*, Princeton University Press, 1970.

[13] E. M. Stein, *Harmonic Analysis: Real-Variable Methods, Orthogonality, and Oscillatory Integrals*, Princeton University Press, 1993.

[14] Companion document `RG-blueprint.pdf`, *Small-field renormalization of YM_3: a blueprint*, version of 29 October 2026 (repository commit `c0df541`), distributed with this note. It records the plan, the domain conventions and the parameter choices that the text refers to.

[15] J. Dimock, *The renormalization group according to Bałaban. I. Small fields*, Rev. Math. Phys. **25** (2013), 1330010. Equation numbers refer to the arXiv version, arXiv:1108.1335v2. Up to (115) they agree with those of the journal version; (116) is the number of Definition 3, which is unnumbered in the journal version, and from (117) on they exceed those of the journal version by one.

[16] J. Dimock, *The renormalization group according to Bałaban. II. Large fields*, J. Math. Phys. **54** (2013), 092301.

[17] J. Dimock, *The renormalization group according to Bałaban. III. Convergence*, Ann. Henri Poincaré **15** (2014), 2133–2175.

[18] R. Kotecký and D. Preiss, Cluster expansion for abstract polymer models, *Comm. Math. Phys.* **103** (1986), 491–498.

[19] W. Dybalski, A. Stottmeister, Y. Tanimoto, *Lattice Green functions for pedestrians: exponential decay*, Rev. Math. Phys. **36** (2024), 2430005.